

HUAWEI NetEngine 8000 M14, M8 and M4 Series V800R023C10SPC500

Hardware Description

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1 Hardware Description

1.1 Chassis

1.1.1 NetEngine 8000 M4

Overview

Table 1-1 Basic information about the NetEngine 8000 M4

Description	Part Number	Model	First supported version
NetEngine 8000 M4 Integrated DC Chassis Components(Including DC Power,Fan Box)	02355FTE	CR8PM4BASDC2	V800R022C00SPC 600
NetEngine 8000 M4 Integrated AC Chassis Components(Including AC Power,Fan Box)	02355FTG	CR8PM4BASAC2	V800R022C00SPC 600
NetEngine 8000 M4 Integrated DC Chassis Components(Including DC Power,Fan Box)	02355JFR	CR8PM4BASDC6	V800R022C10SPC 500

Description	Part Number	Model	First supported version
NetEngine 8000 M4 Integrated AC Chassis Components(Including AC Power,Fan Box)	02355JFT	CR8PM4BASAC6	V800R022C10SPC 500

Appearance

Figure 1-1 Appearance of the NetEngine 8000 M4 (DC)

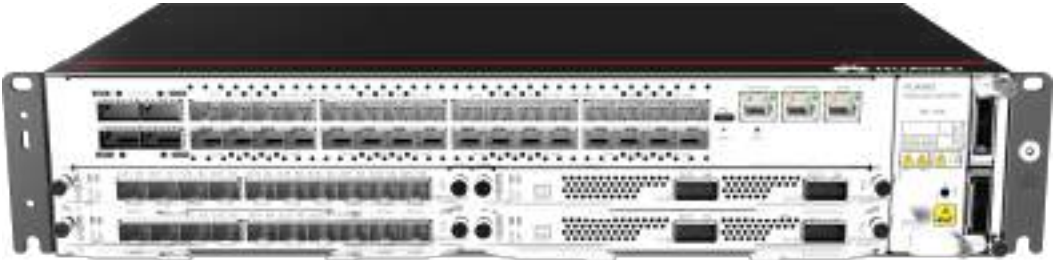


Figure 1-2 Appearance of the NetEngine 8000 M4 (AC)



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter>

Panel

Table 1-2 Indicators on the NetEngine 8000 M4

Silkscreen	Name	Color	Status	Description
STAT	Status indicator	Green	Steady on	The device is working properly.
		Green	Blinking	The board is being registered.
		Red	Steady on	An alarm is generated.
		Orange	Blinking	The password and configuration on the card are being cleared.
ETH/OAM	Management Ethernet port status indicator	Green	Steady on	The link is up.
		Yellow	Blinking	Data is being transmitted.
		-	Steady off	The link is down.
LINK (0-35)	Optical port status indicator	Green	Steady on	The port connection is normal.
		Green	Blinking	Data is being transmitted.
		-	Steady off	The physical connection of the port is abnormal.

Table 1-3 Buttons on the NetEngine 8000 M4

Silkscreen	Name	Description
RST	Reset button	You can press the RST button to reset the NE or restore the factory settings of the NE. The detailed operations are as follows: To reset the NE, press and release the RST button. To restore the factory settings, press and hold the RST button as follows. This operation will clear the NE database and system parameter area configurations, and the cleared configurations cannot be restored. Therefore, perform the following steps with caution. 1. Press the RST button to restart the NE. 2. When the STAT indicator blinks green (frequency: about 250 ms, time window: about 15 seconds), press and hold the RST button. 3. Wait for no more than 2 minutes. When the STAT indicator blinks orange (time window: 5 seconds), release the RST button immediately. NOTE If you click the RST button to clear the configuration file, the original configuration file is cleared. You are advised to periodically back up the configuration file.

Table 1-4 Ports on the NetEngine 8000 M4

Port	Connector Type	Description	Available Components
0 to 3	QSFP28	Interface for inputting and outputting 100GE/50GE/40GE optical signals Description: 1. The 100GE ports on the M4 panel support the 4*10GE/4*25GE breakout function, but do not have breakout indicators. 2. When a 100G 80 km commercial optical module is inserted into ports 0 and 1, the high temperature specification is 50°C. When it is inserted into ports 2 and 3, the high temperature specification is 45°C. Inserting such a module into ports 0 and 1 is recommended. 3. 100G 40 km/10 km commercial optical modules support the highest temperature of 55°C. 4. 100G 10 km industrial optical modules support the highest temperature of 60°C.	100Gbps QSFP28 Optical Module 40Gbps QSFP+ Optical Module 50Gbps QSFP28 BIDI Optical Module 50Gbps QSFP28 Optical Module Optical Amplifier Module

Port	Connector Type	Description	Available Components
4-19	SFP28	Interface for inputting and outputting 25GE/10GE/GE optical signals	155Mbps eSFP Optical Module 155Mbps eSFP BIDI Optical Module 1.25/9.953/10.31 25Gbps SFP+ Optical Module 1.25Gbps eSFP BIDI Optical Module 1.25Gbps eSFP CWDM Optical Module 1.25Gbps eSFP Optical Module 10Gbps SFP+ BIDI Optical Module 10Gbps SFP+ CWDM Optical Module 10Gbps SFP+ Optical Module 125M~2.67Gbps eSFP DWDM Optical Module 1Gbps SyncE and 1588V2 Electrical Module 1Gbps Electrical Module 25Gbps SFP28 Optical Module 25Gbps SFP28 BIDI Optical Module

Port	Connector Type	Description	Available Components
20–35	SFP28	Interface for inputting and outputting 25GE/10GE optical signals	10Gbps SFP+ BIDI Optical Module 10Gbps SFP+ CWDM Optical Module 10Gbps SFP+ Optical Module 25Gbps SFP28 Optical Module 25Gbps SFP28 BIDI Optical Module
ETH/OAM	RJ45	10M/100M interface that can function as an NM interface or console interface NOTE If a service port can be connected to the NMS, it is recommended that the service port be used as the network management port.	ETH/OAM management cable
CLK/TOD/RS-485	RJ45	Clock/Time/RS-485 input/output port RS-485 is a reserved port.	One-to-two clock cable + clock cable (external clock mode-finished cable), applicable to the CLK mode; One-to-two clock cable + clock cable (external clock/external time mode-made on site), applicable to the CLK or TOD mode; One-to-two clock cable + Ethernet cable, applicable to the TOD mode.

Port	Connector Type	Description	Available Components
ALMI	RJ45	Alarm input interface (four inputs)	Alarm input cable
USB	USB TYPE C	USB port (reserved)	-

Components

Figure 1-3 Components of the NetEngine 8000 M4 (DC)



1. Grounding terminal (in the rear of the chassis)	2. PIC	3. Air intake vent	4. IPU	5. Rack-mounting ear
6. Chassis	7. DC Power module	8. Fan module	9. Grounding terminal (on the mounting ear)	10. ESD jack

Figure 1-4 Components of the NetEngine 8000 M4 (AC)



1. Grounding terminal (in the rear of the chassis)	2. PIC	3. Air intake vent	4. IPU	5. AC Power module
6. Chassis	7. Rack-mounting ear	8. Fan module	9. Grounding terminal (on the mounting ear)	10. ESD jack

Slot Layout

Figure 1-5 Slot Layout of the NetEngine 8000 M4

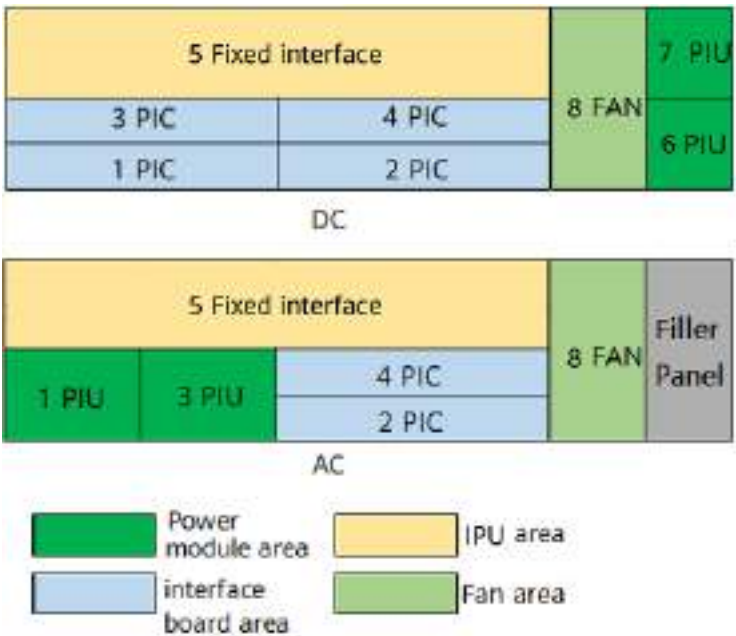


Table 1-5 Slots on the NetEngine 8000 M4

Slot Type	Slot ID	Slot Direction	Remarks
PIC	DC: 1–4 AC: 2 and 4	The top of the card faces upwards.	-
Fixed interface	5	The top of the card faces upwards.	-

Slot Type	Slot ID	Slot Direction	Remarks
PIU	DC: 6 and 7 AC: 1 and 3	-	DC and AC power modules cannot be used together. When AC power modules are installed, insert filler panels into the DC power module slots.
FAN	8	-	-

Labels

Figure 1-6 Label position

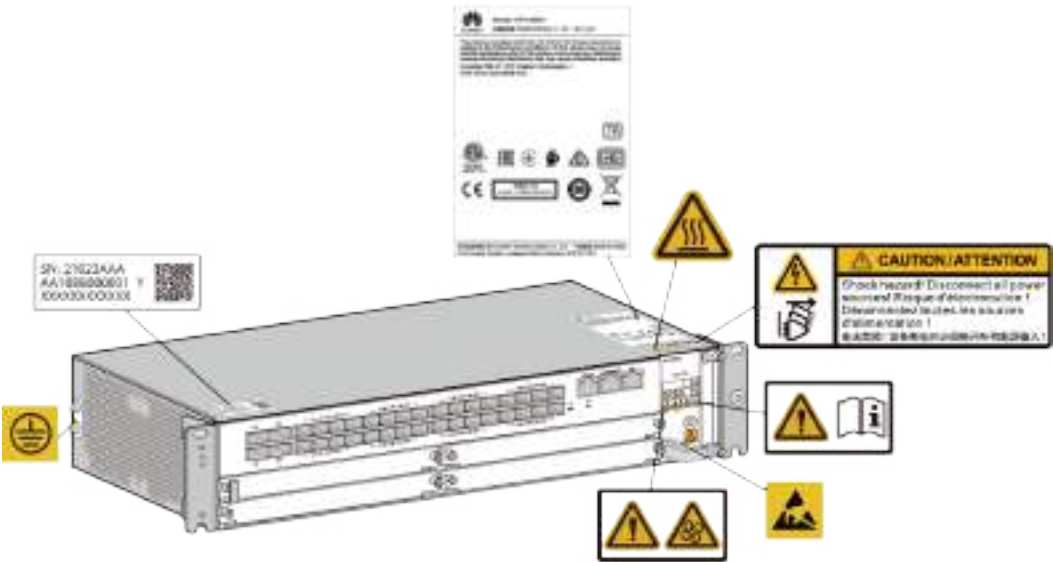


Table 1-6 Label description

Figure	Label Name	Description
	Chassis bar code label	The bar code will be retrieved by the device as the equipment serial number (ESN).
	ESD protection label	The label suggests the electrostatic-sensitive equipment.

Figure	Label Name	Description
	Chassis grounding label	The label indicates the position of the general grounding terminal. When the equipment is installed in a cabinet rather than a 19-inch cabinet, it is recommended that you use the grounding terminal specified in this label.
	Fan warning label	The label suggests that do not touch the fan leaves when the fan is rotating.
	Operation warning label	The label indicates the precaution that should be taken for operations on the PIU board.
	Heat hazard label	This label suggests that the surface temperature is high and it may cause body injury.
	Multi-power input caution	This device has more than one power input. Disconnect all power inputs to power off this device.
	Product nameplate label	The label suggests the product name, certification.

Power Supply System

The power supply system supports 1+1 backup mode. If one power module fails or is removed, the other power module can still supply adequate power to the device.

AC power modules (PIUs) are installed in the two slots in the lower left part of a chassis and DC PIUs are installed in the two slots on the right side of a chassis to supply power to IPU, interface boards, and fan modules.

The following protection functions are provided for the power modules:

- Protection against output overcurrent
- Protection against output overvoltage
- Protection against input undervoltage
- Protection against overtemperature
- Protection against short circuit
- Alarm generation

Figure 1-7 shows the DC power supply architecture. Two power modules provide two -48 V power inputs, which are combined on the backplane and then supplied to the fan module, IPUs, and interface boards. The IPU supplies 3.3 V power to the fan module, power modules, and interface boards.

Figure 1-7 Diagram of the DC power supply architecture

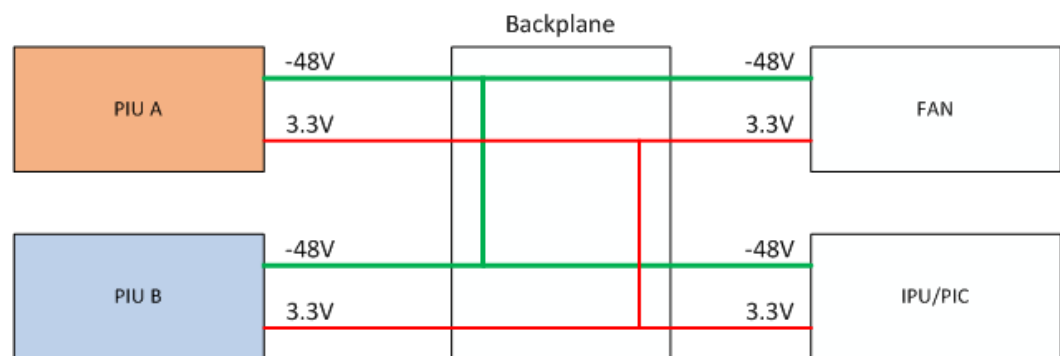
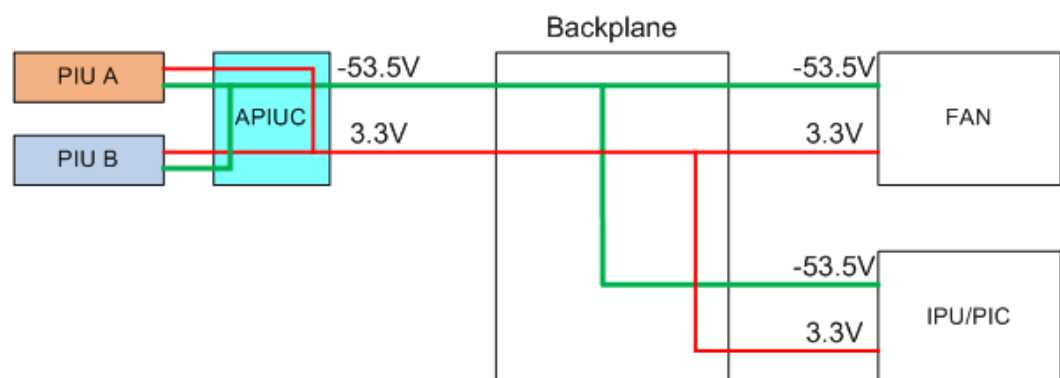


Figure 1-8 shows the AC power supply architecture. Two power modules provide two -53.5 V power inputs, which are combined on the backplane and then supplied to the fan module, IPUs, and interface boards. The IPU supplies 3.3 V power to the fan module, power modules, and interface boards.

Figure 1-8 Diagram of the AC power supply architecture



Heat Dissipation System

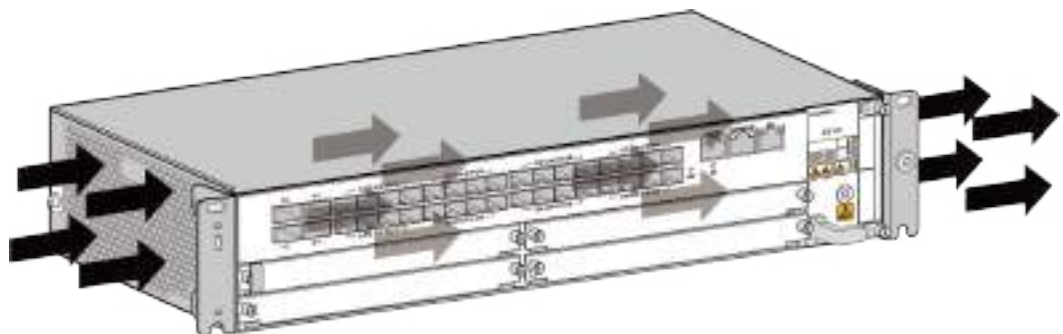
The fans exhaust air from the chassis to dissipate heat. The hot air inside the chassis is sucked out of the subrack to form a ventilation channel from the air intake vent on the left side to the air exhaust vent on the right side. The fan tray assembly consists of two independent fans to dissipate heat for the device and is hot swappable.

NOTICE

- Ensure that heat dissipation holes for the air intake and exhaust vents are clean and not blocked.
- The empty slot should be covered with a filler panel to ensure good electromagnetic compatibility, dustproof, and heat dissipation.
- Keep any objects or obstructions at least 75 mm away from the air intake and exhaust vents to facilitate heat dissipation.
- When installing devices, ensure that the total heat consumption of all devices in the cabinet is less than or equal to the heat dissipation capability of the cabinet.
- The equipment must be stacked in the cabinet or installed at an interval of 2U or more to prevent the return air from affecting heat dissipation.
- When the chassis is installed together with other equipment in the same cabinet, do not install the chassis near the air exhaust vent of other equipment.
- You are required to replace the fan board as soon as possible; otherwise, the active/standby switchover, system reset, or board burning may occur due to high temperature. Exercise caution when performing this operation. The fan board needs to be replaced within one minute at the normal temperature (25°C).

Figure 1-9 shows the air flow in the device.

Figure 1-9 Air flow in the device



The Interface Bandwidth Mode

The NetEngine 8000 M4 supports four groups (groups 1, 2, 3, and 4), as shown in Figure 1-10. Each group consists of one main port and four sub-ports.

Group	Main Port	Sub-port
Group1	0	4, 6, 8, 10
Group2	1	5, 7, 9, 11
Group3	2	12, 14, 16, 18
Group4	3	13, 15, 17, 19

Each group can switch between their six modes independently. In each mode, the capabilities of the four sub-ports vary according to the port capability of the main port, as shown in the following table.

Mode	Main Port	Sub-port
Mode 1	100GE (50GE/40GE auto-sensing)	4 x GE
Mode 2	4 x 25GE	4 x GE
Mode 3	50GE	4 x 10GE/GE ports. Note that 10GE ports must be switched between the LAN and WAN modes at the same time.
Mode 4	40GE	4 x 10GE/GE ports. Note that 10GE ports must be switched between the LAN and WAN modes at the same time.
Mode 5	4 x 10GE	4 x 10GE/GE ports. Note that 10GE ports must be switched between the LAN and WAN modes at the same time.
Mode 6	Not used	4 x 25GE/10GE/GE ports. Note that 10GE ports must be switched between the LAN and WAN modes at the same time.

Figure 1-10 NetEngine 8000 M4's interface bandwidth modes



Technical Specifications

Table 1-7 Technical specifications of the CR8PM4BASDC2

Item	Specification
Cabinet installation standards	ETSI (21-inch); IEC (19-inch)
Dimensions without packaging (H x W x D) [mm(in.)]	88.9 mm x 442 mm x 220 mm (3.50 in. x 17.40 in. x 8.66 in.)
Dimensions with packaging (H x W x D) [mm(in.)]	335 mm x 585 mm x 390 mm (13.19 in. x 23.03 in. x 15.35 in.)
Chassis height [U]	2U
Weight without packaging [kg(lb)]	7.2 kg (15.87 lb)
Weight with packaging [kg(lb)]	9.9 kg (21.83 lb)
Weight without packaging (full configuration) [kg(lb)]	10.4 kg
Weight with packaging (full configuration) [kg(lb)]	13.1 kg
Typical power consumption (with configuration) [W]	283.4 W
Typical heat dissipation (with configuration) [BTU/hour]	919.47 BTU/hour
MTBF [year]	33.40 year
MTTR [hour]	0.5 hour
Availability	0.999998291
CPU	16-core, 2.0 GHz
SDRAM	16GB
Flash memory	64MB
Storage	16 GB
Power supply mode	DC
Rated input voltage [V]	-48 V/-60 V
Input voltage range [V]	-40 V to -72 V
Maximum input current [A]	30 A
Rated output power [W]	1200 W
Maximum input cable size [mm ²]	10 mm ² . Recommended: 4 mm ² (0-14 m), 6 mm ² (15-20 m), 10 mm ² (21-30 m)

Item	Specification
Front-end circuit breaker/fuse [A]	≥ 16 A (Based on long-term evolution requirements and site conditions, the recommended current is greater than or equal to 32 A.)
Heat dissipation mode	Air cooling
Airflow direction	From left to right
Noise at normal temperature (acoustic power) [dB(A)]	< 65 dB(A)
Number of slots	8
Number of service board slots	6
Switching capacity	2.4 Tbit/s (Bidirectional)
Maximum number of physical ports on the entire device	116
Maximum number of 100GE ports	12
Maximum number of 50GE ports	12
Maximum number of 40GE ports	12
Maximum number of 25GE ports	64
Maximum number of 10GE ports	72
Maximum number of GE ports	96
Maximum number of FE ports	96
Maximum number of electrical ports	96
Redundant power supply	1+1
Redundant fans	1+1. Short-term running when a single fan becomes invalid at the ambient temperature of 40°C (104°F) (one fan module contains two fans)
Long-term operating temperature [°C(°F)]	-20°C to +60°C (-4°F to +140°F)
Restriction on the operating temperature variation rate [°C(°F)]	≤ 0.5°C/min (32.9°F/min), non-condensing
Storage temperature [°C(°F)]	-40°C to +70°C (-40°F to +158 °F)
Long-term operating relative humidity [RH]	5% RH to 85% RH, non-condensing
Short-term operating relative humidity [RH]	5% RH to 95% RH, non-condensing

Item	Specification
Storage relative humidity [RH]	5% RH to 95% RH, non-condensing
Long-term operating altitude [m(ft.)]	≤ 4000 m (13123.2 ft.) (For the altitude in the range of 1800 m to 4000 m [5905.44 ft. to 13123.2 ft.], the operating temperature of the device must decrease by 1°C [1.8°F] for every 220 m [721.78 ft.])
Storage altitude [m(ft.)]	< 5000 m (16404.2 ft.)

Table 1-8 Technical specifications of the CR8PM4BASAC2

Item	Specification
Cabinet installation standards	ETSI (21-inch); IEC (19-inch)
Dimensions without packaging (H x W x D) [mm(in.)]	88.9 mm x 442 mm x 220 mm (3.50 in. x 17.40 in. x 8.66 in.)
Dimensions with packaging (H x W x D) [mm(in.)]	335 mm x 585 mm x 390 mm (13.19 in. x 23.03 in. x 15.35 in.)
Chassis height [U]	2U
Weight without packaging [kg(lb)]	9.2 kg (20.28 lb)
Weight with packaging [kg(lb)]	11.9 kg (26.24 lb)
Weight without packaging (full configuration) [kg(lb)]	10.8 kg
Weight with packaging (full configuration) [kg(lb)]	13.5 kg
Typical power consumption (with configuration) [W]	305.1 W
Typical heat dissipation (with configuration) [BTU/hour]	989.87 BTU/hour
MTBF [year]	31.62 year
MTTR [hour]	0.5 hour
Availability	0.999998195
CPU	16-core, 2.0 GHz
SDRAM	16GB
Flash memory	64MB
Storage	16 GB

Item	Specification
Power supply mode	AC
Rated input voltage [V]	200 V–240 V/100 V–127 V dual-live wire, supporting 240 V HVDC
Input voltage range [V]	90 V to 290 V
Maximum input current [A]	10 A
Rated output power [W]	1500 W
Maximum input cable size [mm ²]	Standard C13 cable
Front-end circuit breaker/fuse [A]	≥10 A
Heat dissipation mode	Air cooling
Airflow direction	From left to right
Noise at normal temperature (acoustic power) [dB(A)]	< 65 dB(A)
Number of slots	8
Number of service board slots	2
Switching capacity	2.4 Tbit/s (Bidirectional)
Maximum number of physical ports on the entire device	116
Maximum number of 100GE ports	8
Maximum number of 50GE ports	8
Maximum number of 40GE ports	8
Maximum number of 25GE ports	48
Maximum number of 10GE ports	52
Maximum number of GE ports	56
Maximum number of FE ports	56
Maximum number of electrical ports	56
Redundant power supply	1+1
Redundant fans	1+1. Short-term running when a single fan becomes invalid at the ambient temperature of 40°C (104°F) (one fan module contains two fans)
Long-term operating temperature [°C(°F)]	–20°C to +55°C (–4°F to +131°F)

Item	Specification
Restriction on the operating temperature variation rate [°C(°F)]	≤ 0.5°C/min (32.9°F/min), non-condensing
Storage temperature [°C(°F)]	–40°C to +70°C (–40°F to +158 °F)
Long-term operating relative humidity [RH]	5% RH to 85% RH, non-condensing
Short-term operating relative humidity [RH]	5% RH to 95% RH, non-condensing
Storage relative humidity [RH]	5% RH to 95% RH, non-condensing
Long-term operating altitude [m(ft.)]	≤ 4000 m (13123.2 ft.) (For the altitude in the range of 1800 m to 4000 m [5905.44 ft. to 13123.2 ft.], the operating temperature of the device must decrease by 1°C [1.8°F] for every 220 m [721.78 ft.])
Storage altitude [m(ft.)]	< 5000 m (16404.2 ft.)

Table 1-9 Technical specifications of the CR8PM4BASDC6

Item	Specification
Cabinet installation standards	ETSI (21-inch); IEC (19-inch)
Dimensions without packaging (H x W x D) [mm(in.)]	88.9 mm x 442 mm x 220 mm (3.50 in. x 17.40 in. x 8.66 in.)
Dimensions with packaging (H x W x D) [mm(in.)]	335 mm x 585 mm x 390 mm (13.19 in. x 23.03 in. x 15.35 in.)
Chassis height [U]	2U
Weight without packaging [kg(lb)]	7.2 kg (15.87 lb)
Weight with packaging [kg(lb)]	9.9 kg (21.83 lb)
Weight without packaging (full configuration) [kg(lb)]	10.4 kg
Weight with packaging (full configuration) [kg(lb)]	13.1 kg
Typical power consumption (with configuration) [W]	283.4 W
Typical heat dissipation (with configuration) [BTU/hour]	919.47 BTU/hour
MTBF [year]	33.40 year

Item	Specification
MTTR [hour]	0.5 hour
Availability	0.999998291
CPU	16-core, 2.0 GHz
SDRAM	16GB
Flash memory	64MB
Storage	16 GB
Power supply mode	DC
Rated input voltage [V]	-48 V/-60 V
Input voltage range [V]	-40 V to -72 V
Maximum input current [A]	30 A
Rated output power [W]	1200 W
Maximum input cable size [mm ²]	10 mm ² . Recommended: 4 mm ² (0-14 m), 6 mm ² (15-20 m), 10 mm ² (21-30 m)
Front-end circuit breaker/fuse [A]	≥ 16 A (Based on long-term evolution requirements and site conditions, the recommended current is greater than or equal to 32 A.)
Heat dissipation mode	Air cooling
Airflow direction	From left to right
Noise at normal temperature (acoustic power) [dB(A)]	< 65 dB(A)
Number of slots	8
Number of service board slots	6
Switching capacity	2.4 Tbit/s (Bidirectional)
Maximum number of physical ports on the entire device	116
Maximum number of 100GE ports	12
Maximum number of 50GE ports	12
Maximum number of 40GE ports	12
Maximum number of 25GE ports	64
Maximum number of 10GE ports	72
Maximum number of GE ports	96

Item	Specification
Maximum number of FE ports	96
Maximum number of electrical ports	96
Redundant power supply	1+1
Redundant fans	1+1. Short-term running when a single fan becomes invalid at the ambient temperature of 40°C (104°F) (one fan module contains two fans)
Long-term operating temperature [°C(°F)]	-20°C to +60°C (-4°F to +140°F)
Restriction on the operating temperature variation rate [°C(°F)]	≤ 0.5°C/min (32.9°F/min), non-condensing
Storage temperature [°C(°F)]	-40°C to +70°C (-40°F to +158 °F)
Long-term operating relative humidity [RH]	5% RH to 85% RH, non-condensing
Short-term operating relative humidity [RH]	5% RH to 95% RH, non-condensing
Storage relative humidity [RH]	5% RH to 95% RH, non-condensing
Long-term operating altitude [m(ft.)]	≤ 4000 m (13123.2 ft.) (For the altitude in the range of 1800 m to 4000 m [5905.44 ft. to 13123.2 ft.], the operating temperature of the device must decrease by 1°C [1.8°F] for every 220 m [721.78 ft.])
Storage altitude [m(ft.)]	< 5000 m (16404.2 ft.)

Table 1-10 Technical specifications of the CR8PM4BASAC6

Item	Specification
Cabinet installation standards	ETSI (21-inch); IEC (19-inch)
Dimensions without packaging (H x W x D) [mm(in.)]	88.9 mm x 442 mm x 220 mm (3.50 in. x 17.40 in. x 8.66 in.)
Dimensions with packaging (H x W x D) [mm(in.)]	335 mm x 585 mm x 390 mm (13.19 in. x 23.03 in. x 15.35 in.)
Chassis height [U]	2U
Weight without packaging [kg(lb)]	9.2 kg (20.28 lb)
Weight with packaging [kg(lb)]	11.9 kg (26.24 lb)

Item	Specification
Weight without packaging (full configuration) [kg(lb)]	10.8 kg
Weight with packaging (full configuration) [kg(lb)]	13.5 kg
Typical power consumption (with configuration) [W]	305.1 W
Typical heat dissipation (with configuration) [BTU/hour]	989.87 BTU/hour
MTBF [year]	31.62 year
MTTR [hour]	0.5 hour
Availability	0.999998195
CPU	16-core, 2.0 GHz
SDRAM	16GB
Flash memory	64MB
Storage	16 GB
Power supply mode	AC
Rated input voltage [V]	200 V–240 V/100 V–127 V dual-live wire, supporting 240 V HVDC
Input voltage range [V]	90 V to 290 V
Maximum input current [A]	10 A
Rated output power [W]	1500 W
Maximum input cable size [mm ²]	Standard C13 cable
Front-end circuit breaker/fuse [A]	≥10 A
Heat dissipation mode	Air cooling
Airflow direction	From left to right
Noise at normal temperature (acoustic power) [dB(A)]	< 65 dB(A)
Number of slots	8
Number of service board slots	2
Switching capacity	2.4 Tbit/s (Bidirectional)
Maximum number of physical ports on the entire device	116
Maximum number of 100GE ports	8

Item	Specification
Maximum number of 50GE ports	8
Maximum number of 40GE ports	8
Maximum number of 25GE ports	48
Maximum number of 10GE ports	52
Maximum number of GE ports	56
Maximum number of FE ports	56
Maximum number of electrical ports	56
Redundant power supply	1+1
Redundant fans	1+1. Short-term running when a single fan becomes invalid at the ambient temperature of 40°C (104°F) (one fan module contains two fans)
Long-term operating temperature [°C(°F)]	-20°C to +55°C (-4°F to +131°F)
Restriction on the operating temperature variation rate [°C(°F)]	≤ 0.5°C/min (32.9°F/min), non-condensing
Storage temperature [°C(°F)]	-40°C to +70°C (-40°F to +158 °F)
Long-term operating relative humidity [RH]	5% RH to 85% RH, non-condensing
Short-term operating relative humidity [RH]	5% RH to 95% RH, non-condensing
Storage relative humidity [RH]	5% RH to 95% RH, non-condensing
Long-term operating altitude [m(ft.)]	≤ 4000 m (13123.2 ft.) (For the altitude in the range of 1800 m to 4000 m [5905.44 ft. to 13123.2 ft.], the operating temperature of the device must decrease by 1°C [1.8°F] for every 220 m [721.78 ft.])
Storage altitude [m(ft.)]	< 5000 m (16404.2 ft.)

 NOTE

- About condensation
The ability of air to contain moisture reduces as the temperature decreases. There is a relationship between temperature and relative humidity. As the temperature decreases, the air becomes more saturated with moisture, meaning the relative humidity increases. But at a low enough temperature (called the dew point), the air can no longer contain moisture, which is then precipitated as liquid water. Such water on the surface of relevant equipment is called condensation.
- Hazards of condensation
When condensation mixes with dust inside the equipment, it forms conductive channels that affect the electrical insulation of the equipment. As a result, a non-conductive area of the equipment can become conductive, causing the equipment to malfunction.
- Conditions that cause condensation
Condensation is related to temperature changes and humidity. It occurs when the temperature continues to decrease below the dew point at a given humidity. For example, given that the relative humidity is 50% RH and the temperature is 25°C (77°F), condensation occurs when the temperature decreases below 13.9°C (57.02°F).

1.1.2 NetEngine 8000 M8

Overview

Table 1-11 Basic information about the NetEngine 8000 M8

Description	Part Number	Model	First supported version
NetEngine 8000 M8 Integrated DC Chassis Components (Including 2*DC Power,Fan Box)	02353ANP	CR8BM8BKPDC1	V800R011C10
NetEngine 8000 M8 Integrated AC Chassis Components (Including 2*AC Power,Fan Box)	02353EWU	CR8BM8BKPAC1	V800R012C00
NetEngine 8000 M8-S Integrated DC Chassis Components (Including 2*DC Power,Fan Box)	02356CDA	CR8BM8BKPDC1	V800R023C10SPC 500

Description	Part Number	Model	First supported version
NetEngine 8000 M8-S Integrated AC Chassis Components (Including 2*AC Power,Fan Box)	02356CDX	CR8BM8BKPAC1	V800R023C10SPC 500

Appearance

Figure 1-11 Appearance of the NetEngine 8000 M8 (DC)



Figure 1-12 Appearance of the NetEngine 8000 M8 (AC)



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter>

Components

Figure 1-13 Components of the NetEngine 8000 M8 (DC)



1. Grounding terminal (in the rear of the chassis)	2. PIC	3. Air intake vent	4. IPU	5. Rack-mounting ear
6. Chassis	7. DC Power module	8. Fan module	9. Grounding terminal (on the mounting ear)	10. ESD jack

Figure 1-14 Components of the NetEngine 8000 M8 (AC)



1. Grounding terminal (in the rear of the chassis)	2. PIC	3. Air intake vent	4. IPU	5. AC Power module
6. Chassis	7. DC Power module	8. Fan module	9. Grounding terminal (on the mounting ear)	10. ESD jack

6. Chassis	7. Rack-mounting ear	8. Fan module	9. Grounding terminal (on the mounting ear)	10. ESD jack
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Slot Layout

Figure 1-15 Slot Layout of the NetEngine 8000 M8

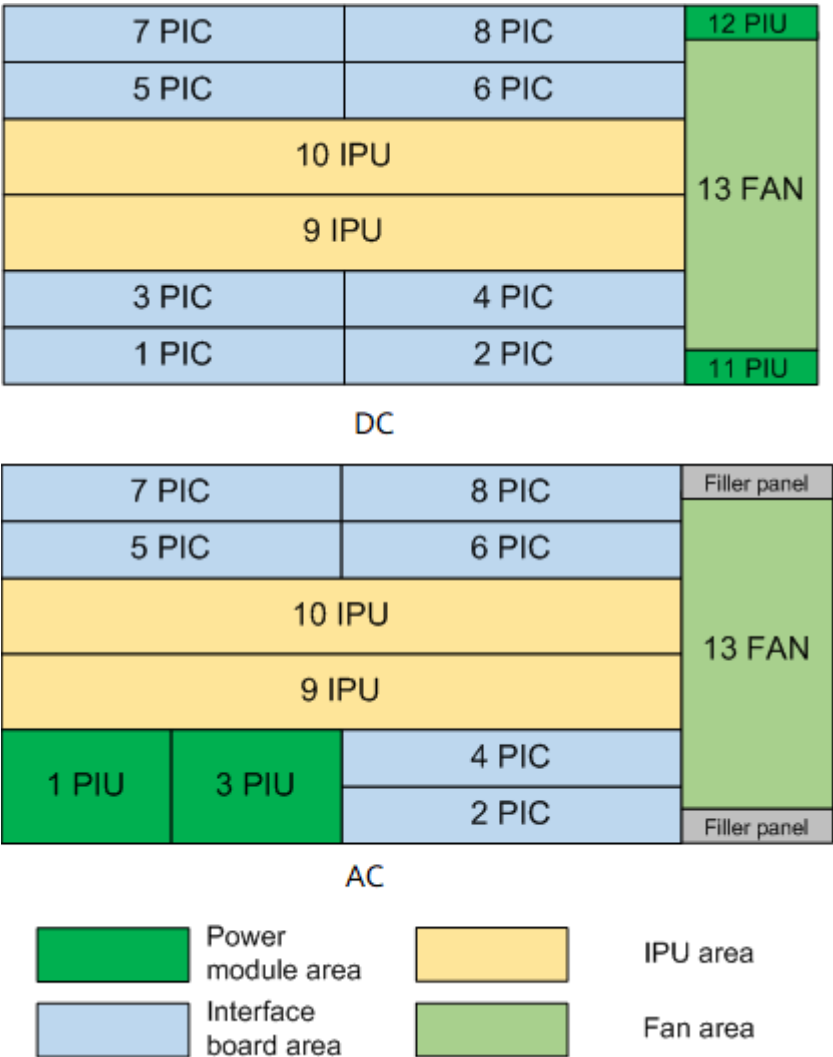


Table 1-12 Slots on the NetEngine 8000 M8

Slot Type	Slot ID	Slot Direction	Remarks
PIC	DC: 1-8 AC: 2, 4-8	The top of the board faces upwards	-

Slot Type	Slot ID	Slot Direction	Remarks
IPU	9,10	The top of the board faces upwards	-
PIU	DC: 11,12 AC: 1,3	-	DC and AC power modules cannot be used together. When AC power modules are installed, insert filler panels into the DC power module slots.
FAN	13	-	-

Labels

Figure 1-16 Label position

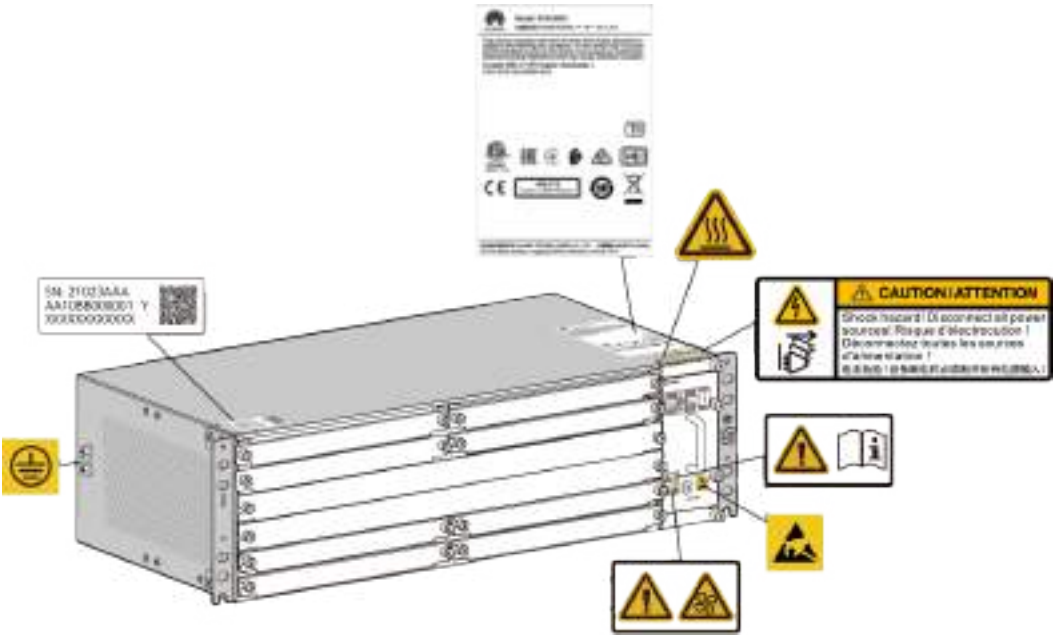


Table 1-13 Label description

Figure	Label Name	Description
	Chassis bar code label	The bar code will be retrieved by the device as the equipment serial number (ESN).

Figure	Label Name	Description
	ESD protection label	The label suggests the electrostatic-sensitive equipment.
	Chassis grounding label	The label indicates the position of the general grounding terminal. When the equipment is installed in a cabinet rather than a 19-inch cabinet, it is recommended that you use the grounding terminal specified in this label.
	Fan warning label	The label suggests that do not touch the fan leaves when the fan is rotating.
	Operation warning label	The label indicates the precaution that should be taken for operations on the PIU board.
	Heat hazard label	This label suggests that the surface temperature is high and it may cause body injury.
	Multi-power input caution	This device has more than one power input. Disconnect all power inputs to power off this device.
	Product nameplate label	The label suggests the product name, certification.

Power Supply System

The power supply system supports 1+1 backup mode. If one power module fails or is removed, the other power module can still supply adequate power to the device.

AC power modules (PIUs) are installed in the two slots in the lower left part of a chassis and DC PIUs are installed in the two slots on the right side of a chassis to supply power to IPU, interface boards, and fan modules.

The following protection functions are provided for the power modules:

- Protection against output overcurrent
- Protection against output overvoltage
- Protection against input undervoltage
- Protection against overtemperature
- Protection against short circuit
- Alarm generation

Figure 1-17 shows the DC power supply architecture. Two power modules provide two 48 V power inputs, which are combined on the backplane and then supplied to the fan module, IPU, and interface boards. The IPU provides two 3.3 V power inputs, which are combined on the backplane and then supplied to the fan module, power modules, and interface boards.

Figure 1-17 Diagram of the DC power supply architecture

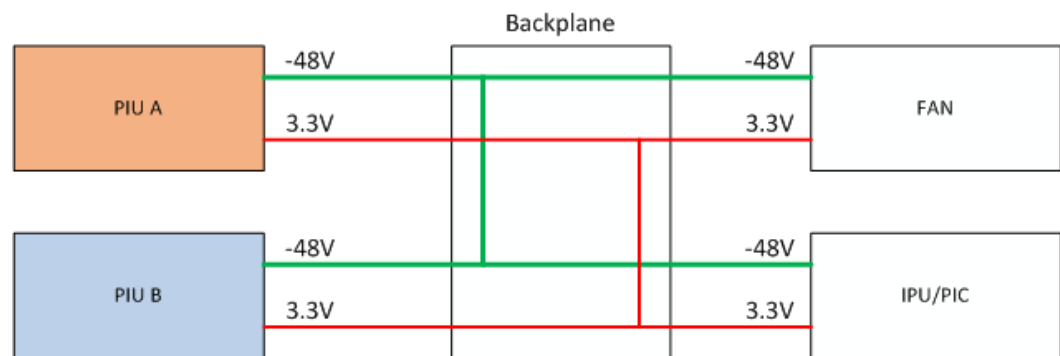
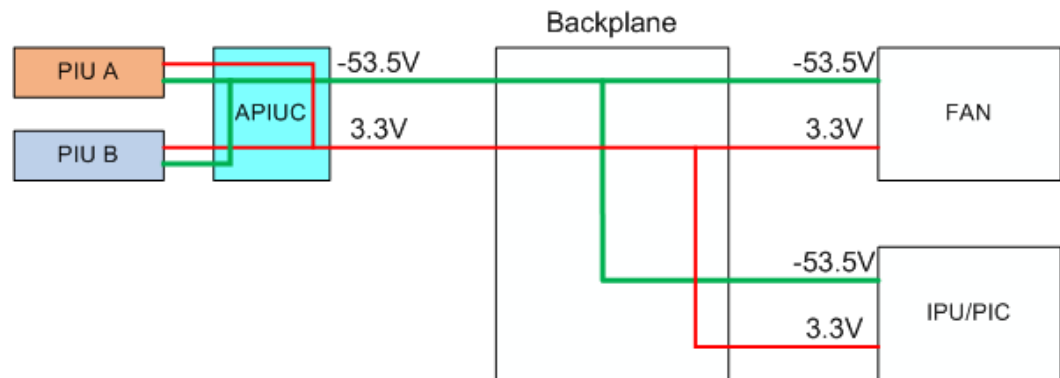


Figure 1-18 shows the AC power supply architecture. Two power modules provide two -53.5 V power inputs, which are combined on the backplane and then supplied to the fan module, IPU, and interface boards. The IPU provides two 3.3 V power inputs, which are combined on the backplane and then supplied to the fan module, power modules, and interface boards.

Figure 1-18 Diagram of the AC power supply architecture



Heat Dissipation System

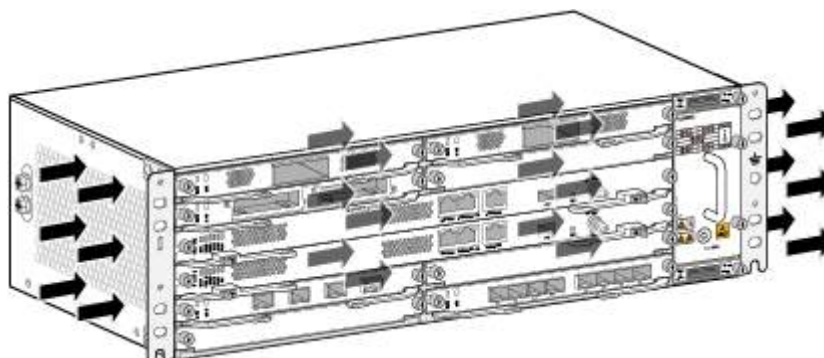
The fans exhaust air from the chassis to dissipate heat. The hot air inside the chassis is sucked out of the subrack to form a ventilation channel from the air intake vent on the left side to the air exhaust vent on the right side. The fan tray assembly consists of two independent fans to dissipate heat for the device and is hot swappable.

NOTICE

- Ensure that heat dissipation holes for the air intake and exhaust vents are clean and not blocked.
- The empty slot should be covered with a filler panel to ensure good electromagnetic compatibility, dustproof, and heat dissipation.
- Keep any objects or obstructions at least 75 mm away from the air intake and exhaust vents to facilitate heat dissipation.
- When installing devices, ensure that the total heat consumption of all devices in the cabinet is less than or equal to the heat dissipation capability of the cabinet.
- The equipment must be stacked in the cabinet or installed at an interval of 2U or more to prevent the return air from affecting heat dissipation.
- When the chassis is installed together with other equipment in the same cabinet, do not install the chassis near the air exhaust vent of other equipment.
- You are required to replace the fan board as soon as possible; otherwise, the active/standby switchover, system reset, or board burning may occur due to high temperature. Exercise caution when performing this operation. The fan board needs to be replaced within one minute at the normal temperature (25°C).

Figure 1-19 shows the air flow in the device.

Figure 1-19 Air flow in the device



Technical Specifications

Table 1-14 Technical specifications of the CR8BM8BKPDC1 (02353ANP)

Item	Specification
Cabinet installation standards	ETSI 21-inch; IEC 19-inch
Dimensions without packaging (H x W x D) [mm(in.)]	132.6 mm x 442 mm x 220 mm (5.22 in. x 17.4 in. x 8.66 in.)
Dimensions with packaging (H x W x D) [mm(in.)]	360 mm x 585 mm x 390 mm (14.17 in. x 23.03 in. x 15.35 in.)
Chassis height [U]	3U
Weight without packaging [kg(lb)]	5.7 kg (12.57 lb)
Weight with packaging [kg(lb)]	9.2 kg (20.28 lb)
Weight without packaging (full configuration) [kg(lb)]	[IPU-1T2/IPU-1T2-B]DC:16.5 kg (36.38 lb) [IPU-480/IPU-480-B/IPU-480-BN]DC: 16.5 kg (36.38 lb)
Weight with packaging (full configuration) [kg(lb)]	[IPU-1T2/IPU-1T2-B]DC:20.0 kg (44.09 lb) [IPU-480/IPU-480-B/IPU-480-BN]DC: 20.0 kg (44.09 lb)
Typical power consumption (with configuration) [W]	[IPU-1T2/IPU-1T2-B]DC: 2*IPU+1*fan +2*DC power +2*CR5D0E5XMF94+1*CR5D00E4XM25 +2*CR5D00LAXF91+2*CR5D00EAGF95= 484.34 W [IPU-480/IPU-480-B/IPU-480-BN]DC: 2*IPU+1*fan+2*DC power +2*CR5D0E5XMF94+2*CR5D00LAXF91 +2*CR5D00EAGF95=428.11 W

Item	Specification
Typical heat dissipation (with configuration) [BTU/hour]	[IPU-1T2/IPU-1T2-B]DC: 2*IPU+1*fan+2*DC power +2*CR5D0E5XMF94+1*CR5D00E4XM25 +2*CR5D00LAXF91+2*CR5D00EAGF95=1532.61 BTU/hour [IPU-480/IPU-480-B/IPU-480-BN]DC: 2*IPU+1*fan+2*DC power +2*CR5D0E5XMF94+2*CR5D00LAXF91 +2*CR5D00EAGF95=1350.19 BTU/hour
MTBF [year]	155.06 year
MTTR [hour]	0.5 hour
Availability	0.999999632
Power supply mode	DC
Rated input voltage [V]	-48 V/-60 V
Input voltage range [V]	-40 V to -72 V
Maximum input current [A]	40 A
Rated output power [W]	1600 W
Maximum input cable size [mm ²]	10mm ² (0-20m)
Front-end circuit breaker/fuse [A]	≥ 32 A (Based on long-term evolution requirements and site conditions, the recommended current is greater than or equal to 40 A.)
Heat dissipation mode	Air cooling
Airflow direction	From left to right
Noise at normal temperature (acoustic power) [dB(A)]	< 72 dB (meeting the ETSI 72 dBA standard)
Number of slots	13
Number of service board slots	8
Switching capacity	IPU-1T2/IPU-1T2-B:2.4 Tbit/s IPU-480/IPU-480-B/IPU-480-BN:960 Gbit/s IPU-240/IPU-240-B:480 Gbit/s
Maximum number of physical ports on the entire device	160

Item	Specification
Maximum number of 100GE ports	IPU-1T2/IPU-1T2-B:12 IPU-480/IPU-480-B/IPU-480-BN:4 IPU-240/IPU-240-B:2
Maximum number of 50GE ports	IPU-1T2/IPU-1T2-B:16 IPU-480/IPU-480-B/IPU-480-BN:8 IPU-240/IPU-240-B:4
Maximum number of 40GE ports	IPU-1T2/IPU-1T2-B:12 IPU-480/IPU-480-B/IPU-480-BN:4 IPU-240/IPU-240-B:2
Maximum number of 25GE ports	IPU-1T2/IPU-1T2-B:32 IPU-480/IPU-480-B/IPU-480-BN:16 IPU-240/IPU-240-B:8
Maximum number of 10GE ports	IPU-1T2/IPU-1T2-B:80 IPU-480/IPU-480-B/IPU-480-BN:48 IPU-240/IPU-240-B:24
Maximum number of GE ports	IPU-1T2/IPU-1T2-B:160 IPU-480/IPU-480-B/IPU-480-BN:160 IPU-240/IPU-240-B:160
Maximum number of FE ports	IPU-1T2/IPU-1T2-B:160 IPU-480/IPU-480-B/IPU-480-BN:160 IPU-240/IPU-240-B:160
Maximum number of electrical ports	IPU-1T2/IPU-1T2-B:80 IPU-480/IPU-480-B/IPU-480-BN:80 IPU-240/IPU-240-B:80
Redundant IPU	1:1
Redundant power supply	1+1
Redundant fans	1+1. Short-term running when a single fan becomes invalid at the ambient temperature of 40°C(104°F) (one fan module contains two fans)
Long-term operating temperature [°C(°F)]	-40°C to +65°C (-40°F to +149°F), startup: ≥ -20°C (-4°F)
Restriction on the operating temperature variation rate [°C(°F)]	≤ 0.5°C/min (32.9°F/min), non-condensing
Storage temperature [°C(°F)]	-40°C to +70°C (-40°F to +158 °F)

Item	Specification
Long-term operating relative humidity [RH]	5% RH to 85% RH, non-condensing
Short-term operating relative humidity [RH]	5% RH to 95% RH, non-condensing
Storage relative humidity [RH]	5% RH to 95% RH, non-condensing
Long-term operating altitude [m(ft.)]	≤ 4000 m (13123.2 ft.) (For the altitude in the range of 1800 m to 4000 m [5905.44 ft. to 13123.2 ft.], the operating temperature of the device must decrease by 1°C [1.8°F] for every 220 m [721.78 ft.])
Storage altitude [m(ft.)]	< 5000 m (16404.2 ft.)

Table 1-15 Technical specifications of the CR8BM8BKPAC1 (02353EWU)

Item	Specification
Cabinet installation standards	ETSI 21-inch; IEC 19-inch
Dimensions without packaging (H x W x D) [mm(in.)]	132.6 mm x 442 mm x 220 mm (5.22 in. x 17.4 in. x 8.66 in.)
Dimensions with packaging (H x W x D) [mm(in.)]	360 mm x 585 mm x 390 mm (14.17 in. x 23.03 in. x 15.35 in.)
Chassis height [U]	3U
Weight without packaging [kg(lb)]	7.8 kg (17.2 lb)
Weight with packaging [kg(lb)]	11.3 kg (24.91 lb)
Weight without packaging (full configuration) [kg(lb)]	[IPU-1T2/IPU-1T2-B]AC:17.1kg(37.71 lb) [IPU-480/IPU-480-B/IPU-480-BN]AC: 17.1kg(37.71 lb)
Weight with packaging (full configuration) [kg(lb)]	[IPU-1T2/IPU-1T2-B]AC:20.6kg(45.42 lb) [IPU-480/IPU-480-B/IPU-480-BN]AC: 20.6kg(45.42 lb)

Item	Specification
Typical power consumption (with configuration) [W]	[IPU-1T2/IPU-1T2-B]AC: 2*IPU+1*fan+2*AC power +2*CR5D0E5XMF94+1*CR5D00E4XM25+2*CR5D00LAXF91+1*CR5D00EAGF95=485.27 W [IPU-480/IPU-480-B/IPU-480-BN]AC: 2*IPU+1*fan+2*AC power +2*CR5D0E5XMF94+2*CR5D00LAXF91+2*CR5D00EAGF95=457.79 W
Typical heat dissipation (with configuration) [BTU/hour]	[IPU-1T2/IPU-1T2-B]AC: 2*IPU+1*fan+2*AC power +2*CR5D0E5XMF94+1*CR5D00E4XM25+2*CR5D00LAXF91+1*CR5D00EAGF95=1532.91 BTU/hour [IPU-480/IPU-480-B/IPU-480-BN]AC: 2*IPU+1*fan+2*AC power +2*CR5D0E5XMF94+2*CR5D00LAXF91+2*CR5D00EAGF95=1443.77 BTU/hour
MTBF [year]	112.71 year
MTTR [hour]	0.5 hour
Availability	0.999999494
Power supply mode	AC
Rated input voltage [V]	200V to 240V/100V to 127V dual live wires, support 240V HVDC
Input voltage range [V]	90 V to 290 V
Maximum input current [A]	10 A
Rated output power [W]	1500 W
Maximum input cable size [mm ²]	Standard C13 cable
Front-end circuit breaker/fuse [A]	≥10 A
Heat dissipation mode	Air cooling
Airflow direction	From left to right
Noise at normal temperature (acoustic power) [dB(A)]	< 72 dB (meeting the ETSI 72 dBA standard)
Number of slots	11
Number of service board slots	6
Switching capacity	IPU-1T2/IPU-1T2-B:2.4 Tbit/s IPU-480/IPU-480-B/IPU-480-BN:960 Gbit/s

Item	Specification
Maximum number of physical ports on the entire device	120
Maximum number of 100GE ports	IPU-1T2/IPU-1T2-B:9 IPU-480/IPU-480-B/IPU-480-BN:4 IPU-240/IPU-240-B:2
Maximum number of 50GE ports	IPU-1T2/IPU-1T2-B:12 IPU-480/IPU-480-B/IPU-480-BN:8 IPU-240/IPU-240-B:4
Maximum number of 40GE ports	IPU-1T2/IPU-1T2-B:9 IPU-480/IPU-480-B/IPU-480-BN:4 IPU-240/IPU-240-B:2
Maximum number of 25GE ports	IPU-1T2/IPU-1T2-B:24 IPU-480/IPU-480-B/IPU-480-BN:16 IPU-240/IPUA-240-B:8
Maximum number of 10GE ports	IPU-1T2/IPU-1T2-B:60 IPU-480/IPU-480-B/IPU-480-BN:48 IPU-240/IPU-240-B:24
Maximum number of GE ports	IPU-1T2/IPU-1T2-B:120 IPU-480/IPU-480-B/IPU-480-BN:120 IPU-240/IPU-240-B:120
Maximum number of FE ports	IPU-1T2/IPU-1T2-B:120 IPU-480/IPU-480-B/IPU-480-BN:120 IPU-240/IPU-240-B:120
Maximum number of electrical ports	IPU-1T2/IPU-1T2-B:60 IPU-480/IPU-480-B/IPU-480-BN:60 IPU-240/IPU-240-B:60
Redundant IPU	1:1
Redundant power supply	1+1
Redundant fans	1+1. Short-term running when a single fan becomes invalid at the ambient temperature of 40°C(104°F) (one fan module contains two fans)
Long-term operating temperature [°C(°F)]	-20°C to +55°C (-4°F to +131°F)
Restriction on the operating temperature variation rate [°C(°F)]	≤ 0.5°C/min (32.9°F/min), non-condensing

Item	Specification
Storage temperature [°C(°F)]	–40°C to +70°C (–40°F to +158 °F)
Long-term operating relative humidity [RH]	5% RH to 85% RH, non-condensing
Short-term operating relative humidity [RH]	5% RH to 95% RH, non-condensing
Storage relative humidity [RH]	5% RH to 95% RH, non-condensing
Long-term operating altitude [m(ft.)]	≤ 4000 m (13123.2 ft.) (For the altitude in the range of 1800 m to 4000 m [5905.44 ft. to 13123.2 ft.], the operating temperature of the device must decrease by 1°C [1.8°F] for every 220 m [721.78 ft.])
Storage altitude [m(ft.)]	< 5000 m (16404.2 ft.)

Table 1-16 Technical specifications of the CR8BM8BKPDC1 (02356CDA)

Item	Specification
Cabinet installation standards	ETSI 21-inch; IEC 19-inch
Dimensions without packaging (H x W x D) [mm(in.)]	132.6 mm x 442 mm x 220 mm (5.22 in. x 17.4 in. x 8.66 in.)
Dimensions with packaging (H x W x D) [mm(in.)]	360 mm x 585 mm x 390 mm (14.17 in. x 23.03 in. x 15.35 in.)
Chassis height [U]	3U
Weight without packaging [kg(lb)]	5.7 kg (12.57 lb)
Weight with packaging [kg(lb)]	9.2 kg (20.28 lb)
Weight without packaging (full configuration) [kg(lb)]	[IPU-1T2]DC:16.5 kg (36.38 lb) [IPU-480/IPU-480-BN]DC:16.5 kg (36.38 lb)
Weight with packaging (full configuration) [kg(lb)]	[IPU-1T2]DC:20.0 kg (44.09 lb) [IPU-480/IPU-480-BN]DC:20.0 kg (44.09 lb)

Item	Specification
Typical power consumption (with configuration) [W]	[IPU-1T2]DC: 2*IPU+1*fan+2*DC power +2*CR5D0E5XMF94+1*CR5D00E4XM25 +2*CR5D00LAXF91+2*CR5D00EAGF95=484.34 W [IPU-480/IPU-480-BN]DC: 2*IPU+1*fan +2*DC power +2*CR5D0E5XMF94+2*CR5D00LAXF91 +2*CR5D00EAGF95=428.11 W
Typical heat dissipation (with configuration) [BTU/hour]	[IPU-1T2]DC: 2*IPU+1*fan+2*DC power +2*CR5D0E5XMF94+1*CR5D00E4XM25 +2*CR5D00LAXF91+2*CR5D00EAGF95=1532.61 BTU/hour [IPU-480/IPU-480-BN]DC: 2*IPU+1*fan +2*DC power +2*CR5D0E5XMF94+2*CR5D00LAXF91 +2*CR5D00EAGF95=1350.19 BTU/hour
MTBF [year]	155.06 year
MTTR [hour]	0.5 hour
Availability	0.999999632
Power supply mode	DC
Rated input voltage [V]	-48 V/-60 V
Input voltage range [V]	-40 V to -72 V
Maximum input current [A]	40 A
Rated output power [W]	1600 W
Maximum input cable size [mm ²]	10mm ² (0-20m)
Front-end circuit breaker/fuse [A]	≥ 32 A (Based on long-term evolution requirements and site conditions, the recommended current is greater than or equal to 40 A.)
Heat dissipation mode	Air cooling
Airflow direction	From left to right
Noise at normal temperature (acoustic power) [dB(A)]	< 72 dB (meeting the ETSI 72 dBA standard)
Number of slots	13
Number of service board slots	8

Item	Specification
Switching capacity	IPU-1T2/IPU-1T2-B:2.4 Tbit/s IPU-480/IPU-480-B/IPU-480-BN:960 Gbit/s IPU-240/IPU-240-B:480 Gbit/s
Maximum number of physical ports on the entire device	160
Maximum number of 100GE ports	IPU-1T2/IPU-1T2-B:12 IPU-480/IPU-480-B/IPU-480-BN:4 IPU-240/IPU-240-B:2
Maximum number of 50GE ports	IPU-1T2/IPU-1T2-B:16 IPU-480/IPU-480-B/IPU-480-BN:8 IPU-240/IPU-240-B:4
Maximum number of 40GE ports	IPU-1T2/IPU-1T2-B:12 IPU-480/IPU-480-B/IPU-480-BN:4 IPU-240/IPU-240-B:2
Maximum number of 25GE ports	IPU-1T2/IPU-1T2-B:32 IPU-480/IPU-480-B/IPU-480-BN:16 IPU-240/IPU-240-B:8
Maximum number of 10GE ports	IPU-1T2/IPU-1T2-B:80 IPU-480/IPU-480-B/IPU-480-BN:48 IPU-240/IPU-240-B:24
Maximum number of GE ports	IPU-1T2/IPU-1T2-B:160 IPU-480/IPU-480-B/IPU-480-BN:160 IPU-240/IPU-240-B:160
Maximum number of FE ports	IPU-1T2/IPU-1T2-B:160 IPU-480/IPU-480-B/IPU-480-BN:160 IPU-240/IPU-240-B:160
Maximum number of electrical ports	IPU-1T2/IPU-1T2-B:80 IPU-480/IPU-480-B/IPU-480-BN:80 IPU-240/IPU-240-B:80
Redundant IPU	1:1
Redundant power supply	1+1
Redundant fans	1+1. Short-term running when a single fan becomes invalid at the ambient temperature of 40°C(104°F) (one fan module contains two fans)

Item	Specification
Long-term operating temperature [°C(°F)]	–40°C to +65°C (–40°F to +149°F), startup: ≥ –20°C (–4°F)
Restriction on the operating temperature variation rate [°C(°F)]	≤ 0.5°C/min (32.9°F/min), non-condensing
Storage temperature [°C(°F)]	–40°C to +70°C (–40°F to +158 °F)
Long-term operating relative humidity [RH]	5% RH to 85% RH, non-condensing
Short-term operating relative humidity [RH]	5% RH to 95% RH, non-condensing
Storage relative humidity [RH]	5% RH to 95% RH, non-condensing
Long-term operating altitude [m(ft.)]	≤ 4000 m (13123.2 ft.) (For the altitude in the range of 1800 m to 4000 m [5905.44 ft. to 13123.2 ft.], the operating temperature of the device must decrease by 1°C [1.8°F] for every 220 m [721.78 ft.])
Storage altitude [m(ft.)]	< 5000 m (16404.2 ft.)

Table 1-17 Technical specifications of the CR8BM8BKPAC1 (02356CDX)

Item	Specification
Cabinet installation standards	ETSI 21-inch; IEC 19-inch
Dimensions without packaging (H x W x D) [mm(in.)]	132.6 mm x 442 mm x 220 mm (5.22 in. x 17.4 in. x 8.66 in.)
Dimensions with packaging (H x W x D) [mm(in.)]	360 mm x 585 mm x 390 mm (14.17 in. x 23.03 in. x 15.35 in.)
Chassis height [U]	3U
Weight without packaging [kg(lb)]	7.8 kg (17.2 lb)
Weight with packaging [kg(lb)]	11.3 kg (24.91 lb)
Weight without packaging (full configuration) [kg(lb)]	[IPU-1T2]AC:17.1kg(37.71 lb) [IPU-480/IPU-480-BN]AC:17.1kg(37.71 lb)
Weight with packaging (full configuration) [kg(lb)]	[IPU-1T2]AC:20.6kg(45.42 lb) [IPU-480/IPU-480-BN]AC:20.6kg(45.42 lb)

Item	Specification
Typical power consumption (with configuration) [W]	[IPU-1T2]AC: 2*IPU+1*fan+2*AC power +2*CR5D0E5XMF94+1*CR5D00E4XM25 +2*CR5D00LAXF91+1*CR5D00EAGF95= 485.27 W [IPU-480/IPU-480-BN]AC: 2*IPU+1*fan +2*AC power +2*CR5D0E5XMF94+2*CR5D00LAXF91 +2*CR5D00EAGF95=457.79 W
Typical heat dissipation (with configuration) [BTU/hour]	[IPU-1T2]AC: 2*IPU+1*fan+2*AC power +2*CR5D0E5XMF94+1*CR5D00E4XM25 +2*CR5D00LAXF91+1*CR5D00EAGF95= 1532.91 BTU/hour [IPU-480/IPU-480-BN]AC: 2*IPU+1*fan +2*AC power +2*CR5D0E5XMF94+2*CR5D00LAXF91 +2*CR5D00EAGF95=1443.77 BTU/hour
MTBF [year]	112.71 year
MTTR [hour]	0.5 hour
Availability	0.999999494
Power supply mode	AC
Rated input voltage [V]	200V to 240V/100V to 127V dual live wires, support 240V HVDC
Input voltage range [V]	90 V to 290 V
Maximum input current [A]	10 A
Rated output power [W]	1500 W
Maximum input cable size [mm ²]	Standard C13 cable
Front-end circuit breaker/fuse [A]	≥10 A
Heat dissipation mode	Air cooling
Airflow direction	From left to right
Noise at normal temperature (acoustic power) [dB(A)]	< 72 dB (meeting the ETSI 72 dBA standard)
Number of slots	11
Number of service board slots	6
Switching capacity	IPU-1T2/IPU-1T2-B:2.4 Tbit/s IPU-480/IPU-480-B/IPU-480-BN:960 Gbit/s

Item	Specification
Maximum number of physical ports on the entire device	120
Maximum number of 100GE ports	IPU-1T2/IPU-1T2-B:9 IPU-480/IPU-480-B/IPU-480-BN:4 IPU-240/IPU-240-B:2
Maximum number of 50GE ports	IPU-1T2/IPU-1T2-B:12 IPU-480/IPU-480-B/IPU-480-BN:8 IPU-240/IPU-240-B:4
Maximum number of 40GE ports	IPU-1T2/IPU-1T2-B:9 IPU-480/IPU-480-B/IPU-480-BN:4 IPU-240/IPU-240-B:2
Maximum number of 25GE ports	IPU-1T2/IPU-1T2-B:24 IPU-480/IPU-480-B/IPU-480-BN:16 IPU-240/IPUA-240-B:8
Maximum number of 10GE ports	IPU-1T2/IPU-1T2-B:60 IPU-480/IPU-480-B/IPU-480-BN:48 IPU-240/IPU-240-B:24
Maximum number of GE ports	IPU-1T2/IPU-1T2-B:120 IPU-480/IPU-480-B/IPU-480-BN:120 IPU-240/IPU-240-B:120
Maximum number of FE ports	IPU-1T2/IPU-1T2-B:120 IPU-480/IPU-480-B/IPU-480-BN:120 IPU-240/IPU-240-B:120
Maximum number of electrical ports	IPU-1T2/IPU-1T2-B:60 IPU-480/IPU-480-B/IPU-480-BN:60 IPU-240/IPU-240-B:60
Redundant IPU	1:1
Redundant power supply	1+1
Redundant fans	1+1. Short-term running when a single fan becomes invalid at the ambient temperature of 40°C(104°F) (one fan module contains two fans)
Long-term operating temperature [°C(°F)]	-20°C to +55°C (-4°F to +131°F)
Restriction on the operating temperature variation rate [°C(°F)]	≤ 0.5°C/min (32.9°F/min), non-condensing

Item	Specification
Storage temperature [°C(°F)]	–40°C to +70°C (–40°F to +158 °F)
Long-term operating relative humidity [RH]	5% RH to 85% RH, non-condensing
Short-term operating relative humidity [RH]	5% RH to 95% RH, non-condensing
Storage relative humidity [RH]	5% RH to 95% RH, non-condensing
Long-term operating altitude [m(ft.)]	≤ 4000 m (13123.2 ft.) (For the altitude in the range of 1800 m to 4000 m [5905.44 ft. to 13123.2 ft.], the operating temperature of the device must decrease by 1°C [1.8°F] for every 220 m [721.78 ft.])
Storage altitude [m(ft.)]	< 5000 m (16404.2 ft.)

 NOTE

- About condensation

The ability of air to contain moisture reduces as the temperature decreases. There is a relationship between temperature and relative humidity. As the temperature decreases, the air becomes more saturated with moisture, meaning the relative humidity increases. But at a low enough temperature (called the dew point), the air can no longer contain moisture, which is then precipitated as liquid water. Such water on the surface of relevant equipment is called condensation.

- Hazards of condensation

When condensation mixes with dust inside the equipment, it forms conductive channels that affect the electrical insulation of the equipment. As a result, a non-conductive area of the equipment can become conductive, causing the equipment to malfunction.

- Conditions that cause condensation

Condensation is related to temperature changes and humidity. It occurs when the temperature continues to decrease below the dew point at a given humidity. For example, given that the relative humidity is 50% RH and the temperature is 25°C (77°F), condensation occurs when the temperature decreases below 13.9°C (57.02°F).

1.1.3 NetEngine 8000 M14

Overview

Table 1-18 Basic information about the NetEngine 8000 M14

Description	Part Number	Model	First supported version
NetEngine 8000 M14 Integrated DC Chassis Components (Including 2*DC Power,Fan Box)	02353ABJ	CR8B0BKPDCC1	V800R012C00
NetEngine 8000 M14 Integrated AC Chassis Components (Including 4*AC Power,Fan Box)	02353ABK	CR8B0BKPACC2	V800R012C00
NetEngine 8000 M14 Integrated AC Chassis Components (Including 2*AC Power,Fan Box)	02353VMM	CR8B0BKPACC1	V800R012C10
NetEngine 8000 M14 Integrated AC Chassis Components(Including 2*AC Power,Fan Box)	02356AQM	CR8B0BKPACC6	V800R023C10SPC 500
NetEngine 8000 M14 Integrated AC Chassis Components(Including 2*AC Power,Fan Box)	02355VUR	CR8B0BKPACM1	V800R023C10SPC 500
NetEngine 8000 M14 Integrated DC Chassis Components(Including 2*DC Power,Fan Box)	02355VUS	CR8B0BKPDCM1	V800R023C10SPC 500

Appearance

Figure 1-20 Appearance of the NetEngine 8000 M14 (DC)



Figure 1-21 Appearance of the NetEngine 8000 M14 (AC Including 4*AC Power)



Figure 1-22 Appearance of the NetEngine 8000 M14 (AC Including 2*AC Power)



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter>

Components

Figure 1-23 Components of the NetEngine 8000 M14 (DC)



1. Fan	2. Mounting ear	3. Ground terminal (on the mounting ear)	4. IPU	5. ESD jack
6. PIC	7. DC Power module	8. Air intake vent	9. Ground terminal (in the rear of the chassis)	-

Figure 1-24 Components of the NetEngine 8000 M14 (AC Including 4*AC Power)



1. Fan	2. Mounting ear	3. Ground terminal (on the mounting ear)	4. IPU	5. ESD jack
6. PIC	7. AC Power module	8. Air intake vent	9. Ground terminal (in the rear of the chassis)	-

Figure 1-25 Components of the NetEngine 8000 M14 (AC Including 2*AC Power)



1. Fan	2. Mounting ear	3. Ground terminal (on the mounting ear)	4. IPU	5. ESD jack
6. PIC	7. AC Power module	8. Air intake vent	9. Ground terminal (in the rear of the chassis)	-

Slot Layout

Figure 1-26 Slot Layout of the NetEngine 8000 M14

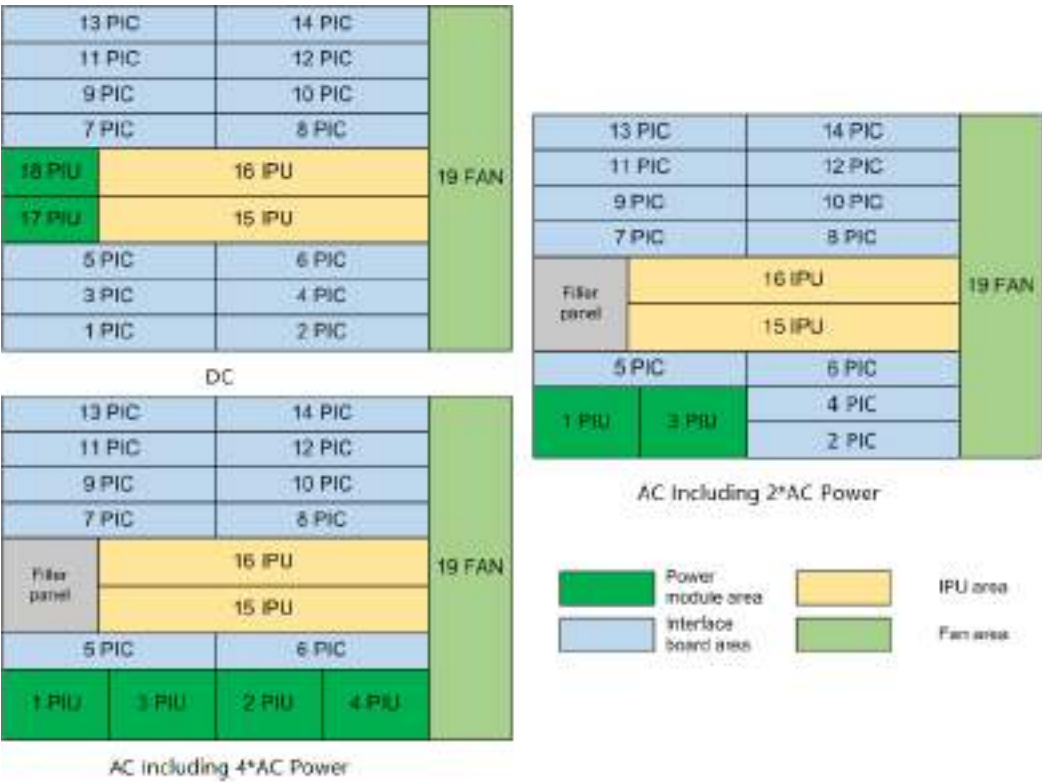


Table 1-19 Slots on the NetEngine 8000 M14

Slot Type	Slot ID	Slot Direction	Remarks
PIC	DC: 1-14 AC (Including 4*AC Power): 5-14 AC (Including 2*AC Power): 2,4-14	The top of the board faces upwards	-
IPU	15,16	The top of the board faces upwards	-

Slot Type	Slot ID	Slot Direction	Remarks
PIU	DC: 17,18 AC (Including 4*AC Power): 1-4 AC (Including 2*AC Power): 1,3	-	DC and AC power modules cannot be used together. When AC power modules are installed, insert filler panels into the DC power module slots. In V800R012C10 and later versions, the NetEngine 8000 M14 equipped with the IPU-1T2-A/ IPU-1T2-BN requires two AC power modules, while the NetEngine 8000 M14 equipped with the IPUA-2T/ IPUA-1T2/ IPUA-2T-B/ IPUA-1T2-B requires four AC power modules.
FAN	19	-	-

Labels

Figure 1-27 Label position

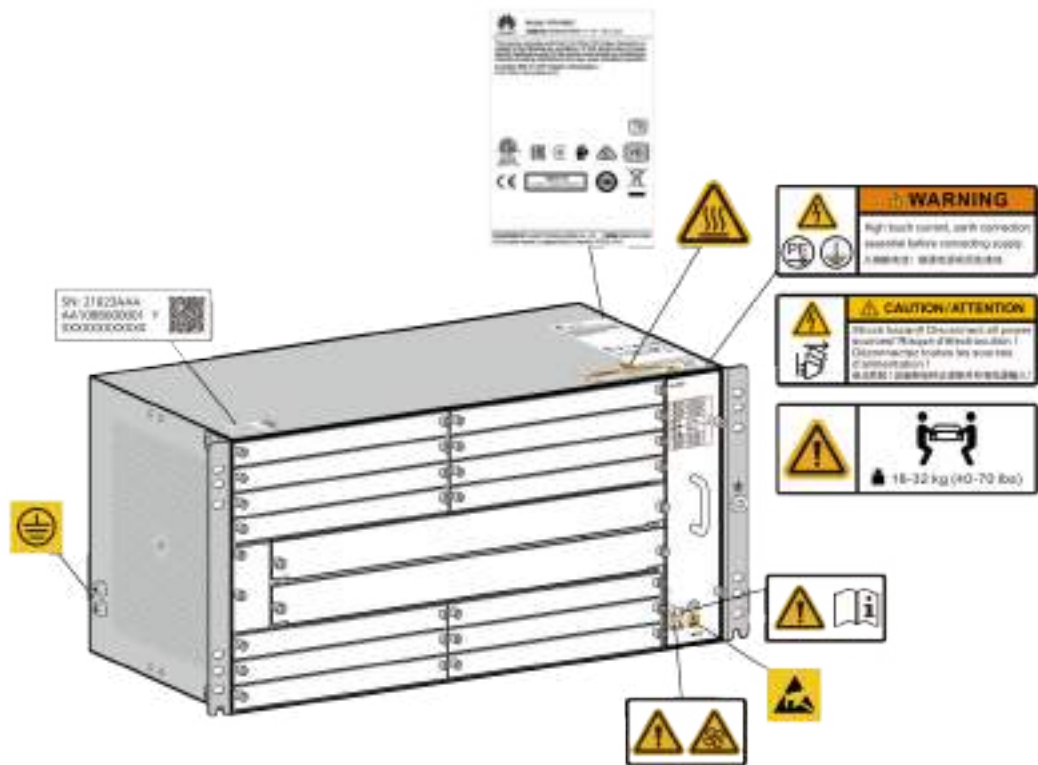


Table 1-20 Label description

Figure	Label Name	Description
	Chassis bar code label	The bar code will be retrieved by the device as the equipment serial number (ESN).
	ESD protection label	The label suggests the electrostatic-sensitive equipment.
	Chassis grounding label	The label indicates the position of the general grounding terminal. When the equipment is installed in a cabinet rather than a 19-inch cabinet, it is recommended that you use the grounding terminal specified in this label.

Figure	Label Name	Description
	Fan warning label	The label suggests that do not touch the fan leaves when the fan is rotating.
	Operation warning label	The label indicates the precaution that should be taken for operations on the PIU board.
	Heat hazard label	This label suggests that the surface temperature is high and it may cause body injury.
	Multi-power input caution	This device has more than one power input. Disconnect all power inputs to power off this device.
	Strong-current label	High touch current, earth connection essential before connecting supply.
	Weight label	18 to 32kg (two persons)
	Product nameplate label	The label suggests the product name, certification.

Power Supply System

The device supports DC and AC power supply.

When DC power is supplied, DC power modules work in 1+1 backup mode. If one power module fails or is removed, the other power module can still supply adequate power to the device. The two DC power modules are installed in slots 17 and 18 on the left of the chassis and supply power to the IPU, PICs and fan module.

When AC power is supplied, AC power modules work in 2+2 or 1+1 backup mode. If one or two power modules fail or are removed, the other power modules can still supply adequate power to the device. The AC power modules at the bottom of the chassis and supply power to the IPU, PICs and fan module.

The following protection functions are provided for the power modules:

- Protection against output overcurrent
- Protection against output overvoltage
- Protection against input undervoltage
- Protection against overtemperature
- Protection against short circuit
- Alarm function

Figure 1-28 shows the DC power supply architecture. Two power modules provide two 48 V power inputs, which are combined on the backplane and then supplied to the fan module, IPU, and PICs. The IPU provides two 3.3 V power inputs, which are combined on the backplane and then supplied to the fan module, power modules, and PICs.

Figure 1-28 DC power system architecture

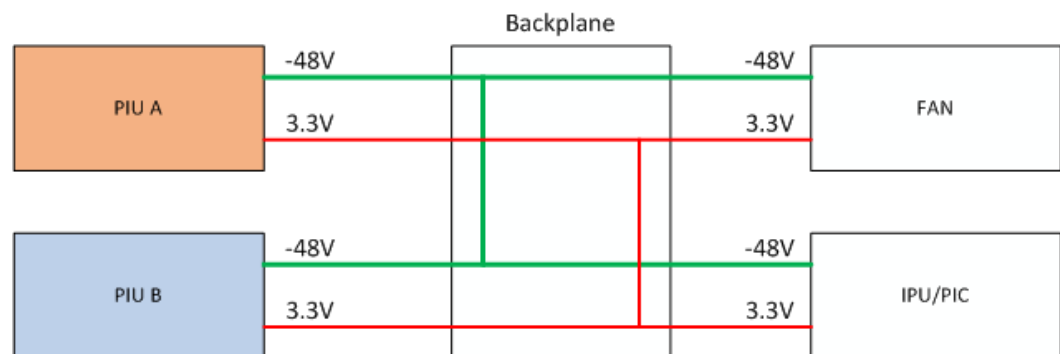


Figure 1-29 shows the 4*AC power supply architecture. Four power modules provide four independent -53.5 V power inputs for the two APIUCs, which combine the four -53.5 V power inputs into two to supply to the fan module, IPU, and PICs. The IPU provides two 3.3 V power inputs, which are combined on the backplane and then supplied to the fan module, power modules, APIUCs, and PICs.

Figure 1-29 AC power system architecture (Including 4*AC Power)

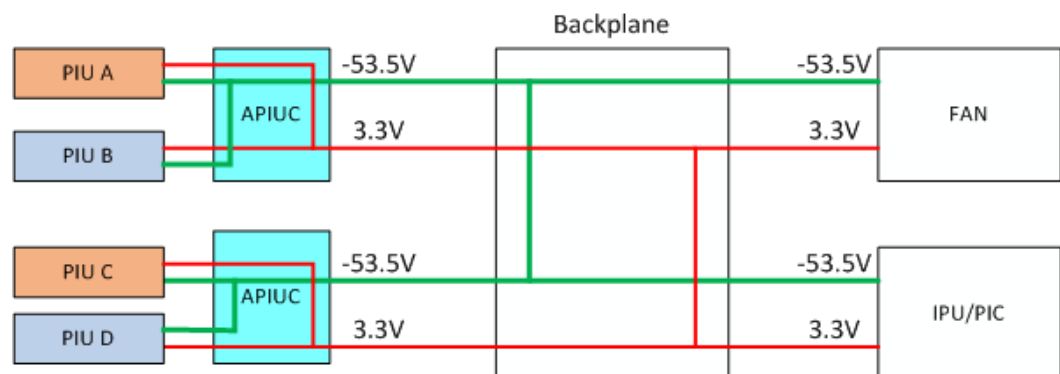
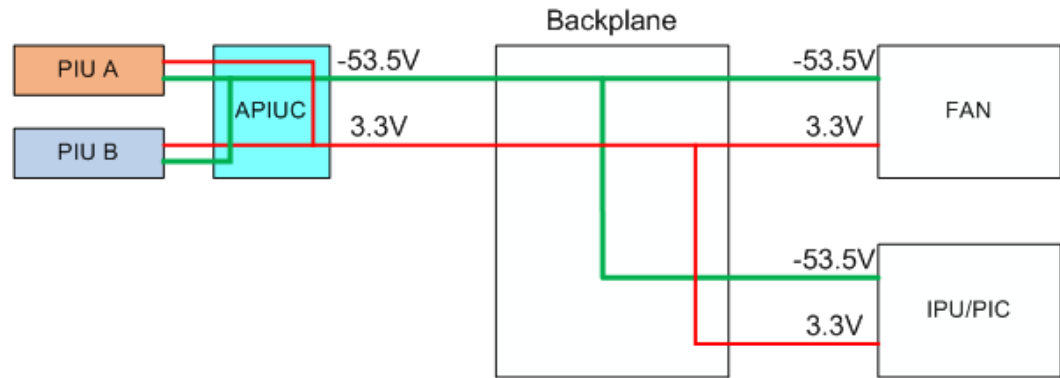


Figure 1-30 shows the 2*AC power supply architecture. Two power modules provide two -53.5 V power inputs, which are combined on the backplane and then supplied to the fan module, IPU, and interface boards. The IPU provides two 3.3 V power inputs, which are combined on the backplane and then supplied to the fan module, power modules, and interface boards.

Figure 1-30 AC power system architecture (Including 2*AC Power)



Heat Dissipation System

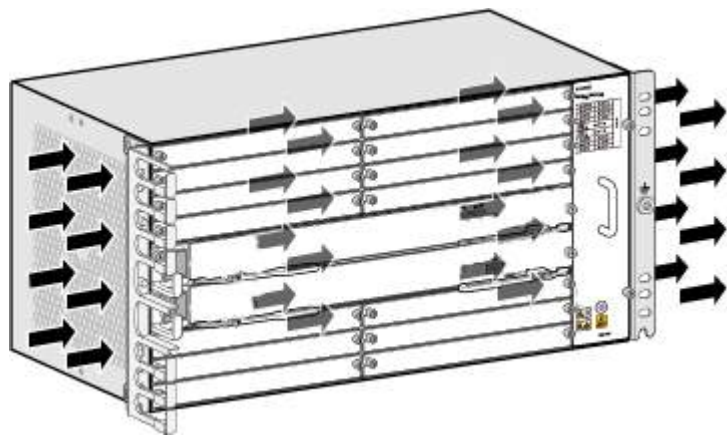
The fans exhaust air from the chassis to dissipate heat. The hot air inside the chassis is sucked out of the subrack to form a ventilation channel from the air intake vent on the left side to the air exhaust vent on the right side. The fan tray assembly consists of four independent fans to dissipate heat for the device and is hot swappable.

NOTICE

- Ensure that heat dissipation holes for the air intake and exhaust vents are clean and not blocked.
- The empty slot should be covered with a filler panel to ensure good electromagnetic compatibility, dustproof, and heat dissipation.
- Keep any objects or obstructions at least 75 mm away from the air intake and exhaust vents to facilitate heat dissipation.
- When installing devices, ensure that the total heat consumption of all devices in the cabinet is less than or equal to the heat dissipation capability of the cabinet.
- The equipment must be stacked in the cabinet or installed at an interval of 2U or more to prevent the return air from affecting heat dissipation.
- When the chassis is installed together with other equipment in the same cabinet, do not install the chassis near the air exhaust vent of other equipment.
- You are required to replace the fan tray as soon as possible; otherwise, the active/standby switchover, system reset, or board burning may occur due to high temperature. Exercise caution when performing this operation. The fan tray needs to be replaced within one minute at the normal temperature (25°C). Note that the four independent fans in the fan tray cannot be replaced separately.

Figure 1-31 shows the air flow in the device.

Figure 1-31 Air flow in the device



Technical Specifications

Table 1-21 Technical specifications of the CR8B0BKPDC1

Item	Specification
Cabinet installation standards	ETSI 21-inch; IEC 19-inch
Dimensions without packaging (H x W x D) [mm(in.)]	222 mm x 442 mm x 220 mm (8.74 in. x 17.40 in. x 8.66 in.)
Dimensions with packaging (H x W x D) [mm(in.)]	505 mm x 590 mm x 390 mm (19.88 in. x 23.23 in. x 15.35 in.)
Chassis height [U]	5U
Weight without packaging [kg(lb)]	10.1 kg (22.27 lb)
Weight with packaging [kg(lb)]	21.1 kg (46.52 lb)
Weight without packaging (full configuration) [kg(lb)]	[IPU-2T/IPU-2T-B]DC:28.5 kg(62.84 lb) [IPU-1T2/IPU-1T2-B]DC:28.5 kg(62.84 lb) [IPU-1T2-A/IPU-1T2-BN]DC:25.6 kg(56.44 lb)
Weight with packaging (full configuration) [kg(lb)]	[IPU-2T/IPU-2T-B]DC:39.5 kg(87.08 lb) [IPU-1T2/IPU-1T2-B]DC:39.5 kg(87.08 lb) [IPU-1T2-A/IPU-1T2-BN]DC:36.6 kg(80.69lb)

Item	Specification
Typical power consumption (with configuration) [W]	[IPU-2T/IPU-2T-B]DC: 1*fan+2*DC power+2*IPU +2*CR5DE2NE4X14+1*CR5D00E8XM25 +3*CR5D00LAXF91+1*CR5D00EAGF95=905.6 W [IPU-1T2/IPU-1T2-B]DC:1*fan+2*DC power+2*IPU +2*CR5DE2NE4X14+3*CR5D00LAXF91+2*CR5D00EAGF95=875.15 W [IPU-1T2-A/IPU-1T2-BN]DC:1*fan+2*DC power+2*IPU +2*CR5D0E5XMF94+2*CR5D00LAXF91+2*CR5D00EAGF70=454.12 W
Typical heat dissipation (with configuration) [BTU/hour]	[IPU-2T/IPU-2T-B]DC: 1*fan+2*DC power+2*IPU +2*CR5DE2NE4X14+1*CR5D00E8XM25 +3*CR5D00LAXF91+1*CR5D00EAGF95=2859.5 BTU/hour [IPU-1T2/IPU-1T2-B]DC:1*fan+2*DC power+2*IPU +2*CR5DE2NE4X14+3*CR5D00LAXF91+2*CR5D00EAGF95=2761.75 BTU/hour [IPU-1T2-A/IPU-1T2-BN]DC:1*fan+2*DC power+2*IPU +2*CR5D0E5XMF94+2*CR5D00LAXF91+2*CR5D00EAGF70=933.94 BTU/hour
MTBF [year]	147.1 year
MTTR [hour]	0.5 hour
Availability	0.999999612
Power supply mode	DC
Rated input voltage [V]	-48 V/-60 V
Input voltage range [V]	-40 V to -72 V
Maximum input current [A]	60 A
Rated output power [W]	2400 W
Maximum input cable size [mm ²]	16mm ² , 10mm ² (0-14meters) recommended, 16mm ² (15-23meters) recommended
Front-end circuit breaker/fuse [A]	≥63 A
Heat dissipation mode	Air cooling
Airflow direction	From left to right

Item	Specification
Noise at normal temperature (acoustic power) [dB(A)]	< 72 dB (meeting the ETSI 72 dBA standard)
Number of slots	19
Number of service board slots	14
Switching capacity	IPU-2T/IPU-2T-B:4 Tbit/s IPU-1T2/IPU-1T2-B:2.4 Tbit/s IPU-1T2-A/IPU-1T2-BN:2.4 Tbit/s
Maximum number of physical ports on the entire device	280
Maximum number of 100GE ports	IPU-2T/IPU-2T-B:20 IPU-1T2/IPU-1T2-B:12 IPU-1T2-A/IPU-1T2-BN:12
Maximum number of 50GE ports	IPU-2T/IPU-2T-B:28 IPU-1T2/IPU-1T2-B:24 IPU-1T2-A/IPU-1T2-BN:20
Maximum number of 40GE ports	IPU-2T/IPU-2T-B:20 IPU-1T2/IPU-1T2-B:12 IPU-1T2-A/IPU-1T2-BN:12
Maximum number of 25GE ports	IPU-2T/IPU-2T-B:80 IPU-1T2/IPU-1T2-B:48 IPU-1T2-A/IPU-1T2-BN:40
Maximum number of 10GE ports	IPU-2T/IPU-2T-B:140 IPU-1T2/IPU-1T2-B:120 IPU-1T2-A/IPU-1T2-BN:100
Maximum number of GE ports	IPU-2T/IPU-2T-B:280 IPU-1T2/IPU-1T2-B:280 IPU-1T2-A/IPU-1T2-BN:240
Maximum number of FE ports	IPU-2T/IPU-2T-B:280 IPU-1T2/IPU-1T2-B:280 IPU-1T2-A/IPU-1T2-BN:240
Maximum number of electrical ports	IPU-2T/IPU-2T-B:140 IPU-1T2/IPU-1T2-B:140 IPU-1T2-A/IPU-1T2-BN:140
Redundant IPU	1:1
Redundant power supply	1+1

Item	Specification
Redundant fans	3+1. Short-term running when a single fan becomes invalid at the ambient temperature of 40°C (104°F)(one fan module contains four fans)
Long-term operating temperature [°C(°F)]	–20°C to +55°C (–4°F to +131°F), startup: ≥ –5°C (23°F)
Restriction on the operating temperature variation rate [°C(°F)]	≤ 0.5°C/min (32.9°F/min), non-condensing
Storage temperature [°C(°F)]	–40°C to +70°C (–40°F to +158 °F)
Long-term operating relative humidity [RH]	5% RH to 85% RH, non-condensing
Short-term operating relative humidity [RH]	5% RH to 95% RH, non-condensing
Storage relative humidity [RH]	5% RH to 95% RH, non-condensing
Long-term operating altitude [m(ft.)]	≤ 4000 m (13123.2 ft.) (For the altitude in the range of 1800 m to 4000 m [5905.44 ft. to 13123.2 ft.], the operating temperature of the device must decrease by 1°C [1.8°F] for every 220 m [721.78 ft.])
Storage altitude [m(ft.)]	< 5000 m (16404.2 ft.)

Table 1-22 Technical specifications of the CR8B0BKPACC2

Item	Specification
Cabinet installation standards	ETSI 21-inch; IEC 19-inch
Dimensions without packaging (H x W x D) [mm(in.)]	222 mm x 442 mm x 220 mm (8.74 in. x 17.40 in. x 8.66 in.)
Dimensions with packaging (H x W x D) [mm(in.)]	505 mm x 590 mm x 390 mm (19.88 in. x 23.23 in. x 15.35 in.)
Chassis height [U]	5U
Weight without packaging [kg(lb)]	14.6 kg (32.19 lb)
Weight with packaging [kg(lb)]	25.6 kg (56.44 lb)
Weight without packaging (full configuration) [kg(lb)]	[IPU-2T/IPU-2T-B]AC:30.0 kg(66.15 lb) [IPU-1T2/IPU-1T2-B]AC:30.0 kg(66.15 lb)

Item	Specification
Weight with packaging (full configuration) [kg(lb)]	[IPU-2T/IPU-2T-B]AC:41.0 kg(90.39 lb) [IPU-1T2/IPU-1T2-B]AC:41.0 kg(90.39 lb)
Typical power consumption (with configuration) [W]	[IPU-2T/IPU-2T-B]AC: 1*fan+4*AC power+2*IPU +2*CR5DE2NE4X14+1*CR5D00E8XM25 +3*CR5D00LAXF91+1*CR5D00EAGF95=968.36 W [IPU-1T2/IPU-1T2-B]AC: 1*fan+2*AC power+2*IPU +2*CR5DE2NE4X14+3*CR5D00LAXF91+2*CR5D00EAGF95=935.81 W
Typical heat dissipation (with configuration) [BTU/hour]	[IPU-2T/IPU-2T-B]AC: 1*fan+4*AC power+2*IPU +2*CR5DE2NE4X14+1*CR5D00E8XM25 +3*CR5D00LAXF91+1*CR5D00EAGF95=3057.69 BTU/hour [IPU-1T2/IPU-1T2-B]AC: 1*fan+2*AC power+2*IPU +2*CR5DE2NE4X14+3*CR5D00LAXF91+2*CR5D00EAGF95=2953.16 BTU/hour
MTBF [year]	147.34 year
MTTR [hour]	0.5 hour
Availability	0.999999613
Power supply mode	AC
Rated input voltage [V]	200V to 240V/100V to 127V dual live wires, support 240V HVDC
Input voltage range [V]	90 V to 290 V
Maximum input current [A]	10 A
Rated output power [W]	2500 W
Maximum input cable size [mm ²]	Standard C13 cable
Front-end circuit breaker/fuse [A]	≥10 A
Heat dissipation mode	Air cooling
Airflow direction	From left to right
Noise at normal temperature (acoustic power) [dB(A)]	< 72 dB (meeting the ETSI 72 dBA standard)
Number of slots	17
Number of service board slots	10

Item	Specification
Switching capacity	IPU-2T/IPU-2T-B:4 Tbit/s IPU-1T2/IPU-1T2-B:2.4 Tbit/s
Maximum number of physical ports on the entire device	200
Maximum number of 100GE ports	IPU-2T/IPU-2T-B:16 IPU-1T2/IPU-1T2-B:12
Maximum number of 50GE ports	IPU-2T/IPU-2T-B:20 IPU-1T2/IPU-1T2-B:20
Maximum number of 40GE ports	IPU-2T/IPU-2T-B:16 IPU-1T2/IPU-1T2-B:12
Maximum number of 25GE ports	IPU-2T/IPU-2T-B:64 IPU-1T2/IPU-1T2-B:48
Maximum number of 10GE ports	IPU-2T/IPU-2T-B:100 IPU-1T2/IPU-1T2-B:100
Maximum number of GE ports	IPU-2T/IPU-2T-B:200 IPU-1T2/IPU-1T2-B:200
Maximum number of FE ports	IPU-2T/IPU-2T-B:200 IPU-1T2/IPU-1T2-B:200
Maximum number of electrical ports	IPU-2T/IPU-2T-B:100 IPU-1T2/IPU-1T2-B:100
Redundant IPU	1:1
Redundant power supply	2+2
Redundant fans	3+1. Short-term running when a single fan becomes invalid at the ambient temperature of 40°C (104°F)(one fan module contains four fans)
Long-term operating temperature [°C(°F)]	-20°C to +55°C (-4°F to +131°F), startup: ≥ -5°C (23°F)
Restriction on the operating temperature variation rate [°C(°F)]	≤ 0.5°C/min (32.9°F/min), non-condensing
Storage temperature [°C(°F)]	-40°C to +70°C (-40°F to +158 °F)
Long-term operating relative humidity [RH]	5% RH to 85% RH, non-condensing
Short-term operating relative humidity [RH]	5% RH to 95% RH, non-condensing

Item	Specification
Storage relative humidity [RH]	5% RH to 95% RH, non-condensing
Long-term operating altitude [m(ft.)]	≤ 4000 m (13123.2 ft.) (For the altitude in the range of 1800 m to 4000 m [5905.44 ft. to 13123.2 ft.], the operating temperature of the device must decrease by 1°C [1.8°F] for every 220 m [721.78 ft.])
Storage altitude [m(ft.)]	< 5000 m (16404.2 ft.)

Table 1-23 Technical specifications of the CR8B0BKPACC1

Item	Specification
Cabinet installation standards	ETSI 21-inch; IEC 19-inch
Dimensions without packaging (H x W x D) [mm(in.)]	222 mm x 442 mm x 220 mm (8.74 in. x 17.40 in. x 8.66 in.)
Dimensions with packaging (H x W x D) [mm(in.)]	505 mm x 590 mm x 390 mm (19.88 in. x 23.23 in. x 15.35 in.)
Chassis height [U]	5U
Weight without packaging [kg(lb)]	12.1 kg (26.68 lb)
Weight with packaging [kg(lb)]	23.1 kg (50.93 lb)
Weight without packaging (full configuration) [kg(lb)]	[IPU-1T2-A/IPU-1T2-BN]AC:26.0 kg(57.32 lb)
Weight with packaging (full configuration) [kg(lb)]	[IPU-1T2-A/IPU-1T2-BN]AC:37.0 kg(81.57 lb)
Typical power consumption (with configuration) [W]	[IPU-1T2-A/IPU-1T2-BN]AC: 1*fan +2*AC power+2*IPU +2*CR5D0E5XMF94+2*CR5D00LAXF91 +2*CR5D00EAGF70=485.59 W
Typical heat dissipation (with configuration) [BTU/hour]	[IPU-1T2-A/IPU-1T2-BN]AC: 1*fan +2*AC power+2*IPU +2*CR5D0E5XMF94+3*CR5D00LAXF91 +2*CR5D00EAGF70=988.03 BTU/hour
MTBF [year]	147.34 year
MTTR [hour]	0.5 hour
Availability	0.999999613
Power supply mode	AC

Item	Specification
Rated input voltage [V]	200V to 240V/100V to 127V dual live wires, support 240V HVDC
Input voltage range [V]	90 V to 290 V
Maximum input current [A]	10 A
Rated output power [W]	1500 W
Maximum input cable size [mm ²]	Standard C13 cable
Front-end circuit breaker/fuse [A]	≥10 A
Heat dissipation mode	Air cooling
Airflow direction	From left to right
Noise at normal temperature (acoustic power) [dB(A)]	< 72 dB (meeting the ETSI 72 dBA standard)
Number of slots	17
Number of service board slots	12
Switching capacity	IPU-1T2-A/IPU-1T2-BN:2.4 Tbit/s
Maximum number of physical ports on the entire device	220
Maximum number of 100GE ports	IPU-1T2-A/IPU-1T2-BN:11
Maximum number of 50GE ports	IPU-1T2-A/IPU-1T2-BN:18
Maximum number of 40GE ports	IPU-1T2-A/IPU-1T2-BN:11
Maximum number of 25GE ports	IPU-1T2-A/IPU-1T2-BN:36
Maximum number of 10GE ports	IPU-1T2-A/IPU-1T2-BN:90
Maximum number of GE ports	IPU-1T2-A/IPU-1T2-BN:220
Maximum number of FE ports	IPU-1T2-A/IPU-1T2-BN:220
Maximum number of electrical ports	IPU-1T2-A/IPU-1T2-BN:120
Redundant IPU	1:1
Redundant power supply	1+1
Redundant fans	3+1. Short-term running when a single fan becomes invalid at the ambient temperature of 40°C (104°F)(one fan module contains four fans)
Long-term operating temperature [°C(°F)]	-20°C to +55°C (-4°F to +131°F), startup: ≥ -5°C (23°F)

Item	Specification
Restriction on the operating temperature variation rate [°C(°F)]	≤ 0.5°C/min (32.9°F/min), non-condensing
Storage temperature [°C(°F)]	–40°C to +70°C (–40°F to +158 °F)
Long-term operating relative humidity [RH]	5% RH to 85% RH, non-condensing
Short-term operating relative humidity [RH]	5% RH to 95% RH, non-condensing
Storage relative humidity [RH]	5% RH to 95% RH, non-condensing
Long-term operating altitude [m(ft.)]	≤ 4000 m (13123.2 ft.) (For the altitude in the range of 1800 m to 4000 m [5905.44 ft. to 13123.2 ft.], the operating temperature of the device must decrease by 1°C [1.8°F] for every 220 m [721.78 ft.])
Storage altitude [m(ft.)]	< 5000 m (16404.2 ft.)

Table 1-24 Technical specifications of the CR8B0BKPACC6

Item	Specification
Cabinet installation standards	ETSI 21-inch; IEC 19-inch
Dimensions without packaging (H x W x D) [mm(in.)]	222 mm x 442 mm x 220 mm (8.74 in. x 17.40 in. x 8.66 in.)
Dimensions with packaging (H x W x D) [mm(in.)]	505 mm x 590 mm x 390 mm (19.88 in. x 23.23 in. x 15.35 in.)
Chassis height [U]	5U
Weight without packaging [kg(lb)]	12.2 kg (26.9 lb)
Weight with packaging [kg(lb)]	23.15 kg (51.04 lb)
Weight without packaging (full configuration) [kg(lb)]	[IPU-2T/IPU-2T-B]AC:29.1 kg(64.15 lb) [IPU-1T2/IPU-1T2-B]AC:29.1 kg(64.15 lb)
Weight with packaging (full configuration) [kg(lb)]	[IPU-2T/IPU-2T-B]AC:40.1 kg(88.41 lb) [IPU-1T2/IPU-1T2-B]AC:40.1 kg(88.41 lb)

Item	Specification
Typical power consumption (with configuration) [W]	[IPU-2T/IPU-2T-B]AC: 1*fan+2*AC power+2*IPU +2*CR5DE2NE4X14+1*CR5D00E8XM25 +3*CR5D00LAXF91+1*CR5D00EAGF95=968.36 W [IPU-1T2/IPU-1T2-B]AC: 1*fan+2*AC power+2*IPU +2*CR5DE2NE4X14+3*CR5D00LAXF91+2*CR5D00EAGF95=935.81 W
Typical heat dissipation (with configuration) [BTU/hour]	[IPU-2T/IPU-2T-B]AC: 1*fan+2*AC power+2*IPU +2*CR5DE2NE4X14+1*CR5D00E8XM25 +3*CR5D00LAXF91+1*CR5D00EAGF95=3057.69 BTU/hour [IPU-1T2/IPU-1T2-B]AC: 1*fan+2*AC power+2*IPU +2*CR5DE2NE4X14+3*CR5D00LAXF91+2*CR5D00EAGF95=2953.16 BTU/hour
MTBF [year]	147.34 year
MTTR [hour]	0.5 hour
Availability	0.999999613
Power supply mode	AC
Rated input voltage [V]	200V to 240V/100V to 127V dual live wires, support 240V HVDC
Input voltage range [V]	90 V to 290 V
Maximum input current [A]	16 A
Rated output power [W]	2400 W
Maximum input cable size [mm ²]	Customized cable
Front-end circuit breaker/fuse [A]	≥16A
Heat dissipation mode	Air cooling
Airflow direction	From left to right
Noise at normal temperature (acoustic power) [dB(A)]	< 72 dB (meeting the ETSI 72 dBA standard)
Number of slots	17
Number of service board slots	12
Switching capacity	IPU-2T/IPU-2T-B:4 Tbit/s IPU-1T2/IPU-1T2-B:2.4 Tbit/s

Item	Specification
Maximum number of physical ports on the entire device	240
Maximum number of 400GE ports	IPU-2T/IPU-2T-B:2 IPU-1T2/IPU-1T2-B:2
Maximum number of 100GE ports	IPU-2T/IPU-2T-B:18 IPU-1T2/IPU-1T2-B:12
Maximum number of 50GE ports	IPU-2T/IPU-2T-B:24 IPU-1T2/IPU-1T2-B:24
Maximum number of 40GE ports	IPU-2T/IPU-2T-B:18 IPU-1T2/IPU-1T2-B:12
Maximum number of 25GE ports	IPU-2T/IPU-2T-B:72 IPU-1T2/IPU-1T2-B:48
Maximum number of 10GE ports	IPU-2T/IPU-2T-B:120 IPU-1T2/IPU-1T2-B:120
Maximum number of GE ports	IPU-2T/IPU-2T-B:240 IPU-1T2/IPU-1T2-B:240
Maximum number of FE ports	IPU-2T/IPU-2T-B:240 IPU-1T2/IPU-1T2-B:240
Maximum number of electrical ports	IPU-2T/IPU-2T-B:240 IPU-1T2/IPU-1T2-B:240
Redundant IPU	1:1
Redundant power supply	1+1
Redundant fans	3+1. Short-term running when a single fan becomes invalid at the ambient temperature of 40°C (104°F)(one fan module contains four fans)
Long-term operating temperature [°C(°F)]	-20°C to +55°C (-4°F to +131°F), startup: ≥ -5°C (23°F)
Restriction on the operating temperature variation rate [°C(°F)]	≤ 0.5°C/min (32.9°F/min), non-condensing
Storage temperature [°C(°F)]	-40°C to +70°C (-40°F to +158 °F)
Long-term operating relative humidity [RH]	5% RH to 85% RH, non-condensing
Short-term operating relative humidity [RH]	5% RH to 95% RH, non-condensing

Item	Specification
Storage relative humidity [RH]	5% RH to 95% RH, non-condensing
Long-term operating altitude [m(ft.)]	≤ 4000 m (13123.2 ft.) (For the altitude in the range of 1800 m to 4000 m [5905.44 ft. to 13123.2 ft.], the operating temperature of the device must decrease by 1°C [1.8°F] for every 220 m [721.78 ft.])
Storage altitude [m(ft.)]	< 5000 m (16404.2 ft.)

Table 1-25 Technical specifications of the CR8B0BKPACM1

Item	Specification
Cabinet installation standards	ETSI 21-inch; IEC 19-inch
Dimensions without packaging (H x W x D) [mm(in.)]	222 mm x 442 mm x 220 mm (8.74 in. x 17.40 in. x 8.66 in.)
Dimensions with packaging (H x W x D) [mm(in.)]	505 mm x 590 mm x 390 mm (19.88 in. x 23.23 in. x 15.35 in.)
Chassis height [U]	5U
Weight without packaging [kg(lb)]	12.3 kg (27.12 lb)
Weight with packaging [kg(lb)]	23.3 kg (51.37 lb)
Weight without packaging (full configuration) [kg(lb)]	[IPU-3T6-A/IPU-3T6-E-B]AC:29.3 kg(64.6 lb) [IPU-2T-AU/IPU-2T-E-BU]AC:29.3 kg(64.6 lb)
Weight with packaging (full configuration) [kg(lb)]	[IPU-3T6-A/IPU-3T6-E-B]AC:40.3 kg(88.85 lb) [IPU-2T-AU/IPU-2T-E-BU]AC:40.3 kg(88.85 lb)
Typical power consumption (with configuration) [W]	[IPU-3T6-A/IPU-3T6-E-B]AC: 1*fan +2*AC power+2*IPU +2*CR8DE1KB00M1+2*CR8DE4NB00M1+2*CR5D00LAXF91+2*CR5D00EAGF95=931.67 W [IPU-2T-AU/IPU-2T-E-BU]AC: 1*fan +2*AC power+2*IPU +2*CR5DE2NE4X14+1*CR5D00E8XM25 +3*CR5D00LAXF91+1*CR5D00EAGF95=791.85 W

Item	Specification
Typical heat dissipation (with configuration) [BTU/hour]	[IPU-3T6-A/IPU-3T6-E-B]AC: 1*fan +2*AC power+2*IPU +2*CR8DE1KB00M1+2*CR8DE4NB00M 1+2*CR5D00LAXF91+2*CR5D00EAGF95 =2950.33 BTU/hour [IPU-2T-AU/IPU-2T-E-BU]AC: 1*fan +2*AC power+2*IPU +2*CR5DE2NE4X14+1*CR5D00E8XM25 +3*CR5D00LAXF91+1*CR5D00EAGF95= 2496.69 BTU/hour
MTBF [year]	147.34 year
MTTR [hour]	0.5 hour
Availability	0.999999613
Power supply mode	AC
Rated input voltage [V]	200V to 240V/100V to 127V dual live wires, support 240V HVDC
Input voltage range [V]	90 V to 290 V
Maximum input current [A]	16 A
Rated output power [W]	2400 W
Maximum input cable size [mm ²]	Customized cable
Front-end circuit breaker/fuse [A]	≥16A
Heat dissipation mode	Air cooling
Airflow direction	From left to right
Noise at normal temperature (acoustic power) [dB(A)]	< 72 dB (meeting the ETSI 72 dBA standard)
Number of slots	17
Number of service board slots	12
Switching capacity	IPU-3T6-A/IPU-3T6-E-B: 7.2 Tbit/s IPU-2T-AU/IPU-2T-E-BU: 4 Tbit/s
Maximum number of physical ports on the entire device	240
Maximum number of 400GE ports	IPU-3T6-A/IPU-3T6-E-B:6 IPU-2T-AU/IPU-2T-E-BU:5
Maximum number of 100GE ports	IPU-3T6-A/IPU-3T6-E-B:33 IPU-2T-AU/IPU-2T-E-BU:20

Item	Specification
Maximum number of 50GE ports	IPU-3T6-A/IPU-3T6-E-B:36 IPU-2T-AU/IPU-2T-E-BU:22
Maximum number of 40GE ports	IPU-3T6-A/IPU-3T6-E-B:33 IPU-2T-AU/IPU-2T-E-BU:20
Maximum number of 25GE ports	IPU-3T6-A/IPU-3T6-E-B:120 IPU-2T-AU/IPU-2T-E-BU:120
Maximum number of 10GE ports	IPU-3T6-A/IPU-3T6-E-B:120 IPU-2T-AU/IPU-2T-E-BU:120
Maximum number of GE ports	IPU-3T6-A/IPU-3T6-E-B:240 IPU-2T-AU/IPU-2T-E-BU:240
Maximum number of FE ports	IPU-3T6-A/IPU-3T6-E-B:240 IPU-2T-AU/IPU-2T-E-BU:240
Maximum number of electrical ports	IPU-3T6-A/IPU-3T6-E-B:240 IPU-2T-AU/IPU-2T-E-BU:240
Redundant IPU	1:1
Redundant power supply	1+1
Redundant fans	3+1. Short-term running when a single fan becomes invalid at the ambient temperature of 40°C (104°F)(one fan module contains four fans)
Long-term operating temperature [°C(°F)]	–20°C to +55°C (–4°F to +131°F), startup: ≥ –5°C (23°F)
Restriction on the operating temperature variation rate [°C(°F)]	≤ 0.5°C/min (32.9°F/min), non-condensing
Storage temperature [°C(°F)]	–40°C to +70°C (–40°F to +158 °F)
Long-term operating relative humidity [RH]	5% RH to 85% RH, non-condensing
Short-term operating relative humidity [RH]	5% RH to 95% RH, non-condensing
Storage relative humidity [RH]	5% RH to 95% RH, non-condensing
Long-term operating altitude [m(ft.)]	≤ 4000 m (13123.2 ft.) (For the altitude in the range of 1800 m to 4000 m [5905.44 ft. to 13123.2 ft.], the operating temperature of the device must decrease by 1°C [1.8°F] for every 220 m [721.78 ft.])

Item	Specification
Storage altitude [m(ft.)]	< 5000 m (16404.2 ft.)

Table 1-26 Technical specifications of the CR8B0BKPDCM1

Item	Specification
Cabinet installation standards	ETSI 21-inch; IEC 19-inch
Dimensions without packaging (H x W x D) [mm(in.)]	222 mm x 442 mm x 220 mm (8.74 in. x 17.40 in. x 8.66 in.)
Dimensions with packaging (H x W x D) [mm(in.)]	505 mm x 590 mm x 390 mm (19.88 in. x 23.23 in. x 15.35 in.)
Chassis height [U]	5U
Weight without packaging [kg(lb)]	10.4 kg (22.93 lb)
Weight with packaging [kg(lb)]	21.4 kg (47.18 lb)
Weight without packaging (full configuration) [kg(lb)]	[IPU-3T6-A/IPU-3T6-E-B]AC:29.0 kg(63.93 lb) [IPU-2T-AU/IPU-2T-E-BU]AC:29.0 kg(63.93 lb)
Weight with packaging (full configuration) [kg(lb)]	[IPU-3T6-A/IPU-3T6-E-B]AC:40.0 kg(88.18 lb) [IPU-2T-AU/IPU-2T-E-BU]AC:40.0 kg(88.18 lb)
Typical power consumption (with configuration) [W]	[IPU-3T6-A/IPU-3T6-E-B]DC: 1*fan +2*DC power+2*IPU +2*CR8DE1KB00M1+2*CR8DE4NB00M1+2*CR5D00LAXF91+2*CR5D00EAGF95=865.79 W [IPU-2T-AU/IPU-2T-E-BU]DC: 1*fan +2*DC power+2*IPU +2*CR5DE2NE4X14+1*CR5D00E8XM25 +3*CR5D00LAXF91+1*CR5D00EAGF95=735.86 W

Item	Specification
Typical heat dissipation (with configuration) [BTU/hour]	[IPU-3T6-A/IPU-3T6-E-B]DC: 1*fan +2*DC power+2*IPU +2*CR8DE1KB00M1+2*CR8DE4NB00M 1+2*CR5D00LAXF91+2*CR5D00EAGF95 =2741.72 BTU/hour [IPU-2T-AU/IPU-2T-E-BU]DC: 1*fan +2*DC power+2*IPU +2*CR5DE2NE4X14+1*CR5D00E8XM25 +3*CR5D00LAXF91+1*CR5D00EAGF95= 2320.16 BTU/hour
MTBF [year]	147.1 year
MTTR [hour]	0.5 hour
Availability	0.999999612
Power supply mode	DC
Rated input voltage [V]	-48 V/-60 V
Input voltage range [V]	-40 V to -72 V
Maximum input current [A]	60 A
Rated output power [W]	2400 W
Maximum input cable size [mm ²]	16mm ² , 10mm ² (0-14meters) recommended, 16mm ² (15-23meters) recommended
Front-end circuit breaker/fuse [A]	≥63 A
Heat dissipation mode	Air cooling
Airflow direction	From left to right
Noise at normal temperature (acoustic power) [dB(A)]	< 72 dB (meeting the ETSI 72 dBA standard)
Number of slots	19
Number of service board slots	14
Switching capacity	IPU-3T6-A/IPU-3T6-E-B: 7.2 Tbit/s IPU-2T-AU/IPU-2T-E-BU: 4 Tbit/s
Maximum number of physical ports on the entire device	280
Maximum number of 400GE ports	IPU-3T6-A/IPU-3T6-E-B:6 IPU-2T-AU/IPU-2T-E-BU:5
Maximum number of 100GE ports	IPU-3T6-A/IPU-3T6-E-B:36 IPU-2T-AU/IPU-2T-E-BU:20

Item	Specification
Maximum number of 50GE ports	IPU-3T6-A/IPU-3T6-E-B:40 IPU-2T-AU/IPU-2T-E-BU:24
Maximum number of 40GE ports	IPU-3T6-A/IPU-3T6-E-B:36 IPU-2T-AU/IPU-2T-E-BU:20
Maximum number of 25GE ports	IPU-3T6-A/IPU-3T6-E-B:140 IPU-2T-AU/IPU-2T-E-BU:140
Maximum number of 10GE ports	IPU-3T6-A/IPU-3T6-E-B:140 IPU-2T-AU/IPU-2T-E-BU:140
Maximum number of GE ports	IPU-3T6-A/IPU-3T6-E-B:280 IPU-2T-AU/IPU-2T-E-BU:280
Maximum number of FE ports	IPU-3T6-A/IPU-3T6-E-B:280 IPU-2T-AU/IPU-2T-E-BU:280
Maximum number of electrical ports	IPU-3T6-A/IPU-3T6-E-B:280 IPU-2T-AU/IPU-2T-E-BU:280
Redundant IPU	1:1
Redundant power supply	1+1
Redundant fans	3+1. Short-term running when a single fan becomes invalid at the ambient temperature of 40°C (104°F)(one fan module contains four fans)
Long-term operating temperature [°C(°F)]	–20°C to +55°C (–4°F to +131°F), startup: ≥ –5°C (23°F)
Restriction on the operating temperature variation rate [°C(°F)]	≤ 0.5°C/min (32.9°F/min), non-condensing
Storage temperature [°C(°F)]	–40°C to +70°C (–40°F to +158 °F)
Long-term operating relative humidity [RH]	5% RH to 85% RH, non-condensing
Short-term operating relative humidity [RH]	5% RH to 95% RH, non-condensing
Storage relative humidity [RH]	5% RH to 95% RH, non-condensing
Long-term operating altitude [m(ft.)]	≤ 4000 m (13123.2 ft.) (For the altitude in the range of 1800 m to 4000 m [5905.44 ft. to 13123.2 ft.], the operating temperature of the device must decrease by 1°C [1.8°F] for every 220 m [721.78 ft.])

Item	Specification
Storage altitude [m(ft.)]	< 5000 m (16404.2 ft.)

 NOTE

- About condensation

The ability of air to contain moisture reduces as the temperature decreases. There is a relationship between temperature and relative humidity. As the temperature decreases, the air becomes more saturated with moisture, meaning the relative humidity increases. But at a low enough temperature (called the dew point), the air can no longer contain moisture, which is then precipitated as liquid water. Such water on the surface of relevant equipment is called condensation.

- Hazards of condensation

When condensation mixes with dust inside the equipment, it forms conductive channels that affect the electrical insulation of the equipment. As a result, a non-conductive area of the equipment can become conductive, causing the equipment to malfunction.

- Conditions that cause condensation

Condensation is related to temperature changes and humidity. It occurs when the temperature continues to decrease below the dew point at a given humidity. For example, given that the relative humidity is 50% RH and the temperature is 25°C (77°F), condensation occurs when the temperature decreases below 13.9°C (57.02°F).

1.2 Power

1.2.1 DP22PDC1K2A1 (Power Interface Board)

The device uses two PIUs, which work in 1+1 backup mode, for power supply.

Overview

Table 1-27 Basic information about the DP22PDC1K2A1

Item	Details
Description	Power Interface Board
Part Number	03025LUL
Model	DP22PDC1K2A1

Item	Details
Remarks	1. To replace a DC power module, disconnect the power input and then remove it. 2. To replace a DC power cable, use the puller on the connector to unlock the cable, and then remove the cable. If the power cable is directly removed, the connector may be damaged. 3. DC power module and AC power module cannot be used at the same time.

Appearance

Figure 1-32 Appearance of the DP22PDC1K2A1



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03025LUL&productModel=DP22PDC1K2A1#matchRelation>

Panel

Figure 1-33 Panel of the DP22PDC1K2A1



1. Status indicator	2. Captive screw	3.NEG(-) port	4.RTN(+) port
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Table 1-28 Indicators on the DP22PDC1K2A1

Silkscreen	Name	Color	Status	Description
PWR	Working status indicator	Green	Steady on	The power input is normal.
		-	Off	The power module is switched off or the hardware of the power module is faulty.

Table 1-29 Ports on the DP22PDC1K2A1

Port	Description
NEG(-)	-48 V/-60 V power input interface
RTN(+)	BGND power input interface

Technical Specifications

Table 1-30 Technical specifications of the DP22PDC1K2A1

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	41.6 mm x 11 mm x 209 mm (1.64 in. x 0.43 in. x 8.23 in.)
Weight without packaging [kg(lb)]	0.13 kg (0.29 lb)
Number of inputs	1
Rated input voltage [V]	-48 V/-60 V
Input voltage range [V]	-40 V to -72 V
Maximum input current [A]	30 A
Rated output voltage [V]	-48 V/-60 V
Rated output current [A]	30 A
Rated output power [W]	1200 W
Power supply efficiency	0.99
Power dissipation Mode	Air cooling

Item	Specification
Type of power cables	4 mm ² (0 to 10 meters)

1.2.2 DP32PDC1K6A1 (Power Interface Board)

The device uses two PIUs, which work in 1+1 backup mode, for power supply.

Overview

Table 1-31 Basic information about the DP32PDC1K6A1

Item	Details
Description	Power Interface Board
Part Number	03025PJF
Model	DP32PDC1K6A1
Remarks	1. To replace a DC power module, disconnect the power input and then remove it. 2. To replace a DC power cable, use the puller on the connector to unlock the cable, and then remove the cable. If the power cable is directly removed, the connector may be damaged. 3. DC power module and AC power module cannot be used at the same time.

Appearance

Figure 1-34 Appearance of the DP32PDC1K6A1



Version Mapping

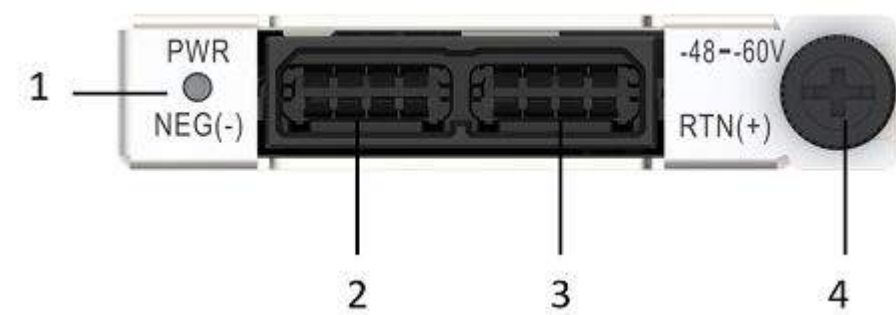
The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03025PJF&productModel=DP32PDC1K6A1#matchRelation>

Panel

Figure 1-35 Panel of the DP32PDC1K6A1



1. Status indicator	2. NEG(-) port	3.RTN(+) port	4.Captive screw
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Table 1-32 Indicators on the DP32PDC1K6A1

Silkscreen	Name	Color	Status	Description
PWR	Working status indicator	Green	Steady on	The power input is normal.
		-	Off	The power module is switched off or the hardware of the power module is faulty.

Table 1-33 Ports on the DP32PDC1K6A1

Port	Description
NEG(-)	-48 V/-60 V power input interface
RTN(+)	BGND power input interface

Technical Specifications

Table 1-34 Technical specifications of the DP32PDC1K6A1

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	11 mm x 53.5 mm x 208.8 mm (0.43 in. x 2.11 in. x 8.22 in.)
Weight without packaging [kg(lb)]	0.2 kg (0.44 lb)
Number of inputs	1
Rated input voltage [V]	-48 V/-60 V
Input voltage range [V]	-40 V to -72 V
Maximum input current [A]	40 A
Rated output voltage [V]	-48 V/-60 V
Rated output current [A]	40 A
Rated output power [W]	1600 W
Power supply efficiency	0.99
Power dissipation Mode	Air cooling

1.2.3 DP52PDC2K4A1 (Power Interface Board)

The device uses two PIUs, which work in 1+1 backup mode, for power supply.

Overview

Table 1-35 Basic information about the DP52PDC2K4A1

Item	Details
Description	Power Interface Board
Part Number	03025PJJ
Model	DP52PDC2K4A1

Item	Details
Remarks	1. Before replacing a DC power module, disconnect the power supply. 2. Before replacing a DC power cable, use the puller on the connector to unlock the power cable and then remove the cable. If you directly remove the power cable, the connector may be damaged. 3. DC power modules and AC power modules cannot be used together.

Appearance

Figure 1-36 Appearance of the DP52PDC2K4A1



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

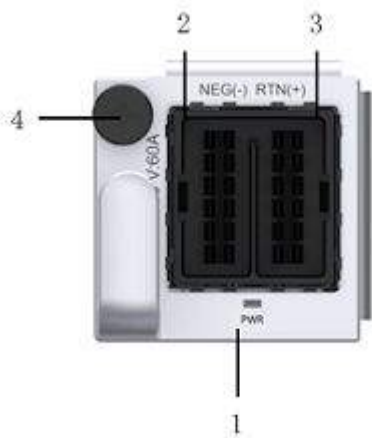
The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find

desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03025PJ&productModel=DP52PDC2K4A1#matchRelation>

Panel

Figure 1-37 Panel of the DP52PDC2K4A1



1. Status indicator	2. NEG(-) port	3.RTN(+) port	4.Captive screw
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Table 1-36 Indicators on the DP52PDC2K4A1

Silkscreen	Name	Color	Status	Description
PWR	Working status indicator	Green	Steady on	The power input is normal.
		-	Off	The power module is switched off or the hardware of the power module is faulty.

Table 1-37 Ports on the DP52PDC2K4A1

Port	Description
NEG(-)	-48 V/-60 V power input interface
RTN(+)	BGND power input interface

Technical Specifications

Table 1-38 Technical specifications of the DP52PDC2K4A1

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	34.8 mm x 35 mm x 209 mm (1.37 in. x 1.38 in. x 8.23 in.)
Weight without packaging [kg(lb)]	0.3 kg (0.44 lb)
Number of inputs	1
Rated input voltage [V]	-48 V/-60 V
Input voltage range [V]	-40 V to -72 V
Maximum input current [A]	60 A
Rated output voltage [V]	-48 V/-60 V
Rated output current [A]	60 A
Rated output power [W]	2400 W
Power supply efficiency	0.99
Power dissipation Mode	Air cooling

1.2.4 PAC1K5S53-AL (1500W AC Power Module)

Overview

Table 1-39 Basic information about the PAC1K5S53-AL

Item	Details
Description	1500W AC Power Module
Part Number	02131663
Model	PAC1K5S53-AL

Item	Details
Remarks	1. Before replacing an AC power module, you are advised to turn off the power input switch. 2. DC power modules and AC power modules cannot be used together.

Appearance

Figure 1-38 Appearance of the PAC1K5S53-AL



NOTE

Before inserting AC power modules into the chassis, install APIUCs (DP52APIUCA00).

Figure 1-39 Appearance of the APIUC



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

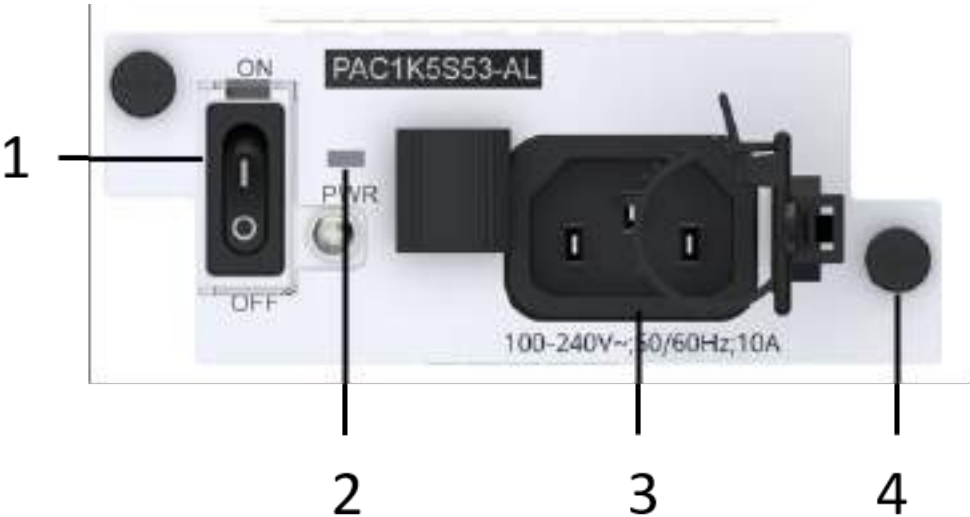
The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find

desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=02131663&productModel=PAC1K5S53-AL#matchRelation>

Panel

Figure 1-40 Panel of the PAC1K5S53-AL



1.ON/OFF button	2.Power supply status indicator	3.AC power cable connector	4.Captive screw
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Table 1-40 Indicators on the PAC1K5S53-AL

Silkscreen	Name	Color	Status	Description
PWR	Working status indicator	Green	Steady on	The power input is normal.
		Green	Blinks (on for 0.1s and off for 0.1s)	A program is being loaded online.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	The power supply is abnormal. An output fault, overtemperature protection alarm, or communication fault occurs.
		-	Off	The power module is switched off or the hardware of the power module is faulty.

Table 1-41 Buttons on the PAC1K5S53-AL

Silkscreen	Name	Description
ON	ON button	Press the ON button to power on the power module.
OFF	OFF button	Press the OFF button to power off the power module.

Table 1-42 Ports on the PAC1K5S53-AL

Port	Description
Power cable connector	C13 power cable interface.

Technical Specifications

Table 1-43 Technical specifications of the PAC1K5S53-AL

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	40 mm x 79.7 mm x 188 mm (1.57 in. x 3.14 in. x 7.4 in.)

Item	Specification
Weight without packaging [kg(lb)]	1 kg (2.2 lb)
Number of inputs	1
Rated input voltage [V]	200V to 240V/100V to 127V dual live wires, support 240V HVDC
Input voltage range [V]	AC: 90 V to 290 V HVDC: 190 V to 290 V
Maximum input current [A]	10 A
Rated output voltage [V]	53.5 V
Rated output current [A]	28.04 A
Rated output power [W]	1500 W. In the 110 V AC scenario, the output power can be 1500 W only when dual live wires are used.
Power supply efficiency	0.92
Power dissipation Mode	Air cooling
Type of power cables	Standard C13 cable

Table 1-44 Technical specifications of the APIUC

Item	Description
Dimensions (H x W x D)	40.1 mm x 193.8 mm x 209.3 mm (1.57 in. x 7.63 in. x 8.24 in.)
Weight	0.5 kg (1.1 lb)

1.2.5 PAH2K4S53-CL (2400W AC Power Module)

Overview

Table 1-45 Basic information about the PAH2K4S53-CL

Item	Details
Description	2400W AC Power Module
Part Number	02131798
Model	PAH2K4S53-CL

Item	Details
Remarks	1. Before replacing an AC power module, you are advised to turn off the power input switch first. 2. DC and AC power modules cannot be used together. 3. It must be used with the CR8XPIU0AC00 AC power transfer board. 4. It cannot be used with the PAC1K5S53-AL AC power module.

Appearance

Figure 1-41 Appearance of the PAH2K4S53-CL



NOTE

Before inserting AC power modules into the chassis, install APIUCs (CR8XPIU0AC00).

Figure 1-42 Appearance of the APIUC



Version Mapping

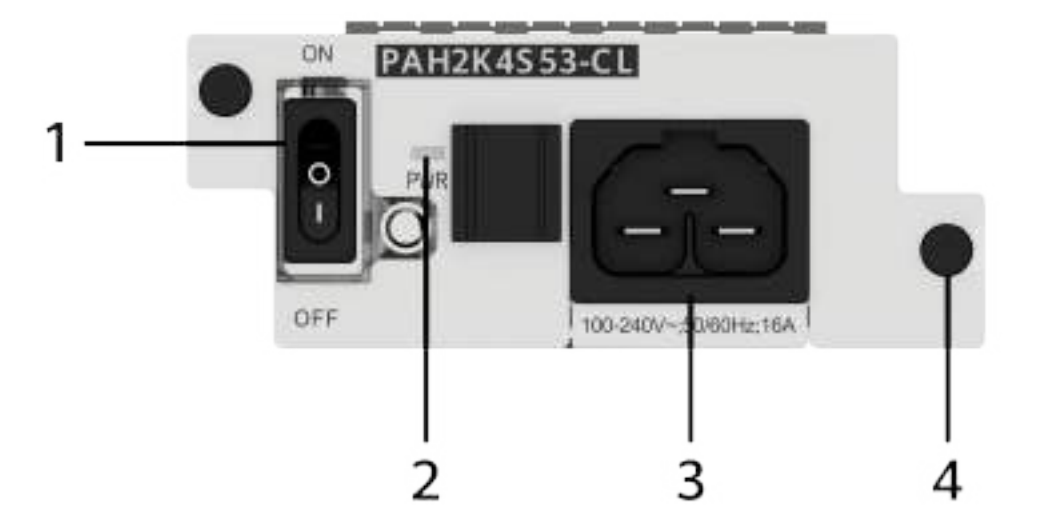
The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=02131798&productModel=PAH2K4S53-CL#matchRelation>

Panel

Figure 1-43 Panel of the PAH2K4S53-CL



1.ON/OFF button	2.Power supply status indicator	3.AC power cable connector	4.Captive screw
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Table 1-46 Indicators on the PAH2K4S53-CL

Silkscreen	Name	Color	Status	Description
PWR	Working status indicator	Green	Steady on	The power input is normal.
		Green	Blinks (on for 0.1s and off for 0.1s)	A program is being loaded online.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	The power supply is abnormal. An output fault, overtemperature protection alarm, or communication fault occurs.
		-	Off	The power module is switched off or the hardware of the power module is faulty.

Table 1-47 Buttons on the PAH2K4S53-CL

Silkscreen	Name	Description
ON	ON button	Press the ON button to power on the power module.
OFF	OFF button	Press the OFF button to power off the power module.

Table 1-48 Ports on the PAH2K4S53-CL

Port	Description
Power cable connector	Customized interface cable

Technical Specifications

Table 1-49 Technical specifications of the PAH2K4S53-CL

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	40 mm x 79.7 mm x 188 mm (1.57 in. x 3.14 in. x 7.4 in.)

Item	Specification
Weight without packaging [kg(lb)]	1 kg (2.20 lb)
Number of inputs	1
Rated input voltage [V]	200 V to 240 V/100 V to 127 V dual live wires, supporting 240 V HVDC
Input voltage range [V]	AC: 90 V to 290 V HVDC: 190 V to 290 V
Maximum input current [A]	16 A
Rated output voltage [V]	53.5 V
Rated output current [A]	44.86 A
Rated output power [W]	190–290 V AC input: 2400 W; 175–190 V AC input: 2200 W; Input voltage below 175 V AC: 1100 W.
Power supply efficiency	0.92
Power dissipation Mode	Air cooling
Type of power cables	Customized cable

Table 1-50 Technical specifications of the APIUC

Item	Description
Dimensions (H x W x D)	40.1 mm x 193.8 mm x 209.3 mm (1.57 in. x 7.63 in. x 8.24 in.)
Weight	0.5 kg (1.1 lb)

1.2.6 ETP4860-2KW-FE (AC Power Box)

Overview

Table 1-51 Basic information about the ETP4860-2KW-FE

Item	Details
Description	AC Power Box
Part Number	02313EAU
Model	ETP4860-2KW-FE

NOTICE

- The external power system can connect to battery strings.
- If a battery string is used, connect the battery string to the battery circuit breaker (do not connect it to other circuit breakers) and set related parameters based on the actual situation of the battery string. Incorrectly setting basic battery parameters may affect the battery charge and discharge management and reduce the battery lifespan. For details about how to power on batteries, see "*Installing Power Boxes (ETP4860-2KW-FE)>Commissioning>Connecting the Battery Supply*".
- If no battery string is connected to the power supply system, you must log in to the power supply system to set **Battery n Connected to No**, and click **Submit**. For details, see "*Installing Power Boxes (ETP4860-2KW-FE)>Commissioning>SMU11B*".
- The BATT circuit breaker can be connected only to batteries and cannot be connected to other devices.

Table 1-52 System configuration

Item	System configuration
Subrack	1 U power system
Power Distribution Unit (PDU)	AC or DC input: M4 OT wiring terminal DC output: supports a maximum of two 63 A circuit breakers, two 40 A circuit breakers, or one 16 A circuit breaker. Battery route: one 50 A circuit breaker
Monitoring module	SMU11B

Appearance

Figure 1-44 Apperance of the ETP4860-2KW-FE



NOTE

The load circuit breaker is red if it is pressed down and green if it is not pressed down.

Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

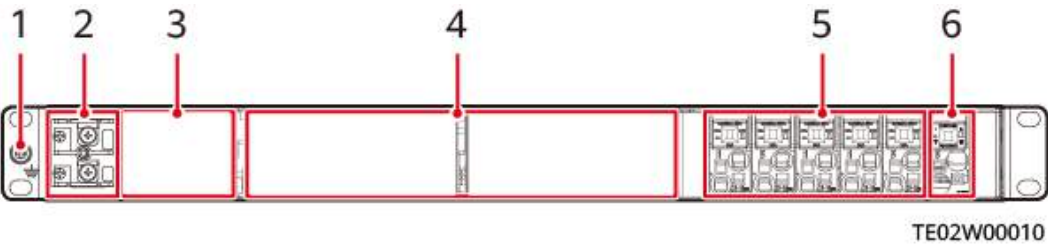
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enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=02313EAU&productModel=ETP4860-2KW-FE#matchRelation>

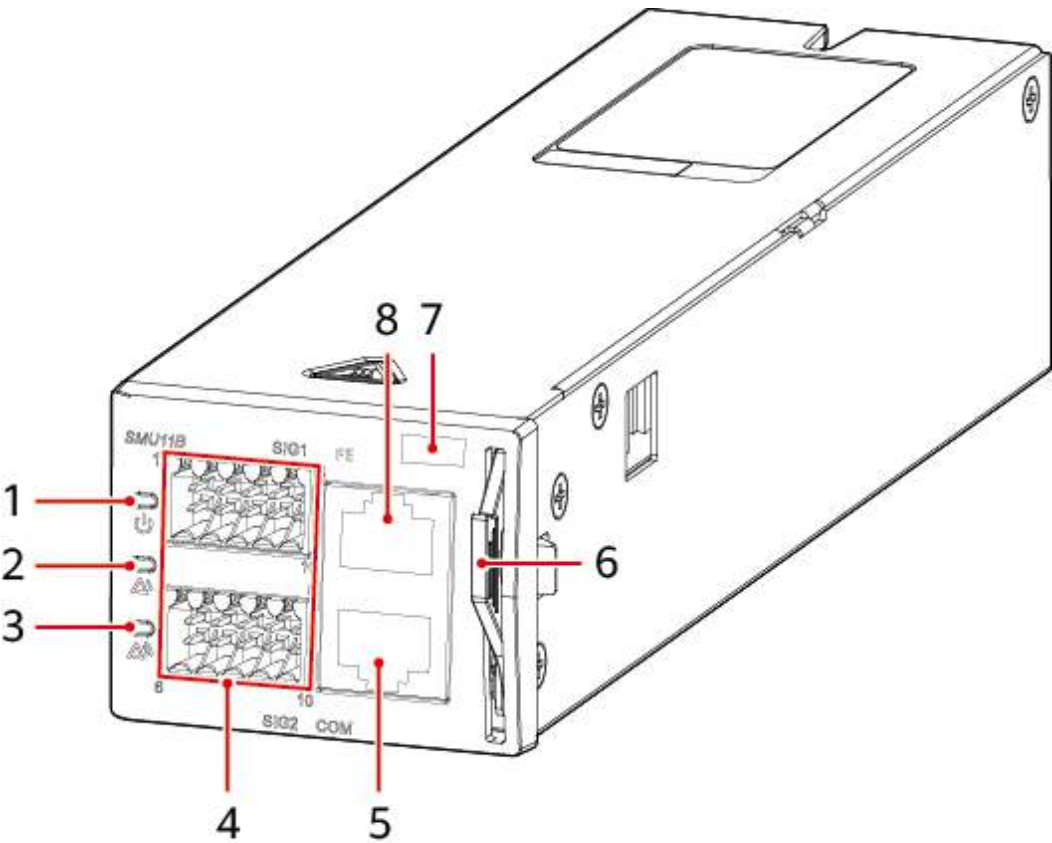
Panel

Figure 1-45 ETP4860-2KW-FE panel



1. Ground screw	2. Input terminals	3. Monitoring module SMU11B (monitoring and managing the power system)	4. Rectifier module (converting AC power into stable DC power)	5. Load circuit breaker	6. Battery circuit breaker
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Figure 1-46 SMU11B monitoring module panel



1. Run indicator	2. Minor alarm indicator	3. Major alarm indicator
4. Wiring terminals	5. Communications port COM	6. Handle
7. Position of the SN code	8. Communications port FE	-

Figure 1-47 Wiring terminals of the SMU11B monitoring module

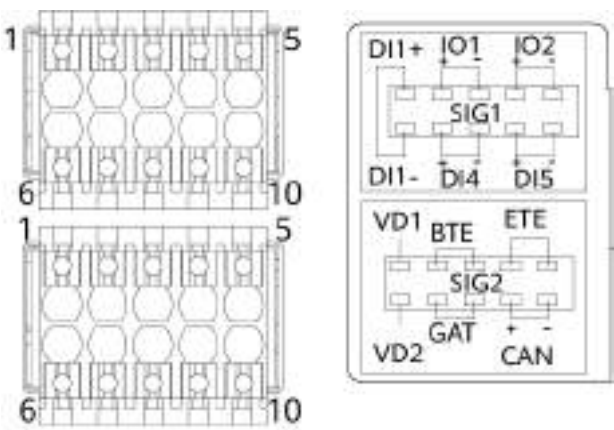


Table 1-53 Pin definitions for SIG1 wiring terminals

Pin	Signal	Description
1	DI1+	Dry contact input
6	DI1-	
2	IO1+	Dry contact input/Dry contact output (When used as a dry contact input, the alarm condition is as follows: normal when open, alarm when closed. When used as a dry contact output, the alarm action is as follows: open when normal, closed when alarm.)
3	IO1-	
4	IO2+	
5	IO2-	
7	DI4+	Dry contact input
8	DI4-	
9	DI5+	
10	DI5-	

Table 1-54 Pin definitions for SIG2 wiring terminals

Pin	Signal	Description
1	VD1	Battery midpoint voltage detection port 1
6	VD2	Battery midpoint voltage detection port 2
2	BTE	Battery temperature sensor port
3		
4	ETE	Ambient temperature sensor port
5		
7	GAT	Door status sensor port
8		
9	CAN+	CAN communications port
10	CAN-	

Table 1-55 Communications port on the SMU11B

Communications Port	Communications Parameter	Communications Protocol	Function
COM	Baud rate: 9600 bit/s, 19200 bit/s, 115200 bit/s autonegotiation	Master/slave protocol	Connects to a Huawei NMS.
		-	Provides 12 V power supply for external devices.
FE	10M/100M autonegotiation	SNMP	Connects to a third-party management system.
		BIN protocol	Connects to a Huawei NMS.
		HTTPS protocol	Connects to a PC and manages the SMU through the WebUI.
NOTE All these ports are protected by a security mechanism.			

Figure 1-48 Pins of the COM port on the SMU11B

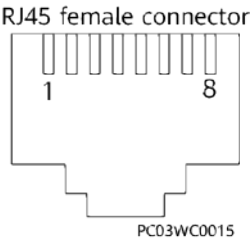


Table 1-56 Pin definitions for the COM port on the SMU11B

Pin	Signal	Description
1	RS485+	RS485 data +
2	RS485-	RS485 data -
3	12V	Supply power
4	RS485+	RS485 data +
5	RS485-	RS485 data -
6	SCL	I ² C clock
7	SDA	I ² C data

Pin	Signal	Description
8	GND	Ground

Figure 1-49 Pins of the FE port on the SMU11B

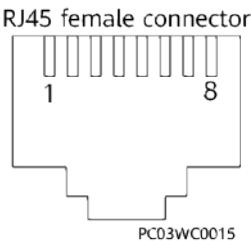
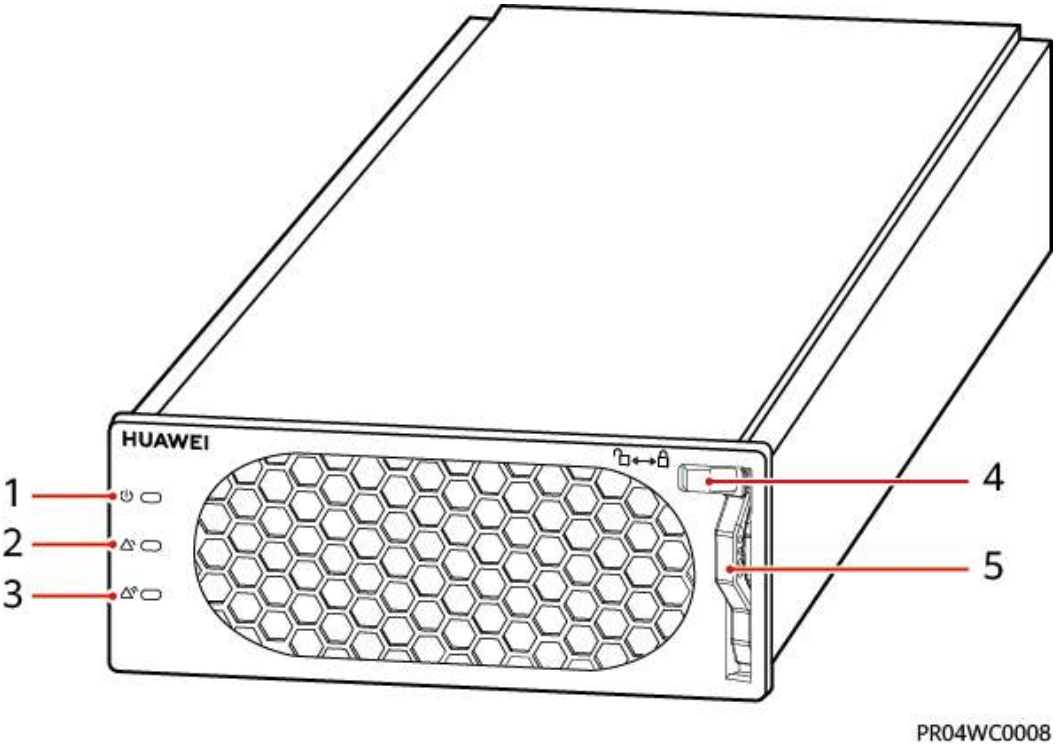


Table 1-57 Pin definitions for the FE port on the SMU11B

Pin	Signal	Description
1	TX+	Transmits data over FE.
2	TX-	
3	RX+	Receives data over FE.
6	RX-	
4, 5, 7 and 8	NA	-

Figure 1-50 Rectifier module panel



1. Power indicator	2. Alarm indicator	3. Fault indicator
4. Locking latch	5. Handle	-

The indicators on the EPTP4860-2KW-FE AC Power Box are classified into rectifier indicators and monitoring module indicators. The following table describes the indicators.

Table 1-58 Indicators on the ETP4860-2KW-FE

Silkscreen	Name	Color	Status	Description
-	Rectifier running indicator	Green	Steady on	The rectifier has an AC power input.
		Green	Off	- The rectifier has no AC power input. - The rectifier is faulty.
		Green	Blinking at 0.5 Hz	The rectifier is being queried.
		Green	Blinking at 4 Hz	The rectifier is loading an application program.
-	Rectifier alarm indicator	Yellow	Off	No alarm is generated.

Silkscreen	Name	Color	Status	Description
		Yellow	Steady on	• The rectifier has generated an alarm due to ambient overtemperature. • The rectifier has generated an alarm for shutdown due to ambient overtemperature or undertemperature. • The rectifier is protecting itself against AC input overvoltage or undervoltage. • The rectifier is hibernating.
		Yellow	Blinking at 0.5 Hz	The communication between the rectifier and the SMU has been interrupted.
-	Rectifier fault indicator	Red	Off	The rectifier is running properly.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	- The rectifier has been locked out due to output overvoltage. - The rectifier has no output due to an internal fault.
-	SMU running indicator	Green	Off	The SMU is faulty or has no power input.
		Green	Blinking slowly (0.5 Hz)	The SMU is running and communicating with the host properly.
		Green	Blinking fast (4 Hz)	The SMU is running properly but fails to communicate with the host properly.
-	SMU minor alarm indicator	Yellow	Off	No minor alarm or warning is generated.
		Yellow	Steady on	A minor alarm or warning is generated.
-	SMU major alarm indicator	Red	Off	No critical or major alarm is generated.
		Red	Steady on	A critical or major alarm is generated.

Technical Specifications

Table 1-59 Technical specifications of the ETP4860-2KW-FE

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	43.6 mm x 442 mm x 255 mm (1.72 in. x 17.4 in. x 10.04 in.)
Dimensions with packaging (H x W x D) [mm(in.)]	340 mm x 615 mm x 465 mm (13.39 in. x 24.21 in. x 18.31 in.)
Weight without packaging [kg(lb)]	7.2 kg (15.87 lb)
Weight with packaging [kg(lb)]	15.6 kg (34.39 lb)
Cabinet installation standards	ETSI 21-inch; IEC 19-inch
Hole distance for installing mounting brackets [mm(in.)]	465 mm (18.31 in.)
Installation type	Vertical or horizontal installation
Rated input voltage [V]	200 to 240V AC,support 240V/380V DC
Input voltage range [V]	85V AC to 300V AC/85V DC to 420V DC
Rated output voltage [V]	53.5 V
Output voltage range [V]	42V DC to 58V DC
Maximum input current [A]	23 A
Maximum output current [A]	37 A
Number of inputs	1
Number of outputs	5
Front-end circuit breaker/fuse [A]	32 A
Input terminal	M4 OT terminal
Output terminal	Cord end terminal
Long-term operating temperature [°C(°F)]	-40°C to 65°C(-40°F to 149°F)
Storage temperature [°C(°F)]	- 40°C to 70°C (- 40°F to 158°F)
Long-term operating relative humidity [RH]	5% RH to 95% RH, non-condensing
Storage relative humidity [RH]	5% RH to 95% RH, non-condensing

Item	Specification
Long-term operating altitude [m(ft.)]	0 – 4000 m (When the altitude ranges from 2000 m to 4000 m, the operating temperature decreases by 1°C for each additional 200 m.)

1.2.7 ETP4860-4KW (AC Power Box)

Overview

Table 1-60 Basic information about the ETP4860-4KW

Item	Details
Description	AC Power Box
Part Number	02312QXK
Model	ETP4860-4KW

NOTICE

- The external power system can connect to battery strings.
- If a battery string is used, connect the battery string to the battery circuit breaker (do not connect it to other circuit breakers) and set related parameters based on the actual situation of the battery string. Incorrectly setting basic battery parameters may affect the battery charge and discharge management and reduce the battery lifespan. For details about how to power on batteries, see "*Installing Power Boxes (ETP4860-4KW)>Commissioning>Connecting the Battery Supply*".
- If no battery string is connected to the power supply system, you must log in to the power supply system to set **Battery n Connected** to **No**, and click **Submit**. For details, see "*Installing Power Boxes (ETP4860-4KW)>Commissioning>SMU11B*".
- The BATT circuit breaker can be connected only to batteries and cannot be connected to other devices.

Table 1-61 System configuration

Item	System configuration
Subrack	1 U power system

Item	System configuration
Power Distribution Unit (PDU)	AC or DC input: M4 OT wiring terminal DC output: supports a maximum of two 63 A circuit breakers, two 40 A circuit breakers, or one 16 A circuit breaker. Battery route: one 50 A circuit breaker
Monitoring module	SMU11B

Appearance

Figure 1-51 Apperance of the ETP4860-4KW



NOTE

The load circuit breaker is red if it is pressed down and green if it is not pressed down.

Version Mapping

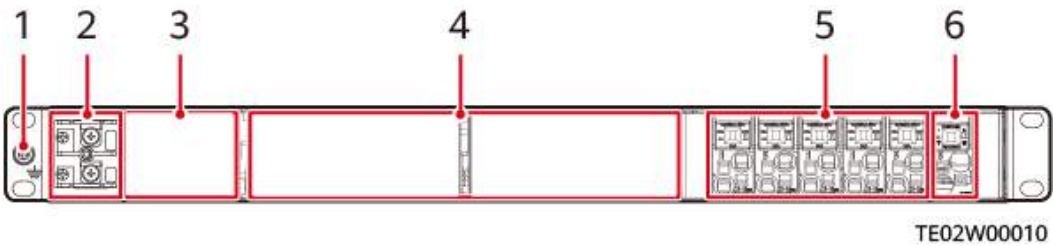
The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

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enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=02312QXK&productModel=ETP4860-4KW#matchRelation>

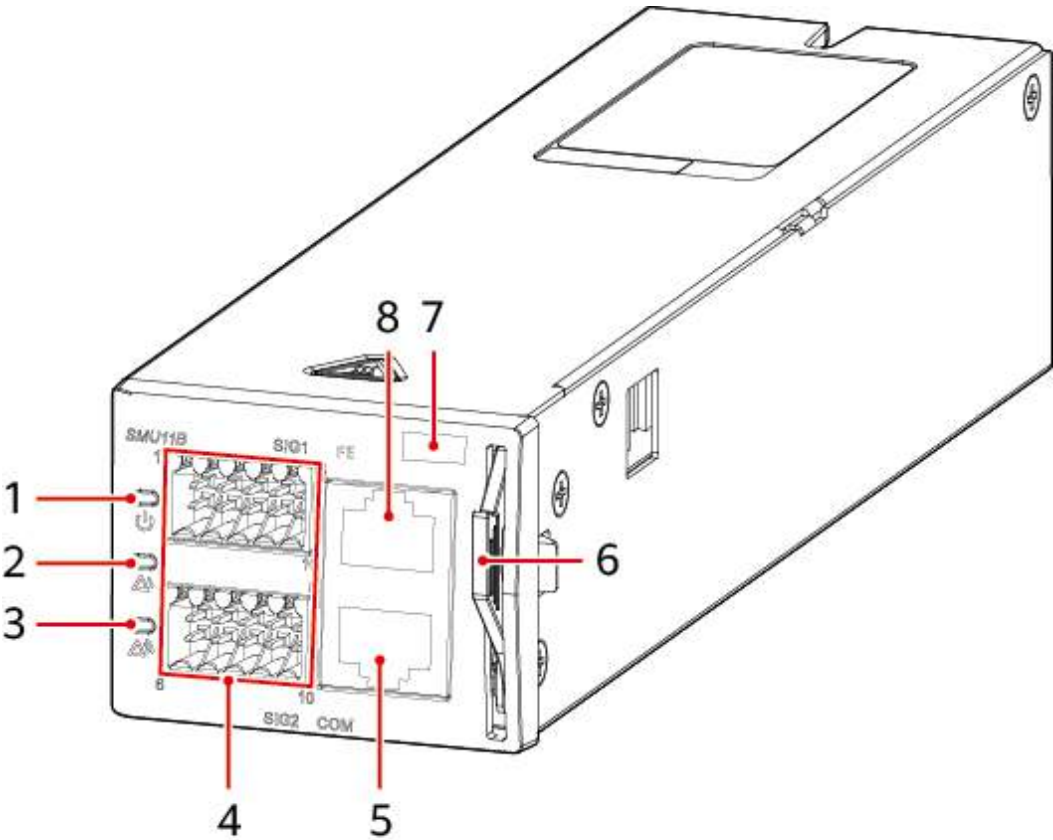
Panel

Figure 1-52 ETP4860-4KW panel



1. Ground screw	2. Input terminals	3. Monitoring module SMU11B (monitoring and managing the power system)	4. Rectifier module (converting AC power into stable DC power)	5. Load circuit breaker	6. Battery circuit breaker
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Figure 1-53 SMU11B monitoring module panel



1. Run indicator	2. Minor alarm indicator	3. Major alarm indicator
4. Wiring terminals	5. Communications port COM	6. Handle
7. Position of the SN code	8. Communications port FE	-

Figure 1-54 Wiring terminals of the SMU11B monitoring module

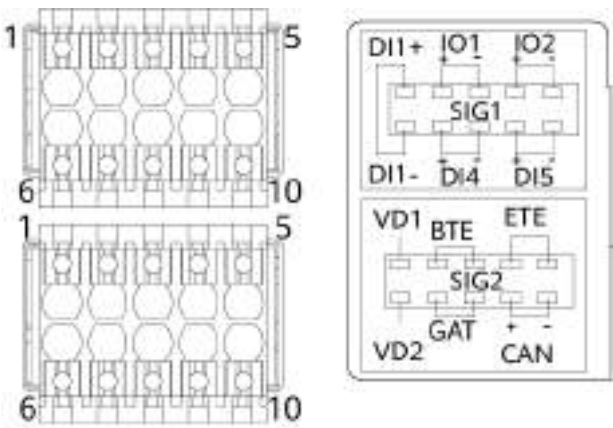


Table 1-62 Pin definitions for SIG1 wiring terminals

Pin	Signal	Description
1	DI1+	Dry contact input
6	DI1-	
2	IO1+	Dry contact input/Dry contact output (When used as a dry contact input, the alarm condition is as follows: normal when open, alarm when closed. When used as a dry contact output, the alarm action is as follows: open when normal, closed when alarm.)
3	IO1-	
4	IO2+	
5	IO2-	
7	DI4+	Dry contact input
8	DI4-	
9	DI5+	
10	DI5-	

Table 1-63 Pin definitions for SIG2 wiring terminals

Pin	Signal	Description
1	VD1	Battery midpoint voltage detection port 1
6	VD2	Battery midpoint voltage detection port 2
2	BTE	Battery temperature sensor port
3		

Pin	Signal	Description
4	ETE	Ambient temperature sensor port
5		
7	GAT	Door status sensor port
8		
9	CAN+	CAN communications port
10	CAN-	

Table 1-64 Communications port on the SMU11B

Communications Port	Communications Parameter	Communications Protocol	Function
COM	Baud rate: 9600 bit/s, 19200 bit/s, 115200 bit/s autonegotiation	Master/slave protocol	Connects to a Huawei NMS.
		-	Provides 12 V power supply for external devices.
FE	10M/100M autonegotiation	SNMP	Connects to a third-party management system.
		BIN protocol	Connects to a Huawei NMS.
		HTTPS protocol	Connects to a PC and manages the SMU through the WebUI.
NOTE All these ports are protected by a security mechanism.			

Figure 1-55 Pins of the COM port on the SMU11B

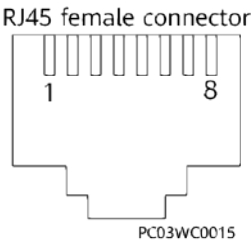


Table 1-65 Pin definitions for the COM port on the SMU11B

Pin	Signal	Description
1	RS485+	RS485 data +
2	RS485-	RS485 data -
3	12V	Supply power
4	RS485+	RS485 data +
5	RS485-	RS485 data -
6	SCL	I ² C clock
7	SDA	I ² C data
8	GND	Ground

Figure 1-56 Pins of the FE port on the SMU11B

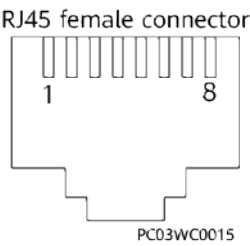
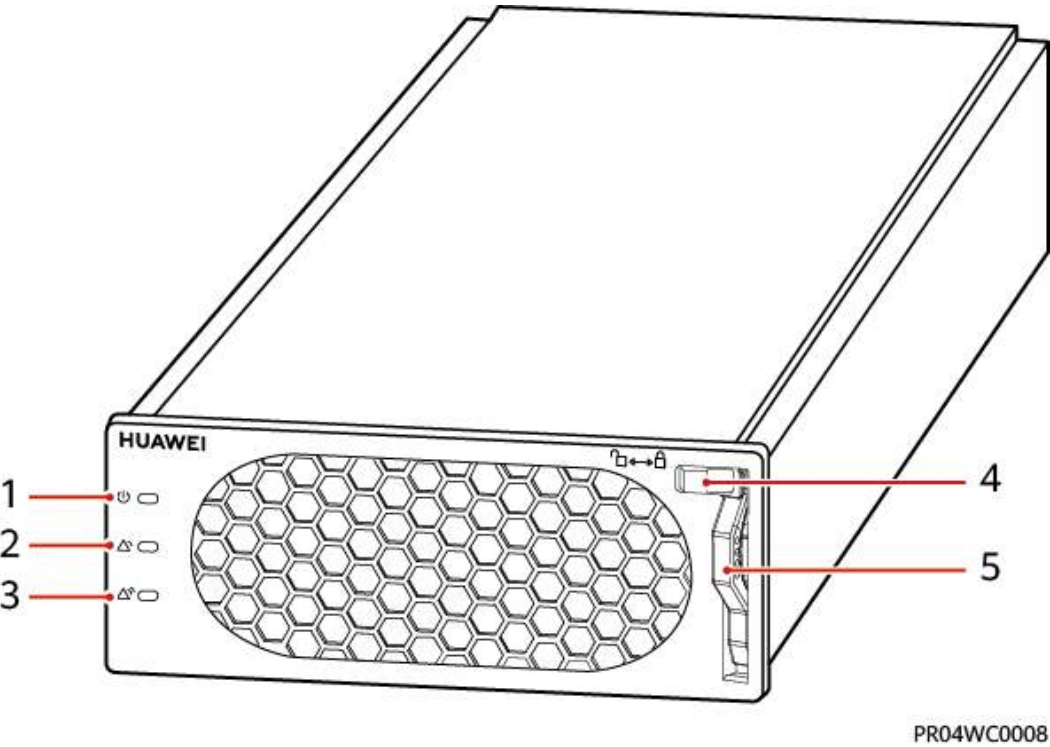


Table 1-66 Pin definitions for the FE port on the SMU11B

Pin	Signal	Description
1	TX+	Transmits data over FE.
2	TX-	
3	RX+	Receives data over FE.
6	RX-	
4, 5, 7 and 8	NA	-

Figure 1-57 Rectifier module panel



1. Power indicator	2. Alarm indicator	3. Fault indicator
4. Locking latch	5. Handle	-

The indicators on the EPTP4860-4KW AC Power Box are classified into rectifier indicators and monitoring module indicators. The following table describes the indicators.

Table 1-67 Indicators on the ETP4860-4KW

Silkscreen	Name	Color	Status	Description
-	Rectifier running indicator	Green	Steady on	The rectifier has an AC power input.
		Green	Off	- The rectifier has no AC power input. - The rectifier is faulty.

Silkscreen	Name	Color	Status	Description
		Green	Blinking at 0.5 Hz	The rectifier is being queried.
		Green	Blinking at 4 Hz	The rectifier is loading an application program.
-	Rectifier alarm indicator	Yellow	Off	No alarm is generated.
		Yellow	Steady on	• The rectifier has generated an alarm due to ambient overtemperature. • The rectifier has generated an alarm for shutdown due to ambient overtemperature or undertemperature. • The rectifier is protecting itself against AC input overvoltage or undervoltage. • The rectifier is hibernating.

Silkscreen	Name	Color	Status	Description
		Yellow	Blinking at 0.5 Hz	The communication between the rectifier and the SMU has been interrupted.
-	Rectifier fault indicator	Red	Off	The rectifier is running properly.
		Red	Steady on	- The rectifier has been locked out due to output overvoltage. - The rectifier has no output due to an internal fault.
-	SMU running indicator	Green	Off	The SMU is faulty or has no power input.
		Green	Blinking slowly (0.5 Hz)	The SMU is running and communicating with the host properly.
		Green	Blinking fast (4 Hz)	The SMU is running properly but fails to communicate with the host properly.
-	SMU minor alarm indicator	Yellow	Off	No minor alarm or warning is generated.

Silkscreen	Name	Color	Status	Description
		Yellow	Steady on	A minor alarm or warning is generated.
-	SMU major alarm indicator	Red	Off	No critical or major alarm is generated.
		Red	Steady on	A critical or major alarm is generated.

Technical Specifications

Table 1-68 Technical specifications of the ETP4860-4KW

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	43.6 mm x 442 mm x 255 mm (1.72 in. x 17.4 in. x 10.04 in.)
Dimensions with packaging (H x W x D) [mm(in.)]	340 mm x 615 mm x 465 mm (13.39 in. x 24.21 in. x 18.31 in.)
Weight without packaging [kg(lb)]	7.2 kg (15.87 lb)
Weight with packaging [kg(lb)]	15.6 kg (34.39 lb)
Cabinet installation standards	ETSI 21-inch; IEC 19-inch
Hole distance for installing mounting brackets [mm(in.)]	465 mm (18.31 in.)
Installation type	Vertical or horizontal installation
Rated input voltage [V]	200 to 240V AC,support 240V/380V DC
Input voltage range [V]	85V AC to 300V AC/85V DC to 420V DC
Rated output voltage [V]	53.5 V
Output voltage range [V]	42V DC to 58V DC
Maximum input current [A]	23 A
Maximum output current [A]	74 A
Number of inputs	1
Number of outputs	5
Front-end circuit breaker/fuse [A]	32 A

Item	Specification
Input terminal	M4 OT terminal
Output terminal	Cord end terminal
Long-term operating temperature [°C(°F)]	-40°C to 65°C(-40°F to 149°F)
Storage temperature [°C(°F)]	- 40°C to 70°C (- 40°F to 158°F)
Long-term operating relative humidity [RH]	5% RH to 95% RH, non-condensing
Storage relative humidity [RH]	5% RH to 95% RH, non-condensing
Long-term operating altitude [m(ft.)]	0 to 4000 m (When the altitude ranges from 2000 m to 4000 m, the operating temperature decreases by 1°C for each additional 200 m.)

1.3 Fan

This section describes the appearance, functions, and technical specifications of the fan module.

1.3.1 CR8M002FBXC1 (NetEngine 8000 M14 Fan Box)

Overview

Table 1-69 Basic information about the CR8M002FBXC1

Item	Details
Description	NetEngine 8000 M14 Fan Box
Part Number	03033JNE
Model	CR8M002FBXC1
Remarks	To ensure that the system provides better heat dissipation performance during main control board startup, the fan speed increases to at least 50% of the full speed as long as one main control board restarts. The system will automatically decrease the fan speed after main control board registration is completed.

Appearance

Figure 1-58 Appearance of the CR8M002FBXC1



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03033JNE&productModel=CR8M002FBXC1#matchRelation>

Panel

Figure 1-59 Panel of the CR8M002FBXC1



1.ESD jack	2.Status indicator
------------	--------------------

Table 1-70 Indicators on the CR8M002FBXC1

Silkscreen	Name	Color	Status	Description
FAN	Working status indicator	Green	Steady on	the fan module is working normally.
		Red	Steady on	the fan module fails.
		Orange	Steady on	the fan module is not registered.
		-	Off	the fan module is not powered on or the fan hardware is faulty.

Table 1-71 Ports on the CR8M002FBXC1

Port	Description
ESD jack	The ESD wrist strap should be connected to this jack for proper grounding of the human body.

Technical Specifications

Table 1-72 Technical specifications of the CR8M002FBXC1

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	218.9 mm x 53.5 mm x 208.7 mm (8.62 in. x 2.11 in. x 8.22 in.)
Weight without packaging [kg(lb)]	1.8 kg (3.97 lb)
Number of fans	4
Typical power consumption [W]	30 W

1.3.2 CR8M002FBXC3 (NetEngine 8000 M14 Fan Box(E))

Overview

Table 1-73 Basic information about the CR8M002FBXC3

Item	Details
Description	NetEngine 8000 M14 Fan Box(E)
Part Number	03035EBQ
Model	CR8M002FBXC3
Remarks	To ensure that the system provides better heat dissipation performance during main control board startup, the fan speed increases to at least 50% of the full speed as long as one main control board restarts. The system will automatically decrease the fan speed after main control board registration is completed.

Appearance

Figure 1-60 Appearance of the CR8M002FBXC3



Version Mapping

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The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03035EBQ&productModel=CR8M002FBXC3#matchRelation>

Panel

Figure 1-61 Panel of the CR8M002FBXC3



1.ESD jack	2.Status indicator
------------	--------------------

Table 1-74 Indicators on the CR8M002FBXC3

Silkscreen	Name	Color	Status	Description
FAN	Working status indicator	Green	Steady on	the fan module is working normally.
		Red	Steady on	the fan module fails.
		Orange	Steady on	the fan module is not registered.
		-	Off	the fan module is not powered on or the fan hardware is faulty.

Table 1-75 Ports on the CR8M002FBXC3

Port	Description
ESD jack	The ESD wrist strap should be connected to this jack for proper grounding of the human body.

Technical Specifications

Table 1-76 Technical specifications of the CR8M002FBXC3

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	218.9 mm x 53.5 mm x 208.7 mm (8.62 in. x 2.11 in. x 8.22 in.)
Weight without packaging [kg(lb)]	1.9 kg (4.19 lb)
Number of fans	4
Typical power consumption [W]	30 W

1.3.3 CR8M004FBXC0 (NetEngine 8000 M4 Fan Box)

Overview

Table 1-77 Basic information about the CR8M004FBXC0

Item	Details
Description	NetEngine 8000 M4 Fan Box
Part Number	03034UQC
Model	CR8M004FBXC0
Remarks	To ensure that the system provides better heat dissipation performance during main control board startup, the fan speed increases to at least 50% of the full speed as long as one main control board restarts. The system will automatically decrease the fan speed after main control board registration is completed.

Appearance

Figure 1-62 Appearance of the CR8M004FBXC0



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03034UQC&productModel=CR8M004FBXC0#matchRelation>

Panel

Table 1-78 Indicators on the CR8M004FBXC0

Silkscreen	Name	Color	Status	Description
FAN	Working status indicator	Green	Steady on	the fan module is working normally.
		Red	Steady on	the fan module fails.
		Orange	Steady on	the fan module is not registered.
		-	Off	the fan module is not powered on or the fan hardware is faulty.

Table 1-79 Ports on the CR8M004FBXC0

Port	Description
ESD	The ESD wrist strap should be connected to this jack for proper grounding of the human body.

Technical Specifications

Table 1-80 Technical specifications of the CR8M004FBXC0

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	85.5 mm x 34.9 mm x 209 mm (3.37 in. x 1.37 in. x 8.23 in.)
Weight without packaging [kg(lb)]	0.5 kg (1.1 lb)
Number of fans	2
Typical power consumption [W]	5 W

1.3.4 CR8MM82FBXC1 (NetEngine 8000 M8 Fan Box)

Overview

Table 1-81 Basic information about the CR8MM82FBXC1

Item	Details
Description	NetEngine 8000 M8 Fan Box
Part Number	03033GGC
Model	CR8MM82FBXC1
Remarks	To ensure that the system provides better heat dissipation performance during main control board startup, the fan speed increases to at least 50% of the full speed as long as one main control board restarts. The system will automatically decrease the fan speed after main control board registration is completed.

Appearance

Figure 1-63 Appearance of the CR8MM82FBXC1



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03033GGC&productModel=CR8MM82FBXC1#matchRelation>

Panel

Figure 1-64 Panel of the CR8MM82FBXC1

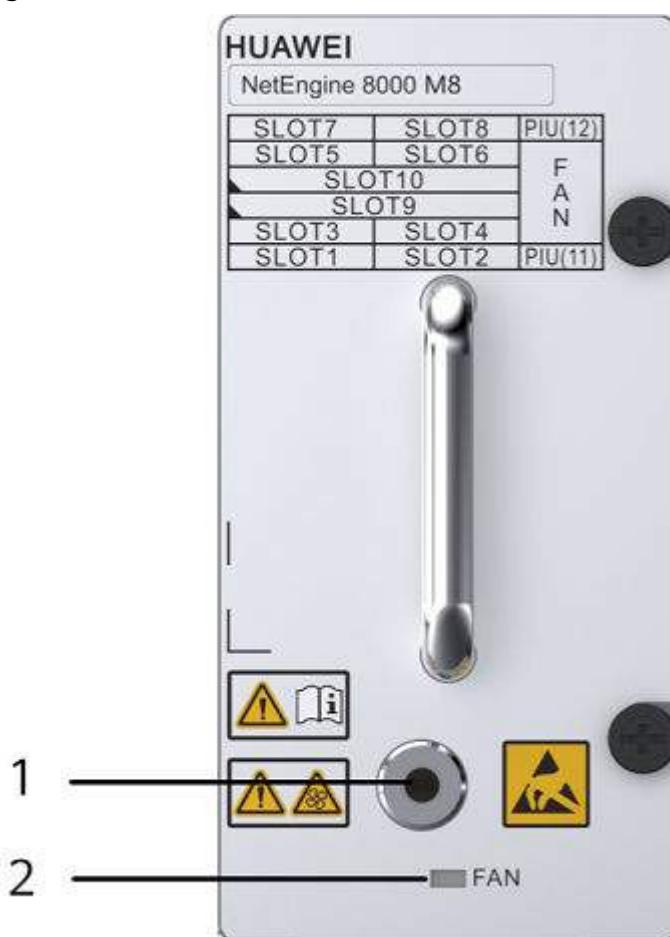


Table 1-82 Indicators on the CR8MM82FBXC1

Silkscreen	Name	Color	Status	Description
FAN	Working status indicator	Green	Steady on	The fan module is working normally.
		Red	Steady on	The fan module fails.
		Orange	Steady on	The fan module is not registered.
		-	Off	The fan module is not powered on or the fan hardware is faulty.

Table 1-83 Ports on the CR8MM82FBXC1

Port	Description
ESD	The ESD wrist strap should be connected to this jack for proper grounding of the human body.

Technical Specifications

Table 1-84 Technical specifications of the CR8MM82FBXC1

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	105.7 mm x 53.5 mm x 209 mm (4.16 in. x 2.11 in. x 8.23 in.)
Weight without packaging [kg(lb)]	1 kg (2.2 lb)
Number of fans	2
Typical power consumption [W]	15 W

1.4 Boards

This chapter describes the boards of the device.

1.4.1 Rules for Numbering Slots and Interfaces

Numbering Rule of Service Interfaces of the NetEngine 8000 M4

- A service interface on a PIC is numbered in the format of 0/PIC slot number/interface number.
- PIC slot number
Slot ID of a PIC where the service interface resides. For DC chassis, the value is an integer ranging from 1 to 5. For AC chassis, the value is 2, 4, 5.
 - Interface number
ID of the service interface on a PIC. The value is an integer ranging from 0 to the maximum number of interfaces on the PIC.

Figure 1-65 shows how a service interface on a PIC is numbered.

Figure 1-65 Numbering rule of service interfaces on the NetEngine 8000 M4

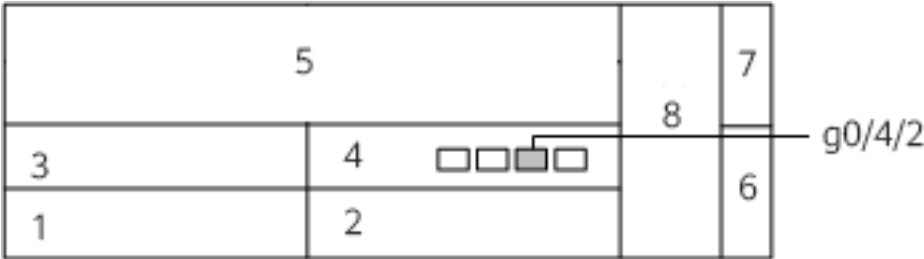


Table 1-85 NetEngine 8000 M4 slot description

Slot	Quantity	Remarks
DC: 1-4 AC: 2, 4	DC: 4 AC: 2	Slots for PICs, which include HICs, FICs and the other subcards.
5	1	Slot for a fixed interface
DC: slot6, slot7 AC: slot1, slot3	2	Slots for power modules in 1+1 backup mode
8	1	Slot for a fan module

Numbering Rule of Service Interfaces of the NetEngine 8000 M8

- A service interface on a PIC is numbered in the format of 0/PIC slot number/interface number.
- PIC slot number
Slot ID of a PIC where the service interface resides. For DC chassis, the value is an integer ranging from 1 to 8. For AC chassis, the value is 2, 4 and 5 to 8.
 - Interface number

ID of the service interface on a PIC. The value is an integer ranging from 0 to the maximum number of interfaces on the PIC.

Figure 1-66 shows how a service interface on a PIC is numbered.

Figure 1-66 Numbering rule of service interfaces on the NetEngine 8000 M8

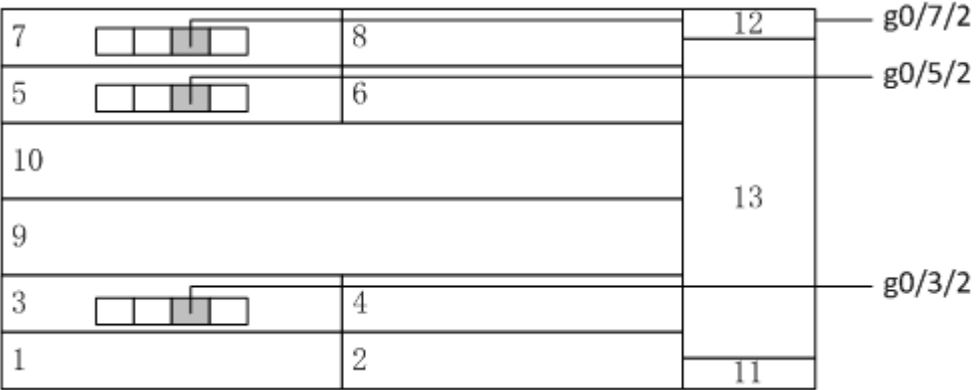


Table 1-86 NetEngine 8000 M8 slot description

Slot	Quant ity	Remarks
DC: 1-8 AC: 2, 4, 5-8	DC: 8 AC: 6	Slots for PICs, which include HICs, FICs and the other subcards.
9 and 10	2	Slot for the IPU's
DC: 11 and 12 AC: 1 and 3	2	Slots for power modules in 1+1 backup mode
13	1	Slot for a fan module

Numbering Rule of Service Interfaces of the NetEngine 8000 M14

A service interface on a PIC is numbered in the format of 0/PIC slot number/ interface number.

- PIC slot number
Slot ID of a PIC where the service interface resides. For DC chassis, the value is an integer ranging from 1 to 14. For AC chassis, the value is an integer ranging from 5 to 14.
- Interface number
ID of the service interface on a PIC. The value is an integer ranging from 0 to the maximum number of interfaces on the PIC.

Figure 1-67 shows how a service interface on a PIC is numbered.

Figure 1-67 Numbering rule of service interfaces on the NetEngine 8000 M14

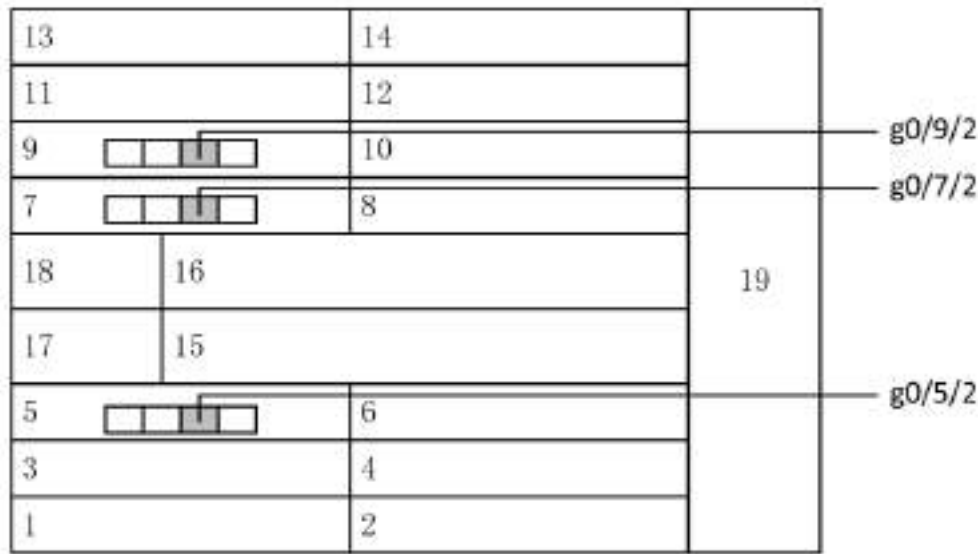


Table 1-87 NetEngine 8000 M14 slot description

Slot	Quantity	Remarks
DC: 1-14 AC: 5-14	DC: 14 AC: 10	Slots for PICs
15, 16	2	Slot for the IPU
DC: 17,18 AC: 1-4	DC: 2 AC: 4	Slots for power modules
19	1	Slot for a fan module

Port Numbering of IPU

When configuring and checking an IPU's MGMT port, its number is 0/0/0. However, in alarms and logs, it is displayed as "IPU slot number/0/0".

1.4.2 Board Code

The board code of a board is provided on its front panel and contains basic information about the board, including the BOM code and manufacture year and month.

A board code indicates the characteristic information of a board and varies by board.

- For boards with the BOM code beginning with 0303, the coding rules are as follows (the Code 128B system is used):
Rightmost six digits of BOM code (six digits) + Manufacturer (two digits) + Year (one digit) + Month (one digit) + Serial number (six digits)

Figure 1-68 Board codes



033FXD: Rightmost six digits of BOM code
10: Manufacturer
L8: Manufacture year and month
000073: Serial number
Y: Environmental protection property
CR8DIPU1T2C1: Board Model

NOTE

The board codes in the figure are used for reference only and may differ from codes on actual devices.

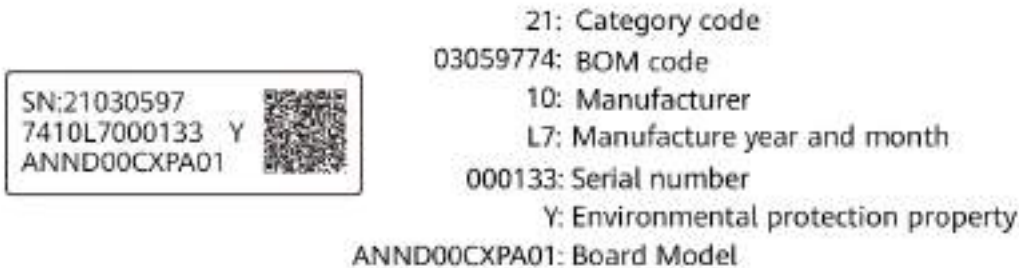
Table 1-88 Description of board codes

Item	Description
Rightmost six digits of BOM code	Indicates the rightmost six digits of a board's eight-digit BOM code. The rightmost six digits plus 03 on the left constitute the BOM code.
Manufacturer	Indicates the vendor of the board. "10" indicates Huawei.
Manufacture year	Indicates the last digit of the year when the board was manufactured. Before 2010, digits were used (for example, "4" indicates 2004). From 2010 onwards, letters (excluding I and O) are used (for example, "A" indicates 2010, "B" indicates 2011, and so on). NOTE The letters A to Z (excluding I and O due to their similarity with digits 1 and 0) are used to indicate years in ascending order. For example, H indicates 2017 and J indicates 2018, with the letter I being skipped.
Manufacture month	Indicates the month when the board was manufactured. The value ranges from 1 to 9 for January to September, and from A to C for October to December.
Serial number	Indicates the production serial number of a board. The value ranges from 000001 to 999999.

Item	Description
Environmental protection property	Indicates whether a board is environmentally friendly. The value Y indicates yes; N indicates no.
Board Model	Indicates the model of the board.

- For boards with the BOM code that does not begin with 0303, the coding rules are as follows (the Code 128B system is used):
Category code (2 digits) + BOM code (8 digits) + Manufacturer (2 digits) + Year (1 digit) + Month (1 digit) + Serial number (6 digits)

Figure 1-69 Board codes



 NOTE

The board codes in the figure are used for reference only and may differ from codes on actual devices.

Table 1-89 Description of board codes

Item	Description
Category code	The value is fixed to 21, indicating the category code of a component or module.
BOM code	Indicates the BOM code of the board.
Manufacturer	Indicates the vendor of the board. "10" indicates Huawei.

Item	Description
Manufacture year	Indicates the last digit of the year when the board was manufactured. Before 2010, digits were used (for example, "4" indicates 2004). From 2010 onwards, letters (excluding I and O) are used (for example, "A" indicates 2010, "B" indicates 2011, and so on). NOTE The letters A to Z (excluding I and O due to their similarity with digits 1 and 0) are used to indicate years in ascending order. For example, H indicates 2017 and J indicates 2018, with the letter I being skipped.
Manufacture month	Indicates the month when the board was manufactured. The value ranges from 1 to 9 for January to September, and from A to C for October to December.
Serial number	Indicates the production serial number of a board. The value ranges from 000001 to 999999.
Environmental protection property	Indicates whether a board is environmentally friendly. The value Y indicates yes; N indicates no.
Board Model	Indicates the model of the board.

1.4.3 IPU

1.4.3.1 CR8D0IPU2TC1 (03033FPV: Integrated Network Processing Unit (IPU-2T))

Overview

Table 1-90 Basic information about the CR8D0IPU2TC1

Item	Details
Description	Integrated Network Processing Unit (IPU-2T)
Part Number	03033FPV

Item	Details
Model	CR8D0IPU2TC1
Card type	IPU
Silkscreen	IPU-2T

Appearance

Figure 1-70 Appearance of the CR8D0IPU2TC1



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03033FPV&productModel=CR8D0IPU2TC1#matchRelation>

Indicators and Buttons

Table 1-91 Indicators on the CR8D0IPU2TC1

Silkscreen	Name	Color	Status	Description
STAT	Working status indicator	Green	Steady on	The board is working normally.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	The device generates a hardware alarm. You can run the display alarm hardware command to view the hardware alarm type.
		-	Off	The board is not running or no power is input.
PROG	Program running indicator	Green	Steady on	The board software is being initialized. The board software is normally initialized, and the board software is running normally.
		Red	Steady on	The memory self-check fails. The board software or the logic file is lost. Loading of the board software fails.
		Green	On for 100 ms and off for 100 ms alternately	Loading of the board software is in process.
		Green	On for 300 ms and off for 300 ms alternately	The BIOS is guiding the upper-layer software.

Silkscreen	Name	Color	Status	Description
		Red	On for 100 ms and off for 100 ms alternately	The BIOS self-check fails.
		-	Off	No power is input.
SYNC	Synchronization status indicator	Green	Steady on	The system clock works in free-run mode, and no priority list is set for the system clocks (only internal clock sources are listed by default). The system clock works in tracing mode and is tracing a clock source other than the internal clock sources in the priority list. The system clock works in time synchronization mode, and the PTP time and system clock are being traced.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	The system clock priority list is set, but all clock sources except the internal clock source are lost. The clock works in holdover or free-run mode. The system clock works in time synchronization mode, but no synchronization source is available. The system clock and PTP time work in holdover or free-run mode.
ACTX	Active/standby status indicator for the cross-connect or clock board	Green	Steady on	The board is in the working switching state.
		-	Off	The board is in the protection switching state.
ACTC	Control status indicator	Green	Steady on	The board is in the working control state.
		Green	On for 100 ms and off for 100 ms alternately	The data of the equipment is backed up in batch.

Silkscreen	Name	Color	Status	Description
		-	Off	The board is in the protection control state.
ALM	Alarm indicator	Red	Steady on	Critical alarms are generated.
		Orange	Steady on	Major, minor alarms are generated.
		-	Off	No alarms are generated in the equipment.
ETH/OAM	ETH/OAM indicator	Green	Steady on	If the indicator is steady on, the link is normal.
		Orange	Blink	If the indicator blinks, data is being transmitted.
		-	Off	If the indicator is off, the link is Down.

Table 1-92 Buttons on the CR8D0IPU2TC1

Silkscreen	Name	Description
RST	Reset button	You can press the RST button to reset the NE or restore the factory settings of the NE. The procedure is as follows: To reset the NE, press the RST button and release it. To restore factory settings, press and hold the RST button. This operation will clear the NE database and system parameter configurations and cannot be restored. Exercise caution when you perform the following operations: 1. Press and release the RST button to restart the NE. 2. When the PROG indicator blinks (the blinking frequency is about 1 second and the time window is about 15 seconds), press and hold the RST button. 3. Wait for no longer than 2 minutes till both the STAT and PROG indicators blink (the time window is 5 seconds). Then release the RST button immediately.

Silkscreen	Name	Description
		NOTE 1. When you press the RST button to clear the configuration file, the original configuration file will be cleared. You are advised to periodically back up the configuration file. 2. When you press the RST button to clear the configuration file, to prevent the cleared configuration from being synchronized again between the active and standby main control boards, remove one main control board first in a dual-main control board chassis to ensure that the configuration file is cleared in a single-main control board scenario. Perform the same operations on the other main control board. 3. When pressing the RST button to clear the configuration file, you only need to pay attention to the STAT and PROG indicators. Other indicators may vary depending on the product model.

Ports

Table 1-93 Ports on the CR8D0IPU2TC1

Port	Connector Type	Description	Available Components
ETH/OAM	RJ45	10M/100M auto-sensing Ethernet network management port or non-standard RS-232 console port. NOTE If a service port can be connected to the NMS, it is recommended that the service port be used as the network management port.	ETH/OAM management cable
CLK/TOD/RS-485	RJ45	Clock/Time/RS-485 input/output port, RS-485 is a reserved port.	1-into-2 clock cable + clock cable (external clock mode, delivery-ready): applicable to the CLK mode 1-into-2 clock cable + clock cable (external clock/external time mode, prepared onsite): applicable to the CLK or TOD mode 1-into-2 clock cable + Ethernet cable: applicable to the TOD mode
USB	Type A	USB port (reserved)	-
CLK/TEST	SMA	Clock test port	Coaxial cable

Port	Connector Type	Description	Available Components
GE0,GE1	SFP	Cascading ports are used for control panel expansion in scenarios like virtual cluster. (reserved)	Optical fiber
HP-GE	SFP	Ultra-high-precision GE port (This first supported version is V800R021C00. This port is reserved in earlier versions.)	1.25Gbps eSFP Optical Module 1.25Gbps eSFP CWDM Optical Module 1.25Gbps eSFP BIDI Optical Module

Table 1-94 Pins of the ETH/OAM interface

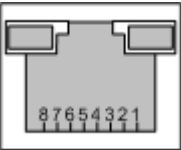
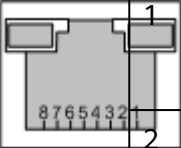
Front View	Pin	Usage
	1	Transmit positive of the NM network port
	2	Transmit negative of the NM network port
	3	Receive positive of the NM network port
	4	Grounding end of the Console interface
	5	Receive end of the Console interface
	6	Receive negative of the NM network port
	7	Unspecified
	8	Transmit end of the Console interface

Table 1-95 Pins of the CLK/TOD/RS-485 interface

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	1	Receive negative of the CLK	Receive negative of the CLK	Unspecified	Unspecified	Unspecified
	2	Receive positive of the CLK	Receive positive of the CLK	Unspecified	Unspecified	Unspecified
	3	Input/Output negative for the 1PPS signal	Unspecified	Input negative for the 1PPS signal (RS422 level)	Output negative for the 1PPS signal (RS422 level)	Transmit negative of the RS-485
	4	Receive negative of the CLK	Receive negative of the CLK	Grounding end	Grounding end	Unspecified
	5	Receive positive of the CLK	Receive positive of the CLK	Grounding end	Grounding end	Unspecified
	6	Input/Output positive for the 1PPS signal	Unspecified	Input positive for the 1PPS signal (RS422 level)	Output positive for the 1PPS signal (RS422 level)	Transmit positive of the RS-485

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	7	Input/Output negative for the time information	Unspecified	Input negative for the time information (RS422 level)	Output negative for the time information (RS422 level)	Receive negative of the RS-485
	8	Input/Output positive for the time information	Unspecified	Input positive for the time information (RS422 level)	Output positive for the time information (RS422 level)	Receive positive of the RS-485
NOTE When this interface is used as a TOD interface, it must be used with a dedicated clock cable. For details, see "Cables" in the Hardware Description.						

Functions and Features

Table 1-96 Functions and features of the CR8D0IPU2TC1

Functions and Features	Description
Basic function	Data forwarding: The IPU is the core of service processing in the entire system, and is connected to all subcards through data channels. Control and management: Through the management channels between IPU and the subcards, the IPU can manage subcards and transmit routing protocol data. The line rate capability is 4T bit/s@256Bytes(Bidirectional).
Reliability and availability	Supports the hot swappable function. Supports the board-level 1:1 backup function.
Restrictions and remarks	1. It cannot be used together with other IPU. 2. When working with the 03033VMC board to support FlexE services, the board supports a maximum of eight FlexE physical interfaces.
Breakout supported	No
Interface rate auto-sensing supported	No

Technical Specifications

Table 1-97 Technical specifications of the CR8D0IPU2TC1

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	34.8 mm x 354.4 mm x 209 mm (1.37 in. x 13.95 in. x 8.23 in.)
Dimensions with packaging (H x W x D) [mm(in.)]	185 mm x 485 mm x 385 mm (7.28 in. x 19.09 in. x 15.16 in.)
Weight without packaging [kg(lb)]	4 kg (8.82 lb)
Weight with packaging [kg(lb)]	5.4 kg (11.9 lb)
Typical power consumption [W]	324.4 W
Typical heat dissipation [BTU/hour]	1052.55 BTU/hour

Item	Specification
Long-term operating temperature [°C(°F)]	–20°C to +55°C (–4°F to +131°F); start: ≥ –5°C (23°F)
CPU	20-core 2.0 GHz
SDRAM	16GB
Storage	4 GB NandFlash

1.4.3.2 CR8D0IPU2TC1 (03033FPV-001: Integrated Network Processing Unit (IPU-2T))

Overview

Table 1-98 Basic information about the CR8D0IPU2TC1

Item	Details
Description	Integrated Network Processing Unit (IPU-2T)
Part Number	03033FPV-001
Model	CR8D0IPU2TC1
Card type	IPU
Silkscreen	IPU-2T

Appearance

Figure 1-71 Appearance of the CR8D0IPU2TC1



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03033FPV-001&productModel=CR8D0IPU2TC1#matchRelation>

Indicators and Buttons

Table 1-99 Indicators on the CR8D0IPU2TC1

Silkscreen	Name	Color	Status	Description
STAT	Working status indicator	Green	Steady on	The board is working normally.
		Red	Steady on	The device generates a hardware alarm. You can run the display alarm hardware command to view the hardware alarm type.
		-	Off	the board is not running or no power is input.
PROG	Program running indicator	Green	Steady on	The board software is being initialized. The board software is normally initialized, and the board software is running normally.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	The memory self-check fails. The board software or logic file is lost. The board software fails to be loaded. The BIOS self-check fails.
		Green	On for 100 ms and off for 100 ms alternately	Loading of the board software is in process.
		Green	On for 300 ms and off for 300 ms alternately	The BIOS is guiding the upper-layer software.
		-	Off	The board is not powered on.

Silkscreen	Name	Color	Status	Description
SYNC	Clock synchronizatio n indicator	Green	Steady on	The system clock works in free-run mode, and no priority list is set for the system clocks (only internal clock sources are listed by default). The system clock works in tracing mode and is tracing a clock source other than the internal clock sources in the priority list. The system clock works in time synchronizatio n mode, and the PTP time and system clock are being traced.

Silkscreen	Name	Color	Status	Description
SYNC	Synchronization status indicator	Red	Steady on	The system clock priority list is set, but all clock sources except the internal clock source are lost. The clock works in holdover or free-run mode. The system clock works in time synchronization mode, but no synchronization source is available. The system clock and PTP time work in holdover or free-run mode.
ACTX	Active/standby status indicator for the cross-connect or clock board	Green	Steady on	The board is in the active state on the service plane.
		-	Off	The board is in the standby state on the service plane.
ACTC	Control status indicator	Green	Steady on	The board is in the active control state.
		Green	Blinks on and off at 100 ms intervals	The data of the device is backed up in batch.
		-	Off	The board is in the standby control state.

Silkscreen	Name	Color	Status	Description
ALM	Alarm indicator	Red	Steady on	Device-level critical alarms exist.
		Orange	Steady on	A device-level major or minor alarm exists.
		-	Off	There is no equipment-level alarm.
ETH/OAM	ETH/OAM indicator	Green	Steady on	The link is up.
		Orange	Blinking	Data is being transmitted or received.
		-	Off	The link is down.

Table 1-100 Buttons on the CR8D0IPU2TC1

Silkscreen	Name	Description
RST	Reset button	You can press the RST button to reset the NE or restore the factory settings of the NE. The procedure is as follows: To reset the NE, press the RST button and release it. To restore factory settings, press and hold the RST button. This operation will clear the NE database and system parameter configurations and cannot be restored. Exercise caution when you perform the following operations: 1. Press and release the RST button to restart the NE. 2. When the PROG indicator blinks (the blinking frequency is about 1 second and the time window is about 15 seconds), press and hold the RST button. 3. Wait for no longer than 2 minutes till both the STAT and PROG indicators blink (the time window is 5 seconds). Then release the RST button immediately.

Silkscreen	Name	Description
		NOTE 1. When you press the RST button to clear the configuration file, the original configuration file will be cleared. You are advised to periodically back up the configuration file. 2. When you press the RST button to clear the configuration file, to prevent the cleared configuration from being synchronized again between the active and standby main control boards, remove one main control board first in a dual-main control board chassis to ensure that the configuration file is cleared in a single-main control board scenario. Perform the same operations on the other main control board. 3. When pressing the RST button to clear the configuration file, you only need to pay attention to the STAT and PROG indicators. Other indicators may vary depending on the product model.

Ports

Table 1-101 Ports on the CR8D0IPU2TC1

Port	Connector Type	Description	Available Components
ETH/OAM	RJ45	10M/100M autosensing Ethernet network management port or non-RS-232-standard console port. NOTE If a service port can be connected to the NMS, it is recommended that the service port be used as the network management port.	ETH/OAM Management Cables
CLK/TOD/RS-485	RJ45	Clock/Time/RS-485 input/output port, RS-485 is a reserved port.	1-into-2 clock cable + clock cable (external clock mode, delivery-ready): applicable to the CLK mode 1-into-2 clock cable + clock cable (external clock/external time mode, prepared onsite): applicable to the CLK or TOD mode 1-into-2 clock cable + Ethernet cable: applicable to the TOD mode
USB	Type A	USB port (reserved)	1-into-2 clock cable + clock cable (external clock/external time mode, prepared onsite): applicable to the CLK or TOD mode

Port	Connector Type	Description	Available Components
CLK/TEST	SMA	Clock test port	1-into-2 clock cable + Ethernet cable: applicable to the TOD mode
HP-GE	SFP	Ultra-high-precision GE port (supported since V800R021C00 and reserved in earlier versions)	1.25Gbps eSFP Optical Module 1.25Gbps eSFP CWDM Optical Module 1.25Gbps eSFP BIDI Optical Module

Table 1-102 Pins of the ETH/OAM interface

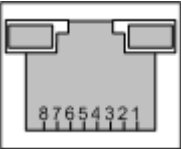
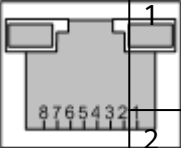
Front View	Pin	Usage
	1	Transmit positive of the NM network port
	2	Transmit negative of the NM network port
	3	Receive positive of the NM network port
	4	Grounding end of the Console interface
	5	Receive end of the Console interface
	6	Receive negative of the NM network port
	7	Unspecified
	8	Transmit end of the Console interface

Table 1-103 Pins of the CLK/TOD/RS-485 interface

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	1	Receive negative of the CLK	Receive negative of the CLK	Unspecified	Unspecified	Unspecified
	2	Receive positive of the CLK	Receive positive of the CLK	Unspecified	Unspecified	Unspecified
	3	Input/Output negative for the 1PPS signal	Unspecified	Input negative for the 1PPS signal (RS422 level)	Output negative for the 1PPS signal (RS422 level)	Transmit negative of the RS-485
	4	Receive negative of the CLK	Receive negative of the CLK	Grounding end	Grounding end	Unspecified
	5	Receive positive of the CLK	Receive positive of the CLK	Grounding end	Grounding end	Unspecified
	6	Input/Output positive for the 1PPS signal	Unspecified	Input positive for the 1PPS signal (RS422 level)	Output positive for the 1PPS signal (RS422 level)	Transmit positive of the RS-485

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	7	Input/Output negative for the time information	Unspecified	Input negative for the time information (RS422 level)	Output negative for the time information (RS422 level)	Receive negative of the RS-485
	8	Input/Output positive for the time information	Unspecified	Input positive for the time information (RS422 level)	Output positive for the time information (RS422 level)	Receive positive of the RS-485
NOTE When this interface is used as a TOD interface, it must be used with a dedicated clock cable. For details, see "Cables" in the Hardware Description.						

Functions and Features

Table 1-104 Functions and features of the CR8D0IPU2TC1

Functions and Features	Description
Basic function	Data forwarding: The IPU is the core of service processing in the entire system, and is connected to all subcards through data channels. Control and management: Through the management channels between IPU and the subcards, the IPU can manage subcards and transmit routing protocol data. The line rate capability is 4T bit/s@256Bytes(Bidirectional).
Reliability and availability	Supports the hot swappable function. Supports the board-level 1:1 backup function.
Restrictions and remarks	1. It cannot be used together with other IPU. 2. When working with the 03033VMC board to support FlexE services, the board supports a maximum of eight FlexE physical interfaces.
Breakout supported	No
Interface rate auto-sensing supported	No

Technical Specifications

Table 1-105 Technical specifications of the CR8D0IPU2TC1

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	34.8 mm x 354.4 mm x 209 mm (1.37 in. x 13.95 in. x 8.23 in.)
Dimensions with packaging (H x W x D) [mm(in.)]	185 mm x 485 mm x 385 mm (7.28 in. x 19.09 in. x 15.16 in.)
Weight without packaging [kg(lb)]	4 kg (8.82 lb)
Weight with packaging [kg(lb)]	5.4 kg (11.9 lb)
Typical power consumption [W]	394.21 W
Typical heat dissipation [BTU/hour]	1278.98 BTU/hour

Item	Specification
Long-term operating temperature [°C(°F)]	–20°C to +55°C (–4°F to +131°F); start: ≥ –5°C (23°F)
CPU	8-core 2.3 GHz
SDRAM	16GB
Storage	16GB M.2

1.4.3.3 CR8D0IPU2TC1 (03033FPV-002: Integrated Network Processing Unit (IPU-2T))

Overview

Table 1-106 Basic information about the CR8D0IPU2TC1

Item	Details
Description	Integrated Network Processing Unit (IPU-2T)
Part Number	03033FPV-002
Model	CR8D0IPU2TC1
Card type	IPU
Silkscreen	IPU-2T

Appearance

Figure 1-72 Appearance of the CR8D0IPU2TC1



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03033FPV-002&productModel=CR8D0IPU2TC1#matchRelation>

Indicators and Buttons

Table 1-107 Indicators on the CR8D0IPU2TC1

Silkscreen	Name	Color	Status	Description
STAT	Working status indicator	Green	Steady on	The board is working normally.
		Red	Steady on	The device generates a hardware alarm. You can run the display alarm hardware command to view the hardware alarm type.
		-	Off	the board is not running or no power is input.
PROG	Program running indicator	Green	Steady on	The board software is being initialized. The board software is normally initialized, and the board software is running normally.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	The memory self-check fails. The board software or the logic file is lost. Loading of the board software fails.
		Green	On for 100 ms and off for 100 ms alternately	Loading of the board software is in process.
		Green	On for 300 ms and off for 300 ms alternately	The BIOS is guiding the upper-layer software.
		Red	On for 100 ms and off for 100 ms alternately	The BIOS self-check fails.
		-	Off	No power is input.

Silkscreen	Name	Color	Status	Description
SYNC	Synchronizati on status indicator	Green	Steady on	The system clock works in free-run mode, and no priority list is set for the system clocks (only internal clock sources are listed by default). The system clock works in tracing mode and is tracing a clock source other than the internal clock sources in the priority list. The system clock works in time synchronizatio n mode, and the PTP time and system clock are being traced.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	The system clock priority list is set, but all clock sources except the internal clock source are lost. The clock works in holdover or free-run mode. The system clock works in time synchronization mode, but no synchronization source is available. The system clock and PTP time work in holdover or free-run mode.
ACTX	Active/standby status indicator for the cross-connect or clock board	Green	Steady on	The board is in the working switching state.
		-	Off	The board is in the protection switching state.
ACTC	Control status indicator	Green	Steady on	The board is in the working control state.
		Green	On for 100 ms and off for 100 ms alternately	The data of the equipment is backed up in batch.

Silkscreen	Name	Color	Status	Description
		-	Off	The board is in the protection control state.
ALM	Alarm indicator	Red	Steady on	Critical alarms are generated.
		Orange	Steady on	Major, minor alarms are generated.
		-	Off	No alarms are generated in the equipment.
ETH/OAM	ETH/OAM indicator	Green	Steady on	If the indicator is steady on, the link is normal.
		Orange	Blink	If the indicator blinks, data is being transmitted.
		-	Off	If the indicator is off, the link is Down.

Table 1-108 Buttons on the CR8D0IPU2TC1

Silkscreen	Name	Description
RST	Reset button	You can press the RST button to reset the NE or restore the factory settings of the NE. The procedure is as follows: To reset the NE, press the RST button and release it. To restore factory settings, press and hold the RST button. This operation will clear the NE database and system parameter configurations and cannot be restored. Exercise caution when you perform the following operations: 1. Press and release the RST button to restart the NE. 2. When the PROG indicator blinks (the blinking frequency is about 1 second and the time window is about 15 seconds), press and hold the RST button. 3. Wait for no longer than 2 minutes till both the STAT and PROG indicators blink (the time window is 5 seconds). Then release the RST button immediately.

Silkscreen	Name	Description
		NOTE 1. When you press the RST button to clear the configuration file, the original configuration file will be cleared. You are advised to periodically back up the configuration file. 2. When you press the RST button to clear the configuration file, to prevent the cleared configuration from being synchronized again between the active and standby main control boards, remove one main control board first in a dual-main control board chassis to ensure that the configuration file is cleared in a single-main control board scenario. Perform the same operations on the other main control board. 3. When pressing the RST button to clear the configuration file, you only need to pay attention to the STAT and PROG indicators. Other indicators may vary depending on the product model.

Ports

Table 1-109 Ports on the CR8D0IPU2TC1

Port	Connector Type	Description	Available Components
ETH/OAM	RJ45	10M/100M autosensing Ethernet network management port or non-RS-232-standard console port. NOTE If a service port can be connected to the NMS, it is recommended that the service port be used as the network management port.	ETH/OAM Management Cables
CLK/TOD/RS-485	RJ45	Clock/Time/RS-485 input/output port, RS-485 is a reserved port.	1-into-2 clock cable + clock cable (external clock mode, delivery-ready): applicable to the CLK mode 1-into-2 clock cable + clock cable (external clock/external time mode, prepared onsite): applicable to the CLK or TOD mode 1-into-2 clock cable + Ethernet cable: applicable to the TOD mode
USB	Type A	USB port (reserved)	1-into-2 clock cable + clock cable (external clock/external time mode, prepared onsite): applicable to the CLK or TOD mode

Port	Connector Type	Description	Available Components
CLK/TEST	SMA	Clock test port	1-into-2 clock cable + Ethernet cable: applicable to the TOD mode
HP-GE	SFP	Ultra-high-precision GE port (supported since V800R021C00 and reserved in earlier versions)	1.25Gbps eSFP Optical Module 1.25Gbps eSFP CWDM Optical Module 1.25Gbps eSFP BIDI Optical Module

Table 1-110 Pins of the ETH/OAM interface

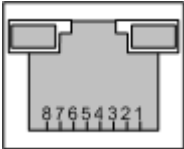
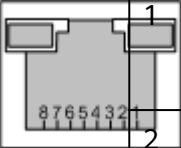
Front View	Pin	Usage
	1	Transmit positive of the NM network port
	2	Transmit negative of the NM network port
	3	Receive positive of the NM network port
	4	Grounding end of the Console interface
	5	Receive end of the Console interface
	6	Receive negative of the NM network port
	7	Unspecified
	8	Transmit end of the Console interface

Table 1-111 Pins of the CLK/TOD/RS-485 interface

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	1	Receive negative of the CLK	Receive negative of the CLK	Unspecified	Unspecified	Unspecified
	2	Receive positive of the CLK	Receive positive of the CLK	Unspecified	Unspecified	Unspecified
	3	Input/Output negative for the 1PPS signal	Unspecified	Input negative for the 1PPS signal (RS422 level)	Output negative for the 1PPS signal (RS422 level)	Transmit negative of the RS-485
	4	Receive negative of the CLK	Receive negative of the CLK	Grounding end	Grounding end	Unspecified
	5	Receive positive of the CLK	Receive positive of the CLK	Grounding end	Grounding end	Unspecified
	6	Input/Output positive for the 1PPS signal	Unspecified	Input positive for the 1PPS signal (RS422 level)	Output positive for the 1PPS signal (RS422 level)	Transmit positive of the RS-485

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	7	Input/Output negative for the time information	Unspecified	Input negative for the time information (RS422 level)	Output negative for the time information (RS422 level)	Receive negative of the RS-485
	8	Input/Output positive for the time information	Unspecified	Input positive for the time information (RS422 level)	Output positive for the time information (RS422 level)	Receive positive of the RS-485
NOTE When this interface is used as a TOD interface, it must be used with a dedicated clock cable. For details, see "Cables" in the Hardware Description.						

Functions and Features

Table 1-112 Functions and features of the CR8D0IPU2TC1

Functions and Features	Description
Basic function	Data forwarding: The IPU is the core of service processing in the entire system, and is connected to all subcards through data channels. Control and management: Through the management channels between IPU and the subcards, the IPU can manage subcards and transmit routing protocol data. The line rate capability is 4T bit/s@256Bytes(Bidirectional).
Reliability and availability	Supports the hot swappable function. Supports the board-level 1:1 backup function.
Restrictions and remarks	1. It cannot be used together with other IPU. 2. When working with the 03033VMC board to support FlexE services, the board supports a maximum of eight FlexE physical interfaces.
Breakout supported	No
Interface rate auto-sensing supported	No

Technical Specifications

Table 1-113 Technical specifications of the CR8D0IPU2TC1

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	34.8 mm x 354.4 mm x 209 mm (1.37 in. x 13.95 in. x 8.23 in.)
Dimensions with packaging (H x W x D) [mm(in.)]	185 mm x 485 mm x 385 mm (7.28 in. x 19.09 in. x 15.16 in.)
Weight without packaging [kg(lb)]	4 kg (8.82 lb)
Weight with packaging [kg(lb)]	5.4 kg (11.9 lb)
Typical power consumption [W]	385.29 W
Typical heat dissipation [BTU/hour]	1250.08 BTU/hour

Item	Specification
Long-term operating temperature [°C(°F)]	–20°C to +55°C (–4°F to +131°F); start: ≥ –5°C (23°F)
CPU	20-core 2.0 GHz
SDRAM	16GB
Storage	4 GB NandFlash

1.4.3.4 CR8D0IPU2TC3 (Integrated Network Processing Unit (IPU-2T-B))

Overview

Table 1-114 Basic information about the CR8D0IPU2TC3

Item	Details
Description	Integrated Network Processing Unit (IPU-2T-B)
Part Number	03034BMA
Model	CR8D0IPU2TC3
Card type	IPU
Silkscreen	IPU-2T-B

Appearance

Figure 1-73 Appearance of the CR8D0IPU2TC3



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03034BMA&productModel=CR8D0IPU2TC3#matchRelation>

Indicators and Buttons

Table 1-115 Indicators on the CR8D0IPU2TC3

Silkscreen	Name	Color	Status	Description
STAT	Working status indicator	Green	Steady on	The board is working normally.
		Red	Steady on	The device generates a hardware alarm. You can run the display alarm hardware command to view the hardware alarm type.
		-	Off	the board is not running or no power is input.
PROG	Program running indicator	Green	Steady on	The board software is being initialized. The board software is normally initialized, and the board software is running normally.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	The memory self-check fails. The board software or the logic file is lost. Loading of the board software fails.
		Green	On for 100 ms and off for 100 ms alternately	Loading of the board software is in process.
		Green	On for 300 ms and off for 300 ms alternately	The BIOS is guiding the upper-layer software.
		Red	On for 100 ms and off for 100 ms alternately	The BIOS self-check fails.
		-	Off	No power is input.

Silkscreen	Name	Color	Status	Description
SYNC	Synchronizati on status indicator	Green	Steady on	The system clock works in free-run mode, and no priority list is set for the system clocks (only internal clock sources are listed by default). The system clock works in tracing mode and is tracing a clock source other than the internal clock sources in the priority list. The system clock works in time synchronizatio n mode, and the PTP time and system clock are being traced.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	The system clock priority list is set, but all clock sources except the internal clock source are lost. The clock works in holdover or free-run mode. The system clock works in time synchronization mode, but no synchronization source is available. The system clock and PTP time work in holdover or free-run mode.
ACTX	Active/standby status indicator for the cross-connect or clock board	Green	Steady on	The board is in the working switching state.
		-	Off	The board is in the protection switching state.
ACTC	Control status indicator	Green	Steady on	The board is in the working control state.
		Green	On for 100 ms and off for 100 ms alternately	The data of the equipment is backed up in batch.

Silkscreen	Name	Color	Status	Description
		-	Off	The board is in the protection control state.
ALM	Alarm indicator	Red	Steady on	Critical alarms are generated.
		Orange	Steady on	Major, minor alarms are generated.
		-	Off	No alarms are generated in the equipment.
ETH/OAM	ETH/OAM indicator	Green	Steady on	If the indicator is steady on, the link is normal.
		Orange	Blink	If the indicator blinks, data is being transmitted.
		-	Off	If the indicator is off, the link is Down.

Table 1-116 Buttons on the CR8D0IPU2TC3

Silkscreen	Name	Description
RST	Reset button	You can press the RST button to reset the NE or restore the factory settings of the NE. The procedure is as follows: To reset the NE, press the RST button and release it. To restore factory settings, press and hold the RST button. This operation will clear the NE database and system parameter configurations and cannot be restored. Exercise caution when you perform the following operations: 1. Press and release the RST button to restart the NE. 2. When the PROG indicator blinks (the blinking frequency is about 1 second and the time window is about 15 seconds), press and hold the RST button. 3. Wait for no longer than 2 minutes till both the STAT and PROG indicators blink (the time window is 5 seconds). Then release the RST button immediately.

Silkscreen	Name	Description
		NOTE 1. When you press the RST button to clear the configuration file, the original configuration file will be cleared. You are advised to periodically back up the configuration file. 2. When you press the RST button to clear the configuration file, to prevent the cleared configuration from being synchronized again between the active and standby main control boards, remove one main control board first in a dual-main control board chassis to ensure that the configuration file is cleared in a single-main control board scenario. Perform the same operations on the other main control board. 3. When pressing the RST button to clear the configuration file, you only need to pay attention to the STAT and PROG indicators. Other indicators may vary depending on the product model.

Ports

Table 1-117 Ports on the CR8D0IPU2TC3

Port	Connector Type	Description	Available Components
ETH/OAM	RJ45	10M/100M autosensing Ethernet network management port or non-RS-232-standard console port. NOTE If a service port can be connected to the NMS, it is recommended that the service port be used as the network management port.	ETH/OAM Management Cables
CLK/TOD/RS-485	RJ45	Clock/Time/RS-485 input/output port, RS-485 is a reserved port.	1-into-2 clock cable + clock cable (external clock mode, delivery-ready): applicable to the CLK mode 1-into-2 clock cable + clock cable (external clock/external time mode, prepared onsite): applicable to the CLK or TOD mode 1-into-2 clock cable + Ethernet cable: applicable to the TOD mode
USB	Type A	USB port (reserved)	1-into-2 clock cable + clock cable (external clock/external time mode, prepared onsite): applicable to the CLK or TOD mode

Port	Connector Type	Description	Available Components
CLK/TEST	SMA	Clock test port	1-into-2 clock cable + Ethernet cable: applicable to the TOD mode
GE0,GE1	SFP	Cascading ports are used for control panel expansion in scenarios like virtual cluster. (reserved)	Optical fiber
HP-GE	SFP	Ultra-high-precision GE port (This first supported version is V800R021C00. This port is reserved in earlier versions.)	1.25Gbps eSFP Optical Module 1.25Gbps eSFP CWDM Optical Module 1.25Gbps eSFP BIDI Optical Module

Table 1-118 Pins of the ETH/OAM interface

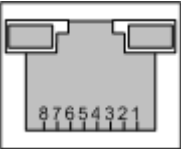
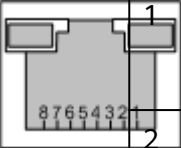
Front View	Pin	Usage
	1	Transmit positive of the NM network port
	2	Transmit negative of the NM network port
	3	Receive positive of the NM network port
	4	Grounding end of the Console interface
	5	Receive end of the Console interface
	6	Receive negative of the NM network port
	7	Unspecified
	8	Transmit end of the Console interface

Table 1-119 Pins of the CLK/TOD/RS-485 interface

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	1	Receive negative of the CLK	Receive negative of the CLK	Unspecified	Unspecified	Unspecified
	2	Receive positive of the CLK	Receive positive of the CLK	Unspecified	Unspecified	Unspecified
	3	Input/Output negative for the 1PPS signal	Unspecified	Input negative for the 1PPS signal (RS422 level)	Output negative for the 1PPS signal (RS422 level)	Transmit negative of the RS-485
	4	Receive negative of the CLK	Receive negative of the CLK	Grounding end	Grounding end	Unspecified
	5	Receive positive of the CLK	Receive positive of the CLK	Grounding end	Grounding end	Unspecified
	6	Input/Output positive for the 1PPS signal	Unspecified	Input positive for the 1PPS signal (RS422 level)	Output positive for the 1PPS signal (RS422 level)	Transmit positive of the RS-485

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	7	Input/Output negative for the time information	Unspecified	Input negative for the time information (RS422 level)	Output negative for the time information (RS422 level)	Receive negative of the RS-485
	8	Input/Output positive for the time information	Unspecified	Input positive for the time information (RS422 level)	Output positive for the time information (RS422 level)	Receive positive of the RS-485
NOTE When this interface is used as a TOD interface, it must be used with a dedicated clock cable. For details, see "Cables" in the Hardware Description.						

Functions and Features

Table 1-120 Functions and features of the CR8D0IPU2TC3

Functions and Features	Description
Basic function	Data forwarding: The IPU is the core of service processing in the entire system, and is connected to all subcards through data channels. Control and management: Through the management channels between IPU and the subcards, the IPU can manage subcards and transmit routing protocol data. The line rate capability is 4T bit/s@256Bytes(Bidirectional).
Reliability and availability	Supports the hot swappable function. Supports the board-level 1:1 backup function.
Restrictions and remarks	1. It cannot be used together with other IPU. 2. When working with the 03033VMC board to support FlexE services, the board supports a maximum of eight FlexE physical interfaces.
Breakout supported	No
Interface rate auto-sensing supported	No

Technical Specifications

Table 1-121 Technical specifications of the CR8D0IPU2TC3

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	34.8 mm x 354.4 mm x 209 mm (1.37 in. x 13.95 in. x 8.23 in.)
Dimensions with packaging (H x W x D) [mm(in.)]	185 mm x 485 mm x 385 mm (7.28 in. x 19.09 in. x 15.16 in.)
Weight without packaging [kg(lb)]	4 kg (8.82 lb)
Weight with packaging [kg(lb)]	5.4 kg (11.9 lb)
Typical power consumption [W]	324.4 W
Typical heat dissipation [BTU/hour]	1052.55 BTU/hour

Item	Specification
Long-term operating temperature [°C(°F)]	–20°C to +55°C (–4°F to +131°F); start: ≥ –5°C (23°F)
CPU	20-core 2.0 GHz
SDRAM	16GB
Storage	4 GB NandFlash

1.4.3.5 CR8D0IPU2TC5 (Integrated Network Processing Unit (IPU-2T-C))

Overview

Table 1-122 Basic information about the CR8D0IPU2TC5

Item	Details
Description	Integrated Network Processing Unit (IPU-2T-C)
Part Number	03034MGL
Model	CR8D0IPU2TC5
Card type	IPU
Silkscreen	IPU-2T-C

Appearance

Figure 1-74 Appearance of the CR8D0IPU2TC5



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03034MGL&productModel=CR8D0IPU2TC5#matchRelation>

Indicators and Buttons

Table 1-123 Indicators on the CR8D0IPU2TC5

Silkscreen	Name	Color	Status	Description
STAT	Working status indicator	Green	Steady on	The board is working properly.
		Red	Steady on	A hardware alarm is generated. You can run the display alarm hardware command to check the hardware alarm type.
		-	Off	The board is not powered on or is not running.
PROG	Program status indicator	Green	Steady on	The board software is being initialized or is successfully initialized and is running properly.
		Red	Steady on	The memory self-check fails. The board software or logic file is lost. The board software fails to be loaded. The BIOS self-check fails.

Silkscreen	Name	Color	Status	Description
		Green	Blinks on and off at 100 ms intervals	The board software is being loaded.
		Green	Blinks on and off at 300 ms intervals	The BIOS is interacting with the upper-layer software.
		-	Off	The board is not powered on.
SYNC	Clock synchronization indicator	Green	Steady on	The system clock works in free-run mode, and no priority list is set for the system clocks (only internal clock sources are listed by default). The system clock works in tracing mode and is tracing a clock source other than the internal clock sources in the priority list. The system clock works in time synchronization mode, and the PTP time and system clock are being traced.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	The system clock priority list is set, but all clock sources except the internal clock source are lost. The clock works in holdover or free-run mode. The system clock works in time synchronization mode, but no synchronization source is available. The system clock and PTP time work in holdover or free-run mode.
ACTX	Active/standby status indicator for the cross-connect or clock board	Green	Steady on	The board is in the active state on the service plane.
		-	Off	The board is in the standby state on the service plane.
ACTC	Control status indicator	Green	Steady on	The board is in the active control state.
		Green	Blinks on and off at 100 ms intervals	The data of the device is backed up in batch.
		-	Off	The board is in the standby control state.

Silkscreen	Name	Color	Status	Description
ALM	Alarm indicator	Red	Steady on	Device-level critical alarms exist.
		Orange	Steady on	A device-level major or minor alarm exists.
		-	Off	There is no equipment-level alarm.
ETH/OAM	ETH/OAM indicator	Green	Steady on	The link is up.
		Orange	Blinking	Data is being transmitted or received.
		-	Off	The link is down.

Table 1-124 Buttons on the CR8D0IPU2TC5

Silkscreen	Name	Description
RST	Board reset button	You can press the RST button to reset the NE or restore the factory settings of the NE. The procedure is as follows: To reset the NE, press the RST button and release it. To restore factory settings, press and hold down the RST button. This operation will clear the NE database and system parameter configurations and cannot be restored. Exercise caution when you perform the following operations: 1.Press the RST button to restart the NE. 2.When the PROG indicator blinks (the blinking frequency is about 1 second and the time window is about 15 seconds), press and hold down the RST button. 3.Wait for no longer than 2 minutes till both the STAT and PROG indicators blink (the time window is 5 seconds). Then release the RST button immediately.

Silkscreen	Name	Description
		NOTE 1. When you press the RST button to clear the configuration file, the original configuration file will be cleared. You are advised to periodically back up the configuration file. 2. When you press the RST button to clear the configuration file, to prevent the cleared configuration from being synchronized again between the active and standby main control boards, remove one main control board first in a dual-main control board chassis to ensure that the configuration file is cleared in a single-main control board scenario. Perform the same operations on the other main control board. 3. When pressing the RST button to clear the configuration file, you only need to pay attention to the STAT and PROG indicators. Other indicators may vary depending on the product model.

Ports

Table 1-125 Ports on the CR8D0IPU2TC5

Port	Connector Type	Description	Available Components
ETH/OAM	RJ45	10M/100M auto-sensing Ethernet network management port or non-standard RS-232 console port NOTE If a service port can be connected to an NMS, you are advised to use the service port as the network management port.	ETH/OAM management cable
CLK/TOD/RS-485	RJ45	Clock/Time/RS-485 input and output ports RS-485 is a reserved port.	1-into-2 clock cable + clock cable (external clock mode, delivery-ready): applicable to the CLK mode 1-into-2 clock cable + clock cable (external clock/external time mode, prepared onsite): applicable to the CLK or TOD mode 1-into-2 clock cable + Ethernet cable: applicable to the TOD mode
USB	Type A	USB port (reserved)	-
CLK/TEST	SMA	Clock test port	Coaxial cable

Port	Connector Type	Description	Available Components
HP-GE	SFP	Ultra-high-precision GE port (supported since V800R021C00 and reserved in earlier versions)	1.25Gbps eSFP Optical Module 1.25Gbps eSFP CWDM Optical Module 1.25Gbps eSFP BIDI Optical Module

Table 1-126 Pins of the ETH/OAM interface

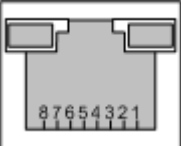
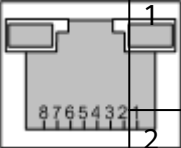
Front View	Pin	Usage
	1	Transmit positive of the NM network port
	2	Transmit negative of the NM network port
	3	Receive positive of the NM network port
	4	Grounding end of the Console interface
	5	Receive end of the Console interface
	6	Receive negative of the NM network port
	7	Unspecified
	8	Transmit end of the Console interface

Table 1-127 Pins of the CLK/TOD/RS-485 interface

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	1	Receive negative of the CLK	Receive negative of the CLK	Unspecified	Unspecified	Unspecified
	2	Receive positive of the CLK	Receive positive of the CLK	Unspecified	Unspecified	Unspecified
	3	Input/Output negative for the 1PPS signal	Unspecified	Input negative for the 1PPS signal (RS422 level)	Output negative for the 1PPS signal (RS422 level)	Transmit negative of the RS-485
	4	Receive negative of the CLK	Receive negative of the CLK	Grounding end	Grounding end	Unspecified
	5	Receive positive of the CLK	Receive positive of the CLK	Grounding end	Grounding end	Unspecified
	6	Input/Output positive for the 1PPS signal	Unspecified	Input positive for the 1PPS signal (RS422 level)	Output positive for the 1PPS signal (RS422 level)	Transmit positive of the RS-485

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	7	Input/Output negative for the time information	Unspecified	Input negative for the time information (RS422 level)	Output negative for the time information (RS422 level)	Receive negative of the RS-485
	8	Input/Output positive for the time information	Unspecified	Input positive for the time information (RS422 level)	Output positive for the time information (RS422 level)	Receive positive of the RS-485
NOTE When this interface is used as a TOD interface, it must be used with a dedicated clock cable. For details, see "Cables" in the Hardware Description.						

Functions and Features

Table 1-128 Functions and features of the CR8D0IPU2TC5

Functions and Features	Description
Basic function	Data forwarding: The IPU is the core of service processing in the entire system, and is connected to all subcards through data channels. Control and management: The MPU manages subcards and transmits routing protocol data through management channels. Supports 4 Tbit/s@256 bytes (bidirectional).
Reliability and availability	Hot swappable. 1:1 backup is supported.
Restrictions and remarks	1. It cannot be used together with other IPU. 2. When working with the 03033VMC board to support FlexE services, the board supports a maximum of eight FlexE physical interfaces.
Breakout supported	No
Interface rate auto-sensing supported	No

Technical Specifications

Table 1-129 Technical specifications of the CR8D0IPU2TC5

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	34.8 mm x 354.4 mm x 209 mm (1.37 in. x 13.95 in. x 8.23 in.)
Dimensions with packaging (H x W x D) [mm(in.)]	185 mm x 485 mm x 385 mm (7.28 in. x 19.09 in. x 15.16 in.)
Weight without packaging [kg(lb)]	4 kg (8.82 lb)
Weight with packaging [kg(lb)]	5.4 kg (11.9 lb)
Typical power consumption [W]	394.21 W
Typical heat dissipation [BTU/hour]	1278.98 BTU/hour

Item	Specification
Long-term operating temperature [°C(°F)]	–20°C to +55°C, startup: ≥ –5°C
CPU	8-core 2.3 GHz
SDRAM	16GB
Storage	16GB M.2

1.4.3.6 CR8D0IPU2TM1 (Integrated Network Processing Unit (IPU-2T-E))

Overview

Table 1-130 Basic information about the CR8D0IPU2TM1

Item	Details
Description	Integrated Network Processing Unit (IPU-2T-E)
Part Number	03034VAT
Model	CR8D0IPU2TM1
Card type	IPU
Silkscreen	IPU-2T-E

Appearance

Figure 1-75 Appearance of the CR8D0IPU2TM1



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03034VAT&productModel=CR8D0IPU2TM1#matchRelation>

Indicators and Buttons

Table 1-131 Indicators on the CR8D0IPU2TM1

Silkscreen	Name	Color	Status	Description
STAT	Working status indicator	Green	Steady on	The board is working properly.
		Red	Steady on	A hardware alarm is generated. You can run the display alarm hardware command to check the hardware alarm type.
		-	Off	The board is not powered on or is not running.
PROG	Program status indicator	Green	Steady on	The board software is being initialized or is successfully initialized and is running properly.
		Red	Steady on	The memory self-check fails. The board software or logic file is lost. The board software fails to be loaded.

Silkscreen	Name	Color	Status	Description
		Green	Blinks on and off at 100 ms intervals	The board software is being loaded.
		Green	Blinks on and off at 300 ms intervals	The BIOS is interacting with the upper-layer software.
		Red	Blinks on and off at 100 ms intervals	The BIOS self-check fails.
		-	Off	The board is not powered on.
SYNC	Clock synchronization indicator	Green	Steady on	The system clock works in free-run mode, and no priority list is set for the system clocks (only internal clock sources are listed by default). The system clock works in tracing mode and is tracing a clock source other than the internal clock sources in the priority list. The system clock works in time synchronization mode, and the PTP time and system clock are being traced.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	The system clock priority list is set, but all clock sources except the internal clock source are lost. The clock works in holdover or free-run mode. The system clock works in time synchronization mode, but no synchronization source is available. The system clock and PTP time work in holdover or free-run mode.
ACTX	Active/standby status indicator for the cross-connect or clock board	Green	Steady on	The board is in the active state on the service plane.
		-	Off	The board is in the standby state on the service plane.
ACTC	Control status indicator	Green	Steady on	The board is in the active control state.
		Green	Blinks on and off at 100 ms intervals	The data of the device is backed up in batch.
		-	Off	The board is in the standby control state.

Silkscreen	Name	Color	Status	Description
ALM	Alarm indicator	Red	Steady on	Device-level critical alarms exist.
		Orange	Steady on	A device-level major or minor alarm exists.
		-	Off	There is no equipment-level alarm.
ETH/OAM	ETH/OAM indicator	Green	Steady on	The link is up.
		Orange	Blinking	Data is being transmitted or received.
		-	Off	The link is down.

Table 1-132 Buttons on the CR8D0IPU2TM1

Silkscreen	Name	Description
RST	Board reset button	You can press the RST button to reset the NE or restore the factory settings of the NE. The procedure is as follows: To reset the NE, press the RST button and release it. To restore factory settings, press and hold down the RST button. This operation will clear the NE database and system parameter configurations and cannot be restored. Exercise caution when you perform the following operations: 1.Press the RST button to restart the NE. 2.When the PROG indicator blinks (the blinking frequency is about 1 second and the time window is about 15 seconds), press and hold down the RST button. 3.Wait for no longer than 2 minutes till both the STAT and PROG indicators blink (the time window is 5 seconds). Then release the RST button immediately.

Silkscreen	Name	Description
		NOTE 1. When you press the RST button to clear the configuration file, the original configuration file will be cleared. You are advised to periodically back up the configuration file. 2. When you press the RST button to clear the configuration file, to prevent the cleared configuration from being synchronized again between the active and standby main control boards, remove one main control board first in a dual-main control board chassis to ensure that the configuration file is cleared in a single-main control board scenario. Perform the same operations on the other main control board. 3. When pressing the RST button to clear the configuration file, you only need to pay attention to the STAT and PROG indicators. Other indicators may vary depending on the product model.

Ports

Table 1-133 Ports on the CR8D0IPU2TM1

Port	Connector Type	Description	Available Components
ETH/OAM	RJ45	10M/100M auto-sensing Ethernet network management port or non-standard RS-232 console port NOTE If a service port can be connected to an NMS, you are advised to use the service port as the network management port.	ETH/OAM management cable
CLK/TOD/RS-485	RJ45	Clock/Time/RS-485 input and output ports RS-485 is a reserved port.	1-into-2 clock cable + clock cable (external clock mode, delivery-ready): applicable to the CLK mode 1-into-2 clock cable + clock cable (external clock/external time mode, prepared onsite): applicable to the CLK or TOD mode 1-into-2 clock cable + Ethernet cable: applicable to the TOD mode
USB	Type A	USB port (reserved)	-
CLK/TEST	SMA	Clock test port	Coaxial cable

Port	Connector Type	Description	Available Components
HP-GE	SFP	Ultra-high-precision GE port (supported since V800R021C00 and reserved in earlier versions)	1.25Gbps eSFP Optical Module 1.25Gbps eSFP CWDM Optical Module 1.25Gbps eSFP BIDI Optical Module

Table 1-134 Pins of the ETH/OAM interface

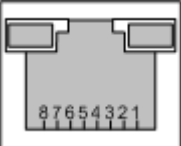
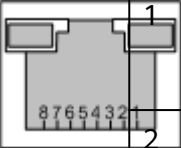
Front View	Pin	Usage
	1	Transmit positive of the NM network port
	2	Transmit negative of the NM network port
	3	Receive positive of the NM network port
	4	Grounding end of the Console interface
	5	Receive end of the Console interface
	6	Receive negative of the NM network port
	7	Unspecified
	8	Transmit end of the Console interface

Table 1-135 Pins of the CLK/TOD/RS-485 interface

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	1	Receive negative of the CLK	Receive negative of the CLK	Unspecified	Unspecified	Unspecified
	2	Receive positive of the CLK	Receive positive of the CLK	Unspecified	Unspecified	Unspecified
	3	Input/Output negative for the 1PPS signal	Unspecified	Input negative for the 1PPS signal (RS422 level)	Output negative for the 1PPS signal (RS422 level)	Transmit negative of the RS-485
	4	Receive negative of the CLK	Receive negative of the CLK	Grounding end	Grounding end	Unspecified
	5	Receive positive of the CLK	Receive positive of the CLK	Grounding end	Grounding end	Unspecified
	6	Input/Output positive for the 1PPS signal	Unspecified	Input positive for the 1PPS signal (RS422 level)	Output positive for the 1PPS signal (RS422 level)	Transmit positive of the RS-485

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	7	Input/Output negative for the time information	Unspecified	Input negative for the time information (RS422 level)	Output negative for the time information (RS422 level)	Receive negative of the RS-485
	8	Input/Output positive for the time information	Unspecified	Input positive for the time information (RS422 level)	Output positive for the time information (RS422 level)	Receive positive of the RS-485
NOTE When this interface is used as a TOD interface, it must be used with a dedicated clock cable. For details, see "Cables" in the Hardware Description.						

Functions and Features

Table 1-136 Functions and features of the CR8D0IPU2TM1

Functions and Features	Description
Basic function	Data forwarding: The IPU is the core of service processing in the entire system, and is connected to all subcards through data channels. Control and management: The MPU manages subcards and transmits routing protocol data through management channels. Supports 4 Tbit/s@256 bytes (bidirectional).
Reliability and availability	Hot swappable. 1:1 backup is supported.
Restrictions and remarks	1. It cannot be used together with other IPU. 2. When working with the 03033VMC board to support FlexE services, the board supports a maximum of eight FlexE physical interfaces.
Breakout supported	No
Interface rate auto-sensing supported	No

Technical Specifications

Table 1-137 Technical specifications of the CR8D0IPU2TM1

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	34.8 mm x 354.4 mm x 209 mm (1.37 in. x 13.95 in. x 8.23 in.)
Dimensions with packaging (H x W x D) [mm(in.)]	185 mm x 485 mm x 385 mm (7.28 in. x 19.09 in. x 15.16 in.)
Weight without packaging [kg(lb)]	4 kg (8.82 lb)
Weight with packaging [kg(lb)]	5.4 kg (11.9 lb)
Typical power consumption [W]	385.3W
Typical heat dissipation [BTU/hour]	1250.08 BTU/hour

Item	Specification
Long-term operating temperature [°C(°F)]	–20°C to +55°C, startup: ≥ –5°C
CPU	20-core 2.0 GHz
SDRAM	16GB
Storage	4 GB NandFlash

1.4.3.7 CR8D0IPU3TM1 (Integrated Network Processing Unit (IPU-3T6-A))

Overview

Table 1-138 Basic information about the CR8D0IPU3TM1

Item	Details
Description	Integrated Network Processing Unit (IPU-3T6-A)
Part Number	03034YNL
Model	CR8D0IPU3TM1
Card type	IPU
Silkscreen	IPU-3T6-A
Remarks	The CR8M002FBXC3 fan is required.

Appearance

Figure 1-76 Appearance of the CR8D0IPU3TM1



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find

desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03034YNL&productModel=CR8D0IPU3TM1#matchRelation>

Indicators and Buttons

Table 1-139 Indicators on the CR8D0IPU3TM1

Silkscreen	Name	Color	Status	Description
STAT	Working status indicator	Green	Steady on	The board is working normally.
		Red	Steady on	A hardware alarm is generated. You can run the display alarm hardware command to check the hardware alarm type.
		-	Steady off	The board is not running or no power is input.
PROG	Program status indicator	Green	Steady on	The board software is being initialized or is successfully initialized and is running properly.
		Red	Steady on	The memory self-check fails. The board software or logic file is lost. The board software fails to be loaded.

Silkscreen	Name	Color	Status	Description
		Green	Blinks on and off at 100 ms intervals	The board software is being loaded.
		Green	On for 300 ms and off for 300 ms repeatedly	The BIOS is interacting with the upper-layer software.
		Red	Blinks on and off at 100 ms intervals	The BIOS self-check fails.
		-	Steady off	The board is not powered on.
SYNC	Clock synchronization indicator	Green	Steady on	The system clock works in free-run mode, and no priority list is set for the system clocks (only internal clock sources are listed by default). The system clock works in tracing mode and is tracing a clock source other than the internal clock sources in the priority list. The system clock works in time synchronization mode, and the PTP time and system clock are being traced.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	The system clock priority list is set, but all clock sources except the internal clock source are lost. The clock works in holdover or free-run mode. The system clock works in time synchronization mode, but no synchronization source is available. The system clock and PTP time work in holdover or free-run mode.
ACTX	Active/standby status indicator for the cross-connect or clock board	Green	Steady on	The board is in the active state on the service plane.
		-	Steady off	The board is in the standby state on the service plane.
ACTC	Control status indicator	Green	Steady on	The board is in the active control state.
		Green	Blinks on and off at 100 ms intervals	The data of the device is backed up in batch.
		-	Steady off	The board is in the standby control state.

Silkscreen	Name	Color	Status	Description
ALM	Alarm indicator	Red	Steady on	Device-level critical alarms exist.
		Orange	Steady on	A device-level major or minor alarm exists.
		-	Steady off	There is no equipment-level alarm.
ETH/OAM	ETH/OAM indicator	Green	Steady on	The link is up.
		Orange	Blinking	Data is being transmitted and received.
		-	Steady off	The link is Down.

Table 1-140 Buttons on the CR8D0IPU3TM1

Silkscreen	Name	Description
RST	Board reset button	You can press the RST button to reset the NE or restore the factory settings of the NE. The procedure is as follows: To reset the NE, press the RST button and release it. To restore factory settings, press and hold down the RST button. This operation will clear the NE database and system parameter configurations and cannot be restored. Exercise caution when you perform the following operations: 1.Press the RST button to restart the NE. 2.When the PROG indicator blinks (the blinking frequency is about 1 second and the time window is about 15 seconds), press and hold down the RST button. 3.Wait for no longer than 2 minutes till both the STAT and PROG indicators blink (the time window is 5 seconds). Then release the RST button immediately.

Silkscreen	Name	Description
		NOTE 1. When you press the RST button to clear the configuration file, the original configuration file will be cleared. You are advised to periodically back up the configuration file. 2. When you press the RST button to clear the configuration file, to prevent the cleared configuration from being synchronized again between the active and standby main control boards, remove one main control board first in a dual-main control board chassis to ensure that the configuration file is cleared in a single-main control board scenario. Perform the same operations on the other main control board. 3. When pressing the RST button to clear the configuration file, you only need to pay attention to the STAT and PROG indicators. Other indicators may vary depending on the product model.

Ports

Table 1-141 Ports on the CR8D0IPU3TM1

Port	Connector Type	Description	Available Components
ETH/OAM	RJ45	10M/100M auto-sensing Ethernet network management port and non-standard RS-232 console port NOTE If a service port can be connected to the NMS, it is recommended that the service port be used as the network management port.	ETH/OAM management cable
CLK/TOD/RS-485	RJ45	Clock/Time/RS-485 input and output ports RS-485 is a reserved port.	1-into-2 clock cable + clock cable (external clock mode, delivery-ready): applicable to the CLK mode 1-into-2 clock cable + clock cable (external clock/external time mode, prepared onsite): applicable to the CLK or TOD mode 1-into-2 clock cable + Ethernet cable: applicable to the TOD mode
USB	Type A	USB port (reserved)	-
CLK/TEST	SMA	Clock test port	Coaxial cable

Port	Connector Type	Description	Available Components
HP-GE	SFP	Ultra-high-precision GE interface	1.25Gbps eSFP Optical Module 1.25Gbps eSFP CWDM Optical Module 1.25Gbps eSFP BIDI Optical Module

Table 1-142 Pins of the ETH/OAM interface

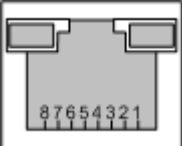
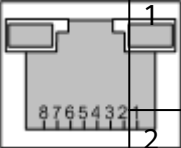
Front View	Pin	Usage
	1	Transmit positive of the NM network port
	2	Transmit negative of the NM network port
	3	Receive positive of the NM network port
	4	Grounding end of the Console interface
	5	Receive end of the Console interface
	6	Receive negative of the NM network port
	7	Unspecified
	8	Transmit end of the Console interface

Table 1-143 Pins of the CLK/TOD/RS-485 interface

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	1	Receive negative of the CLK	Receive negative of the CLK	Unspecified	Unspecified	Unspecified
	2	Receive positive of the CLK	Receive positive of the CLK	Unspecified	Unspecified	Unspecified
	3	Input/Output negative for the 1PPS signal	Unspecified	Input negative for the 1PPS signal (RS422 level)	Output negative for the 1PPS signal (RS422 level)	Transmit negative of the RS-485
	4	Receive negative of the CLK	Receive negative of the CLK	Grounding end	Grounding end	Unspecified
	5	Receive positive of the CLK	Receive positive of the CLK	Grounding end	Grounding end	Unspecified
	6	Input/Output positive for the 1PPS signal	Unspecified	Input positive for the 1PPS signal (RS422 level)	Output positive for the 1PPS signal (RS422 level)	Transmit positive of the RS-485

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	7	Input/Output negative for the time information	Unspecified	Input negative for the time information (RS422 level)	Output negative for the time information (RS422 level)	Receive negative of the RS-485
	8	Input/Output positive for the time information	Unspecified	Input positive for the time information (RS422 level)	Output positive for the time information (RS422 level)	Receive positive of the RS-485
NOTE When this interface is used as a TOD interface, it must be used with a dedicated clock cable. For details, see "Cables" in the Hardware Description.						

Functions and Features

Table 1-144 Functions and features of the CR8D0IPU3TM1

Functions and Features	Description
Basic function	Data forwarding: The IPU is the core of service processing in the entire system, and is connected to all subcards through data channels. Control and management: The main control board manages subcards and transmits routing protocol data through management channels. 7.2 Tbit/s@544 bytes (bidirectional)
Reliability and availability	Hot swappable. 1:1 backup is supported.
Restrictions and remarks	This board cannot be used together with other IPU.
Breakout supported	No
Interface rate auto-sensing supported	No

Technical Specifications

Table 1-145 Technical specifications of the CR8D0IPU3TM1

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	34.8 mm x 354.4 mm x 209 mm (1.37 in. x 13.95 in. x 8.23 in.)
Dimensions with packaging (H x W x D) [mm(in.)]	185 mm x 485 mm x 385 mm (7.28 in. x 19.09 in. x 15.16 in.)
Weight without packaging [kg(lb)]	3.7 kg (8.16 lb)
Weight with packaging [kg(lb)]	5.1 kg (11.24 lb)
Typical power consumption [W]	253.1 W
Typical heat dissipation [BTU/hour]	821.05 BTU/hour
Long-term operating temperature [°C(°F)]	-20°C to +55°C, startup: ≥ -5°C
CPU	16-core 2.0 GHz
SDRAM	16GB
Storage	16GB M.2

1.4.3.8 CR8D0IPU3TM2 (Integrated Network Processing Unit (IPU-3T6-E-B))

Overview

Table 1-146 Basic information about the CR8D0IPU3TM2

Item	Details
Description	Integrated Network Processing Unit (IPU-3T6-E-B)
Part Number	03035FBB
Model	CR8D0IPU3TM2
Card type	IPU
Silkscreen	IPU-3T6-E-B
Remarks	The CR8M002FBXC3 fan is required.

Appearance

Figure 1-77 Appearance of the CR8D0IPU3TM2



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03035FBB&productModel=CR8D0IPU3TM2#matchRelation>

Indicators and Buttons

Table 1-147 Indicators on the CR8D0IPU3TM2

Silkscreen	Name	Color	Status	Description
STAT	Working status indicator	Green	Steady on	The board is working normally.
		Red	Steady on	A hardware alarm is generated. You can run the display alarm hardware command to check the hardware alarm type.
		-	Steady off	The board is not running or no power is input.
PROG	Program status indicator	Green	Steady on	The board software is being initialized or is successfully initialized and is running properly.
		Red	Steady on	The memory self-check fails. The board software or logic file is lost. The board software fails to be loaded.
		Green	Blinks on and off at 100 ms intervals	The board software is being loaded.

Silkscreen	Name	Color	Status	Description
		Green	On for 300 ms and off for 300 ms repeatedly	The BIOS is interacting with the upper-layer software.
		Red	Blinks on and off at 100 ms intervals	The BIOS self-check fails.
		-	Steady off	The board is not powered on.
SYNC	Clock synchronization indicator	Green	Steady on	The system clock works in free-run mode, and no priority list is set for the system clocks (only internal clock sources are listed by default). The system clock works in tracing mode and is tracing a clock source other than the internal clock sources in the priority list. The system clock works in time synchronization mode, and the PTP time and system clock are being traced.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	The system clock priority list is set, but all clock sources except the internal clock source are lost. The clock works in holdover or free-run mode. The system clock works in time synchronization mode, but no synchronization source is available. The system clock and PTP time work in holdover or free-run mode.
ACTX	Active/standby status indicator for the cross-connect or clock board	Green	Steady on	The board is in the active state on the service plane.
		-	Steady off	The board is in the standby state on the service plane.
ACTC	Control status indicator	Green	Steady on	The board is in the active control state.
		Green	Blinks on and off at 100 ms intervals	The data of the device is backed up in batch.
		-	Steady off	The board is in the standby control state.

Silkscreen	Name	Color	Status	Description
ALM	Alarm indicator	Red	Steady on	Device-level critical alarms exist.
		Orange	Steady on	A device-level major or minor alarm exists.
		-	Steady off	There is no equipment-level alarm.
ETH/OAM	ETH/OAM indicator	Green	Steady on	The link is up.
		Orange	Blinking	Data is being transmitted and received.
		-	Steady off	The link is Down.

Table 1-148 Buttons on the CR8D0IPU3TM2

Silkscreen	Name	Description
RST	Board reset button	You can press the RST button to reset the NE or restore the factory settings of the NE. The procedure is as follows: To reset the NE, press the RST button and release it. To restore factory settings, press and hold down the RST button. This operation will clear the NE database and system parameter configurations and cannot be restored. Exercise caution when you perform the following operations: 1.Press the RST button to restart the NE. 2.When the PROG indicator blinks (the blinking frequency is about 1 second and the time window is about 15 seconds), press and hold down the RST button. 3.Wait for no longer than 2 minutes till both the STAT and PROG indicators blink (the time window is 5 seconds). Then release the RST button immediately.

Silkscreen	Name	Description
		NOTE 1. When you press the RST button to clear the configuration file, the original configuration file will be cleared. You are advised to periodically back up the configuration file. 2. When you press the RST button to clear the configuration file, to prevent the cleared configuration from being synchronized again between the active and standby main control boards, remove one main control board first in a dual-main control board chassis to ensure that the configuration file is cleared in a single-main control board scenario. Perform the same operations on the other main control board. 3. When pressing the RST button to clear the configuration file, you only need to pay attention to the STAT and PROG indicators. Other indicators may vary depending on the product model.

Ports

Table 1-149 Ports on the CR8D0IPU3TM2

Port	Connector Type	Description	Available Components
ETH/OAM	RJ45	10M/100M auto-sensing Ethernet network management port and non-standard RS-232 console port NOTE If a service port can be connected to the NMS, it is recommended that the service port be used as the network management port.	ETH/OAM management cable
CLK/TOD/RS-485	RJ45	Clock/Time/RS-485 input and output ports RS-485 is a reserved port.	1-into-2 clock cable + clock cable (external clock mode, delivery-ready): applicable to the CLK mode 1-into-2 clock cable + clock cable (external clock/external time mode, prepared onsite): applicable to the CLK or TOD mode 1-into-2 clock cable + Ethernet cable: applicable to the TOD mode
USB	Type A	USB port (reserved)	-
CLK/TEST	SMA	Clock test port	Coaxial cable

Port	Connector Type	Description	Available Components
HP-GE	SFP	Ultra-high-precision GE interface	1.25Gbps eSFP Optical Module 1.25Gbps eSFP CWDM Optical Module 1.25Gbps eSFP BIDI Optical Module

Table 1-150 Pins of the ETH/OAM interface

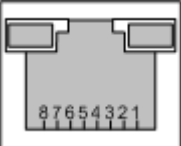
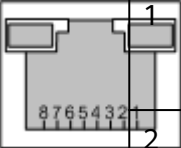
Front View	Pin	Usage
	1	Transmit positive of the NM network port
	2	Transmit negative of the NM network port
	3	Receive positive of the NM network port
	4	Grounding end of the Console interface
	5	Receive end of the Console interface
	6	Receive negative of the NM network port
	7	Unspecified
	8	Transmit end of the Console interface

Table 1-151 Pins of the CLK/TOD/RS-485 interface

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	1	Receive negative of the CLK	Receive negative of the CLK	Unspecified	Unspecified	Unspecified
	2	Receive positive of the CLK	Receive positive of the CLK	Unspecified	Unspecified	Unspecified
	3	Input/Output negative for the 1PPS signal	Unspecified	Input negative for the 1PPS signal (RS422 level)	Output negative for the 1PPS signal (RS422 level)	Transmit negative of the RS-485
	4	Receive negative of the CLK	Receive negative of the CLK	Grounding end	Grounding end	Unspecified
	5	Receive positive of the CLK	Receive positive of the CLK	Grounding end	Grounding end	Unspecified
	6	Input/Output positive for the 1PPS signal	Unspecified	Input positive for the 1PPS signal (RS422 level)	Output positive for the 1PPS signal (RS422 level)	Transmit positive of the RS-485

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	7	Input/Output negative for the time information	Unspecified	Input negative for the time information (RS422 level)	Output negative for the time information (RS422 level)	Receive negative of the RS-485
	8	Input/Output positive for the time information	Unspecified	Input positive for the time information (RS422 level)	Output positive for the time information (RS422 level)	Receive positive of the RS-485
NOTE When this interface is used as a TOD interface, it must be used with a dedicated clock cable. For details, see "Cables" in the Hardware Description.						

Functions and Features

Table 1-152 Functions and features of the CR8D0IPU3TM2

Functions and Features	Description
Basic function	Data forwarding: The IPU is the core of service processing in the entire system, and is connected to all subcards through data channels. Control and management: The main control board manages subcards and transmits routing protocol data through management channels. 7.2 Tbit/s@544 bytes (bidirectional)
Reliability and availability	Hot swappable. 1:1 backup is supported.
Restrictions and remarks	This board cannot be used together with other IPU.
Breakout supported	No
Interface rate auto-sensing supported	No

Technical Specifications

Table 1-153 Technical specifications of the CR8D0IPU3TM2

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	34.8 mm x 354.4 mm x 209 mm (1.37 in. x 13.95 in. x 8.23 in.)
Dimensions with packaging (H x W x D) [mm(in.)]	185 mm x 485 mm x 385 mm (7.28 in. x 19.09 in. x 15.16 in.)
Weight without packaging [kg(lb)]	3.7 kg (8.16 lb)
Weight with packaging [kg(lb)]	5.1 kg (11.24 lb)
Typical power consumption [W]	253.1 W
Typical heat dissipation [BTU/hour]	821.05 BTU/hour
Long-term operating temperature [°C(°F)]	-20°C to +55°C, startup: ≥ -5°C
CPU	16-core 2.0 GHz
SDRAM	32GB
Storage	16GB M.2

1.4.3.9 CR8DIPU1T2C1 (03033FXD: Integrated Network Processing Unit (IPU-1T2))

Overview

Table 1-154 Basic information about the CR8DIPU1T2C1

Item	Details
Description	Integrated Network Processing Unit (IPU-1T2)
Part Number	03033FXD
Model	CR8DIPU1T2C1
Card type	IPU
Silkscreen	IPU-1T2

Appearance

Figure 1-78 Appearance of the CR8DIPU1T2C1



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03033FXD&productModel=CR8DIPU1T2C1#matchRelation>

Indicators and Buttons

Table 1-155 Indicators on the CR8DIPU1T2C1

Silkscreen	Name	Color	Status	Description
STAT	Working status indicator	Green	Steady on	The board is working normally.
		Red	Steady on	The device generates a hardware alarm. You can run the display alarm hardware command to view the hardware alarm type.
		-	Off	The board is not running or no power is input.
PROG	Program running indicator	Green	Steady on	The board software is being initialized. The board software is normally initialized, and the board software is running normally.
		Red	Steady on	The memory self-check fails. The board software or the logic file is lost. Loading of the board software fails.

Silkscreen	Name	Color	Status	Description
		Green	On for 100 ms and off for 100 ms alternately	Loading of the board software is in process.
		Green	On for 300 ms and off for 300 ms alternately	The BIOS is guiding the upper-layer software.
		Red	On for 100 ms and off for 100 ms alternately	The BIOS self-check fails.
		-	Off	No power is input.
SYNC	Synchronization status indicator	Green	Steady on	The system clock works in free-run mode, and no priority list is set for the system clocks (only internal clock sources are listed by default). The system clock works in tracing mode and is tracing a clock source other than the internal clock sources in the priority list. The system clock works in time synchronization mode, and the PTP time and system clock are being traced.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	The system clock priority list is set, but all clock sources except the internal clock source are lost. The clock works in holdover or free-run mode. The system clock works in time synchronization mode, but no synchronization source is available. The system clock and PTP time work in holdover or free-run mode.
ACTX	Active/standby status indicator for the cross-connect or clock board	Green	Steady on	The board is in the working switching state.
		-	Off	The board is in the protection switching state.
ACTC	Control status indicator	Green	Steady on	The board is in the working control state.
		Green	On for 100 ms and off for 100 ms alternately	The data of the equipment is backed up in batch.

Silkscreen	Name	Color	Status	Description
		-	Off	The board is in the protection control state.
ALM	Alarm indicator	Red	Steady on	Critical alarms are generated.
		Orange	Steady on	Major, minor alarms are generated.
		-	Off	No alarms are generated in the equipment.
ETH/OAM	ETH/OAM indicator	Green	Steady on	If the indicator is steady on, the link is normal.
		Orange	Blink	If the indicator blinks, data is being transmitted.
		-	Off	If the indicator is off, the link is Down.

Table 1-156 Buttons on the CR8DIPU1T2C1

Silkscreen	Name	Description
RST	Reset button	You can press the RST button to reset the NE or restore the factory settings of the NE. The procedure is as follows: To reset the NE, press the RST button and release it. To restore factory settings, press and hold the RST button. This operation will clear the NE database and system parameter configurations and cannot be restored. Exercise caution when you perform the following operations: 1. Press and release the RST button to restart the NE. 2. When the PROG indicator blinks (the blinking frequency is about 1 second and the time window is about 15 seconds), press and hold the RST button. 3. Wait for no longer than 2 minutes till both the STAT and PROG indicators blink (the time window is 5 seconds). Then release the RST button immediately.

Silkscreen	Name	Description
		NOTE 1. When you press the RST button to clear the configuration file, the original configuration file will be cleared. You are advised to periodically back up the configuration file. 2. When you press the RST button to clear the configuration file, to prevent the cleared configuration from being synchronized again between the active and standby main control boards, remove one main control board first in a dual-main control board chassis to ensure that the configuration file is cleared in a single-main control board scenario. Perform the same operations on the other main control board. 3. When pressing the RST button to clear the configuration file, you only need to pay attention to the STAT and PROG indicators. Other indicators may vary depending on the product model.

Ports

Table 1-157 Ports on the CR8DIPU1T2C1

Port	Connector Type	Description	Available Components
ETH/OAM	RJ45	10M/100M autosensing Ethernet network management port or non-RS-232-standard console port. NOTE If a service port can be connected to the NMS, it is recommended that the service port be used as the network management port.	ETH/OAM Management Cables
CLK/TOD/RS-485	RJ45	Clock/Time/RS-485 input/output port, RS-485 is a reserved port.	1-into-2 clock cable + clock cable (external clock mode, delivery-ready): applicable to the CLK mode 1-into-2 clock cable + clock cable (external clock/external time mode, prepared onsite): applicable to the CLK or TOD mode 1-into-2 clock cable + Ethernet cable: applicable to the TOD mode
ALMI/ALMO	RJ45	Alarm input/output port	Alarm Input/Output Cables
USB	Type A	USB port (reserved)	-
CLK/TEST	SMA	Clock test port	Coaxial Cable

Table 1-158 Pins of the ETH/OAM interface

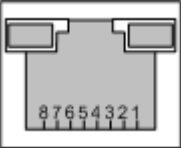
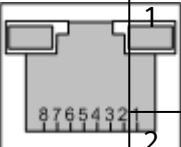
Front View	Pin	Usage
	1	Transmit positive of the NM network port
	2	Transmit negative of the NM network port
	3	Receive positive of the NM network port
	4	Grounding end of the Console interface
	5	Receive end of the Console interface
	6	Receive negative of the NM network port
	7	Unspecified
	8	Transmit end of the Console interface

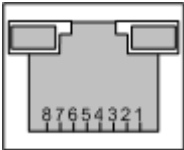
Table 1-159 Pins of the CLK/TOD/RS-485 interface

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	1	Receive negative of the CLK	Receive negative of the CLK	Unspecified	Unspecified	Unspecified
	2	Receive positive of the CLK	Receive positive of the CLK	Unspecified	Unspecified	Unspecified

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	3	Input/Output negative for the 1PPS signal	Unspecified	Input negative for the 1PPS signal (RS422 level)	Output negative for the 1PPS signal (RS422 level)	Transmit negative of the RS-485
	4	Receive negative of the CLK	Receive negative of the CLK	Grounding end	Grounding end	Unspecified
	5	Receive positive of the CLK	Receive positive of the CLK	Grounding end	Grounding end	Unspecified
	6	Input/Output positive for the 1PPS signal	Unspecified	Input positive for the 1PPS signal (RS422 level)	Output positive for the 1PPS signal (RS422 level)	Transmit positive of the RS-485

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	7	Input/Output negative for the time information	Unspecified	Input negative for the time information (RS422 level)	Output negative for the time information (RS422 level)	Receive negative of the RS-485
	8	Input/Output positive for the time information	Unspecified	Input positive for the time information (RS422 level)	Output positive for the time information (RS422 level)	Receive positive of the RS-485
NOTE When this interface is used as a TOD interface, it must be used with a dedicated clock cable. For details, see "Cables" in the Hardware Description.						

Table 1-160 Pins of the ALMI/ALMO interface

Front View	Pin	Usage
	1	Alarm input 1
	2	Grounding end for alarm input 1
	3	Alarm input 2

Front View	Pin	Usage
	4	Alarm input 3
	5	Grounding end for alarm input 3
	6	Grounding end for alarm input 2
	7	Positive for alarm signal output
	8	Negative for alarm signal output

Functions and Features

Table 1-161 Functions and features of the CR8DIPU1T2C1

Functions and Features	Description
Basic function	Data forwarding: The IPU is the core of service processing in the entire system, and is connected to all subcards through data channels. Control and management: Through the management channels between IPU and the subcards, the IPU can manage subcards and transmit routing protocol data. The line rate capability is 2.4T bit/s@310Bytes(Bidirectional).
Reliability and availability	Supports the hot swappable function. Supports the board-level 1:1 backup function.
Restrictions and remarks	This board cannot be used together with other IPU.
Breakout supported	No
Interface rate auto-sensing supported	No

Technical Specifications

Table 1-162 Technical specifications of the CR8DIPU1T2C1

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	22.4 mm x 388.4 mm x 209.3 mm (0.88 in. x 15.29 in. x 8.24 in.)
Dimensions with packaging (H x W x D) [mm(in.)]	170 mm x 560 mm x 385 mm (6.69 in. x 22.05 in. x 15.16 in.)
Weight without packaging [kg(lb)]	2.3 kg (5.07 lb)
Weight with packaging [kg(lb)]	3.7 kg (8.16 lb)
Typical power consumption [W]	144.2 W
Typical heat dissipation [BTU/hour]	467.94 BTU/hour
Long-term operating temperature [°C(°F)]	–40°C to +65°C (–40°F to +149°F), start: ≥ –20°C (–4°F)
CPU	8-core 1.5 GHz
SDRAM	8GB
Storage	4GB NandFlash

1.4.3.10 CR8DIPU1T2C1 (03033FXD-001: Integrated Network Processing Unit (IPU-1T2))

Overview

Table 1-163 Basic information about the CR8DIPU1T2C1

Item	Details
Description	Integrated Network Processing Unit (IPU-1T2)
Part Number	03033FXD-001
Model	CR8DIPU1T2C1
Card type	IPU
Silkscreen	IPU-1T2

Appearance

Figure 1-79 Appearance of the CR8DIPU1T2C1



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03033FXD-001&productModel=CR8DIPU1T2C1#matchRelation>

Indicators and Buttons

Table 1-164 Indicators on the CR8DIPU1T2C1

Silkscreen	Name	Color	Status	Description
STAT	Working status indicator	Green	Steady on	The board is working normally.
		Red	Steady on	The device generates a hardware alarm. You can run the display alarm hardware command to view the hardware alarm type.
		-	Off	The board is not running or no power is input.

Silkscreen	Name	Color	Status	Description
PROG	Program running indicator	Green	Steady on	The board software is being initialized. The board software is normally initialized, and the board software is running normally.
		Red	Steady on	The memory self-check fails. The board software or the logic file is lost. Loading of the board software fails.
		Green	On for 100 ms and off for 100 ms alternately	Loading of the board software is in process.
		Green	On for 300 ms and off for 300 ms alternately	The BIOS is guiding the upper-layer software.
		Red	On for 100 ms and off for 100 ms alternately	The BIOS self-check fails.
		-	Off	No power is input.

Silkscreen	Name	Color	Status	Description
SYNC	Synchronizati on status indicator	Green	Steady on	The system clock works in free-run mode, and no priority list is set for the system clocks (only internal clock sources are listed by default). The system clock works in tracing mode and is tracing a clock source other than the internal clock sources in the priority list. The system clock works in time synchronizatio n mode, and the PTP time and system clock are being traced.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	The system clock priority list is set, but all clock sources except the internal clock source are lost. The clock works in holdover or free-run mode. The system clock works in time synchronization mode, but no synchronization source is available. The system clock and PTP time work in holdover or free-run mode.
ACTX	Active/standby status indicator for the cross-connect or clock board	Green	Steady on	The board is in the working switching state.
		-	Off	The board is in the protection switching state.
ACTC	Control status indicator	Green	Steady on	The board is in the working control state.
		Green	On for 100 ms and off for 100 ms alternately	The data of the equipment is backed up in batch.

Silkscreen	Name	Color	Status	Description
		-	Off	The board is in the protection control state.
ALM	Alarm indicator	Red	Steady on	Critical alarms are generated.
		Orange	Steady on	Major, minor alarms are generated.
		-	Off	No alarms are generated in the equipment.
ETH/OAM	ETH/OAM indicator	Green	Steady on	If the indicator is steady on, the link is normal.
		Orange	Blink	If the indicator blinks, data is being transmitted.
		-	Off	If the indicator is off, the link is Down.

Table 1-165 Buttons on the CR8DIPU1T2C1

Silkscreen	Name	Description
RST	Reset button	You can press the RST button to reset the NE or restore the factory settings of the NE. The procedure is as follows: To reset the NE, press the RST button and release it. To restore factory settings, press and hold the RST button. This operation will clear the NE database and system parameter configurations and cannot be restored. Exercise caution when you perform the following operations: 1. Press and release the RST button to restart the NE. 2. When the PROG indicator blinks (the blinking frequency is about 1 second and the time window is about 15 seconds), press and hold the RST button. 3. Wait for no longer than 2 minutes till both the STAT and PROG indicators blink (the time window is 5 seconds). Then release the RST button immediately.

Silkscreen	Name	Description
		NOTE 1. When you press the RST button to clear the configuration file, the original configuration file will be cleared. You are advised to periodically back up the configuration file. 2. When you press the RST button to clear the configuration file, to prevent the cleared configuration from being synchronized again between the active and standby main control boards, remove one main control board first in a dual-main control board chassis to ensure that the configuration file is cleared in a single-main control board scenario. Perform the same operations on the other main control board. 3. When pressing the RST button to clear the configuration file, you only need to pay attention to the STAT and PROG indicators. Other indicators may vary depending on the product model.

Ports

Table 1-166 Ports on the CR8DIPU1T2C1

Port	Connector Type	Description	Available Components
ETH/OAM	RJ45	10M/100M autosensing Ethernet network management port or non-RS-232-standard console port. NOTE If a service port can be connected to the NMS, it is recommended that the service port be used as the network management port.	ETH/OAM Management Cables
CLK/TOD/RS-485	RJ45	Clock/Time/RS-485 input/output port, RS-485 is a reserved port.	1-into-2 clock cable + clock cable (external clock mode, delivery-ready): applicable to the CLK mode 1-into-2 clock cable + clock cable (external clock/external time mode, prepared onsite): applicable to the CLK or TOD mode 1-into-2 clock cable + Ethernet cable: applicable to the TOD mode
ALMI/ALMO	RJ45	Alarm input/output port	Alarm Input/Output Cables
USB	Type A	USB port (reserved)	-
CLK/TEST	SMA	Clock test port	Coaxial Cable

Table 1-167 Pins of the ETH/OAM interface

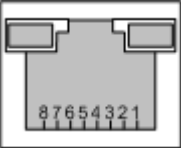
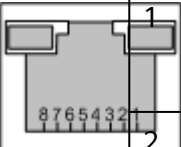
Front View	Pin	Usage
	1	Transmit positive of the NM network port
	2	Transmit negative of the NM network port
	3	Receive positive of the NM network port
	4	Grounding end of the Console interface
	5	Receive end of the Console interface
	6	Receive negative of the NM network port
	7	Unspecified
	8	Transmit end of the Console interface

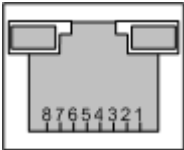
Table 1-168 Pins of the CLK/TOD/RS-485 interface

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	1	Receive negative of the CLK	Receive negative of the CLK	Unspecified	Unspecified	Unspecified
	2	Receive positive of the CLK	Receive positive of the CLK	Unspecified	Unspecified	Unspecified

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	3	Input/Output negative for the 1PPS signal	Unspecified	Input negative for the 1PPS signal (RS422 level)	Output negative for the 1PPS signal (RS422 level)	Transmit negative of the RS-485
	4	Receive negative of the CLK	Receive negative of the CLK	Grounding end	Grounding end	Unspecified
	5	Receive positive of the CLK	Receive positive of the CLK	Grounding end	Grounding end	Unspecified
	6	Input/Output positive for the 1PPS signal	Unspecified	Input positive for the 1PPS signal (RS422 level)	Output positive for the 1PPS signal (RS422 level)	Transmit positive of the RS-485

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	7	Input/Output negative for the time information	Unspecified	Input negative for the time information (RS422 level)	Output negative for the time information (RS422 level)	Receive negative of the RS-485
	8	Input/Output positive for the time information	Unspecified	Input positive for the time information (RS422 level)	Output positive for the time information (RS422 level)	Receive positive of the RS-485
NOTE When this interface is used as a TOD interface, it must be used with a dedicated clock cable. For details, see "Cables" in the Hardware Description.						

Table 1-169 Pins of the ALMI/ALMO interface

Front View	Pin	Usage
	1	Alarm input 1
	2	Grounding end for alarm input 1
	3	Alarm input 2

Front View	Pin	Usage
	4	Alarm input 3
	5	Grounding end for alarm input 3
	6	Grounding end for alarm input 2
	7	Positive for alarm signal output
	8	Negative for alarm signal output

Functions and Features

Table 1-170 Functions and features of the CR8DIPU1T2C1

Functions and Features	Description
Basic function	Data forwarding: The IPU is the core of service processing in the entire system, and is connected to all subcards through data channels. Control and management: Through the management channels between IPU's and the subcards, the IPU's can manage subcards and transmit routing protocol data. The line rate capability is 2.4T bit/s@310Bytes(Bidirectional).
Reliability and availability	Supports the hot swappable function. Supports the board-level 1:1 backup function.
Restrictions and remarks	This board cannot be used together with other IPU's.
Breakout supported	No
Interface rate auto-sensing supported	No

Technical Specifications

Table 1-171 Technical specifications of the CR8DIPU1T2C1

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	22.4 mm x 388.4 mm x 209.3 mm (0.88 in. x 15.29 in. x 8.24 in.)
Dimensions with packaging (H x W x D) [mm(in.)]	170 mm x 560 mm x 385 mm (6.69 in. x 22.05 in. x 15.16 in.)
Weight without packaging [kg(lb)]	2.3 kg (5.07 lb)
Weight with packaging [kg(lb)]	3.7 kg (8.16 lb)
Typical power consumption [W]	144.2 W
Typical heat dissipation [BTU/hour]	467.94 BTU/hour
Long-term operating temperature [°C(°F)]	–40°C to +65°C (–40°F to +149°F), start: ≥ –20°C (–4°F)
CPU	8-core 1.5 GHz
SDRAM	8GB
Storage	4GB NandFlash

1.4.3.11 CR8DIPU1T2C1 (03033FXD-002: Integrated Network Processing Unit (IPU-1T2))

Overview

Table 1-172 Basic information about the CR8DIPU1T2C1

Item	Details
Description	Integrated Network Processing Unit (IPU-1T2)
Part Number	03033FXD-002
Model	CR8DIPU1T2C1
Card type	IPU
Silkscreen	IPU-1T2

Appearance

Figure 1-80 Appearance of the CR8DIPU1T2C1



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03033FXD-002&productModel=CR8DIPU1T2C1#matchRelation>

Indicators and Buttons

Table 1-173 Indicators on the CR8DIPU1T2C1

Silkscreen	Name	Color	Status	Description
STAT	Working status indicator	Green	Steady on	The board is working properly.
		Red	Steady on	A hardware alarm is generated. You can run the display alarm hardware command to check the hardware alarm type.
		-	Off	The board is not powered on or is not running.

Silkscreen	Name	Color	Status	Description
PROG	Program status indicator	Green	Steady on	The board software is being initialized or is successfully initialized and is running properly.
		Red	Steady on	The memory self-check fails. The board software or logic file is lost. The board software fails to be loaded.
		Green	Blinks on and off at 100 ms intervals	The board software is being loaded.
		Green	Blinks on and off at 300 ms intervals	The BIOS is interacting with the upper-layer software.
		Red	Blinks on and off at 100 ms intervals	The BIOS self-check fails.
		-	Off	The board is not powered on.

Silkscreen	Name	Color	Status	Description
SYNC	Clock synchronizatio n indicator	Green	Steady on	The system clock works in free-run mode, and no priority list is set for the system clocks (only internal clock sources are listed by default). The system clock works in tracing mode and is tracing a clock source other than the internal clock sources in the priority list. The system clock works in time synchronizatio n mode, and the PTP time and system clock are being traced.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	The system clock priority list is set, but all clock sources except the internal clock source are lost. The clock works in holdover or free-run mode. The system clock works in time synchronization mode, but no synchronization source is available. The system clock and PTP time work in holdover or free-run mode.
ACTX	Active/standby status indicator for the cross-connect or clock board	Green	Steady on	The board is in the active state on the service plane.
		-	Off	The board is in the standby state on the service plane.
ACTC	Control status indicator	Green	Steady on	The board is in the active control state.
		Green	Blinks on and off at 100 ms intervals	The data of the device is backed up in batch.
		-	Off	The board is in the standby control state.

Silkscreen	Name	Color	Status	Description
ALM	Alarm indicator	Red	Steady on	Device-level critical alarms exist.
		Orange	Steady on	A device-level major or minor alarm exists.
		-	Off	There is no equipment-level alarm.
ETH/OAM	ETH/OAM indicator	Green	Steady on	The link is up.
		Orange	Blinking	Data is being transmitted or received.
		-	Off	The link is down.

Table 1-174 Buttons on the CR8DIPU1T2C1

Silkscreen	Name	Description
RST	Board reset button	You can press the RST button to reset the NE or restore the factory settings of the NE. The procedure is as follows: To reset the NE, press the RST button and release it. To restore factory settings, press and hold down the RST button. This operation will clear the NE database and system parameter configurations and cannot be restored. Exercise caution when you perform the following operations: 1.Press the RST button to restart the NE. 2.When the PROG indicator blinks (the blinking frequency is about 1 second and the time window is about 15 seconds), press and hold down the RST button. 3.Wait for no longer than 2 minutes till both the STAT and PROG indicators blink (the time window is 5 seconds). Then release the RST button immediately.

Silkscreen	Name	Description
		NOTE 1. When you press the RST button to clear the configuration file, the original configuration file will be cleared. You are advised to periodically back up the configuration file. 2. When you press the RST button to clear the configuration file, to prevent the cleared configuration from being synchronized again between the active and standby main control boards, remove one main control board first in a dual-main control board chassis to ensure that the configuration file is cleared in a single-main control board scenario. Perform the same operations on the other main control board. 3. When pressing the RST button to clear the configuration file, you only need to pay attention to the STAT and PROG indicators. Other indicators may vary depending on the product model.

Ports

Table 1-175 Ports on the CR8DIPU1T2C1

Port	Connector Type	Description	Available Components
ETH/OAM	RJ45	10M/100M auto-sensing Ethernet network management port or non-standard RS-232 console port NOTE If a service port can be connected to an NMS, you are advised to use the service port as the network management port.	ETH/OAM management cable
CLK/TOD/RS-485	RJ45	Clock/Time/RS-485 input and output ports RS-485 is a reserved port.	1-into-2 clock cable + clock cable (external clock mode, delivery-ready): applicable to the CLK mode 1-into-2 clock cable + clock cable (external clock/external time mode, prepared onsite): applicable to the CLK or TOD mode 1-into-2 clock cable + Ethernet cable: applicable to the TOD mode
ALMI/ALMO	RJ45	Alarm input/output interface	Alarm input/output cable
USB	Type A	USB port (reserved)	-
CLK/TEST	SMA	Clock test port	Coaxial cable

Table 1-176 Pins of the ETH/OAM interface

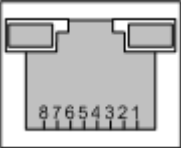
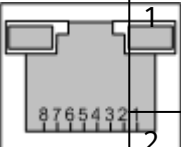
Front View	Pin	Usage
	1	Transmit positive of the NM network port
	2	Transmit negative of the NM network port
	3	Receive positive of the NM network port
	4	Grounding end of the Console interface
	5	Receive end of the Console interface
	6	Receive negative of the NM network port
	7	Unspecified
	8	Transmit end of the Console interface

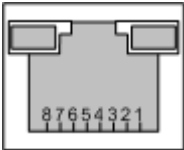
Table 1-177 Pins of the CLK/TOD/RS-485 interface

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	1	Receive negative of the CLK	Receive negative of the CLK	Unspecified	Unspecified	Unspecified
	2	Receive positive of the CLK	Receive positive of the CLK	Unspecified	Unspecified	Unspecified

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	3	Input/Output negative for the 1PPS signal	Unspecified	Input negative for the 1PPS signal (RS422 level)	Output negative for the 1PPS signal (RS422 level)	Transmit negative of the RS-485
	4	Receive negative of the CLK	Receive negative of the CLK	Grounding end	Grounding end	Unspecified
	5	Receive positive of the CLK	Receive positive of the CLK	Grounding end	Grounding end	Unspecified
	6	Input/Output positive for the 1PPS signal	Unspecified	Input positive for the 1PPS signal (RS422 level)	Output positive for the 1PPS signal (RS422 level)	Transmit positive of the RS-485

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	7	Input/Output negative for the time information	Unspecified	Input negative for the time information (RS422 level)	Output negative for the time information (RS422 level)	Receive negative of the RS-485
	8	Input/Output positive for the time information	Unspecified	Input positive for the time information (RS422 level)	Output positive for the time information (RS422 level)	Receive positive of the RS-485
NOTE When this interface is used as a TOD interface, it must be used with a dedicated clock cable. For details, see "Cables" in the Hardware Description.						

Table 1-178 Pins of the ALMI/ALMO interface

Front View	Pin	Usage
	1	Alarm input 1
	2	Grounding end for alarm input 1
	3	Alarm input 2

Front View	Pin	Usage
	4	Alarm input 3
	5	Grounding end for alarm input 3
	6	Grounding end for alarm input 2
	7	Positive for alarm signal output
	8	Negative for alarm signal output

Functions and Features

Table 1-179 Functions and features of the CR8DIPU1T2C1

Functions and Features	Description
Basic function	Data forwarding: The IPU is the core of service processing in the entire system, and is connected to all subcards through data channels. Control and management: The IPU manages subcards and transmits routing protocol data through management channels. 2.4 Tbit/s@310 bytes (bidirectional)
Reliability and availability	Hot swappable. 1:1 backup is supported.
Restrictions and remarks	This board cannot be used together with other IPU.
Breakout supported	No
Interface rate auto-sensing supported	No

Technical Specifications

Table 1-180 Technical specifications of the CR8DIPU1T2C1

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	22.4 mm x 388.4 mm x 209.3 mm (0.88 in. x 15.29 in. x 8.24 in.)

Item	Specification
Dimensions with packaging (H x W x D) [mm(in.)]	170 mm x 560 mm x 385 mm (6.69 in. x 22.05 in. x 15.16 in.)
Weight without packaging [kg(lb)]	2.3 kg (5.07 lb)
Weight with packaging [kg(lb)]	3.7 kg (8.16 lb)
Typical power consumption [W]	144.2 W
Typical heat dissipation [BTU/hour]	467.94 BTU/hour
Long-term operating temperature [°C(°F)]	–40°C to +65°C (–40°F to +149°F), start: ≥ –20°C (–4°F)
CPU	8-core 1.5 GHz
SDRAM	8GB
Storage	4 GB NandFlash

1.4.3.12 CR8DIPU1T2C1 (03034VCX: Integrated Network Processing Unit (IPU-1T2))

Overview

Table 1-181 Basic information about the CR8DIPU1T2C1

Item	Details
Description	Integrated Network Processing Unit (IPU-1T2)
Part Number	03034VCX
Model	CR8DIPU1T2C1
Card type	IPU
Silkscreen	IPU-1T2

Appearance

Figure 1-81 Appearance of the CR8DIPU1T2C1



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03034VCX&productModel=CR8DIPU1T2C1#matchRelation>

Indicators and Buttons

Table 1-182 Indicators on the CR8DIPU1T2C1

Silkscreen	Name	Color	Status	Description
STAT	Working status indicator	Green	Steady on	The board is working properly.
		Red	Steady on	A hardware alarm is generated. You can run the display alarm hardware command to check the hardware alarm type.
		-	Off	The board is not powered on or is not running.
PROG	Program status indicator	Green	Steady on	The board software is being initialized or is successfully initialized and is running properly.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	The memory self-check fails. The board software or logic file is lost. The board software fails to be loaded.
		Green	Blinks on and off at 100 ms intervals	The board software is being loaded.
		Green	Blinks on and off at 300 ms intervals	The BIOS is interacting with the upper-layer software.
		Red	Blinks on and off at 100 ms intervals	The BIOS self-check fails.
		-	Off	The board is not powered on.

Silkscreen	Name	Color	Status	Description
SYNC	Clock synchronizatio n indicator	Green	Steady on	The system clock works in free-run mode, and no priority list is set for the system clocks (only internal clock sources are listed by default). The system clock works in tracing mode and is tracing a clock source other than the internal clock sources in the priority list. The system clock works in time synchronizatio n mode, and the PTP time and system clock are being traced.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	The system clock priority list is set, but all clock sources except the internal clock source are lost. The clock works in holdover or free-run mode. The system clock works in time synchronization mode, but no synchronization source is available. The system clock and PTP time work in holdover or free-run mode.
ACTX	Active/standby status indicator for the cross-connect or clock board	Green	Steady on	The board is in the active state on the service plane.
		-	Off	The board is in the standby state on the service plane.
ACTC	Control status indicator	Green	Steady on	The board is in the active control state.
		Green	Blinks on and off at 100 ms intervals	The data of the device is backed up in batch.
		-	Off	The board is in the standby control state.

Silkscreen	Name	Color	Status	Description
ALM	Alarm indicator	Red	Steady on	Device-level critical alarms exist.
		Orange	Steady on	A device-level major or minor alarm exists.
		-	Off	There is no equipment-level alarm.
ETH/OAM	ETH/OAM indicator	Green	Steady on	The link is up.
		Orange	Blinking	Data is being transmitted or received.
		-	Off	The link is down.

Table 1-183 Buttons on the CR8DIPU1T2C1

Silkscreen	Name	Description
RST	Board reset button	You can press the RST button to reset the NE or restore the factory settings of the NE. The procedure is as follows: To reset the NE, press the RST button and release it. To restore factory settings, press and hold down the RST button. This operation will clear the NE database and system parameter configurations and cannot be restored. Exercise caution when you perform the following operations: 1.Press the RST button to restart the NE. 2.When the PROG indicator blinks (the blinking frequency is about 1 second and the time window is about 15 seconds), press and hold down the RST button. 3.Wait for no longer than 2 minutes till both the STAT and PROG indicators blink (the time window is 5 seconds). Then release the RST button immediately.

Silkscreen	Name	Description
		NOTE 1. When you press the RST button to clear the configuration file, the original configuration file will be cleared. You are advised to periodically back up the configuration file. 2. When you press the RST button to clear the configuration file, to prevent the cleared configuration from being synchronized again between the active and standby main control boards, remove one main control board first in a dual-main control board chassis to ensure that the configuration file is cleared in a single-main control board scenario. Perform the same operations on the other main control board. 3. When pressing the RST button to clear the configuration file, you only need to pay attention to the STAT and PROG indicators. Other indicators may vary depending on the product model.

Ports

Table 1-184 Ports on the CR8DIPU1T2C1

Port	Connector Type	Description	Available Components
ETH/OAM	RJ45	10M/100M auto-sensing Ethernet network management port or non-standard RS-232 console port NOTE If a service port can be connected to an NMS, you are advised to use the service port as the network management port.	ETH/OAM management cable
CLK/TOD/RS-485	RJ45	Clock/Time/RS-485 input and output ports RS-485 is a reserved port.	1-into-2 clock cable + clock cable (external clock mode, delivery-ready): applicable to the CLK mode 1-into-2 clock cable + clock cable (external clock/external time mode, prepared onsite): applicable to the CLK or TOD mode 1-into-2 clock cable + Ethernet cable: applicable to the TOD mode
ALMI/ALMO	RJ45	Alarm input/output interface	Alarm input/output cable
USB	Type A	USB port (reserved)	-
CLK/TEST	SMA	Clock test port	Coaxial cable

Table 1-185 Pins of the ETH/OAM interface

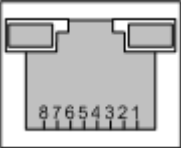
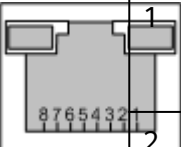
Front View	Pin	Usage
	1	Transmit positive of the NM network port
	2	Transmit negative of the NM network port
	3	Receive positive of the NM network port
	4	Grounding end of the Console interface
	5	Receive end of the Console interface
	6	Receive negative of the NM network port
	7	Unspecified
	8	Transmit end of the Console interface

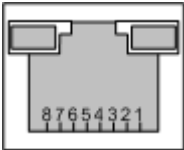
Table 1-186 Pins of the CLK/TOD/RS-485 interface

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	1	Receive negative of the CLK	Receive negative of the CLK	Unspecified	Unspecified	Unspecified
	2	Receive positive of the CLK	Receive positive of the CLK	Unspecified	Unspecified	Unspecified

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	3	Input/Output negative for the 1PPS signal	Unspecified	Input negative for the 1PPS signal (RS422 level)	Output negative for the 1PPS signal (RS422 level)	Transmit negative of the RS-485
	4	Receive negative of the CLK	Receive negative of the CLK	Grounding end	Grounding end	Unspecified
	5	Receive positive of the CLK	Receive positive of the CLK	Grounding end	Grounding end	Unspecified
	6	Input/Output positive for the 1PPS signal	Unspecified	Input positive for the 1PPS signal (RS422 level)	Output positive for the 1PPS signal (RS422 level)	Transmit positive of the RS-485

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	7	Input/Output negative for the time information	Unspecified	Input negative for the time information (RS422 level)	Output negative for the time information (RS422 level)	Receive negative of the RS-485
	8	Input/Output positive for the time information	Unspecified	Input positive for the time information (RS422 level)	Output positive for the time information (RS422 level)	Receive positive of the RS-485
NOTE When this interface is used as a TOD interface, it must be used with a dedicated clock cable. For details, see "Cables" in the Hardware Description.						

Table 1-187 Pins of the ALMI/ALMO interface

Front View	Pin	Usage
	1	Alarm input 1
	2	Grounding end for alarm input 1
	3	Alarm input 2

Front View	Pin	Usage
	4	Alarm input 3
	5	Grounding end for alarm input 3
	6	Grounding end for alarm input 2
	7	Positive for alarm signal output
	8	Negative for alarm signal output

Functions and Features

Table 1-188 Functions and features of the CR8DIPU1T2C1

Functions and Features	Description
Basic function	Data forwarding: The IPU is the core of service processing in the entire system, and is connected to all subcards through data channels. Control and management: The IPU manages subcards and transmits routing protocol data through management channels. 2.4 Tbit/s@310 bytes (bidirectional)
Reliability and availability	Hot swappable. 1:1 backup is supported.
Restrictions and remarks	This board cannot be used together with other IPU.
Breakout supported	No
Interface rate auto-sensing supported	No

Technical Specifications

Table 1-189 Technical specifications of the CR8DIPU1T2C1

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	22.4 mm x 388.4 mm x 209.3 mm (0.88 in. x 15.29 in. x 8.24 in.)

Item	Specification
Dimensions with packaging (H x W x D) [mm(in.)]	170 mm x 560 mm x 385 mm (6.69 in. x 22.05 in. x 15.16 in.)
Weight without packaging [kg(lb)]	2.3 kg (5.07 lb)
Weight with packaging [kg(lb)]	3.7 kg (8.16 lb)
Typical power consumption [W]	144.2 W
Typical heat dissipation [BTU/hour]	467.94 BTU/hour
Long-term operating temperature [°C(°F)]	–40°C to +65°C (–40°F to +149°F), start: ≥ –20°C (–4°F)
CPU	8-core 1.5 GHz
SDRAM	8GB
Storage	4 GB NandFlash

1.4.3.13 CR8DIPU1T2C1 (03035JVD: Integrated Network Processing Unit (IPU-1T2) for SME)

Overview

Table 1-190 Basic information about the CR8DIPU1T2C1

Item	Details
Description	Integrated Network Processing Unit (IPU-1T2) for SME
Part Number	03035JVD
Model	CR8DIPU1T2C1
Card type	IPU
Silkscreen	IPU-1T2

Appearance

Figure 1-82 Appearance of the CR8DIPU1T2C1



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03035JVD&productModel=CR8DIPU1T2C1#matchRelation>

Indicators and Buttons

Table 1-191 Indicators on the CR8DIPU1T2C1

Silkscreen	Name	Color	Status	Description
STAT	Working status indicator	Green	Steady on	The board is working normally.
		Red	Steady on	A hardware alarm is generated. You can run the display alarm hardware command to check the hardware alarm type.
		-	Steady off	The board is not running or no power is input.
PROG	Program status indicator	Green	Steady on	The board software is being initialized or is successfully initialized and is running properly.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	The memory self-check fails. The board software or logic file is lost. The board software fails to be loaded.
		Green	Blinks on and off at 100 ms intervals	The board software is being loaded.
		Green	On for 300 ms and off for 300 ms repeatedly	The BIOS is interacting with the upper-layer software.
		Red	Blinks on and off at 100 ms intervals	The BIOS self-check fails.
		-	Steady off	The board is not powered on.

Silkscreen	Name	Color	Status	Description
SYNC	Clock synchronizatio n indicator	Green	Steady on	The system clock works in free-run mode, and no priority list is set for the system clocks (only internal clock sources are listed by default). The system clock works in tracing mode and is tracing a clock source other than the internal clock sources in the priority list. The system clock works in time synchronizatio n mode, and the PTP time and system clock are being traced.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	The system clock priority list is set, but all clock sources except the internal clock source are lost. The clock works in holdover or free-run mode. The system clock works in time synchronization mode, but no synchronization source is available. The system clock and PTP time work in holdover or free-run mode.
ACTX	Active/standby status indicator for the cross-connect or clock board	Green	Steady on	The board is in the active state on the service plane.
		-	Steady off	The board is in the standby state on the service plane.
ACTC	Control status indicator	Green	Steady on	The board is in the active control state.
		Green	Blinks on and off at 100 ms intervals	The data of the device is backed up in batch.
		-	Steady off	The board is in the standby control state.

Silkscreen	Name	Color	Status	Description
ALM	Alarm indicator	Red	Steady on	Device-level critical alarms exist.
		Orange	Steady on	A device-level major or minor alarm exists.
		-	Steady off	There is no equipment-level alarm.
ETH/OAM	ETH/OAM indicator	Green	Steady on	The link is up.
		Orange	Blinking	Data is being transmitted and received.
		-	Steady off	The link is Down.

Table 1-192 Buttons on the CR8DIPU1T2C1

Silkscreen	Name	Description
RST	Board reset button	You can press the RST button to reset the NE or restore the factory settings of the NE. The procedure is as follows: To reset the NE, press the RST button and release it. To restore factory settings, press and hold down the RST button. This operation will clear the NE database and system parameter configurations and cannot be restored. Exercise caution when you perform the following operations: 1.Press the RST button to restart the NE. 2.When the PROG indicator blinks (the blinking frequency is about 1 second and the time window is about 15 seconds), press and hold down the RST button. 3.Wait for no longer than 2 minutes till both the STAT and PROG indicators blink (the time window is 5 seconds). Then release the RST button immediately.

Silkscreen	Name	Description
		NOTE 1. When you press the RST button to clear the configuration file, the original configuration file will be cleared. You are advised to periodically back up the configuration file. 2. When you press the RST button to clear the configuration file, to prevent the cleared configuration from being synchronized again between the active and standby main control boards, remove one main control board first in a dual-main control board chassis to ensure that the configuration file is cleared in a single-main control board scenario. Perform the same operations on the other main control board. 3. When pressing the RST button to clear the configuration file, you only need to pay attention to the STAT and PROG indicators. Other indicators may vary depending on the product model.

Ports

Table 1-193 Ports on the CR8DIPU1T2C1

Port	Connector Type	Description	Available Components
ETH/OAM	RJ45	10M/100M auto-sensing Ethernet network management port and non-standard RS-232 console port NOTE If a service port can be connected to the NMS, it is recommended that the service port be used as the network management port.	ETH/OAM management cable
CLK/TOD/RS-485	RJ45	Clock/Time/RS-485 input and output ports RS-485 is a reserved port.	1-into-2 clock cable + clock cable (external clock mode, delivery-ready): applicable to the CLK mode 1-into-2 clock cable + clock cable (external clock/external time mode, prepared onsite): applicable to the CLK or TOD mode 1-into-2 clock cable + Ethernet cable: applicable to the TOD mode
ALMI/ALMO	RJ45	Alarm input/output interface	Alarm input/output cable
USB	Type A	USB port (reserved)	-
CLK/TEST	SMA	Clock test port	Coaxial cable

Table 1-194 Pins of the ETH/OAM interface

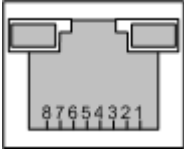
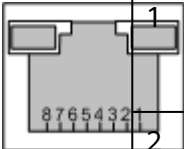
Front View	Pin	Usage
	1	Transmit positive of the NM network port
	2	Transmit negative of the NM network port
	3	Receive positive of the NM network port
	4	Grounding end of the Console interface
	5	Receive end of the Console interface
	6	Receive negative of the NM network port
	7	Unspecified
	8	Transmit end of the Console interface

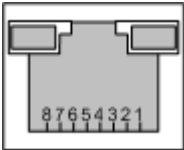
Table 1-195 Pins of the CLK/TOD/RS-485 interface

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	1	Receive negative of the CLK	Receive negative of the CLK	Unspecified	Unspecified	Unspecified
	2	Receive positive of the CLK	Receive positive of the CLK	Unspecified	Unspecified	Unspecified

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	3	Input/Output negative for the 1PPS signal	Unspecified	Input negative for the 1PPS signal (RS422 level)	Output negative for the 1PPS signal (RS422 level)	Transmit negative of the RS-485
	4	Receive negative of the CLK	Receive negative of the CLK	Grounding end	Grounding end	Unspecified
	5	Receive positive of the CLK	Receive positive of the CLK	Grounding end	Grounding end	Unspecified
	6	Input/Output positive for the 1PPS signal	Unspecified	Input positive for the 1PPS signal (RS422 level)	Output positive for the 1PPS signal (RS422 level)	Transmit positive of the RS-485

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	7	Input/Output negative for the time information	Unspecified	Input negative for the time information (RS422 level)	Output negative for the time information (RS422 level)	Receive negative of the RS-485
	8	Input/Output positive for the time information	Unspecified	Input positive for the time information (RS422 level)	Output positive for the time information (RS422 level)	Receive positive of the RS-485
NOTE When this interface is used as a TOD interface, it must be used with a dedicated clock cable. For details, see "Cables" in the Hardware Description.						

Table 1-196 Pins of the ALMI/ALMO interface

Front View	Pin	Usage
	1	Alarm input 1
	2	Grounding end for alarm input 1
	3	Alarm input 2

Front View	Pin	Usage
	4	Alarm input 3
	5	Grounding end for alarm input 3
	6	Grounding end for alarm input 2
	7	Positive for alarm signal output
	8	Negative for alarm signal output

Functions and Features

Table 1-197 Functions and features of the CR8DIPU1T2C1

Functions and Features	Description
Basic function	Data forwarding: The IPU is the core of service processing in the entire system, and is connected to all subcards through data channels. Control and management: The IPU manages subcards and transmits routing protocol data through management channels. 2.4 Tbit/s@310 bytes (bidirectional)
Reliability and availability	Hot swappable. 1:1 backup is supported.
Restrictions and remarks	This board cannot be used together with other IPU.
Breakout supported	No
Interface rate auto-sensing supported	No

Technical Specifications

Table 1-198 Technical specifications of the CR8DIPU1T2C1

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	22.4 mm x 388.4 mm x 209.3 mm (0.88 in. x 15.29 in. x 8.24 in.)

Item	Specification
Dimensions with packaging (H x W x D) [mm(in.)]	170 mm x 560 mm x 385 mm (6.69 in. x 22.05 in. x 15.16 in.)
Weight without packaging [kg(lb)]	2.3 kg (5.07 lb)
Weight with packaging [kg(lb)]	3.7 kg (8.16 lb)
Typical power consumption [W]	144.2 W
Typical heat dissipation [BTU/hour]	467.94 BTU/hour
Long-term operating temperature [°C(°F)]	–40°C to +65°C (–40°F to +149°F), start: ≥ –20°C (–4°F)
CPU	8-core 1.5 GHz
SDRAM	8GB
Storage	4 GB NandFlash

1.4.3.14 CR8DIPU1T2C3 (03033GRW: Integrated Network Processing Unit (IPU-1T2))

Overview

Table 1-199 Basic information about the CR8DIPU1T2C3

Item	Details
Description	Integrated Network Processing Unit (IPU-1T2)
Part Number	03033GRW
Model	CR8DIPU1T2C3
Card type	IPU
Silkscreen	IPU-1T2

Appearance

Figure 1-83 Appearance of the CR8DIPU1T2C3



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03033GRW&productModel=CR8DIPU1T2C3#matchRelation>

Indicators and Buttons

Table 1-200 Indicators on the CR8DIPU1T2C3

Silkscreen	Name	Color	Status	Description
STAT	Working status indicator	Green	Steady on	The board is working normally.
		Red	Steady on	The device generates a hardware alarm. You can run the display alarm hardware command to view the hardware alarm type.
		-	Off	The board is not running or no power is input.
PROG	Program running indicator	Green	Steady on	The board software is being initialized. The board software is normally initialized, and the board software is running normally.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	The memory self-check fails. The board software or the logic file is lost. Loading of the board software fails.
		Green	On for 100 ms and off for 100 ms alternately	Loading of the board software is in process.
		Green	On for 300 ms and off for 300 ms alternately	The BIOS is guiding the upper-layer software.
		Red	On for 100 ms and off for 100 ms alternately	The BIOS self-check fails.
		-	Off	No power is input.

Silkscreen	Name	Color	Status	Description
SYNC	Synchronizati on status indicator	Green	Steady on	The system clock works in free-run mode, and no priority list is set for the system clocks (only internal clock sources are listed by default). The system clock works in tracing mode and is tracing a clock source other than the internal clock sources in the priority list. The system clock works in time synchronizatio n mode, and the PTP time and system clock are being traced.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	The system clock priority list is set, but all clock sources except the internal clock source are lost. The clock works in holdover or free-run mode. The system clock works in time synchronization mode, but no synchronization source is available. The system clock and PTP time work in holdover or free-run mode.
ACTX	Active/standby status indicator for the cross-connect or clock board	Green	Steady on	The board is in the working switching state.
		-	Off	The board is in the protection switching state.
ACTC	Control status indicator	Green	Steady on	The board is in the working control state.
		Green	On for 100 ms and off for 100 ms alternately	The data of the equipment is backed up in batch.

Silkscreen	Name	Color	Status	Description
		-	Off	The board is in the protection control state.
ALM	Alarm indicator	Red	Steady on	Critical alarms are generated.
		Orange	Steady on	Major, minor alarms are generated.
		-	Off	No alarms are generated in the equipment.
ETH/OAM	ETH/OAM indicator	Green	Steady on	If the indicator is steady on, the link is normal.
		Orange	Blink	If the indicator blinks, data is being transmitted.
		-	Off	If the indicator is off, the link is Down.

Table 1-201 Buttons on the CR8DIPU1T2C3

Silkscreen	Name	Description
RST	Reset button	You can press the RST button to reset the NE or restore the factory settings of the NE. The procedure is as follows: To reset the NE, press the RST button and release it. To restore factory settings, press and hold the RST button. This operation will clear the NE database and system parameter configurations and cannot be restored. Exercise caution when you perform the following operations: 1. Press and release the RST button to restart the NE. 2. When the PROG indicator blinks (the blinking frequency is about 1 second and the time window is about 15 seconds), press and hold the RST button. 3. Wait for no longer than 2 minutes till both the STAT and PROG indicators blink (the time window is 5 seconds). Then release the RST button immediately.

Silkscreen	Name	Description
		NOTE 1. When you press the RST button to clear the configuration file, the original configuration file will be cleared. You are advised to periodically back up the configuration file. 2. When you press the RST button to clear the configuration file, to prevent the cleared configuration from being synchronized again between the active and standby main control boards, remove one main control board first in a dual-main control board chassis to ensure that the configuration file is cleared in a single-main control board scenario. Perform the same operations on the other main control board. 3. When pressing the RST button to clear the configuration file, you only need to pay attention to the STAT and PROG indicators. Other indicators may vary depending on the product model.

Ports

Table 1-202 Ports on the CR8DIPU1T2C3

Port	Connector Type	Description	Available Components
ETH/OAM	RJ45	10M/100M autosensing Ethernet network management port or non-RS-232-standard console port. NOTE If a service port can be connected to the NMS, it is recommended that the service port be used as the network management port.	ETH/OAM Management Cables
CLK/TOD/RS-485	RJ45	Clock/Time/RS-485 input/output port, RS-485 is a reserved port.	1-into-2 clock cable + clock cable (external clock mode, delivery-ready): applicable to the CLK mode 1-into-2 clock cable + clock cable (external clock/external time mode, prepared onsite): applicable to the CLK or TOD mode 1-into-2 clock cable + Ethernet cable: applicable to the TOD mode
USB	Type A	USB port (reserved)	-
CLK/TEST	SMA	Clock test port	Coaxial Cable

Port	Connector Type	Description	Available Components
GE0,GE1	SFP	Cascading ports are used for control panel expansion in scenarios like virtual cluster. (reserved)	Optical fiber
HP-GE	SFP	Ultra-high-precision GE port (This first supported version is V800R021C00. This port is reserved in earlier versions.)	1.25Gbps eSFP Optical Module 1.25Gbps eSFP CWDM Optical Module 1.25Gbps eSFP BIDI Optical Module

Table 1-203 Pins of the ETH/OAM interface

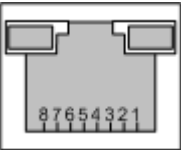
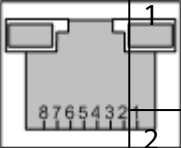
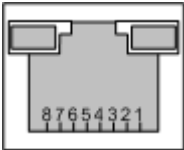
Front View	Pin	Usage
	1	Transmit positive of the NM network port
	2	Transmit negative of the NM network port
	3	Receive positive of the NM network port
	4	Grounding end of the Console interface
	5	Receive end of the Console interface
	6	Receive negative of the NM network port
	7	Unspecified
	8	Transmit end of the Console interface

Table 1-204 Pins of the CLK/TOD/RS-485 interface

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	1	Receive negative of the CLK	Receive negative of the CLK	Unspecified	Unspecified	Unspecified
	2	Receive positive of the CLK	Receive positive of the CLK	Unspecified	Unspecified	Unspecified
	3	Input/Output negative for the 1PPS signal	Unspecified	Input negative for the 1PPS signal (RS422 level)	Output negative for the 1PPS signal (RS422 level)	Transmit negative of the RS-485
	4	Receive negative of the CLK	Receive negative of the CLK	Grounding end	Grounding end	Unspecified
	5	Receive positive of the CLK	Receive positive of the CLK	Grounding end	Grounding end	Unspecified
	6	Input/Output positive for the 1PPS signal	Unspecified	Input positive for the 1PPS signal (RS422 level)	Output positive for the 1PPS signal (RS422 level)	Transmit positive of the RS-485

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	7	Input/Output negative for the time information	Unspecified	Input negative for the time information (RS422 level)	Output negative for the time information (RS422 level)	Receive negative of the RS-485
	8	Input/Output positive for the time information	Unspecified	Input positive for the time information (RS422 level)	Output positive for the time information (RS422 level)	Receive positive of the RS-485
NOTE When this interface is used as a TOD interface, it must be used with a dedicated clock cable. For details, see "Cables" in the Hardware Description.						

Table 1-205 Pins of the ALMI/ALMO interface

Front View	Pin	Usage
	1	Alarm input 1
	2	Grounding end for alarm input 1
	3	Alarm input 2

Front View	Pin	Usage
	4	Alarm input 3
	5	Grounding end for alarm input 3
	6	Grounding end for alarm input 2
	7	Positive for alarm signal output
	8	Negative for alarm signal output

Functions and Features

Table 1-206 Functions and features of the CR8DIPU1T2C3

Functions and Features	Description
Basic function	Data forwarding: The IPU is the core of service processing in the entire system, and is connected to all subcards through data channels. Control and management: Through the management channels between IPU's and the subcards, the IPU's can manage subcards and transmit routing protocol data. The line rate capability is 2.4T bit/s@310Bytes(Bidirectional).
Reliability and availability	Supports the hot swappable function. Supports the board-level 1:1 backup function.
Restrictions and remarks	1. It cannot be used together with other IPU's. 2. When working with the 03033VMC board to support FlexE services, the board supports a maximum of eight FlexE physical interfaces.
Breakout supported	No
Interface rate auto-sensing supported	No

Technical Specifications

Table 1-207 Technical specifications of the CR8DIPU1T2C3

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	34.8 mm x 354.4 mm x 209 mm (1.37 in. x 13.95 in. x 8.23 in.)
Dimensions with packaging (H x W x D) [mm(in.)]	185 mm x 485 mm x 385 mm (7.28 in. x 19.09 in. x 15.16 in.)
Weight without packaging [kg(lb)]	4 kg (8.82 lb)
Weight with packaging [kg(lb)]	5.4 kg (11.9 lb)
Typical power consumption [W]	315.5 W
Typical heat dissipation [BTU/hour]	1023.58 BTU/hour
Long-term operating temperature [°C(°F)]	-20°C to +55°C (-4°F to +131°F); start: ≥ -5°C (23°F)
CPU	20-core 2.0 GHz
SDRAM	16GB
Storage	4GB NandFlash

1.4.3.15 CR8DIPU1T2C3 (03033GRW-001: Integrated Network Processing Unit (IPU-1T2))

Overview

Table 1-208 Basic information about the CR8DIPU1T2C3

Item	Details
Description	Integrated Network Processing Unit (IPU-1T2)
Part Number	03033GRW-001
Model	CR8DIPU1T2C3
Card type	IPU
Silkscreen	IPU-1T2

Appearance

Figure 1-84 Appearance of the CR8DIPU1T2C3



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03033GRW-001&productModel=CR8DIPU1T2C3#matchRelation>

Indicators and Buttons

Table 1-209 Indicators on the CR8DIPU1T2C3

Silkscreen	Name	Color	Status	Description
STAT	Working status indicator	Green	Steady on	The board is working normally.
		Red	Steady on	The device generates a hardware alarm. You can run the display alarm hardware command to view the hardware alarm type.
		-	Off	The board is not running or no power is input.

Silkscreen	Name	Color	Status	Description
PROG	Program running indicator	Green	Steady on	The board software is being initialized. The board software is normally initialized, and the board software is running normally.
		Red	Steady on	The memory self-check fails. The board software or logic file is lost. The board software fails to be loaded. The BIOS self-check fails.
		Green	On for 100 ms and off for 100 ms alternately	Loading of the board software is in process.
		Green	On for 300 ms and off for 300 ms alternately	The BIOS is guiding the upper-layer software.
		-	Off	The board is not powered on.

Silkscreen	Name	Color	Status	Description
SYNC	Clock synchronizatio n indicator	Green	Steady on	The system clock works in free-run mode, and no priority list is set for the system clocks (only internal clock sources are listed by default). The system clock works in tracing mode and is tracing a clock source other than the internal clock sources in the priority list. The system clock works in time synchronizatio n mode, and the PTP time and system clock are being traced.

Silkscreen	Name	Color	Status	Description
SYNC	Synchronization status indicator	Red	Steady on	The system clock priority list is set, but all clock sources except the internal clock source are lost. The clock works in holdover or free-run mode. The system clock works in time synchronization mode, but no synchronization source is available. The system clock and PTP time work in holdover or free-run mode.
ACTX	Active/standby status indicator for the cross-connect or clock board	Green	Steady on	The board is in the active state on the service plane.
		-	Off	The board is in the standby state on the service plane.
ACTC	Control status indicator	Green	Steady on	The board is in the active control state.
		Green	Blinks on and off at 100 ms intervals	The data of the device is backed up in batch.
		-	Off	The board is in the standby control state.

Silkscreen	Name	Color	Status	Description
ALM	Alarm indicator	Red	Steady on	Device-level critical alarms exist.
		Orange	Steady on	A device-level major or minor alarm exists.
		-	Off	There is no equipment-level alarm.
ETH/OAM	ETH/OAM indicator	Green	Steady on	The link is up.
		Orange	Blinking	Data is being transmitted or received.
		-	Off	The link is down.

Table 1-210 Buttons on the CR8DIPU1T2C3

Silkscreen	Name	Description
RST	Reset button	You can press the RST button to reset the NE or restore the factory settings of the NE. The procedure is as follows: To reset the NE, press the RST button and release it. To restore factory settings, press and hold the RST button. This operation will clear the NE database and system parameter configurations and cannot be restored. Exercise caution when you perform the following operations: 1. Press and release the RST button to restart the NE. 2. When the PROG indicator blinks (the blinking frequency is about 1 second and the time window is about 15 seconds), press and hold the RST button. 3. Wait for no longer than 2 minutes till both the STAT and PROG indicators blink (the time window is 5 seconds). Then release the RST button immediately.

Silkscreen	Name	Description
		NOTE 1. When you press the RST button to clear the configuration file, the original configuration file will be cleared. You are advised to periodically back up the configuration file. 2. When you press the RST button to clear the configuration file, to prevent the cleared configuration from being synchronized again between the active and standby main control boards, remove one main control board first in a dual-main control board chassis to ensure that the configuration file is cleared in a single-main control board scenario. Perform the same operations on the other main control board. 3. When pressing the RST button to clear the configuration file, you only need to pay attention to the STAT and PROG indicators. Other indicators may vary depending on the product model.

Ports

Table 1-211 Ports on the CR8DIPU1T2C3

Port	Connector Type	Description	Available Components
ETH/OAM	RJ45	10M/100M autosensing Ethernet network management port or non-RS-232-standard console port. NOTE If a service port can be connected to the NMS, it is recommended that the service port be used as the network management port.	ETH/OAM Management Cables
CLK/TOD/RS-485	RJ45	Clock/Time/RS-485 input/output port, RS-485 is a reserved port.	1-into-2 clock cable + clock cable (external clock mode, delivery-ready): applicable to the CLK mode 1-into-2 clock cable + clock cable (external clock/external time mode, prepared onsite): applicable to the CLK or TOD mode 1-into-2 clock cable + Ethernet cable: applicable to the TOD mode
USB	Type A	USB port (reserved)	-
CLK/TEST	SMA	Clock test port	Coaxial Cable

Port	Connector Type	Description	Available Components
HP-GE	SFP	Ultra-high-precision GE port (supported since V800R021C00 and reserved in earlier versions)	1.25Gbps eSFP Optical Module 1.25Gbps eSFP CWDM Optical Module 1.25Gbps eSFP BIDI Optical Module

Table 1-212 Pins of the ETH/OAM interface

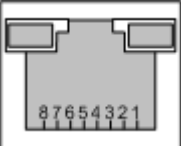
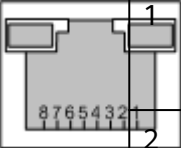
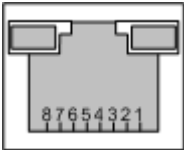
Front View	Pin	Usage
	1	Transmit positive of the NM network port
	2	Transmit negative of the NM network port
	3	Receive positive of the NM network port
	4	Grounding end of the Console interface
	5	Receive end of the Console interface
	6	Receive negative of the NM network port
	7	Unspecified
	8	Transmit end of the Console interface

Table 1-213 Pins of the CLK/TOD/RS-485 interface

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	1	Receive negative of the CLK	Receive negative of the CLK	Unspecified	Unspecified	Unspecified
	2	Receive positive of the CLK	Receive positive of the CLK	Unspecified	Unspecified	Unspecified
	3	Input/Output negative for the 1PPS signal	Unspecified	Input negative for the 1PPS signal (RS422 level)	Output negative for the 1PPS signal (RS422 level)	Transmit negative of the RS-485
	4	Receive negative of the CLK	Receive negative of the CLK	Grounding end	Grounding end	Unspecified
	5	Receive positive of the CLK	Receive positive of the CLK	Grounding end	Grounding end	Unspecified
	6	Input/Output positive for the 1PPS signal	Unspecified	Input positive for the 1PPS signal (RS422 level)	Output positive for the 1PPS signal (RS422 level)	Transmit positive of the RS-485

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	7	Input/Output negative for the time information	Unspecified	Input negative for the time information (RS422 level)	Output negative for the time information (RS422 level)	Receive negative of the RS-485
	8	Input/Output positive for the time information	Unspecified	Input positive for the time information (RS422 level)	Output positive for the time information (RS422 level)	Receive positive of the RS-485
NOTE When this interface is used as a TOD interface, it must be used with a dedicated clock cable. For details, see "Cables" in the Hardware Description.						

Table 1-214 Pins of the ALMI/ALMO interface

Front View	Pin	Usage
	1	Alarm input 1
	2	Grounding end for alarm input 1
	3	Alarm input 2

Front View	Pin	Usage
	4	Alarm input 3
	5	Grounding end for alarm input 3
	6	Grounding end for alarm input 2
	7	Positive for alarm signal output
	8	Negative for alarm signal output

Functions and Features

Table 1-215 Functions and features of the CR8DIPU1T2C3

Functions and Features	Description
Basic function	Data forwarding: The IPU is the core of service processing in the entire system, and is connected to all subcards through data channels. Control and management: Through the management channels between IPU and the subcards, the IPU can manage subcards and transmit routing protocol data. The line rate capability is 2.4T bit/s@310Bytes(Bidirectional).
Reliability and availability	Supports the hot swappable function. Supports the board-level 1:1 backup function.
Restrictions and remarks	1. It cannot be used together with other IPU. 2. When working with the 03033VMC board to support FlexE services, the board supports a maximum of eight FlexE physical interfaces.
Breakout supported	No
Interface rate auto-sensing supported	No

Technical Specifications

Table 1-216 Technical specifications of the CR8DIPU1T2C3

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	34.8 mm x 354.4 mm x 209 mm (1.37 in. x 13.95 in. x 8.23 in.)
Dimensions with packaging (H x W x D) [mm(in.)]	185 mm x 485 mm x 385 mm (7.28 in. x 19.09 in. x 15.16 in.)
Weight without packaging [kg(lb)]	4 kg (8.82 lb)
Weight with packaging [kg(lb)]	5.4 kg (11.9 lb)
Typical power consumption [W]	366.62 W
Typical heat dissipation [BTU/hour]	1189.45 BTU/hour
Long-term operating temperature [°C(°F)]	-20°C to +55°C (-4°F to +131°F); start: ≥ -5°C (23°F)
CPU	8-core 2.3 GHz
SDRAM	16GB
Storage	16GB M.2

1.4.3.16 CR8DIPU1T2C3 (03033GRW-002: Integrated Network Processing Unit (IPU-1T2))

Overview

Table 1-217 Basic information about the CR8DIPU1T2C3

Item	Details
Description	Integrated Network Processing Unit (IPU-1T2)
Part Number	03033GRW-002
Model	CR8DIPU1T2C3
Card type	IPU
Silkscreen	IPU-1T2

Appearance

Figure 1-85 Appearance of the CR8DIPU1T2C3



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03033GRW-002&productModel=CR8DIPU1T2C3#matchRelation>

Indicators and Buttons

Table 1-218 Indicators on the CR8DIPU1T2C3

Silkscreen	Name	Color	Status	Description
STAT	Working status indicator	Green	Steady on	The board is working normally.
		Red	Steady on	The device generates a hardware alarm. You can run the display alarm hardware command to view the hardware alarm type.
		-	Off	The board is not running or no power is input.

Silkscreen	Name	Color	Status	Description
PROG	Program running indicator	Green	Steady on	The board software is being initialized. The board software is normally initialized, and the board software is running normally.
		Red	Steady on	The memory self-check fails. The board software or the logic file is lost. Loading of the board software fails.
		Green	On for 100 ms and off for 100 ms alternately	Loading of the board software is in process.
		Green	On for 300 ms and off for 300 ms alternately	The BIOS is guiding the upper-layer software.
		Red	On for 100 ms and off for 100 ms alternately	The BIOS self-check fails.
		-	Off	No power is input.

Silkscreen	Name	Color	Status	Description
SYNC	Synchronizati on status indicator	Green	Steady on	The system clock works in free-run mode, and no priority list is set for the system clocks (only internal clock sources are listed by default). The system clock works in tracing mode and is tracing a clock source other than the internal clock sources in the priority list. The system clock works in time synchronizatio n mode, and the PTP time and system clock are being traced.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	The system clock priority list is set, but all clock sources except the internal clock source are lost. The clock works in holdover or free-run mode. The system clock works in time synchronization mode, but no synchronization source is available. The system clock and PTP time work in holdover or free-run mode.
ACTX	Active/standby status indicator for the cross-connect or clock board	Green	Steady on	The board is in the working switching state.
		-	Off	The board is in the protection switching state.
ACTC	Control status indicator	Green	Steady on	The board is in the working control state.
		Green	On for 100 ms and off for 100 ms alternately	The data of the equipment is backed up in batch.

Silkscreen	Name	Color	Status	Description
		-	Off	The board is in the protection control state.
ALM	Alarm indicator	Red	Steady on	Critical alarms are generated.
		Orange	Steady on	Major, minor alarms are generated.
		-	Off	No alarms are generated in the equipment.
ETH/OAM	ETH/OAM indicator	Green	Steady on	If the indicator is steady on, the link is normal.
		Orange	Blink	If the indicator blinks, data is being transmitted.
		-	Off	If the indicator is off, the link is Down.

Table 1-219 Buttons on the CR8DIPU1T2C3

Silkscreen	Name	Description
RST	Reset button	You can press the RST button to reset the NE or restore the factory settings of the NE. The procedure is as follows: To reset the NE, press the RST button and release it. To restore factory settings, press and hold the RST button. This operation will clear the NE database and system parameter configurations and cannot be restored. Exercise caution when you perform the following operations: 1. Press and release the RST button to restart the NE. 2. When the PROG indicator blinks (the blinking frequency is about 1 second and the time window is about 15 seconds), press and hold the RST button. 3. Wait for no longer than 2 minutes till both the STAT and PROG indicators blink (the time window is 5 seconds). Then release the RST button immediately.

Silkscreen	Name	Description
		NOTE 1. When you press the RST button to clear the configuration file, the original configuration file will be cleared. You are advised to periodically back up the configuration file. 2. When you press the RST button to clear the configuration file, to prevent the cleared configuration from being synchronized again between the active and standby main control boards, remove one main control board first in a dual-main control board chassis to ensure that the configuration file is cleared in a single-main control board scenario. Perform the same operations on the other main control board. 3. When pressing the RST button to clear the configuration file, you only need to pay attention to the STAT and PROG indicators. Other indicators may vary depending on the product model.

Ports

Table 1-220 Ports on the CR8DIPU1T2C3

Port	Connector Type	Description	Available Components
ETH/OAM	RJ45	10M/100M autosensing Ethernet network management port or non-RS-232-standard console port. NOTE If a service port can be connected to the NMS, it is recommended that the service port be used as the network management port.	ETH/OAM Management Cables
CLK/TOD/RS-485	RJ45	Clock/Time/RS-485 input/output port, RS-485 is a reserved port.	1-into-2 clock cable + clock cable (external clock mode, delivery-ready): applicable to the CLK mode 1-into-2 clock cable + clock cable (external clock/external time mode, prepared onsite): applicable to the CLK or TOD mode 1-into-2 clock cable + Ethernet cable: applicable to the TOD mode
USB	Type A	USB port (reserved)	-
CLK/TEST	SMA	Clock test port	Coaxial Cable

Port	Connector Type	Description	Available Components
HP-GE	SFP	Ultra-high-precision GE port (supported since V800R021C00 and reserved in earlier versions)	1.25Gbps eSFP Optical Module 1.25Gbps eSFP CWDM Optical Module 1.25Gbps eSFP BIDI Optical Module

Table 1-221 Pins of the ETH/OAM interface

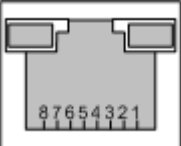
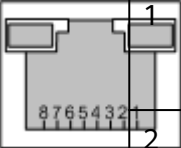
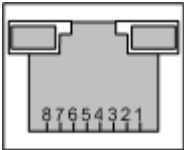
Front View	Pin	Usage
	1	Transmit positive of the NM network port
	2	Transmit negative of the NM network port
	3	Receive positive of the NM network port
	4	Grounding end of the Console interface
	5	Receive end of the Console interface
	6	Receive negative of the NM network port
	7	Unspecified
	8	Transmit end of the Console interface

Table 1-222 Pins of the CLK/TOD/RS-485 interface

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	1	Receive negative of the CLK	Receive negative of the CLK	Unspecified	Unspecified	Unspecified
	2	Receive positive of the CLK	Receive positive of the CLK	Unspecified	Unspecified	Unspecified
	3	Input/Output negative for the 1PPS signal	Unspecified	Input negative for the 1PPS signal (RS422 level)	Output negative for the 1PPS signal (RS422 level)	Transmit negative of the RS-485
	4	Receive negative of the CLK	Receive negative of the CLK	Grounding end	Grounding end	Unspecified
	5	Receive positive of the CLK	Receive positive of the CLK	Grounding end	Grounding end	Unspecified
	6	Input/Output positive for the 1PPS signal	Unspecified	Input positive for the 1PPS signal (RS422 level)	Output positive for the 1PPS signal (RS422 level)	Transmit positive of the RS-485

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	7	Input/Output negative for the time information	Unspecified	Input negative for the time information (RS422 level)	Output negative for the time information (RS422 level)	Receive negative of the RS-485
	8	Input/Output positive for the time information	Unspecified	Input positive for the time information (RS422 level)	Output positive for the time information (RS422 level)	Receive positive of the RS-485
NOTE When this interface is used as a TOD interface, it must be used with a dedicated clock cable. For details, see "Cables" in the Hardware Description.						

Table 1-223 Pins of the ALMI/ALMO interface

Front View	Pin	Usage
	1	Alarm input 1
	2	Grounding end for alarm input 1
	3	Alarm input 2

Front View	Pin	Usage
	4	Alarm input 3
	5	Grounding end for alarm input 3
	6	Grounding end for alarm input 2
	7	Positive for alarm signal output
	8	Negative for alarm signal output

Functions and Features

Table 1-224 Functions and features of the CR8DIPU1T2C3

Functions and Features	Description
Basic function	Data forwarding: The IPU is the core of service processing in the entire system, and is connected to all subcards through data channels. Control and management: Through the management channels between IPU and the subcards, the IPU can manage subcards and transmit routing protocol data. The line rate capability is 2.4T bit/s@310Bytes(Bidirectional).
Reliability and availability	Supports the hot swappable function. Supports the board-level 1:1 backup function.
Restrictions and remarks	1. It cannot be used together with other IPU. 2. When working with the 03033VMC board to support FlexE services, the board supports a maximum of eight FlexE physical interfaces.
Breakout supported	No
Interface rate auto-sensing supported	No

Technical Specifications

Table 1-225 Technical specifications of the CR8DIPU1T2C3

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	34.8 mm x 354.4 mm x 209 mm (1.37 in. x 13.95 in. x 8.23 in.)
Dimensions with packaging (H x W x D) [mm(in.)]	185 mm x 485 mm x 385 mm (7.28 in. x 19.09 in. x 15.16 in.)
Weight without packaging [kg(lb)]	4 kg (8.82 lb)
Weight with packaging [kg(lb)]	5.4 kg (11.9 lb)
Typical power consumption [W]	359.1 W
Typical heat dissipation [BTU/hour]	1165.07 BTU/hour
Long-term operating temperature [°C(°F)]	-20°C to +55°C (-4°F to +131°F); start: ≥ -5°C (23°F)
CPU	20-core 2.0 GHz
SDRAM	16GB
Storage	4GB NandFlash

1.4.3.17 CR8DIPU1T2C5 (03033NRF: Integrated Network Processing Unit (IPU-1T2-A))

Overview

Table 1-226 Basic information about the CR8DIPU1T2C5

Item	Details
Description	Integrated Network Processing Unit (IPU-1T2-A)
Part Number	03033NRF
Model	CR8DIPU1T2C5
Card type	IPU
Silkscreen	IPU-1T2-A

Appearance

Figure 1-86 Appearance of the CR8DIPU1T2C5



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03033NRF&productModel=CR8DIPU1T2C5#matchRelation>

Indicators and Buttons

Table 1-227 Indicators on the CR8DIPU1T2C5

Silkscreen	Name	Color	Status	Description
STAT	Working Status Indicator	Green	Steady on	The board is working properly.
		Red	Steady on	A board hardware fault has occurred.
		Orange	Steady on	The interface board is not loaded with the logic.
		-	Off	The board is not powered on or the board is not started.

Silkscreen	Name	Color	Status	Description
PROG	Program Status Indicator	Green	Steady on	Board software is being initialized. Board software has been initialized, or board software is running properly.
		Red	Steady on	The memory self-check fails. Board software or a logic file is lost. Loading of the board software fails.
		Green	On for 100 ms and off for 100 ms alternately	Board software is being loaded.
		Green	On for 300 ms and off for 300 ms alternately	The BIOS is guiding upper-layer software.
		Red	On for 100 ms and off for 100 ms alternately	The BIOS fails self-check.
		-	Off	The board is not powered on.

Silkscreen	Name	Color	Status	Description
SYNC	Synchronizati on Status Indicator	Green	Steady on	The system clock works in free-run mode and no priority list is configured for the system clock. (By default, the list contains only internal clock sources.) The system clock is working in tracing mode and is tracing a clock source rather than the internal sources in the priority list. The system clock is working in time synchronization mode, and the PTP time and system clock are in the tracing state.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	The system clock priority list is set. All the clock sources, however, are lost except for the internal clock sources. The clock works in holdover mode or free-run mode. The system clock is working in time synchronization mode, but no synchronization source is available. The system clock and PTP time are working in holdover or free-run mode.
ACTX	Switching Status Indicator	Green	Steady on	The board is in the working switching state.
		-	Off	The board is in the protection switching state.
ACTC	Control Status Indicator	Green	Steady on	The board is in the working control state.

Silkscreen	Name	Color	Status	Description
		Green	On for 100 ms and off for 100 ms alternately	he data in the database of the device is backed up in batch.
		-	Off	The board is in the protection control state.
ALM	Alarm Indicator	Red	Steady on	A critical alarm is generated on the device.
		Orange	Steady on	A major alarm or a minor alarm is generated on the device.
		-	Off	No alarm is generated on the device.
ETH/OAM	ETH/OAM indicator	Green	Steady on	If the indicator is steady on, the link is normal.
		Orange	Blink	If the indicator blinks, data is being transmitted.
		-	Off	If the indicator is off, the link is Down.

Table 1-228 Buttons on the CR8DIPU1T2C5

Silkscreen	Name	Description
RST	RST	You can press the RST button to reset the NE or restore the factory settings of the NE. The procedure is as follows: To reset the NE, press the RST button and release it. To restore factory settings, press and hold the RST button. This operation will clear the NE database and system parameter configurations and cannot be restored. Exercise caution when you perform the following operations: 1. Press and release the RST button to restart the NE. 2. When the PROG indicator blinks (the blinking frequency is about 1 second and the time window is about 15 seconds), press and hold the RST button. 3. Wait for no longer than 2 minutes till both the STAT and PROG indicators blink (the time window is 5 seconds). Then release the RST button immediately.

Silkscreen	Name	Description
		NOTE 1. When you press the RST button to clear the configuration file, the original configuration file will be cleared. You are advised to periodically back up the configuration file. 2. When you press the RST button to clear the configuration file, to prevent the cleared configuration from being synchronized again between the active and standby main control boards, remove one main control board first in a dual-main control board chassis to ensure that the configuration file is cleared in a single-main control board scenario. Perform the same operations on the other main control board. 3. When pressing the RST button to clear the configuration file, you only need to pay attention to the STAT and PROG indicators. Other indicators may vary depending on the product model.

Ports

Table 1-229 Ports on the CR8DIPU1T2C5

Port	Connector Type	Description	Available Components
CLK/TEST	SMA	Clock test port	Coaxial Cable
USB	Type A	USB port (reserved)	-

Port	Connector Type	Description	Available Components
ETH/OAM	RJ45	10M/100M autosensing Ethernet network management port or non-RS-232-standard console port. NOTE If a service port can be connected to the NMS, it is recommended that the service port be used as the network management port.	ETH/OAM Management Cables
CLK/TOD/RS-485	RJ45	Clock/Time/RS-485 input/output port, RS-485 is a reserved port.	1-into-2 clock cable + clock cable (external clock mode, delivery-ready): applicable to the CLK mode 1-into-2 clock cable + clock cable (external clock/external time mode, prepared onsite): applicable to the CLK or TOD mode 1-into-2 clock cable + Ethernet cable: applicable to the TOD mode
ALMI/ALMO	RJ45	Alarm input/output port	Alarm Input/Output Cables
HP-GE	SFP	Ultra-high-precision GE port (reserved)	Optical fiber

Table 1-230 Pins of the ETH/OAM interface

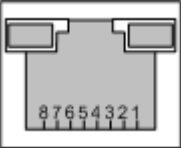
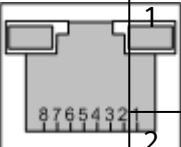
Front View	Pin	Usage
	1	Transmit positive of the NM network port
	2	Transmit negative of the NM network port
	3	Receive positive of the NM network port
	4	Grounding end of the Console interface
	5	Receive end of the Console interface
	6	Receive negative of the NM network port
	7	Unspecified
	8	Transmit end of the Console interface

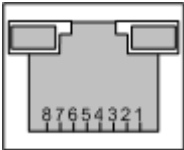
Table 1-231 Pins of the CLK/TOD/RS-485 interface

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	1	Receive negative of the CLK	Receive negative of the CLK	Unspecified	Unspecified	Unspecified
	2	Receive positive of the CLK	Receive positive of the CLK	Unspecified	Unspecified	Unspecified

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	3	Input/Output negative for the 1PPS signal	Unspecified	Input negative for the 1PPS signal (RS422 level)	Output negative for the 1PPS signal (RS422 level)	Transmit negative of the RS-485
	4	Receive negative of the CLK	Receive negative of the CLK	Grounding end	Grounding end	Unspecified
	5	Receive positive of the CLK	Receive positive of the CLK	Grounding end	Grounding end	Unspecified
	6	Input/Output positive for the 1PPS signal	Unspecified	Input positive for the 1PPS signal (RS422 level)	Output positive for the 1PPS signal (RS422 level)	Transmit positive of the RS-485

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	7	Input/Output negative for the time information	Unspecified	Input negative for the time information (RS422 level)	Output negative for the time information (RS422 level)	Receive negative of the RS-485
	8	Input/Output positive for the time information	Unspecified	Input positive for the time information (RS422 level)	Output positive for the time information (RS422 level)	Receive positive of the RS-485
NOTE When this interface is used as a TOD interface, it must be used with a dedicated clock cable. For details, see "Cables" in the Hardware Description.						

Table 1-232 Pins of the ALMI/ALMO interface

Front View	Pin	Usage
	1	Alarm input 1
	2	Grounding end for alarm input 1
	3	Alarm input 2

Front View	Pin	Usage
	4	Alarm input 3
	5	Grounding end for alarm input 3
	6	Grounding end for alarm input 2
	7	Positive for alarm signal output
	8	Negative for alarm signal output

Functions and Features

Table 1-233 Functions and features of the CR8DIPU1T2C5

Functions and Features	Description
Basic function	Line-Rate capability: 1200 Gbit/s. Supports switching, control, and clock management. Supports the board-level 1:1 backup function. Supports ultra-high-precision clock.
Reliability and availability	Supports the hot swappable function. Supports the board-level 1:1 backup function.

Technical Specifications

Table 1-234 Technical specifications of the CR8DIPU1T2C5

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	34.8 mm x 354.4 mm x 209 mm (1.37 in. x 13.95 in. x 8.23 in.)
Dimensions with packaging (H x W x D) [mm(in.)]	185 mm x 485 mm x 385 mm (7.28 in. x 19.09 in. x 15.16 in.)
Weight without packaging [kg(lb)]	2.55 kg (5.62 lb)
Weight with packaging [kg(lb)]	3.95 kg (8.71 lb)
Typical power consumption [W]	135.6 W

Item	Specification
Typical heat dissipation [BTU/hour]	440 BTU/hour
Long-term operating temperature [°C(°F)]	-20°C to +55°C (-4°F to +131°F); start: ≥ -5°C (23°F)
Switching capacity	2.4 Tbit/s
CPU	8-core 1.5 GHz
SDRAM	8GB
Storage	4GB NandFlash

1.4.3.18 CR8DIPU1T2C5 (03033NRF-001: Integrated Network Processing Unit (IPU-1T2-A))

Overview

Table 1-235 Basic information about the CR8DIPU1T2C5

Item	Details
Description	Integrated Network Processing Unit (IPU-1T2-A)
Part Number	03033NRF-001
Model	CR8DIPU1T2C5
Card type	IPU
Silkscreen	IPU-1T2-A

Appearance

Figure 1-87 Appearance of the CR8DIPU1T2C5



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find

desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03033NRF-001&productModel=CR8DIPU1T2C5#matchRelation>

Indicators and Buttons

Table 1-236 Indicators on the CR8DIPU1T2C5

Silkscreen	Name	Color	Status	Description
STAT	Working Status Indicator	Green	Steady on	The board is working properly.
		Red	Steady on	A board hardware fault has occurred.
		Orange	Steady on	The interface board is not loaded with the logic.
		-	Off	The board is not powered on or the board is not started.
PROG	Program Status Indicator	Green	Steady on	Board software is being initialized. Board software has been initialized, or board software is running properly.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	The memory self-check fails. Board software or a logic file is lost. Loading of the board software fails.
		Green	On for 100 ms and off for 100 ms alternately	Board software is being loaded.
		Green	On for 300 ms and off for 300 ms alternately	The BIOS is guiding upper-layer software.
		Red	On for 100 ms and off for 100 ms alternately	The BIOS fails self-check.
		-	Off	The board is not powered on.

Silkscreen	Name	Color	Status	Description
SYNC	Synchronizati on Status Indicator	Green	Steady on	The system clock works in free-run mode and no priority list is configured for the system clock. (By default, the list contains only internal clock sources.) The system clock is working in tracing mode and is tracing a clock source rather than the internal sources in the priority list. The system clock is working in time synchronization mode, and the PTP time and system clock are in the tracing state.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	The system clock priority list is set. All the clock sources, however, are lost except for the internal clock sources. The clock works in holdover mode or free-run mode. The system clock is working in time synchronization mode, but no synchronization source is available. The system clock and PTP time are working in holdover or free-run mode.
ACTX	Switching Status Indicator	Green	Steady on	The board is in the working switching state.
		-	Off	The board is in the protection switching state.
ACTC	Control Status Indicator	Green	Steady on	The board is in the working control state.

Silkscreen	Name	Color	Status	Description
		Green	On for 100 ms and off for 100 ms alternately	he data in the database of the device is backed up in batch.
		-	Off	The board is in the protection control state.
ALM	Alarm Indicator	Red	Steady on	A critical alarm is generated on the device.
		Orange	Steady on	A major alarm or a minor alarm is generated on the device.
		-	Off	No alarm is generated on the device.
ETH/OAM	ETH/OAM indicator	Green	Steady on	If the indicator is steady on, the link is normal.
		Orange	Blink	If the indicator blinks, data is being transmitted.
		-	Off	If the indicator is off, the link is Down.

Table 1-237 Buttons on the CR8DIPU1T2C5

Silkscreen	Name	Description
RST	RST	You can press the RST button to reset the NE or restore the factory settings of the NE. The procedure is as follows: To reset the NE, press the RST button and release it. To restore factory settings, press and hold the RST button. This operation will clear the NE database and system parameter configurations and cannot be restored. Exercise caution when you perform the following operations: 1. Press and release the RST button to restart the NE. 2. When the PROG indicator blinks (the blinking frequency is about 1 second and the time window is about 15 seconds), press and hold the RST button. 3. Wait for no longer than 2 minutes till both the STAT and PROG indicators blink (the time window is 5 seconds). Then release the RST button immediately.

Silkscreen	Name	Description
		NOTE 1. When you press the RST button to clear the configuration file, the original configuration file will be cleared. You are advised to periodically back up the configuration file. 2. When you press the RST button to clear the configuration file, to prevent the cleared configuration from being synchronized again between the active and standby main control boards, remove one main control board first in a dual-main control board chassis to ensure that the configuration file is cleared in a single-main control board scenario. Perform the same operations on the other main control board. 3. When pressing the RST button to clear the configuration file, you only need to pay attention to the STAT and PROG indicators. Other indicators may vary depending on the product model.

Ports

Table 1-238 Ports on the CR8DIPU1T2C5

Port	Connector Type	Description	Available Components
CLK/TEST	SMA	Clock test port	Coaxial Cable
USB	Type A	USB port (reserved)	-

Port	Connector Type	Description	Available Components
ETH/OAM	RJ45	10M/100M autosensing Ethernet network management port or non-RS-232-standard console port. NOTE If a service port can be connected to the NMS, it is recommended that the service port be used as the network management port.	ETH/OAM Management Cables
CLK/TOD/RS-485	RJ45	Clock/Time/RS-485 input/output port, RS-485 is a reserved port.	1-into-2 clock cable + clock cable (external clock mode, delivery-ready): applicable to the CLK mode 1-into-2 clock cable + clock cable (external clock/external time mode, prepared onsite): applicable to the CLK or TOD mode 1-into-2 clock cable + Ethernet cable: applicable to the TOD mode
ALMI/ALMO	RJ45	Alarm input/output port	Alarm Input/Output Cables
HP-GE	SFP	Ultra-high-precision GE port (reserved)	Optical fiber

Table 1-239 Pins of the ETH/OAM interface

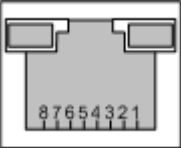
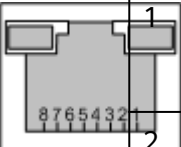
Front View	Pin	Usage
	1	Transmit positive of the NM network port
	2	Transmit negative of the NM network port
	3	Receive positive of the NM network port
	4	Grounding end of the Console interface
	5	Receive end of the Console interface
	6	Receive negative of the NM network port
	7	Unspecified
	8	Transmit end of the Console interface

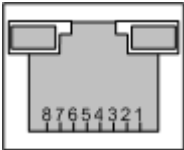
Table 1-240 Pins of the CLK/TOD/RS-485 interface

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	1	Receive negative of the CLK	Receive negative of the CLK	Unspecified	Unspecified	Unspecified
	2	Receive positive of the CLK	Receive positive of the CLK	Unspecified	Unspecified	Unspecified

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	3	Input/Output negative for the 1PPS signal	Unspecified	Input negative for the 1PPS signal (RS422 level)	Output negative for the 1PPS signal (RS422 level)	Transmit negative of the RS-485
	4	Receive negative of the CLK	Receive negative of the CLK	Grounding end	Grounding end	Unspecified
	5	Receive positive of the CLK	Receive positive of the CLK	Grounding end	Grounding end	Unspecified
	6	Input/Output positive for the 1PPS signal	Unspecified	Input positive for the 1PPS signal (RS422 level)	Output positive for the 1PPS signal (RS422 level)	Transmit positive of the RS-485

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	7	Input/Output negative for the time information	Unspecified	Input negative for the time information (RS422 level)	Output negative for the time information (RS422 level)	Receive negative of the RS-485
	8	Input/Output positive for the time information	Unspecified	Input positive for the time information (RS422 level)	Output positive for the time information (RS422 level)	Receive positive of the RS-485
NOTE When this interface is used as a TOD interface, it must be used with a dedicated clock cable. For details, see "Cables" in the Hardware Description.						

Table 1-241 Pins of the ALMI/ALMO interface

Front View	Pin	Usage
	1	Alarm input 1
	2	Grounding end for alarm input 1
	3	Alarm input 2

Front View	Pin	Usage
	4	Alarm input 3
	5	Grounding end for alarm input 3
	6	Grounding end for alarm input 2
	7	Positive for alarm signal output
	8	Negative for alarm signal output

Functions and Features

Table 1-242 Functions and features of the CR8DIPU1T2C5

Functions and Features	Description
Basic function	Line-Rate capability: 1200 Gbit/s. Supports switching, control, and clock management. Supports the board-level 1:1 backup function. Supports ultra-high-precision clock.
Reliability and availability	Supports the hot swappable function. Supports the board-level 1:1 backup function.

Technical Specifications

Table 1-243 Technical specifications of the CR8DIPU1T2C5

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	34.8 mm x 354.4 mm x 209 mm (1.37 in. x 13.95 in. x 8.23 in.)
Dimensions with packaging (H x W x D) [mm(in.)]	185 mm x 485 mm x 385 mm (7.28 in. x 19.09 in. x 15.16 in.)
Weight without packaging [kg(lb)]	2.55 kg (5.62 lb)
Weight with packaging [kg(lb)]	3.95 kg (8.71 lb)
Typical power consumption [W]	135.6 W

Item	Specification
Typical heat dissipation [BTU/hour]	440 BTU/hour
Long-term operating temperature [°C(°F)]	-20°C to +55°C (-4°F to +131°F); start: ≥ -5°C (23°F)
Switching capacity	2.4 Tbit/s
CPU	8-core 1.5 GHz
SDRAM	8GB
Storage	4GB NandFlash

1.4.3.19 CR8DIPU1T2C5 (03033NRF-002: Integrated Network Processing Unit (IPU-1T2-A))

Overview

Table 1-244 Basic information about the CR8DIPU1T2C5

Item	Details
Description	Integrated Network Processing Unit (IPU-1T2-A)
Part Number	03033NRF-002
Model	CR8DIPU1T2C5
Card type	IPU
Silkscreen	IPU-1T2-A

Appearance

Figure 1-88 Appearance of the CR8DIPU1T2C5



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find

desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03033NRF-002&productModel=CR8DIPU1T2C5#matchRelation>

Indicators and Buttons

Table 1-245 Indicators on the CR8DIPU1T2C5

Silkscreen	Name	Color	Status	Description
STAT	Working status indicator	Green	Steady on	The board is working properly.
		Red	Steady on	The board hardware is faulty.
		Orange	Steady on	The interface board is not loaded with logic.
		-	Off	The board is not powered on or is not running.
PROG	Program status indicator	Green	Steady on	The board software is being initialized or is successfully initialized and is running properly.
		Red	Steady on	The memory self-check fails. The board software or logic file is lost. The board software fails to be loaded.
		Green	Blinks on and off at 100 ms intervals	The board software is being loaded.

Silkscreen	Name	Color	Status	Description
		Green	Blinks on and off at 300 ms intervals	The BIOS is booting the upper-layer software.
		Red	Blinks on and off at 100 ms intervals	The BIOS self-check fails.
		-	Off	The board is not powered on.
SYNC	Clock synchronization indicator	Green	Steady on	The system clock works in free-run mode and no priority list is configured for the system clock. (By default, the list contains only internal clock sources.) The system clock is working in tracing mode and is tracing a clock source rather than the internal sources in the priority list. The system clock is working in time synchronization mode, and the PTP time and system clock are in the tracing state.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	The system clock priority list is set. All the clock sources, however, are lost except for the internal clock sources. The clock works in holdover mode or free-run mode. The system clock is working in time synchronization mode, but no synchronization source is available. The system clock and PTP time are working in holdover or free-run mode.
ACTX	Active/standby status indicator for the cross-connect or clock board	Green	Steady on	The board is in the active switching state.
		-	Off	The board is in the standby switching state.
ACTC	Control status indicator	Green	Steady on	The board is in the active control state.
		Green	Blinks on and off at 100 ms intervals	The data of the device is backed up in batch.

Silkscreen	Name	Color	Status	Description
		-	Off	The board is in the standby control state.
ALM	Device alarm indicator	Red	Steady on	A device-level critical alarm is generated.
		Orange	Steady on	There is an equipment-level major or minor alarm.
		-	Off	There is no equipment-level alarm.
ETH/OAM	ETH/OAM indicator	Green	Steady on	The link is up.
		Orange	Blinking	Data is being transmitted or received.
		-	Off	The link is down.

Table 1-246 Buttons on the CR8DIPU1T2C5

Silkscreen	Name	Description
RST	Board reset button	You can press the RST button to reset the NE or restore the factory settings of the NE. The procedure is as follows: To reset the NE, press the RST button and release it. To restore factory settings, press and hold down the RST button. This operation will clear the NE database and system parameter configurations and cannot be restored. Exercise caution when you perform the following operations: 1.Press the RST button to restart the NE. 2.When the PROG indicator blinks (the blinking frequency is about 1 second and the time window is about 15 seconds), press and hold down the RST button. 3.Wait for no longer than 2 minutes till both the STAT and PROG indicators blink (the time window is 5 seconds). Then release the RST button immediately.

Silkscreen	Name	Description
		NOTE 1. When you press the RST button to clear the configuration file, the original configuration file will be cleared. You are advised to periodically back up the configuration file. 2. When you press the RST button to clear the configuration file, to prevent the cleared configuration from being synchronized again between the active and standby main control boards, remove one main control board first in a dual-main control board chassis to ensure that the configuration file is cleared in a single-main control board scenario. Perform the same operations on the other main control board. 3. When pressing the RST button to clear the configuration file, you only need to pay attention to the STAT and PROG indicators. Other indicators may vary depending on the product model.

Ports

Table 1-247 Ports on the CR8DIPU1T2C5

Port	Connector Type	Description	Available Components
CLK/TEST	SMA	Clock test port	Coaxial cable
USB	Type A	USB port (reserved)	-

Port	Connector Type	Description	Available Components
ETH/OAM	RJ45	10M/100M auto-sensing Ethernet NM interface and non-standard RS-232 Console interface. NOTE If a service port can be connected to an NMS, you are advised to use the service port as the network management port.	ETH/OAM management cable
CLK/TOD/RS-485	RJ45	Clock/Time/RS-485 input and output ports RS-485 is a reserved port.	1-into-2 clock cable + clock cable (external clock mode, delivery-ready): applicable to the CLK mode 1-into-2 clock cable + clock cable (external clock/external time mode, prepared onsite): applicable to the CLK or TOD mode 1-into-2 clock cable + Ethernet cable: applicable to the TOD mode
ALMI/ALMO	RJ45	Alarm input/output interface	Alarm input/output cable
HP-GE	SFP	Ultra-high-precision GE port (reserved)	Optical fiber

Table 1-248 Pins of the ETH/OAM interface

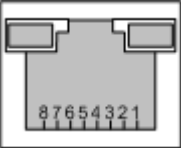
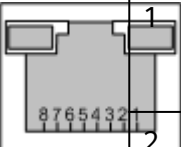
Front View	Pin	Usage
	1	Transmit positive of the NM network port
	2	Transmit negative of the NM network port
	3	Receive positive of the NM network port
	4	Grounding end of the Console interface
	5	Receive end of the Console interface
	6	Receive negative of the NM network port
	7	Unspecified
	8	Transmit end of the Console interface

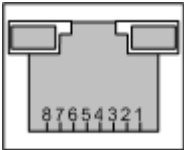
Table 1-249 Pins of the CLK/TOD/RS-485 interface

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	1	Receive negative of the CLK	Receive negative of the CLK	Unspecified	Unspecified	Unspecified
	2	Receive positive of the CLK	Receive positive of the CLK	Unspecified	Unspecified	Unspecified

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	3	Input/Output negative for the 1PPS signal	Unspecified	Input negative for the 1PPS signal (RS422 level)	Output negative for the 1PPS signal (RS422 level)	Transmit negative of the RS-485
	4	Receive negative of the CLK	Receive negative of the CLK	Grounding end	Grounding end	Unspecified
	5	Receive positive of the CLK	Receive positive of the CLK	Grounding end	Grounding end	Unspecified
	6	Input/Output positive for the 1PPS signal	Unspecified	Input positive for the 1PPS signal (RS422 level)	Output positive for the 1PPS signal (RS422 level)	Transmit positive of the RS-485

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	7	Input/Output negative for the time information	Unspecified	Input negative for the time information (RS422 level)	Output negative for the time information (RS422 level)	Receive negative of the RS-485
	8	Input/Output positive for the time information	Unspecified	Input positive for the time information (RS422 level)	Output positive for the time information (RS422 level)	Receive positive of the RS-485
NOTE When this interface is used as a TOD interface, it must be used with a dedicated clock cable. For details, see "Cables" in the Hardware Description.						

Table 1-250 Pins of the ALMI/ALMO interface

Front View	Pin	Usage
	1	Alarm input 1
	2	Grounding end for alarm input 1
	3	Alarm input 2

Front View	Pin	Usage
	4	Alarm input 3
	5	Grounding end for alarm input 3
	6	Grounding end for alarm input 2
	7	Positive for alarm signal output
	8	Negative for alarm signal output

Functions and Features

Table 1-251 Functions and features of the CR8DIPU1T2C5

Functions and Features	Description
Basic function	Line rate: 1200 Gbit/s. Supports switching, control, and clock management. Supports board-level 1:1 backup. Supports the ultra-high-precision clock.
Reliability and availability	Hot swappable. 1:1 backup is supported.

Technical Specifications

Table 1-252 Technical specifications of the CR8DIPU1T2C5

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	34.8 mm x 354.4 mm x 209 mm (1.37 in. x 13.95 in. x 8.23 in.)
Dimensions with packaging (H x W x D) [mm(in.)]	185 mm x 485 mm x 385 mm (7.28 in. x 19.09 in. x 15.16 in.)
Weight without packaging [kg(lb)]	2.55 kg (5.62 lb)
Weight with packaging [kg(lb)]	3.95 kg (8.71 lb)
Typical power consumption [W]	135.6 W
Typical heat dissipation [BTU/hour]	440 BTU/hour

Item	Specification
Long-term operating temperature [°C(°F)]	–20°C to +55°C, startup: ≥ –5°C
Switching capacity	2.4 Tbit/s
CPU	8-core 1.5 GHz
SDRAM	8GB
Storage	4 GB NandFlash

1.4.3.20 CR8DIPU1T2C5 (03034VCU: Integrated Network Processing Unit (IPU-1T2-A))

Overview

Table 1-253 Basic information about the CR8DIPU1T2C5

Item	Details
Description	Integrated Network Processing Unit (IPU-1T2-A)
Part Number	03034VCU
Model	CR8DIPU1T2C5
Card type	IPU
Silkscreen	IPU-1T2-A

Appearance

Figure 1-89 Appearance of the CR8DIPU1T2C5



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find

desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03034VCU&productModel=CR8DIPU1T2C5#matchRelation>

Indicators and Buttons

Table 1-254 Indicators on the CR8DIPU1T2C5

Silkscreen	Name	Color	Status	Description
STAT	Working status indicator	Green	Steady on	The board is working properly.
		Red	Steady on	The board hardware is faulty.
		Orange	Steady on	The interface board is not loaded with logic.
		-	Off	The board is not powered on or is not running.
PROG	Program status indicator	Green	Steady on	The board software is being initialized or is successfully initialized and is running properly.
		Red	Steady on	The memory self-check fails. The board software or logic file is lost. The board software fails to be loaded.
		Green	Blinks on and off at 100 ms intervals	The board software is being loaded.

Silkscreen	Name	Color	Status	Description
		Green	Blinks on and off at 300 ms intervals	The BIOS is booting the upper-layer software.
		Red	Blinks on and off at 100 ms intervals	The BIOS self-check fails.
		-	Off	The board is not powered on.
SYNC	Clock synchronization indicator	Green	Steady on	The system clock works in free-run mode and no priority list is configured for the system clock. (By default, the list contains only internal clock sources.) The system clock is working in tracing mode and is tracing a clock source rather than the internal sources in the priority list. The system clock is working in time synchronization mode, and the PTP time and system clock are in the tracing state.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	The system clock priority list is set. All the clock sources, however, are lost except for the internal clock sources. The clock works in holdover mode or free-run mode. The system clock is working in time synchronization mode, but no synchronization source is available. The system clock and PTP time are working in holdover or free-run mode.
ACTX	Active/standby status indicator for the cross-connect or clock board	Green	Steady on	The board is in the active switching state.
		-	Off	The board is in the standby switching state.
ACTC	Control status indicator	Green	Steady on	The board is in the active control state.
		Green	Blinks on and off at 100 ms intervals	The data of the device is backed up in batch.

Silkscreen	Name	Color	Status	Description
		-	Off	The board is in the standby control state.
ALM	Device alarm indicator	Red	Steady on	A device-level critical alarm is generated.
		Orange	Steady on	There is an equipment-level major or minor alarm.
		-	Off	There is no equipment-level alarm.
ETH/OAM	ETH/OAM indicator	Green	Steady on	The link is up.
		Orange	Blinking	Data is being transmitted or received.
		-	Off	The link is down.

Table 1-255 Buttons on the CR8DIPU1T2C5

Silkscreen	Name	Description
RST	Board reset button	You can press the RST button to reset the NE or restore the factory settings of the NE. The procedure is as follows: To reset the NE, press the RST button and release it. To restore factory settings, press and hold down the RST button. This operation will clear the NE database and system parameter configurations and cannot be restored. Exercise caution when you perform the following operations: 1.Press the RST button to restart the NE. 2.When the PROG indicator blinks (the blinking frequency is about 1 second and the time window is about 15 seconds), press and hold down the RST button. 3.Wait for no longer than 2 minutes till both the STAT and PROG indicators blink (the time window is 5 seconds). Then release the RST button immediately.

Silkscreen	Name	Description
		NOTE 1. When you press the RST button to clear the configuration file, the original configuration file will be cleared. You are advised to periodically back up the configuration file. 2. When you press the RST button to clear the configuration file, to prevent the cleared configuration from being synchronized again between the active and standby main control boards, remove one main control board first in a dual-main control board chassis to ensure that the configuration file is cleared in a single-main control board scenario. Perform the same operations on the other main control board. 3. When pressing the RST button to clear the configuration file, you only need to pay attention to the STAT and PROG indicators. Other indicators may vary depending on the product model.

Ports

Table 1-256 Ports on the CR8DIPU1T2C5

Port	Connector Type	Description	Available Components
CLK/TEST	SMA	Clock test port	Coaxial cable
USB	Type A	USB port (reserved)	-

Port	Connector Type	Description	Available Components
ETH/OAM	RJ45	10M/100M auto-sensing Ethernet NM interface and non-standard RS-232 Console interface. NOTE If a service port can be connected to an NMS, you are advised to use the service port as the network management port.	ETH/OAM management cable
CLK/TOD/RS-485	RJ45	Clock/Time/RS-485 input and output ports RS-485 is a reserved port.	1-into-2 clock cable + clock cable (external clock mode, delivery-ready): applicable to the CLK mode 1-into-2 clock cable + clock cable (external clock/external time mode, prepared onsite): applicable to the CLK or TOD mode 1-into-2 clock cable + Ethernet cable: applicable to the TOD mode
ALMI/ALMO	RJ45	Alarm input/output interface	Alarm input/output cable
HP-GE	SFP	Ultra-high-precision GE port (reserved)	Optical fiber

Table 1-257 Pins of the ETH/OAM interface

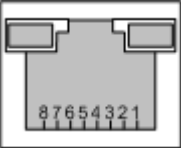
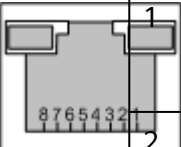
Front View	Pin	Usage
	1	Transmit positive of the NM network port
	2	Transmit negative of the NM network port
	3	Receive positive of the NM network port
	4	Grounding end of the Console interface
	5	Receive end of the Console interface
	6	Receive negative of the NM network port
	7	Unspecified
	8	Transmit end of the Console interface

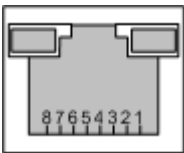
Table 1-258 Pins of the CLK/TOD/RS-485 interface

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	1	Receive negative of the CLK	Receive negative of the CLK	Unspecified	Unspecified	Unspecified
	2	Receive positive of the CLK	Receive positive of the CLK	Unspecified	Unspecified	Unspecified

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	3	Input/Output negative for the 1PPS signal	Unspecified	Input negative for the 1PPS signal (RS422 level)	Output negative for the 1PPS signal (RS422 level)	Transmit negative of the RS-485
	4	Receive negative of the CLK	Receive negative of the CLK	Grounding end	Grounding end	Unspecified
	5	Receive positive of the CLK	Receive positive of the CLK	Grounding end	Grounding end	Unspecified
	6	Input/Output positive for the 1PPS signal	Unspecified	Input positive for the 1PPS signal (RS422 level)	Output positive for the 1PPS signal (RS422 level)	Transmit positive of the RS-485

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	7	Input/Output negative for the time information	Unspecified	Input negative for the time information (RS422 level)	Output negative for the time information (RS422 level)	Receive negative of the RS-485
	8	Input/Output positive for the time information	Unspecified	Input positive for the time information (RS422 level)	Output positive for the time information (RS422 level)	Receive positive of the RS-485
NOTE When this interface is used as a TOD interface, it must be used with a dedicated clock cable. For details, see "Cables" in the Hardware Description.						

Table 1-259 Pins of the ALMI/ALMO interface

Front View	Pin	Usage
	1	Alarm input 1
	2	Grounding end for alarm input 1
	3	Alarm input 2

Front View	Pin	Usage
	4	Alarm input 3
	5	Grounding end for alarm input 3
	6	Grounding end for alarm input 2
	7	Positive for alarm signal output
	8	Negative for alarm signal output

Functions and Features

Table 1-260 Functions and features of the CR8DIPU1T2C5

Functions and Features	Description
Basic function	Line rate: 1200 Gbit/s. Supports switching, control, and clock management. Supports board-level 1:1 backup. Supports the ultra-high-precision clock.
Reliability and availability	Hot swappable. 1:1 backup is supported.

Technical Specifications

Table 1-261 Technical specifications of the CR8DIPU1T2C5

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	34.8 mm x 354.4 mm x 209 mm (1.37 in. x 13.95 in. x 8.23 in.)
Dimensions with packaging (H x W x D) [mm(in.)]	185 mm x 485 mm x 385 mm (7.28 in. x 19.09 in. x 15.16 in.)
Weight without packaging [kg(lb)]	2.55 kg (5.62 lb)
Weight with packaging [kg(lb)]	3.95 kg (8.71 lb)
Typical power consumption [W]	135.6 W
Typical heat dissipation [BTU/hour]	440 BTU/hour

Item	Specification
Long-term operating temperature [°C(°F)]	–20°C to +55°C, startup: ≥ –5°C
Switching capacity	2.4 Tbit/s
CPU	8-core 1.5 GHz
SDRAM	8GB
Storage	4 GB NandFlash

1.4.3.21 CR8DIPU1T2C7 (Integrated Network Processing Unit (IPU-1T2-B))

Overview

Table 1-262 Basic information about the CR8DIPU1T2C7

Item	Details
Description	Integrated Network Processing Unit (IPU-1T2-B)
Part Number	03034BMD
Model	CR8DIPU1T2C7
Card type	IPU
Silkscreen	IPU-1T2-B

Appearance

Figure 1-90 Appearance of the CR8DIPU1T2C7



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03034BMD&productModel=CR8DIPU1T2C7#matchRelation>

Indicators and Buttons

Table 1-263 Indicators on the CR8DIPU1T2C7

Silkscreen	Name	Color	Status	Description
STAT	Working status indicator	Green	Steady on	The board is working normally.
		Red	Steady on	The device generates a hardware alarm. You can run the display alarm hardware command to view the hardware alarm type.
		-	Off	The board is not running or no power is input.
PROG	Program running indicator	Green	Steady on	The board software is being initialized. The board software is normally initialized, and the board software is running normally.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	The memory self-check fails. The board software or the logic file is lost. Loading of the board software fails.
		Green	On for 100 ms and off for 100 ms alternately	Loading of the board software is in process.
		Green	On for 300 ms and off for 300 ms alternately	The BIOS is guiding the upper-layer software.
		Red	On for 100 ms and off for 100 ms alternately	The BIOS self-check fails.
		-	Off	No power is input.

Silkscreen	Name	Color	Status	Description
SYNC	Synchronizati on status indicator	Green	Steady on	The system clock works in free-run mode, and no priority list is set for the system clocks (only internal clock sources are listed by default). The system clock works in tracing mode and is tracing a clock source other than the internal clock sources in the priority list. The system clock works in time synchronizatio n mode, and the PTP time and system clock are being traced.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	The system clock priority list is set, but all clock sources except the internal clock source are lost. The clock works in holdover or free-run mode. The system clock works in time synchronization mode, but no synchronization source is available. The system clock and PTP time work in holdover or free-run mode.
ACTX	Active/standby status indicator for the cross-connect or clock board	Green	Steady on	The board is in the working switching state.
		-	Off	The board is in the protection switching state.
ACTC	Control status indicator	Green	Steady on	The board is in the working control state.
		Green	On for 100 ms and off for 100 ms alternately	The data of the equipment is backed up in batch.

Silkscreen	Name	Color	Status	Description
		-	Off	The board is in the protection control state.
ALM	Alarm indicator	Red	Steady on	Critical alarms are generated.
		Orange	Steady on	Major, minor alarms are generated.
		-	Off	No alarms are generated in the equipment.
ETH/OAM	ETH/OAM indicator	Green	Steady on	If the indicator is steady on, the link is normal.
		Orange	Blink	If the indicator blinks, data is being transmitted.
		-	Off	If the indicator is off, the link is Down.

Table 1-264 Buttons on the CR8DIPU1T2C7

Silkscreen	Name	Description
RST	Reset button	You can press the RST button to reset the NE or restore the factory settings of the NE. The procedure is as follows: To reset the NE, press the RST button and release it. To restore factory settings, press and hold the RST button. This operation will clear the NE database and system parameter configurations and cannot be restored. Exercise caution when you perform the following operations: 1. Press and release the RST button to restart the NE. 2. When the PROG indicator blinks (the blinking frequency is about 1 second and the time window is about 15 seconds), press and hold the RST button. 3. Wait for no longer than 2 minutes till both the STAT and PROG indicators blink (the time window is 5 seconds). Then release the RST button immediately.

Silkscreen	Name	Description
		NOTE 1. When you press the RST button to clear the configuration file, the original configuration file will be cleared. You are advised to periodically back up the configuration file. 2. When you press the RST button to clear the configuration file, to prevent the cleared configuration from being synchronized again between the active and standby main control boards, remove one main control board first in a dual-main control board chassis to ensure that the configuration file is cleared in a single-main control board scenario. Perform the same operations on the other main control board. 3. When pressing the RST button to clear the configuration file, you only need to pay attention to the STAT and PROG indicators. Other indicators may vary depending on the product model.

Ports

Table 1-265 Ports on the CR8DIPU1T2C7

Port	Connector Type	Description	Available Components
ETH/OAM	RJ45	10M/100M autosensing Ethernet network management port or non-RS-232-standard console port. NOTE If a service port can be connected to the NMS, it is recommended that the service port be used as the network management port.	ETH/OAM Management Cables
CLK/TOD/RS-485	RJ45	Clock/Time/RS-485 input/output port, RS-485 is a reserved port.	1-into-2 clock cable + clock cable (external clock mode, delivery-ready): applicable to the CLK mode 1-into-2 clock cable + clock cable (external clock/external time mode, prepared onsite): applicable to the CLK or TOD mode 1-into-2 clock cable + Ethernet cable: applicable to the TOD mode
USB	Type A	USB port (reserved)	-
CLK/TEST	SMA	Clock test port	Coaxial Cable

Port	Connector Type	Description	Available Components
GE0,GE1	SFP	Cascading ports are used for control panel expansion in scenarios like virtual cluster. (reserved)	Optical fiber
HP-GE	SFP	Ultra-high-precision GE port (This first supported version is V800R021C00. This port is reserved in earlier versions.)	1.25Gbps eSFP Optical Module 1.25Gbps eSFP CWDM Optical Module 1.25Gbps eSFP BIDI Optical Module

Table 1-266 Pins of the ETH/OAM interface

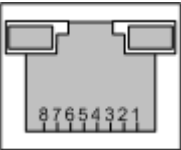
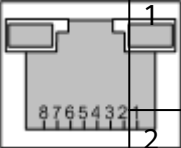
Front View	Pin	Usage
	1	Transmit positive of the NM network port
	2	Transmit negative of the NM network port
	3	Receive positive of the NM network port
	4	Grounding end of the Console interface
	5	Receive end of the Console interface
	6	Receive negative of the NM network port
	7	Unspecified
	8	Transmit end of the Console interface

Table 1-267 Pins of the CLK/TOD/RS-485 interface

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	1	Receive negative of the CLK	Receive negative of the CLK	Unspecified	Unspecified	Unspecified
	2	Receive positive of the CLK	Receive positive of the CLK	Unspecified	Unspecified	Unspecified
	3	Input/Output negative for the 1PPS signal	Unspecified	Input negative for the 1PPS signal (RS422 level)	Output negative for the 1PPS signal (RS422 level)	Transmit negative of the RS-485
	4	Receive negative of the CLK	Receive negative of the CLK	Grounding end	Grounding end	Unspecified
	5	Receive positive of the CLK	Receive positive of the CLK	Grounding end	Grounding end	Unspecified
	6	Input/Output positive for the 1PPS signal	Unspecified	Input positive for the 1PPS signal (RS422 level)	Output positive for the 1PPS signal (RS422 level)	Transmit positive of the RS-485

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	7	Input/Output negative for the time information	Unspecified	Input negative for the time information (RS422 level)	Output negative for the time information (RS422 level)	Receive negative of the RS-485
	8	Input/Output positive for the time information	Unspecified	Input positive for the time information (RS422 level)	Output positive for the time information (RS422 level)	Receive positive of the RS-485
NOTE When this interface is used as a TOD interface, it must be used with a dedicated clock cable. For details, see "Cables" in the Hardware Description.						

Functions and Features

Table 1-268 Functions and features of the CR8DIPU1T2C7

Functions and Features	Description
Basic function	Data forwarding: The IPU is the core of service processing in the entire system, and is connected to all subcards through data channels. Control and management: Through the management channels between IPU and the subcards, the IPU can manage subcards and transmit routing protocol data. The line rate capability is 2.4T bit/s@310Bytes(Bidirectional).
Reliability and availability	Supports the hot swappable function. Supports the board-level 1:1 backup function.
Restrictions and remarks	1. It cannot be used together with other IPU. 2. When working with the 03033VMC board to support FlexE services, the board supports a maximum of eight FlexE physical interfaces.
Breakout supported	No
Interface rate auto-sensing supported	No

Technical Specifications

Table 1-269 Technical specifications of the CR8DIPU1T2C7

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	34.8 mm x 354.4 mm x 209 mm (1.37 in. x 13.95 in. x 8.23 in.)
Dimensions with packaging (H x W x D) [mm(in.)]	185 mm x 485 mm x 385 mm (7.28 in. x 19.09 in. x 15.16 in.)
Weight without packaging [kg(lb)]	4 kg (8.82 lb)
Weight with packaging [kg(lb)]	5.4 kg (11.9 lb)
Typical power consumption [W]	315.5 W
Typical heat dissipation [BTU/hour]	1023.58 BTU/hour

Item	Specification
Long-term operating temperature [°C(°F)]	–20°C to +55°C (–4°F to +131°F); start: ≥ –5°C (23°F)
CPU	20-core 2.0 GHz
SDRAM	16GB
Storage	4GB NandFlash

1.4.3.22 CR8DIPU1T2C9 (03034BMF: Integrated Network Processing Unit (IPU-1T2-BN))

Overview

Table 1-270 Basic information about the CR8DIPU1T2C9

Item	Details
Description	Integrated Network Processing Unit (IPU-1T2-BN)
Part Number	03034BMF
Model	CR8DIPU1T2C9
Card type	IPU
Silkscreen	IPU-1T2-BN

Appearance

Figure 1-91 Appearance of the CR8DIPU1T2C9



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03034BMF&productModel=CR8DIPU1T2C9#matchRelation>

Indicators and Buttons

Table 1-271 Indicators on the CR8DIPU1T2C9

Silkscreen	Name	Color	Status	Description
STAT	Working Status Indicator	Green	Steady on	The board is working properly.
		Red	Steady on	A board hardware fault has occurred.
		Orange	Steady on	The interface board is not loaded with the logic.
		-	Off	The board is not powered on or the board is not started.
PROG	Program Status Indicator	Green	Steady on	Board software is being initialized. Board software has been initialized, or board software is running properly.
		Red	Steady on	The memory self-check fails. Board software or a logic file is lost. Loading of the board software fails.

Silkscreen	Name	Color	Status	Description
		Green	On for 100 ms and off for 100 ms alternately	Board software is being loaded.
		Green	On for 300 ms and off for 300 ms alternately	The BIOS is guiding upper-layer software.
		Red	On for 100 ms and off for 100 ms alternately	The BIOS fails self-check.
		-	Off	The board is not powered on.

Silkscreen	Name	Color	Status	Description
SYNC	Synchronizati on Status Indicator	Green	Steady on	The system clock works in free-run mode and no priority list is configured for the system clock. (By default, the list contains only internal clock sources.) The system clock is working in tracing mode and is tracing a clock source rather than the internal sources in the priority list. The system clock is working in time synchronizatio n mode, and the PTP time and system clock are in the tracing state.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	The system clock priority list is set. All the clock sources, however, are lost except for the internal clock sources. The clock works in holdover mode or free-run mode. The system clock is working in time synchronization mode, but no synchronization source is available. The system clock and PTP time are working in holdover or free-run mode.
ACTX	Switching Status Indicator	Green	Steady on	The board is in the working switching state.
		-	Off	The board is in the protection switching state.
ACTC	Control Status Indicator	Green	Steady on	The board is in the working control state.

Silkscreen	Name	Color	Status	Description
		Green	On for 100 ms and off for 100 ms alternately	he data in the database of the device is backed up in batch.
		-	Off	The board is in the protection control state.
ALM	Alarm Indicator	Red	Steady on	A critical alarm is generated on the device.
		Orange	Steady on	A major alarm or a minor alarm is generated on the device.
		-	Off	No alarm is generated on the device.
ETH/OAM	ETH/OAM indicator	Green	Steady on	If the indicator is steady on, the link is normal.
		Orange	Blink	If the indicator blinks, data is being transmitted.
		-	Off	If the indicator is off, the link is Down.

Table 1-272 Buttons on the CR8DIPU1T2C9

Silkscreen	Name	Description
RST	RST	You can press the RST button to reset the NE or restore the factory settings of the NE. The procedure is as follows: To reset the NE, press the RST button and release it. To restore factory settings, press and hold the RST button. This operation will clear the NE database and system parameter configurations and cannot be restored. Exercise caution when you perform the following operations: 1. Press and release the RST button to restart the NE. 2. When the PROG indicator blinks (the blinking frequency is about 1 second and the time window is about 15 seconds), press and hold the RST button. 3. Wait for no longer than 2 minutes till both the STAT and PROG indicators blink (the time window is 5 seconds). Then release the RST button immediately.

Silkscreen	Name	Description
		NOTE 1. When you press the RST button to clear the configuration file, the original configuration file will be cleared. You are advised to periodically back up the configuration file. 2. When you press the RST button to clear the configuration file, to prevent the cleared configuration from being synchronized again between the active and standby main control boards, remove one main control board first in a dual-main control board chassis to ensure that the configuration file is cleared in a single-main control board scenario. Perform the same operations on the other main control board. 3. When pressing the RST button to clear the configuration file, you only need to pay attention to the STAT and PROG indicators. Other indicators may vary depending on the product model.

Ports

Table 1-273 Ports on the CR8DIPU1T2C9

Port	Connector Type	Description	Available Components
CLK/TEST	SMA	Clock test port	Coaxial Cable
USB	Type A	USB port (reserved)	-

Port	Connector Type	Description	Available Components
ETH/OAM	RJ45	10M/100M autosensing Ethernet network management port or non-RS-232-standard console port. NOTE If a service port can be connected to the NMS, it is recommended that the service port be used as the network management port.	ETH/OAM Management Cables
CLK/TOD/RS-485	RJ45	Clock/Time/RS-485 input/output port, RS-485 is a reserved port.	1-into-2 clock cable + clock cable (external clock mode, delivery-ready): applicable to the CLK mode 1-into-2 clock cable + clock cable (external clock/external time mode, prepared onsite): applicable to the CLK or TOD mode 1-into-2 clock cable + Ethernet cable: applicable to the TOD mode
ALMI/ALMO	RJ45	Alarm input/output port	Alarm Input/Output Cables
HP-GE	SFP	Ultra-high-precision GE port (reserved)	Optical fiber

Table 1-274 Pins of the ETH/OAM interface

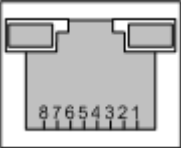
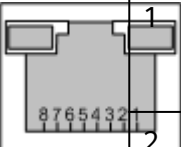
Front View	Pin	Usage
	1	Transmit positive of the NM network port
	2	Transmit negative of the NM network port
	3	Receive positive of the NM network port
	4	Grounding end of the Console interface
	5	Receive end of the Console interface
	6	Receive negative of the NM network port
	7	Unspecified
	8	Transmit end of the Console interface

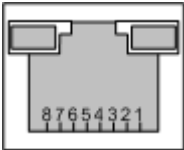
Table 1-275 Pins of the CLK/TOD/RS-485 interface

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	1	Receive negative of the CLK	Receive negative of the CLK	Unspecified	Unspecified	Unspecified
	2	Receive positive of the CLK	Receive positive of the CLK	Unspecified	Unspecified	Unspecified

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	3	Input/Output negative for the 1PPS signal	Unspecified	Input negative for the 1PPS signal (RS422 level)	Output negative for the 1PPS signal (RS422 level)	Transmit negative of the RS-485
	4	Receive negative of the CLK	Receive negative of the CLK	Grounding end	Grounding end	Unspecified
	5	Receive positive of the CLK	Receive positive of the CLK	Grounding end	Grounding end	Unspecified
	6	Input/Output positive for the 1PPS signal	Unspecified	Input positive for the 1PPS signal (RS422 level)	Output positive for the 1PPS signal (RS422 level)	Transmit positive of the RS-485

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	7	Input/Output negative for the time information	Unspecified	Input negative for the time information (RS422 level)	Output negative for the time information (RS422 level)	Receive negative of the RS-485
	8	Input/Output positive for the time information	Unspecified	Input positive for the time information (RS422 level)	Output positive for the time information (RS422 level)	Receive positive of the RS-485
NOTE When this interface is used as a TOD interface, it must be used with a dedicated clock cable. For details, see "Cables" in the Hardware Description.						

Table 1-276 Pins of the ALMI/ALMO interface

Front View	Pin	Usage
	1	Alarm input 1
	2	Grounding end for alarm input 1
	3	Alarm input 2

Front View	Pin	Usage
	4	Alarm input 3
	5	Grounding end for alarm input 3
	6	Grounding end for alarm input 2
	7	Positive for alarm signal output
	8	Negative for alarm signal output

Functions and Features

Table 1-277 Functions and features of the CR8DIPU1T2C9

Functions and Features	Description
Basic function	Line-Rate capability: 1200 Gbit/s. Supports switching, control, and clock management. Supports the board-level 1:1 backup function. Supports ultra-high-precision clock.
Reliability and availability	Supports the hot swappable function. Supports the board-level 1:1 backup function.

Technical Specifications

Table 1-278 Technical specifications of the CR8DIPU1T2C9

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	34.8 mm x 354.4 mm x 209 mm (1.37 in. x 13.95 in. x 8.23 in.)
Dimensions with packaging (H x W x D) [mm(in.)]	185 mm x 485 mm x 385 mm (7.28 in. x 19.09 in. x 15.16 in.)
Weight without packaging [kg(lb)]	2.55 kg (5.62 lb)
Weight with packaging [kg(lb)]	3.95 kg (8.71 lb)
Typical power consumption [W]	135.6 W

Item	Specification
Typical heat dissipation [BTU/hour]	440 BTU/hour
Long-term operating temperature [°C(°F)]	-20°C to +55°C (-4°F to +131°F); start: ≥ -5°C (23°F)
Switching capacity	2.4 Tbit/s
CPU	8-core 1.5 GHz
SDRAM	8GB
Storage	4GB NandFlash

1.4.3.23 CR8DIPU1T2C9 (03034BMF-001: Integrated Network Processing Unit (IPU-1T2-BN))

Overview

Table 1-279 Basic information about the CR8DIPU1T2C9

Item	Details
Description	Integrated Network Processing Unit (IPU-1T2-BN)
Part Number	03034BMF-001
Model	CR8DIPU1T2C9
Card type	IPU
Silkscreen	IPU-1T2-BN

Appearance

Figure 1-92 Appearance of the CR8DIPU1T2C9



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find

desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03034BMF-001&productModel=CR8DIPU1T2C9#matchRelation>

Indicators and Buttons

Table 1-280 Indicators on the CR8DIPU1T2C9

Silkscreen	Name	Color	Status	Description
STAT	Working Status Indicator	Green	Steady on	The board is working properly.
		Red	Steady on	A board hardware fault has occurred.
		Orange	Steady on	The interface board is not loaded with the logic.
		-	Off	The board is not powered on or the board is not started.
PROG	Program Status Indicator	Green	Steady on	Board software is being initialized. Board software has been initialized, or board software is running properly.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	The memory self-check fails. Board software or a logic file is lost. Loading of the board software fails.
		Green	On for 100 ms and off for 100 ms alternately	Board software is being loaded.
		Green	On for 300 ms and off for 300 ms alternately	The BIOS is guiding upper-layer software.
		Red	On for 100 ms and off for 100 ms alternately	The BIOS fails self-check.
		-	Off	The board is not powered on.

Silkscreen	Name	Color	Status	Description
SYNC	Synchronizati on Status Indicator	Green	Steady on	The system clock works in free-run mode and no priority list is configured for the system clock. (By default, the list contains only internal clock sources.) The system clock is working in tracing mode and is tracing a clock source rather than the internal sources in the priority list. The system clock is working in time synchronization mode, and the PTP time and system clock are in the tracing state.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	The system clock priority list is set. All the clock sources, however, are lost except for the internal clock sources. The clock works in holdover mode or free-run mode. The system clock is working in time synchronization mode, but no synchronization source is available. The system clock and PTP time are working in holdover or free-run mode.
ACTX	Switching Status Indicator	Green	Steady on	The board is in the working switching state.
		-	Off	The board is in the protection switching state.
ACTC	Control Status Indicator	Green	Steady on	The board is in the working control state.

Silkscreen	Name	Color	Status	Description
		Green	On for 100 ms and off for 100 ms alternately	he data in the database of the device is backed up in batch.
		-	Off	The board is in the protection control state.
ALM	Alarm Indicator	Red	Steady on	A critical alarm is generated on the device.
		Orange	Steady on	A major alarm or a minor alarm is generated on the device.
		-	Off	No alarm is generated on the device.
ETH/OAM	ETH/OAM indicator	Green	Steady on	If the indicator is steady on, the link is normal.
		Orange	Blink	If the indicator blinks, data is being transmitted.
		-	Off	If the indicator is off, the link is Down.

Table 1-281 Buttons on the CR8DIPU1T2C9

Silkscreen	Name	Description
RST	RST	You can press the RST button to reset the NE or restore the factory settings of the NE. The procedure is as follows: To reset the NE, press the RST button and release it. To restore factory settings, press and hold the RST button. This operation will clear the NE database and system parameter configurations and cannot be restored. Exercise caution when you perform the following operations: 1. Press and release the RST button to restart the NE. 2. When the PROG indicator blinks (the blinking frequency is about 1 second and the time window is about 15 seconds), press and hold the RST button. 3. Wait for no longer than 2 minutes till both the STAT and PROG indicators blink (the time window is 5 seconds). Then release the RST button immediately.

Silkscreen	Name	Description
		NOTE 1. When you press the RST button to clear the configuration file, the original configuration file will be cleared. You are advised to periodically back up the configuration file. 2. When you press the RST button to clear the configuration file, to prevent the cleared configuration from being synchronized again between the active and standby main control boards, remove one main control board first in a dual-main control board chassis to ensure that the configuration file is cleared in a single-main control board scenario. Perform the same operations on the other main control board. 3. When pressing the RST button to clear the configuration file, you only need to pay attention to the STAT and PROG indicators. Other indicators may vary depending on the product model.

Ports

Table 1-282 Ports on the CR8DIPU1T2C9

Port	Connector Type	Description	Available Components
CLK/TEST	SMA	Clock test port	Coaxial Cable
USB	Type A	USB port (reserved)	-

Port	Connector Type	Description	Available Components
ETH/OAM	RJ45	10M/100M autosensing Ethernet network management port or non-RS-232-standard console port. NOTE If a service port can be connected to the NMS, it is recommended that the service port be used as the network management port.	ETH/OAM Management Cables
CLK/TOD/RS-485	RJ45	Clock/Time/RS-485 input/output port, RS-485 is a reserved port.	1-into-2 clock cable + clock cable (external clock mode, delivery-ready): applicable to the CLK mode 1-into-2 clock cable + clock cable (external clock/external time mode, prepared onsite): applicable to the CLK or TOD mode 1-into-2 clock cable + Ethernet cable: applicable to the TOD mode
ALMI/ALMO	RJ45	Alarm input/output port	Alarm Input/Output Cables
HP-GE	SFP	Ultra-high-precision GE port (reserved)	Optical fiber

Table 1-283 Pins of the ETH/OAM interface

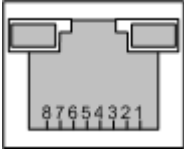
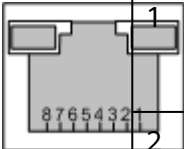
Front View	Pin	Usage
	1	Transmit positive of the NM network port
	2	Transmit negative of the NM network port
	3	Receive positive of the NM network port
	4	Grounding end of the Console interface
	5	Receive end of the Console interface
	6	Receive negative of the NM network port
	7	Unspecified
	8	Transmit end of the Console interface

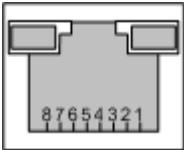
Table 1-284 Pins of the CLK/TOD/RS-485 interface

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	1	Receive negative of the CLK	Receive negative of the CLK	Unspecified	Unspecified	Unspecified
	2	Receive positive of the CLK	Receive positive of the CLK	Unspecified	Unspecified	Unspecified

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	3	Input/Output negative for the 1PPS signal	Unspecified	Input negative for the 1PPS signal (RS422 level)	Output negative for the 1PPS signal (RS422 level)	Transmit negative of the RS-485
	4	Receive negative of the CLK	Receive negative of the CLK	Grounding end	Grounding end	Unspecified
	5	Receive positive of the CLK	Receive positive of the CLK	Grounding end	Grounding end	Unspecified
	6	Input/Output positive for the 1PPS signal	Unspecified	Input positive for the 1PPS signal (RS422 level)	Output positive for the 1PPS signal (RS422 level)	Transmit positive of the RS-485

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	7	Input/Output negative for the time information	Unspecified	Input negative for the time information (RS422 level)	Output negative for the time information (RS422 level)	Receive negative of the RS-485
	8	Input/Output positive for the time information	Unspecified	Input positive for the time information (RS422 level)	Output positive for the time information (RS422 level)	Receive positive of the RS-485
NOTE When this interface is used as a TOD interface, it must be used with a dedicated clock cable. For details, see "Cables" in the Hardware Description.						

Table 1-285 Pins of the ALMI/ALMO interface

Front View	Pin	Usage
	1	Alarm input 1
	2	Grounding end for alarm input 1
	3	Alarm input 2

Front View	Pin	Usage
	4	Alarm input 3
	5	Grounding end for alarm input 3
	6	Grounding end for alarm input 2
	7	Positive for alarm signal output
	8	Negative for alarm signal output

Functions and Features

Table 1-286 Functions and features of the CR8DIPU1T2C9

Functions and Features	Description
Basic function	Line-Rate capability: 1200 Gbit/s. Supports switching, control, and clock management. Supports the board-level 1:1 backup function. Supports ultra-high-precision clock.
Reliability and availability	Supports the hot swappable function. Supports the board-level 1:1 backup function.

Technical Specifications

Table 1-287 Technical specifications of the CR8DIPU1T2C9

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	34.8 mm x 354.4 mm x 209 mm (1.37 in. x 13.95 in. x 8.23 in.)
Dimensions with packaging (H x W x D) [mm(in.)]	185 mm x 485 mm x 385 mm (7.28 in. x 19.09 in. x 15.16 in.)
Weight without packaging [kg(lb)]	2.55 kg (5.62 lb)
Weight with packaging [kg(lb)]	3.95 kg (8.71 lb)
Typical power consumption [W]	135.6 W

Item	Specification
Typical heat dissipation [BTU/hour]	440 BTU/hour
Long-term operating temperature [°C(°F)]	-20°C to +55°C (-4°F to +131°F); start: ≥ -5°C (23°F)
Switching capacity	2.4 Tbit/s
CPU	8-core 1.5 GHz
SDRAM	8GB
Storage	4GB NandFlash

1.4.3.24 CR8DIPU1T2CB (Integrated Network Processing Unit (IPU-1T2-B))

Overview

Table 1-288 Basic information about the CR8DIPU1T2CB

Item	Details
Description	Integrated Network Processing Unit (IPU-1T2-B)
Part Number	03034BMH
Model	CR8DIPU1T2CB
Card type	IPU
Silkscreen	IPU-1T2-B

Appearance

Figure 1-93 Appearance of the CR8DIPU1T2CB



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find

desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03034BMH&productModel=CR8DIPU1T2CB#matchRelation>

Indicators and Buttons

Table 1-289 Indicators on the CR8DIPU1T2CB

Silkscreen	Name	Color	Status	Description
STAT	Working status indicator	Green	Steady on	The board is working normally.
		Red	Steady on	The device generates a hardware alarm. You can run the display alarm hardware command to view the hardware alarm type.
		-	Off	The board is not running or no power is input.
PROG	Program running indicator	Green	Steady on	The board software is being initialized. The board software is normally initialized, and the board software is running normally.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	The memory self-check fails. The board software or the logic file is lost. Loading of the board software fails.
		Green	On for 100 ms and off for 100 ms alternately	Loading of the board software is in process.
		Green	On for 300 ms and off for 300 ms alternately	The BIOS is guiding the upper-layer software.
		Red	On for 100 ms and off for 100 ms alternately	The BIOS self-check fails.
		-	Off	No power is input.

Silkscreen	Name	Color	Status	Description
SYNC	Synchronizati on status indicator	Green	Steady on	The system clock works in free-run mode, and no priority list is set for the system clocks (only internal clock sources are listed by default). The system clock works in tracing mode and is tracing a clock source other than the internal clock sources in the priority list. The system clock works in time synchronizatio n mode, and the PTP time and system clock are being traced.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	The system clock priority list is set, but all clock sources except the internal clock source are lost. The clock works in holdover or free-run mode. The system clock works in time synchronization mode, but no synchronization source is available. The system clock and PTP time work in holdover or free-run mode.
ACTX	Active/standby status indicator for the cross-connect or clock board	Green	Steady on	The board is in the working switching state.
		-	Off	The board is in the protection switching state.
ACTC	Control status indicator	Green	Steady on	The board is in the working control state.
		Green	On for 100 ms and off for 100 ms alternately	The data of the equipment is backed up in batch.

Silkscreen	Name	Color	Status	Description
		-	Off	The board is in the protection control state.
ALM	Alarm indicator	Red	Steady on	Critical alarms are generated.
		Orange	Steady on	Major, minor alarms are generated.
		-	Off	No alarms are generated in the equipment.
ETH/OAM	ETH/OAM indicator	Green	Steady on	If the indicator is steady on, the link is normal.
		Orange	Blink	If the indicator blinks, data is being transmitted.
		-	Off	If the indicator is off, the link is Down.

Table 1-290 Buttons on the CR8DIPU1T2CB

Silkscreen	Name	Description
RST	Reset button	You can press the RST button to reset the NE or restore the factory settings of the NE. The procedure is as follows: To reset the NE, press the RST button and release it. To restore factory settings, press and hold the RST button. This operation will clear the NE database and system parameter configurations and cannot be restored. Exercise caution when you perform the following operations: 1. Press and release the RST button to restart the NE. 2. When the PROG indicator blinks (the blinking frequency is about 1 second and the time window is about 15 seconds), press and hold the RST button. 3. Wait for no longer than 2 minutes till both the STAT and PROG indicators blink (the time window is 5 seconds). Then release the RST button immediately.

Silkscreen	Name	Description
		NOTE 1. When you press the RST button to clear the configuration file, the original configuration file will be cleared. You are advised to periodically back up the configuration file. 2. When you press the RST button to clear the configuration file, to prevent the cleared configuration from being synchronized again between the active and standby main control boards, remove one main control board first in a dual-main control board chassis to ensure that the configuration file is cleared in a single-main control board scenario. Perform the same operations on the other main control board. 3. When pressing the RST button to clear the configuration file, you only need to pay attention to the STAT and PROG indicators. Other indicators may vary depending on the product model.

Ports

Table 1-291 Ports on the CR8DIPU1T2CB

Port	Connector Type	Description	Available Components
ETH/OAM	RJ45	10M/100M autosensing Ethernet network management port or non-RS-232-standard console port. NOTE If a service port can be connected to the NMS, it is recommended that the service port be used as the network management port.	ETH/OAM Management Cables
CLK/TOD/RS-485	RJ45	Clock/Time/RS-485 input/output port, RS-485 is a reserved port.	1-into-2 clock cable + clock cable (external clock mode, delivery-ready): applicable to the CLK mode 1-into-2 clock cable + clock cable (external clock/external time mode, prepared onsite): applicable to the CLK or TOD mode 1-into-2 clock cable + Ethernet cable: applicable to the TOD mode
ALMI/ALMO	RJ45	Alarm input/output port	Alarm Input/Output Cables
USB	Type A	USB port (reserved)	-
CLK/TEST	SMA	Clock test port	Coaxial Cable

Table 1-292 Pins of the ETH/OAM interface

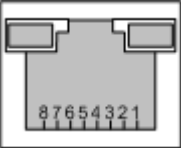
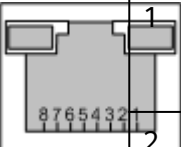
Front View	Pin	Usage
	1	Transmit positive of the NM network port
	2	Transmit negative of the NM network port
	3	Receive positive of the NM network port
	4	Grounding end of the Console interface
	5	Receive end of the Console interface
	6	Receive negative of the NM network port
	7	Unspecified
	8	Transmit end of the Console interface

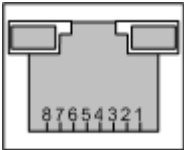
Table 1-293 Pins of the CLK/TOD/RS-485 interface

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	1	Receive negative of the CLK	Receive negative of the CLK	Unspecified	Unspecified	Unspecified
	2	Receive positive of the CLK	Receive positive of the CLK	Unspecified	Unspecified	Unspecified

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	3	Input/Output negative for the 1PPS signal	Unspecified	Input negative for the 1PPS signal (RS422 level)	Output negative for the 1PPS signal (RS422 level)	Transmit negative of the RS-485
	4	Receive negative of the CLK	Receive negative of the CLK	Grounding end	Grounding end	Unspecified
	5	Receive positive of the CLK	Receive positive of the CLK	Grounding end	Grounding end	Unspecified
	6	Input/Output positive for the 1PPS signal	Unspecified	Input positive for the 1PPS signal (RS422 level)	Output positive for the 1PPS signal (RS422 level)	Transmit positive of the RS-485

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	7	Input/Output negative for the time information	Unspecified	Input negative for the time information (RS422 level)	Output negative for the time information (RS422 level)	Receive negative of the RS-485
	8	Input/Output positive for the time information	Unspecified	Input positive for the time information (RS422 level)	Output positive for the time information (RS422 level)	Receive positive of the RS-485
NOTE When this interface is used as a TOD interface, it must be used with a dedicated clock cable. For details, see "Cables" in the Hardware Description.						

Table 1-294 Pins of the ALMI/ALMO interface

Front View	Pin	Usage
	1	Alarm input 1
	2	Grounding end for alarm input 1
	3	Alarm input 2

Front View	Pin	Usage
	4	Alarm input 3
	5	Grounding end for alarm input 3
	6	Grounding end for alarm input 2
	7	Positive for alarm signal output
	8	Negative for alarm signal output

Functions and Features

Table 1-295 Functions and features of the CR8DIPU1T2CB

Functions and Features	Description
Basic function	Data forwarding: The IPU is the core of service processing in the entire system, and is connected to all subcards through data channels. Control and management: Through the management channels between IPU and the subcards, the IPU can manage subcards and transmit routing protocol data. The line rate capability is 2.4T bit/s@310Bytes(Bidirectional).
Reliability and availability	Supports the hot swappable function. Supports the board-level 1:1 backup function.
Restrictions and remarks	This board cannot be used together with other IPU.
Breakout supported	No
Interface rate auto-sensing supported	No

Technical Specifications

Table 1-296 Technical specifications of the CR8DIPU1T2CB

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	22.4 mm x 388.4 mm x 209.3 mm (0.88 in. x 15.29 in. x 8.24 in.)
Dimensions with packaging (H x W x D) [mm(in.)]	170 mm x 560 mm x 385 mm (6.69 in. x 22.05 in. x 15.16 in.)
Weight without packaging [kg(lb)]	2.3 kg (5.07 lb)
Weight with packaging [kg(lb)]	3.7 kg (8.16 lb)
Typical power consumption [W]	144.2 W
Typical heat dissipation [BTU/hour]	467.94 BTU/hour
Long-term operating temperature [°C(°F)]	–40°C to +65°C (–40°F to +149°F), start: ≥ –20°C (–4°F)
CPU	8-core 1.5 GHz
SDRAM	8GB
Storage	4GB NandFlash

1.4.3.25 CR8DIPU1T2CE (Integrated Network Processing Unit (IPU-1T2-CN))

Overview

Table 1-297 Basic information about the CR8DIPU1T2CE

Item	Details
Description	Integrated Network Processing Unit (IPU-1T2-CN)
Part Number	03034NWT
Model	CR8DIPU1T2CE
Card type	IPU
Silkscreen	IPU-1T2-CN

Appearance

Figure 1-94 Appearance of the CR8DIPU1T2CE



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03034NWT&productModel=CR8DIPU1T2CE#matchRelation>

Indicators and Buttons

Table 1-298 Indicators on the CR8DIPU1T2CE

Silkscreen	Name	Color	Status	Description
STAT	Working status indicator	Green	Steady on	The board is working properly.
		Red	Steady on	The board hardware is faulty.
		Orange	Steady on	The interface board is not loaded with logic.
		-	Off	The board is not powered on or is not running.

Silkscreen	Name	Color	Status	Description
PROG	Program status indicator	Green	Steady on	The board software is being initialized or is successfully initialized and is running properly.
		Red	Steady on	The memory self-check fails. The board software or logic file is lost. The board software fails to be loaded.
		Green	Blinks on and off at 100 ms intervals	The board software is being loaded.
		Green	Blinks on and off at 300 ms intervals	The BIOS is booting the upper-layer software.
		Red	Blinks on and off at 100 ms intervals	The BIOS self-check fails.
		-	Off	The board is not powered on.

Silkscreen	Name	Color	Status	Description
SYNC	Clock synchronizatio n indicator	Green	Steady on	The system clock works in free-run mode and no priority list is configured for the system clock. (By default, the list contains only internal clock sources.) The system clock is working in tracing mode and is tracing a clock source rather than the internal sources in the priority list. The system clock is working in time synchronization mode, and the PTP time and system clock are in the tracing state.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	The system clock priority list is set. All the clock sources, however, are lost except for the internal clock sources. The clock works in holdover mode or free-run mode. The system clock is working in time synchronization mode, but no synchronization source is available. The system clock and PTP time are working in holdover or free-run mode.
ACTX	Active/standby status indicator for the cross-connect or clock board	Green	Steady on	The board is in the active switching state.
		-	Off	The board is in the standby switching state.
ACTC	Control status indicator	Green	Steady on	The board is in the active control state.
		Green	Blinks on and off at 100 ms intervals	The data of the device is backed up in batch.

Silkscreen	Name	Color	Status	Description
		-	Off	The board is in the standby control state.
ALM	Device alarm indicator	Red	Steady on	A device-level critical alarm is generated.
		Orange	Steady on	There is an equipment-level major or minor alarm.
		-	Off	There is no equipment-level alarm.
ETH/OAM	ETH/OAM indicator	Green	Steady on	The link is up.
		Orange	Blinking	Data is being transmitted or received.
		-	Off	The link is down.

Table 1-299 Buttons on the CR8DIPU1T2CE

Silkscreen	Name	Description
RST	Board reset button	You can press the RST button to reset the NE or restore the factory settings of the NE. The procedure is as follows: To reset the NE, press the RST button and release it. To restore factory settings, press and hold down the RST button. This operation will clear the NE database and system parameter configurations and cannot be restored. Exercise caution when you perform the following operations: 1.Press the RST button to restart the NE. 2.When the PROG indicator blinks (the blinking frequency is about 1 second and the time window is about 15 seconds), press and hold down the RST button. 3.Wait for no longer than 2 minutes till both the STAT and PROG indicators blink (the time window is 5 seconds). Then release the RST button immediately.

Silkscreen	Name	Description
		NOTE 1. When you press the RST button to clear the configuration file, the original configuration file will be cleared. You are advised to periodically back up the configuration file. 2. When you press the RST button to clear the configuration file, to prevent the cleared configuration from being synchronized again between the active and standby main control boards, remove one main control board first in a dual-main control board chassis to ensure that the configuration file is cleared in a single-main control board scenario. Perform the same operations on the other main control board. 3. When pressing the RST button to clear the configuration file, you only need to pay attention to the STAT and PROG indicators. Other indicators may vary depending on the product model.

Ports

Table 1-300 Ports on the CR8DIPU1T2CE

Port	Connector Type	Description	Available Components
CLK/TEST	SMA	Clock test port	Coaxial cable
USB	Type A	USB port (reserved)	-

Port	Connector Type	Description	Available Components
ETH/OAM	RJ45	10M/100M auto-sensing Ethernet NM interface and non-standard RS-232 Console interface. NOTE If a service port can be connected to an NMS, you are advised to use the service port as the network management port.	ETH/OAM management cable
CLK/TOD/RS-485	RJ45	Clock/Time/RS-485 input and output ports RS-485 is a reserved port.	1-into-2 clock cable + clock cable (external clock mode, delivery-ready): applicable to the CLK mode 1-into-2 clock cable + clock cable (external clock/external time mode, prepared onsite): applicable to the CLK or TOD mode 1-into-2 clock cable + Ethernet cable: applicable to the TOD mode
ALMI/ALMO	RJ45	Alarm input/output interface	Alarm input/output cable
HP-GE	SFP	Ultra-high-precision GE port (reserved)	Optical fiber

Table 1-301 Pins of the ETH/OAM interface

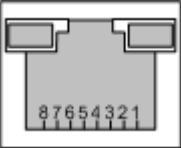
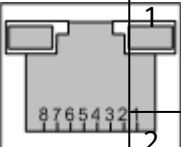
Front View	Pin	Usage
	1	Transmit positive of the NM network port
	2	Transmit negative of the NM network port
	3	Receive positive of the NM network port
	4	Grounding end of the Console interface
	5	Receive end of the Console interface
	6	Receive negative of the NM network port
	7	Unspecified
	8	Transmit end of the Console interface

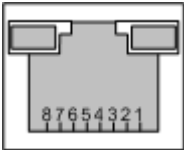
Table 1-302 Pins of the CLK/TOD/RS-485 interface

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	1	Receive negative of the CLK	Receive negative of the CLK	Unspecified	Unspecified	Unspecified
	2	Receive positive of the CLK	Receive positive of the CLK	Unspecified	Unspecified	Unspecified

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	3	Input/Output negative for the 1PPS signal	Unspecified	Input negative for the 1PPS signal (RS422 level)	Output negative for the 1PPS signal (RS422 level)	Transmit negative of the RS-485
	4	Receive negative of the CLK	Receive negative of the CLK	Grounding end	Grounding end	Unspecified
	5	Receive positive of the CLK	Receive positive of the CLK	Grounding end	Grounding end	Unspecified
	6	Input/Output positive for the 1PPS signal	Unspecified	Input positive for the 1PPS signal (RS422 level)	Output positive for the 1PPS signal (RS422 level)	Transmit positive of the RS-485

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	7	Input/Output negative for the time information	Unspecified	Input negative for the time information (RS422 level)	Output negative for the time information (RS422 level)	Receive negative of the RS-485
	8	Input/Output positive for the time information	Unspecified	Input positive for the time information (RS422 level)	Output positive for the time information (RS422 level)	Receive positive of the RS-485
NOTE When this interface is used as a TOD interface, it must be used with a dedicated clock cable. For details, see "Cables" in the Hardware Description.						

Table 1-303 Pins of the ALMI/ALMO interface

Front View	Pin	Usage
	1	Alarm input 1
	2	Grounding end for alarm input 1
	3	Alarm input 2

Front View	Pin	Usage
	4	Alarm input 3
	5	Grounding end for alarm input 3
	6	Grounding end for alarm input 2
	7	Positive for alarm signal output
	8	Negative for alarm signal output

Functions and Features

Table 1-304 Functions and features of the CR8DIPU1T2CE

Functions and Features	Description
Basic function	Line rate: 1200 Gbit/s. Supports switching, control, and clock management. Supports board-level 1:1 backup. Supports the ultra-high-precision clock.
Reliability and availability	Hot swappable. 1:1 backup is supported.

Technical Specifications

Table 1-305 Technical specifications of the CR8DIPU1T2CE

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	34.8 mm x 354.4 mm x 209 mm (1.37 in. x 13.95 in. x 8.23 in.)
Dimensions with packaging (H x W x D) [mm(in.)]	185 mm x 485 mm x 385 mm (7.28 in. x 19.09 in. x 15.16 in.)
Weight without packaging [kg(lb)]	2.55 kg (5.62 lb)
Weight with packaging [kg(lb)]	3.95 kg (8.71 lb)
Typical power consumption [W]	135.6 W
Typical heat dissipation [BTU/hour]	440 BTU/hour

Item	Specification
Long-term operating temperature [°C(°F)]	–20°C to +55°C, startup: ≥ –5°C
Switching capacity	2.4 Tbit/s
CPU	8-core 1.5 GHz
SDRAM	8GB
Storage	4 GB NandFlash

1.4.3.26 CR8DIPU1T2CG (Integrated Network Processing Unit (IPU-1T2-C))

Overview

Table 1-306 Basic information about the CR8DIPU1T2CG

Item	Details
Description	Integrated Network Processing Unit (IPU-1T2-C)
Part Number	03034NWX
Model	CR8DIPU1T2CG
Card type	IPU
Silkscreen	IPU-1T2-C

Appearance

Figure 1-95 Appearance of the CR8DIPU1T2CG



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03034NWY&productModel=CR8DIPU1T2CG#matchRelation>

Indicators and Buttons

Table 1-307 Indicators on the CR8DIPU1T2CG

Silkscreen	Name	Color	Status	Description
STAT	Working status indicator	Green	Steady on	The board is working properly.
		Red	Steady on	A hardware alarm is generated. You can run the display alarm hardware command to check the hardware alarm type.
		-	Off	The board is not powered on or is not running.
PROG	Program status indicator	Green	Steady on	The board software is being initialized or is successfully initialized and is running properly.
		Red	Steady on	The memory self-check fails. The board software or logic file is lost. The board software fails to be loaded.

Silkscreen	Name	Color	Status	Description
		Green	Blinks on and off at 100 ms intervals	The board software is being loaded.
		Green	Blinks on and off at 300 ms intervals	The BIOS is interacting with the upper-layer software.
		Red	Blinks on and off at 100 ms intervals	The BIOS self-check fails.
		-	Off	The board is not powered on.
SYNC	Clock synchronization indicator	Green	Steady on	The system clock works in free-run mode, and no priority list is set for the system clocks (only internal clock sources are listed by default). The system clock works in tracing mode and is tracing a clock source other than the internal clock sources in the priority list. The system clock works in time synchronization mode, and the PTP time and system clock are being traced.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	The system clock priority list is set, but all clock sources except the internal clock source are lost. The clock works in holdover or free-run mode. The system clock works in time synchronization mode, but no synchronization source is available. The system clock and PTP time work in holdover or free-run mode.
ACTX	Active/standby status indicator for the cross-connect or clock board	Green	Steady on	The board is in the active state on the service plane.
		-	Off	The board is in the standby state on the service plane.
ACTC	Control status indicator	Green	Steady on	The board is in the active control state.
		Green	Blinks on and off at 100 ms intervals	The data of the device is backed up in batch.
		-	Off	The board is in the standby control state.

Silkscreen	Name	Color	Status	Description
ALM	Alarm indicator	Red	Steady on	Device-level critical alarms exist.
		Orange	Steady on	A device-level major or minor alarm exists.
		-	Off	There is no equipment-level alarm.
ETH/OAM	ETH/OAM indicator	Green	Steady on	The link is up.
		Orange	Blinking	Data is being transmitted or received.
		-	Off	The link is down.

Table 1-308 Buttons on the CR8DIPU1T2CG

Silkscreen	Name	Description
RST	Board reset button	You can press the RST button to reset the NE or restore the factory settings of the NE. The procedure is as follows: To reset the NE, press the RST button and release it. To restore factory settings, press and hold down the RST button. This operation will clear the NE database and system parameter configurations and cannot be restored. Exercise caution when you perform the following operations: 1.Press the RST button to restart the NE. 2.When the PROG indicator blinks (the blinking frequency is about 1 second and the time window is about 15 seconds), press and hold down the RST button. 3.Wait for no longer than 2 minutes till both the STAT and PROG indicators blink (the time window is 5 seconds). Then release the RST button immediately.

Silkscreen	Name	Description
		NOTE 1. When you press the RST button to clear the configuration file, the original configuration file will be cleared. You are advised to periodically back up the configuration file. 2. When you press the RST button to clear the configuration file, to prevent the cleared configuration from being synchronized again between the active and standby main control boards, remove one main control board first in a dual-main control board chassis to ensure that the configuration file is cleared in a single-main control board scenario. Perform the same operations on the other main control board. 3. When pressing the RST button to clear the configuration file, you only need to pay attention to the STAT and PROG indicators. Other indicators may vary depending on the product model.

Ports

Table 1-309 Ports on the CR8DIPU1T2CG

Port	Connector Type	Description	Available Components
ETH/OAM	RJ45	10M/100M auto-sensing Ethernet network management port or non-standard RS-232 console port NOTE If a service port can be connected to an NMS, you are advised to use the service port as the network management port.	ETH/OAM management cable
CLK/TOD/RS-485	RJ45	Clock/Time/RS-485 input and output ports RS-485 is a reserved port.	1-into-2 clock cable + clock cable (external clock mode, delivery-ready): applicable to the CLK mode 1-into-2 clock cable + clock cable (external clock/external time mode, prepared onsite): applicable to the CLK or TOD mode 1-into-2 clock cable + Ethernet cable: applicable to the TOD mode
ALMI/ALMO	RJ45	Alarm input/output interface	Alarm input/output cable
USB	Type A	USB port (reserved)	-
CLK/TEST	SMA	Clock test port	Coaxial cable

Table 1-310 Pins of the ETH/OAM interface

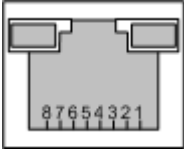
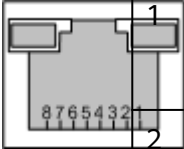
Front View	Pin	Usage
	1	Transmit positive of the NM network port
	2	Transmit negative of the NM network port
	3	Receive positive of the NM network port
	4	Grounding end of the Console interface
	5	Receive end of the Console interface
	6	Receive negative of the NM network port
	7	Unspecified
	8	Transmit end of the Console interface

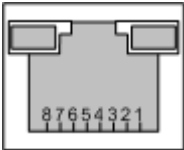
Table 1-311 Pins of the CLK/TOD/RS-485 interface

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	1	Receive negative of the CLK	Receive negative of the CLK	Unspecified	Unspecified	Unspecified
	2	Receive positive of the CLK	Receive positive of the CLK	Unspecified	Unspecified	Unspecified

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	3	Input/Output negative for the 1PPS signal	Unspecified	Input negative for the 1PPS signal (RS422 level)	Output negative for the 1PPS signal (RS422 level)	Transmit negative of the RS-485
	4	Receive negative of the CLK	Receive negative of the CLK	Grounding end	Grounding end	Unspecified
	5	Receive positive of the CLK	Receive positive of the CLK	Grounding end	Grounding end	Unspecified
	6	Input/Output positive for the 1PPS signal	Unspecified	Input positive for the 1PPS signal (RS422 level)	Output positive for the 1PPS signal (RS422 level)	Transmit positive of the RS-485

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	7	Input/Output negative for the time information	Unspecified	Input negative for the time information (RS422 level)	Output negative for the time information (RS422 level)	Receive negative of the RS-485
	8	Input/Output positive for the time information	Unspecified	Input positive for the time information (RS422 level)	Output positive for the time information (RS422 level)	Receive positive of the RS-485
NOTE When this interface is used as a TOD interface, it must be used with a dedicated clock cable. For details, see "Cables" in the Hardware Description.						

Table 1-312 Pins of the ALMI/ALMO interface

Front View	Pin	Usage
	1	Alarm input 1
	2	Grounding end for alarm input 1
	3	Alarm input 2

Front View	Pin	Usage
	4	Alarm input 3
	5	Grounding end for alarm input 3
	6	Grounding end for alarm input 2
	7	Positive for alarm signal output
	8	Negative for alarm signal output

Functions and Features

Table 1-313 Functions and features of the CR8DIPU1T2CG

Functions and Features	Description
Basic function	Data forwarding: The IPU is the core of service processing in the entire system, and is connected to all subcards through data channels. Control and management: The IPU manages subcards and transmits routing protocol data through management channels. 2.4 Tbit/s@310 bytes (bidirectional)
Reliability and availability	Hot swappable. 1:1 backup is supported.
Restrictions and remarks	This board cannot be used together with other IPU.
Breakout supported	No
Interface rate auto-sensing supported	No

Technical Specifications

Table 1-314 Technical specifications of the CR8DIPU1T2CG

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	22.4 mm x 388.4 mm x 209.3 mm (0.88 in. x 15.29 in. x 8.24 in.)

Item	Specification
Dimensions with packaging (H x W x D) [mm(in.)]	170 mm x 560 mm x 385 mm (6.69 in. x 22.05 in. x 15.16 in.)
Weight without packaging [kg(lb)]	2.3 kg (5.07 lb)
Weight with packaging [kg(lb)]	3.7 kg (8.16 lb)
Typical power consumption [W]	144.2 W
Typical heat dissipation [BTU/hour]	467.94 BTU/hour
Long-term operating temperature [°C(°F)]	–40°C to +65°C (–40°F to +149°F), start: ≥ –20°C (–4°F)
CPU	8-core 1.5 GHz
SDRAM	8GB
Storage	4 GB NandFlash

1.4.3.27 CR8DIPU1T2CJ (Integrated Network Processing Unit (IPU-1T2-C))

Overview

Table 1-315 Basic information about the CR8DIPU1T2CJ

Item	Details
Description	Integrated Network Processing Unit (IPU-1T2-C)
Part Number	03034UNG
Model	CR8DIPU1T2CJ
Card type	IPU
Silkscreen	IPU-1T2-C

Appearance

Figure 1-96 Appearance of the CR8DIPU1T2CJ



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03034UNG&productModel=CR8DIPU1T2CJ#matchRelation>

Indicators and Buttons

Table 1-316 Indicators on the CR8DIPU1T2CJ

Silkscreen	Name	Color	Status	Description
STAT	Working status indicator	Green	Steady on	The board is working properly.
		Red	Steady on	A hardware alarm is generated. You can run the display alarm hardware command to check the hardware alarm type.
		-	Off	The board is not powered on or is not running.
PROG	Program status indicator	Green	Steady on	The board software is being initialized or is successfully initialized and is running properly.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	The memory self-check fails. The board software or logic file is lost. The board software fails to be loaded. The BIOS self-check fails.
		Green	Blinks on and off at 100 ms intervals	The board software is being loaded.
		Green	Blinks on and off at 300 ms intervals	The BIOS is interacting with the upper-layer software.
		-	Off	The board is not powered on.

Silkscreen	Name	Color	Status	Description
SYNC	Clock synchronizatio n indicator	Green	Steady on	The system clock works in free-run mode, and no priority list is set for the system clocks (only internal clock sources are listed by default). The system clock works in tracing mode and is tracing a clock source other than the internal clock sources in the priority list. The system clock works in time synchronizatio n mode, and the PTP time and system clock are being traced.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	The system clock priority list is set, but all clock sources except the internal clock source are lost. The clock works in holdover or free-run mode. The system clock works in time synchronization mode, but no synchronization source is available. The system clock and PTP time work in holdover or free-run mode.
ACTX	Active/standby status indicator for the cross-connect or clock board	Green	Steady on	The board is in the active state on the service plane.
		-	Off	The board is in the standby state on the service plane.
ACTC	Control status indicator	Green	Steady on	The board is in the active control state.
		Green	Blinks on and off at 100 ms intervals	The data of the device is backed up in batch.
		-	Off	The board is in the standby control state.

Silkscreen	Name	Color	Status	Description
ALM	Alarm indicator	Red	Steady on	Device-level critical alarms exist.
		Orange	Steady on	A device-level major or minor alarm exists.
		-	Off	There is no equipment-level alarm.
ETH/OAM	ETH/OAM indicator	Green	Steady on	The link is up.
		Orange	Blinking	Data is being transmitted or received.
		-	Off	The link is down.

Table 1-317 Buttons on the CR8DIPU1T2CJ

Silkscreen	Name	Description
RST	Board reset button	You can press the RST button to reset the NE or restore the factory settings of the NE. The procedure is as follows: To reset the NE, press the RST button and release it. To restore factory settings, press and hold down the RST button. This operation will clear the NE database and system parameter configurations and cannot be restored. Exercise caution when you perform the following operations: 1.Press the RST button to restart the NE. 2.When the PROG indicator blinks (the blinking frequency is about 1 second and the time window is about 15 seconds), press and hold down the RST button. 3.Wait for no longer than 2 minutes till both the STAT and PROG indicators blink (the time window is 5 seconds). Then release the RST button immediately.

Silkscreen	Name	Description
		NOTE 1. When you press the RST button to clear the configuration file, the original configuration file will be cleared. You are advised to periodically back up the configuration file. 2. When you press the RST button to clear the configuration file, to prevent the cleared configuration from being synchronized again between the active and standby main control boards, remove one main control board first in a dual-main control board chassis to ensure that the configuration file is cleared in a single-main control board scenario. Perform the same operations on the other main control board. 3. When pressing the RST button to clear the configuration file, you only need to pay attention to the STAT and PROG indicators. Other indicators may vary depending on the product model.

Ports

Table 1-318 Ports on the CR8DIPU1T2CJ

Port	Connector Type	Description	Available Components
ETH/OAM	RJ45	10M/100M auto-sensing Ethernet network management port or non-standard RS-232 console port NOTE If a service port can be connected to an NMS, you are advised to use the service port as the network management port.	ETH/OAM management cable
CLK/TOD/RS-485	RJ45	Clock/Time/RS-485 input and output ports RS-485 is a reserved port.	1-into-2 clock cable + clock cable (external clock mode, delivery-ready): applicable to the CLK mode 1-into-2 clock cable + clock cable (external clock/external time mode, prepared onsite): applicable to the CLK or TOD mode 1-into-2 clock cable + Ethernet cable: applicable to the TOD mode
USB	Type A	USB port (reserved)	-
CLK/TEST	SMA	Clock test port	Coaxial cable

Port	Connector Type	Description	Available Components
HP-GE	SFP	Ultra-high-precision GE port (supported since V800R021C00 and reserved in earlier versions)	1.25Gbps eSFP Optical Module 1.25Gbps eSFP CWDM Optical Module 1.25Gbps eSFP BIDI Optical Module

Table 1-319 Pins of the ETH/OAM interface

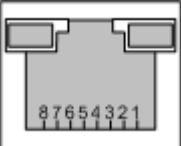
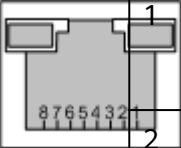
Front View	Pin	Usage
	1	Transmit positive of the NM network port
	2	Transmit negative of the NM network port
	3	Receive positive of the NM network port
	4	Grounding end of the Console interface
	5	Receive end of the Console interface
	6	Receive negative of the NM network port
	7	Unspecified
	8	Transmit end of the Console interface

Table 1-320 Pins of the CLK/TOD/RS-485 interface

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	1	Receive negative of the CLK	Receive negative of the CLK	Unspecified	Unspecified	Unspecified
	2	Receive positive of the CLK	Receive positive of the CLK	Unspecified	Unspecified	Unspecified
	3	Input/Output negative for the 1PPS signal	Unspecified	Input negative for the 1PPS signal (RS422 level)	Output negative for the 1PPS signal (RS422 level)	Transmit negative of the RS-485
	4	Receive negative of the CLK	Receive negative of the CLK	Grounding end	Grounding end	Unspecified
	5	Receive positive of the CLK	Receive positive of the CLK	Grounding end	Grounding end	Unspecified
	6	Input/Output positive for the 1PPS signal	Unspecified	Input positive for the 1PPS signal (RS422 level)	Output positive for the 1PPS signal (RS422 level)	Transmit positive of the RS-485

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	7	Input/Output negative for the time information	Unspecified	Input negative for the time information (RS422 level)	Output negative for the time information (RS422 level)	Receive negative of the RS-485
	8	Input/Output positive for the time information	Unspecified	Input positive for the time information (RS422 level)	Output positive for the time information (RS422 level)	Receive positive of the RS-485
NOTE When this interface is used as a TOD interface, it must be used with a dedicated clock cable. For details, see "Cables" in the Hardware Description.						

Functions and Features

Table 1-321 Functions and features of the CR8DIPU1T2CJ

Functions and Features	Description
Basic function	Data forwarding: The IPU is the core of service processing in the entire system, and is connected to all subcards through data channels. Control and management: The IPU manages subcards and transmits routing protocol data through management channels. 2.4 Tbit/s@310 bytes (bidirectional)
Reliability and availability	Hot swappable. 1:1 backup is supported.
Restrictions and remarks	1. It cannot be used together with other IPU's. 2. When working with the 03033VMC board to support FlexE services, the board supports a maximum of eight FlexE physical interfaces.
Breakout supported	No
Interface rate auto-sensing supported	No

Technical Specifications

Table 1-322 Technical specifications of the CR8DIPU1T2CJ

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	34.8 mm x 354.4 mm x 209 mm (1.37 in. x 13.95 in. x 8.23 in.)
Dimensions with packaging (H x W x D) [mm(in.)]	185 mm x 485 mm x 385 mm (7.28 in. x 19.09 in. x 15.16 in.)
Weight without packaging [kg(lb)]	4 kg (8.82 lb)
Weight with packaging [kg(lb)]	5.4 kg (11.9 lb)
Typical power consumption [W]	366.62 W
Typical heat dissipation [BTU/hour]	1189.45 BTU/hour
Long-term operating temperature [°C(°F)]	-20°C to +55°C, startup: ≥ -5°C

Item	Specification
CPU	8-core 2.3 GHz
SDRAM	16GB
Storage	16GB M.2

1.4.3.28 CR8DIPU1T2CN (Integrated Network Processing Unit (IPU-1T2-D))

Overview

Table 1-323 Basic information about the CR8DIPU1T2CN

Item	Details
Description	Integrated Network Processing Unit (IPU-1T2-D)
Part Number	03034UST
Model	CR8DIPU1T2CN
Card type	IPU
Silkscreen	IPU-1T2-D

Appearance

Figure 1-97 Appearance of the CR8DIPU1T2CN



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03034UST&productModel=CR8DIPU1T2CN#matchRelation>

Indicators and Buttons

Table 1-324 Indicators on the CR8DIPU1T2CN

Silkscreen	Name	Color	Status	Description
STAT	Working status indicator	Green	Steady on	The board is working properly.
		Red	Steady on	The board hardware is faulty.
		Orange	Steady on	The interface board is not loaded with logic.
		-	Off	The board is not powered on or is not running.
PROG	Program status indicator	Green	Steady on	The board software is being initialized or is successfully initialized and is running properly.
		Red	Steady on	The memory self-check fails. The board software or logic file is lost. The board software fails to be loaded.
		Green	Blinks on and off at 100 ms intervals	The board software is being loaded.
		Green	Blinks on and off at 300 ms intervals	The BIOS is booting the upper-layer software.

Silkscreen	Name	Color	Status	Description
		Red	Blinks on and off at 100 ms intervals	The BIOS self-check fails.
		-	Off	The board is not powered on.
SYNC	Clock synchronization indicator	Green	Steady on	The system clock works in free-run mode and no priority list is configured for the system clock. (By default, the list contains only internal clock sources.) The system clock is working in tracing mode and is tracing a clock source rather than the internal sources in the priority list. The system clock is working in time synchronization mode, and the PTP time and system clock are in the tracing state.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	The system clock priority list is set. All the clock sources, however, are lost except for the internal clock sources. The clock works in holdover mode or free-run mode. The system clock is working in time synchronization mode, but no synchronization source is available. The system clock and PTP time are working in holdover or free-run mode.
ACTX	Active/standby status indicator for the cross-connect or clock board	Green	Steady on	The board is in the active switching state.
		-	Off	The board is in the standby switching state.
ACTC	Control status indicator	Green	Steady on	The board is in the active control state.
		Green	Blinks on and off at 100 ms intervals	The data of the device is backed up in batch.

Silkscreen	Name	Color	Status	Description
		-	Off	The board is in the standby control state.
ALM	Device alarm indicator	Red	Steady on	A device-level critical alarm is generated.
		Orange	Steady on	There is an equipment-level major or minor alarm.
		-	Off	There is no equipment-level alarm.
ETH/OAM	ETH/OAM indicator	Green	Steady on	The link is up.
		Orange	Blinking	Data is being transmitted or received.
		-	Off	The link is down.

Table 1-325 Buttons on the CR8DIPU1T2CN

Silkscreen	Name	Description
RST	Board reset button	You can press the RST button to reset the NE or restore the factory settings of the NE. The procedure is as follows: To reset the NE, press the RST button and release it. To restore factory settings, press and hold down the RST button. This operation will clear the NE database and system parameter configurations and cannot be restored. Exercise caution when you perform the following operations: 1.Press the RST button to restart the NE. 2.When the PROG indicator blinks (the blinking frequency is about 1 second and the time window is about 15 seconds), press and hold down the RST button. 3.Wait for no longer than 2 minutes till both the STAT and PROG indicators blink (the time window is 5 seconds). Then release the RST button immediately.

Silkscreen	Name	Description
		NOTE 1. When you press the RST button to clear the configuration file, the original configuration file will be cleared. You are advised to periodically back up the configuration file. 2. When you press the RST button to clear the configuration file, to prevent the cleared configuration from being synchronized again between the active and standby main control boards, remove one main control board first in a dual-main control board chassis to ensure that the configuration file is cleared in a single-main control board scenario. Perform the same operations on the other main control board. 3. When pressing the RST button to clear the configuration file, you only need to pay attention to the STAT and PROG indicators. Other indicators may vary depending on the product model.

Ports

Table 1-326 Ports on the CR8DIPU1T2CN

Port	Connector Type	Description	Available Components
CLK/TEST	SMA	Clock test port	Coaxial cable
USB	Type A	USB port (reserved)	-

Port	Connector Type	Description	Available Components
ETH/OAM	RJ45	10M/100M auto-sensing Ethernet NM interface and non-standard RS-232 Console interface. NOTE If a service port can be connected to an NMS, you are advised to use the service port as the network management port.	ETH/OAM management cable
CLK/TOD/RS-485	RJ45	Clock/Time/RS-485 input and output ports RS-485 is a reserved port.	1-into-2 clock cable + clock cable (external clock mode, delivery-ready): applicable to the CLK mode 1-into-2 clock cable + clock cable (external clock/external time mode, prepared onsite): applicable to the CLK or TOD mode 1-into-2 clock cable + Ethernet cable: applicable to the TOD mode
ALMI/ALMO	RJ45	Alarm input/output interface	Alarm input/output cable
HP-GE	SFP	Ultra-high-precision GE port (reserved)	Optical fiber

Table 1-327 Pins of the ETH/OAM interface

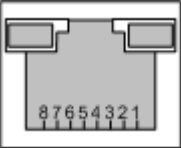
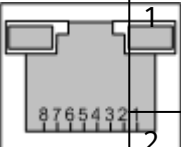
Front View	Pin	Usage
	1	Transmit positive of the NM network port
	2	Transmit negative of the NM network port
	3	Receive positive of the NM network port
	4	Grounding end of the Console interface
	5	Receive end of the Console interface
	6	Receive negative of the NM network port
	7	Unspecified
	8	Transmit end of the Console interface

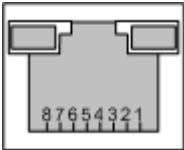
Table 1-328 Pins of the CLK/TOD/RS-485 interface

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	1	Receive negative of the CLK	Receive negative of the CLK	Unspecified	Unspecified	Unspecified
	2	Receive positive of the CLK	Receive positive of the CLK	Unspecified	Unspecified	Unspecified

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	3	Input/Output negative for the 1PPS signal	Unspecified	Input negative for the 1PPS signal (RS422 level)	Output negative for the 1PPS signal (RS422 level)	Transmit negative of the RS-485
	4	Receive negative of the CLK	Receive negative of the CLK	Grounding end	Grounding end	Unspecified
	5	Receive positive of the CLK	Receive positive of the CLK	Grounding end	Grounding end	Unspecified
	6	Input/Output positive for the 1PPS signal	Unspecified	Input positive for the 1PPS signal (RS422 level)	Output positive for the 1PPS signal (RS422 level)	Transmit positive of the RS-485

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	7	Input/Output negative for the time information	Unspecified	Input negative for the time information (RS422 level)	Output negative for the time information (RS422 level)	Receive negative of the RS-485
	8	Input/Output positive for the time information	Unspecified	Input positive for the time information (RS422 level)	Output positive for the time information (RS422 level)	Receive positive of the RS-485
NOTE When this interface is used as a TOD interface, it must be used with a dedicated clock cable. For details, see "Cables" in the Hardware Description.						

Table 1-329 Pins of the ALMI/ALMO interface

Front View	Pin	Usage
	1	Alarm input 1
	2	Grounding end for alarm input 1
	3	Alarm input 2

Front View	Pin	Usage
	4	Alarm input 3
	5	Grounding end for alarm input 3
	6	Grounding end for alarm input 2
	7	Positive for alarm signal output
	8	Negative for alarm signal output

Functions and Features

Table 1-330 Functions and features of the CR8DIPU1T2CN

Functions and Features	Description
Basic function	Line rate: 1200 Gbit/s. Supports switching, control, and clock management. Supports board-level 1:1 backup. Supports the ultra-high-precision clock.
Reliability and availability	Hot swappable. 1:1 backup is supported.

Technical Specifications

Table 1-331 Technical specifications of the CR8DIPU1T2CN

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	34.8 mm x 354.4 mm x 209 mm (1.37 in. x 13.95 in. x 8.23 in.)
Dimensions with packaging (H x W x D) [mm(in.)]	185 mm x 485 mm x 385 mm (7.28 in. x 19.09 in. x 15.16 in.)
Weight without packaging [kg(lb)]	2.55 kg (5.62 lb)
Weight with packaging [kg(lb)]	3.95 kg (8.71 lb)
Typical power consumption [W]	135.6 W
Typical heat dissipation [BTU/hour]	440 BTU/hour

Item	Specification
Long-term operating temperature [°C(°F)]	–20°C to +55°C, startup: ≥ –5°C
Switching capacity	2.4 Tbit/s
CPU	8-core 1.5 GHz
SDRAM	8GB
Storage	4 GB NandFlash

1.4.3.29 CR8DIPU1T2CU (Integrated Network Processing Unit (IPU-1T2-D))

Overview

Table 1-332 Basic information about the CR8DIPU1T2CU

Item	Details
Description	Integrated Network Processing Unit (IPU-1T2-D)
Part Number	03034USU
Model	CR8DIPU1T2CU
Card type	IPU
Silkscreen	IPU-1T2-D

Appearance

Figure 1-98 Appearance of the CR8DIPU1T2CU



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03034USU&productModel=CR8DIPU1T2CU#matchRelation>

Indicators and Buttons

Table 1-333 Indicators on the CR8DIPU1T2CU

Silkscreen	Name	Color	Status	Description
STAT	Working status indicator	Green	Steady on	The board is working properly.
		Red	Steady on	A hardware alarm is generated. You can run the display alarm hardware command to check the hardware alarm type.
		-	Off	The board is not powered on or is not running.
PROG	Program status indicator	Green	Steady on	The board software is being initialized or is successfully initialized and is running properly.
		Red	Steady on	The memory self-check fails. The board software or logic file is lost. The board software fails to be loaded.

Silkscreen	Name	Color	Status	Description
		Green	Blinks on and off at 100 ms intervals	The board software is being loaded.
		Green	Blinks on and off at 300 ms intervals	The BIOS is interacting with the upper-layer software.
		Red	Blinks on and off at 100 ms intervals	The BIOS self-check fails.
		-	Off	The board is not powered on.
SYNC	Clock synchronization indicator	Green	Steady on	The system clock works in free-run mode, and no priority list is set for the system clocks (only internal clock sources are listed by default). The system clock works in tracing mode and is tracing a clock source other than the internal clock sources in the priority list. The system clock works in time synchronization mode, and the PTP time and system clock are being traced.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	The system clock priority list is set, but all clock sources except the internal clock source are lost. The clock works in holdover or free-run mode. The system clock works in time synchronization mode, but no synchronization source is available. The system clock and PTP time work in holdover or free-run mode.
ACTX	Active/standby status indicator for the cross-connect or clock board	Green	Steady on	The board is in the active state on the service plane.
		-	Off	The board is in the standby state on the service plane.
ACTC	Control status indicator	Green	Steady on	The board is in the active control state.
		Green	Blinks on and off at 100 ms intervals	The data of the device is backed up in batch.
		-	Off	The board is in the standby control state.

Silkscreen	Name	Color	Status	Description
ALM	Alarm indicator	Red	Steady on	Device-level critical alarms exist.
		Orange	Steady on	A device-level major or minor alarm exists.
		-	Off	There is no equipment-level alarm.
ETH/OAM	ETH/OAM indicator	Green	Steady on	The link is up.
		Orange	Blinking	Data is being transmitted or received.
		-	Off	The link is down.

Table 1-334 Buttons on the CR8DIPU1T2CU

Silkscreen	Name	Description
RST	Board reset button	You can press the RST button to reset the NE or restore the factory settings of the NE. The procedure is as follows: To reset the NE, press the RST button and release it. To restore factory settings, press and hold down the RST button. This operation will clear the NE database and system parameter configurations and cannot be restored. Exercise caution when you perform the following operations: 1.Press the RST button to restart the NE. 2.When the PROG indicator blinks (the blinking frequency is about 1 second and the time window is about 15 seconds), press and hold down the RST button. 3.Wait for no longer than 2 minutes till both the STAT and PROG indicators blink (the time window is 5 seconds). Then release the RST button immediately.

Silkscreen	Name	Description
		NOTE 1. When you press the RST button to clear the configuration file, the original configuration file will be cleared. You are advised to periodically back up the configuration file. 2. When you press the RST button to clear the configuration file, to prevent the cleared configuration from being synchronized again between the active and standby main control boards, remove one main control board first in a dual-main control board chassis to ensure that the configuration file is cleared in a single-main control board scenario. Perform the same operations on the other main control board. 3. When pressing the RST button to clear the configuration file, you only need to pay attention to the STAT and PROG indicators. Other indicators may vary depending on the product model.

Ports

Table 1-335 Ports on the CR8DIPU1T2CU

Port	Connector Type	Description	Available Components
ETH/OAM	RJ45	10M/100M auto-sensing Ethernet network management port or non-standard RS-232 console port NOTE If a service port can be connected to an NMS, you are advised to use the service port as the network management port.	ETH/OAM management cable
CLK/TOD/RS-485	RJ45	Clock/Time/RS-485 input and output ports RS-485 is a reserved port.	1-into-2 clock cable + clock cable (external clock mode, delivery-ready): applicable to the CLK mode 1-into-2 clock cable + clock cable (external clock/external time mode, prepared onsite): applicable to the CLK or TOD mode 1-into-2 clock cable + Ethernet cable: applicable to the TOD mode
ALMI/ALMO	RJ45	Alarm input/output interface	Alarm input/output cable
USB	Type A	USB port (reserved)	-
CLK/TEST	SMA	Clock test port	Coaxial cable

Table 1-336 Pins of the ETH/OAM interface

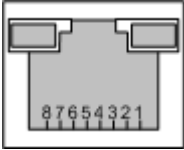
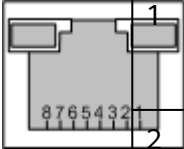
Front View	Pin	Usage
	1	Transmit positive of the NM network port
	2	Transmit negative of the NM network port
	3	Receive positive of the NM network port
	4	Grounding end of the Console interface
	5	Receive end of the Console interface
	6	Receive negative of the NM network port
	7	Unspecified
	8	Transmit end of the Console interface

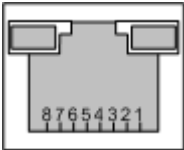
Table 1-337 Pins of the CLK/TOD/RS-485 interface

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	1	Receive negative of the CLK	Receive negative of the CLK	Unspecified	Unspecified	Unspecified
	2	Receive positive of the CLK	Receive positive of the CLK	Unspecified	Unspecified	Unspecified

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	3	Input/Output negative for the 1PPS signal	Unspecified	Input negative for the 1PPS signal (RS422 level)	Output negative for the 1PPS signal (RS422 level)	Transmit negative of the RS-485
	4	Receive negative of the CLK	Receive negative of the CLK	Grounding end	Grounding end	Unspecified
	5	Receive positive of the CLK	Receive positive of the CLK	Grounding end	Grounding end	Unspecified
	6	Input/Output positive for the 1PPS signal	Unspecified	Input positive for the 1PPS signal (RS422 level)	Output positive for the 1PPS signal (RS422 level)	Transmit positive of the RS-485

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	7	Input/Output negative for the time information	Unspecified	Input negative for the time information (RS422 level)	Output negative for the time information (RS422 level)	Receive negative of the RS-485
	8	Input/Output positive for the time information	Unspecified	Input positive for the time information (RS422 level)	Output positive for the time information (RS422 level)	Receive positive of the RS-485
NOTE When this interface is used as a TOD interface, it must be used with a dedicated clock cable. For details, see "Cables" in the Hardware Description.						

Table 1-338 Pins of the ALMI/ALMO interface

Front View	Pin	Usage
	1	Alarm input 1
	2	Grounding end for alarm input 1
	3	Alarm input 2

Front View	Pin	Usage
	4	Alarm input 3
	5	Grounding end for alarm input 3
	6	Grounding end for alarm input 2
	7	Positive for alarm signal output
	8	Negative for alarm signal output

Functions and Features

Table 1-339 Functions and features of the CR8DIPU1T2CU

Functions and Features	Description
Basic function	Data forwarding: The IPU is the core of service processing in the entire system, and is connected to all subcards through data channels. Control and management: The IPU manages subcards and transmits routing protocol data through management channels. 2.4 Tbit/s@310 bytes (bidirectional)
Reliability and availability	Hot swappable. 1:1 backup is supported.
Restrictions and remarks	This board cannot be used together with other IPU.
Breakout supported	No
Interface rate auto-sensing supported	No

Technical Specifications

Table 1-340 Technical specifications of the CR8DIPU1T2CU

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	22.4 mm x 388.4 mm x 209.3 mm (0.88 in. x 15.29 in. x 8.24 in.)

Item	Specification
Dimensions with packaging (H x W x D) [mm(in.)]	170 mm x 560 mm x 385 mm (6.69 in. x 22.05 in. x 15.16 in.)
Weight without packaging [kg(lb)]	2.3 kg (5.07 lb)
Weight with packaging [kg(lb)]	3.7 kg (8.16 lb)
Typical power consumption [W]	144.2 W
Typical heat dissipation [BTU/hour]	467.94 BTU/hour
Long-term operating temperature [°C(°F)]	–40°C to +65°C (–40°F to +149°F), start: ≥ –20°C (–4°F)
CPU	8-core 1.5 GHz
SDRAM	8GB
Storage	4 GB NandFlash

1.4.3.30 CR8DIPU1T2M3 (Integrated Network Processing Unit (IPU-1T2-EN))

Overview

Table 1-341 Basic information about the CR8DIPU1T2M3

Item	Details
Description	Integrated Network Processing Unit (IPU-1T2-EN)
Part Number	03034VAV
Model	CR8DIPU1T2M3
Card type	IPU
Silkscreen	IPU-1T2-EN

Appearance

Figure 1-99 Appearance of the CR8DIPU1T2M3



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03034VAV&productModel=CR8DIPU1T2M3#matchRelation>

Indicators and Buttons

Table 1-342 Indicators on the CR8DIPU1T2M3

Silkscreen	Name	Color	Status	Description
STAT	Working status indicator	Green	Steady on	The board is working properly.
		Red	Steady on	The board hardware is faulty.
		Orange	Steady on	The interface board is not loaded with logic.
		-	Off	The board is not powered on or is not running.
PROG	Program status indicator	Green	Steady on	The board software is being initialized or is successfully initialized and is running properly.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	The memory self-check fails. The board software or logic file is lost. The board software fails to be loaded.
		Green	Blinks on and off at 100 ms intervals	The board software is being loaded.
		Green	Blinks on and off at 300 ms intervals	The BIOS is booting the upper-layer software.
		Red	Blinks on and off at 100 ms intervals	The BIOS self-check fails.
		-	Off	The board is not powered on.

Silkscreen	Name	Color	Status	Description
SYNC	Clock synchronizatio n indicator	Green	Steady on	The system clock works in free-run mode and no priority list is configured for the system clock. (By default, the list contains only internal clock sources.) The system clock is working in tracing mode and is tracing a clock source rather than the internal sources in the priority list. The system clock is working in time synchronization mode, and the PTP time and system clock are in the tracing state.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	The system clock priority list is set. All the clock sources, however, are lost except for the internal clock sources. The clock works in holdover mode or free-run mode. The system clock is working in time synchronization mode, but no synchronization source is available. The system clock and PTP time are working in holdover or free-run mode.
ACTX	Active/standby status indicator for the cross-connect or clock board	Green	Steady on	The board is in the active switching state.
		-	Off	The board is in the standby switching state.
ACTC	Control status indicator	Green	Steady on	The board is in the active control state.
		Green	Blinks on and off at 100 ms intervals	The data of the device is backed up in batch.

Silkscreen	Name	Color	Status	Description
		-	Off	The board is in the standby control state.
ALM	Device alarm indicator	Red	Steady on	A device-level critical alarm is generated.
		Orange	Steady on	There is an equipment-level major or minor alarm.
		-	Off	There is no equipment-level alarm.
ETH/OAM	ETH/OAM indicator	Green	Steady on	The link is up.
		Orange	Blinking	Data is being transmitted or received.
		-	Off	The link is down.

Table 1-343 Buttons on the CR8DIPU1T2M3

Silkscreen	Name	Description
RST	Board reset button	You can press the RST button to reset the NE or restore the factory settings of the NE. The procedure is as follows: To reset the NE, press the RST button and release it. To restore factory settings, press and hold down the RST button. This operation will clear the NE database and system parameter configurations and cannot be restored. Exercise caution when you perform the following operations: 1.Press the RST button to restart the NE. 2.When the PROG indicator blinks (the blinking frequency is about 1 second and the time window is about 15 seconds), press and hold down the RST button. 3.Wait for no longer than 2 minutes till both the STAT and PROG indicators blink (the time window is 5 seconds). Then release the RST button immediately.

Silkscreen	Name	Description
		NOTE 1. When you press the RST button to clear the configuration file, the original configuration file will be cleared. You are advised to periodically back up the configuration file. 2. When you press the RST button to clear the configuration file, to prevent the cleared configuration from being synchronized again between the active and standby main control boards, remove one main control board first in a dual-main control board chassis to ensure that the configuration file is cleared in a single-main control board scenario. Perform the same operations on the other main control board. 3. When pressing the RST button to clear the configuration file, you only need to pay attention to the STAT and PROG indicators. Other indicators may vary depending on the product model.

Ports

Table 1-344 Ports on the CR8DIPU1T2M3

Port	Connector Type	Description	Available Components
CLK/TEST	SMA	Clock test port	Coaxial cable
USB	Type A	USB port (reserved)	-

Port	Connector Type	Description	Available Components
ETH/OAM	RJ45	10M/100M auto-sensing Ethernet NM interface and non-standard RS-232 Console interface. NOTE If a service port can be connected to an NMS, you are advised to use the service port as the network management port.	ETH/OAM management cable
CLK/TOD/RS-485	RJ45	Clock/Time/RS-485 input and output ports RS-485 is a reserved port.	1-into-2 clock cable + clock cable (external clock mode, delivery-ready): applicable to the CLK mode 1-into-2 clock cable + clock cable (external clock/external time mode, prepared onsite): applicable to the CLK or TOD mode 1-into-2 clock cable + Ethernet cable: applicable to the TOD mode
ALMI/ALMO	RJ45	Alarm input/output interface	Alarm input/output cable
HP-GE	SFP	Ultra-high-precision GE port (reserved)	Optical fiber

Table 1-345 Pins of the ETH/OAM interface

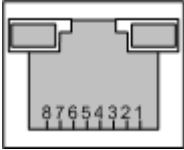
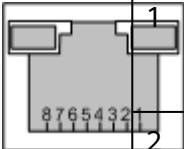
Front View	Pin	Usage
	1	Transmit positive of the NM network port
	2	Transmit negative of the NM network port
	3	Receive positive of the NM network port
	4	Grounding end of the Console interface
	5	Receive end of the Console interface
	6	Receive negative of the NM network port
	7	Unspecified
	8	Transmit end of the Console interface

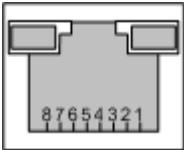
Table 1-346 Pins of the CLK/TOD/RS-485 interface

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	1	Receive negative of the CLK	Receive negative of the CLK	Unspecified	Unspecified	Unspecified
	2	Receive positive of the CLK	Receive positive of the CLK	Unspecified	Unspecified	Unspecified

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	3	Input/Output negative for the 1PPS signal	Unspecified	Input negative for the 1PPS signal (RS422 level)	Output negative for the 1PPS signal (RS422 level)	Transmit negative of the RS-485
	4	Receive negative of the CLK	Receive negative of the CLK	Grounding end	Grounding end	Unspecified
	5	Receive positive of the CLK	Receive positive of the CLK	Grounding end	Grounding end	Unspecified
	6	Input/Output positive for the 1PPS signal	Unspecified	Input positive for the 1PPS signal (RS422 level)	Output positive for the 1PPS signal (RS422 level)	Transmit positive of the RS-485

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	7	Input/Output negative for the time information	Unspecified	Input negative for the time information (RS422 level)	Output negative for the time information (RS422 level)	Receive negative of the RS-485
	8	Input/Output positive for the time information	Unspecified	Input positive for the time information (RS422 level)	Output positive for the time information (RS422 level)	Receive positive of the RS-485
NOTE When this interface is used as a TOD interface, it must be used with a dedicated clock cable. For details, see "Cables" in the Hardware Description.						

Table 1-347 Pins of the ALMI/ALMO interface

Front View	Pin	Usage
	1	Alarm input 1
	2	Grounding end for alarm input 1
	3	Alarm input 2

Front View	Pin	Usage
	4	Alarm input 3
	5	Grounding end for alarm input 3
	6	Grounding end for alarm input 2
	7	Positive for alarm signal output
	8	Negative for alarm signal output

Functions and Features

Table 1-348 Functions and features of the CR8DIPU1T2M3

Functions and Features	Description
Basic function	Line rate: 1200 Gbit/s. Supports switching, control, and clock management. Supports board-level 1:1 backup. Supports the ultra-high-precision clock.
Reliability and availability	Hot swappable. 1:1 backup is supported.

Technical Specifications

Table 1-349 Technical specifications of the CR8DIPU1T2M3

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	34.8 mm x 354.4 mm x 209 mm (1.37 in. x 13.95 in. x 8.23 in.)
Dimensions with packaging (H x W x D) [mm(in.)]	185 mm x 485 mm x 385 mm (7.28 in. x 19.09 in. x 15.16 in.)
Weight without packaging [kg(lb)]	2.55 kg (5.62 lb)
Weight with packaging [kg(lb)]	3.95 kg (8.71 lb)
Typical power consumption [W]	135.6 W
Typical heat dissipation [BTU/hour]	440 BTU/hour

Item	Specification
Long-term operating temperature [°C(°F)]	–20°C to +55°C, startup: ≥ –5°C
Switching capacity	2.4 Tbit/s
CPU	8-core 1.5 GHz
SDRAM	8GB
Storage	4 GB NandFlash

1.4.3.31 CR8DIPU1T2M5 (Integrated Network Processing Unit (IPU-1T2-E))

Overview

Table 1-350 Basic information about the CR8DIPU1T2M5

Item	Details
Description	Integrated Network Processing Unit (IPU-1T2-E)
Part Number	03034VPH
Model	CR8DIPU1T2M5
Card type	IPU
Silkscreen	IPU-1T2-E

Appearance

Figure 1-100 Appearance of the CR8DIPU1T2M5



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03034VPH&productModel=CR8DIPU1T2M5#matchRelation>

Indicators and Buttons

Table 1-351 Indicators on the CR8DIPU1T2M5

Silkscreen	Name	Color	Status	Description
STAT	Working status indicator	Green	Steady on	The board is working properly.
		Red	Steady on	A hardware alarm is generated. You can run the display alarm hardware command to check the hardware alarm type.
		-	Off	The board is not powered on or is not running.
PROG	Program status indicator	Green	Steady on	The board software is being initialized or is successfully initialized and is running properly.
		Red	Steady on	The memory self-check fails. The board software or logic file is lost. The board software fails to be loaded.

Silkscreen	Name	Color	Status	Description
		Green	Blinks on and off at 100 ms intervals	The board software is being loaded.
		Green	Blinks on and off at 300 ms intervals	The BIOS is interacting with the upper-layer software.
		Red	Blinks on and off at 100 ms intervals	The BIOS self-check fails.
		-	Off	The board is not powered on.
SYNC	Clock synchronization indicator	Green	Steady on	The system clock works in free-run mode, and no priority list is set for the system clocks (only internal clock sources are listed by default). The system clock works in tracing mode and is tracing a clock source other than the internal clock sources in the priority list. The system clock works in time synchronization mode, and the PTP time and system clock are being traced.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	The system clock priority list is set, but all clock sources except the internal clock source are lost. The clock works in holdover or free-run mode. The system clock works in time synchronization mode, but no synchronization source is available. The system clock and PTP time work in holdover or free-run mode.
ACTX	Active/standby status indicator for the cross-connect or clock board	Green	Steady on	The board is in the active state on the service plane.
		-	Off	The board is in the standby state on the service plane.
ACTC	Control status indicator	Green	Steady on	The board is in the active control state.
		Green	Blinks on and off at 100 ms intervals	The data of the device is backed up in batch.
		-	Off	The board is in the standby control state.

Silkscreen	Name	Color	Status	Description
ALM	Alarm indicator	Red	Steady on	Device-level critical alarms exist.
		Orange	Steady on	A device-level major or minor alarm exists.
		-	Off	There is no equipment-level alarm.
ETH/OAM	ETH/OAM indicator	Green	Steady on	The link is up.
		Orange	Blinking	Data is being transmitted or received.
		-	Off	The link is down.

Table 1-352 Buttons on the CR8DIPU1T2M5

Silkscreen	Name	Description
RST	Board reset button	You can press the RST button to reset the NE or restore the factory settings of the NE. The procedure is as follows: To reset the NE, press the RST button and release it. To restore factory settings, press and hold down the RST button. This operation will clear the NE database and system parameter configurations and cannot be restored. Exercise caution when you perform the following operations: 1.Press the RST button to restart the NE. 2.When the PROG indicator blinks (the blinking frequency is about 1 second and the time window is about 15 seconds), press and hold down the RST button. 3.Wait for no longer than 2 minutes till both the STAT and PROG indicators blink (the time window is 5 seconds). Then release the RST button immediately.

Silkscreen	Name	Description
		NOTE 1. When you press the RST button to clear the configuration file, the original configuration file will be cleared. You are advised to periodically back up the configuration file. 2. When you press the RST button to clear the configuration file, to prevent the cleared configuration from being synchronized again between the active and standby main control boards, remove one main control board first in a dual-main control board chassis to ensure that the configuration file is cleared in a single-main control board scenario. Perform the same operations on the other main control board. 3. When pressing the RST button to clear the configuration file, you only need to pay attention to the STAT and PROG indicators. Other indicators may vary depending on the product model.

Ports

Table 1-353 Ports on the CR8DIPU1T2M5

Port	Connector Type	Description	Available Components
ETH/OAM	RJ45	10M/100M auto-sensing Ethernet network management port or non-standard RS-232 console port NOTE If a service port can be connected to an NMS, you are advised to use the service port as the network management port.	ETH/OAM management cable
CLK/TOD/RS-485	RJ45	Clock/Time/RS-485 input and output ports RS-485 is a reserved port.	1-into-2 clock cable + clock cable (external clock mode, delivery-ready): applicable to the CLK mode 1-into-2 clock cable + clock cable (external clock/external time mode, prepared onsite): applicable to the CLK or TOD mode 1-into-2 clock cable + Ethernet cable: applicable to the TOD mode
ALMI/ALMO	RJ45	Alarm input/output interface	Alarm input/output cable
USB	Type A	USB port (reserved)	-
CLK/TEST	SMA	Clock test port	Coaxial cable

Table 1-354 Pins of the ETH/OAM interface

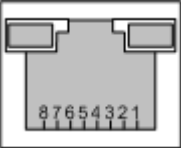
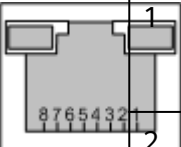
Front View	Pin	Usage
	1	Transmit positive of the NM network port
	2	Transmit negative of the NM network port
	3	Receive positive of the NM network port
	4	Grounding end of the Console interface
	5	Receive end of the Console interface
	6	Receive negative of the NM network port
	7	Unspecified
	8	Transmit end of the Console interface

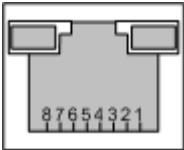
Table 1-355 Pins of the CLK/TOD/RS-485 interface

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	1	Receive negative of the CLK	Receive negative of the CLK	Unspecified	Unspecified	Unspecified
	2	Receive positive of the CLK	Receive positive of the CLK	Unspecified	Unspecified	Unspecified

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	3	Input/Output negative for the 1PPS signal	Unspecified	Input negative for the 1PPS signal (RS422 level)	Output negative for the 1PPS signal (RS422 level)	Transmit negative of the RS-485
	4	Receive negative of the CLK	Receive negative of the CLK	Grounding end	Grounding end	Unspecified
	5	Receive positive of the CLK	Receive positive of the CLK	Grounding end	Grounding end	Unspecified
	6	Input/Output positive for the 1PPS signal	Unspecified	Input positive for the 1PPS signal (RS422 level)	Output positive for the 1PPS signal (RS422 level)	Transmit positive of the RS-485

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	7	Input/Output negative for the time information	Unspecified	Input negative for the time information (RS422 level)	Output negative for the time information (RS422 level)	Receive negative of the RS-485
	8	Input/Output positive for the time information	Unspecified	Input positive for the time information (RS422 level)	Output positive for the time information (RS422 level)	Receive positive of the RS-485
NOTE When this interface is used as a TOD interface, it must be used with a dedicated clock cable. For details, see "Cables" in the Hardware Description.						

Table 1-356 Pins of the ALMI/ALMO interface

Front View	Pin	Usage
	1	Alarm input 1
	2	Grounding end for alarm input 1
	3	Alarm input 2

Front View	Pin	Usage
	4	Alarm input 3
	5	Grounding end for alarm input 3
	6	Grounding end for alarm input 2
	7	Positive for alarm signal output
	8	Negative for alarm signal output

Functions and Features

Table 1-357 Functions and features of the CR8DIPU1T2M5

Functions and Features	Description
Basic function	Data forwarding: The IPU is the core of service processing in the entire system, and is connected to all subcards through data channels. Control and management: The IPU manages subcards and transmits routing protocol data through management channels. 2.4 Tbit/s@310 bytes (bidirectional)
Reliability and availability	Hot swappable. 1:1 backup is supported.
Restrictions and remarks	This board cannot be used together with other IPU.
Breakout supported	No
Interface rate auto-sensing supported	No

Technical Specifications

Table 1-358 Technical specifications of the CR8DIPU1T2M5

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	22.4 mm x 388.4 mm x 209.3 mm (0.88 in. x 15.29 in. x 8.24 in.)

Item	Specification
Dimensions with packaging (H x W x D) [mm(in.)]	170 mm x 560 mm x 385 mm (6.69 in. x 22.05 in. x 15.16 in.)
Weight without packaging [kg(lb)]	2.3 kg (5.07 lb)
Weight with packaging [kg(lb)]	3.7 kg (8.16 lb)
Typical power consumption [W]	144.2 W
Typical heat dissipation [BTU/hour]	467.94 BTU/hour
Long-term operating temperature [°C(°F)]	–40°C to +65°C (–40°F to +149°F), start: ≥ –20°C (–4°F)
CPU	8-core 1.5 GHz
SDRAM	8GB
Storage	4 GB NandFlash

1.4.3.32 CR8DIPU1T2M6 (Integrated Network Processing Unit (IPU-1T2-E))

Overview

Table 1-359 Basic information about the CR8DIPU1T2M6

Item	Details
Description	Integrated Network Processing Unit (IPU-1T2-E)
Part Number	03034VAU
Model	CR8DIPU1T2M6
Card type	IPU
Silkscreen	IPU-1T2-E

Appearance

Figure 1-101 Appearance of the CR8DIPU1T2M6



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03034VAU&productModel=CR8DIPU1T2M6#matchRelation>

Indicators and Buttons

Table 1-360 Indicators on the CR8DIPU1T2M6

Silkscreen	Name	Color	Status	Description
STAT	Working status indicator	Green	Steady on	The board is working properly.
		Red	Steady on	A hardware alarm is generated. You can run the display alarm hardware command to check the hardware alarm type.
		-	Off	The board is not powered on or is not running.
PROG	Program status indicator	Green	Steady on	The board software is being initialized or is successfully initialized and is running properly.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	The memory self-check fails. The board software or logic file is lost. The board software fails to be loaded.
		Green	Blinks on and off at 100 ms intervals	The board software is being loaded.
		Green	Blinks on and off at 300 ms intervals	The BIOS is interacting with the upper-layer software.
		Red	Blinks on and off at 100 ms intervals	The BIOS self-check fails.
		-	Off	The board is not powered on.

Silkscreen	Name	Color	Status	Description
SYNC	Clock synchronizatio n indicator	Green	Steady on	The system clock works in free-run mode, and no priority list is set for the system clocks (only internal clock sources are listed by default). The system clock works in tracing mode and is tracing a clock source other than the internal clock sources in the priority list. The system clock works in time synchronizatio n mode, and the PTP time and system clock are being traced.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	The system clock priority list is set, but all clock sources except the internal clock source are lost. The clock works in holdover or free-run mode. The system clock works in time synchronization mode, but no synchronization source is available. The system clock and PTP time work in holdover or free-run mode.
ACTX	Active/standby status indicator for the cross-connect or clock board	Green	Steady on	The board is in the active state on the service plane.
		-	Off	The board is in the standby state on the service plane.
ACTC	Control status indicator	Green	Steady on	The board is in the active control state.
		Green	Blinks on and off at 100 ms intervals	The data of the device is backed up in batch.
		-	Off	The board is in the standby control state.

Silkscreen	Name	Color	Status	Description
ALM	Alarm indicator	Red	Steady on	Device-level critical alarms exist.
		Orange	Steady on	A device-level major or minor alarm exists.
		-	Off	There is no equipment-level alarm.
ETH/OAM	ETH/OAM indicator	Green	Steady on	The link is up.
		Orange	Blinking	Data is being transmitted or received.
		-	Off	The link is down.

Table 1-361 Buttons on the CR8DIPU1T2M6

Silkscreen	Name	Description
RST	Board reset button	You can press the RST button to reset the NE or restore the factory settings of the NE. The procedure is as follows: To reset the NE, press the RST button and release it. To restore factory settings, press and hold down the RST button. This operation will clear the NE database and system parameter configurations and cannot be restored. Exercise caution when you perform the following operations: 1.Press the RST button to restart the NE. 2.When the PROG indicator blinks (the blinking frequency is about 1 second and the time window is about 15 seconds), press and hold down the RST button. 3.Wait for no longer than 2 minutes till both the STAT and PROG indicators blink (the time window is 5 seconds). Then release the RST button immediately.

Silkscreen	Name	Description
		NOTE 1. When you press the RST button to clear the configuration file, the original configuration file will be cleared. You are advised to periodically back up the configuration file. 2. When you press the RST button to clear the configuration file, to prevent the cleared configuration from being synchronized again between the active and standby main control boards, remove one main control board first in a dual-main control board chassis to ensure that the configuration file is cleared in a single-main control board scenario. Perform the same operations on the other main control board. 3. When pressing the RST button to clear the configuration file, you only need to pay attention to the STAT and PROG indicators. Other indicators may vary depending on the product model.

Ports

Table 1-362 Ports on the CR8DIPU1T2M6

Port	Connector Type	Description	Available Components
ETH/OAM	RJ45	10M/100M auto-sensing Ethernet network management port or non-standard RS-232 console port NOTE If a service port can be connected to an NMS, you are advised to use the service port as the network management port.	ETH/OAM management cable
CLK/TOD/RS-485	RJ45	Clock/Time/RS-485 input and output ports RS-485 is a reserved port.	1-into-2 clock cable + clock cable (external clock mode, delivery-ready): applicable to the CLK mode 1-into-2 clock cable + clock cable (external clock/external time mode, prepared onsite): applicable to the CLK or TOD mode 1-into-2 clock cable + Ethernet cable: applicable to the TOD mode
USB	Type A	USB port (reserved)	-
CLK/TEST	SMA	Clock test port	Coaxial cable

Port	Connector Type	Description	Available Components
HP-GE	SFP	Ultra-high-precision GE port (supported since V800R021C00 and reserved in earlier versions)	1.25Gbps eSFP Optical Module 1.25Gbps eSFP CWDM Optical Module 1.25Gbps eSFP BIDI Optical Module

Table 1-363 Pins of the ETH/OAM interface

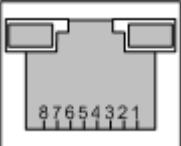
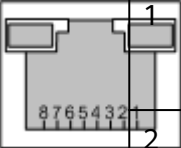
Front View	Pin	Usage
	1	Transmit positive of the NM network port
	2	Transmit negative of the NM network port
	3	Receive positive of the NM network port
	4	Grounding end of the Console interface
	5	Receive end of the Console interface
	6	Receive negative of the NM network port
	7	Unspecified
	8	Transmit end of the Console interface

Table 1-364 Pins of the CLK/TOD/RS-485 interface

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	1	Receive negative of the CLK	Receive negative of the CLK	Unspecified	Unspecified	Unspecified
	2	Receive positive of the CLK	Receive positive of the CLK	Unspecified	Unspecified	Unspecified
	3	Input/Output negative for the 1PPS signal	Unspecified	Input negative for the 1PPS signal (RS422 level)	Output negative for the 1PPS signal (RS422 level)	Transmit negative of the RS-485
	4	Receive negative of the CLK	Receive negative of the CLK	Grounding end	Grounding end	Unspecified
	5	Receive positive of the CLK	Receive positive of the CLK	Grounding end	Grounding end	Unspecified
	6	Input/Output positive for the 1PPS signal	Unspecified	Input positive for the 1PPS signal (RS422 level)	Output positive for the 1PPS signal (RS422 level)	Transmit positive of the RS-485

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	7	Input/Output negative for the time information	Unspecified	Input negative for the time information (RS422 level)	Output negative for the time information (RS422 level)	Receive negative of the RS-485
	8	Input/Output positive for the time information	Unspecified	Input positive for the time information (RS422 level)	Output positive for the time information (RS422 level)	Receive positive of the RS-485
NOTE When this interface is used as a TOD interface, it must be used with a dedicated clock cable. For details, see "Cables" in the Hardware Description.						

Functions and Features

Table 1-365 Functions and features of the CR8DIPU1T2M6

Functions and Features	Description
Basic function	Data forwarding: The IPU is the core of service processing in the entire system, and is connected to all subcards through data channels. Control and management: The IPU manages subcards and transmits routing protocol data through management channels. 2.4 Tbit/s@310 bytes (bidirectional)
Reliability and availability	Hot swappable. 1:1 backup is supported.
Restrictions and remarks	1. It cannot be used together with other IPU's. 2. When working with the 03033VMC board to support FlexE services, the board supports a maximum of eight FlexE physical interfaces.
Breakout supported	No
Interface rate auto-sensing supported	No

Technical Specifications

Table 1-366 Technical specifications of the CR8DIPU1T2M6

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	34.8 mm x 354.4 mm x 209 mm (1.37 in. x 13.95 in. x 8.23 in.)
Dimensions with packaging (H x W x D) [mm(in.)]	185 mm x 485 mm x 385 mm (7.28 in. x 19.09 in. x 15.16 in.)
Weight without packaging [kg(lb)]	4 kg (8.82 lb)
Weight with packaging [kg(lb)]	5.4 kg (11.9 lb)
Typical power consumption [W]	359.1 W
Typical heat dissipation [BTU/hour]	1165.07 BTU/hour
Long-term operating temperature [°C(°F)]	-20°C to +55°C, startup: ≥ -5°C

Item	Specification
CPU	20-core 2.0 GHz
SDRAM	16GB
Storage	4 GB NandFlash

1.4.3.33 CR8DIPU1T2MS (Integrated Network Processing Unit (IPU-1T2-DA))

Overview

Table 1-367 Basic information about the CR8DIPU1T2MS

Item	Details
Description	Integrated Network Processing Unit (IPU-1T2-DA)
Part Number	03034WFB
Model	CR8DIPU1T2MS
Card type	IPU
Silkscreen	IPU-1T2-DA

Appearance

Figure 1-102 Appearance of the CR8DIPU1T2MS



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03034WFB&productModel=CR8DIPU1T2MS#matchRelation>

Indicators and Buttons

Table 1-368 Indicators on the CR8DIPU1T2MS

Silkscreen	Name	Color	Status	Description
STAT	Working status indicator	Green	Steady on	The board is working properly.
		Red	Steady on	The board hardware is faulty.
		Orange	Steady on	The interface board is not loaded with logic.
		-	Off	The board is not powered on or is not running.
PROG	Program status indicator	Green	Steady on	The board software is being initialized or is successfully initialized and is running properly.
		Red	Steady on	The memory self-check fails. The board software or logic file is lost. The board software fails to be loaded.
		Green	Blinks on and off at 100 ms intervals	The board software is being loaded.
		Green	Blinks on and off at 300 ms intervals	The BIOS is booting the upper-layer software.

Silkscreen	Name	Color	Status	Description
		Red	Blinks on and off at 100 ms intervals	The BIOS self-check fails.
		-	Off	The board is not powered on.
SYNC	Clock synchronization indicator	Green	Steady on	The system clock works in free-run mode and no priority list is configured for the system clock. (By default, the list contains only internal clock sources.) The system clock is working in tracing mode and is tracing a clock source rather than the internal sources in the priority list. The system clock is working in time synchronization mode, and the PTP time and system clock are in the tracing state.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	The system clock priority list is set. All the clock sources, however, are lost except for the internal clock sources. The clock works in holdover mode or free-run mode. The system clock is working in time synchronization mode, but no synchronization source is available. The system clock and PTP time are working in holdover or free-run mode.
ACTX	Active/standby status indicator for the cross-connect or clock board	Green	Steady on	The board is in the active switching state.
		-	Off	The board is in the standby switching state.
ACTC	Control status indicator	Green	Steady on	The board is in the active control state.
		Green	Blinks on and off at 100 ms intervals	The data of the device is backed up in batch.

Silkscreen	Name	Color	Status	Description
		-	Off	The board is in the standby control state.
ALM	Device alarm indicator	Red	Steady on	A device-level critical alarm is generated.
		Orange	Steady on	There is an equipment-level major or minor alarm.
		-	Off	There is no equipment-level alarm.
ETH/OAM	ETH/OAM indicator	Green	Steady on	The link is up.
		Orange	Blinking	Data is being transmitted or received.
		-	Off	The link is down.

Table 1-369 Buttons on the CR8DIPU1T2MS

Silkscreen	Name	Description
RST	Board reset button	You can press the RST button to reset the NE or restore the factory settings of the NE. The procedure is as follows: To reset the NE, press the RST button and release it. To restore factory settings, press and hold down the RST button. This operation will clear the NE database and system parameter configurations and cannot be restored. Exercise caution when you perform the following operations: 1.Press the RST button to restart the NE. 2.When the PROG indicator blinks (the blinking frequency is about 1 second and the time window is about 15 seconds), press and hold down the RST button. 3.Wait for no longer than 2 minutes till both the STAT and PROG indicators blink (the time window is 5 seconds). Then release the RST button immediately.

Silkscreen	Name	Description
		NOTE 1. When you press the RST button to clear the configuration file, the original configuration file will be cleared. You are advised to periodically back up the configuration file. 2. When you press the RST button to clear the configuration file, to prevent the cleared configuration from being synchronized again between the active and standby main control boards, remove one main control board first in a dual-main control board chassis to ensure that the configuration file is cleared in a single-main control board scenario. Perform the same operations on the other main control board. 3. When pressing the RST button to clear the configuration file, you only need to pay attention to the STAT and PROG indicators. Other indicators may vary depending on the product model.

Ports

Table 1-370 Ports on the CR8DIPU1T2MS

Port	Connector Type	Description	Available Components
CLK/TEST	SMA	Clock test port	Coaxial cable
USB	Type A	USB port (reserved)	-

Port	Connector Type	Description	Available Components
ETH/OAM	RJ45	10M/100M auto-sensing Ethernet NM interface and non-standard RS-232 Console interface. NOTE If a service port can be connected to an NMS, you are advised to use the service port as the network management port.	ETH/OAM management cable
CLK/TOD/RS-485	RJ45	Clock/Time/RS-485 input and output ports RS-485 is a reserved port.	1-into-2 clock cable + clock cable (external clock mode, delivery-ready): applicable to the CLK mode 1-into-2 clock cable + clock cable (external clock/external time mode, prepared onsite): applicable to the CLK or TOD mode 1-into-2 clock cable + Ethernet cable: applicable to the TOD mode
ALMI/ALMO	RJ45	Alarm input/output interface	Alarm input/output cable
HP-GE	SFP	Ultra-high-precision GE port (reserved)	Optical fiber

Table 1-371 Pins of the ETH/OAM interface

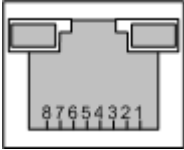
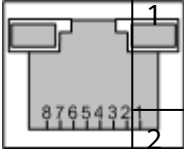
Front View	Pin	Usage
	1	Transmit positive of the NM network port
	2	Transmit negative of the NM network port
	3	Receive positive of the NM network port
	4	Grounding end of the Console interface
	5	Receive end of the Console interface
	6	Receive negative of the NM network port
	7	Unspecified
	8	Transmit end of the Console interface

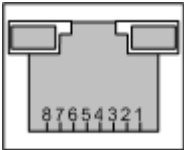
Table 1-372 Pins of the CLK/TOD/RS-485 interface

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	1	Receive negative of the CLK	Receive negative of the CLK	Unspecified	Unspecified	Unspecified
	2	Receive positive of the CLK	Receive positive of the CLK	Unspecified	Unspecified	Unspecified

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	3	Input/Output negative for the 1PPS signal	Unspecified	Input negative for the 1PPS signal (RS422 level)	Output negative for the 1PPS signal (RS422 level)	Transmit negative of the RS-485
	4	Receive negative of the CLK	Receive negative of the CLK	Grounding end	Grounding end	Unspecified
	5	Receive positive of the CLK	Receive positive of the CLK	Grounding end	Grounding end	Unspecified
	6	Input/Output positive for the 1PPS signal	Unspecified	Input positive for the 1PPS signal (RS422 level)	Output positive for the 1PPS signal (RS422 level)	Transmit positive of the RS-485

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	7	Input/Output negative for the time information	Unspecified	Input negative for the time information (RS422 level)	Output negative for the time information (RS422 level)	Receive negative of the RS-485
	8	Input/Output positive for the time information	Unspecified	Input positive for the time information (RS422 level)	Output positive for the time information (RS422 level)	Receive positive of the RS-485
NOTE When this interface is used as a TOD interface, it must be used with a dedicated clock cable. For details, see "Cables" in the Hardware Description.						

Table 1-373 Pins of the ALMI/ALMO interface

Front View	Pin	Usage
	1	Alarm input 1
	2	Grounding end for alarm input 1
	3	Alarm input 2

Front View	Pin	Usage
	4	Alarm input 3
	5	Grounding end for alarm input 3
	6	Grounding end for alarm input 2
	7	Positive for alarm signal output
	8	Negative for alarm signal output

Functions and Features

Table 1-374 Functions and features of the CR8DIPU1T2MS

Functions and Features	Description
Basic function	Line rate: 1200 Gbit/s. Supports switching, control, and clock management. Supports board-level 1:1 backup. Supports the ultra-high-precision clock.
Reliability and availability	Hot swappable. 1:1 backup is supported.

Technical Specifications

Table 1-375 Technical specifications of the CR8DIPU1T2MS

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	34.8 mm x 354.4 mm x 209 mm (1.37 in. x 13.95 in. x 8.23 in.)
Dimensions with packaging (H x W x D) [mm(in.)]	185 mm x 485 mm x 385 mm (7.28 in. x 19.09 in. x 15.16 in.)
Weight without packaging [kg(lb)]	2.55 kg (5.62 lb)
Weight with packaging [kg(lb)]	3.95 kg (8.71 lb)
Typical power consumption [W]	135.6 W
Typical heat dissipation [BTU/hour]	440 BTU/hour

Item	Specification
Long-term operating temperature [°C(°F)]	–20°C to +55°C, startup: ≥ –5°C
Switching capacity	2.4 Tbit/s
CPU	8-core 1.5 GHz
SDRAM	8GB
Storage	4 GB NandFlash

1.4.3.34 CR8DIPU1T2MT (Integrated Network Processing Unit (IPU-1T2-DA))

Overview

Table 1-376 Basic information about the CR8DIPU1T2MT

Item	Details
Description	Integrated Network Processing Unit (IPU-1T2-DA)
Part Number	03034WFC
Model	CR8DIPU1T2MT
Card type	IPU
Silkscreen	IPU-1T2-DA

Appearance

Figure 1-103 Appearance of the CR8DIPU1T2MT



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03034WFC&productModel=CR8DIPU1T2MT#matchRelation>

Indicators and Buttons

Table 1-377 Indicators on the CR8DIPU1T2MT

Silkscreen	Name	Color	Status	Description
STAT	Working status indicator	Green	Steady on	The board is working properly.
		Red	Steady on	A hardware alarm is generated. You can run the display alarm hardware command to check the hardware alarm type.
		-	Off	The board is not powered on or is not running.
PROG	Program status indicator	Green	Steady on	The board software is being initialized or is successfully initialized and is running properly.
		Red	Steady on	The memory self-check fails. The board software or logic file is lost. The board software fails to be loaded.

Silkscreen	Name	Color	Status	Description
		Green	Blinks on and off at 100 ms intervals	The board software is being loaded.
		Green	Blinks on and off at 300 ms intervals	The BIOS is interacting with the upper-layer software.
		Red	Blinks on and off at 100 ms intervals	The BIOS self-check fails.
		-	Off	The board is not powered on.
SYNC	Clock synchronization indicator	Green	Steady on	The system clock works in free-run mode, and no priority list is set for the system clocks (only internal clock sources are listed by default). The system clock works in tracing mode and is tracing a clock source other than the internal clock sources in the priority list. The system clock works in time synchronization mode, and the PTP time and system clock are being traced.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	The system clock priority list is set, but all clock sources except the internal clock source are lost. The clock works in holdover or free-run mode. The system clock works in time synchronization mode, but no synchronization source is available. The system clock and PTP time work in holdover or free-run mode.
ACTX	Active/standby status indicator for the cross-connect or clock board	Green	Steady on	The board is in the active state on the service plane.
		-	Off	The board is in the standby state on the service plane.
ACTC	Control status indicator	Green	Steady on	The board is in the active control state.
		Green	Blinks on and off at 100 ms intervals	The data of the device is backed up in batch.
		-	Off	The board is in the standby control state.

Silkscreen	Name	Color	Status	Description
ALM	Alarm indicator	Red	Steady on	Device-level critical alarms exist.
		Orange	Steady on	A device-level major or minor alarm exists.
		-	Off	There is no equipment-level alarm.
ETH/OAM	ETH/OAM indicator	Green	Steady on	The link is up.
		Orange	Blinking	Data is being transmitted or received.
		-	Off	The link is down.

Table 1-378 Buttons on the CR8DIPU1T2MT

Silkscreen	Name	Description
RST	Board reset button	You can press the RST button to reset the NE or restore the factory settings of the NE. The procedure is as follows: To reset the NE, press the RST button and release it. To restore factory settings, press and hold down the RST button. This operation will clear the NE database and system parameter configurations and cannot be restored. Exercise caution when you perform the following operations: 1.Press the RST button to restart the NE. 2.When the PROG indicator blinks (the blinking frequency is about 1 second and the time window is about 15 seconds), press and hold down the RST button. 3.Wait for no longer than 2 minutes till both the STAT and PROG indicators blink (the time window is 5 seconds). Then release the RST button immediately.

Silkscreen	Name	Description
		NOTE 1. When you press the RST button to clear the configuration file, the original configuration file will be cleared. You are advised to periodically back up the configuration file. 2. When you press the RST button to clear the configuration file, to prevent the cleared configuration from being synchronized again between the active and standby main control boards, remove one main control board first in a dual-main control board chassis to ensure that the configuration file is cleared in a single-main control board scenario. Perform the same operations on the other main control board. 3. When pressing the RST button to clear the configuration file, you only need to pay attention to the STAT and PROG indicators. Other indicators may vary depending on the product model.

Ports

Table 1-379 Ports on the CR8DIPU1T2MT

Port	Connector Type	Description	Available Components
ETH/OAM	RJ45	10M/100M auto-sensing Ethernet network management port or non-standard RS-232 console port NOTE If a service port can be connected to an NMS, you are advised to use the service port as the network management port.	ETH/OAM management cable
CLK/TOD/RS-485	RJ45	Clock/Time/RS-485 input and output ports RS-485 is a reserved port.	1-into-2 clock cable + clock cable (external clock mode, delivery-ready): applicable to the CLK mode 1-into-2 clock cable + clock cable (external clock/external time mode, prepared onsite): applicable to the CLK or TOD mode 1-into-2 clock cable + Ethernet cable: applicable to the TOD mode
ALMI/ALMO	RJ45	Alarm input/output interface	Alarm input/output cable
USB	Type A	USB port (reserved)	-
CLK/TEST	SMA	Clock test port	Coaxial cable

Table 1-380 Pins of the ETH/OAM interface

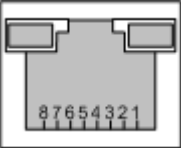
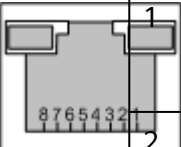
Front View	Pin	Usage
	1	Transmit positive of the NM network port
	2	Transmit negative of the NM network port
	3	Receive positive of the NM network port
	4	Grounding end of the Console interface
	5	Receive end of the Console interface
	6	Receive negative of the NM network port
	7	Unspecified
	8	Transmit end of the Console interface

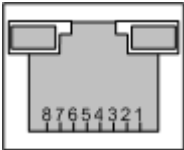
Table 1-381 Pins of the CLK/TOD/RS-485 interface

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	1	Receive negative of the CLK	Receive negative of the CLK	Unspecified	Unspecified	Unspecified
	2	Receive positive of the CLK	Receive positive of the CLK	Unspecified	Unspecified	Unspecified

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	3	Input/Output negative for the 1PPS signal	Unspecified	Input negative for the 1PPS signal (RS422 level)	Output negative for the 1PPS signal (RS422 level)	Transmit negative of the RS-485
	4	Receive negative of the CLK	Receive negative of the CLK	Grounding end	Grounding end	Unspecified
	5	Receive positive of the CLK	Receive positive of the CLK	Grounding end	Grounding end	Unspecified
	6	Input/Output positive for the 1PPS signal	Unspecified	Input positive for the 1PPS signal (RS422 level)	Output positive for the 1PPS signal (RS422 level)	Transmit positive of the RS-485

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	7	Input/Output negative for the time information	Unspecified	Input negative for the time information (RS422 level)	Output negative for the time information (RS422 level)	Receive negative of the RS-485
	8	Input/Output positive for the time information	Unspecified	Input positive for the time information (RS422 level)	Output positive for the time information (RS422 level)	Receive positive of the RS-485
NOTE When this interface is used as a TOD interface, it must be used with a dedicated clock cable. For details, see "Cables" in the Hardware Description.						

Table 1-382 Pins of the ALMI/ALMO interface

Front View	Pin	Usage
	1	Alarm input 1
	2	Grounding end for alarm input 1
	3	Alarm input 2

Front View	Pin	Usage
	4	Alarm input 3
	5	Grounding end for alarm input 3
	6	Grounding end for alarm input 2
	7	Positive for alarm signal output
	8	Negative for alarm signal output

Functions and Features

Table 1-383 Functions and features of the CR8DIPU1T2MT

Functions and Features	Description
Basic function	Data forwarding: The IPU is the core of service processing in the entire system, and is connected to all subcards through data channels. Control and management: The IPU manages subcards and transmits routing protocol data through management channels. 2.4 Tbit/s@310 bytes (bidirectional)
Reliability and availability	Hot swappable. 1:1 backup is supported.
Restrictions and remarks	This board cannot be used together with other IPU.
Breakout supported	No
Interface rate auto-sensing supported	No

Technical Specifications

Table 1-384 Technical specifications of the CR8DIPU1T2MT

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	22.4 mm x 388.4 mm x 209.3 mm (0.88 in. x 15.29 in. x 8.24 in.)

Item	Specification
Dimensions with packaging (H x W x D) [mm(in.)]	170 mm x 560 mm x 385 mm (6.69 in. x 22.05 in. x 15.16 in.)
Weight without packaging [kg(lb)]	2.3 kg (5.07 lb)
Weight with packaging [kg(lb)]	3.7 kg (8.16 lb)
Typical power consumption [W]	144.2 W
Typical heat dissipation [BTU/hour]	467.94 BTU/hour
Long-term operating temperature [°C(°F)]	–40°C to +65°C (–40°F to +149°F), start: ≥ –20°C (–4°F)
CPU	8-core 1.5 GHz
SDRAM	8GB
Storage	4 GB NandFlash

1.4.3.35 CR8DIPU240C1 (Integrated Network Processing Unit (IPU-240))

Overview

Table 1-385 Basic information about the CR8DIPU240C1

Item	Details
Description	Integrated Network Processing Unit (IPU-240)
Part Number	03033NNR
Model	CR8DIPU240C1
Card type	IPU
Silkscreen	IPU-240

Appearance

Figure 1-104 Appearance of the CR8DIPU240C1



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03033NNR&productModel=CR8DIPU240C1#matchRelation>

Indicators and Buttons

Table 1-386 Indicators on the CR8DIPU240C1

Silkscreen	Name	Color	Status	Description
STAT	Working status indicator	Green	Steady on	The board is working normally.
		Red	Steady on	The device generates a hardware alarm. You can run the display alarm hardware command to view the hardware alarm type.
		-	Off	The board is not running or no power is input.
PROG	Program running indicator	Green	Steady on	The board software is being initialized. The board software is normally initialized, and the board software is running normally.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	The memory self-check fails. The board software or the logic file is lost. Loading of the board software fails.
		Green	On for 100 ms and off for 100 ms alternately	Loading of the board software is in process.
		Green	On for 300 ms and off for 300 ms alternately	The BIOS is guiding the upper-layer software.
		Red	On for 100 ms and off for 100 ms alternately	The BIOS self-check fails.
		-	Off	No power is input.

Silkscreen	Name	Color	Status	Description
SYNC	Synchronizati on status indicator	Green	Steady on	The system clock works in free-run mode, and no priority list is set for the system clocks (only internal clock sources are listed by default). The system clock works in tracing mode and is tracing a clock source other than the internal clock sources in the priority list. The system clock works in time synchronizatio n mode, and the PTP time and system clock are being traced.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	The system clock priority list is set, but all clock sources except the internal clock source are lost. The clock works in holdover or free-run mode. The system clock works in time synchronization mode, but no synchronization source is available. The system clock and PTP time work in holdover or free-run mode.
ACTX	Active/standby status indicator for the cross-connect or clock board	Green	Steady on	The board is in the working switching state.
		-	Off	The board is in the protection switching state.
ACTC	Control status indicator	Green	Steady on	The board is in the working control state.
		Green	On for 100 ms and off for 100 ms alternately	The data of the equipment is backed up in batch.

Silkscreen	Name	Color	Status	Description
		-	Off	The board is in the protection control state.
ALM	Alarm indicator	Red	Steady on	Critical alarms are generated.
		Orange	Steady on	Major, minor alarms are generated.
		-	Off	No alarms are generated in the equipment.
ETH/OAM	ETH/OAM indicator	Green	Steady on	If the indicator is steady on, the link is normal.
		Orange	Blink	If the indicator blinks, data is being transmitted.
		-	Off	If the indicator is off, the link is Down.

Table 1-387 Buttons on the CR8DIPU240C1

Silkscreen	Name	Description
RST	Reset button	You can press the RST button to reset the NE or restore the factory settings of the NE. The procedure is as follows: To reset the NE, press the RST button and release it. To restore factory settings, press and hold the RST button. This operation will clear the NE database and system parameter configurations and cannot be restored. Exercise caution when you perform the following operations: 1. Press and release the RST button to restart the NE. 2. When the PROG indicator blinks (the blinking frequency is about 1 second and the time window is about 15 seconds), press and hold the RST button. 3. Wait for no longer than 2 minutes till both the STAT and PROG indicators blink (the time window is 5 seconds). Then release the RST button immediately.

Silkscreen	Name	Description
		NOTE 1. When you press the RST button to clear the configuration file, the original configuration file will be cleared. You are advised to periodically back up the configuration file. 2. When you press the RST button to clear the configuration file, to prevent the cleared configuration from being synchronized again between the active and standby main control boards, remove one main control board first in a dual-main control board chassis to ensure that the configuration file is cleared in a single-main control board scenario. Perform the same operations on the other main control board. 3. When pressing the RST button to clear the configuration file, you only need to pay attention to the STAT and PROG indicators. Other indicators may vary depending on the product model.

Ports

Table 1-388 Ports on the CR8DIPU240C1

Port	Connector Type	Description	Available Components
ETH/OAM	RJ45	10M/100M autosensing Ethernet network management port or non-RS-232-standard console port. NOTE If a service port can be connected to the NMS, it is recommended that the service port be used as the network management port.	ETH/OAM Management Cables
CLK/TOD/RS-485	RJ45	Clock/Time/RS-485 input/output port, RS-485 is a reserved port.	1-into-2 clock cable + clock cable (external clock mode, delivery-ready): applicable to the CLK mode 1-into-2 clock cable + clock cable (external clock/external time mode, prepared onsite): applicable to the CLK or TOD mode 1-into-2 clock cable + Ethernet cable: applicable to the TOD mode
ALMI/ALMO(Reserved)	RJ45	Alarm input/output port	Alarm Input/Output Cables
USB	Type A	USB port (reserved)	-
CLK/TEST	SMA	Clock test port	Coaxial Cable

Table 1-389 Pins of the ETH/OAM interface

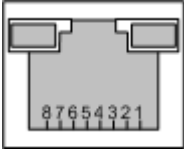
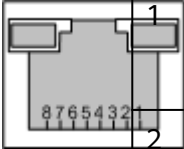
Front View	Pin	Usage
	1	Transmit positive of the NM network port
	2	Transmit negative of the NM network port
	3	Receive positive of the NM network port
	4	Grounding end of the Console interface
	5	Receive end of the Console interface
	6	Receive negative of the NM network port
	7	Unspecified
	8	Transmit end of the Console interface

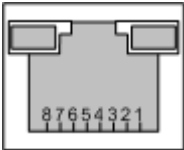
Table 1-390 Pins of the CLK/TOD/RS-485 interface

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	1	Receive negative of the CLK	Receive negative of the CLK	Unspecified	Unspecified	Unspecified
	2	Receive positive of the CLK	Receive positive of the CLK	Unspecified	Unspecified	Unspecified

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	3	Input/Output negative for the 1PPS signal	Unspecified	Input negative for the 1PPS signal (RS422 level)	Output negative for the 1PPS signal (RS422 level)	Transmit negative of the RS-485
	4	Receive negative of the CLK	Receive negative of the CLK	Grounding end	Grounding end	Unspecified
	5	Receive positive of the CLK	Receive positive of the CLK	Grounding end	Grounding end	Unspecified
	6	Input/Output positive for the 1PPS signal	Unspecified	Input positive for the 1PPS signal (RS422 level)	Output positive for the 1PPS signal (RS422 level)	Transmit positive of the RS-485

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	7	Input/Output negative for the time information	Unspecified	Input negative for the time information (RS422 level)	Output negative for the time information (RS422 level)	Receive negative of the RS-485
	8	Input/Output positive for the time information	Unspecified	Input positive for the time information (RS422 level)	Output positive for the time information (RS422 level)	Receive positive of the RS-485
NOTE When this interface is used as a TOD interface, it must be used with a dedicated clock cable. For details, see "Cables" in the Hardware Description.						

Table 1-391 Pins of the ALMI/ALMO interface

Front View	Pin	Usage
	1	Alarm input 1
	2	Grounding end for alarm input 1
	3	Alarm input 2

Front View	Pin	Usage
	4	Alarm input 3
	5	Grounding end for alarm input 3
	6	Grounding end for alarm input 2
	7	Positive for alarm signal output
	8	Negative for alarm signal output

Functions and Features

Table 1-392 Functions and features of the CR8DIPU240C1

Functions and Features	Description
Basic function	Data forwarding: The IPU is the core of service processing in the entire system, and is connected to all subcards through data channels. Control and management: Through the management channels between IPU and the subcards, the IPU can manage subcards and transmit routing protocol data. The line rate capability is 480G bit/s@100Bytes(Bidirectional).
Reliability and availability	Supports the hot swappable function. Supports the board-level 1:1 backup function.
Restrictions and remarks	This board cannot be used together with other IPU.
Breakout supported	No
Interface rate auto-sensing supported	No

Technical Specifications

Table 1-393 Technical specifications of the CR8DIPU240C1

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	22.4 mm x 388.4 mm x 209.3 mm (0.88 in. x 15.29 in. x 8.24 in.)
Dimensions with packaging (H x W x D) [mm(in.)]	170 mm x 560 mm x 385 mm (6.69 in. x 22.05 in. x 15.16 in.)
Weight without packaging [kg(lb)]	2.3 kg (5.07 lb)
Weight with packaging [kg(lb)]	3.7 kg (8.16 lb)
Typical power consumption [W]	113.3 W
Typical heat dissipation [BTU/hour]	367.66 BTU/hour
Long-term operating temperature [°C(°F)]	–40°C to +65°C (–40°F to +149°F), start: ≥ –20°C (–4°F)
CPU	8-core 1.5 GHz
SDRAM	8GB
Storage	4GB NandFlash

1.4.3.36 CR8DIPU480C1 (03033GLB: Integrated Network Processing Unit (IPU-480))

Overview

Table 1-394 Basic information about the CR8DIPU480C1

Item	Details
Description	Integrated Network Processing Unit (IPU-480)
Part Number	03033GLB
Model	CR8DIPU480C1
Card type	IPU
Silkscreen	IPU-480

Appearance

Figure 1-105 Appearance of the CR8DIPU480C1



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03033GLB&productModel=CR8DIPU480C1#matchRelation>

Indicators and Buttons

Table 1-395 Indicators on the CR8DIPU480C1

Silkscreen	Name	Color	Status	Description
STAT	Working status indicator	Green	Steady on	The board is working normally.
		Red	Steady on	The device generates a hardware alarm. You can run the display alarm hardware command to view the hardware alarm type.
		-	Off	The board is not running or no power is input.

Silkscreen	Name	Color	Status	Description
PROG	Program running indicator	Green	Steady on	The board software is being initialized. The board software is normally initialized, and the board software is running normally.
		Red	Steady on	The memory self-check fails. The board software or the logic file is lost. Loading of the board software fails.
		Green	On for 100 ms and off for 100 ms alternately	Loading of the board software is in process.
		Green	On for 300 ms and off for 300 ms alternately	The BIOS is guiding the upper-layer software.
		Red	On for 100 ms and off for 100 ms alternately	The BIOS self-check fails.
		-	Off	No power is input.

Silkscreen	Name	Color	Status	Description
SYNC	Synchronizati on status indicator	Green	Steady on	The system clock works in free-run mode, and no priority list is set for the system clocks (only internal clock sources are listed by default). The system clock works in tracing mode and is tracing a clock source other than the internal clock sources in the priority list. The system clock works in time synchronizatio n mode, and the PTP time and system clock are being traced.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	The system clock priority list is set, but all clock sources except the internal clock source are lost. The clock works in holdover or free-run mode. The system clock works in time synchronization mode, but no synchronization source is available. The system clock and PTP time work in holdover or free-run mode.
ACTX	Active/standby status indicator for the cross-connect or clock board	Green	Steady on	The board is in the working switching state.
		-	Off	The board is in the protection switching state.
ACTC	Control status indicator	Green	Steady on	The board is in the working control state.
		Green	On for 100 ms and off for 100 ms alternately	The data of the equipment is backed up in batch.

Silkscreen	Name	Color	Status	Description
		-	Off	The board is in the protection control state.
ALM	Alarm indicator	Red	Steady on	Critical alarms are generated.
		Orange	Steady on	Major, minor alarms are generated.
		-	Off	No alarms are generated in the equipment.
ETH/OAM	ETH/OAM indicator	Green	Steady on	If the indicator is steady on, the link is normal.
		Orange	Blink	If the indicator blinks, data is being transmitted.
		-	Off	If the indicator is off, the link is Down.

Table 1-396 Buttons on the CR8DIPU480C1

Silkscreen	Name	Description
RST	Reset button	You can press the RST button to reset the NE or restore the factory settings of the NE. The procedure is as follows: To reset the NE, press the RST button and release it. To restore factory settings, press and hold the RST button. This operation will clear the NE database and system parameter configurations and cannot be restored. Exercise caution when you perform the following operations: 1. Press and release the RST button to restart the NE. 2. When the PROG indicator blinks (the blinking frequency is about 1 second and the time window is about 15 seconds), press and hold the RST button. 3. Wait for no longer than 2 minutes till both the STAT and PROG indicators blink (the time window is 5 seconds). Then release the RST button immediately.

Silkscreen	Name	Description
		NOTE 1. When you press the RST button to clear the configuration file, the original configuration file will be cleared. You are advised to periodically back up the configuration file. 2. When you press the RST button to clear the configuration file, to prevent the cleared configuration from being synchronized again between the active and standby main control boards, remove one main control board first in a dual-main control board chassis to ensure that the configuration file is cleared in a single-main control board scenario. Perform the same operations on the other main control board. 3. When pressing the RST button to clear the configuration file, you only need to pay attention to the STAT and PROG indicators. Other indicators may vary depending on the product model.

Ports

Table 1-397 Ports on the CR8DIPU480C1

Port	Connector Type	Description	Available Components
ETH/OAM	RJ45	10M/100M autosensing Ethernet network management port or non-RS-232-standard console port. NOTE If a service port can be connected to the NMS, it is recommended that the service port be used as the network management port.	ETH/OAM Management Cables
CLK/TOD/RS-485	RJ45	Clock/Time/RS-485 input/output port, RS-485 is a reserved port.	1-into-2 clock cable + clock cable (external clock mode, delivery-ready): applicable to the CLK mode 1-into-2 clock cable + clock cable (external clock/external time mode, prepared onsite): applicable to the CLK or TOD mode 1-into-2 clock cable + Ethernet cable: applicable to the TOD mode
ALMI/ALMO	RJ45	Alarm input/output port	Alarm Input/Output Cables
USB	Type A	USB port (reserved)	-
CLK/TEST	SMA	Clock test port	Coaxial Cable

Table 1-398 Pins of the ETH/OAM interface

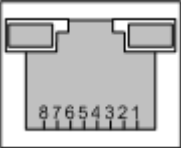
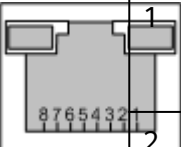
Front View	Pin	Usage
	1	Transmit positive of the NM network port
	2	Transmit negative of the NM network port
	3	Receive positive of the NM network port
	4	Grounding end of the Console interface
	5	Receive end of the Console interface
	6	Receive negative of the NM network port
	7	Unspecified
	8	Transmit end of the Console interface

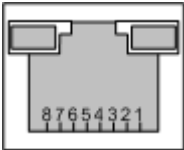
Table 1-399 Pins of the CLK/TOD/RS-485 interface

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	1	Receive negative of the CLK	Receive negative of the CLK	Unspecified	Unspecified	Unspecified
	2	Receive positive of the CLK	Receive positive of the CLK	Unspecified	Unspecified	Unspecified

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	3	Input/Output negative for the 1PPS signal	Unspecified	Input negative for the 1PPS signal (RS422 level)	Output negative for the 1PPS signal (RS422 level)	Transmit negative of the RS-485
	4	Receive negative of the CLK	Receive negative of the CLK	Grounding end	Grounding end	Unspecified
	5	Receive positive of the CLK	Receive positive of the CLK	Grounding end	Grounding end	Unspecified
	6	Input/Output positive for the 1PPS signal	Unspecified	Input positive for the 1PPS signal (RS422 level)	Output positive for the 1PPS signal (RS422 level)	Transmit positive of the RS-485

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	7	Input/Output negative for the time information	Unspecified	Input negative for the time information (RS422 level)	Output negative for the time information (RS422 level)	Receive negative of the RS-485
	8	Input/Output positive for the time information	Unspecified	Input positive for the time information (RS422 level)	Output positive for the time information (RS422 level)	Receive positive of the RS-485
NOTE When this interface is used as a TOD interface, it must be used with a dedicated clock cable. For details, see "Cables" in the Hardware Description.						

Table 1-400 Pins of the ALMI/ALMO interface

Front View	Pin	Usage
	1	Alarm input 1
	2	Grounding end for alarm input 1
	3	Alarm input 2

Front View	Pin	Usage
	4	Alarm input 3
	5	Grounding end for alarm input 3
	6	Grounding end for alarm input 2
	7	Positive for alarm signal output
	8	Negative for alarm signal output

Functions and Features

Table 1-401 Functions and features of the CR8DIPU480C1

Functions and Features	Description
Basic function	Data forwarding: The IPU is the core of service processing in the entire system, and is connected to all subcards through data channels. Control and management: Through the management channels between IPU's and the subcards, the IPU's can manage subcards and transmit routing protocol data. The line rate capability is 960G bit/s@190Bytes(Bidirectional).
Reliability and availability	Supports the hot swappable function. Supports the board-level 1:1 backup function.
Restrictions and remarks	This board cannot be used together with other IPU's.
Breakout supported	No
Interface rate auto-sensing supported	No

Technical Specifications

Table 1-402 Technical specifications of the CR8DIPU480C1

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	22.4 mm x 388.4 mm x 209.3 mm (0.88 in. x 15.29 in. x 8.24 in.)
Dimensions with packaging (H x W x D) [mm(in.)]	170 mm x 560 mm x 385 mm (6.69 in. x 22.05 in. x 15.16 in.)
Weight without packaging [kg(lb)]	2.3 kg (5.07 lb)
Weight with packaging [kg(lb)]	3.7 kg (8.16 lb)
Typical power consumption [W]	113.3 W
Typical heat dissipation [BTU/hour]	367.66 BTU/hour
Long-term operating temperature [°C(°F)]	–40°C to +65°C (–40°F to +149°F), start: ≥ –20°C (–4°F)
CPU	8-core 1.5 GHz
SDRAM	8GB
Storage	4GB NandFlash

1.4.3.37 CR8DIPU480C1 (03033GLB-001: Integrated Network Processing Unit (IPU-480))

Overview

Table 1-403 Basic information about the CR8DIPU480C1

Item	Details
Description	Integrated Network Processing Unit (IPU-480)
Part Number	03033GLB-001
Model	CR8DIPU480C1
Card type	IPU
Silkscreen	IPU-480

Appearance

Figure 1-106 Appearance of the CR8DIPU480C1



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03033GLB-001&productModel=CR8DIPU480C1#matchRelation>

Indicators and Buttons

Table 1-404 Indicators on the CR8DIPU480C1

Silkscreen	Name	Color	Status	Description
STAT	Working status indicator	Green	Steady on	The board is working normally.
		Red	Steady on	The device generates a hardware alarm. You can run the display alarm hardware command to view the hardware alarm type.
		-	Off	The board is not running or no power is input.

Silkscreen	Name	Color	Status	Description
PROG	Program running indicator	Green	Steady on	The board software is being initialized. The board software is normally initialized, and the board software is running normally.
		Red	Steady on	The memory self-check fails. The board software or the logic file is lost. Loading of the board software fails.
		Green	On for 100 ms and off for 100 ms alternately	Loading of the board software is in process.
		Green	On for 300 ms and off for 300 ms alternately	The BIOS is guiding the upper-layer software.
		Red	On for 100 ms and off for 100 ms alternately	The BIOS self-check fails.
		-	Off	No power is input.

Silkscreen	Name	Color	Status	Description
SYNC	Synchronizati on status indicator	Green	Steady on	The system clock works in free-run mode, and no priority list is set for the system clocks (only internal clock sources are listed by default). The system clock works in tracing mode and is tracing a clock source other than the internal clock sources in the priority list. The system clock works in time synchronizatio n mode, and the PTP time and system clock are being traced.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	The system clock priority list is set, but all clock sources except the internal clock source are lost. The clock works in holdover or free-run mode. The system clock works in time synchronization mode, but no synchronization source is available. The system clock and PTP time work in holdover or free-run mode.
ACTX	Active/standby status indicator for the cross-connect or clock board	Green	Steady on	The board is in the working switching state.
		-	Off	The board is in the protection switching state.
ACTC	Control status indicator	Green	Steady on	The board is in the working control state.
		Green	On for 100 ms and off for 100 ms alternately	The data of the equipment is backed up in batch.

Silkscreen	Name	Color	Status	Description
		-	Off	The board is in the protection control state.
ALM	Alarm indicator	Red	Steady on	Critical alarms are generated.
		Orange	Steady on	Major, minor alarms are generated.
		-	Off	No alarms are generated in the equipment.
ETH/OAM	ETH/OAM indicator	Green	Steady on	If the indicator is steady on, the link is normal.
		Orange	Blink	If the indicator blinks, data is being transmitted.
		-	Off	If the indicator is off, the link is Down.

Table 1-405 Buttons on the CR8DIPU480C1

Silkscreen	Name	Description
RST	Reset button	You can press the RST button to reset the NE or restore the factory settings of the NE. The procedure is as follows: To reset the NE, press the RST button and release it. To restore factory settings, press and hold the RST button. This operation will clear the NE database and system parameter configurations and cannot be restored. Exercise caution when you perform the following operations: 1. Press and release the RST button to restart the NE. 2. When the PROG indicator blinks (the blinking frequency is about 1 second and the time window is about 15 seconds), press and hold the RST button. 3. Wait for no longer than 2 minutes till both the STAT and PROG indicators blink (the time window is 5 seconds). Then release the RST button immediately.

Silkscreen	Name	Description
		NOTE 1. When you press the RST button to clear the configuration file, the original configuration file will be cleared. You are advised to periodically back up the configuration file. 2. When you press the RST button to clear the configuration file, to prevent the cleared configuration from being synchronized again between the active and standby main control boards, remove one main control board first in a dual-main control board chassis to ensure that the configuration file is cleared in a single-main control board scenario. Perform the same operations on the other main control board. 3. When pressing the RST button to clear the configuration file, you only need to pay attention to the STAT and PROG indicators. Other indicators may vary depending on the product model.

Ports

Table 1-406 Ports on the CR8DIPU480C1

Port	Connector Type	Description	Available Components
ETH/OAM	RJ45	10M/100M autosensing Ethernet network management port or non-RS-232-standard console port. NOTE If a service port can be connected to the NMS, it is recommended that the service port be used as the network management port.	ETH/OAM Management Cables
CLK/TOD/RS-485	RJ45	Clock/Time/RS-485 input/output port, RS-485 is a reserved port.	1-into-2 clock cable + clock cable (external clock mode, delivery-ready): applicable to the CLK mode 1-into-2 clock cable + clock cable (external clock/external time mode, prepared onsite): applicable to the CLK or TOD mode 1-into-2 clock cable + Ethernet cable: applicable to the TOD mode
ALMI/ALMO	RJ45	Alarm input/output port	Alarm Input/Output Cables
USB	Type A	USB port (reserved)	-
CLK/TEST	SMA	Clock test port	Coaxial Cable

Table 1-407 Pins of the ETH/OAM interface

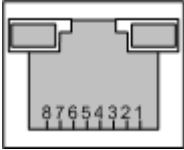
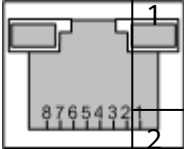
Front View	Pin	Usage
	1	Transmit positive of the NM network port
	2	Transmit negative of the NM network port
	3	Receive positive of the NM network port
	4	Grounding end of the Console interface
	5	Receive end of the Console interface
	6	Receive negative of the NM network port
	7	Unspecified
	8	Transmit end of the Console interface

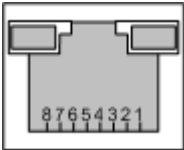
Table 1-408 Pins of the CLK/TOD/RS-485 interface

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	1	Receive negative of the CLK	Receive negative of the CLK	Unspecified	Unspecified	Unspecified
	2	Receive positive of the CLK	Receive positive of the CLK	Unspecified	Unspecified	Unspecified

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	3	Input/Output negative for the 1PPS signal	Unspecified	Input negative for the 1PPS signal (RS422 level)	Output negative for the 1PPS signal (RS422 level)	Transmit negative of the RS-485
	4	Receive negative of the CLK	Receive negative of the CLK	Grounding end	Grounding end	Unspecified
	5	Receive positive of the CLK	Receive positive of the CLK	Grounding end	Grounding end	Unspecified
	6	Input/Output positive for the 1PPS signal	Unspecified	Input positive for the 1PPS signal (RS422 level)	Output positive for the 1PPS signal (RS422 level)	Transmit positive of the RS-485

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	7	Input/Output negative for the time information	Unspecified	Input negative for the time information (RS422 level)	Output negative for the time information (RS422 level)	Receive negative of the RS-485
	8	Input/Output positive for the time information	Unspecified	Input positive for the time information (RS422 level)	Output positive for the time information (RS422 level)	Receive positive of the RS-485
NOTE When this interface is used as a TOD interface, it must be used with a dedicated clock cable. For details, see "Cables" in the Hardware Description.						

Table 1-409 Pins of the ALMI/ALMO interface

Front View	Pin	Usage
	1	Alarm input 1
	2	Grounding end for alarm input 1
	3	Alarm input 2

Front View	Pin	Usage
	4	Alarm input 3
	5	Grounding end for alarm input 3
	6	Grounding end for alarm input 2
	7	Positive for alarm signal output
	8	Negative for alarm signal output

Functions and Features

Table 1-410 Functions and features of the CR8DIPU480C1

Functions and Features	Description
Basic function	Data forwarding: The IPU is the core of service processing in the entire system, and is connected to all subcards through data channels. Control and management: Through the management channels between IPUs and the subcards, the IPUs can manage subcards and transmit routing protocol data. The line rate capability is 960G bit/s@190Bytes(Bidirectional).
Reliability and availability	Supports the hot swappable function. Supports the board-level 1:1 backup function.
Restrictions and remarks	This board cannot be used together with other IPU.
Breakout supported	No
Interface rate auto-sensing supported	No

Technical Specifications

Table 1-411 Technical specifications of the CR8DIPU480C1

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	22.4 mm x 388.4 mm x 209.3 mm (0.88 in. x 15.29 in. x 8.24 in.)
Dimensions with packaging (H x W x D) [mm(in.)]	170 mm x 560 mm x 385 mm (6.69 in. x 22.05 in. x 15.16 in.)
Weight without packaging [kg(lb)]	2.3 kg (5.07 lb)
Weight with packaging [kg(lb)]	3.7 kg (8.16 lb)
Typical power consumption [W]	113.3 W
Typical heat dissipation [BTU/hour]	367.66 BTU/hour
Long-term operating temperature [°C(°F)]	–40°C to +65°C (–40°F to +149°F), start: ≥ –20°C (–4°F)
CPU	8-core 1.5 GHz
SDRAM	8GB
Storage	4GB NandFlash

1.4.3.38 CR8DIPU480C1 (03033GLB-002: Integrated Network Processing Unit (IPU-480))

Overview

Table 1-412 Basic information about the CR8DIPU480C1

Item	Details
Description	Integrated Network Processing Unit (IPU-480)
Part Number	03033GLB-002
Model	CR8DIPU480C1
Card type	IPU
Silkscreen	IPU-480

Appearance

Figure 1-107 Appearance of the CR8DIPU480C1



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03033GLB-002&productModel=CR8DIPU480C1#matchRelation>

Indicators and Buttons

Table 1-413 Indicators on the CR8DIPU480C1

Silkscreen	Name	Color	Status	Description
STAT	Working status indicator	Green	Steady on	The board is working properly.
		Red	Steady on	A hardware alarm is generated. You can run the display alarm hardware command to check the hardware alarm type.
		-	Off	The board is not powered on or is not running.

Silkscreen	Name	Color	Status	Description
PROG	Program status indicator	Green	Steady on	The board software is being initialized or is successfully initialized and is running properly.
		Red	Steady on	The memory self-check fails. The board software or logic file is lost. The board software fails to be loaded.
		Green	Blinks on and off at 100 ms intervals	The board software is being loaded.
		Green	Blinks on and off at 300 ms intervals	The BIOS is interacting with the upper-layer software.
		Red	Blinks on and off at 100 ms intervals	The BIOS self-check fails.
		-	Off	The board is not powered on.

Silkscreen	Name	Color	Status	Description
SYNC	Clock synchronizatio n indicator	Green	Steady on	The system clock works in free-run mode, and no priority list is set for the system clocks (only internal clock sources are listed by default). The system clock works in tracing mode and is tracing a clock source other than the internal clock sources in the priority list. The system clock works in time synchronizatio n mode, and the PTP time and system clock are being traced.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	The system clock priority list is set, but all clock sources except the internal clock source are lost. The clock works in holdover or free-run mode. The system clock works in time synchronization mode, but no synchronization source is available. The system clock and PTP time work in holdover or free-run mode.
ACTX	Active/standby status indicator for the cross-connect or clock board	Green	Steady on	The board is in the active state on the service plane.
		-	Off	The board is in the standby state on the service plane.
ACTC	Control status indicator	Green	Steady on	The board is in the active control state.
		Green	Blinks on and off at 100 ms intervals	The data of the device is backed up in batch.
		-	Off	The board is in the standby control state.

Silkscreen	Name	Color	Status	Description
ALM	Alarm indicator	Red	Steady on	Device-level critical alarms exist.
		Orange	Steady on	A device-level major or minor alarm exists.
		-	Off	There is no equipment-level alarm.
ETH/OAM	ETH/OAM indicator	Green	Steady on	The link is up.
		Orange	Blinking	Data is being transmitted or received.
		-	Off	The link is down.

Table 1-414 Buttons on the CR8DIPU480C1

Silkscreen	Name	Description
RST	Board reset button	You can press the RST button to reset the NE or restore the factory settings of the NE. The procedure is as follows: To reset the NE, press the RST button and release it. To restore factory settings, press and hold down the RST button. This operation will clear the NE database and system parameter configurations and cannot be restored. Exercise caution when you perform the following operations: 1.Press the RST button to restart the NE. 2.When the PROG indicator blinks (the blinking frequency is about 1 second and the time window is about 15 seconds), press and hold down the RST button. 3.Wait for no longer than 2 minutes till both the STAT and PROG indicators blink (the time window is 5 seconds). Then release the RST button immediately.

Silkscreen	Name	Description
		NOTE 1. When you press the RST button to clear the configuration file, the original configuration file will be cleared. You are advised to periodically back up the configuration file. 2. When you press the RST button to clear the configuration file, to prevent the cleared configuration from being synchronized again between the active and standby main control boards, remove one main control board first in a dual-main control board chassis to ensure that the configuration file is cleared in a single-main control board scenario. Perform the same operations on the other main control board. 3. When pressing the RST button to clear the configuration file, you only need to pay attention to the STAT and PROG indicators. Other indicators may vary depending on the product model.

Ports

Table 1-415 Ports on the CR8DIPU480C1

Port	Connector Type	Description	Available Components
ETH/OAM	RJ45	10M/100M auto-sensing Ethernet network management port or non-standard RS-232 console port NOTE If a service port can be connected to an NMS, you are advised to use the service port as the network management port.	ETH/OAM management cable
CLK/TOD/RS-485	RJ45	Clock/Time/RS-485 input and output ports RS-485 is a reserved port.	1-into-2 clock cable + clock cable (external clock mode, delivery-ready): applicable to the CLK mode 1-into-2 clock cable + clock cable (external clock/external time mode, prepared onsite): applicable to the CLK or TOD mode 1-into-2 clock cable + Ethernet cable: applicable to the TOD mode
ALMI/ALMO	RJ45	Alarm input/output interface	Alarm input/output cable
USB	Type A	USB port (reserved)	-
CLK/TEST	SMA	Clock test port	Coaxial cable

Table 1-416 Pins of the ETH/OAM interface

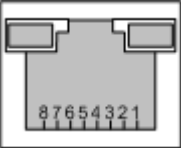
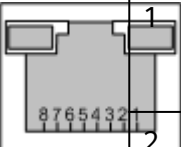
Front View	Pin	Usage
	1	Transmit positive of the NM network port
	2	Transmit negative of the NM network port
	3	Receive positive of the NM network port
	4	Grounding end of the Console interface
	5	Receive end of the Console interface
	6	Receive negative of the NM network port
	7	Unspecified
	8	Transmit end of the Console interface

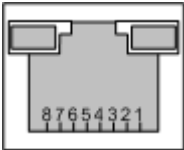
Table 1-417 Pins of the CLK/TOD/RS-485 interface

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	1	Receive negative of the CLK	Receive negative of the CLK	Unspecified	Unspecified	Unspecified
	2	Receive positive of the CLK	Receive positive of the CLK	Unspecified	Unspecified	Unspecified

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	3	Input/Output negative for the 1PPS signal	Unspecified	Input negative for the 1PPS signal (RS422 level)	Output negative for the 1PPS signal (RS422 level)	Transmit negative of the RS-485
	4	Receive negative of the CLK	Receive negative of the CLK	Grounding end	Grounding end	Unspecified
	5	Receive positive of the CLK	Receive positive of the CLK	Grounding end	Grounding end	Unspecified
	6	Input/Output positive for the 1PPS signal	Unspecified	Input positive for the 1PPS signal (RS422 level)	Output positive for the 1PPS signal (RS422 level)	Transmit positive of the RS-485

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	7	Input/Output negative for the time information	Unspecified	Input negative for the time information (RS422 level)	Output negative for the time information (RS422 level)	Receive negative of the RS-485
	8	Input/Output positive for the time information	Unspecified	Input positive for the time information (RS422 level)	Output positive for the time information (RS422 level)	Receive positive of the RS-485
NOTE When this interface is used as a TOD interface, it must be used with a dedicated clock cable. For details, see "Cables" in the Hardware Description.						

Table 1-418 Pins of the ALMI/ALMO interface

Front View	Pin	Usage
	1	Alarm input 1
	2	Grounding end for alarm input 1
	3	Alarm input 2

Front View	Pin	Usage
	4	Alarm input 3
	5	Grounding end for alarm input 3
	6	Grounding end for alarm input 2
	7	Positive for alarm signal output
	8	Negative for alarm signal output

Functions and Features

Table 1-419 Functions and features of the CR8DIPU480C1

Functions and Features	Description
Basic function	Data forwarding: The IPU is the core of service processing in the entire system, and is connected to all subcards through data channels. Control and management: The IPU manages subcards and transmits routing protocol data through management channels. 960 Gbit/s@190 bytes (bidirectional)
Reliability and availability	Hot swappable. 1:1 backup is supported.
Restrictions and remarks	This board cannot be used together with other IPU.
Breakout supported	No
Interface rate auto-sensing supported	No

Technical Specifications

Table 1-420 Technical specifications of the CR8DIPU480C1

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	22.4 mm x 388.4 mm x 209.3 mm (0.88 in. x 15.29 in. x 8.24 in.)

Item	Specification
Dimensions with packaging (H x W x D) [mm(in.)]	170 mm x 560 mm x 385 mm (6.69 in. x 22.05 in. x 15.16 in.)
Weight without packaging [kg(lb)]	2.3 kg (5.07 lb)
Weight with packaging [kg(lb)]	3.7 kg (8.16 lb)
Typical power consumption [W]	113.3 W
Typical heat dissipation [BTU/hour]	367.66 BTU/hour
Long-term operating temperature [°C(°F)]	–40°C to +65°C (–40°F to +149°F), start: ≥ –20°C (–4°F)
CPU	8-core 1.5 GHz
SDRAM	8GB
Storage	4 GB NandFlash

1.4.3.39 CR8DIPU480C1 (03035JVA: Integrated Network Processing Unit (IPU-480) for SME)

Overview

Table 1-421 Basic information about the CR8DIPU480C1

Item	Details
Description	Integrated Network Processing Unit (IPU-480) for SME
Part Number	03035JVA
Model	CR8DIPU480C1
Card type	IPU
Silkscreen	IPU-480

Appearance

Figure 1-108 Appearance of the CR8DIPU480C1



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03035JVA&productModel=CR8DIPU480C1#matchRelation>

Indicators and Buttons

Table 1-422 Indicators on the CR8DIPU480C1

Silkscreen	Name	Color	Status	Description
STAT	Working status indicator	Green	Steady on	The board is working properly.
		Red	Steady on	A hardware alarm is generated. You can run the display alarm hardware command to check the hardware alarm type.
		-	Off	The board is not powered on or is not running.
PROG	Program status indicator	Green	Steady on	The board software is being initialized or is successfully initialized and is running properly.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	The memory self-check fails. The board software or logic file is lost. The board software fails to be loaded.
		Green	Blinks on and off at 100 ms intervals	The board software is being loaded.
		Green	Blinks on and off at 300 ms intervals	The BIOS is interacting with the upper-layer software.
		Red	Blinks on and off at 100 ms intervals	The BIOS self-check fails.
		-	Off	The board is not powered on.

Silkscreen	Name	Color	Status	Description
SYNC	Clock synchronizatio n indicator	Green	Steady on	The system clock works in free-run mode, and no priority list is set for the system clocks (only internal clock sources are listed by default). The system clock works in tracing mode and is tracing a clock source other than the internal clock sources in the priority list. The system clock works in time synchronizatio n mode, and the PTP time and system clock are being traced.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	The system clock priority list is set, but all clock sources except the internal clock source are lost. The clock works in holdover or free-run mode. The system clock works in time synchronization mode, but no synchronization source is available. The system clock and PTP time work in holdover or free-run mode.
ACTX	Active/standby status indicator for the cross-connect or clock board	Green	Steady on	The board is in the active state on the service plane.
		-	Off	The board is in the standby state on the service plane.
ACTC	Control status indicator	Green	Steady on	The board is in the active control state.
		Green	Blinks on and off at 100 ms intervals	The data of the device is backed up in batch.
		-	Off	The board is in the standby control state.

Silkscreen	Name	Color	Status	Description
ALM	Alarm indicator	Red	Steady on	Device-level critical alarms exist.
		Orange	Steady on	A device-level major or minor alarm exists.
		-	Off	There is no equipment-level alarm.
ETH/OAM	ETH/OAM indicator	Green	Steady on	The link is up.
		Orange	Blinking	Data is being transmitted or received.
		-	Off	The link is down.

Table 1-423 Buttons on the CR8DIPU480C1

Silkscreen	Name	Description
RST	Board reset button	You can press the RST button to reset the NE or restore the factory settings of the NE. The procedure is as follows: To reset the NE, press the RST button and release it. To restore factory settings, press and hold down the RST button. This operation will clear the NE database and system parameter configurations and cannot be restored. Exercise caution when you perform the following operations: 1.Press the RST button to restart the NE. 2.When the PROG indicator blinks (the blinking frequency is about 1 second and the time window is about 15 seconds), press and hold down the RST button. 3.Wait for no longer than 2 minutes till both the STAT and PROG indicators blink (the time window is 5 seconds). Then release the RST button immediately.

Silkscreen	Name	Description
		NOTE 1. When you press the RST button to clear the configuration file, the original configuration file will be cleared. You are advised to periodically back up the configuration file. 2. When you press the RST button to clear the configuration file, to prevent the cleared configuration from being synchronized again between the active and standby main control boards, remove one main control board first in a dual-main control board chassis to ensure that the configuration file is cleared in a single-main control board scenario. Perform the same operations on the other main control board. 3. When pressing the RST button to clear the configuration file, you only need to pay attention to the STAT and PROG indicators. Other indicators may vary depending on the product model.

Ports

Table 1-424 Ports on the CR8DIPU480C1

Port	Connector Type	Description	Available Components
ETH/OAM	RJ45	10M/100M auto-sensing Ethernet network management port or non-standard RS-232 console port NOTE If a service port can be connected to an NMS, you are advised to use the service port as the network management port.	ETH/OAM management cable
CLK/TOD/RS-485	RJ45	Clock/Time/RS-485 input and output ports RS-485 is a reserved port.	1-into-2 clock cable + clock cable (external clock mode, delivery-ready): applicable to the CLK mode 1-into-2 clock cable + clock cable (external clock/external time mode, prepared onsite): applicable to the CLK or TOD mode 1-into-2 clock cable + Ethernet cable: applicable to the TOD mode
ALMI/ALMO	RJ45	Alarm input/output interface	Alarm input/output cable
USB	Type A	USB port (reserved)	-
CLK/TEST	SMA	Clock test port	Coaxial cable

Table 1-425 Pins of the ETH/OAM interface

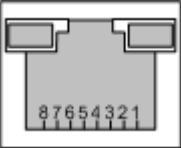
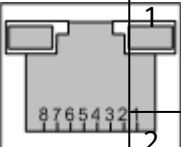
Front View	Pin	Usage
	1	Transmit positive of the NM network port
	2	Transmit negative of the NM network port
	3	Receive positive of the NM network port
	4	Grounding end of the Console interface
	5	Receive end of the Console interface
	6	Receive negative of the NM network port
	7	Unspecified
	8	Transmit end of the Console interface

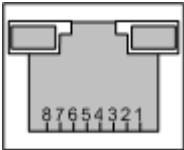
Table 1-426 Pins of the CLK/TOD/RS-485 interface

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	1	Receive negative of the CLK	Receive negative of the CLK	Unspecified	Unspecified	Unspecified
	2	Receive positive of the CLK	Receive positive of the CLK	Unspecified	Unspecified	Unspecified

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	3	Input/Output negative for the 1PPS signal	Unspecified	Input negative for the 1PPS signal (RS422 level)	Output negative for the 1PPS signal (RS422 level)	Transmit negative of the RS-485
	4	Receive negative of the CLK	Receive negative of the CLK	Grounding end	Grounding end	Unspecified
	5	Receive positive of the CLK	Receive positive of the CLK	Grounding end	Grounding end	Unspecified
	6	Input/Output positive for the 1PPS signal	Unspecified	Input positive for the 1PPS signal (RS422 level)	Output positive for the 1PPS signal (RS422 level)	Transmit positive of the RS-485

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	7	Input/Output negative for the time information	Unspecified	Input negative for the time information (RS422 level)	Output negative for the time information (RS422 level)	Receive negative of the RS-485
	8	Input/Output positive for the time information	Unspecified	Input positive for the time information (RS422 level)	Output positive for the time information (RS422 level)	Receive positive of the RS-485
NOTE When this interface is used as a TOD interface, it must be used with a dedicated clock cable. For details, see "Cables" in the Hardware Description.						

Table 1-427 Pins of the ALMI/ALMO interface

Front View	Pin	Usage
	1	Alarm input 1
	2	Grounding end for alarm input 1
	3	Alarm input 2

Front View	Pin	Usage
	4	Alarm input 3
	5	Grounding end for alarm input 3
	6	Grounding end for alarm input 2
	7	Positive for alarm signal output
	8	Negative for alarm signal output

Functions and Features

Table 1-428 Functions and features of the CR8DIPU480C1

Functions and Features	Description
Basic function	Data forwarding: The IPU is the core of service processing in the entire system, and is connected to all subcards through data channels. Control and management: The IPU manages subcards and transmits routing protocol data through management channels. 960 Gbit/s@190 bytes (bidirectional)
Reliability and availability	Hot swappable. 1:1 backup is supported.
Restrictions and remarks	This board cannot be used together with other IPU.
Breakout supported	No
Interface rate auto-sensing supported	No

Technical Specifications

Table 1-429 Technical specifications of the CR8DIPU480C1

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	22.4 mm x 388.4 mm x 209.3 mm (0.88 in. x 15.29 in. x 8.24 in.)

Item	Specification
Dimensions with packaging (H x W x D) [mm(in.)]	170 mm x 560 mm x 385 mm (6.69 in. x 22.05 in. x 15.16 in.)
Weight without packaging [kg(lb)]	2.3 kg (5.07 lb)
Weight with packaging [kg(lb)]	3.7 kg (8.16 lb)
Typical power consumption [W]	113.3 W
Typical heat dissipation [BTU/hour]	367.66 BTU/hour
Long-term operating temperature [°C(°F)]	–40°C to +65°C (–40°F to +149°F), start: ≥ –20°C (–4°F)
CPU	8-core 1.5 GHz
SDRAM	8GB
Storage	4 GB NandFlash

1.4.3.40 CR8DIPU480C3 (Integrated Network Processing Unit (IPU-480-B))

Overview

Table 1-430 Basic information about the CR8DIPU480C3

Item	Details
Description	Integrated Network Processing Unit (IPU-480-B)
Part Number	03034BMK
Model	CR8DIPU480C3
Card type	IPU
Silkscreen	IPU-480-B

Appearance

Figure 1-109 Appearance of the CR8DIPU480C3



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03034BMK&productModel=CR8DIPU480C3#matchRelation>

Indicators and Buttons

Table 1-431 Indicators on the CR8DIPU480C3

Silkscreen	Name	Color	Status	Description
STAT	Working status indicator	Green	Steady on	The board is working normally.
		Red	Steady on	The device generates a hardware alarm. You can run the display alarm hardware command to view the hardware alarm type.
		-	Off	The board is not running or no power is input.
PROG	Program running indicator	Green	Steady on	The board software is being initialized. The board software is normally initialized, and the board software is running normally.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	The memory self-check fails. The board software or the logic file is lost. Loading of the board software fails.
		Green	On for 100 ms and off for 100 ms alternately	Loading of the board software is in process.
		Green	On for 300 ms and off for 300 ms alternately	The BIOS is guiding the upper-layer software.
		Red	On for 100 ms and off for 100 ms alternately	The BIOS self-check fails.
		-	Off	No power is input.

Silkscreen	Name	Color	Status	Description
SYNC	Synchronizati on status indicator	Green	Steady on	The system clock works in free-run mode, and no priority list is set for the system clocks (only internal clock sources are listed by default). The system clock works in tracing mode and is tracing a clock source other than the internal clock sources in the priority list. The system clock works in time synchronizatio n mode, and the PTP time and system clock are being traced.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	The system clock priority list is set, but all clock sources except the internal clock source are lost. The clock works in holdover or free-run mode. The system clock works in time synchronization mode, but no synchronization source is available. The system clock and PTP time work in holdover or free-run mode.
ACTX	Active/standby status indicator for the cross-connect or clock board	Green	Steady on	The board is in the working switching state.
		-	Off	The board is in the protection switching state.
ACTC	Control status indicator	Green	Steady on	The board is in the working control state.
		Green	On for 100 ms and off for 100 ms alternately	The data of the equipment is backed up in batch.

Silkscreen	Name	Color	Status	Description
		-	Off	The board is in the protection control state.
ALM	Alarm indicator	Red	Steady on	Critical alarms are generated.
		Orange	Steady on	Major, minor alarms are generated.
		-	Off	No alarms are generated in the equipment.
ETH/OAM	ETH/OAM indicator	Green	Steady on	If the indicator is steady on, the link is normal.
		Orange	Blink	If the indicator blinks, data is being transmitted.
		-	Off	If the indicator is off, the link is Down.

Table 1-432 Buttons on the CR8DIPU480C3

Silkscreen	Name	Description
RST	Reset button	You can press the RST button to reset the NE or restore the factory settings of the NE. The procedure is as follows: To reset the NE, press the RST button and release it. To restore factory settings, press and hold the RST button. This operation will clear the NE database and system parameter configurations and cannot be restored. Exercise caution when you perform the following operations: 1. Press and release the RST button to restart the NE. 2. When the PROG indicator blinks (the blinking frequency is about 1 second and the time window is about 15 seconds), press and hold the RST button. 3. Wait for no longer than 2 minutes till both the STAT and PROG indicators blink (the time window is 5 seconds). Then release the RST button immediately.

Silkscreen	Name	Description
		NOTE 1. When you press the RST button to clear the configuration file, the original configuration file will be cleared. You are advised to periodically back up the configuration file. 2. When you press the RST button to clear the configuration file, to prevent the cleared configuration from being synchronized again between the active and standby main control boards, remove one main control board first in a dual-main control board chassis to ensure that the configuration file is cleared in a single-main control board scenario. Perform the same operations on the other main control board. 3. When pressing the RST button to clear the configuration file, you only need to pay attention to the STAT and PROG indicators. Other indicators may vary depending on the product model.

Ports

Table 1-433 Ports on the CR8DIPU480C3

Port	Connector Type	Description	Available Components
ETH/OAM	RJ45	10M/100M autosensing Ethernet network management port or non-RS-232-standard console port. NOTE If a service port can be connected to the NMS, it is recommended that the service port be used as the network management port.	ETH/OAM Management Cables
CLK/TOD/RS-485	RJ45	Clock/Time/RS-485 input/output port, RS-485 is a reserved port.	1-into-2 clock cable + clock cable (external clock mode, delivery-ready): applicable to the CLK mode 1-into-2 clock cable + clock cable (external clock/external time mode, prepared onsite): applicable to the CLK or TOD mode 1-into-2 clock cable + Ethernet cable: applicable to the TOD mode
ALMI/ALMO	RJ45	Alarm input/output port	Alarm Input/Output Cables
USB	Type A	USB port (reserved)	-
CLK/TEST	SMA	Clock test port	Coaxial Cable

Table 1-434 Pins of the ETH/OAM interface

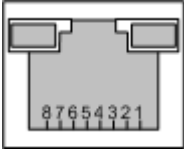
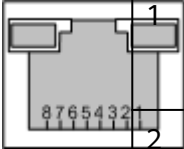
Front View	Pin	Usage
	1	Transmit positive of the NM network port
	2	Transmit negative of the NM network port
	3	Receive positive of the NM network port
	4	Grounding end of the Console interface
	5	Receive end of the Console interface
	6	Receive negative of the NM network port
	7	Unspecified
	8	Transmit end of the Console interface

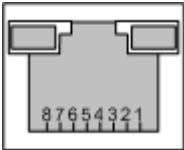
Table 1-435 Pins of the CLK/TOD/RS-485 interface

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	1	Receive negative of the CLK	Receive negative of the CLK	Unspecified	Unspecified	Unspecified
	2	Receive positive of the CLK	Receive positive of the CLK	Unspecified	Unspecified	Unspecified

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	3	Input/Output negative for the 1PPS signal	Unspecified	Input negative for the 1PPS signal (RS422 level)	Output negative for the 1PPS signal (RS422 level)	Transmit negative of the RS-485
	4	Receive negative of the CLK	Receive negative of the CLK	Grounding end	Grounding end	Unspecified
	5	Receive positive of the CLK	Receive positive of the CLK	Grounding end	Grounding end	Unspecified
	6	Input/Output positive for the 1PPS signal	Unspecified	Input positive for the 1PPS signal (RS422 level)	Output positive for the 1PPS signal (RS422 level)	Transmit positive of the RS-485

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	7	Input/Output negative for the time information	Unspecified	Input negative for the time information (RS422 level)	Output negative for the time information (RS422 level)	Receive negative of the RS-485
	8	Input/Output positive for the time information	Unspecified	Input positive for the time information (RS422 level)	Output positive for the time information (RS422 level)	Receive positive of the RS-485
NOTE When this interface is used as a TOD interface, it must be used with a dedicated clock cable. For details, see "Cables" in the Hardware Description.						

Table 1-436 Pins of the ALMI/ALMO interface

Front View	Pin	Usage
	1	Alarm input 1
	2	Grounding end for alarm input 1
	3	Alarm input 2

Front View	Pin	Usage
	4	Alarm input 3
	5	Grounding end for alarm input 3
	6	Grounding end for alarm input 2
	7	Positive for alarm signal output
	8	Negative for alarm signal output

Functions and Features

Table 1-437 Functions and features of the CR8DIPU480C3

Functions and Features	Description
Basic function	Data forwarding: The IPU is the core of service processing in the entire system, and is connected to all subcards through data channels. Control and management: Through the management channels between IPU and the subcards, the IPU can manage subcards and transmit routing protocol data. The line rate capability is 960G bit/s@190Bytes(Bidirectional).
Reliability and availability	Supports the hot swappable function. Supports the board-level 1:1 backup function.
Restrictions and remarks	This board cannot be used together with other IPU.
Breakout supported	No
Interface rate auto-sensing supported	No

Technical Specifications

Table 1-438 Technical specifications of the CR8DIPU480C3

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	22.4 mm x 388.4 mm x 209.3 mm (0.88 in. x 15.29 in. x 8.24 in.)
Dimensions with packaging (H x W x D) [mm(in.)]	170 mm x 560 mm x 385 mm (6.69 in. x 22.05 in. x 15.16 in.)
Weight without packaging [kg(lb)]	2.3 kg (5.07 lb)
Weight with packaging [kg(lb)]	3.7 kg (8.16 lb)
Typical power consumption [W]	113.3 W
Typical heat dissipation [BTU/hour]	367.66 BTU/hour
Long-term operating temperature [°C(°F)]	–40°C to +65°C (–40°F to +149°F), start: ≥ –20°C (–4°F)
CPU	8-core 1.5 GHz
SDRAM	8GB
Storage	4GB NandFlash

1.4.3.41 CR8DIPU480C5 (03034JTH: Integrated Network Processing Unit (IPU-480-BN))

Overview

Table 1-439 Basic information about the CR8DIPU480C5

Item	Details
Description	Integrated Network Processing Unit (IPU-480-BN)
Part Number	03034JTH
Model	CR8DIPU480C5
Card type	IPU
Silkscreen	IPU-480-BN

Appearance

Figure 1-110 Appearance of the CR8DIPU480C5



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03034JTH&productModel=CR8DIPU480C5#matchRelation>

Indicators and Buttons

Table 1-440 Indicators on the CR8DIPU480C5

Silkscreen	Name	Color	Status	Description
STAT	Working status indicator	Green	Steady on	The board is working normally.
		Red	Steady on	The device generates a hardware alarm. You can run the display alarm hardware command to view the hardware alarm type.
		-	Off	The board is not running or no power is input.

Silkscreen	Name	Color	Status	Description
PROG	Program running indicator	Green	Steady on	The board software is being initialized. The board software is normally initialized, and the board software is running normally.
		Red	Steady on	The memory self-check fails. The board software or the logic file is lost. Loading of the board software fails.
		Green	On for 100 ms and off for 100 ms alternately	Loading of the board software is in process.
		Green	On for 300 ms and off for 300 ms alternately	The BIOS is guiding the upper-layer software.
		Red	On for 100 ms and off for 100 ms alternately	The BIOS self-check fails.
		-	Off	No power is input.

Silkscreen	Name	Color	Status	Description
SYNC	Synchronizati on status indicator	Green	Steady on	The system clock works in free-run mode, and no priority list is set for the system clocks (only internal clock sources are listed by default). The system clock works in tracing mode and is tracing a clock source other than the internal clock sources in the priority list. The system clock works in time synchronizatio n mode, and the PTP time and system clock are being traced.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	The system clock priority list is set, but all clock sources except the internal clock source are lost. The clock works in holdover or free-run mode. The system clock works in time synchronization mode, but no synchronization source is available. The system clock and PTP time work in holdover or free-run mode.
ACTX	Active/standby status indicator for the cross-connect or clock board	Green	Steady on	The board is in the working switching state.
		-	Off	The board is in the protection switching state.
ACTC	Control status indicator	Green	Steady on	The board is in the working control state.
		Green	On for 100 ms and off for 100 ms alternately	The data of the equipment is backed up in batch.

Silkscreen	Name	Color	Status	Description
		-	Off	The board is in the protection control state.
ALM	Alarm indicator	Red	Steady on	Critical alarms are generated.
		Orange	Steady on	Major, minor alarms are generated.
		-	Off	No alarms are generated in the equipment.
ETH/OAM	ETH/OAM indicator	Green	Steady on	If the indicator is steady on, the link is normal.
		Orange	Blink	If the indicator blinks, data is being transmitted.
		-	Off	If the indicator is off, the link is Down.

Table 1-441 Buttons on the CR8DIPU480C5

Silkscreen	Name	Description
RST	Reset button	You can press the RST button to reset the NE or restore the factory settings of the NE. The procedure is as follows: To reset the NE, press the RST button and release it. To restore factory settings, press and hold the RST button. This operation will clear the NE database and system parameter configurations and cannot be restored. Exercise caution when you perform the following operations: 1. Press and release the RST button to restart the NE. 2. When the PROG indicator blinks (the blinking frequency is about 1 second and the time window is about 15 seconds), press and hold the RST button. 3. Wait for no longer than 2 minutes till both the STAT and PROG indicators blink (the time window is 5 seconds). Then release the RST button immediately.

Silkscreen	Name	Description
		NOTE 1. When you press the RST button to clear the configuration file, the original configuration file will be cleared. You are advised to periodically back up the configuration file. 2. When you press the RST button to clear the configuration file, to prevent the cleared configuration from being synchronized again between the active and standby main control boards, remove one main control board first in a dual-main control board chassis to ensure that the configuration file is cleared in a single-main control board scenario. Perform the same operations on the other main control board. 3. When pressing the RST button to clear the configuration file, you only need to pay attention to the STAT and PROG indicators. Other indicators may vary depending on the product model.

Ports

Table 1-442 Ports on the CR8DIPU480C5

Port	Connector Type	Description	Available Components
ETH/OAM	RJ45	10M/100M autosensing Ethernet network management port or non-RS-232-standard console port. NOTE If a service port can be connected to the NMS, it is recommended that the service port be used as the network management port.	ETH/OAM Management Cables
CLK/TOD/RS-485	RJ45	Clock/Time/RS-485 input/output port, RS-485 is a reserved port.	1-into-2 clock cable + clock cable (external clock mode, delivery-ready): applicable to the CLK mode 1-into-2 clock cable + clock cable (external clock/external time mode, prepared onsite): applicable to the CLK or TOD mode 1-into-2 clock cable + Ethernet cable: applicable to the TOD mode
ALMI/ALMO	RJ45	Alarm input/output port	Alarm Input/Output Cables
USB	Type A	USB port (reserved)	-
CLK/TEST	SMA	Clock test port	Coaxial Cable

Table 1-443 Pins of the ETH/OAM interface

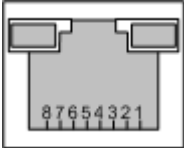
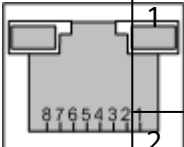
Front View	Pin	Usage
	1	Transmit positive of the NM network port
	2	Transmit negative of the NM network port
	3	Receive positive of the NM network port
	4	Grounding end of the Console interface
	5	Receive end of the Console interface
	6	Receive negative of the NM network port
	7	Unspecified
	8	Transmit end of the Console interface

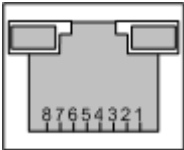
Table 1-444 Pins of the CLK/TOD/RS-485 interface

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	1	Receive negative of the CLK	Receive negative of the CLK	Unspecified	Unspecified	Unspecified
	2	Receive positive of the CLK	Receive positive of the CLK	Unspecified	Unspecified	Unspecified

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	3	Input/Output negative for the 1PPS signal	Unspecified	Input negative for the 1PPS signal (RS422 level)	Output negative for the 1PPS signal (RS422 level)	Transmit negative of the RS-485
	4	Receive negative of the CLK	Receive negative of the CLK	Grounding end	Grounding end	Unspecified
	5	Receive positive of the CLK	Receive positive of the CLK	Grounding end	Grounding end	Unspecified
	6	Input/Output positive for the 1PPS signal	Unspecified	Input positive for the 1PPS signal (RS422 level)	Output positive for the 1PPS signal (RS422 level)	Transmit positive of the RS-485

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	7	Input/Output negative for the time information	Unspecified	Input negative for the time information (RS422 level)	Output negative for the time information (RS422 level)	Receive negative of the RS-485
	8	Input/Output positive for the time information	Unspecified	Input positive for the time information (RS422 level)	Output positive for the time information (RS422 level)	Receive positive of the RS-485
NOTE When this interface is used as a TOD interface, it must be used with a dedicated clock cable. For details, see "Cables" in the Hardware Description.						

Table 1-445 Pins of the ALMI/ALMO interface

Front View	Pin	Usage
	1	Alarm input 1
	2	Grounding end for alarm input 1
	3	Alarm input 2

Front View	Pin	Usage
	4	Alarm input 3
	5	Grounding end for alarm input 3
	6	Grounding end for alarm input 2
	7	Positive for alarm signal output
	8	Negative for alarm signal output

Functions and Features

Table 1-446 Functions and features of the CR8DIPU480C5

Functions and Features	Description
Basic function	Data forwarding: The IPU is the core of service processing in the entire system, and is connected to all subcards through data channels. Control and management: Through the management channels between IPU and the subcards, the IPU can manage subcards and transmit routing protocol data. The line rate capability is 960G bit/s@190Bytes(Bidirectional).
Reliability and availability	Supports the hot swappable function. Supports the board-level 1:1 backup function.
Restrictions and remarks	This board cannot be used together with other IPU.
Breakout supported	No
Interface rate auto-sensing supported	No

Technical Specifications

Table 1-447 Technical specifications of the CR8DIPU480C5

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	22.4 mm x 388.4 mm x 209.3 mm (0.88 in. x 15.29 in. x 8.24 in.)
Dimensions with packaging (H x W x D) [mm(in.)]	170 mm x 560 mm x 385 mm (6.69 in. x 22.05 in. x 15.16 in.)
Weight without packaging [kg(lb)]	2.3 kg (5.07 lb)
Weight with packaging [kg(lb)]	3.7 kg (8.16 lb)
Typical power consumption [W]	113.3 W
Typical heat dissipation [BTU/hour]	367.66 BTU/hour
Long-term operating temperature [°C(°F)]	–40°C to +65°C (–40°F to +149°F), start: ≥ –20°C (–4°F)
CPU	8-core 1.5 GHz
SDRAM	8GB
Storage	4GB NandFlash

1.4.3.42 CR8DIPU480C5 (03034JTH-001: Integrated Network Processing Unit (IPU-480-BN))

Overview

Table 1-448 Basic information about the CR8DIPU480C5

Item	Details
Description	Integrated Network Processing Unit (IPU-480-BN)
Part Number	03034JTH-001
Model	CR8DIPU480C5
Card type	IPU
Silkscreen	IPU-480-BN

Appearance

Figure 1-111 Appearance of the CR8DIPU480C5



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03034JTH-001&productModel=CR8DIPU480C5#matchRelation>

Indicators and Buttons

Table 1-449 Indicators on the CR8DIPU480C5

Silkscreen	Name	Color	Status	Description
STAT	Working status indicator	Green	Steady on	The board is working normally.
		Red	Steady on	The device generates a hardware alarm. You can run the display alarm hardware command to view the hardware alarm type.
		-	Off	The board is not running or no power is input.

Silkscreen	Name	Color	Status	Description
PROG	Program running indicator	Green	Steady on	The board software is being initialized. The board software is normally initialized, and the board software is running normally.
		Red	Steady on	The memory self-check fails. The board software or the logic file is lost. Loading of the board software fails.
		Green	On for 100 ms and off for 100 ms alternately	Loading of the board software is in process.
		Green	On for 300 ms and off for 300 ms alternately	The BIOS is guiding the upper-layer software.
		Red	On for 100 ms and off for 100 ms alternately	The BIOS self-check fails.
		-	Off	No power is input.

Silkscreen	Name	Color	Status	Description
SYNC	Synchronizati on status indicator	Green	Steady on	The system clock works in free-run mode, and no priority list is set for the system clocks (only internal clock sources are listed by default). The system clock works in tracing mode and is tracing a clock source other than the internal clock sources in the priority list. The system clock works in time synchronizatio n mode, and the PTP time and system clock are being traced.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	The system clock priority list is set, but all clock sources except the internal clock source are lost. The clock works in holdover or free-run mode. The system clock works in time synchronization mode, but no synchronization source is available. The system clock and PTP time work in holdover or free-run mode.
ACTX	Active/standby status indicator for the cross-connect or clock board	Green	Steady on	The board is in the working switching state.
		-	Off	The board is in the protection switching state.
ACTC	Control status indicator	Green	Steady on	The board is in the working control state.
		Green	On for 100 ms and off for 100 ms alternately	The data of the equipment is backed up in batch.

Silkscreen	Name	Color	Status	Description
		-	Off	The board is in the protection control state.
ALM	Alarm indicator	Red	Steady on	Critical alarms are generated.
		Orange	Steady on	Major, minor alarms are generated.
		-	Off	No alarms are generated in the equipment.
ETH/OAM	ETH/OAM indicator	Green	Steady on	If the indicator is steady on, the link is normal.
		Orange	Blink	If the indicator blinks, data is being transmitted.
		-	Off	If the indicator is off, the link is Down.

Table 1-450 Buttons on the CR8DIPU480C5

Silkscreen	Name	Description
RST	Reset button	You can press the RST button to reset the NE or restore the factory settings of the NE. The procedure is as follows: To reset the NE, press the RST button and release it. To restore factory settings, press and hold the RST button. This operation will clear the NE database and system parameter configurations and cannot be restored. Exercise caution when you perform the following operations: 1. Press and release the RST button to restart the NE. 2. When the PROG indicator blinks (the blinking frequency is about 1 second and the time window is about 15 seconds), press and hold the RST button. 3. Wait for no longer than 2 minutes till both the STAT and PROG indicators blink (the time window is 5 seconds). Then release the RST button immediately.

Silkscreen	Name	Description
		NOTE 1. When you press the RST button to clear the configuration file, the original configuration file will be cleared. You are advised to periodically back up the configuration file. 2. When you press the RST button to clear the configuration file, to prevent the cleared configuration from being synchronized again between the active and standby main control boards, remove one main control board first in a dual-main control board chassis to ensure that the configuration file is cleared in a single-main control board scenario. Perform the same operations on the other main control board. 3. When pressing the RST button to clear the configuration file, you only need to pay attention to the STAT and PROG indicators. Other indicators may vary depending on the product model.

Ports

Table 1-451 Ports on the CR8DIPU480C5

Port	Connector Type	Description	Available Components
ETH/OAM	RJ45	10M/100M autosensing Ethernet network management port or non-RS-232-standard console port. NOTE If a service port can be connected to the NMS, it is recommended that the service port be used as the network management port.	ETH/OAM Management Cables
CLK/TOD/RS-485	RJ45	Clock/Time/RS-485 input/output port, RS-485 is a reserved port.	1-into-2 clock cable + clock cable (external clock mode, delivery-ready): applicable to the CLK mode 1-into-2 clock cable + clock cable (external clock/external time mode, prepared onsite): applicable to the CLK or TOD mode 1-into-2 clock cable + Ethernet cable: applicable to the TOD mode
ALMI/ALMO	RJ45	Alarm input/output port	Alarm Input/Output Cables
USB	Type A	USB port (reserved)	-
CLK/TEST	SMA	Clock test port	Coaxial Cable

Table 1-452 Pins of the ETH/OAM interface

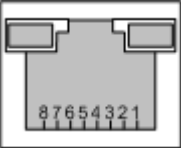
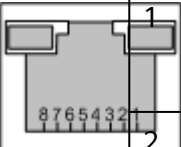
Front View	Pin	Usage
	1	Transmit positive of the NM network port
	2	Transmit negative of the NM network port
	3	Receive positive of the NM network port
	4	Grounding end of the Console interface
	5	Receive end of the Console interface
	6	Receive negative of the NM network port
	7	Unspecified
	8	Transmit end of the Console interface

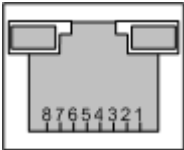
Table 1-453 Pins of the CLK/TOD/RS-485 interface

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	1	Receive negative of the CLK	Receive negative of the CLK	Unspecified	Unspecified	Unspecified
	2	Receive positive of the CLK	Receive positive of the CLK	Unspecified	Unspecified	Unspecified

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	3	Input/Output negative for the 1PPS signal	Unspecified	Input negative for the 1PPS signal (RS422 level)	Output negative for the 1PPS signal (RS422 level)	Transmit negative of the RS-485
	4	Receive negative of the CLK	Receive negative of the CLK	Grounding end	Grounding end	Unspecified
	5	Receive positive of the CLK	Receive positive of the CLK	Grounding end	Grounding end	Unspecified
	6	Input/Output positive for the 1PPS signal	Unspecified	Input positive for the 1PPS signal (RS422 level)	Output positive for the 1PPS signal (RS422 level)	Transmit positive of the RS-485

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	7	Input/Output negative for the time information	Unspecified	Input negative for the time information (RS422 level)	Output negative for the time information (RS422 level)	Receive negative of the RS-485
	8	Input/Output positive for the time information	Unspecified	Input positive for the time information (RS422 level)	Output positive for the time information (RS422 level)	Receive positive of the RS-485
NOTE When this interface is used as a TOD interface, it must be used with a dedicated clock cable. For details, see "Cables" in the Hardware Description.						

Table 1-454 Pins of the ALMI/ALMO interface

Front View	Pin	Usage
	1	Alarm input 1
	2	Grounding end for alarm input 1
	3	Alarm input 2

Front View	Pin	Usage
	4	Alarm input 3
	5	Grounding end for alarm input 3
	6	Grounding end for alarm input 2
	7	Positive for alarm signal output
	8	Negative for alarm signal output

Functions and Features

Table 1-455 Functions and features of the CR8DIPU480C5

Functions and Features	Description
Basic function	Data forwarding: The IPU is the core of service processing in the entire system, and is connected to all subcards through data channels. Control and management: Through the management channels between IPU and the subcards, the IPU can manage subcards and transmit routing protocol data. The line rate capability is 960G bit/s@190Bytes(Bidirectional).
Reliability and availability	Supports the hot swappable function. Supports the board-level 1:1 backup function.
Restrictions and remarks	This board cannot be used together with other IPU.
Breakout supported	No
Interface rate auto-sensing supported	No

Technical Specifications

Table 1-456 Technical specifications of the CR8DIPU480C5

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	22.4 mm x 388.4 mm x 209.3 mm (0.88 in. x 15.29 in. x 8.24 in.)
Dimensions with packaging (H x W x D) [mm(in.)]	170 mm x 560 mm x 385 mm (6.69 in. x 22.05 in. x 15.16 in.)
Weight without packaging [kg(lb)]	2.3 kg (5.07 lb)
Weight with packaging [kg(lb)]	3.7 kg (8.16 lb)
Typical power consumption [W]	113.3 W
Typical heat dissipation [BTU/hour]	367.66 BTU/hour
Long-term operating temperature [°C(°F)]	–40°C to +65°C (–40°F to +149°F), start: ≥ –20°C (–4°F)
CPU	8-core 1.5 GHz
SDRAM	8GB
Storage	4GB NandFlash

1.4.3.43 CR8DIPU480C5 (03034JTH-002: Integrated Network Processing Unit (IPU-480-BN))

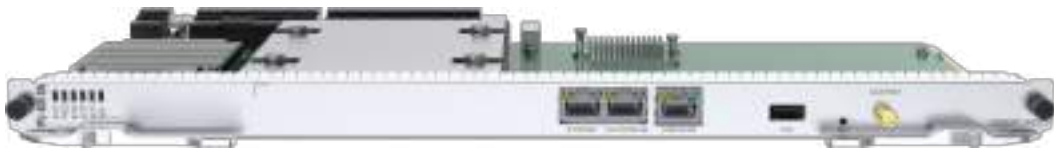
Overview

Table 1-457 Basic information about the CR8DIPU480C5

Item	Details
Description	Integrated Network Processing Unit (IPU-480-BN)
Part Number	03034JTH-002
Model	CR8DIPU480C5
Card type	IPU
Silkscreen	IPU-480-BN

Appearance

Figure 1-112 Appearance of the CR8DIPU480C5



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03034JTH-002&productModel=CR8DIPU480C5#matchRelation>

Indicators and Buttons

Table 1-458 Indicators on the CR8DIPU480C5

Silkscreen	Name	Color	Status	Description
STAT	Working status indicator	Green	Steady on	The board is working properly.
		Red	Steady on	A hardware alarm is generated. You can run the display alarm hardware command to check the hardware alarm type.
		-	Off	The board is not powered on or is not running.

Silkscreen	Name	Color	Status	Description
PROG	Program status indicator	Green	Steady on	The board software is being initialized or is successfully initialized and is running properly.
		Red	Steady on	The memory self-check fails. The board software or logic file is lost. The board software fails to be loaded.
		Green	Blinks on and off at 100 ms intervals	The board software is being loaded.
		Green	Blinks on and off at 300 ms intervals	The BIOS is interacting with the upper-layer software.
		Red	Blinks on and off at 100 ms intervals	The BIOS self-check fails.
		-	Off	The board is not powered on.

Silkscreen	Name	Color	Status	Description
SYNC	Clock synchronizatio n indicator	Green	Steady on	The system clock works in free-run mode, and no priority list is set for the system clocks (only internal clock sources are listed by default). The system clock works in tracing mode and is tracing a clock source other than the internal clock sources in the priority list. The system clock works in time synchronizatio n mode, and the PTP time and system clock are being traced.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	The system clock priority list is set, but all clock sources except the internal clock source are lost. The clock works in holdover or free-run mode. The system clock works in time synchronization mode, but no synchronization source is available. The system clock and PTP time work in holdover or free-run mode.
ACTX	Active/standby status indicator for the cross-connect or clock board	Green	Steady on	The board is in the active state on the service plane.
		-	Off	The board is in the standby state on the service plane.
ACTC	Control status indicator	Green	Steady on	The board is in the active control state.
		Green	Blinks on and off at 100 ms intervals	The data of the device is backed up in batch.
		-	Off	The board is in the standby control state.

Silkscreen	Name	Color	Status	Description
ALM	Alarm indicator	Red	Steady on	Device-level critical alarms exist.
		Orange	Steady on	A device-level major or minor alarm exists.
		-	Off	There is no equipment-level alarm.
ETH/OAM	ETH/OAM indicator	Green	Steady on	The link is up.
		Orange	Blinking	Data is being transmitted or received.
		-	Off	The link is down.

Table 1-459 Buttons on the CR8DIPU480C5

Silkscreen	Name	Description
RST	Board reset button	You can press the RST button to reset the NE or restore the factory settings of the NE. The procedure is as follows: To reset the NE, press the RST button and release it. To restore factory settings, press and hold down the RST button. This operation will clear the NE database and system parameter configurations and cannot be restored. Exercise caution when you perform the following operations: 1.Press the RST button to restart the NE. 2.When the PROG indicator blinks (the blinking frequency is about 1 second and the time window is about 15 seconds), press and hold down the RST button. 3.Wait for no longer than 2 minutes till both the STAT and PROG indicators blink (the time window is 5 seconds). Then release the RST button immediately.

Silkscreen	Name	Description
		NOTE 1. When you press the RST button to clear the configuration file, the original configuration file will be cleared. You are advised to periodically back up the configuration file. 2. When you press the RST button to clear the configuration file, to prevent the cleared configuration from being synchronized again between the active and standby main control boards, remove one main control board first in a dual-main control board chassis to ensure that the configuration file is cleared in a single-main control board scenario. Perform the same operations on the other main control board. 3. When pressing the RST button to clear the configuration file, you only need to pay attention to the STAT and PROG indicators. Other indicators may vary depending on the product model.

Ports

Table 1-460 Ports on the CR8DIPU480C5

Port	Connector Type	Description	Available Components
ETH/OAM	RJ45	10M/100M auto-sensing Ethernet network management port or non-standard RS-232 console port NOTE If a service port can be connected to an NMS, you are advised to use the service port as the network management port.	ETH/OAM management cable
CLK/TOD/RS-485	RJ45	Clock/Time/RS-485 input and output ports RS-485 is a reserved port.	1-into-2 clock cable + clock cable (external clock mode, delivery-ready): applicable to the CLK mode 1-into-2 clock cable + clock cable (external clock/external time mode, prepared onsite): applicable to the CLK or TOD mode 1-into-2 clock cable + Ethernet cable: applicable to the TOD mode
ALMI/ALMO	RJ45	Alarm input/output interface	Alarm input/output cable
USB	Type A	USB port (reserved)	-
CLK/TEST	SMA	Clock test port	Coaxial cable

Table 1-461 Pins of the ETH/OAM interface

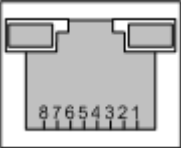
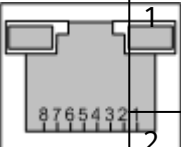
Front View	Pin	Usage
	1	Transmit positive of the NM network port
	2	Transmit negative of the NM network port
	3	Receive positive of the NM network port
	4	Grounding end of the Console interface
	5	Receive end of the Console interface
	6	Receive negative of the NM network port
	7	Unspecified
	8	Transmit end of the Console interface

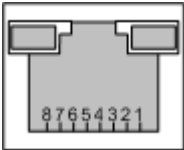
Table 1-462 Pins of the CLK/TOD/RS-485 interface

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	1	Receive negative of the CLK	Receive negative of the CLK	Unspecified	Unspecified	Unspecified
	2	Receive positive of the CLK	Receive positive of the CLK	Unspecified	Unspecified	Unspecified

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	3	Input/Output negative for the 1PPS signal	Unspecified	Input negative for the 1PPS signal (RS422 level)	Output negative for the 1PPS signal (RS422 level)	Transmit negative of the RS-485
	4	Receive negative of the CLK	Receive negative of the CLK	Grounding end	Grounding end	Unspecified
	5	Receive positive of the CLK	Receive positive of the CLK	Grounding end	Grounding end	Unspecified
	6	Input/Output positive for the 1PPS signal	Unspecified	Input positive for the 1PPS signal (RS422 level)	Output positive for the 1PPS signal (RS422 level)	Transmit positive of the RS-485

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	7	Input/Output negative for the time information	Unspecified	Input negative for the time information (RS422 level)	Output negative for the time information (RS422 level)	Receive negative of the RS-485
	8	Input/Output positive for the time information	Unspecified	Input positive for the time information (RS422 level)	Output positive for the time information (RS422 level)	Receive positive of the RS-485
NOTE When this interface is used as a TOD interface, it must be used with a dedicated clock cable. For details, see "Cables" in the Hardware Description.						

Table 1-463 Pins of the ALMI/ALMO interface

Front View	Pin	Usage
	1	Alarm input 1
	2	Grounding end for alarm input 1
	3	Alarm input 2

Front View	Pin	Usage
	4	Alarm input 3
	5	Grounding end for alarm input 3
	6	Grounding end for alarm input 2
	7	Positive for alarm signal output
	8	Negative for alarm signal output

Functions and Features

Table 1-464 Functions and features of the CR8DIPU480C5

Functions and Features	Description
Basic function	Data forwarding: The IPU is the core of service processing in the entire system, and is connected to all subcards through data channels. Control and management: The IPU manages subcards and transmits routing protocol data through management channels. 960 Gbit/s@190 bytes (bidirectional)
Reliability and availability	Hot swappable. 1:1 backup is supported.
Restrictions and remarks	This board cannot be used together with other IPU.
Breakout supported	No
Interface rate auto-sensing supported	No

Technical Specifications

Table 1-465 Technical specifications of the CR8DIPU480C5

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	22.4 mm x 388.4 mm x 209.3 mm (0.88 in. x 15.29 in. x 8.24 in.)

Item	Specification
Dimensions with packaging (H x W x D) [mm(in.)]	170 mm x 560 mm x 385 mm (6.69 in. x 22.05 in. x 15.16 in.)
Weight without packaging [kg(lb)]	2.3 kg (5.07 lb)
Weight with packaging [kg(lb)]	3.7 kg (8.16 lb)
Typical power consumption [W]	113.3 W
Typical heat dissipation [BTU/hour]	367.66 BTU/hour
Long-term operating temperature [°C(°F)]	–40°C to +65°C (–40°F to +149°F), start: ≥ –20°C (–4°F)
CPU	8-core 1.5 GHz
SDRAM	8GB
Storage	4 GB NandFlash

1.4.3.44 CR8DIPU480C5 (03034VDC: Integrated Network Processing Unit (IPU-480-BN))

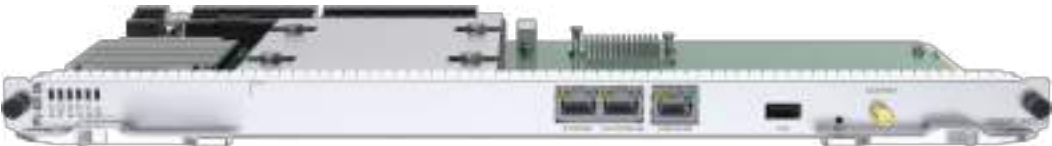
Overview

Table 1-466 Basic information about the CR8DIPU480C5

Item	Details
Description	Integrated Network Processing Unit (IPU-480-BN)
Part Number	03034VDC
Model	CR8DIPU480C5
Card type	IPU
Silkscreen	IPU-480-BN

Appearance

Figure 1-113 Appearance of the CR8DIPU480C5



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03034VDC&productModel=CR8DIPU480C5#matchRelation>

Indicators and Buttons

Table 1-467 Indicators on the CR8DIPU480C5

Silkscreen	Name	Color	Status	Description
STAT	Working status indicator	Green	Steady on	The board is working properly.
		Red	Steady on	A hardware alarm is generated. You can run the display alarm hardware command to check the hardware alarm type.
		-	Off	The board is not powered on or is not running.
PROG	Program status indicator	Green	Steady on	The board software is being initialized or is successfully initialized and is running properly.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	The memory self-check fails. The board software or logic file is lost. The board software fails to be loaded.
		Green	Blinks on and off at 100 ms intervals	The board software is being loaded.
		Green	Blinks on and off at 300 ms intervals	The BIOS is interacting with the upper-layer software.
		Red	Blinks on and off at 100 ms intervals	The BIOS self-check fails.
		-	Off	The board is not powered on.

Silkscreen	Name	Color	Status	Description
SYNC	Clock synchronizatio n indicator	Green	Steady on	The system clock works in free-run mode, and no priority list is set for the system clocks (only internal clock sources are listed by default). The system clock works in tracing mode and is tracing a clock source other than the internal clock sources in the priority list. The system clock works in time synchronizatio n mode, and the PTP time and system clock are being traced.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	The system clock priority list is set, but all clock sources except the internal clock source are lost. The clock works in holdover or free-run mode. The system clock works in time synchronization mode, but no synchronization source is available. The system clock and PTP time work in holdover or free-run mode.
ACTX	Active/standby status indicator for the cross-connect or clock board	Green	Steady on	The board is in the active state on the service plane.
		-	Off	The board is in the standby state on the service plane.
ACTC	Control status indicator	Green	Steady on	The board is in the active control state.
		Green	Blinks on and off at 100 ms intervals	The data of the device is backed up in batch.
		-	Off	The board is in the standby control state.

Silkscreen	Name	Color	Status	Description
ALM	Alarm indicator	Red	Steady on	Device-level critical alarms exist.
		Orange	Steady on	A device-level major or minor alarm exists.
		-	Off	There is no equipment-level alarm.
ETH/OAM	ETH/OAM indicator	Green	Steady on	The link is up.
		Orange	Blinking	Data is being transmitted or received.
		-	Off	The link is down.

Table 1-468 Buttons on the CR8DIPU480C5

Silkscreen	Name	Description
RST	Board reset button	You can press the RST button to reset the NE or restore the factory settings of the NE. The procedure is as follows: To reset the NE, press the RST button and release it. To restore factory settings, press and hold down the RST button. This operation will clear the NE database and system parameter configurations and cannot be restored. Exercise caution when you perform the following operations: 1.Press the RST button to restart the NE. 2.When the PROG indicator blinks (the blinking frequency is about 1 second and the time window is about 15 seconds), press and hold down the RST button. 3.Wait for no longer than 2 minutes till both the STAT and PROG indicators blink (the time window is 5 seconds). Then release the RST button immediately.

Silkscreen	Name	Description
		NOTE 1. When you press the RST button to clear the configuration file, the original configuration file will be cleared. You are advised to periodically back up the configuration file. 2. When you press the RST button to clear the configuration file, to prevent the cleared configuration from being synchronized again between the active and standby main control boards, remove one main control board first in a dual-main control board chassis to ensure that the configuration file is cleared in a single-main control board scenario. Perform the same operations on the other main control board. 3. When pressing the RST button to clear the configuration file, you only need to pay attention to the STAT and PROG indicators. Other indicators may vary depending on the product model.

Ports

Table 1-469 Ports on the CR8DIPU480C5

Port	Connector Type	Description	Available Components
ETH/OAM	RJ45	10M/100M auto-sensing Ethernet network management port or non-standard RS-232 console port NOTE If a service port can be connected to an NMS, you are advised to use the service port as the network management port.	ETH/OAM management cable
CLK/TOD/RS-485	RJ45	Clock/Time/RS-485 input and output ports RS-485 is a reserved port.	1-into-2 clock cable + clock cable (external clock mode, delivery-ready): applicable to the CLK mode 1-into-2 clock cable + clock cable (external clock/external time mode, prepared onsite): applicable to the CLK or TOD mode 1-into-2 clock cable + Ethernet cable: applicable to the TOD mode
ALMI/ALMO	RJ45	Alarm input/output interface	Alarm input/output cable
USB	Type A	USB port (reserved)	-
CLK/TEST	SMA	Clock test port	Coaxial cable

Table 1-470 Pins of the ETH/OAM interface

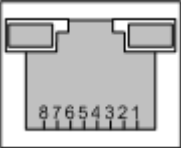
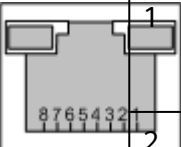
Front View	Pin	Usage
	1	Transmit positive of the NM network port
	2	Transmit negative of the NM network port
	3	Receive positive of the NM network port
	4	Grounding end of the Console interface
	5	Receive end of the Console interface
	6	Receive negative of the NM network port
	7	Unspecified
	8	Transmit end of the Console interface

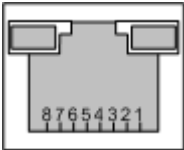
Table 1-471 Pins of the CLK/TOD/RS-485 interface

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	1	Receive negative of the CLK	Receive negative of the CLK	Unspecified	Unspecified	Unspecified
	2	Receive positive of the CLK	Receive positive of the CLK	Unspecified	Unspecified	Unspecified

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	3	Input/Output negative for the 1PPS signal	Unspecified	Input negative for the 1PPS signal (RS422 level)	Output negative for the 1PPS signal (RS422 level)	Transmit negative of the RS-485
	4	Receive negative of the CLK	Receive negative of the CLK	Grounding end	Grounding end	Unspecified
	5	Receive positive of the CLK	Receive positive of the CLK	Grounding end	Grounding end	Unspecified
	6	Input/Output positive for the 1PPS signal	Unspecified	Input positive for the 1PPS signal (RS422 level)	Output positive for the 1PPS signal (RS422 level)	Transmit positive of the RS-485

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	7	Input/Output negative for the time information	Unspecified	Input negative for the time information (RS422 level)	Output negative for the time information (RS422 level)	Receive negative of the RS-485
	8	Input/Output positive for the time information	Unspecified	Input positive for the time information (RS422 level)	Output positive for the time information (RS422 level)	Receive positive of the RS-485
NOTE When this interface is used as a TOD interface, it must be used with a dedicated clock cable. For details, see "Cables" in the Hardware Description.						

Table 1-472 Pins of the ALMI/ALMO interface

Front View	Pin	Usage
	1	Alarm input 1
	2	Grounding end for alarm input 1
	3	Alarm input 2

Front View	Pin	Usage
	4	Alarm input 3
	5	Grounding end for alarm input 3
	6	Grounding end for alarm input 2
	7	Positive for alarm signal output
	8	Negative for alarm signal output

Functions and Features

Table 1-473 Functions and features of the CR8DIPU480C5

Functions and Features	Description
Basic function	Data forwarding: The IPU is the core of service processing in the entire system, and is connected to all subcards through data channels. Control and management: The IPU manages subcards and transmits routing protocol data through management channels. 960 Gbit/s@190 bytes (bidirectional)
Reliability and availability	Hot swappable. 1:1 backup is supported.
Restrictions and remarks	This board cannot be used together with other IPU.
Breakout supported	No
Interface rate auto-sensing supported	No

Technical Specifications

Table 1-474 Technical specifications of the CR8DIPU480C5

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	22.4 mm x 388.4 mm x 209.3 mm (0.88 in. x 15.29 in. x 8.24 in.)

Item	Specification
Dimensions with packaging (H x W x D) [mm(in.)]	170 mm x 560 mm x 385 mm (6.69 in. x 22.05 in. x 15.16 in.)
Weight without packaging [kg(lb)]	2.3 kg (5.07 lb)
Weight with packaging [kg(lb)]	3.7 kg (8.16 lb)
Typical power consumption [W]	113.3 W
Typical heat dissipation [BTU/hour]	367.66 BTU/hour
Long-term operating temperature [°C(°F)]	–40°C to +65°C (–40°F to +149°F), start: ≥ –20°C (–4°F)
CPU	8-core 1.5 GHz
SDRAM	8GB
Storage	4 GB NandFlash

1.4.3.45 CR8DIPU480C5 (03035JVB: Integrated Network Processing Unit (IPU-480-BN) for SME)

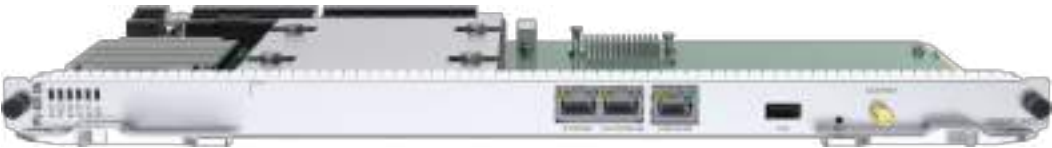
Overview

Table 1-475 Basic information about the CR8DIPU480C5

Item	Details
Description	Integrated Network Processing Unit (IPU-480-BN) for SME
Part Number	03035JVB
Model	CR8DIPU480C5
Card type	IPU
Silkscreen	IPU-480-BN

Appearance

Figure 1-114 Appearance of the CR8DIPU480C5



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03035JVB&productModel=CR8DIPU480C5#matchRelation>

Indicators and Buttons

Table 1-476 Indicators on the CR8DIPU480C5

Silkscreen	Name	Color	Status	Description
STAT	Working status indicator	Green	Steady on	The board is working normally.
		Red	Steady on	A hardware alarm is generated. You can run the display alarm hardware command to check the hardware alarm type.
		-	Steady off	The board is not running or no power is input.
PROG	Program status indicator	Green	Steady on	The board software is being initialized or is successfully initialized and is running properly.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	The memory self-check fails. The board software or logic file is lost. The board software fails to be loaded.
		Green	Blinks on and off at 100 ms intervals	The board software is being loaded.
		Green	On for 300 ms and off for 300 ms repeatedly	The BIOS is interacting with the upper-layer software.
		Red	Blinks on and off at 100 ms intervals	The BIOS self-check fails.
		-	Steady off	The board is not powered on.

Silkscreen	Name	Color	Status	Description
SYNC	Clock synchronizatio n indicator	Green	Steady on	The system clock works in free-run mode, and no priority list is set for the system clocks (only internal clock sources are listed by default). The system clock works in tracing mode and is tracing a clock source other than the internal clock sources in the priority list. The system clock works in time synchronizatio n mode, and the PTP time and system clock are being traced.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	The system clock priority list is set, but all clock sources except the internal clock source are lost. The clock works in holdover or free-run mode. The system clock works in time synchronization mode, but no synchronization source is available. The system clock and PTP time work in holdover or free-run mode.
ACTX	Active/standby status indicator for the cross-connect or clock board	Green	Steady on	The board is in the active state on the service plane.
		-	Steady off	The board is in the standby state on the service plane.
ACTC	Control status indicator	Green	Steady on	The board is in the active control state.
		Green	Blinks on and off at 100 ms intervals	The data of the device is backed up in batch.
		-	Steady off	The board is in the standby control state.

Silkscreen	Name	Color	Status	Description
ALM	Alarm indicator	Red	Steady on	Device-level critical alarms exist.
		Orange	Steady on	A device-level major or minor alarm exists.
		-	Steady off	There is no equipment-level alarm.
ETH/OAM	ETH/OAM indicator	Green	Steady on	The link is up.
		Orange	Blinking	Data is being transmitted and received.
		-	Steady off	The link is Down.

Table 1-477 Buttons on the CR8DIPU480C5

Silkscreen	Name	Description
RST	Board reset button	You can press the RST button to reset the NE or restore the factory settings of the NE. The procedure is as follows: To reset the NE, press the RST button and release it. To restore factory settings, press and hold down the RST button. This operation will clear the NE database and system parameter configurations and cannot be restored. Exercise caution when you perform the following operations: 1.Press the RST button to restart the NE. 2.When the PROG indicator blinks (the blinking frequency is about 1 second and the time window is about 15 seconds), press and hold down the RST button. 3.Wait for no longer than 2 minutes till both the STAT and PROG indicators blink (the time window is 5 seconds). Then release the RST button immediately.

Silkscreen	Name	Description
		NOTE 1. When you press the RST button to clear the configuration file, the original configuration file will be cleared. You are advised to periodically back up the configuration file. 2. When you press the RST button to clear the configuration file, to prevent the cleared configuration from being synchronized again between the active and standby main control boards, remove one main control board first in a dual-main control board chassis to ensure that the configuration file is cleared in a single-main control board scenario. Perform the same operations on the other main control board. 3. When pressing the RST button to clear the configuration file, you only need to pay attention to the STAT and PROG indicators. Other indicators may vary depending on the product model.

Ports

Table 1-478 Ports on the CR8DIPU480C5

Port	Connector Type	Description	Available Components
ETH/OAM	RJ45	10M/100M auto-sensing Ethernet network management port and non-standard RS-232 console port NOTE If a service port can be connected to the NMS, it is recommended that the service port be used as the network management port.	ETH/OAM management cable
CLK/TOD/RS-485	RJ45	Clock/Time/RS-485 input and output ports RS-485 is a reserved port.	1-into-2 clock cable + clock cable (external clock mode, delivery-ready): applicable to the CLK mode 1-into-2 clock cable + clock cable (external clock/external time mode, prepared onsite): applicable to the CLK or TOD mode 1-into-2 clock cable + Ethernet cable: applicable to the TOD mode
ALMI/ALMO	RJ45	Alarm input/output interface	Alarm input/output cable
USB	Type A	USB port (reserved)	-
CLK/TEST	SMA	Clock test port	Coaxial cable

Table 1-479 Pins of the ETH/OAM interface

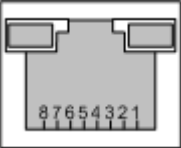
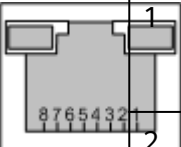
Front View	Pin	Usage
	1	Transmit positive of the NM network port
	2	Transmit negative of the NM network port
	3	Receive positive of the NM network port
	4	Grounding end of the Console interface
	5	Receive end of the Console interface
	6	Receive negative of the NM network port
	7	Unspecified
	8	Transmit end of the Console interface

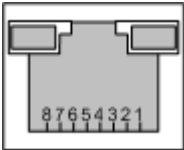
Table 1-480 Pins of the CLK/TOD/RS-485 interface

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	1	Receive negative of the CLK	Receive negative of the CLK	Unspecified	Unspecified	Unspecified
	2	Receive positive of the CLK	Receive positive of the CLK	Unspecified	Unspecified	Unspecified

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	3	Input/Output negative for the 1PPS signal	Unspecified	Input negative for the 1PPS signal (RS422 level)	Output negative for the 1PPS signal (RS422 level)	Transmit negative of the RS-485
	4	Receive negative of the CLK	Receive negative of the CLK	Grounding end	Grounding end	Unspecified
	5	Receive positive of the CLK	Receive positive of the CLK	Grounding end	Grounding end	Unspecified
	6	Input/Output positive for the 1PPS signal	Unspecified	Input positive for the 1PPS signal (RS422 level)	Output positive for the 1PPS signal (RS422 level)	Transmit positive of the RS-485

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	7	Input/Output negative for the time information	Unspecified	Input negative for the time information (RS422 level)	Output negative for the time information (RS422 level)	Receive negative of the RS-485
	8	Input/Output positive for the time information	Unspecified	Input positive for the time information (RS422 level)	Output positive for the time information (RS422 level)	Receive positive of the RS-485
NOTE When this interface is used as a TOD interface, it must be used with a dedicated clock cable. For details, see "Cables" in the Hardware Description.						

Table 1-481 Pins of the ALMI/ALMO interface

Front View	Pin	Usage
	1	Alarm input 1
	2	Grounding end for alarm input 1
	3	Alarm input 2

Front View	Pin	Usage
	4	Alarm input 3
	5	Grounding end for alarm input 3
	6	Grounding end for alarm input 2
	7	Positive for alarm signal output
	8	Negative for alarm signal output

Functions and Features

Table 1-482 Functions and features of the CR8DIPU480C5

Functions and Features	Description
Basic function	Data forwarding: The IPU is the core of service processing in the entire system, and is connected to all subcards through data channels. Control and management: The IPU manages subcards and transmits routing protocol data through management channels. 960 Gbit/s@190 bytes (bidirectional)
Reliability and availability	Hot swappable. 1:1 backup is supported.
Restrictions and remarks	This board cannot be used together with other IPU.
Breakout supported	No
Interface rate auto-sensing supported	No

Technical Specifications

Table 1-483 Technical specifications of the CR8DIPU480C5

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	22.4 mm x 388.4 mm x 209.3 mm (0.88 in. x 15.29 in. x 8.24 in.)

Item	Specification
Dimensions with packaging (H x W x D) [mm(in.)]	170 mm x 560 mm x 385 mm (6.69 in. x 22.05 in. x 15.16 in.)
Weight without packaging [kg(lb)]	2.3 kg (5.07 lb)
Weight with packaging [kg(lb)]	3.7 kg (8.16 lb)
Typical power consumption [W]	113.3 W
Typical heat dissipation [BTU/hour]	367.66 BTU/hour
Long-term operating temperature [°C(°F)]	–40°C to +65°C (–40°F to +149°F), start: ≥ –20°C (–4°F)
CPU	8-core 1.5 GHz
SDRAM	8GB
Storage	4 GB NandFlash

1.4.3.46 CR8DIPU480C7 (Integrated Network Processing Unit (IPU-480-C))

Overview

Table 1-484 Basic information about the CR8DIPU480C7

Item	Details
Description	Integrated Network Processing Unit (IPU-480-C)
Part Number	03034NXA
Model	CR8DIPU480C7
Card type	IPU
Silkscreen	IPU-480-C

Appearance

Figure 1-115 Appearance of the CR8DIPU480C7



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03034NXA&productModel=CR8DIPU480C7#matchRelation>

Indicators and Buttons

Table 1-485 Indicators on the CR8DIPU480C7

Silkscreen	Name	Color	Status	Description
STAT	Working status indicator	Green	Steady on	The board is working properly.
		Red	Steady on	A hardware alarm is generated. You can run the display alarm hardware command to check the hardware alarm type.
		-	Off	The board is not powered on or is not running.
PROG	Program status indicator	Green	Steady on	The board software is being initialized or is successfully initialized and is running properly.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	The memory self-check fails. The board software or logic file is lost. The board software fails to be loaded.
		Green	Blinks on and off at 100 ms intervals	The board software is being loaded.
		Green	Blinks on and off at 300 ms intervals	The BIOS is interacting with the upper-layer software.
		Red	Blinks on and off at 100 ms intervals	The BIOS self-check fails.
		-	Off	The board is not powered on.

Silkscreen	Name	Color	Status	Description
SYNC	Clock synchronizatio n indicator	Green	Steady on	The system clock works in free-run mode, and no priority list is set for the system clocks (only internal clock sources are listed by default). The system clock works in tracing mode and is tracing a clock source other than the internal clock sources in the priority list. The system clock works in time synchronizatio n mode, and the PTP time and system clock are being traced.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	The system clock priority list is set, but all clock sources except the internal clock source are lost. The clock works in holdover or free-run mode. The system clock works in time synchronization mode, but no synchronization source is available. The system clock and PTP time work in holdover or free-run mode.
ACTX	Active/standby status indicator for the cross-connect or clock board	Green	Steady on	The board is in the active state on the service plane.
		-	Off	The board is in the standby state on the service plane.
ACTC	Control status indicator	Green	Steady on	The board is in the active control state.
		Green	Blinks on and off at 100 ms intervals	The data of the device is backed up in batch.
		-	Off	The board is in the standby control state.

Silkscreen	Name	Color	Status	Description
ALM	Alarm indicator	Red	Steady on	Device-level critical alarms exist.
		Orange	Steady on	A device-level major or minor alarm exists.
		-	Off	There is no equipment-level alarm.
ETH/OAM	ETH/OAM indicator	Green	Steady on	The link is up.
		Orange	Blinking	Data is being transmitted or received.
		-	Off	The link is down.

Table 1-486 Buttons on the CR8DIPU480C7

Silkscreen	Name	Description
RST	Board reset button	You can press the RST button to reset the NE or restore the factory settings of the NE. The procedure is as follows: To reset the NE, press the RST button and release it. To restore factory settings, press and hold down the RST button. This operation will clear the NE database and system parameter configurations and cannot be restored. Exercise caution when you perform the following operations: 1.Press the RST button to restart the NE. 2.When the PROG indicator blinks (the blinking frequency is about 1 second and the time window is about 15 seconds), press and hold down the RST button. 3.Wait for no longer than 2 minutes till both the STAT and PROG indicators blink (the time window is 5 seconds). Then release the RST button immediately.

Silkscreen	Name	Description
		NOTE 1. When you press the RST button to clear the configuration file, the original configuration file will be cleared. You are advised to periodically back up the configuration file. 2. When you press the RST button to clear the configuration file, to prevent the cleared configuration from being synchronized again between the active and standby main control boards, remove one main control board first in a dual-main control board chassis to ensure that the configuration file is cleared in a single-main control board scenario. Perform the same operations on the other main control board. 3. When pressing the RST button to clear the configuration file, you only need to pay attention to the STAT and PROG indicators. Other indicators may vary depending on the product model.

Ports

Table 1-487 Ports on the CR8DIPU480C7

Port	Connector Type	Description	Available Components
ETH/OAM	RJ45	10M/100M auto-sensing Ethernet network management port or non-standard RS-232 console port NOTE If a service port can be connected to an NMS, you are advised to use the service port as the network management port.	ETH/OAM management cable
CLK/TOD/RS-485	RJ45	Clock/Time/RS-485 input and output ports RS-485 is a reserved port.	1-into-2 clock cable + clock cable (external clock mode, delivery-ready): applicable to the CLK mode 1-into-2 clock cable + clock cable (external clock/external time mode, prepared onsite): applicable to the CLK or TOD mode 1-into-2 clock cable + Ethernet cable: applicable to the TOD mode
ALMI/ALMO	RJ45	Alarm input/output interface	Alarm input/output cable
USB	Type A	USB port (reserved)	-
CLK/TEST	SMA	Clock test port	Coaxial cable

Table 1-488 Pins of the ETH/OAM interface

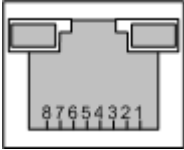
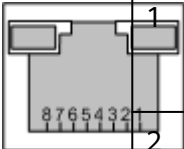
Front View	Pin	Usage
	1	Transmit positive of the NM network port
	2	Transmit negative of the NM network port
	3	Receive positive of the NM network port
	4	Grounding end of the Console interface
	5	Receive end of the Console interface
	6	Receive negative of the NM network port
	7	Unspecified
	8	Transmit end of the Console interface

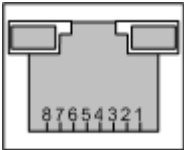
Table 1-489 Pins of the CLK/TOD/RS-485 interface

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	1	Receive negative of the CLK	Receive negative of the CLK	Unspecified	Unspecified	Unspecified
	2	Receive positive of the CLK	Receive positive of the CLK	Unspecified	Unspecified	Unspecified

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	3	Input/Output negative for the 1PPS signal	Unspecified	Input negative for the 1PPS signal (RS422 level)	Output negative for the 1PPS signal (RS422 level)	Transmit negative of the RS-485
	4	Receive negative of the CLK	Receive negative of the CLK	Grounding end	Grounding end	Unspecified
	5	Receive positive of the CLK	Receive positive of the CLK	Grounding end	Grounding end	Unspecified
	6	Input/Output positive for the 1PPS signal	Unspecified	Input positive for the 1PPS signal (RS422 level)	Output positive for the 1PPS signal (RS422 level)	Transmit positive of the RS-485

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	7	Input/Output negative for the time information	Unspecified	Input negative for the time information (RS422 level)	Output negative for the time information (RS422 level)	Receive negative of the RS-485
	8	Input/Output positive for the time information	Unspecified	Input positive for the time information (RS422 level)	Output positive for the time information (RS422 level)	Receive positive of the RS-485
NOTE When this interface is used as a TOD interface, it must be used with a dedicated clock cable. For details, see "Cables" in the Hardware Description.						

Table 1-490 Pins of the ALMI/ALMO interface

Front View	Pin	Usage
	1	Alarm input 1
	2	Grounding end for alarm input 1
	3	Alarm input 2

Front View	Pin	Usage
	4	Alarm input 3
	5	Grounding end for alarm input 3
	6	Grounding end for alarm input 2
	7	Positive for alarm signal output
	8	Negative for alarm signal output

Functions and Features

Table 1-491 Functions and features of the CR8DIPU480C7

Functions and Features	Description
Basic function	Data forwarding: The IPU is the core of service processing in the entire system, and is connected to all subcards through data channels. Control and management: The IPU manages subcards and transmits routing protocol data through management channels. 960 Gbit/s@190 bytes (bidirectional)
Reliability and availability	Hot swappable. 1:1 backup is supported.
Restrictions and remarks	This board cannot be used together with other IPU.
Breakout supported	No
Interface rate auto-sensing supported	No

Technical Specifications

Table 1-492 Technical specifications of the CR8DIPU480C7

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	22.4 mm x 388.4 mm x 209.3 mm (0.88 in. x 15.29 in. x 8.24 in.)

Item	Specification
Dimensions with packaging (H x W x D) [mm(in.)]	170 mm x 560 mm x 385 mm (6.69 in. x 22.05 in. x 15.16 in.)
Weight without packaging [kg(lb)]	2.3 kg (5.07 lb)
Weight with packaging [kg(lb)]	3.7 kg (8.16 lb)
Typical power consumption [W]	113.3 W
Typical heat dissipation [BTU/hour]	367.66 BTU/hour
Long-term operating temperature [°C(°F)]	–40°C to +65°C (–40°F to +149°F), start: ≥ –20°C (–4°F)
CPU	8-core 1.5 GHz
SDRAM	8GB
Storage	4 GB NandFlash

1.4.3.47 CR8DIPU480C9 (Integrated Network Processing Unit (IPU-480-D))

Overview

Table 1-493 Basic information about the CR8DIPU480C9

Item	Details
Description	Integrated Network Processing Unit (IPU-480-D)
Part Number	03034USV
Model	CR8DIPU480C9
Card type	IPU
Silkscreen	IPU-480-D

Appearance

Figure 1-116 Appearance of the CR8DIPU480C9



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03034USV&productModel=CR8DIPU480C9#matchRelation>

Indicators and Buttons

Table 1-494 Indicators on the CR8DIPU480C9

Silkscreen	Name	Color	Status	Description
STAT	Working status indicator	Green	Steady on	The board is working properly.
		Red	Steady on	A hardware alarm is generated. You can run the display alarm hardware command to check the hardware alarm type.
		-	Off	The board is not powered on or is not running.
PROG	Program status indicator	Green	Steady on	The board software is being initialized or is successfully initialized and is running properly.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	The memory self-check fails. The board software or logic file is lost. The board software fails to be loaded.
		Green	Blinks on and off at 100 ms intervals	The board software is being loaded.
		Green	Blinks on and off at 300 ms intervals	The BIOS is interacting with the upper-layer software.
		Red	Blinks on and off at 100 ms intervals	The BIOS self-check fails.
		-	Off	The board is not powered on.

Silkscreen	Name	Color	Status	Description
SYNC	Clock synchronizatio n indicator	Green	Steady on	The system clock works in free-run mode, and no priority list is set for the system clocks (only internal clock sources are listed by default). The system clock works in tracing mode and is tracing a clock source other than the internal clock sources in the priority list. The system clock works in time synchronizatio n mode, and the PTP time and system clock are being traced.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	The system clock priority list is set, but all clock sources except the internal clock source are lost. The clock works in holdover or free-run mode. The system clock works in time synchronization mode, but no synchronization source is available. The system clock and PTP time work in holdover or free-run mode.
ACTX	Active/standby status indicator for the cross-connect or clock board	Green	Steady on	The board is in the active state on the service plane.
		-	Off	The board is in the standby state on the service plane.
ACTC	Control status indicator	Green	Steady on	The board is in the active control state.
		Green	Blinks on and off at 100 ms intervals	The data of the device is backed up in batch.
		-	Off	The board is in the standby control state.

Silkscreen	Name	Color	Status	Description
ALM	Alarm indicator	Red	Steady on	Device-level critical alarms exist.
		Orange	Steady on	A device-level major or minor alarm exists.
		-	Off	There is no equipment-level alarm.
ETH/OAM	ETH/OAM indicator	Green	Steady on	The link is up.
		Orange	Blinking	Data is being transmitted or received.
		-	Off	The link is down.

Table 1-495 Buttons on the CR8DIPU480C9

Silkscreen	Name	Description
RST	Board reset button	You can press the RST button to reset the NE or restore the factory settings of the NE. The procedure is as follows: To reset the NE, press the RST button and release it. To restore factory settings, press and hold down the RST button. This operation will clear the NE database and system parameter configurations and cannot be restored. Exercise caution when you perform the following operations: 1.Press the RST button to restart the NE. 2.When the PROG indicator blinks (the blinking frequency is about 1 second and the time window is about 15 seconds), press and hold down the RST button. 3.Wait for no longer than 2 minutes till both the STAT and PROG indicators blink (the time window is 5 seconds). Then release the RST button immediately.

Silkscreen	Name	Description
		NOTE 1. When you press the RST button to clear the configuration file, the original configuration file will be cleared. You are advised to periodically back up the configuration file. 2. When you press the RST button to clear the configuration file, to prevent the cleared configuration from being synchronized again between the active and standby main control boards, remove one main control board first in a dual-main control board chassis to ensure that the configuration file is cleared in a single-main control board scenario. Perform the same operations on the other main control board. 3. When pressing the RST button to clear the configuration file, you only need to pay attention to the STAT and PROG indicators. Other indicators may vary depending on the product model.

Ports

Table 1-496 Ports on the CR8DIPU480C9

Port	Connector Type	Description	Available Components
ETH/OAM	RJ45	10M/100M auto-sensing Ethernet network management port or non-standard RS-232 console port NOTE If a service port can be connected to an NMS, you are advised to use the service port as the network management port.	ETH/OAM management cable
CLK/TOD/RS-485	RJ45	Clock/Time/RS-485 input and output ports RS-485 is a reserved port.	1-into-2 clock cable + clock cable (external clock mode, delivery-ready): applicable to the CLK mode 1-into-2 clock cable + clock cable (external clock/external time mode, prepared onsite): applicable to the CLK or TOD mode 1-into-2 clock cable + Ethernet cable: applicable to the TOD mode
ALMI/ALMO	RJ45	Alarm input/output interface	Alarm input/output cable
USB	Type A	USB port (reserved)	-
CLK/TEST	SMA	Clock test port	Coaxial cable

Table 1-497 Pins of the ETH/OAM interface

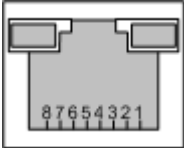
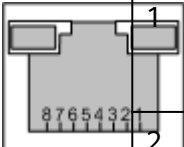
Front View	Pin	Usage
	1	Transmit positive of the NM network port
	2	Transmit negative of the NM network port
	3	Receive positive of the NM network port
	4	Grounding end of the Console interface
	5	Receive end of the Console interface
	6	Receive negative of the NM network port
	7	Unspecified
	8	Transmit end of the Console interface

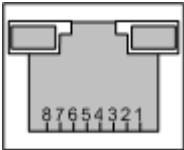
Table 1-498 Pins of the CLK/TOD/RS-485 interface

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	1	Receive negative of the CLK	Receive negative of the CLK	Unspecified	Unspecified	Unspecified
	2	Receive positive of the CLK	Receive positive of the CLK	Unspecified	Unspecified	Unspecified

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	3	Input/Output negative for the 1PPS signal	Unspecified	Input negative for the 1PPS signal (RS422 level)	Output negative for the 1PPS signal (RS422 level)	Transmit negative of the RS-485
	4	Receive negative of the CLK	Receive negative of the CLK	Grounding end	Grounding end	Unspecified
	5	Receive positive of the CLK	Receive positive of the CLK	Grounding end	Grounding end	Unspecified
	6	Input/Output positive for the 1PPS signal	Unspecified	Input positive for the 1PPS signal (RS422 level)	Output positive for the 1PPS signal (RS422 level)	Transmit positive of the RS-485

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	7	Input/Output negative for the time information	Unspecified	Input negative for the time information (RS422 level)	Output negative for the time information (RS422 level)	Receive negative of the RS-485
	8	Input/Output positive for the time information	Unspecified	Input positive for the time information (RS422 level)	Output positive for the time information (RS422 level)	Receive positive of the RS-485
NOTE When this interface is used as a TOD interface, it must be used with a dedicated clock cable. For details, see "Cables" in the Hardware Description.						

Table 1-499 Pins of the ALMI/ALMO interface

Front View	Pin	Usage
	1	Alarm input 1
	2	Grounding end for alarm input 1
	3	Alarm input 2

Front View	Pin	Usage
	4	Alarm input 3
	5	Grounding end for alarm input 3
	6	Grounding end for alarm input 2
	7	Positive for alarm signal output
	8	Negative for alarm signal output

Functions and Features

Table 1-500 Functions and features of the CR8DIPU480C9

Functions and Features	Description
Basic function	Data forwarding: The IPU is the core of service processing in the entire system, and is connected to all subcards through data channels. Control and management: The IPU manages subcards and transmits routing protocol data through management channels. 960 Gbit/s@190 bytes (bidirectional)
Reliability and availability	Hot swappable. 1:1 backup is supported.
Restrictions and remarks	This board cannot be used together with other IPU.
Breakout supported	No
Interface rate auto-sensing supported	No

Technical Specifications

Table 1-501 Technical specifications of the CR8DIPU480C9

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	22.4 mm x 388.4 mm x 209.3 mm (0.88 in. x 15.29 in. x 8.24 in.)

Item	Specification
Dimensions with packaging (H x W x D) [mm(in.)]	170 mm x 560 mm x 385 mm (6.69 in. x 22.05 in. x 15.16 in.)
Weight without packaging [kg(lb)]	2.3 kg (5.07 lb)
Weight with packaging [kg(lb)]	3.7 kg (8.16 lb)
Typical power consumption [W]	113.3 W
Typical heat dissipation [BTU/hour]	367.66 BTU/hour
Long-term operating temperature [°C(°F)]	–40°C to +65°C (–40°F to +149°F), start: ≥ –20°C (–4°F)
CPU	8-core 1.5 GHz
SDRAM	8GB
Storage	4 GB NandFlash

1.4.3.48 CR8DIPU480M3 (Integrated Network Processing Unit (IPU-480-E))

Overview

Table 1-502 Basic information about the CR8DIPU480M3

Item	Details
Description	Integrated Network Processing Unit (IPU-480-E)
Part Number	03034VPJ
Model	CR8DIPU480M3
Card type	IPU
Silkscreen	IPU-480-E

Appearance

Figure 1-117 Appearance of the CR8DIPU480M3



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03034VPJ&productModel=CR8DIPU480M3#matchRelation>

Indicators and Buttons

Table 1-503 Indicators on the CR8DIPU480M3

Silkscreen	Name	Color	Status	Description
STAT	Working status indicator	Green	Steady on	The board is working properly.
		Red	Steady on	A hardware alarm is generated. You can run the display alarm hardware command to check the hardware alarm type.
		-	Off	The board is not powered on or is not running.
PROG	Program status indicator	Green	Steady on	The board software is being initialized or is successfully initialized and is running properly.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	The memory self-check fails. The board software or logic file is lost. The board software fails to be loaded.
		Green	Blinks on and off at 100 ms intervals	The board software is being loaded.
		Green	Blinks on and off at 300 ms intervals	The BIOS is interacting with the upper-layer software.
		Red	Blinks on and off at 100 ms intervals	The BIOS self-check fails.
		-	Off	The board is not powered on.

Silkscreen	Name	Color	Status	Description
SYNC	Clock synchronizatio n indicator	Green	Steady on	The system clock works in free-run mode, and no priority list is set for the system clocks (only internal clock sources are listed by default). The system clock works in tracing mode and is tracing a clock source other than the internal clock sources in the priority list. The system clock works in time synchronizatio n mode, and the PTP time and system clock are being traced.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	The system clock priority list is set, but all clock sources except the internal clock source are lost. The clock works in holdover or free-run mode. The system clock works in time synchronization mode, but no synchronization source is available. The system clock and PTP time work in holdover or free-run mode.
ACTX	Active/standby status indicator for the cross-connect or clock board	Green	Steady on	The board is in the active state on the service plane.
		-	Off	The board is in the standby state on the service plane.
ACTC	Control status indicator	Green	Steady on	The board is in the active control state.
		Green	Blinks on and off at 100 ms intervals	The data of the device is backed up in batch.
		-	Off	The board is in the standby control state.

Silkscreen	Name	Color	Status	Description
ALM	Alarm indicator	Red	Steady on	Device-level critical alarms exist.
		Orange	Steady on	A device-level major or minor alarm exists.
		-	Off	There is no equipment-level alarm.
ETH/OAM	ETH/OAM indicator	Green	Steady on	The link is up.
		Orange	Blinking	Data is being transmitted or received.
		-	Off	The link is down.

Table 1-504 Buttons on the CR8DIPU480M3

Silkscreen	Name	Description
RST	Board reset button	You can press the RST button to reset the NE or restore the factory settings of the NE. The procedure is as follows: To reset the NE, press the RST button and release it. To restore factory settings, press and hold down the RST button. This operation will clear the NE database and system parameter configurations and cannot be restored. Exercise caution when you perform the following operations: 1.Press the RST button to restart the NE. 2.When the PROG indicator blinks (the blinking frequency is about 1 second and the time window is about 15 seconds), press and hold down the RST button. 3.Wait for no longer than 2 minutes till both the STAT and PROG indicators blink (the time window is 5 seconds). Then release the RST button immediately.

Silkscreen	Name	Description
		NOTE 1. When you press the RST button to clear the configuration file, the original configuration file will be cleared. You are advised to periodically back up the configuration file. 2. When you press the RST button to clear the configuration file, to prevent the cleared configuration from being synchronized again between the active and standby main control boards, remove one main control board first in a dual-main control board chassis to ensure that the configuration file is cleared in a single-main control board scenario. Perform the same operations on the other main control board. 3. When pressing the RST button to clear the configuration file, you only need to pay attention to the STAT and PROG indicators. Other indicators may vary depending on the product model.

Ports

Table 1-505 Ports on the CR8DIPU480M3

Port	Connector Type	Description	Available Components
ETH/OAM	RJ45	10M/100M auto-sensing Ethernet network management port or non-standard RS-232 console port NOTE If a service port can be connected to an NMS, you are advised to use the service port as the network management port.	ETH/OAM management cable
CLK/TOD/RS-485	RJ45	Clock/Time/RS-485 input and output ports RS-485 is a reserved port.	1-into-2 clock cable + clock cable (external clock mode, delivery-ready): applicable to the CLK mode 1-into-2 clock cable + clock cable (external clock/external time mode, prepared onsite): applicable to the CLK or TOD mode 1-into-2 clock cable + Ethernet cable: applicable to the TOD mode
ALMI/ALMO	RJ45	Alarm input/output interface	Alarm input/output cable
USB	Type A	USB port (reserved)	-
CLK/TEST	SMA	Clock test port	Coaxial cable

Table 1-506 Pins of the ETH/OAM interface

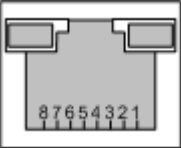
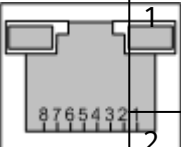
Front View	Pin	Usage
	1	Transmit positive of the NM network port
	2	Transmit negative of the NM network port
	3	Receive positive of the NM network port
	4	Grounding end of the Console interface
	5	Receive end of the Console interface
	6	Receive negative of the NM network port
	7	Unspecified
	8	Transmit end of the Console interface

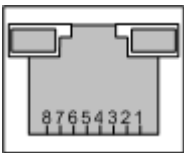
Table 1-507 Pins of the CLK/TOD/RS-485 interface

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	1	Receive negative of the CLK	Receive negative of the CLK	Unspecified	Unspecified	Unspecified
	2	Receive positive of the CLK	Receive positive of the CLK	Unspecified	Unspecified	Unspecified

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	3	Input/Output negative for the 1PPS signal	Unspecified	Input negative for the 1PPS signal (RS422 level)	Output negative for the 1PPS signal (RS422 level)	Transmit negative of the RS-485
	4	Receive negative of the CLK	Receive negative of the CLK	Grounding end	Grounding end	Unspecified
	5	Receive positive of the CLK	Receive positive of the CLK	Grounding end	Grounding end	Unspecified
	6	Input/Output positive for the 1PPS signal	Unspecified	Input positive for the 1PPS signal (RS422 level)	Output positive for the 1PPS signal (RS422 level)	Transmit positive of the RS-485

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	7	Input/Output negative for the time information	Unspecified	Input negative for the time information (RS422 level)	Output negative for the time information (RS422 level)	Receive negative of the RS-485
	8	Input/Output positive for the time information	Unspecified	Input positive for the time information (RS422 level)	Output positive for the time information (RS422 level)	Receive positive of the RS-485
NOTE When this interface is used as a TOD interface, it must be used with a dedicated clock cable. For details, see "Cables" in the Hardware Description.						

Table 1-508 Pins of the ALMI/ALMO interface

Front View	Pin	Usage
	1	Alarm input 1
	2	Grounding end for alarm input 1
	3	Alarm input 2

Front View	Pin	Usage
	4	Alarm input 3
	5	Grounding end for alarm input 3
	6	Grounding end for alarm input 2
	7	Positive for alarm signal output
	8	Negative for alarm signal output

Functions and Features

Table 1-509 Functions and features of the CR8DIPU480M3

Functions and Features	Description
Basic function	Data forwarding: The IPU is the core of service processing in the entire system, and is connected to all subcards through data channels. Control and management: The IPU manages subcards and transmits routing protocol data through management channels. 960 Gbit/s@190 bytes (bidirectional)
Reliability and availability	Hot swappable. 1:1 backup is supported.
Restrictions and remarks	This board cannot be used together with other IPU.
Breakout supported	No
Interface rate auto-sensing supported	No

Technical Specifications

Table 1-510 Technical specifications of the CR8DIPU480M3

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	22.4 mm x 388.4 mm x 209.3 mm (0.88 in. x 15.29 in. x 8.24 in.)

Item	Specification
Dimensions with packaging (H x W x D) [mm(in.)]	170 mm x 560 mm x 385 mm (6.69 in. x 22.05 in. x 15.16 in.)
Weight without packaging [kg(lb)]	2.3 kg (5.07 lb)
Weight with packaging [kg(lb)]	3.7 kg (8.16 lb)
Typical power consumption [W]	113.3 W
Typical heat dissipation [BTU/hour]	367.66 BTU/hour
Long-term operating temperature [°C(°F)]	–40°C to +65°C (–40°F to +149°F), start: ≥ –20°C (–4°F)
CPU	8-core 1.5 GHz
SDRAM	8GB
Storage	4 GB NandFlash

1.4.3.49 CR8DIPU480MA (Integrated Network Processing Unit (IPU-480-CN))

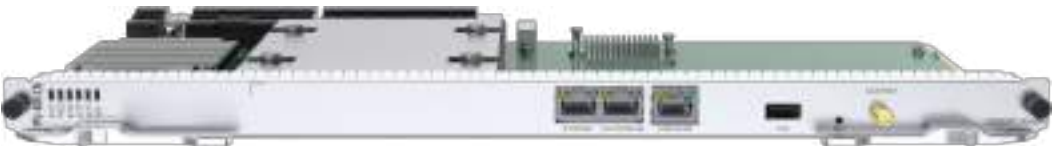
Overview

Table 1-511 Basic information about the CR8DIPU480MA

Item	Details
Description	Integrated Network Processing Unit (IPU-480-CN)
Part Number	03034WGL
Model	CR8DIPU480MA
Card type	IPU
Silkscreen	IPU-480-CN

Appearance

Figure 1-118 Appearance of the CR8DIPU480MA



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03034WGL&productModel=CR8DIPU480MA#matchRelation>

Indicators and Buttons

Table 1-512 Indicators on the CR8DIPU480MA

Silkscreen	Name	Color	Status	Description
STAT	Working status indicator	Green	Steady on	The board is working properly.
		Red	Steady on	A hardware alarm is generated. You can run the display alarm hardware command to check the hardware alarm type.
		-	Off	The board is not powered on or is not running.
PROG	Program status indicator	Green	Steady on	The board software is being initialized or is successfully initialized and is running properly.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	The memory self-check fails. The board software or logic file is lost. The board software fails to be loaded.
		Green	Blinks on and off at 100 ms intervals	The board software is being loaded.
		Green	Blinks on and off at 300 ms intervals	The BIOS is interacting with the upper-layer software.
		Red	Blinks on and off at 100 ms intervals	The BIOS self-check fails.
		-	Off	The board is not powered on.

Silkscreen	Name	Color	Status	Description
SYNC	Clock synchronizatio n indicator	Green	Steady on	The system clock works in free-run mode, and no priority list is set for the system clocks (only internal clock sources are listed by default). The system clock works in tracing mode and is tracing a clock source other than the internal clock sources in the priority list. The system clock works in time synchronizatio n mode, and the PTP time and system clock are being traced.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	The system clock priority list is set, but all clock sources except the internal clock source are lost. The clock works in holdover or free-run mode. The system clock works in time synchronization mode, but no synchronization source is available. The system clock and PTP time work in holdover or free-run mode.
ACTX	Active/standby status indicator for the cross-connect or clock board	Green	Steady on	The board is in the active state on the service plane.
		-	Off	The board is in the standby state on the service plane.
ACTC	Control status indicator	Green	Steady on	The board is in the active control state.
		Green	Blinks on and off at 100 ms intervals	The data of the device is backed up in batch.
		-	Off	The board is in the standby control state.

Silkscreen	Name	Color	Status	Description
ALM	Alarm indicator	Red	Steady on	Device-level critical alarms exist.
		Orange	Steady on	A device-level major or minor alarm exists.
		-	Off	There is no equipment-level alarm.
ETH/OAM	ETH/OAM indicator	Green	Steady on	The link is up.
		Orange	Blinking	Data is being transmitted or received.
		-	Off	The link is down.

Table 1-513 Buttons on the CR8DIPU480MA

Silkscreen	Name	Description
RST	Board reset button	You can press the RST button to reset the NE or restore the factory settings of the NE. The procedure is as follows: To reset the NE, press the RST button and release it. To restore factory settings, press and hold down the RST button. This operation will clear the NE database and system parameter configurations and cannot be restored. Exercise caution when you perform the following operations: 1.Press the RST button to restart the NE. 2.When the PROG indicator blinks (the blinking frequency is about 1 second and the time window is about 15 seconds), press and hold down the RST button. 3.Wait for no longer than 2 minutes till both the STAT and PROG indicators blink (the time window is 5 seconds). Then release the RST button immediately.

Silkscreen	Name	Description
		NOTE 1. When you press the RST button to clear the configuration file, the original configuration file will be cleared. You are advised to periodically back up the configuration file. 2. When you press the RST button to clear the configuration file, to prevent the cleared configuration from being synchronized again between the active and standby main control boards, remove one main control board first in a dual-main control board chassis to ensure that the configuration file is cleared in a single-main control board scenario. Perform the same operations on the other main control board. 3. When pressing the RST button to clear the configuration file, you only need to pay attention to the STAT and PROG indicators. Other indicators may vary depending on the product model.

Ports

Table 1-514 Ports on the CR8DIPU480MA

Port	Connector Type	Description	Available Components
ETH/OAM	RJ45	10M/100M auto-sensing Ethernet network management port or non-standard RS-232 console port NOTE If a service port can be connected to an NMS, you are advised to use the service port as the network management port.	ETH/OAM management cable
CLK/TOD/RS-485	RJ45	Clock/Time/RS-485 input and output ports RS-485 is a reserved port.	1-into-2 clock cable + clock cable (external clock mode, delivery-ready): applicable to the CLK mode 1-into-2 clock cable + clock cable (external clock/external time mode, prepared onsite): applicable to the CLK or TOD mode 1-into-2 clock cable + Ethernet cable: applicable to the TOD mode
ALMI/ALMO	RJ45	Alarm input/output interface	Alarm input/output cable
USB	Type A	USB port (reserved)	-
CLK/TEST	SMA	Clock test port	Coaxial cable

Table 1-515 Pins of the ETH/OAM interface

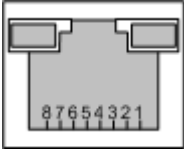
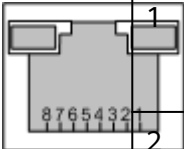
Front View	Pin	Usage
	1	Transmit positive of the NM network port
	2	Transmit negative of the NM network port
	3	Receive positive of the NM network port
	4	Grounding end of the Console interface
	5	Receive end of the Console interface
	6	Receive negative of the NM network port
	7	Unspecified
	8	Transmit end of the Console interface

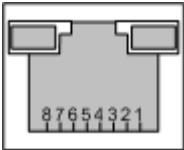
Table 1-516 Pins of the CLK/TOD/RS-485 interface

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	1	Receive negative of the CLK	Receive negative of the CLK	Unspecified	Unspecified	Unspecified
	2	Receive positive of the CLK	Receive positive of the CLK	Unspecified	Unspecified	Unspecified

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	3	Input/Output negative for the 1PPS signal	Unspecified	Input negative for the 1PPS signal (RS422 level)	Output negative for the 1PPS signal (RS422 level)	Transmit negative of the RS-485
	4	Receive negative of the CLK	Receive negative of the CLK	Grounding end	Grounding end	Unspecified
	5	Receive positive of the CLK	Receive positive of the CLK	Grounding end	Grounding end	Unspecified
	6	Input/Output positive for the 1PPS signal	Unspecified	Input positive for the 1PPS signal (RS422 level)	Output positive for the 1PPS signal (RS422 level)	Transmit positive of the RS-485

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	7	Input/Output negative for the time information	Unspecified	Input negative for the time information (RS422 level)	Output negative for the time information (RS422 level)	Receive negative of the RS-485
	8	Input/Output positive for the time information	Unspecified	Input positive for the time information (RS422 level)	Output positive for the time information (RS422 level)	Receive positive of the RS-485
NOTE When this interface is used as a TOD interface, it must be used with a dedicated clock cable. For details, see "Cables" in the Hardware Description.						

Table 1-517 Pins of the ALMI/ALMO interface

Front View	Pin	Usage
	1	Alarm input 1
	2	Grounding end for alarm input 1
	3	Alarm input 2

Front View	Pin	Usage
	4	Alarm input 3
	5	Grounding end for alarm input 3
	6	Grounding end for alarm input 2
	7	Positive for alarm signal output
	8	Negative for alarm signal output

Functions and Features

Table 1-518 Functions and features of the CR8DIPU480MA

Functions and Features	Description
Basic function	Data forwarding: The IPU is the core of service processing in the entire system, and is connected to all subcards through data channels. Control and management: The IPU manages subcards and transmits routing protocol data through management channels. 960 Gbit/s@190 bytes (bidirectional)
Reliability and availability	Hot swappable. 1:1 backup is supported.
Restrictions and remarks	This board cannot be used together with other IPU.
Breakout supported	No
Interface rate auto-sensing supported	No

Technical Specifications

Table 1-519 Technical specifications of the CR8DIPU480MA

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	22.4 mm x 388.4 mm x 209.3 mm (0.88 in. x 15.29 in. x 8.24 in.)

Item	Specification
Dimensions with packaging (H x W x D) [mm(in.)]	170 mm x 560 mm x 385 mm (6.69 in. x 22.05 in. x 15.16 in.)
Weight without packaging [kg(lb)]	2.3 kg (5.07 lb)
Weight with packaging [kg(lb)]	3.7 kg (8.16 lb)
Typical power consumption [W]	113.3 W
Typical heat dissipation [BTU/hour]	367.66 BTU/hour
Long-term operating temperature [°C(°F)]	–40°C to +65°C (–40°F to +149°F), start: ≥ –20°C (–4°F)
CPU	8-core 1.5 GHz
SDRAM	8GB
Storage	4 GB NandFlash

1.4.3.50 CR8DIPU480MG (Integrated Network Processing Unit (IPU-480-DA))

Overview

Table 1-520 Basic information about the CR8DIPU480MG

Item	Details
Description	Integrated Network Processing Unit (IPU-480-DA)
Part Number	03034WFD
Model	CR8DIPU480MG
Card type	IPU
Silkscreen	IPU-480-DA

Appearance

Figure 1-119 Appearance of the CR8DIPU480MG



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03034WFD&productModel=CR8DIPU480MG#matchRelation>

Indicators and Buttons

Table 1-521 Indicators on the CR8DIPU480MG

Silkscreen	Name	Color	Status	Description
STAT	Working status indicator	Green	Steady on	The board is working properly.
		Red	Steady on	A hardware alarm is generated. You can run the display alarm hardware command to check the hardware alarm type.
		-	Off	The board is not powered on or is not running.
PROG	Program status indicator	Green	Steady on	The board software is being initialized or is successfully initialized and is running properly.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	The memory self-check fails. The board software or logic file is lost. The board software fails to be loaded.
		Green	Blinks on and off at 100 ms intervals	The board software is being loaded.
		Green	Blinks on and off at 300 ms intervals	The BIOS is interacting with the upper-layer software.
		Red	Blinks on and off at 100 ms intervals	The BIOS self-check fails.
		-	Off	The board is not powered on.

Silkscreen	Name	Color	Status	Description
SYNC	Clock synchronizatio n indicator	Green	Steady on	The system clock works in free-run mode, and no priority list is set for the system clocks (only internal clock sources are listed by default). The system clock works in tracing mode and is tracing a clock source other than the internal clock sources in the priority list. The system clock works in time synchronizatio n mode, and the PTP time and system clock are being traced.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	The system clock priority list is set, but all clock sources except the internal clock source are lost. The clock works in holdover or free-run mode. The system clock works in time synchronization mode, but no synchronization source is available. The system clock and PTP time work in holdover or free-run mode.
ACTX	Active/standby status indicator for the cross-connect or clock board	Green	Steady on	The board is in the active state on the service plane.
		-	Off	The board is in the standby state on the service plane.
ACTC	Control status indicator	Green	Steady on	The board is in the active control state.
		Green	Blinks on and off at 100 ms intervals	The data of the device is backed up in batch.
		-	Off	The board is in the standby control state.

Silkscreen	Name	Color	Status	Description
ALM	Alarm indicator	Red	Steady on	Device-level critical alarms exist.
		Orange	Steady on	A device-level major or minor alarm exists.
		-	Off	There is no equipment-level alarm.
ETH/OAM	ETH/OAM indicator	Green	Steady on	The link is up.
		Orange	Blinking	Data is being transmitted or received.
		-	Off	The link is down.

Table 1-522 Buttons on the CR8DIPU480MG

Silkscreen	Name	Description
RST	Board reset button	You can press the RST button to reset the NE or restore the factory settings of the NE. The procedure is as follows: To reset the NE, press the RST button and release it. To restore factory settings, press and hold down the RST button. This operation will clear the NE database and system parameter configurations and cannot be restored. Exercise caution when you perform the following operations: 1.Press the RST button to restart the NE. 2.When the PROG indicator blinks (the blinking frequency is about 1 second and the time window is about 15 seconds), press and hold down the RST button. 3.Wait for no longer than 2 minutes till both the STAT and PROG indicators blink (the time window is 5 seconds). Then release the RST button immediately.

Silkscreen	Name	Description
		NOTE 1. When you press the RST button to clear the configuration file, the original configuration file will be cleared. You are advised to periodically back up the configuration file. 2. When you press the RST button to clear the configuration file, to prevent the cleared configuration from being synchronized again between the active and standby main control boards, remove one main control board first in a dual-main control board chassis to ensure that the configuration file is cleared in a single-main control board scenario. Perform the same operations on the other main control board. 3. When pressing the RST button to clear the configuration file, you only need to pay attention to the STAT and PROG indicators. Other indicators may vary depending on the product model.

Ports

Table 1-523 Ports on the CR8DIPU480MG

Port	Connector Type	Description	Available Components
ETH/OAM	RJ45	10M/100M auto-sensing Ethernet network management port or non-standard RS-232 console port NOTE If a service port can be connected to an NMS, you are advised to use the service port as the network management port.	ETH/OAM management cable
CLK/TOD/RS-485	RJ45	Clock/Time/RS-485 input and output ports RS-485 is a reserved port.	1-into-2 clock cable + clock cable (external clock mode, delivery-ready): applicable to the CLK mode 1-into-2 clock cable + clock cable (external clock/external time mode, prepared onsite): applicable to the CLK or TOD mode 1-into-2 clock cable + Ethernet cable: applicable to the TOD mode
ALMI/ALMO	RJ45	Alarm input/output interface	Alarm input/output cable
USB	Type A	USB port (reserved)	-
CLK/TEST	SMA	Clock test port	Coaxial cable

Table 1-524 Pins of the ETH/OAM interface

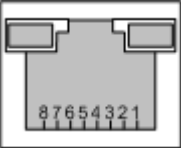
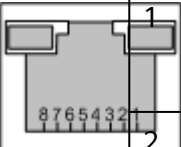
Front View	Pin	Usage
	1	Transmit positive of the NM network port
	2	Transmit negative of the NM network port
	3	Receive positive of the NM network port
	4	Grounding end of the Console interface
	5	Receive end of the Console interface
	6	Receive negative of the NM network port
	7	Unspecified
	8	Transmit end of the Console interface

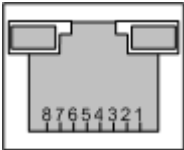
Table 1-525 Pins of the CLK/TOD/RS-485 interface

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	1	Receive negative of the CLK	Receive negative of the CLK	Unspecified	Unspecified	Unspecified
	2	Receive positive of the CLK	Receive positive of the CLK	Unspecified	Unspecified	Unspecified

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	3	Input/Output negative for the 1PPS signal	Unspecified	Input negative for the 1PPS signal (RS422 level)	Output negative for the 1PPS signal (RS422 level)	Transmit negative of the RS-485
	4	Receive negative of the CLK	Receive negative of the CLK	Grounding end	Grounding end	Unspecified
	5	Receive positive of the CLK	Receive positive of the CLK	Grounding end	Grounding end	Unspecified
	6	Input/Output positive for the 1PPS signal	Unspecified	Input positive for the 1PPS signal (RS422 level)	Output positive for the 1PPS signal (RS422 level)	Transmit positive of the RS-485

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	7	Input/Output negative for the time information	Unspecified	Input negative for the time information (RS422 level)	Output negative for the time information (RS422 level)	Receive negative of the RS-485
	8	Input/Output positive for the time information	Unspecified	Input positive for the time information (RS422 level)	Output positive for the time information (RS422 level)	Receive positive of the RS-485
NOTE When this interface is used as a TOD interface, it must be used with a dedicated clock cable. For details, see "Cables" in the Hardware Description.						

Table 1-526 Pins of the ALMI/ALMO interface

Front View	Pin	Usage
	1	Alarm input 1
	2	Grounding end for alarm input 1
	3	Alarm input 2

Front View	Pin	Usage
	4	Alarm input 3
	5	Grounding end for alarm input 3
	6	Grounding end for alarm input 2
	7	Positive for alarm signal output
	8	Negative for alarm signal output

Functions and Features

Table 1-527 Functions and features of the CR8DIPU480MG

Functions and Features	Description
Basic function	Data forwarding: The IPU is the core of service processing in the entire system, and is connected to all subcards through data channels. Control and management: The IPU manages subcards and transmits routing protocol data through management channels. 960 Gbit/s@190 bytes (bidirectional)
Reliability and availability	Hot swappable. 1:1 backup is supported.
Restrictions and remarks	This board cannot be used together with other IPU.
Breakout supported	No
Interface rate auto-sensing supported	No

Technical Specifications

Table 1-528 Technical specifications of the CR8DIPU480MG

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	22.4 mm x 388.4 mm x 209.3 mm (0.88 in. x 15.29 in. x 8.24 in.)

Item	Specification
Dimensions with packaging (H x W x D) [mm(in.)]	170 mm x 560 mm x 385 mm (6.69 in. x 22.05 in. x 15.16 in.)
Weight without packaging [kg(lb)]	2.3 kg (5.07 lb)
Weight with packaging [kg(lb)]	3.7 kg (8.16 lb)
Typical power consumption [W]	113.3 W
Typical heat dissipation [BTU/hour]	367.66 BTU/hour
Long-term operating temperature [°C(°F)]	–40°C to +65°C (–40°F to +149°F), start: ≥ –20°C (–4°F)
CPU	8-core 1.5 GHz
SDRAM	8GB
Storage	4 GB NandFlash

1.4.3.51 CR8DIPU800CG (03034MGN: Integrated Network Processing Unit (IPU-800-BN))

Overview

Table 1-529 Basic information about the CR8DIPU800CG

Item	Details
Description	Integrated Network Processing Unit (IPU-800-BN)
Part Number	03034MGN
Model	CR8DIPU800CG
Card type	IPU
Silkscreen	IPU-800-BN

Appearance

Figure 1-120 Appearance of the CR8DIPU800CG



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03034MGN&productModel=CR8DIPU800CG#matchRelation>

Indicators and Buttons

Table 1-530 Indicators on the CR8DIPU800CG

Silkscreen	Name	Color	Status	Description
STAT	Working status indicator	Green	Steady on	The board is working properly.
		Red	Steady on	The board hardware is faulty.
		Orange	Steady on	The interface board is not loaded with logic.
		-	Steady off	The board is not powered on or is not running.
PROG	Program status indicator	Green	Steady on	The board software is being initialized or is successfully initialized and is running properly.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	The memory self-check fails. The board software or logic file is lost. The board software fails to be loaded.
		Green	Blinks on and off at 100 ms intervals	The board software is being loaded.
		Green	Blinks on and off at 300 ms intervals	The BIOS is booting the upper-layer software.
		Red	Blinks on and off at 100 ms intervals	The BIOS self-check fails.
		-	Off	The board is not powered on.

Silkscreen	Name	Color	Status	Description
SYNC	Clock synchronizatio n indicator	Green	Steady on	The system clock works in free-run mode and no priority list is configured for the system clock. (By default, the list contains only internal clock sources.) The system clock is working in tracing mode and is tracing a clock source rather than the internal sources in the priority list. The system clock is working in time synchronization mode, and the PTP time and system clock are in the tracing state.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	The system clock priority list is set. All the clock sources, however, are lost except for the internal clock sources. The clock works in holdover mode or free-run mode. The system clock is working in time synchronization mode, but no synchronization source is available. The system clock and PTP time are working in holdover or free-run mode.
ACTX	Active/standby status indicator for the cross-connect or clock board	Green	Steady on	The board is in the active switching state.
		-	Off	The board is in the standby switching state.
ACTC	Control status indicator	Green	Steady on	The board is in the active control state.
		Green	Blinks on and off at 100 ms intervals	The data of the device is backed up in batch.

Silkscreen	Name	Color	Status	Description
		-	Off	The board is in the standby control state.
ALM	Device alarm indicator	Red	Steady on	There is an equipment-level critical alarm.
		Orange	Steady on	There is an equipment-level major or minor alarm.
		-	Off	There is no equipment-level alarm.
ETH/OAM	ETH/OAM indicator	Green	Steady on	The link is up.
		Orange	Blinking	Data is being transmitted or received.
		-	Off	The link is down.

Table 1-531 Buttons on the CR8DIPU800CG

Silkscreen	Name	Description
RST	Board reset button	You can press the RST button to reset the NE or restore the factory settings of the NE. The procedure is as follows: To reset the NE, press the RST button and release it. To restore factory settings, press and hold the RST button. This operation will clear the NE database and system parameter configurations and cannot be restored. Exercise caution when you perform the following operations: 1. Press and release the RST button to restart the NE. 2. When the PROG indicator blinks (the blinking frequency is about 1 second and the time window is about 15 seconds), press and hold the RST button. 3. Wait for no longer than 2 minutes till both the STAT and PROG indicators blink (the time window is 5 seconds). Then release the RST button immediately.

Silkscreen	Name	Description
		NOTE 1. When you press the RST button to clear the configuration file, the original configuration file will be cleared. You are advised to periodically back up the configuration file. 2. When you press the RST button to clear the configuration file, to prevent the cleared configuration from being synchronized again between the active and standby main control boards, remove one main control board first in a dual-main control board chassis to ensure that the configuration file is cleared in a single-main control board scenario. Perform the same operations on the other main control board. 3. When pressing the RST button to clear the configuration file, you only need to pay attention to the STAT and PROG indicators. Other indicators may vary depending on the product model.

Ports

Table 1-532 Ports on the CR8DIPU800CG

Port	Connector Type	Description	Available Components
CLK/TEST	SMA	Clock test port	Coaxial cable
USB	Type A	USB port (reserved)	-

Port	Connector Type	Description	Available Components
ETH/OAM	RJ45	10M/100M auto-sensing Ethernet NM interface and non-standard RS-232 Console interface. NOTE If a service port can be connected to an NMS, you are advised to use the service port as the network management port.	ETH/OAM management cable
CLK/TOD/RS-485	RJ45	Clock/Time/RS-485 input/output port RS-485 is a reserved port.	1-into-2 clock cable + clock cable (external clock mode, delivery-ready): applicable to the CLK mode 1-into-2 clock cable + clock cable (external clock/external time mode, prepared onsite): applicable to the CLK or TOD mode 1-into-2 clock cable + Ethernet cable: applicable to the TOD mode
ALMI/ALMO	RJ45	Alarm input/output interface	Alarm input/output cable
HP-GE	SFP	Ultra-high precision GE port (reserved)	Optical fiber

Table 1-533 Pins of the ETH/OAM interface

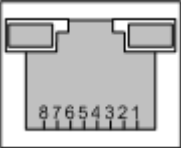
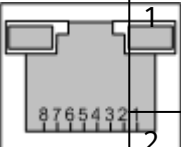
Front View	Pin	Usage
	1	Transmit positive of the NM network port
	2	Transmit negative of the NM network port
	3	Receive positive of the NM network port
	4	Grounding end of the Console interface
	5	Receive end of the Console interface
	6	Receive negative of the NM network port
	7	Unspecified
	8	Transmit end of the Console interface

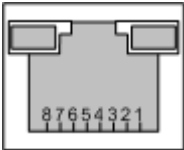
Table 1-534 Pins of the CLK/TOD/RS-485 interface

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	1	Receive negative of the CLK	Receive negative of the CLK	Unspecified	Unspecified	Unspecified
	2	Receive positive of the CLK	Receive positive of the CLK	Unspecified	Unspecified	Unspecified

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	3	Input/Output negative for the 1PPS signal	Unspecified	Input negative for the 1PPS signal (RS422 level)	Output negative for the 1PPS signal (RS422 level)	Transmit negative of the RS-485
	4	Receive negative of the CLK	Receive negative of the CLK	Grounding end	Grounding end	Unspecified
	5	Receive positive of the CLK	Receive positive of the CLK	Grounding end	Grounding end	Unspecified
	6	Input/Output positive for the 1PPS signal	Unspecified	Input positive for the 1PPS signal (RS422 level)	Output positive for the 1PPS signal (RS422 level)	Transmit positive of the RS-485

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	7	Input/Output negative for the time information	Unspecified	Input negative for the time information (RS422 level)	Output negative for the time information (RS422 level)	Receive negative of the RS-485
	8	Input/Output positive for the time information	Unspecified	Input positive for the time information (RS422 level)	Output positive for the time information (RS422 level)	Receive positive of the RS-485
NOTE When this interface is used as a TOD interface, it must be used with a dedicated clock cable. For details, see "Cables" in the Hardware Description.						

Table 1-535 Pins of the ALMI/ALMO interface

Front View	Pin	Usage
	1	Alarm input 1
	2	Grounding end for alarm input 1
	3	Alarm input 2

Front View	Pin	Usage
	4	Alarm input 3
	5	Grounding end for alarm input 3
	6	Grounding end for alarm input 2
	7	Positive for alarm signal output
	8	Negative for alarm signal output

Functions and Features

Table 1-536 Functions and features of the CR8DIPU800CG

Functions and Features	Description
Basic function	Line rate: 1200 Gbit/s. Supports switching, control, and clock management. Supports board-level 1:1 backup. Supports the ultra-high-precision clock.
Reliability and availability	Hot swappable. 1:1 backup is supported.

Technical Specifications

Table 1-537 Technical specifications of the CR8DIPU800CG

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	34.8 mm x 354.4 mm x 209 mm (1.37 in. x 13.95 in. x 8.23 in.)
Dimensions with packaging (H x W x D) [mm(in.)]	185 mm x 485 mm x 385 mm (7.28 in. x 19.09 in. x 15.16 in.)
Weight without packaging [kg(lb)]	2 kg (4.41 lb)
Weight with packaging [kg(lb)]	3.4 kg (7.5 lb)
Typical power consumption [W]	93.6 W
Typical heat dissipation [BTU/hour]	303.6 BTU/hour

Item	Specification
Long-term operating temperature [°C(°F)]	–20°C to +55°C (–4°F to +131°F); start: ≥ –5°C (23°F)
Switching capacity	1.6 Tbit/s
CPU	8-core 1.5 GHz
SDRAM	8GB
Storage	8GB M.2

1.4.3.52 CR8DIPU800CG (03034MGN-001: Integrated Network Processing Unit (IPU-800-BN))

Overview

Table 1-538 Basic information about the CR8DIPU800CG

Item	Details
Description	Integrated Network Processing Unit (IPU-800-BN)
Part Number	03034MGN-001
Model	CR8DIPU800CG
Card type	IPU
Silkscreen	IPU-800-BN

Appearance

Figure 1-121 Appearance of the CR8DIPU800CG



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find

desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03034MGN-001&productModel=CR8DIPU800CG#matchRelation>

Indicators and Buttons

Table 1-539 Indicators on the CR8DIPU800CG

Silkscreen	Name	Color	Status	Description
STAT	Working status indicator	Green	Steady on	The board is working properly.
		Red	Steady on	The board hardware is faulty.
		Orange	Steady on	The interface board is not loaded with logic.
		-	Steady off	The board is not powered on or is not running.
PROG	Program status indicator	Green	Steady on	The board software is being initialized or is successfully initialized and is running properly.
		Red	Steady on	The memory self-check fails. The board software or logic file is lost. The board software fails to be loaded.
		Green	Blinks on and off at 100 ms intervals	The board software is being loaded.

Silkscreen	Name	Color	Status	Description
		Green	Blinks on and off at 300 ms intervals	The BIOS is booting the upper-layer software.
		Red	Blinks on and off at 100 ms intervals	The BIOS self-check fails.
		-	Off	The board is not powered on.
SYNC	Clock synchronization indicator	Green	Steady on	The system clock works in free-run mode and no priority list is configured for the system clock. (By default, the list contains only internal clock sources.) The system clock is working in tracing mode and is tracing a clock source rather than the internal sources in the priority list. The system clock is working in time synchronization mode, and the PTP time and system clock are in the tracing state.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	The system clock priority list is set. All the clock sources, however, are lost except for the internal clock sources. The clock works in holdover mode or free-run mode. The system clock is working in time synchronization mode, but no synchronization source is available. The system clock and PTP time are working in holdover or free-run mode.
ACTX	Active/standby status indicator for the cross-connect or clock board	Green	Steady on	The board is in the active switching state.
		-	Off	The board is in the standby switching state.
ACTC	Control status indicator	Green	Steady on	The board is in the active control state.
		Green	Blinks on and off at 100 ms intervals	The data of the device is backed up in batch.

Silkscreen	Name	Color	Status	Description
		-	Off	The board is in the standby control state.
ALM	Device alarm indicator	Red	Steady on	There is an equipment-level critical alarm.
		Orange	Steady on	There is an equipment-level major or minor alarm.
		-	Off	There is no equipment-level alarm.
ETH/OAM	ETH/OAM indicator	Green	Steady on	The link is up.
		Orange	Blinking	Data is being transmitted or received.
		-	Off	The link is down.

Table 1-540 Buttons on the CR8DIPU800CG

Silkscreen	Name	Description
RST	Board reset button	You can press the RST button to reset the NE or restore the factory settings of the NE. The procedure is as follows: To reset the NE, press the RST button and release it. To restore factory settings, press and hold the RST button. This operation will clear the NE database and system parameter configurations and cannot be restored. Exercise caution when you perform the following operations: 1. Press and release the RST button to restart the NE. 2. When the PROG indicator blinks (the blinking frequency is about 1 second and the time window is about 15 seconds), press and hold the RST button. 3. Wait for no longer than 2 minutes till both the STAT and PROG indicators blink (the time window is 5 seconds). Then release the RST button immediately.

Silkscreen	Name	Description
		NOTE 1. When you press the RST button to clear the configuration file, the original configuration file will be cleared. You are advised to periodically back up the configuration file. 2. When you press the RST button to clear the configuration file, to prevent the cleared configuration from being synchronized again between the active and standby main control boards, remove one main control board first in a dual-main control board chassis to ensure that the configuration file is cleared in a single-main control board scenario. Perform the same operations on the other main control board. 3. When pressing the RST button to clear the configuration file, you only need to pay attention to the STAT and PROG indicators. Other indicators may vary depending on the product model.

Ports

Table 1-541 Ports on the CR8DIPU800CG

Port	Connector Type	Description	Available Components
CLK/TEST	SMA	Clock test port	Coaxial cable
USB	Type A	USB port (reserved)	-

Port	Connector Type	Description	Available Components
ETH/OAM	RJ45	10M/100M auto-sensing Ethernet NM interface and non-standard RS-232 Console interface. NOTE If a service port can be connected to an NMS, you are advised to use the service port as the network management port.	ETH/OAM management cable
CLK/TOD/RS-485	RJ45	Clock/Time/RS-485 input/output port RS-485 is a reserved port.	1-into-2 clock cable + clock cable (external clock mode, delivery-ready): applicable to the CLK mode 1-into-2 clock cable + clock cable (external clock/external time mode, prepared onsite): applicable to the CLK or TOD mode 1-into-2 clock cable + Ethernet cable: applicable to the TOD mode
ALMI/ALMO	RJ45	Alarm input/output interface	Alarm input/output cable
HP-GE	SFP	Ultra-high precision GE port (reserved)	Optical fiber

Table 1-542 Pins of the ETH/OAM interface

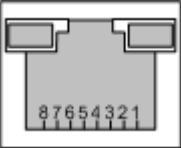
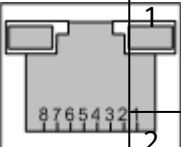
Front View	Pin	Usage
	1	Transmit positive of the NM network port
	2	Transmit negative of the NM network port
	3	Receive positive of the NM network port
	4	Grounding end of the Console interface
	5	Receive end of the Console interface
	6	Receive negative of the NM network port
	7	Unspecified
	8	Transmit end of the Console interface

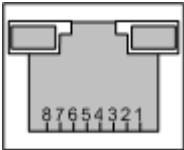
Table 1-543 Pins of the CLK/TOD/RS-485 interface

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	1	Receive negative of the CLK	Receive negative of the CLK	Unspecified	Unspecified	Unspecified
	2	Receive positive of the CLK	Receive positive of the CLK	Unspecified	Unspecified	Unspecified

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	3	Input/Output negative for the 1PPS signal	Unspecified	Input negative for the 1PPS signal (RS422 level)	Output negative for the 1PPS signal (RS422 level)	Transmit negative of the RS-485
	4	Receive negative of the CLK	Receive negative of the CLK	Grounding end	Grounding end	Unspecified
	5	Receive positive of the CLK	Receive positive of the CLK	Grounding end	Grounding end	Unspecified
	6	Input/Output positive for the 1PPS signal	Unspecified	Input positive for the 1PPS signal (RS422 level)	Output positive for the 1PPS signal (RS422 level)	Transmit positive of the RS-485

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	7	Input/Output negative for the time information	Unspecified	Input negative for the time information (RS422 level)	Output negative for the time information (RS422 level)	Receive negative of the RS-485
	8	Input/Output positive for the time information	Unspecified	Input positive for the time information (RS422 level)	Output positive for the time information (RS422 level)	Receive positive of the RS-485
NOTE When this interface is used as a TOD interface, it must be used with a dedicated clock cable. For details, see "Cables" in the Hardware Description.						

Table 1-544 Pins of the ALMI/ALMO interface

Front View	Pin	Usage
	1	Alarm input 1
	2	Grounding end for alarm input 1
	3	Alarm input 2

Front View	Pin	Usage
	4	Alarm input 3
	5	Grounding end for alarm input 3
	6	Grounding end for alarm input 2
	7	Positive for alarm signal output
	8	Negative for alarm signal output

Functions and Features

Table 1-545 Functions and features of the CR8DIPU800CG

Functions and Features	Description
Basic function	Line rate: 1200 Gbit/s. Supports switching, control, and clock management. Supports board-level 1:1 backup. Supports the ultra-high-precision clock.
Reliability and availability	Hot swappable. 1:1 backup is supported.

Technical Specifications

Table 1-546 Technical specifications of the CR8DIPU800CG

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	34.8 mm x 354.4 mm x 209 mm (1.37 in. x 13.95 in. x 8.23 in.)
Dimensions with packaging (H x W x D) [mm(in.)]	185 mm x 485 mm x 385 mm (7.28 in. x 19.09 in. x 15.16 in.)
Weight without packaging [kg(lb)]	2 kg (4.41 lb)
Weight with packaging [kg(lb)]	3.4 kg (7.5 lb)
Typical power consumption [W]	93.6 W
Typical heat dissipation [BTU/hour]	303.6 BTU/hour

Item	Specification
Long-term operating temperature [°C(°F)]	–20°C to +55°C (–4°F to +131°F); start: ≥ –5°C (23°F)
Switching capacity	1.6 Tbit/s
CPU	8-core 1.5 GHz
SDRAM	8GB
Storage	8GB M.2

1.4.3.53 CR8DUIPU2TM1 (Integrated Network Processing Unit (IPU-2T-AU))

Overview

Table 1-547 Basic information about the CR8DUIPU2TM1

Item	Details
Description	Integrated Network Processing Unit (IPU-2T-AU)
Part Number	03035FAV
Model	CR8DUIPU2TM1
Card type	IPU
Silkscreen	IPU-2T-AU
Remarks	The CR8M002FBXC3 fan is required.

Appearance

Figure 1-122 Appearance of the CR8DUIPU2TM1



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find

desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03035FAV&productModel=CR8DUIPU2TM1#matchRelation>

Indicators and Buttons

Table 1-548 Indicators on the CR8DUIPU2TM1

Silkscreen	Name	Color	Status	Description
STAT	Working status indicator	Green	Steady on	The board is working normally.
		Red	Steady on	A hardware alarm is generated. You can run the display alarm hardware command to check the hardware alarm type.
		-	Steady off	The board is not running or no power is input.
PROG	Program status indicator	Green	Steady on	The board software is being initialized or is successfully initialized and is running properly.
		Red	Steady on	The memory self-check fails. The board software or logic file is lost. The board software fails to be loaded.

Silkscreen	Name	Color	Status	Description
		Green	Blinks on and off at 100 ms intervals	The board software is being loaded.
		Green	On for 300 ms and off for 300 ms repeatedly	The BIOS is interacting with the upper-layer software.
		Red	Blinks on and off at 100 ms intervals	The BIOS self-check fails.
		-	Steady off	The board is not powered on.
SYNC	Clock synchronization indicator	Green	Steady on	The system clock works in free-run mode, and no priority list is set for the system clocks (only internal clock sources are listed by default). The system clock works in tracing mode and is tracing a clock source other than the internal clock sources in the priority list. The system clock works in time synchronization mode, and the PTP time and system clock are being traced.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	The system clock priority list is set, but all clock sources except the internal clock source are lost. The clock works in holdover or free-run mode. The system clock works in time synchronization mode, but no synchronization source is available. The system clock and PTP time work in holdover or free-run mode.
ACTX	Active/standby status indicator for the cross-connect or clock board	Green	Steady on	The board is in the active state on the service plane.
		-	Steady off	The board is in the standby state on the service plane.
ACTC	Control status indicator	Green	Steady on	The board is in the active control state.
		Green	Blinks on and off at 100 ms intervals	The data of the device is backed up in batch.
		-	Steady off	The board is in the standby control state.

Silkscreen	Name	Color	Status	Description
ALM	Alarm indicator	Red	Steady on	Device-level critical alarms exist.
		Orange	Steady on	A device-level major or minor alarm exists.
		-	Steady off	There is no equipment-level alarm.
ETH/OAM	ETH/OAM indicator	Green	Steady on	The link is up.
		Orange	Blinking	Data is being transmitted and received.
		-	Steady off	The link is Down.

Table 1-549 Buttons on the CR8DUIPU2TM1

Silkscreen	Name	Description
RST	Board reset button	You can press the RST button to reset the NE or restore the factory settings of the NE. The procedure is as follows: To reset the NE, press the RST button and release it. To restore factory settings, press and hold down the RST button. This operation will clear the NE database and system parameter configurations and cannot be restored. Exercise caution when you perform the following operations: 1.Press the RST button to restart the NE. 2.When the PROG indicator blinks (the blinking frequency is about 1 second and the time window is about 15 seconds), press and hold down the RST button. 3.Wait for no longer than 2 minutes till both the STAT and PROG indicators blink (the time window is 5 seconds). Then release the RST button immediately.

Silkscreen	Name	Description
		NOTE 1. When you press the RST button to clear the configuration file, the original configuration file will be cleared. You are advised to periodically back up the configuration file. 2. When you press the RST button to clear the configuration file, to prevent the cleared configuration from being synchronized again between the active and standby main control boards, remove one main control board first in a dual-main control board chassis to ensure that the configuration file is cleared in a single-main control board scenario. Perform the same operations on the other main control board. 3. When pressing the RST button to clear the configuration file, you only need to pay attention to the STAT and PROG indicators. Other indicators may vary depending on the product model.

Ports

Table 1-550 Ports on the CR8DUIPU2TM1

Port	Connector Type	Description	Available Components
ETH/OAM	RJ45	10M/100M auto-sensing Ethernet network management port and non-standard RS-232 console port NOTE If a service port can be connected to the NMS, it is recommended that the service port be used as the network management port.	ETH/OAM management cable
CLK/TOD/RS-485	RJ45	Clock/Time/RS-485 input and output ports RS-485 is a reserved port.	1-into-2 clock cable + clock cable (external clock mode, delivery-ready): applicable to the CLK mode 1-into-2 clock cable + clock cable (external clock/external time mode, prepared onsite): applicable to the CLK or TOD mode 1-into-2 clock cable + Ethernet cable: applicable to the TOD mode
USB	Type A	USB port (reserved)	-
CLK/TEST	SMA	Clock test port	Coaxial cable

Port	Connector Type	Description	Available Components
HP-GE	SFP	Ultra-high-precision GE interface	1.25Gbps eSFP Optical Module 1.25Gbps eSFP CWDM Optical Module 1.25Gbps eSFP BIDI Optical Module

Table 1-551 Pins of the ETH/OAM interface

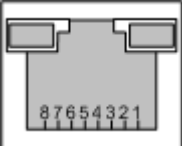
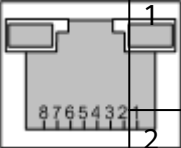
Front View	Pin	Usage
	1	Transmit positive of the NM network port
	2	Transmit negative of the NM network port
	3	Receive positive of the NM network port
	4	Grounding end of the Console interface
	5	Receive end of the Console interface
	6	Receive negative of the NM network port
	7	Unspecified
	8	Transmit end of the Console interface

Table 1-552 Pins of the CLK/TOD/RS-485 interface

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	1	Receive negative of the CLK	Receive negative of the CLK	Unspecified	Unspecified	Unspecified
	2	Receive positive of the CLK	Receive positive of the CLK	Unspecified	Unspecified	Unspecified
	3	Input/Output negative for the 1PPS signal	Unspecified	Input negative for the 1PPS signal (RS422 level)	Output negative for the 1PPS signal (RS422 level)	Transmit negative of the RS-485
	4	Receive negative of the CLK	Receive negative of the CLK	Grounding end	Grounding end	Unspecified
	5	Receive positive of the CLK	Receive positive of the CLK	Grounding end	Grounding end	Unspecified
	6	Input/Output positive for the 1PPS signal	Unspecified	Input positive for the 1PPS signal (RS422 level)	Output positive for the 1PPS signal (RS422 level)	Transmit positive of the RS-485

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	7	Input/Output negative for the time information	Unspecified	Input negative for the time information (RS422 level)	Output negative for the time information (RS422 level)	Receive negative of the RS-485
	8	Input/Output positive for the time information	Unspecified	Input positive for the time information (RS422 level)	Output positive for the time information (RS422 level)	Receive positive of the RS-485
NOTE When this interface is used as a TOD interface, it must be used with a dedicated clock cable. For details, see "Cables" in the Hardware Description.						

Functions and Features

Table 1-553 Functions and features of the CR8DUIPU2TM1

Functions and Features	Description
Basic function	Data forwarding: The IPU is the core of service processing in the entire system, and is connected to all subcards through data channels. Control and management: The main control board manages subcards and transmits routing protocol data through management channels. 4 Tbit/s@544 bytes (bidirectional)
Reliability and availability	Hot swappable. 1:1 backup is supported.
Restrictions and remarks	1. It cannot be used together with other IPU.
Breakout supported	No
Interface rate auto-sensing supported	No

Technical Specifications

Table 1-554 Technical specifications of the CR8DUIPU2TM1

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	34.8 mm x 354.4 mm x 209 mm (1.37 in. x 13.95 in. x 8.23 in.)
Dimensions with packaging (H x W x D) [mm(in.)]	185 mm x 485 mm x 385 mm (7.28 in. x 19.09 in. x 15.16 in.)
Weight without packaging [kg(lb)]	3.7 kg (8.16 lb)
Weight with packaging [kg(lb)]	5.1 kg (11.24 lb)
Typical power consumption [W]	240.4 W
Typical heat dissipation [BTU/hour]	779.81 BTU/hour
Long-term operating temperature [°C(°F)]	-20°C to +55°C, startup: ≥ -5°C
CPU	16-core 2.0 GHz
SDRAM	16GB
Storage	16GB M.2

1.4.3.54 CR8DUIPU2TM2 (Integrated Network Processing Unit (IPU-2T-E-BU))

Overview

Table 1-555 Basic information about the CR8DUIPU2TM2

Item	Details
Description	Integrated Network Processing Unit (IPU-2T-E-BU)
Part Number	03035FBC
Model	CR8DUIPU2TM2
Card type	IPU
Silkscreen	IPU-2T-E-BU
Remarks	The CR8M002FBXC3 fan is required.

Appearance

Figure 1-123 Appearance of the CR8DUIPU2TM2



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03035FBC&productModel=CR8DUIPU2TM2#matchRelation>

Indicators and Buttons

Table 1-556 Indicators on the CR8DUIPU2TM2

Silkscreen	Name	Color	Status	Description
STAT	Working status indicator	Green	Steady on	The board is working normally.
		Red	Steady on	A hardware alarm is generated. You can run the display alarm hardware command to check the hardware alarm type.
		-	Steady off	The board is not running or no power is input.
PROG	Program status indicator	Green	Steady on	The board software is being initialized or is successfully initialized and is running properly.
		Red	Steady on	The memory self-check fails. The board software or logic file is lost. The board software fails to be loaded.
		Green	Blinks on and off at 100 ms intervals	The board software is being loaded.

Silkscreen	Name	Color	Status	Description
		Green	On for 300 ms and off for 300 ms repeatedly	The BIOS is interacting with the upper-layer software.
		Red	Blinks on and off at 100 ms intervals	The BIOS self-check fails.
		-	Steady off	The board is not powered on.
SYNC	Clock synchronization indicator	Green	Steady on	The system clock works in free-run mode, and no priority list is set for the system clocks (only internal clock sources are listed by default). The system clock works in tracing mode and is tracing a clock source other than the internal clock sources in the priority list. The system clock works in time synchronization mode, and the PTP time and system clock are being traced.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	The system clock priority list is set, but all clock sources except the internal clock source are lost. The clock works in holdover or free-run mode. The system clock works in time synchronization mode, but no synchronization source is available. The system clock and PTP time work in holdover or free-run mode.
ACTX	Active/standby status indicator for the cross-connect or clock board	Green	Steady on	The board is in the active state on the service plane.
		-	Steady off	The board is in the standby state on the service plane.
ACTC	Control status indicator	Green	Steady on	The board is in the active control state.
		Green	Blinks on and off at 100 ms intervals	The data of the device is backed up in batch.
		-	Steady off	The board is in the standby control state.

Silkscreen	Name	Color	Status	Description
ALM	Alarm indicator	Red	Steady on	Device-level critical alarms exist.
		Orange	Steady on	A device-level major or minor alarm exists.
		-	Steady off	There is no equipment-level alarm.
ETH/OAM	ETH/OAM indicator	Green	Steady on	The link is up.
		Orange	Blinking	Data is being transmitted and received.
		-	Steady off	The link is Down.

Table 1-557 Buttons on the CR8DUIPU2TM2

Silkscreen	Name	Description
RST	Board reset button	You can press the RST button to reset the NE or restore the factory settings of the NE. The procedure is as follows: To reset the NE, press the RST button and release it. To restore factory settings, press and hold down the RST button. This operation will clear the NE database and system parameter configurations and cannot be restored. Exercise caution when you perform the following operations: 1.Press the RST button to restart the NE. 2.When the PROG indicator blinks (the blinking frequency is about 1 second and the time window is about 15 seconds), press and hold down the RST button. 3.Wait for no longer than 2 minutes till both the STAT and PROG indicators blink (the time window is 5 seconds). Then release the RST button immediately.

Silkscreen	Name	Description
		NOTE 1. When you press the RST button to clear the configuration file, the original configuration file will be cleared. You are advised to periodically back up the configuration file. 2. When you press the RST button to clear the configuration file, to prevent the cleared configuration from being synchronized again between the active and standby main control boards, remove one main control board first in a dual-main control board chassis to ensure that the configuration file is cleared in a single-main control board scenario. Perform the same operations on the other main control board. 3. When pressing the RST button to clear the configuration file, you only need to pay attention to the STAT and PROG indicators. Other indicators may vary depending on the product model.

Ports

Table 1-558 Ports on the CR8DUIPU2TM2

Port	Connector Type	Description	Available Components
ETH/OAM	RJ45	10M/100M auto-sensing Ethernet network management port and non-standard RS-232 console port NOTE If a service port can be connected to the NMS, it is recommended that the service port be used as the network management port.	ETH/OAM management cable
CLK/TOD/RS-485	RJ45	Clock/Time/RS-485 input and output ports RS-485 is a reserved port.	1-into-2 clock cable + clock cable (external clock mode, delivery-ready): applicable to the CLK mode 1-into-2 clock cable + clock cable (external clock/external time mode, prepared onsite): applicable to the CLK or TOD mode 1-into-2 clock cable + Ethernet cable: applicable to the TOD mode
USB	Type A	USB port (reserved)	-
CLK/TEST	SMA	Clock test port	Coaxial cable

Port	Connector Type	Description	Available Components
HP-GE	SFP	Ultra-high-precision GE interface	1.25Gbps eSFP Optical Module 1.25Gbps eSFP CWDM Optical Module 1.25Gbps eSFP BIDI Optical Module

Table 1-559 Pins of the ETH/OAM interface

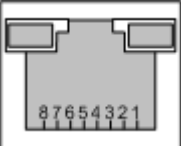
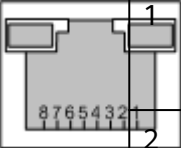
Front View	Pin	Usage
	1	Transmit positive of the NM network port
	2	Transmit negative of the NM network port
	3	Receive positive of the NM network port
	4	Grounding end of the Console interface
	5	Receive end of the Console interface
	6	Receive negative of the NM network port
	7	Unspecified
	8	Transmit end of the Console interface

Table 1-560 Pins of the CLK/TOD/RS-485 interface

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	1	Receive negative of the CLK	Receive negative of the CLK	Unspecified	Unspecified	Unspecified
	2	Receive positive of the CLK	Receive positive of the CLK	Unspecified	Unspecified	Unspecified
	3	Input/Output negative for the 1PPS signal	Unspecified	Input negative for the 1PPS signal (RS422 level)	Output negative for the 1PPS signal (RS422 level)	Transmit negative of the RS-485
	4	Receive negative of the CLK	Receive negative of the CLK	Grounding end	Grounding end	Unspecified
	5	Receive positive of the CLK	Receive positive of the CLK	Grounding end	Grounding end	Unspecified
	6	Input/Output positive for the 1PPS signal	Unspecified	Input positive for the 1PPS signal (RS422 level)	Output positive for the 1PPS signal (RS422 level)	Transmit positive of the RS-485

Front View	Pin	Definition of the RJ45 Interface on the Panel	Working Mode			
			External Clock	External Time Input (1PPS + Time Information)	External Time Output (1PPS + Time Information)	RS-485
	7	Input/Output negative for the time information	Unspecified	Input negative for the time information (RS422 level)	Output negative for the time information (RS422 level)	Receive negative of the RS-485
	8	Input/Output positive for the time information	Unspecified	Input positive for the time information (RS422 level)	Output positive for the time information (RS422 level)	Receive positive of the RS-485
NOTE When this interface is used as a TOD interface, it must be used with a dedicated clock cable. For details, see "Cables" in the Hardware Description.						

Functions and Features

Table 1-561 Functions and features of the CR8DUIPU2TM2

Functions and Features	Description
Basic function	Data forwarding: The IPU is the core of service processing in the entire system, and is connected to all subcards through data channels. Control and management: The main control board manages subcards and transmits routing protocol data through management channels. 4 Tbit/s@293 bytes (bidirectional)
Reliability and availability	Hot swappable. 1:1 backup is supported.
Restrictions and remarks	This board cannot be used together with other IPU.
Breakout supported	No
Interface rate auto-sensing supported	No

Technical Specifications

Table 1-562 Technical specifications of the CR8DUIPU2TM2

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	34.8 mm x 354.4 mm x 209 mm (1.37 in. x 13.95 in. x 8.23 in.)
Dimensions with packaging (H x W x D) [mm(in.)]	185 mm x 485 mm x 385 mm (7.28 in. x 19.09 in. x 15.16 in.)
Weight without packaging [kg(lb)]	3.7 kg (8.16 lb)
Weight with packaging [kg(lb)]	5.1 kg (11.24 lb)
Typical power consumption [W]	240.4 W
Typical heat dissipation [BTU/hour]	779.81 BTU/hour
Long-term operating temperature [°C(°F)]	-20°C to +55°C, startup: ≥ -5°C
CPU	16-core 2.0 GHz
SDRAM	32GB
Storage	16GB M.2

1.4.4 Interface Card

1.4.4.1 CR2D000DE110 (16-Port E1 Physical Interface Card(75ohm))

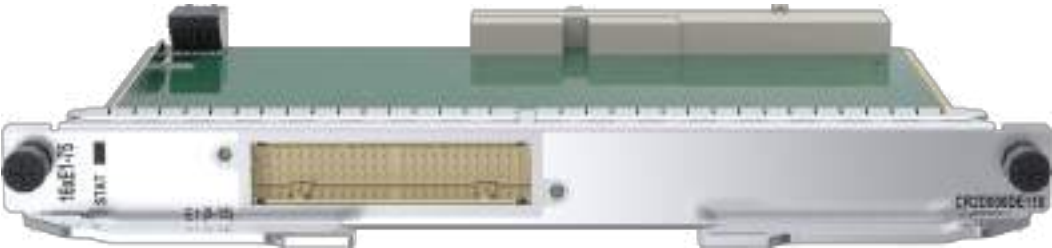
Overview

Table 1-563 Basic information about the CR2D000DE110

Item	Details
Description	16-Port E1 Physical Interface Card(75ohm)
Part Number	03030RFA
Model	CR2D000DE110
Card type	Interface Card
Silkscreen	16xE1-75

Appearance

Figure 1-124 Appearance of the CR2D000DE110



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03030RFA&productModel=CR2D000DE110#matchRelation>

Indicators

Table 1-564 Indicators on the CR2D000DE110

Silkscreen	Name	Color	Status	Description
STAT	Working status indicator	Green	Steady on	The board is working properly.
		Red	Steady on	The hardware on the PIC is faulty.
		Yellow	Steady on	A high-speed interface card is unexpectedly inserted into a low-speed interface card slot and an alarm is generated, or the logic is not loaded to the board.
		-	Off	The board is not powered on or is not registered.

Ports

Table 1-565 Ports on the CR2D000DE110

Port	Connector Type	Description	Available Components
E1(0-15)	Anea 96	16 E1 interfaces	75-Ohm 16 x E1 Cables

Table 1-566 Port attributes about the CR2D000DE110

Attributes	Details
Working mode	Full-duplex

Attributes	Details
Standards compliance	IMA 1.1, RFC4717, RFC4385, RFC4816, RFC5086, RFC4553_SATOP, and RFC1662
Frame format	Non-framed, CRC4 and NO-CRC4
Interface code	HDB3
Stated bit rate	2048 kbit/s

Functions and Features

Table 1-567 Functions and features of the CR2D000DE110

Functions and Features	Description
Basic function	Supports a maximum of 16 E1 interfaces. Services on each interface can be configured.
Reliability and availability	Hot swappable.
Breakout supported	No
Interface rate auto-sensing supported	No
FlexE supported	No
Link protocol	TDM, PPP, MP, ATM, IMA

Technical Specifications

Table 1-568 Technical specifications of the CR2D000DE110

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	19.8 mm x 193.8 mm x 209.3 mm (0.78 in. x 7.63 in. x 8.24 in.)
Dimensions with packaging (H x W x D) [mm(in.)]	70 mm x 285 mm x 300 mm (2.76 in. x 11.22 in. x 11.81 in.)
Weight without packaging [kg(lb)]	0.5 kg (1.1 lb)
Weight with packaging [kg(lb)]	0.9 kg (1.98 lb)
Typical power consumption [W]	9.5 W
Typical heat dissipation [BTU/hour]	30.66 BTU/hour

Item	Specification
Long-term operating temperature [°C(°F)]	–40°C to +65°C (–40°F to +149°F)

1.4.4.2 CR2D000DE111 (16-Port E1 Physical Interface Card(120ohm))

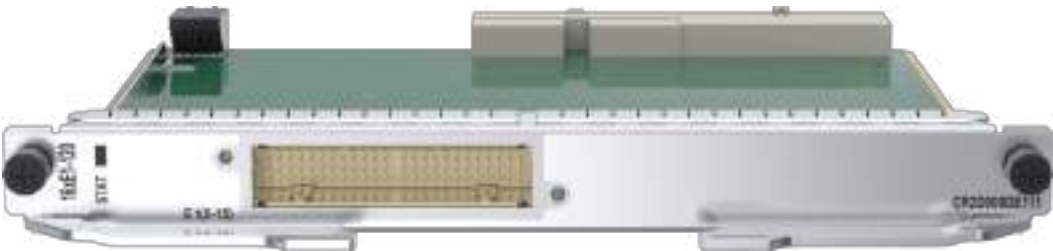
Overview

Table 1-569 Basic information about the CR2D000DE111

Item	Details
Description	16-Port E1 Physical Interface Card(120ohm)
Part Number	03030REY
Model	CR2D000DE111
Card type	Interface Card
Silkscreen	16xE1-120

Appearance

Figure 1-125 Appearance of the CR2D000DE111



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03030REY&productModel=CR2D000DE111#matchRelation>

Indicators

Table 1-570 Indicators on the CR2D000DE111

Silkscreen	Name	Color	Status	Description
STAT	Working status indicator	Green	Steady on	The board is working properly.
		Red	Steady on	The hardware on the PIC is faulty.
		Yellow	Steady on	A high-speed interface card is unexpectedly inserted into a low-speed interface card slot and an alarm is generated, or the logic is not loaded to the board.
		-	Off	The board is not powered on or is not registered.

Ports

Table 1-571 Ports on the CR2D000DE111

Port	Connector Type	Description	Available Components
E1(0-15)	Anea 96	16 E1 interfaces	120-Ohm 16 x E1 Cables

Table 1-572 Port attributes about the CR2D000DE111

Attributes	Details
Working mode	Full-duplex

Attributes	Details
Standards compliance	IMA 1.1, RFC4717, RFC4385, RFC4816, RFC5086, RFC4553_SATOP, and RFC1662
Frame format	Non-framed, CRC4 and NO-CRC4
Interface code	HDB3
Stated bit rate	2048 kbit/s

Functions and Features

Table 1-573 Functions and features of the CR2D000DE111

Functions and Features	Description
Basic function	Supports a maximum of 16 E1 interfaces. Services on each interface can be configured.
Reliability and availability	Hot swappable.
Breakout supported	No
Interface rate auto-sensing supported	No
FlexE supported	No
Link protocol	TDM, PPP, MP, ATM, IMA

Technical Specifications

Table 1-574 Technical specifications of the CR2D000DE111

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	19.8 mm x 193.8 mm x 209.3 mm (0.78 in. x 7.63 in. x 8.24 in.)
Dimensions with packaging (H x W x D) [mm(in.)]	70 mm x 285 mm x 300 mm (2.76 in. x 11.22 in. x 11.81 in.)
Weight without packaging [kg(lb)]	0.5 kg (1.1 lb)
Weight with packaging [kg(lb)]	0.9 kg (1.98 lb)
Typical power consumption [W]	9.5 W
Typical heat dissipation [BTU/hour]	30.66 BTU/hour

Item	Specification
Long-term operating temperature [°C(°F)]	–40°C to +65°C (–40°F to +149°F)

1.4.4.3 CR2D000EM890 (6 Channels E&M Interface Board)

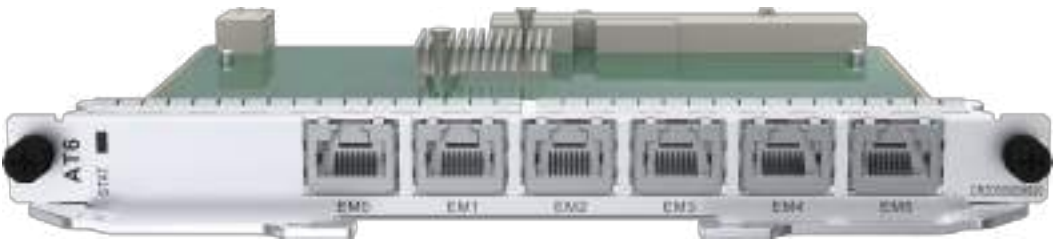
Overview

Table 1-575 Basic information about the CR2D000EM890

Item	Details
Description	6 Channels E&M Interface Board
Part Number	03032VFA
Model	CR2D000EM890
Card type	Interface Card
Silkscreen	AT6
Remarks	-

Appearance

Figure 1-126 Appearance of the CR2D000EM890



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03032VFA&productModel=CR2D000EM890#matchRelation>

Indicators

Table 1-576 Indicators on the CR2D000EM890

Silkscreen	Name	Color	Status	Description
STAT	Working status indicator	Green	Steady on	The PIC is working properly.
		Red	Steady on	The hardware on the PIC is faulty.
		Orange	Steady on	The PIC is not loaded with the logic.
		-	Off	The PIC is powered off or is not registered.

Ports

Table 1-577 Ports on the CR2D000EM890

Port	Connector Type	Description	Available Components
EM0 to EM5	RJ45	6-channel E&M	Prepare cables based on the actual interface type of the peer device. Shielded cables are recommended.

Table 1-578 Port attributes about the CR2D000EM890

Attributes	Details
Working mode	Full-duplex
Frame format	PWE3

Functions and Features

Table 1-579 Functions and features of the CR2D000EM890

Functions and Features	Description
Basic function	Occupies one flexible card slot. Supports the line-rate receiving and transmitting on six E&M interfaces.
Reliability and availability	Hot swappable.
Restrictions and remarks	Not applicable to outdoor scenarios.
Breakout supported	No
Interface rate auto-sensing supported	No
FlexE supported	No

Technical Specifications

Table 1-580 Technical specifications of the CR2D000EM890

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	19.8 mm x 193.8 mm x 209.3 mm (0.78 in. x 7.63 in. x 8.24 in.)
Dimensions with packaging (H x W x D) [mm(in.)]	70 mm x 285 mm x 300 mm (2.76 in. x 11.22 in. x 11.81 in.)
Weight without packaging [kg(lb)]	0.5 kg (1.1 lb)
Weight with packaging [kg(lb)]	0.9 kg (1.98 lb)
Typical power consumption [W]	10 W
Typical heat dissipation [BTU/hour]	32.44 BTU/hour
Long-term operating temperature [°C(°F)]	-5°C to +55°C (23°F to 131°F)

1.4.4.4 CR2D000IE110 (32-Port E1 Physical Interface Card(75ohm))

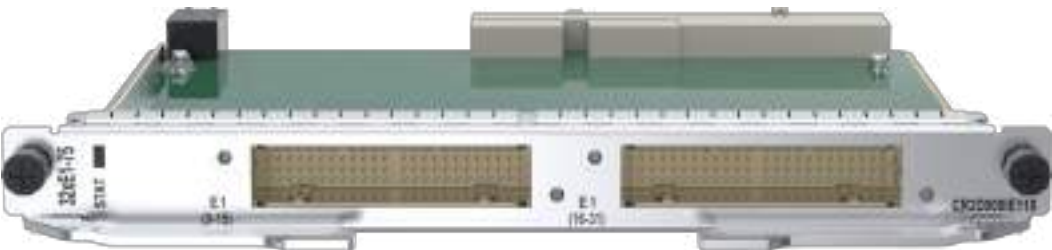
Overview

Table 1-581 Basic information about the CR2D000IE110

Item	Details
Description	32-Port E1 Physical Interface Card(75ohm)
Part Number	03030QCQ
Model	CR2D000IE110
Card type	Interface Card
Silkscreen	32xE1-75

Appearance

Figure 1-127 Appearance of the CR2D000IE110



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03030QCQ&productModel=CR2D000IE110#matchRelation>

Indicators

Table 1-582 Indicators on the CR2D000IE110

Silkscreen	Name	Color	Status	Description
STAT	Working status indicator	Green	Steady on	The board is working properly.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	The hardware on the PIC is faulty.
		Yellow	Steady on	A high-speed interface card is unexpectedly inserted into a low-speed interface card slot and an alarm is generated, or the logic is not loaded to the board.
		-	Off	The board is not powered on or is not registered.

Ports

Table 1-583 Ports on the CR2D000IE110

Port	Connector Type	Description	Available Components
E1(0-15) E1(16-31)	Anea 96	32 E1 interfaces	75-Ohm 16 x E1 Cables

Table 1-584 Port attributes about the CR2D000IE110

Attributes	Details
Working mode	Full-duplex
Standards compliance	IMA 1.1, RFC4717, RFC4385, RFC4816, RFC5086, RFC4553_SATOP, and RFC1662
Frame format	Non-framed, CRC4 and NO-CRC4
Interface code	HDB3
Stated bit rate	2048 kbit/s

Functions and Features

Table 1-585 Functions and features of the CR2D000IE110

Functions and Features	Description
Basic function	Supports a maximum of 32 E1 interfaces. Services on each interface can be configured.
Reliability and availability	Hot swappable.
Breakout supported	No
Interface rate auto-sensing supported	No
FlexE supported	No
Link protocol	TDM, PPP, MP, ATM, IMA

Technical Specifications

Table 1-586 Technical specifications of the CR2D000IE110

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	19.8 mm x 193.8 mm x 209.3 mm (0.78 in. x 7.63 in. x 8.24 in.)
Dimensions with packaging (H x W x D) [mm(in.)]	70 mm x 285 mm x 300 mm (2.76 in. x 11.22 in. x 11.81 in.)
Weight without packaging [kg(lb)]	0.5 kg (1.1 lb)
Weight with packaging [kg(lb)]	0.9 kg (1.98 lb)
Typical power consumption [W]	13.5 W
Typical heat dissipation [BTU/hour]	43.8 BTU/hour
Long-term operating temperature [°C(°F)]	-40°C to +65°C (-40°F to +149°F)

1.4.4.5 CR2D000IE111 (32-Port E1 Physical Interface Card(120ohm))

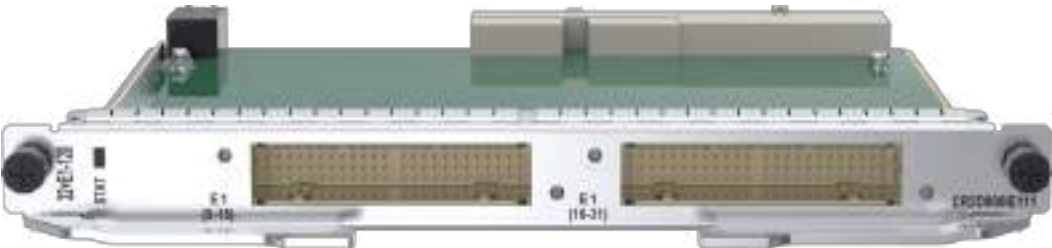
Overview

Table 1-587 Basic information about the CR2D000IE111

Item	Details
Description	32-Port E1 Physical Interface Card(120ohm)
Part Number	03030QCP
Model	CR2D000IE111
Card type	Interface Card
Silkscreen	32xE1-120

Appearance

Figure 1-128 Appearance of the CR2D000IE111



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03030QCP&productModel=CR2D000IE111#matchRelation>

Indicators

Table 1-588 Indicators on the CR2D000IE111

Silkscreen	Name	Color	Status	Description
STAT	Working status indicator	Green	Steady on	The board is working properly.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	The hardware on the PIC is faulty.
		Yellow	Steady on	A high-speed interface card is unexpectedly inserted into a low-speed interface card slot and an alarm is generated, or the logic is not loaded to the board.
		-	Off	The board is not powered on or is not registered.

Ports

Table 1-589 Ports on the CR2D000IE111

Port	Connector Type	Description	Available Components
E1(0-15) E1(16-31)	Anea 96	32 E1 interfaces	120-Ohm 16 x E1 Cables

Table 1-590 Port attributes about the CR2D000IE111

Attributes	Details
Working mode	Full-duplex
Standards compliance	IMA 1.1, RFC4717, RFC4385, RFC4816, RFC5086, RFC4553_SATOP, and RFC1662
Frame format	Non-framed, CRC4 and NO-CRC4
Interface code	HDB3
Stated bit rate	2048 kbit/s

Functions and Features

Table 1-591 Functions and features of the CR2D000IE111

Functions and Features	Description
Basic function	Supports a maximum of 32 E1 interfaces. Services on each interface can be configured.
Reliability and availability	Hot swappable.
Breakout supported	No
Interface rate auto-sensing supported	No
FlexE supported	No
Link protocol	TDM, PPP, MP, ATM, IMA

Technical Specifications

Table 1-592 Technical specifications of the CR2D000IE111

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	19.8 mm x 193.8 mm x 209.3 mm (0.78 in. x 7.63 in. x 8.24 in.)
Dimensions with packaging (H x W x D) [mm(in.)]	70 mm x 285 mm x 300 mm (2.76 in. x 11.22 in. x 11.81 in.)
Weight without packaging [kg(lb)]	0.5 kg (1.1 lb)
Weight with packaging [kg(lb)]	0.9 kg (1.98 lb)
Typical power consumption [W]	13.5 W
Typical heat dissipation [BTU/hour]	43.8 BTU/hour
Long-term operating temperature [°C(°F)]	-40°C to +65°C (-40°F to +149°F)

1.4.4.6 CR2D000SA890 (8-Channel V.35/X.21/V.24 Board)

Overview

Table 1-593 Basic information about the CR2D000SA890

Item	Details
Description	8-Channel V.35/X.21/V.24 Board
Part Number	03032TGR
Model	CR2D000SA890
Card type	Interface Card
Silkscreen	SA8
Remarks	-

Appearance

Figure 1-129 Appearance of the CR2D000SA890



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03032TGR&productModel=CR2D000SA890#matchRelation>

Indicators

Table 1-594 Indicators on the CR2D000SA890

Silkscreen	Name	Color	Status	Description
STAT	Working status indicator	Green	Steady on	The PIC is working properly.
		Red	Steady on	The hardware on the PIC is faulty.
		Yellow	Steady on	The PIC is installed in a slot for a HIC and an alarm is reported or the PIC is not loaded with the logic.
		-	Off	The PIC is powered off or is not registered.
L/A(0 to 7)	Running status indicator	Green	Steady on	The link is normal
		Green	Blinking	Data is being transmitted.
		-	Off	The link is Down.

Ports

Table 1-595 Ports on the CR2D000SA890

Port	Connector Type	Description	Available Components
0 to 7	DB28 serial interface	8-channel V.35/X.21/V.24 interface	V.35 DTE, DCE cable /X.21 DTE, DCE cable /V.24 DTE, DCE cable

Table 1-596 Port attributes about the CR2D000SA890

Attributes	Details
Working mode	Full-duplex
Frame format	PWE3

Functions and Features

Table 1-597 Functions and features of the CR2D000SA890

Functions and Features	Description
Basic function	Supports 8-channel V.35/X.21/V.24 interface features.
Reliability and availability	Hot swappable.
Restrictions and remarks	Not applicable to outdoor scenarios.
Breakout supported	No
Interface rate auto-sensing supported	No
FlexE supported	No

Technical Specifications

Table 1-598 Technical specifications of the CR2D000SA890

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	19.8 mm x 193.8 mm x 209.3 mm (0.78 in. x 7.63 in. x 8.24 in.)
Dimensions with packaging (H x W x D) [mm(in.)]	70 mm x 285 mm x 300 mm (2.76 in. x 11.22 in. x 11.81 in.)
Weight without packaging [kg(lb)]	0.6 kg (1.32 lb)
Weight with packaging [kg(lb)]	1.0 kg (2.2 lb)
Typical power consumption [W]	18 W
Typical heat dissipation [BTU/hour]	58.40 BTU/hour
Long-term operating temperature [°C(°F)]	0°C to 45°C (32°F to 113°F)
Short-term operating temperature [°C(°F)]	-5°C to +55°C (23°F to 131°F)

1.4.4.7 CR2D00E8GE12 (8-Port 100/1000Base-RJ45 Physical Interface Card)

Overview

Table 1-599 Basic information about the CR2D00E8GE12

Item	Details
Description	8-Port 100/1000Base-RJ45 Physical Interface Card
Part Number	03031DHB
Model	CR2D00E8GE12
Card type	Interface Card
Silkscreen	8xFE/GE-RJ

Appearance

Figure 1-130 Appearance of the CR2D00E8GE12



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03031DHB&productModel=CR2D00E8GE12#matchRelation>

Indicators

Table 1-600 Indicators on the CR2D00E8GE12

Silkscreen	Name	Color	Status	Description
STAT	Working status indicator	Green	Steady on	The board is working properly.
		Red	Steady on	The hardware on the PIC is faulty.
		Orange	Steady on	A high-speed interface card is unexpectedly inserted into a low-speed interface card slot and an alarm is generated, or the logic is not loaded to the board.
		-	Off	The board is not powered on or is not registered.
LINK	Connection status indicator	Green	Steady on	The LINK is located above each FE/GE electrical port. The data interface is transmitting or receiving data.
		-	Off	The physical port connection fails.

Silkscreen	Name	Color	Status	Description
ACT	Transmitting/ receiving status indicator	Orange	Blinking	The ACT is located above each FE/GE electrical port. The data interface is not transmitting or receiving data.
		-	Off	The data interface is not transmitting or receiving data.

Ports

Table 1-601 Ports on the CR2D00E8GE12

Port	Connector Type	Description	Available Components
FE/GE0-FE/GE7	RJ45	GE/FE service signal input and output interface	Ethernet Cables

Table 1-602 Port attributes about the CR2D00E8GE12

Attributes	Details
Working mode	Full-duplex
Standards compliance	IEEE 802.3
Frame format	Ethernet_II, Ethernet_SAP and Ethernet_SNAP

Functions and Features

Table 1-603 Functions and features of the CR2D00E8GE12

Functions and Features	Description
Basic function	Occupies one flexible interface card slot. Provides eight GE/FE interfaces for line-rate transmitting and receiving, and supports 10M/100M/1000M electrical interfaces.
Reliability and availability	Hot swappable.
Breakout supported	No
Interface rate auto-sensing supported	No
FlexE supported	No

Technical Specifications

Table 1-604 Technical specifications of the CR2D00E8GE12

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	19.8 mm x 193.8 mm x 209.3 mm (0.78 in. x 7.63 in. x 8.24 in.)
Dimensions with packaging (H x W x D) [mm(in.)]	70 mm x 285 mm x 300 mm (2.76 in. x 11.22 in. x 11.81 in.)
Weight without packaging [kg(lb)]	0.5 kg (1.1 lb)
Weight with packaging [kg(lb)]	0.9 kg (1.98 lb)
Typical power consumption [W]	11.7 W
Typical heat dissipation [BTU/hour]	37.96 BTU/hour
Long-term operating temperature [°C(°F)]	–40°C to +65°C (–40°F to +149°F)

1.4.4.8 CR2D00EAGF10 (10-Port 100/1000Base-X-SFP Physical Interface Card)

Overview

Table 1-605 Basic information about the CR2D00EAGF10

Item	Details
Description	10-Port 100/1000Base-X-SFP Physical Interface Card
Part Number	03031DJK
Model	CR2D00EAGF10
Card type	Interface Card
Silkscreen	10xFE/GE-SFP

Appearance

Figure 1-131 Appearance of the CR2D00EAGF10



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03031DJK&productModel=CR2D00EAGF10#matchRelation>

Indicators

Table 1-606 Indicators on the CR2D00EAGF10

Silkscreen	Name	Color	Status	Description
STAT	Working status indicator	Green	Steady on	The board is working properly.
		Red	Steady on	The hardware on the PIC is faulty.
		Orange	Steady on	A high-speed interface card is unexpectedly inserted into a low-speed interface card slot and an alarm is generated, or the logic is not loaded to the board.
		-	Off	The board is not powered on or is not registered.
L/A (0 to 9)	Interface status indicator	Green	Steady on	The link is Up.
		Green	Blinking	Data is being transmitted or received.
		-	Off	The link is Down.

Ports

Table 1-607 Ports on the CR2D00EAGF10

Port	Connector Type	Description	Available Components
OUT0 IN0-OUT9 IN9	SFP	GE/FE service signal input and output interface	155Mbps SFP Electrical Module 155Mbps eSFP Optical Module 155Mbps eSFP BIDI Optical Module 1Gbps Electrical Module 1.25Gbps eSFP Optical Module 1.25Gbps eSFP CWDM Optical Module 1.25Gbps eSFP BIDI Optical Module 125M~2.67Gbps eSFP DWDM Optical Module

Table 1-608 Port attributes about the CR2D00EAGF10

Attributes	Details
Working mode	Full-duplex
Standards compliance	IEEE 802.3
Frame format	Ethernet_II, Ethernet_SAP and Ethernet_SNAP

Functions and Features

Table 1-609 Functions and features of the CR2D00EAGF10

Functions and Features	Description
Basic function	Occupies one flexible interface card slot. Provides ten GE/FE interfaces for line-rate transmitting and receiving, and supports SFP optical modules.
Reliability and availability	Hot swappable.
Breakout supported	No
Interface rate auto-sensing supported	No
FlexE supported	No

Technical Specifications

Table 1-610 Technical specifications of the CR2D00EAGF10

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	19.8 mm x 193.8 mm x 209.3 mm (0.78 in. x 7.63 in. x 8.24 in.)
Dimensions with packaging (H x W x D) [mm(in.)]	70 mm x 285 mm x 300 mm (2.76 in. x 11.22 in. x 11.81 in.)
Weight without packaging [kg(lb)]	0.6 kg (1.32 lb)
Weight with packaging [kg(lb)]	1.0 kg (2.2 lb)
Typical power consumption [W]	13.3 W
Typical heat dissipation [BTU/hour]	43.15 BTU/hour
Long-term operating temperature [°C(°F)]	–40°C to +65°C (–40°F to +149°F)

1.4.4.9 CR2D00L4XF11 (4-Port 10GBase LAN/WAN-SFP+ Physical Interface Card)

Overview

Table 1-611 Basic information about the CR2D00L4XF11

Item	Details
Description	4-Port 10GBase LAN/WAN-SFP+ Physical Interface Card
Part Number	03031DJP
Model	CR2D00L4XF11
Card type	Interface Card
Silkscreen	4×10GE-SFP+

Appearance

Figure 1-132 Appearance of the CR2D00L4XF11



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03031DJP&productModel=CR2D00L4XF11#matchRelation>

Indicators

Table 1-612 Indicators on the CR2D00L4XF11

Silkscreen	Name	Color	Status	Description
STAT	Working status indicator	Green	Steady on	The board is working properly.
		Red	Steady on	The hardware on the PIC is faulty.
		Orange	Steady on	A high-speed interface card is unexpectedly inserted into a low-speed interface card slot and an alarm is generated, or the logic is not loaded to the board.
		-	Off	The board is not powered on or is not registered.
L/A (0 to 3)	Interface status indicator	Green	Steady on	The link is Up.
		Green	Blinking	Data is being transmitted or received.
		-	Off	The link is Down.

Ports

Table 1-613 Ports on the CR2D00L4XF11

Port	Connector Type	Description	Available Components
OUT0 IN0-OUT3 IN3	SFP+	10 GE service signal input and output interface	10Gbps SFP+ Optical Module 10Gbps SFP+ BIDI Optical Module 10Gbps SFP+ CWDM Optical Module 1.25/9.953/10.31 25Gbps SFP+ Optical Module 1.25Gbps eSFP BIDI Optical Module 1.25Gbps eSFP CWDM Optical Module 1.25Gbps eSFP Optical Module 125M~2.67Gbps eSFP DWDM Optical Module 1Gbps Electrical Module 155Mbps eSFP Optical Module 155Mbps eSFP BIDI Optical Module

Table 1-614 Port attributes about the CR2D00L4XF11

Attributes	Details
Working mode	Full-duplex
Standards compliance	IEEE 802.3
Frame format	Ethernet_II, Ethernet_SAP and Ethernet_SNAP

Functions and Features

Table 1-615 Functions and features of the CR2D00L4XF11

Functions and Features	Description
Basic function	Occupies one flexible interface card slot. Provides four 10GE interfaces for line-rate transmitting and receiving, and supports SFP+ optical modules.
Reliability and availability	Hot swappable.
Restrictions and remarks	The board does not support the OTN mode or FEC function. If a colored optical module is used and does not support optical amplifier insertion, the board supports only point-to-point optical transmission.
Breakout supported	No
Interface rate auto-sensing supported	Supports GE/10GE auto-sensing.
FlexE supported	No

Technical Specifications

Table 1-616 Technical specifications of the CR2D00L4XF11

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	19.8 mm x 193.8 mm x 209.3 mm (0.78 in. x 7.63 in. x 8.24 in.)
Dimensions with packaging (H x W x D) [mm(in.)]	70 mm x 285 mm x 300 mm (2.76 in. x 11.22 in. x 11.81 in.)
Weight without packaging [kg(lb)]	0.7 kg (1.54 lb)
Weight with packaging [kg(lb)]	1.1 kg (2.43 lb)
Typical power consumption [W]	25 W
Typical heat dissipation [BTU/hour]	81.11 BTU/hour
Long-term operating temperature [°C(°F)]	–40°C to +65°C (–40°F to +149°F)

1.4.4.10 CR2D00MP8A90 (4 Channels C37.94 Optical Interface and 4 Channels CoDir64K Electric Interface Board)

Overview

Table 1-617 Basic information about the CR2D00MP8A90

Item	Details
Description	4 Channels C37.94 Optical Interface and 4 Channels CoDir64K Electric Interface Board
Part Number	03032TGS
Model	CR2D00MP8A90
Card type	Interface Card
Silkscreen	MP8A

Appearance

Figure 1-133 Appearance of the CR2D00MP8A90



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03032TGS&productModel=CR2D00MP8A90#matchRelation>

Indicators

Table 1-618 Indicators on the CR2D00MP8A90

Silkscreen	Name	Color	Status	Description
STAT	Working status indicator	Green	Steady on	The PIC is working properly.
		Red	Steady on	The hardware on the PIC is faulty.
		Yellow	Steady on	The PIC is installed in a slot for a HIC and an alarm is reported or the PIC is not loaded with the logic.
		-	Off	The PIC is powered off or is not registered.
LOS(0 to 3)	Optical Interface LOS Indicator	Green	Steady on	The optical signal is normal.
		Red	Steady on	Signals on an optical port are lost.
		Orange	Steady on	A LOS alarm is generated on the port.
		-	Off	The optical module cannot be detected.
LINK (Codir)	Connection/data transmission status indicator	green	Steady on	The data interface is transmitting or receiving data.
		-	Off	The physical port connection fails.

Silkscreen	Name	Color	Status	Description
LOS (Codir)		orange	Steady on	LOS alarms are generated on the port.
		-	Off	The device is not powered on or has not been initialized.

Ports

Table 1-619 Ports on the CR2D00MP8A90

Port	Connector Type	Description	Available Components
OUT0 IN0 to OUT3 IN3	SFP	4-channel C37.94 optical port	2.048Mbps SFP Optical Module
4 to 7	RJ45	4-channel CoDir64K electrical port	Codirectional 64K shielded cable

Table 1-620 Port attributes about the CR2D00MP8A90

Attributes	Details
Working mode	Full-duplex
Standards compliance	ITU-T G.703(codirectional 64K)
Frame format	PWE3
Stated bit rate	2048 kbit/s(C37.94)
Center wavelength [nm]	1310 nm

Functions and Features

Table 1-621 Functions and features of the CR2D00MP8A90

Functions and Features	Description
Basic function	Supports 4-channel C37.94 optical port and 4-channel CoDir64K electrical interface features.
Reliability and availability	Hot swappable.
Restrictions and remarks	Not applicable to outdoor scenarios.
Breakout supported	No
Interface rate auto-sensing supported	No
FlexE supported	No

Technical Specifications

Table 1-622 Technical specifications of the CR2D00MP8A90

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	19.8 mm x 193.8 mm x 209.3 mm (0.78 in. x 7.63 in. x 8.24 in.)
Dimensions with packaging (H x W x D) [mm(in.)]	70 mm x 285 mm x 300 mm (2.76 in. x 11.22 in. x 11.81 in.)
Weight without packaging [kg(lb)]	0.5 kg (1.1 lb)
Weight with packaging [kg(lb)]	0.9 kg (1.98 lb)
Typical power consumption [W]	6 W
Typical heat dissipation [BTU/hour]	19.47 BTU/hour
Long-term operating temperature [°C(°F)]	0°C to 45°C (32°F to 113°F)
Short-term operating temperature [°C(°F)]	-5°C to +55°C (23°F to 131°F)

1.4.4.11 CR2D00MPAA90 (4-Channel FXS/FXO+2-Channel E&M+2-Channel RS232+2-Channel RS485 Board)

Overview

Table 1-623 Basic information about the CR2D00MPAA90

Item	Details
Description	4-Channel FXS/FXO+2-Channel E&M +2-Channel RS232+2-Channel RS485 Board
Part Number	03032TGP
Model	CR2D00MPAA90
Card type	Interface Card
Silkscreen	MPAA
Remarks	-

Appearance

Figure 1-134 Appearance of the CR2D00MPAA90



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03032TGP&productModel=CR2D00MPAA90#matchRelation>

Indicators

Table 1-624 Indicators on the CR2D00MPAA90

Silkscreen	Name	Color	Status	Description
STAT	Working status indicator	Green	Steady on	The PIC is working properly.
		Red	Steady on	The hardware on the PIC is faulty.
		Yellow	Steady on	The PIC is installed in a slot for a HIC and an alarm is reported or the PIC is not loaded with the logic.
		-	Off	The PIC is powered off or is not registered.

Ports

Table 1-625 Ports on the CR2D00MPAA90

Port	Connector Type	Description	Available Components
RS232 0/2	RJ45	2-channel RS232	Serial interface cable
RS485 1/3	RJ45	2-channel RS485/RS422 interface	RS-485 /RS-422 cables
EM 4/5	RJ45	2-channel E&M	E&M cable
FXSO 6 to 9	Telephone interface	4-channel FXS/FXO	Telephone line

Table 1-626 Port attributes about the CR2D00MPAA90

Attributes	Details
Working mode	Full-duplex

Attributes	Details
Frame format	PWE3

Functions and Features

Table 1-627 Functions and features of the CR2D00MPAA90

Functions and Features	Description
Basic function	Supports 2-channel RS232, 2-channel RS485&RS422, 2-channel E&M, and 4-channel FXS/FXO interface features.
Reliability and availability	Hot swappable.
Restrictions and remarks	Not applicable to outdoor scenarios.
Breakout supported	No
Interface rate auto-sensing supported	No
FlexE supported	No

Technical Specifications

Table 1-628 Technical specifications of the CR2D00MPAA90

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	19.8 mm x 193.8 mm x 209.3 mm (0.78 in. x 7.63 in. x 8.24 in.)
Dimensions with packaging (H x W x D) [mm(in.)]	70 mm x 285 mm x 300 mm (2.76 in. x 11.22 in. x 11.81 in.)
Weight without packaging [kg(lb)]	0.6 kg (1.32 lb)
Weight with packaging [kg(lb)]	1.0 kg (2.2 lb)
Typical power consumption [W]	12 W
Typical heat dissipation [BTU/hour]	38.93 BTU/hour
Long-term operating temperature [°C(°F)]	0°C to 45°C (32°F to 113°F)
Short-term operating temperature [°C(°F)]	-5°C to +55°C (23°F to 131°F)

1.4.4.12 CR5D000DE170 (16-Port E1 Physical Interface Card(PIC,75ohm))

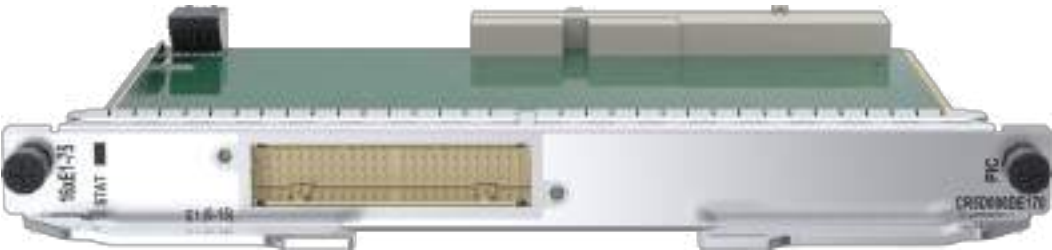
Overview

Table 1-629 Basic information about the CR5D000DE170

Item	Details
Description	16-Port E1 Physical Interface Card(PIC, 75ohm)
Part Number	03030REX
Model	CR5D000DE170
Card type	Interface Card
Silkscreen	16xE1-75

Appearance

Figure 1-135 Appearance of the CR5D000DE170



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03030REX&productModel=CR5D000DE170#matchRelation>

Indicators

Table 1-630 Indicators on the CR5D000DE170

Silkscreen	Name	Color	Status	Description
STAT	Working status indicator	Green	Steady on	The board is working properly.
		Red	Steady on	The hardware on the board is faulty.
		Yellow	Steady on	A high-speed interface card is unexpectedly inserted into a low-speed interface card slot and an alarm is generated, or the logic is not loaded to the board.
		-	Off	The board is not powered on or is not registered.

Ports

Table 1-631 Ports on the CR5D000DE170

Port	Connector Type	Description	Available Components
E1(0-15)	Anea 96	16 E1 interfaces	75-Ohm 16 x E1 Cables

Table 1-632 Port attributes about the CR5D000DE170

Attributes	Details
Working mode	Full-duplex

Attributes	Details
Standards compliance	IMA 1.1, RFC4717, RFC4385, RFC4816, RFC5086, RFC4553_SATOP, and RFC1662
Frame format	Non-framed, CRC4 and NO-CRC4
Interface code	HDB3
Stated bit rate	2048 kbit/s

Functions and Features

Table 1-633 Functions and features of the CR5D000DE170

Functions and Features	Description
Basic function	Supports a maximum of 16 E1 interfaces. Services on each interface can be configured.
Reliability and availability	Hot swappable.
Breakout supported	No
Interface rate auto-sensing supported	No
FlexE supported	No
Link protocol	TDM, PPP, MP, ATM, IMA

Technical Specifications

Table 1-634 Technical specifications of the CR5D000DE170

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	19.8 mm x 193.8 mm x 209.3 mm (0.78 in. x 7.63 in. x 8.24 in.)
Dimensions with packaging (H x W x D) [mm(in.)]	70 mm x 285 mm x 300 mm (2.76 in. x 11.22 in. x 11.81 in.)
Weight without packaging [kg(lb)]	0.5 kg (1.1 lb)
Weight with packaging [kg(lb)]	0.9 kg (1.98 lb)
Typical power consumption [W]	9.5 W
Typical heat dissipation [BTU/hour]	30.66 BTU/hour

Item	Specification
Long-term operating temperature [°C(°F)]	–40°C to +65°C (–40°F to +149°F)

1.4.4.13 CR5D000DE171 (16-Port E1 Physical Interface Card(PIC,120ohm))

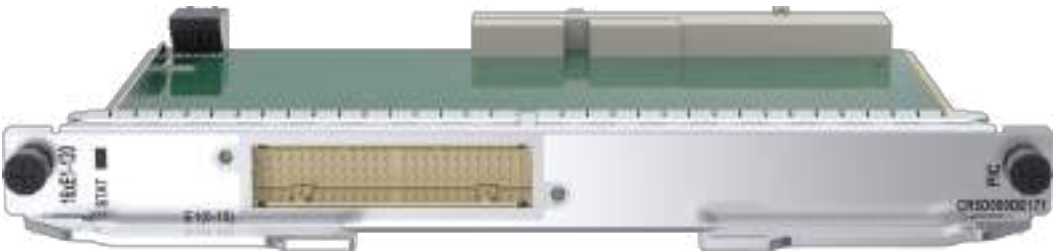
Overview

Table 1-635 Basic information about the CR5D000DE171

Item	Details
Description	16-Port E1 Physical Interface Card(PIC, 120ohm)
Part Number	03030REW
Model	CR5D000DE171
Card type	Interface Card
Silkscreen	16xE1-120

Appearance

Figure 1-136 Appearance of the CR5D000DE171



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03030REW&productModel=CR5D000DE171#matchRelation>

Indicators

Table 1-636 Indicators on the CR5D000DE171

Silkscreen	Name	Color	Status	Description
STAT	Working status indicator	Green	Steady on	The board is working properly.
		Red	Steady on	The hardware on the board is faulty.
		Yellow	Steady on	A high-speed interface card is unexpectedly inserted into a low-speed interface card slot and an alarm is generated, or the logic is not loaded to the board.
		-	Off	The board is not powered on or is not registered.

Ports

Table 1-637 Ports on the CR5D000DE171

Port	Connector Type	Description	Available Components
E1(0-15)	Anea 96	16 E1 interfaces	120-Ohm 16 x E1 Cables

Table 1-638 Port attributes about the CR5D000DE171

Attributes	Details
Working mode	Full-duplex

Attributes	Details
Standards compliance	IMA 1.1, RFC4717, RFC4385, RFC4816, RFC5086, RFC4553_SATOP, and RFC1662
Frame format	Non-framed, CRC4 and NO-CRC4
Interface code	HDB3
Stated bit rate	2048 kbit/s

Functions and Features

Table 1-639 Functions and features of the CR5D000DE171

Functions and Features	Description
Basic function	Supports a maximum of 16 E1 interfaces. Services on each interface can be configured.
Reliability and availability	Hot swappable.
Breakout supported	No
Interface rate auto-sensing supported	No
FlexE supported	No
Link protocol	TDM, PPP, MP, ATM, IMA

Technical Specifications

Table 1-640 Technical specifications of the CR5D000DE171

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	19.8 mm x 193.8 mm x 209.3 mm (0.78 in. x 7.63 in. x 8.24 in.)
Dimensions with packaging (H x W x D) [mm(in.)]	70 mm x 285 mm x 300 mm (2.76 in. x 11.22 in. x 11.81 in.)
Weight without packaging [kg(lb)]	0.5 kg (1.1 lb)
Weight with packaging [kg(lb)]	0.9 kg (1.98 lb)
Typical power consumption [W]	9.5 W
Typical heat dissipation [BTU/hour]	30.66 BTU/hour

Item	Specification
Long-term operating temperature [°C(°F)]	–40°C to +65°C (–40°F to +149°F)

1.4.4.14 CR5D000DE1C1 (16-Port E1 Physical Interface Card(PIC,75ohm/120ohm))

Overview

Table 1-641 Basic information about the CR5D000DE1C1

Item	Details
Description	16-Port E1 Physical Interface Card(PIC, 75ohm/120ohm)
Part Number	03033GGD
Model	CR5D000DE1C1
Card type	Interface Card
Silkscreen	16xE1-75/120

Appearance

Figure 1-137 Appearance of the CR5D000DE1C1



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03033GGD&productModel=CR5D000DE1C1#matchRelation>

Indicators

Table 1-642 Indicators on the CR5D000DE1C1

Silkscreen	Name	Color	Status	Description
STAT	Working status indicator	Green	Steady on	If the indicator is steady on, the PIC is working properly.
		Red	Steady on	If the indicator is steady on, the hardware on the PIC is faulty.
		Orange	Steady on	If the indicator is steady on, the PIC is installed in a slot for a HIC and an alarm is reported or the PIC is not loaded with the logic.
		-	Off	If the indicator is off, the PIC is powered off or is not registered.
SRV	Service status indicator	Green	Steady on	The board has no CRC alarm.
		Orange	Steady on	The board has CRC alarms.
		-	Off	The board is not running or no power is input.

Ports

Table 1-643 Ports on the CR5D000DE1C1

Port	Connector Type	Description	Available Components
E1(0-15)	ANEA 96	16 E1 interfaces	75/120 ohm 16xE1 cable

Table 1-644 Port attributes about the CR5D000DE1C1

Attributes	Details
Working mode	Full-duplex
Standards compliance	IMA 1.1, RFC4385, RFC5086, RFC4553_SATOP, RFC1662
Frame format	Unframed mode, CRC4, and NO-CRC4

Functions and Features

Table 1-645 Functions and features of the CR5D000DE1C1

Functions and Features	Description
Basic function	Supports a maximum of 16 E1 interfaces. Services on each interface can be configured.
Reliability and availability	Hot swappable.
Breakout supported	No
Interface rate auto-sensing supported	No
FlexE supported	No
Link protocol	TDM, PPP, MP, ATM, IMA

Technical Specifications

Table 1-646 Technical specifications of the CR5D000DE1C1

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	19.8 mm x 193.8 mm x 209.3 mm (0.78 in. x 7.63 in. x 8.24 in.)

Item	Specification
Dimensions with packaging (H x W x D) [mm(in.)]	70 mm x 285 mm x 300 mm (2.76 in. x 11.22 in. x 11.81 in.)
Weight without packaging [kg(lb)]	0.6 kg (1.32 lb)
Weight with packaging [kg(lb)]	1.0 kg (2.2 lb)
Typical power consumption [W]	15.1 W
Typical heat dissipation [BTU/hour]	48.86 BTU/hour
Long-term operating temperature [°C(°F)]	–40°C to +65°C (–40°F to +149°F)

1.4.4.15 CR5D000IE170 (32-Port E1 Physical Interface Card(PIC,75ohm))

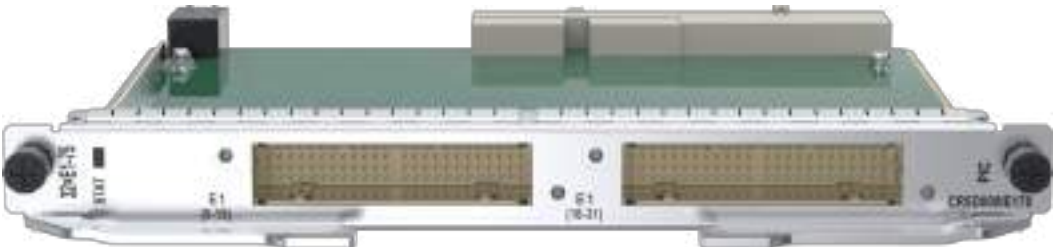
Overview

Table 1-647 Basic information about the CR5D000IE170

Item	Details
Description	32-Port E1 Physical Interface Card(PIC, 75ohm)
Part Number	03030QBN
Model	CR5D000IE170
Card type	Interface Card
Silkscreen	32xE1-75

Appearance

Figure 1-138 Appearance of the CR5D000IE170



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03030QBN&productModel=CR5D000IE170#matchRelation>

Indicators

Table 1-648 Indicators on the CR5D000IE170

Silkscreen	Name	Color	Status	Description
STAT	Working status indicator	Green	Steady on	The board is working properly.
		Red	Steady on	The hardware on the board is faulty.
		Yellow	Steady on	A high-speed interface card is unexpectedly inserted into a low-speed interface card slot and an alarm is generated, or the logic is not loaded to the board.
		-	Off	The board is not powered on or is not registered.

Ports

Table 1-649 Ports on the CR5D000IE170

Port	Connector Type	Description	Available Components
E1(0-15) E1(16-31)	Anea 96	32 E1 interfaces	75-Ohm 16 x E1 Cables

Table 1-650 Port attributes about the CR5D000IE170

Attributes	Details
Working mode	Full-duplex
Standards compliance	IMA 1.1, RFC4717, RFC4385, RFC4816, RFC5086, RFC4553_SATOP, and RFC1662
Frame format	Non-framed, CRC4 and NO-CRC4
Interface code	HDB3
Stated bit rate	2048 kbit/s

Functions and Features

Table 1-651 Functions and features of the CR5D000IE170

Functions and Features	Description
Basic function	Supports a maximum of 32 E1 interfaces. Services on each interface can be configured.
Reliability and availability	Hot swappable.
Breakout supported	No
Interface rate auto-sensing supported	No
FlexE supported	No
Link protocol	TDM, PPP, MP, ATM, IMA

Technical Specifications

Table 1-652 Technical specifications of the CR5D000IE170

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	19.8 mm x 193.8 mm x 209.3 mm (0.78 in. x 7.63 in. x 8.24 in.)
Dimensions with packaging (H x W x D) [mm(in.)]	70 mm x 285 mm x 300 mm (2.76 in. x 11.22 in. x 11.81 in.)
Weight without packaging [kg(lb)]	0.5 kg (1.1 lb)
Weight with packaging [kg(lb)]	0.9 kg (1.98 lb)
Typical power consumption [W]	13.5 W

Item	Specification
Typical heat dissipation [BTU/hour]	43.8 BTU/hour
Long-term operating temperature [°C(°F)]	–40°C to +65°C (–40°F to +149°F)

1.4.4.16 CR5D000IE171 (32-Port E1 Physical Interface Card(PIC,120ohm))

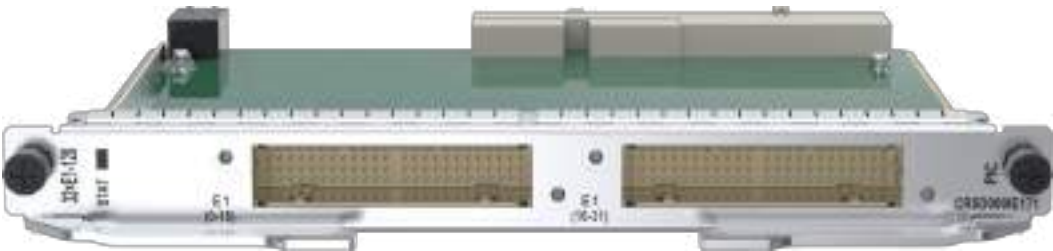
Overview

Table 1-653 Basic information about the CR5D000IE171

Item	Details
Description	32-Port E1 Physical Interface Card(PIC, 120ohm)
Part Number	03030QBP
Model	CR5D000IE171
Card type	Interface Card
Silkscreen	32xE1-120

Appearance

Figure 1-139 Appearance of the CR5D000IE171



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03030QBP&productModel=CR5D000IE171#matchRelation>

Indicators

Table 1-654 Indicators on the CR5D000IE171

Silkscreen	Name	Color	Status	Description
STAT	Working status indicator	Green	Steady on	The board is working properly.
		Red	Steady on	The hardware on the board is faulty.
		Yellow	Steady on	A high-speed interface card is unexpectedly inserted into a low-speed interface card slot and an alarm is generated, or the logic is not loaded to the board.
		-	Off	The board is not powered on or is not registered.

Ports

Table 1-655 Ports on the CR5D000IE171

Port	Connector Type	Description	Available Components
E1(0-15) E1(16-31)	Anea 96	32 E1 interfaces	120-Ohm 16 x E1 Cables

Table 1-656 Port attributes about the CR5D000IE171

Attributes	Details
Working mode	Full-duplex
Standards compliance	IMA 1.1, RFC4717, RFC4385, RFC4816, RFC5086, RFC4553_SATOP, and RFC1662
Frame format	Non-framed, CRC4 and NO-CRC4
Interface code	HDB3
Stated bit rate	2048 kbit/s

Functions and Features

Table 1-657 Functions and features of the CR5D000IE171

Functions and Features	Description
Basic function	Supports a maximum of 32 E1 interfaces. Services on each interface can be configured.
Reliability and availability	Hot swappable.
Breakout supported	No
Interface rate auto-sensing supported	No
FlexE supported	No
Link protocol	TDM, PPP, MP, ATM, IMA

Technical Specifications

Table 1-658 Technical specifications of the CR5D000IE171

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	19.8 mm x 193.8 mm x 209.3 mm (0.78 in. x 7.63 in. x 8.24 in.)
Dimensions with packaging (H x W x D) [mm(in.)]	70 mm x 285 mm x 300 mm (2.76 in. x 11.22 in. x 11.81 in.)
Weight without packaging [kg(lb)]	0.5 kg (1.1 lb)
Weight with packaging [kg(lb)]	0.9 kg (1.98 lb)
Typical power consumption [W]	13.5 W

Item	Specification
Typical heat dissipation [BTU/hour]	43.8 BTU/hour
Long-term operating temperature [°C(°F)]	–40°C to +65°C (–40°F to +149°F)

1.4.4.17 CR5D00C4CFC1 (4-Port Channelized STM-1c POS-SFP Physical Interface Card(PIC))

Overview

Table 1-659 Basic information about the CR5D00C4CFC1

Item	Details
Description	4-Port Channelized STM-1c POS-SFP Physical Interface Card(PIC)
Part Number	03033GGF
Model	CR5D00C4CFC1
Card type	Interface Card
Silkscreen	4xSTM1-cPOS

Appearance

Figure 1-140 Appearance of the CR5D00C4CFC1



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03033GGF&productModel=CR5D00C4CFC1#matchRelation>

Indicators

Table 1-660 Indicators on the CR5D00C4CFC1

Silkscreen	Name	Color	Status	Description
STAT	Working status indicator	Green	Steady on	The board is working normally.
		Red	Steady on	The board hardware is faulty.
		Orange	Steady on	The board is not loaded with the logic.
		-	Off	The board is not running or no power is input.
SRV	Service status indicator	Green	Steady on	The board has no CRC alarm.
		Orange	Steady on	The board has CRC alarms.
		-	Off	The board is not running or no power is input.
LOS(0 to 3)	Connection/ data transmission status indicator	Green	Steady on	The connection on the physical port is normal.
		Green	Blink	The connection on the physical port is normal, and data is received or transmitted on the port.

Silkscreen	Name	Color	Status	Description
		-	Off	The physical connection fails.

Ports

Table 1-661 Ports on the CR5D00C4CFC1

Port	Connector Type	Description	Available Components
OUT0 IN0 to OUT3 IN3	SFP	STM-1 service signal input and output interface	155Mbps SFP Electrical Module 155Mbps eSFP Optical Module 155Mbps eSFP BIDI Optical Module

Table 1-662 Port attributes about the CR5D00C4CFC1

Attributes	Details
Working mode	Full-duplex
Standards compliance	IMA 1.1, RFC4385, RFC5086, RFC4553_SATOP, RFC1662
Frame format	155M interfaces support SDH. Channelized E1 interfaces support the framed mode (channelized mode), unframed mode (clear channel mode), CRC4, and NO-CRC4.

Functions and Features

Table 1-663 Functions and features of the CR5D00C4CFC1

Functions and Features	Description
Basic function	Supports 63 E1 links over four 155M SDH links.
Reliability and availability	Hot swappable.

Functions and Features	Description
Restrictions and remarks	Ports 0 to 3 support clock master mode.
Breakout supported	No
Interface rate auto-sensing supported	No
FlexE supported	No
Link protocol	TDM, PPP, MP, ATM, IMA

Technical Specifications

Table 1-664 Technical specifications of the CR5D00C4CFC1

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	19.8 mm x 193.8 mm x 209.3 mm (0.78 in. x 7.63 in. x 8.24 in.)
Dimensions with packaging (H x W x D) [mm(in.)]	70 mm x 285 mm x 300 mm (2.76 in. x 11.22 in. x 11.81 in.)
Weight without packaging [kg(lb)]	0.5 kg (1.1 lb)
Weight with packaging [kg(lb)]	0.9 kg (1.98 lb)
Typical power consumption [W]	16.8 W
Typical heat dissipation [BTU/hour]	54.5 BTU/hour
Long-term operating temperature [°C(°F)]	-20°C to +65°C (-4°F to +149°F)

1.4.4.18 CR5D00E1NC97 (03033HAF: 1-Port 100GE/ 1-Port 200GE CFP2 Physical Interface Card(PIC) (Enable 1*100GE by default))

Overview

Table 1-665 Basic information about the CR5D00E1NC97

Item	Details
Description	1-Port 100GE/ 1-Port 200GE CFP2 Physical Interface Card(PIC) (Enable 1*100GE by default)
Part Number	03033HAF

Item	Details
Model	CR5D00E1NC97
Card type	Interface Card
Silkscreen	1x100/200GE-CFP2

Appearance

Figure 1-141 Appearance of the CR5D00E1NC97



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03033HAF&productModel=CR5D00E1NC97#matchRelation>

Indicators

Table 1-666 Indicators on the CR5D00E1NC97

Silkscreen	Name	Color	Status	Description
STAT	Working status indicator	Green	Steady on	If the indicator is steady on, the PIC is working properly.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	If the indicator is steady on, the hardware on the PIC is faulty.
		Orange	Steady on	If the indicator is steady on, the PIC is installed in a slot for a HIC and an alarm is reported or the PIC is not loaded with the logic.
		-	Off	If the indicator is off, the PIC is powered off or is not registered.
SRV	Service status indicator	Green	Steady on	The board has no CRC alarm.
		Orange	Steady on	The board has CRC alarms.
		-	Off	The board is not running or no power is input.
L/A 0	Running status indicator	Green	Steady on	The link is Up.
		Green	Blink	If the indicator blinks, data is being transmitted.
		-	Off	If the indicator is off, the link is Down.

Ports

Table 1-667 Ports on the CR5D00E1NC97

Port	Connector Type	Description	Available Components
OUT0 IN0	CFP2	Interfaces for 1-channel CFP2 optical signal input and output	100Gbps CFP2 Optical Module 100Gbps CFP2 DWDM Optical Module

Table 1-668 Port attributes about the CR5D00E1NC97

Attributes	Details
Working mode	Full-duplex
Standards compliance	IEEE 802.3
Frame format	Ethernet_II, Ethernet_SAP, and Ethernet_SNAP

Functions and Features

Table 1-669 Functions and features of the CR5D00E1NC97

Functions and Features	Description
Basic function	Provides one 100GEBase CFP2 interface. The 200GE interface capability is reserved on the hardware. MACsec is supported.
Reliability and availability	Hot swappable.
Breakout supported	No
Interface rate auto-sensing supported	No
FlexE supported	No

Technical Specifications

Table 1-670 Technical specifications of the CR5D00E1NC97

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	19.8 mm x 193.8 mm x 209.3 mm (0.78 in. x 7.63 in. x 8.24 in.)
Dimensions with packaging (H x W x D) [mm(in.)]	70 mm x 285 mm x 300 mm (2.76 in. x 11.22 in. x 11.81 in.)
Weight without packaging [kg(lb)]	0.8 kg (1.76 lb)
Weight with packaging [kg(lb)]	1.2 kg (2.65 lb)
Typical power consumption [W]	43.2 W
Typical heat dissipation [BTU/hour]	140.16 BTU/hour
Long-term operating temperature [°C(°F)]	-5°C to +50°C (23°F to 122°F)

1.4.4.19 CR5D00E1NC97 (03033HAF-001: 1-Port 100GE/ 1-Port 200GE CFP2 Physical Interface Card(PIC) (Enable 1*100GE by default))

Overview

Table 1-671 Basic information about the CR5D00E1NC97

Item	Details
Description	1-Port 100GE/ 1-Port 200GE CFP2 Physical Interface Card(PIC) (Enable 1*100GE by default)
Part Number	03033HAF-001
Model	CR5D00E1NC97
Card type	Interface Card
Silkscreen	1x100/200GE-CFP2

Appearance

Figure 1-142 Appearance of the CR5D00E1NC97



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03033HAF-001&productModel=CR5D00E1NC97#matchRelation>

Indicators

Table 1-672 Indicators on the CR5D00E1NC97

Silkscreen	Name	Color	Status	Description
STAT	Working status indicator	Green	Steady on	If the indicator is steady on, the PIC is working properly.
		Red	Steady on	If the indicator is steady on, the hardware on the PIC is faulty.

Silkscreen	Name	Color	Status	Description
		Orange	Steady on	If the indicator is steady on, the PIC is installed in a slot for a HIC and an alarm is reported or the PIC is not loaded with the logic.
		-	Off	If the indicator is off, the PIC is powered off or is not registered.
SRV	Service status indicator	Green	Steady on	The board has no CRC alarm.
		Orange	Steady on	The board has CRC alarms.
		-	Off	The board is not running or no power is input.
L/A 0	Running status indicator	Green	Steady on	The link is Up.
		Green	Blink	If the indicator blinks, data is being transmitted.
		-	Off	If the indicator is off, the link is Down.

Ports

Table 1-673 Ports on the CR5D00E1NC97

Port	Connector Type	Description	Available Components
OUT0 IN0	CFP2	Interfaces for 1-channel CFP2 optical signal input and output	100Gbps CFP2 Optical Module 100Gbps CFP2 DWDM Optical Module

Table 1-674 Port attributes about the CR5D00E1NC97

Attributes	Details
Working mode	Full-duplex
Standards compliance	IEEE 802.3
Frame format	Ethernet_II, Ethernet_SAP, and Ethernet_SNAP

Functions and Features

Table 1-675 Functions and features of the CR5D00E1NC97

Functions and Features	Description
Basic function	Provides one 100GEBase CFP2 interface. The 200GE interface capability is reserved on the hardware. MACsec is supported.
Reliability and availability	Hot swappable.
Breakout supported	No
Interface rate auto-sensing supported	No
FlexE supported	No

Technical Specifications

Table 1-676 Technical specifications of the CR5D00E1NC97

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	19.8 mm x 193.8 mm x 209.3 mm (0.78 in. x 7.63 in. x 8.24 in.)
Dimensions with packaging (H x W x D) [mm(in.)]	70 mm x 285 mm x 300 mm (2.76 in. x 11.22 in. x 11.81 in.)
Weight without packaging [kg(lb)]	0.8 kg (1.76 lb)
Weight with packaging [kg(lb)]	1.2 kg (2.65 lb)
Typical power consumption [W]	43.2 W
Typical heat dissipation [BTU/hour]	140.16 BTU/hour
Long-term operating temperature [°C(°F)]	-5°C to +50°C (23°F to 122°F)

1.4.4.20 CR5D00E4XM25 (03033BLK: 4-Port 25GE(SFP28)/10GE(SFP+) MACsec Physical Interface Card (PIC))

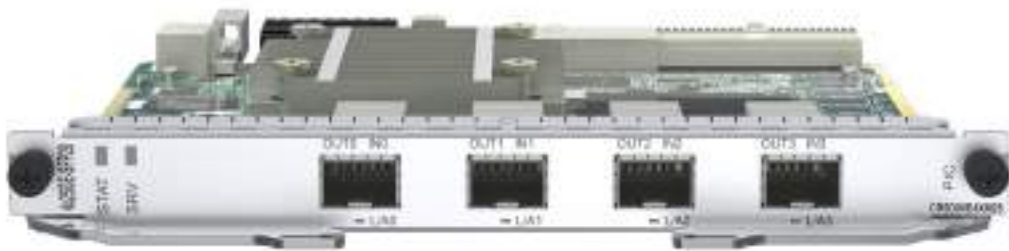
Overview

Table 1-677 Basic information about the CR5D00E4XM25

Item	Details
Description	4-Port 25GE(SFP28)/10GE(SFP+) MACsec Physical Interface Card (PIC)
Part Number	03033BLK
Model	CR5D00E4XM25
Card type	Interface Card
Silkscreen	4x25GE-SFP28

Appearance

Figure 1-143 Appearance of the CR5D00E4XM25



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03033BLK&productModel=CR5D00E4XM25#matchRelation>

Indicators

Table 1-678 Indicators on the CR5D00E4XM25

Silkscreen	Name	Color	Status	Description
STAT	Working status indicator	Green	Steady on	The board is working properly.
		Red	Steady on	The hardware on the board is faulty.
		Orange	Steady on	A high-speed interface card is unexpectedly inserted into a low-speed interface card slot and an alarm is generated, or the logic is not loaded to the board.
		-	Off	The board is not powered on or is not registered.
SRV	Service Status Indicator	Green	Steady on	The board has no CRC alarm.
		Orange	Steady on	The board has CRC alarm.

Silkscreen	Name	Color	Status	Description
		-	Off	The board is not running or no power is input.
L/A (0-3)	Interface status indicator	Green	Steady on	The link is Up.
		Green	Blinking	Data is being transmitted or received.
		-	Off	The link is Down.

Ports

Table 1-679 Ports on the CR5D00E4XM25

Port	Connector Type	Description	Available Components
OUT0 IN0-OUT3 IN3	SFP28	25GE/10GE service signal input and output interface	155Mbps eSFP Optical Module 155Mbps eSFP BIDI Optical Module 1.25/9.953/10.31 25Gbps SFP+ Optical Module 1.25Gbps eSFP BIDI Optical Module 1.25Gbps eSFP CWDM Optical Module 1.25Gbps eSFP Optical Module 10Gbps SFP+ BIDI Optical Module 10Gbps SFP+ CWDM Optical Module 10Gbps SFP+ Optical Module 125M~2.67Gbps eSFP DWDM Optical Module 1Gbps SyncE and 1588V2 Electrical Module 1Gbps Electrical Module 25Gbps SFP28 Optical Module 25Gbps SFP28 BIDI Optical Module

Table 1-680 Port attributes about the CR5D00E4XM25

Attributes	Details
Working mode	Full-duplex
Standards compliance	IEEE 802.3
Frame format	Ethernet_II, Ethernet_SAP and Ethernet_SNAP

Functions and Features

Table 1-681 Functions and features of the CR5D00E4XM25

Functions and Features	Description
Basic function	Occupies one flexible interface card slot. Provides four 25GE/10GE interfaces for line-rate transmitting and receiving, and supports SFP28/SFP+ optical modules. MACsec is supported.
Reliability and availability	Hot swappable.
Restrictions and remarks	By default, 25GE/10GE auto-negotiation is supported. You can run the port-mode group-id command in the slot view to switch the mode to 10GE/GE auto-negotiation.
Breakout supported	No
Interface rate auto-sensing supported	In 25GE mode, 10GE/25GE auto-sensing supported; In 10GE mode, GE/10GE auto-sensing supported.
FlexE supported	No

Technical Specifications

Table 1-682 Technical specifications of the CR5D00E4XM25

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	19.8 mm x 193.8 mm x 209.3 mm (0.78 in. x 7.63 in. x 8.24 in.)

Item	Specification
Dimensions with packaging (H x W x D) [mm(in.)]	70 mm x 285 mm x 300 mm (2.76 in. x 11.22 in. x 11.81 in.)
Weight without packaging [kg(lb)]	0.7 kg (1.54 lb)
Weight with packaging [kg(lb)]	1.1 kg (2.43 lb)
Typical power consumption [W]	25.3 W
Typical heat dissipation [BTU/hour]	82.15 BTU/hour
Long-term operating temperature [°C(°F)]	–40°C to +65°C (–40°F to +149°F)

1.4.4.21 CR5D00E4XM25 (03033BLK-001: 4-Port 25GE(SFP28)/10GE(SFP+) MACsec Physical Interface Card (PIC))

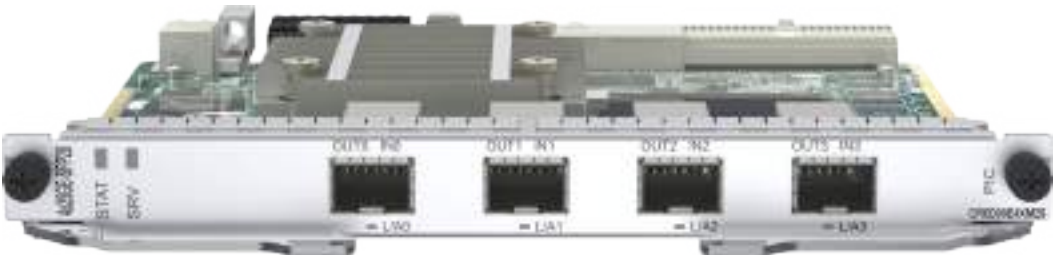
Overview

Table 1-683 Basic information about the CR5D00E4XM25

Item	Details
Description	4-Port 25GE(SFP28)/10GE(SFP+) MACsec Physical Interface Card (PIC)
Part Number	03033BLK-001
Model	CR5D00E4XM25
Card type	Interface Card
Silkscreen	4x25GE-SFP28

Appearance

Figure 1-144 Appearance of the CR5D00E4XM25



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03033BLK-001&productModel=CR5D00E4XM25#matchRelation>

Indicators

Table 1-684 Indicators on the CR5D00E4XM25

Silkscreen	Name	Color	Status	Description
STAT	Working status indicator	Green	Steady on	The board is working properly.
		Red	Steady on	The hardware on the board is faulty.
		Orange	Steady on	A high-speed interface card is unexpectedly inserted into a low-speed interface card slot and an alarm is generated, or the logic is not loaded to the board.
		-	Off	The board is not powered on or is not registered.
SRV	Service Status Indicator	Green	Steady on	The board has no CRC alarm.
		Orange	Steady on	The board has CRC alarm.

Silkscreen	Name	Color	Status	Description
		-	Off	The board is not running or no power is input.
L/A (0-3)	Interface status indicator	Green	Steady on	The link is Up.
		Green	Blinking	Data is being transmitted or received.
		-	Off	The link is Down.

Ports

Table 1-685 Ports on the CR5D00E4XM25

Port	Connector Type	Description	Available Components
OUT0 IN0-OUT3 IN3	SFP28	25GE/10GE service signal input and output interface	155Mbps eSFP Optical Module 155Mbps eSFP BIDI Optical Module 1.25/9.953/10.31 25Gbps SFP+ Optical Module 1.25Gbps eSFP BIDI Optical Module 1.25Gbps eSFP CWDM Optical Module 1.25Gbps eSFP Optical Module 10Gbps SFP+ BIDI Optical Module 10Gbps SFP+ CWDM Optical Module 10Gbps SFP+ Optical Module 125M~2.67Gbps eSFP DWDM Optical Module 1Gbps SyncE and 1588V2 Electrical Module 1Gbps Electrical Module 25Gbps SFP28 Optical Module 25Gbps SFP28 BIDI Optical Module

Table 1-686 Port attributes about the CR5D00E4XM25

Attributes	Details
Working mode	Full-duplex
Standards compliance	IEEE 802.3
Frame format	Ethernet_II, Ethernet_SAP and Ethernet_SNAP

Functions and Features

Table 1-687 Functions and features of the CR5D00E4XM25

Functions and Features	Description
Basic function	Occupies one flexible interface card slot. Provides four 25GE/10GE interfaces for line-rate transmitting and receiving, and supports SFP28/SFP+ optical modules. MACsec is supported.
Reliability and availability	Hot swappable.
Restrictions and remarks	By default, 25GE/10GE auto-negotiation is supported. You can run the port-mode group-id command in the slot view to switch the mode to 10GE/GE auto-negotiation.
Breakout supported	No
Interface rate auto-sensing supported	In 25GE mode, 10GE/25GE auto-sensing supported; In 10GE mode, GE/10GE auto-sensing supported.
FlexE supported	No

Technical Specifications

Table 1-688 Technical specifications of the CR5D00E4XM25

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	19.8 mm x 193.8 mm x 209.3 mm (0.78 in. x 7.63 in. x 8.24 in.)

Item	Specification
Dimensions with packaging (H x W x D) [mm(in.)]	70 mm x 285 mm x 300 mm (2.76 in. x 11.22 in. x 11.81 in.)
Weight without packaging [kg(lb)]	0.7 kg (1.54 lb)
Weight with packaging [kg(lb)]	1.1 kg (2.43 lb)
Typical power consumption [W]	25.3 W
Typical heat dissipation [BTU/hour]	82.15 BTU/hour
Long-term operating temperature [°C(°F)]	–40°C to +65°C (–40°F to +149°F)

1.4.4.22 CR5D00E8GE71 (8-Port 100/1000Base-RJ45 Physical Interface Card(PIC))

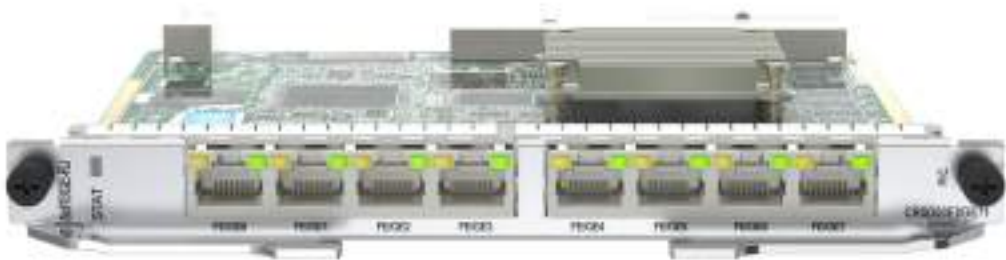
Overview

Table 1-689 Basic information about the CR5D00E8GE71

Item	Details
Description	8-Port 100/1000Base-RJ45 Physical Interface Card(PIC)
Part Number	03031DGY
Model	CR5D00E8GE71
Card type	Interface Card
Silkscreen	8xFE/GE-RJ

Appearance

Figure 1-145 Appearance of the CR5D00E8GE71



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03031DGY&productModel=CR5D00E8GE71#matchRelation>

Indicators

Table 1-690 Indicators on the CR5D00E8GE71

Silkscreen	Name	Color	Status	Description
STAT	Working status indicator	Green	Steady on	The board is working properly.
		Red	Steady on	The hardware on the board is faulty.
		Orange	Steady on	A high-speed interface card is unexpectedly inserted into a low-speed interface card slot and an alarm is generated, or the logic is not loaded to the board.
		-	Off	The board is not powered on or is not registered.
LINK	Connection status indicator	Green	Steady on	The physical connection of the port is normal.

Silkscreen	Name	Color	Status	Description
		-	Off	The physical port connection fails.
ACT	Transmitting/receiving status indicator	Orange	Blinking	The data interface is not transmitting or receiving data.
		-	Off	The data interface is not transmitting or receiving data.

Ports

Table 1-691 Ports on the CR5D00E8GE71

Port	Connector Type	Description	Available Components
FE/GE0-FE/GE7	RJ45	GE/FE service signal input and output interface	Ethernet Cables

Table 1-692 Port attributes about the CR5D00E8GE71

Attributes	Details
Working mode	Full-duplex
Standards compliance	IEEE 802.3
Frame format	Ethernet_II, Ethernet_SAP and Ethernet_SNAP

Functions and Features

Table 1-693 Functions and features of the CR5D00E8GE71

Functions and Features	Description
Basic function	Occupies one flexible interface card slot. Provides eight GE/FE interfaces for line-rate transmitting and receiving, and supports 10M/100M/1000M electrical interfaces.
Reliability and availability	Hot swappable.
Breakout supported	No
Interface rate auto-sensing supported	No
FlexE supported	No

Technical Specifications

Table 1-694 Technical specifications of the CR5D00E8GE71

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	19.8 mm x 193.8 mm x 209.3 mm (0.78 in. x 7.63 in. x 8.24 in.)
Dimensions with packaging (H x W x D) [mm(in.)]	70 mm x 285 mm x 300 mm (2.76 in. x 11.22 in. x 11.81 in.)
Weight without packaging [kg(lb)]	0.5 kg (1.1 lb)
Weight with packaging [kg(lb)]	0.9 kg (1.98 lb)
Typical power consumption [W]	11.7 W
Typical heat dissipation [BTU/hour]	37.96 BTU/hour
Long-term operating temperature [°C(°F)]	–40°C to +65°C (–40°F to +149°F)

1.4.4.23 CR5D00E8XM25 (03033BLM: 8-Port 25GE(SFP28)/10GE(SFP+) MACsec Physical Interface Card (PIC))

Overview

Table 1-695 Basic information about the CR5D00E8XM25

Item	Details
Description	8-Port 25GE(SFP28)/10GE(SFP+) MACsec Physical Interface Card (PIC)
Part Number	03033BLM
Model	CR5D00E8XM25
Card type	Interface Card
Silkscreen	8x25GE-SFP28

Appearance

Figure 1-146 Appearance of the CR5D00E8XM25



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03033BLM&productModel=CR5D00E8XM25#matchRelation>

Indicators

Table 1-696 Indicators on the CR5D00E8XM25

Silkscreen	Name	Color	Status	Description
STAT	Working status indicator	Green	Steady on	The card is working properly.
		Red	Steady on	If the indicator is steady on, the hardware on the PIC is faulty.
		Orange	Steady on	If the indicator is steady on, the PIC is installed in a slot for a HIC and an alarm is reported or the PIC is not loaded with the logic.
		-	Off	If the indicator is off, the PIC is powered off or is not registered.
SRV	Service status indicator	Green	Steady on	the board has no CRC alarm.
		Orange	Steady on	the board has CRC alarms.
		-	Off	the board is not running or no power is input.
L/A (0-7)	Running status indicator	Green	Steady on	The link is up.
		Green	Blinking	Data is being transmitted or received.
		-	Off	The link is down.

Ports

Table 1-697 Ports on the CR5D00E8XM25

Port	Connector Type	Description	Available Components
OUT0 IN0 to OUT7 IN7	SFP28	25GE/10GE service signal input and output interface	155Mbps eSFP Optical Module 155Mbps eSFP BIDI Optical Module 1.25/9.953/10.31 25Gbps SFP+ Optical Module 1.25Gbps eSFP BIDI Optical Module 1.25Gbps eSFP CWDM Optical Module 1.25Gbps eSFP Optical Module 10Gbps SFP+ BIDI Optical Module 10Gbps SFP+ CWDM Optical Module 10Gbps SFP+ Optical Module 125M~2.67Gbps eSFP DWDM Optical Module 1Gbps Electrical Module 25Gbps SFP28 Optical Module

Table 1-698 Port attributes about the CR5D00E8XM25

Attributes	Details
Working mode	Full-duplex

Attributes	Details
Standards compliance	IEEE 802.3
Frame format	Ethernet_II, Ethernet_SAP, and Ethernet_SNAP

Functions and Features

Table 1-699 Functions and features of the CR5D00E8XM25

Functions and Features	Description
Basic function	Supports optical signal input and output of eight 25GE/10GE ports. Supports MACSEC.
Reliability and availability	Supports the hot swappable function.
Restrictions and remarks	Ports 0 to 3 belong to a port group, and ports 4 to 7 belong to another one. By default, 25GE/10GE auto-negotiation is supported. You can run the port-mode group-id command in the slot view to switch the mode of a port group to 10GE/GE auto-negotiation. In LAN/WAN switching, the mode of ports in a port group must be switched at the same time.
Breakout supported	No
Interface rate auto-sensing supported	In 25GE mode, 10GE/25GE auto-sensing supported; In 10GE mode, GE/10GE auto-sensing supported.
FlexE supported	No

Technical Specifications

Table 1-700 Technical specifications of the CR5D00E8XM25

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	19.8 mm x 193.8 mm x 209.3 mm (0.78 in. x 7.63 in. x 8.24 in.)
Dimensions with packaging (H x W x D) [mm(in.)]	70 mm x 285 mm x 300 mm (2.76 in. x 11.22 in. x 11.81 in.)

Item	Specification
Weight without packaging [kg(lb)]	0.8 kg (1.76 lb)
Weight with packaging [kg(lb)]	1.2 kg (2.65 lb)
Typical power consumption [W]	37.5 W
Typical heat dissipation [BTU/hour]	121.57 BTU/hour
Long-term operating temperature [°C(°F)]	–40°C to +65°C (–40°F to +149°F)

1.4.4.24 CR5D00E8XM25 (03033BLM-001: 8-Port 25GE(SFP28)/10GE(SFP+) MACsec Physical Interface Card (PIC))

Overview

Table 1-701 Basic information about the CR5D00E8XM25

Item	Details
Description	8-Port 25GE(SFP28)/10GE(SFP+) MACsec Physical Interface Card (PIC)
Part Number	03033BLM-001
Model	CR5D00E8XM25
Card type	Interface Card
Silkscreen	8x25GE-SFP28

Appearance

Figure 1-147 Appearance of the CR5D00E8XM25



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03033BLM-001&productModel=CR5D00E8XM25#matchRelation>

Indicators

Table 1-702 Indicators on the CR5D00E8XM25

Silkscreen	Name	Color	Status	Description
STAT	Working status indicator	Green	Steady on	The card is working properly.
		Red	Steady on	If the indicator is steady on, the hardware on the PIC is faulty.
		Orange	Steady on	If the indicator is steady on, the PIC is installed in a slot for a HIC and an alarm is reported or the PIC is not loaded with the logic.
		-	Off	If the indicator is off, the PIC is powered off or is not registered.
SRV	Service status indicator	Green	Steady on	the board has no CRC alarm.
		Orange	Steady on	the board has CRC alarms.
		-	Off	the board is not running or no power is input.

Silkscreen	Name	Color	Status	Description
L/A (0-7)	Running status indicator	Green	Steady on	The link is up.
		Green	Blinking	Data is being transmitted or received.
		-	Off	The link is down.

Ports

Table 1-703 Ports on the CR5D00E8XM25

Port	Connector Type	Description	Available Components
OUT0 IN0 to OUT7 IN7	SFP28	25GE/10GE service signal input and output interface	155Mbps eSFP Optical Module 155Mbps eSFP BIDI Optical Module 1.25/9.953/10.31 25Gbps SFP+ Optical Module 1.25Gbps eSFP BIDI Optical Module 1.25Gbps eSFP CWDM Optical Module 1.25Gbps eSFP Optical Module 10Gbps SFP+ BIDI Optical Module 10Gbps SFP+ CWDM Optical Module 10Gbps SFP+ Optical Module 125M~2.67Gbps eSFP DWDM Optical Module 1Gbps Electrical Module 25Gbps SFP28 Optical Module

Table 1-704 Port attributes about the CR5D00E8XM25

Attributes	Details
Working mode	Full-duplex
Standards compliance	IEEE 802.3

Attributes	Details
Frame format	Ethernet_II, Ethernet_SAP, and Ethernet_SNAP

Functions and Features

Table 1-705 Functions and features of the CR5D00E8XM25

Functions and Features	Description
Basic function	Supports optical signal input and output of eight 25GE/10GE ports. Supports MACSEC.
Reliability and availability	Supports the hot swappable function.
Restrictions and remarks	Ports 0 to 3 belong to a port group, and ports 4 to 7 belong to another one. By default, 25GE/10GE auto-negotiation is supported. You can run the port-mode group-id command in the slot view to switch the mode of a port group to 10GE/GE auto-negotiation. In LAN/WAN switching, the mode of ports in a port group must be switched at the same time.
Breakout supported	No
Interface rate auto-sensing supported	In 25GE mode, 10GE/25GE auto-sensing supported; In 10GE mode, GE/10GE auto-sensing supported.
FlexE supported	No

Technical Specifications

Table 1-706 Technical specifications of the CR5D00E8XM25

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	19.8 mm x 193.8 mm x 209.3 mm (0.78 in. x 7.63 in. x 8.24 in.)
Dimensions with packaging (H x W x D) [mm(in.)]	70 mm x 285 mm x 300 mm (2.76 in. x 11.22 in. x 11.81 in.)
Weight without packaging [kg(lb)]	0.8 kg (1.76 lb)

Item	Specification
Weight with packaging [kg(lb)]	1.2 kg (2.65 lb)
Typical power consumption [W]	37.5 W
Typical heat dissipation [BTU/hour]	121.57 BTU/hour
Long-term operating temperature [°C(°F)]	–40°C to +65°C (–40°F to +149°F)

1.4.4.25 CR5D00EAGF70 (10-Port 100/1000Base-X-SFP Physical Interface Card(PIC))

Overview

Table 1-707 Basic information about the CR5D00EAGF70

Item	Details
Description	10-Port 100/1000Base-X-SFP Physical Interface Card(PIC)
Part Number	03031DJR
Model	CR5D00EAGF70
Card type	Interface Card
Silkscreen	10xFE/GE-SFP

Appearance

Figure 1-148 Appearance of the CR5D00EAGF70



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find

desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03031DJR&productModel=CR5D00EAGF70#matchRelation>

Indicators

Table 1-708 Indicators on the CR5D00EAGF70

Silkscreen	Name	Color	Status	Description
STAT	Working status indicator	Green	Steady on	The board is working properly.
		Red	Steady on	The hardware on the board is faulty.
		Orange	Steady on	A high-speed interface card is unexpectedly inserted into a low-speed interface card slot and an alarm is generated, or the logic is not loaded to the board.
		-	Off	The board is not powered on or is not registered.
L/A (0-9)	Interface status indicator	Green	Steady on	The link is Up.
		Green	Blinking	Data is being transmitted or received.
		-	Off	The link is Down.

Ports

Table 1-709 Ports on the CR5D00EAGF70

Port	Connector Type	Description	Available Components
OUT0 IN0-OUT9 IN9	SFP	GE/FE service signal input and output interface	155Mbps SFP Electrical Module 155Mbps eSFP Optical Module 155Mbps eSFP BIDI Optical Module 1Gbps Electrical Module 1.25Gbps eSFP Optical Module 1.25Gbps eSFP CWDM Optical Module 1.25Gbps eSFP BIDI Optical Module 125M~2.67Gbps eSFP DWDM Optical Module

Table 1-710 Port attributes about the CR5D00EAGF70

Attributes	Details
Working mode	Full-duplex
Standards compliance	IEEE 802.3
Frame format	Ethernet_II, Ethernet_SAP and Ethernet_SNAP

Functions and Features

Table 1-711 Functions and features of the CR5D00EAGF70

Functions and Features	Description
Basic function	Occupies one flexible interface card slot. Provides 10 GE interfaces for line-rate transmitting and receiving, and supports SFP optical modules.
Reliability and availability	Hot swappable.
Breakout supported	No
Interface rate auto-sensing supported	No
FlexE supported	No

Technical Specifications

Table 1-712 Technical specifications of the CR5D00EAGF70

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	19.8 mm x 193.8 mm x 209.3 mm (0.78 in. x 7.63 in. x 8.24 in.)
Dimensions with packaging (H x W x D) [mm(in.)]	70 mm x 285 mm x 300 mm (2.76 in. x 11.22 in. x 11.81 in.)
Weight without packaging [kg(lb)]	0.6 kg (1.32 lb)
Weight with packaging [kg(lb)]	1.0 kg (2.2 lb)
Typical power consumption [W]	13.3 W
Typical heat dissipation [BTU/hour]	43.15 BTU/hour
Long-term operating temperature [°C(°F)]	–40°C to +65°C (–40°F to +149°F)

1.4.4.26 CR5D00EAGF95 (03033FXE: 10-port 100/1000Base-X-SFP MACsec Physical Interface Card(PIC))

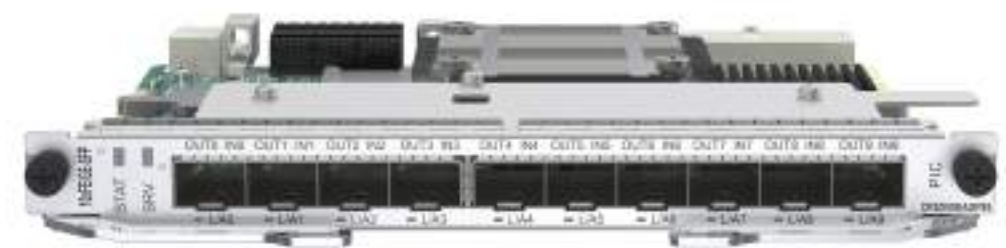
Overview

Table 1-713 Basic information about the CR5D00EAGF95

Item	Details
Description	10-port 100/1000Base-X-SFP MACsec Physical Interface Card(PIC)
Part Number	03033FXE
Model	CR5D00EAGF95
Card type	Interface Card
Silkscreen	10xFE/GE-SFP

Appearance

Figure 1-149 Appearance of the CR5D00EAGF95



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03033FXE&productModel=CR5D00EAGF95#matchRelation>

Indicators

Table 1-714 Indicators on the CR5D00EAGF95

Silkscreen	Name	Color	Status	Description
STAT	Working status indicator	Green	Steady on	If the indicator is steady on, the PIC is working properly.
		Red	Steady on	If the indicator is steady on, the hardware on the PIC is faulty.
		Orange	Steady on	If the indicator is steady on, the PIC is installed in a slot for a HIC and an alarm is reported or the PIC is not loaded with the logic.
		-	Off	If the indicator is off, the PIC is powered off or is not registered.
SRV	Service status indicator	Green	Steady on	The board has no CRC alarm.
		Orange	Steady on	The board has CRC alarms.
		-	Off	The board is not running or no power is input.
L/A(0 to 9)	Running status indicator	Green	Steady on	The link is Up.

Silkscreen	Name	Color	Status	Description
		Green	Blink	If the indicator blinks, data is being transmitted.
		-	Off	If the indicator is off, the link is Down.

Ports

Table 1-715 Ports on the CR5D00EAGF95

Port	Connector Type	Description	Available Components
OUT0 IN0 to OUT9 IN9	SFP	GE/FE service signal input and output interface	155Mbps eSFP Optical Module 155Mbps eSFP BIDI Optical Module 1Gbps Electrical Module 1.25Gbps eSFP Optical Module 1.25Gbps eSFP CWDM Optical Module 1.25Gbps eSFP BIDI Optical Module 125M~2.67Gbps eSFP DWDM Optical Module

Table 1-716 Port attributes about the CR5D00EAGF95

Attributes	Details
Working mode	Full-duplex
Standards compliance	IEEE 802.3

Attributes	Details
Frame format	Ethernet_II, Ethernet_SAP, and Ethernet_SNAP

Functions and Features

Table 1-717 Functions and features of the CR5D00EAGF95

Functions and Features	Description
Basic function	Occupies one flexible interface card slot. Provides ten FE/GE interfaces for line-rate transmitting and receiving, and supports SFP optical modules. Supports MACsec.
Reliability and availability	Hot swappable.
Breakout supported	No
Interface rate auto-sensing supported	No
FlexE supported	No

Technical Specifications

Table 1-718 Technical specifications of the CR5D00EAGF95

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	19.8 mm x 193.8 mm x 209.3 mm (0.78 in. x 7.63 in. x 8.24 in.)
Dimensions with packaging (H x W x D) [mm(in.)]	70 mm x 285 mm x 300 mm (2.76 in. x 11.22 in. x 11.81 in.)
Weight without packaging [kg(lb)]	0.7 kg (1.54 lb)
Weight with packaging [kg(lb)]	1.1 kg (2.43 lb)
Typical power consumption [W]	19.3 W
Typical heat dissipation [BTU/hour]	62.54 BTU/hour
Long-term operating temperature [°C(°F)]	–40°C to +65°C (–40°F to +149°F)

1.4.4.27 CR5D00EAGF95 (03033FXE-001: 10-port 100/1000Base-X-SFP MACsec Physical Interface Card(PIC))

Overview

Table 1-719 Basic information about the CR5D00EAGF95

Item	Details
Description	10-port 100/1000Base-X-SFP MACsec Physical Interface Card(PIC)
Part Number	03033FXE-001
Model	CR5D00EAGF95
Card type	Interface Card
Silkscreen	10xFE/GE-SFP

Appearance

Figure 1-150 Appearance of the CR5D00EAGF95



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03033FXE-001&productModel=CR5D00EAGF95#matchRelation>

Indicators

Table 1-720 Indicators on the CR5D00EAGF95

Silkscreen	Name	Color	Status	Description
STAT	Working status indicator	Green	Steady on	If the indicator is steady on, the PIC is working properly.
		Red	Steady on	If the indicator is steady on, the hardware on the PIC is faulty.
		Orange	Steady on	If the indicator is steady on, the PIC is installed in a slot for a HIC and an alarm is reported or the PIC is not loaded with the logic.
		-	Off	If the indicator is off, the PIC is powered off or is not registered.
SRV	Service status indicator	Green	Steady on	The board has no CRC alarm.
		Orange	Steady on	The board has CRC alarms.
		-	Off	The board is not running or no power is input.
L/A(0 to 9)	Running status indicator	Green	Steady on	The link is Up.

Silkscreen	Name	Color	Status	Description
		Green	Blink	If the indicator blinks, data is being transmitted.
		-	Off	If the indicator is off, the link is Down.

Ports

Table 1-721 Ports on the CR5D00EAGF95

Port	Connector Type	Description	Available Components
OUT0 IN0 to OUT9 IN9	SFP	GE/FE service signal input and output interface	155Mbps eSFP Optical Module 155Mbps eSFP BIDI Optical Module 1Gbps Electrical Module 1.25Gbps eSFP Optical Module 1.25Gbps eSFP CWDM Optical Module 1.25Gbps eSFP BIDI Optical Module 125M~2.67Gbps eSFP DWDM Optical Module

Table 1-722 Port attributes about the CR5D00EAGF95

Attributes	Details
Working mode	Full-duplex
Standards compliance	IEEE 802.3

Attributes	Details
Frame format	Ethernet_II, Ethernet_SAP, and Ethernet_SNAP

Functions and Features

Table 1-723 Functions and features of the CR5D00EAGF95

Functions and Features	Description
Basic function	Occupies one flexible interface card slot. Provides ten FE/GE interfaces for line-rate transmitting and receiving, and supports SFP optical modules. Supports MACsec.
Reliability and availability	Hot swappable.
Breakout supported	No
Interface rate auto-sensing supported	No
FlexE supported	No

Technical Specifications

Table 1-724 Technical specifications of the CR5D00EAGF95

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	19.8 mm x 193.8 mm x 209.3 mm (0.78 in. x 7.63 in. x 8.24 in.)
Dimensions with packaging (H x W x D) [mm(in.)]	70 mm x 285 mm x 300 mm (2.76 in. x 11.22 in. x 11.81 in.)
Weight without packaging [kg(lb)]	0.7 kg (1.54 lb)
Weight with packaging [kg(lb)]	1.1 kg (2.43 lb)
Typical power consumption [W]	19.3 W
Typical heat dissipation [BTU/hour]	62.54 BTU/hour
Long-term operating temperature [°C(°F)]	–40°C to +65°C (–40°F to +149°F)

1.4.4.28 CR5D00EAGF96 (10-port 100/1000Base-X-SFP MACsec Physical Interface Card(PIC))

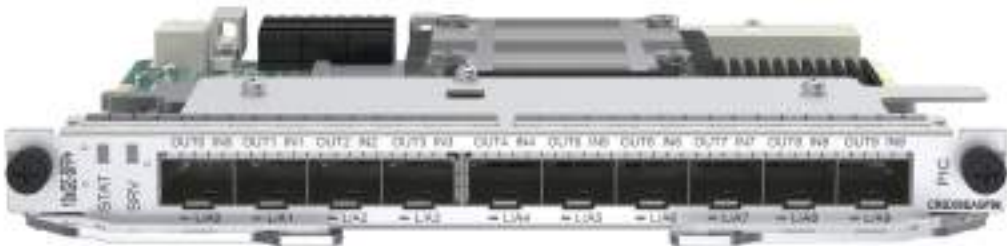
Overview

Table 1-725 Basic information about the CR5D00EAGF96

Item	Details
Description	10-port 100/1000Base-X-SFP MACsec Physical Interface Card(PIC)
Part Number	03033GFJ
Model	CR5D00EAGF96
Card type	Interface Card
Silkscreen	10xGE-SFP

Appearance

Figure 1-151 Appearance of the CR5D00EAGF96



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03033GFJ&productModel=CR5D00EAGF96#matchRelation>

Indicators

Table 1-726 Indicators on the CR5D00EAGF96

Silkscreen	Name	Color	Status	Description
STAT	Working status indicator	Green	Steady on	If the indicator is steady on, the PIC is working properly.
		Red	Steady on	If the indicator is steady on, the hardware on the PIC is faulty.
		Orange	Steady on	If the indicator is steady on, the PIC is installed in a slot for a HIC and an alarm is reported or the PIC is not loaded with the logic.
		-	Off	If the indicator is off, the PIC is powered off or is not registered.
SRV	Service status indicator	Green	Steady on	The board has no CRC alarm.
		Orange	Steady on	The board has CRC alarms.
		-	Off	The board is not running or no power is input.
L/A(0-9)	Running status indicator	Green	Steady on	The link is Up.

Silkscreen	Name	Color	Status	Description
		Green	Blink	If the indicator blinks, data is being transmitted.
		-	Off	If the indicator is off, the link is Down.

Ports

Table 1-727 Ports on the CR5D00EAGF96

Port	Connector Type	Description	Available Components
OUT0 IN0 to OUT9 IN9	SFP	Interfaces for 10-channel SFP optical signal input and output	155Mbps eSFP Optical Module 155Mbps eSFP BIDI Optical Module 1Gbps Electrical Module 1.25Gbps eSFP Optical Module 1.25Gbps eSFP CWDM Optical Module 1.25Gbps eSFP BIDI Optical Module 125M~2.67Gbps eSFP DWDM Optical Module

Table 1-728 Port attributes about the CR5D00EAGF96

Attributes	Details
Working mode	Full-duplex
Standards compliance	IEEE 802.3

Attributes	Details
Frame format	Ethernet_II, Ethernet_SAP, and Ethernet_SNAP

Functions and Features

Table 1-729 Functions and features of the CR5D00EAGF96

Functions and Features	Description
Basic function	Occupies one flexible card slot. Supports the line-rate receiving and transmitting on ten FE/GE interfaces and supports SFP optical modules on the interfaces. Supports MACSEC.
Reliability and availability	Supports the hot swappable function.
Breakout supported	No
Interface rate auto-sensing supported	No
FlexE supported	No

Technical Specifications

Table 1-730 Technical specifications of the CR5D00EAGF96

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	19.8 mm x 193.8 mm x 209.3 mm (0.78 in. x 7.63 in. x 8.24 in.)
Dimensions with packaging (H x W x D) [mm(in.)]	70 mm x 285 mm x 300 mm (2.76 in. x 11.22 in. x 11.81 in.)
Weight without packaging [kg(lb)]	0.8 kg (1.76 lb)
Weight with packaging [kg(lb)]	1.2 kg (2.65 lb)
Typical power consumption [W]	30.2 W
Typical heat dissipation [BTU/hour]	98.08 BTU/hour
Long-term operating temperature [°C(°F)]	–40°C to +65°C (–40°F to +149°F)

1.4.4.29 CR5D00EEGF73 (03033GMU: 20-Port 100/1000Base-X-CSFP/10-Port 100/1000Base-X-SFP Physical Interface Card(PIC))

Overview

Table 1-731 Basic information about the CR5D00EEGF73

Item	Details
Description	20-Port 100/1000Base-X-CSFP/10-Port 100/1000Base-X-SFP Physical Interface Card(PIC)
Part Number	03033GMU
Model	CR5D00EEGF73
Card type	Interface Card
Silkscreen	20xGE-CSFP

Appearance

Figure 1-152 Appearance of the CR5D00EEGF73



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03033GMU&productModel=CR5D00EEGF73#matchRelation>

Indicators

Table 1-732 Indicators on the CR5D00EEGF73

Silkscreen	Name	Color	Status	Description
STAT	Working status indicator	Green	Steady on	The board is working properly.
		Red	Steady on	The hardware on the PIC is faulty.
		Orange	Steady on	A high-speed interface card is unexpectedly inserted into a low-speed interface card slot and an alarm is generated, or the logic is not loaded to the board.
		-	Off	The board is not powered on or is not registered.
OUT0 IN0 to OUT19 IN19	Interface status indicator	Green	Steady on	The link is Up.
		Green	Blinking	Data is being transmitted or received.
		-	Off	The link is Down.
SRV	Service Status Indicator	Green	Steady on	The board has no CRC alarm.
		Orange	Steady on	The board has CRC alarm.
		-	Off	The board is not running or no power is input.

Ports

Table 1-733 Ports on the CR5D00EEGF73

Port	Connector Type	Description	Available Components
OUT0 IN0-OUT19 IN19	CSFP/SFP	GE/FE service signal input and output interface	1.25Gbps eSFP Optical Module 1.25Gbps eSFP CWDM Optical Module 1.25Gbps eSFP BIDI Optical Module 1.25Gbps CSFP BIDI Optical Module 125M~1.25Gbps CSFP BIDI Optical Module 125M~2.67Gbps eSFP DWDM Optical Module 155Mbps eSFP Optical Module 155Mbps eSFP BIDI Optical Module 1Gbps Electrical Module

Table 1-734 Port attributes about the CR5D00EEGF73

Attributes	Details
Working mode	Full-duplex
Standards compliance	IEEE 802.3
Frame format	Ethernet_II, Ethernet_SAP and Ethernet_SNAP

Functions and Features

Table 1-735 Functions and features of the CR5D00EEGF73

Functions and Features	Description
Basic function	Occupies one flexible interface card slot. Provides 20 GE interfaces for line-rate transmitting and receiving, and supports CSFP optical modules on interfaces 0 to 19. Provides 10 GE interfaces for line-rate transmitting and receiving, and supports SFP modules on even-numbered interfaces (0, 2, 4, ..., 18). Supports CSFP and SFP modules used separately and in combination. Supports MACsec.
Reliability and availability	Hot swappable.
Breakout supported	No
Interface rate auto-sensing supported	No
FlexE supported	No

Technical Specifications

Table 1-736 Technical specifications of the CR5D00EEGF73

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	19.8 mm x 193.8 mm x 209.3 mm (0.78 in. x 7.63 in. x 8.24 in.)
Dimensions with packaging (H x W x D) [mm(in.)]	70 mm x 285 mm x 300 mm (2.76 in. x 11.22 in. x 11.81 in.)
Weight without packaging [kg(lb)]	0.8 kg (1.76 lb)
Weight with packaging [kg(lb)]	1.3 kg (2.87 lb)
Typical power consumption [W]	32.2 W
Typical heat dissipation [BTU/hour]	104.34 BTU/hour
Long-term operating temperature [°C(°F)]	-20°C to +65°C (-4°F to +149°F)

1.4.4.30 CR5D00EEGF73 (03033GMU-001: 20-Port 100/1000Base-X-CSFP/10-Port 100/1000Base-X-SFP Physical Interface Card(PIC))

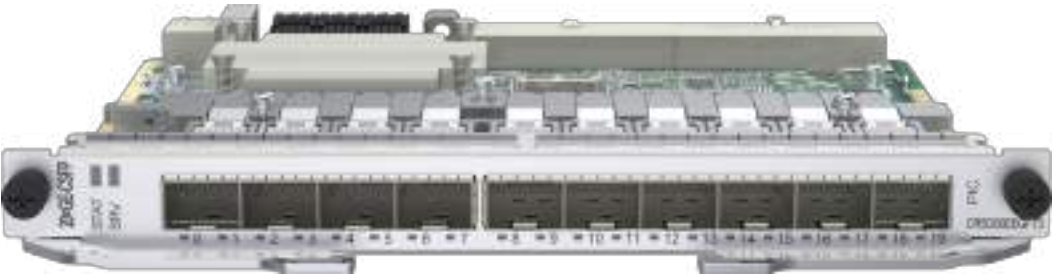
Overview

Table 1-737 Basic information about the CR5D00EEGF73

Item	Details
Description	20-Port 100/1000Base-X-CSFP/10-Port 100/1000Base-X-SFP Physical Interface Card(PIC)
Part Number	03033GMU-001
Model	CR5D00EEGF73
Card type	Interface Card
Silkscreen	20xGE-CSFP

Appearance

Figure 1-153 Appearance of the CR5D00EEGF73



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03033GMU-001&productModel=CR5D00EEGF73#matchRelation>

Indicators

Table 1-738 Indicators on the CR5D00EEGF73

Silkscreen	Name	Color	Status	Description
STAT	Working status indicator	Green	Steady on	The board is working properly.
		Red	Steady on	The hardware on the PIC is faulty.
		Orange	Steady on	A high-speed interface card is unexpectedly inserted into a low-speed interface card slot and an alarm is generated, or the logic is not loaded to the board.
		-	Off	The board is not powered on or is not registered.
OUT0 IN0 to OUT19 IN19	Interface status indicator	Green	Steady on	The link is Up.
		Green	Blinking	Data is being transmitted or received.
		-	Off	The link is Down.
SRV	Service Status Indicator	Green	Steady on	The board has no CRC alarm.
		Orange	Steady on	The board has CRC alarm.
		-	Off	The board is not running or no power is input.

Ports

Table 1-739 Ports on the CR5D00EEGF73

Port	Connector Type	Description	Available Components
OUT0 IN0-OUT19 IN19	CSFP/SFP	GE/FE service signal input and output interface	1.25Gbps eSFP Optical Module 1.25Gbps eSFP CWDM Optical Module 1.25Gbps eSFP BIDI Optical Module 1.25Gbps CSFP BIDI Optical Module 125M~1.25Gbps CSFP BIDI Optical Module 125M~2.67Gbps eSFP DWDM Optical Module 155Mbps eSFP Optical Module 155Mbps eSFP BIDI Optical Module 1Gbps Electrical Module

Table 1-740 Port attributes about the CR5D00EEGF73

Attributes	Details
Working mode	Full-duplex
Standards compliance	IEEE 802.3
Frame format	Ethernet_II, Ethernet_SAP and Ethernet_SNAP

Functions and Features

Table 1-741 Functions and features of the CR5D00EEGF73

Functions and Features	Description
Basic function	Occupies one flexible interface card slot. Provides 20 GE interfaces for line-rate transmitting and receiving, and supports CSFP optical modules on interfaces 0 to 19. Provides 10 GE interfaces for line-rate transmitting and receiving, and supports SFP modules on even-numbered interfaces (0, 2, 4, ..., 18). Supports CSFP and SFP modules used separately and in combination. Supports MACsec.
Reliability and availability	Hot swappable.
Breakout supported	No
Interface rate auto-sensing supported	No
FlexE supported	No

Technical Specifications

Table 1-742 Technical specifications of the CR5D00EEGF73

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	19.8 mm x 193.8 mm x 209.3 mm (0.78 in. x 7.63 in. x 8.24 in.)
Dimensions with packaging (H x W x D) [mm(in.)]	70 mm x 285 mm x 300 mm (2.76 in. x 11.22 in. x 11.81 in.)
Weight without packaging [kg(lb)]	0.8 kg (1.76 lb)
Weight with packaging [kg(lb)]	1.3 kg (2.87 lb)
Typical power consumption [W]	32.2 W
Typical heat dissipation [BTU/hour]	104.34 BTU/hour
Long-term operating temperature [°C(°F)]	-20°C to +65°C (-4°F to +149°F)

1.4.4.31 CR5D00L4XF72 (4-Port 10GBase LAN/WAN-SFP+ Physical Interface Card(PIC))

Overview

Table 1-743 Basic information about the CR5D00L4XF72

Item	Details
Description	4-Port 10GBase LAN/WAN-SFP+ Physical Interface Card(PIC)
Part Number	03031DJM
Model	CR5D00L4XF72
Card type	Interface Card
Silkscreen	4×10GE-SFP+

Appearance

Figure 1-154 Appearance of the CR5D00L4XF72



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03031DJM&productModel=CR5D00L4XF72#matchRelation>

Indicators

Table 1-744 Indicators on the CR5D00L4XF72

Silkscreen	Name	Color	Status	Description
STAT	Working status indicator	Green	Steady on	The board is working properly.
		Red	Steady on	The hardware on the board is faulty.
		Orange	Steady on	A high-speed interface card is unexpectedly inserted into a low-speed interface card slot and an alarm is generated, or the logic is not loaded to the board.
		-	Off	The board is not powered on or is not registered.
L/A (0-3)	Interface status indicator	Green	Steady on	The link is Up.
		Green	Blinking	Data is being transmitted or received.
		-	Off	The link is Down.

Ports

Table 1-745 Ports on the CR5D00L4XF72

Port	Connector Type	Description	Available Components
OUT0 IN0-OUT3 IN3	SFP+	10 GE service signal input and output interface	10Gbps SFP+ Optical Module 10Gbps SFP+ CWDM Optical Module 10Gbps SFP+ BIDI Optical Module 155Mbps eSFP Optical Module 155Mbps eSFP BIDI Optical Module 1Gbps Electrical Module 1.25Gbps eSFP Optical Module 1.25Gbps eSFP CWDM Optical Module 1.25Gbps eSFP BIDI Optical Module 125M~2.67Gbps eSFP DWDM Optical Module

Table 1-746 Port attributes about the CR5D00L4XF72

Attributes	Details
Working mode	Full-duplex
Standards compliance	IEEE 802.3
Frame format	Ethernet_II, Ethernet_SAP and Ethernet_SNAP

Functions and Features

Table 1-747 Functions and features of the CR5D00L4XF72

Functions and Features	Description
Basic function	Occupies one flexible interface card slot. Provides four 10GE interfaces for line-rate transmitting and receiving, and supports SFP+ optical modules.
Reliability and availability	Hot swappable.
Restrictions and remarks	The board does not support the OTN mode or FEC function. If a colored optical module is used and does not support optical amplifier insertion, the board supports only point-to-point optical transmission.
Breakout supported	No
Interface rate auto-sensing supported	Supports GE/10GE auto-sensing.
FlexE supported	No

Technical Specifications

Table 1-748 Technical specifications of the CR5D00L4XF72

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	19.8 mm x 193.8 mm x 209.3 mm (0.78 in. x 7.63 in. x 8.24 in.)
Dimensions with packaging (H x W x D) [mm(in.)]	70 mm x 285 mm x 300 mm (2.76 in. x 11.22 in. x 11.81 in.)
Weight without packaging [kg(lb)]	0.7 kg (1.54 lb)
Weight with packaging [kg(lb)]	1.1 kg (2.43 lb)
Typical power consumption [W]	25 W
Typical heat dissipation [BTU/hour]	81.11 BTU/hour
Long-term operating temperature [°C(°F)]	–40°C to +65°C (–40°F to +149°F)

1.4.4.32 CR5D00L4XF91 (03033FXF: 4-Port 10GE(SFP+)/GE(SFP) MACsec Physical Interface Card(PIC))

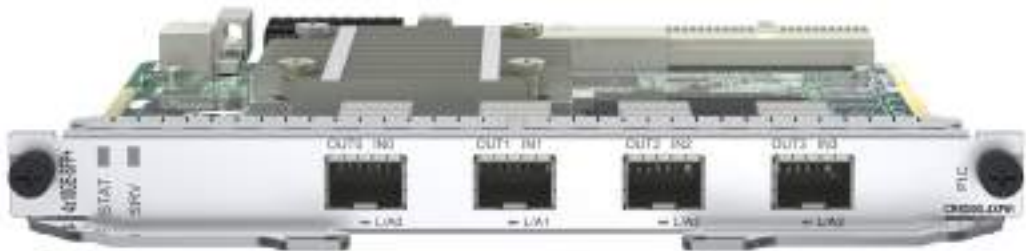
Overview

Table 1-749 Basic information about the CR5D00L4XF91

Item	Details
Description	4-Port 10GE(SFP+)/GE(SFP) MACsec Physical Interface Card(PIC)
Part Number	03033FXF
Model	CR5D00L4XF91
Card type	Interface Card
Silkscreen	4x10GE-SFP+

Appearance

Figure 1-155 Appearance of the CR5D00L4XF91



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03033FXF&productModel=CR5D00L4XF91#matchRelation>

Indicators

Table 1-750 Indicators on the CR5D00L4XF91

Silkscreen	Name	Color	Status	Description
STAT	Working status indicator	Green	Steady on	If the indicator is steady on, the PIC is working properly.
		Red	Steady on	If the indicator is steady on, the hardware on the PIC is faulty.
		Orange	Steady on	If the indicator is steady on, the PIC is installed in a slot for a HIC and an alarm is reported or the PIC is not loaded with the logic.
		-	Off	If the indicator is off, the PIC is powered off or is not registered.
SRV	Service status indicator	Green	Steady on	The board has no CRC alarm.
		Orange	Steady on	The board has CRC alarms.
		-	Off	The board is not running or no power is input.
L/A(0 to 3)	Running status indicator	Green	Steady on	The link is Up.

Silkscreen	Name	Color	Status	Description
		Green	Blink	If the indicator blinks, data is being transmitted.
		-	Off	If the indicator is off, the link is Down.

Ports

Table 1-751 Ports on the CR5D00L4XF91

Port	Connector Type	Description	Available Components
OUT0 IN0 to OUT3 IN3	SFP+	Interfaces for 4-channel SFP+/SFP optical signal input and output	1.25/9.953/10.31 25Gbps SFP+ Optical Module 1.25Gbps eSFP BIDI Optical Module 1.25Gbps eSFP CWDM Optical Module 1.25Gbps eSFP Optical Module 10Gbps SFP+ BIDI Optical Module 10Gbps SFP+ CWDM Optical Module 10Gbps SFP+ Optical Module 125M~2.67Gbps eSFP DWDM Optical Module 1Gbps Electrical Module

Table 1-752 Port attributes about the CR5D00L4XF91

Attributes	Details
Working mode	Full-duplex
Standards compliance	IEEE 802.1AE MACsec
Frame format	Ethernet_II, Ethernet_SAP, and Ethernet_SNAP

Functions and Features

Table 1-753 Functions and features of the CR5D00L4XF91

Functions and Features	Description
Basic function	Occupies one flexible interface card slot. Provides four 10GE/GE interfaces for line-rate transmitting and receiving, and supports SFP+/SFP optical modules. Supports MACsec.
Reliability and availability	Hot swappable.
Restrictions and remarks	Ports 0 to 3 belong to a port group. In LAN/WAN switching, the mode of ports in a port group must be switched at the same time.
Breakout supported	No
Interface rate auto-sensing supported	Supports GE/10GE auto-sensing.
FlexE supported	No

Technical Specifications

Table 1-754 Technical specifications of the CR5D00L4XF91

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	19.8 mm x 193.8 mm x 209.3 mm (0.78 in. x 7.63 in. x 8.24 in.)
Dimensions with packaging (H x W x D) [mm(in.)]	70 mm x 285 mm x 300 mm (2.76 in. x 11.22 in. x 11.81 in.)
Weight without packaging [kg(lb)]	0.7 kg (1.54 lb)

Item	Specification
Weight with packaging [kg(lb)]	1.1 kg (2.43 lb)
Typical power consumption [W]	23.9 W
Typical heat dissipation [BTU/hour]	77.38 BTU/hour
Long-term operating temperature [°C(°F)]	–40°C to +65°C (–40°F to +149°F)

1.4.4.33 CR5D00L4XF91 (03033FXF-001: 4-Port 10GE(SFP+)/GE(SFP) MACsec Physical Interface Card(PIC))

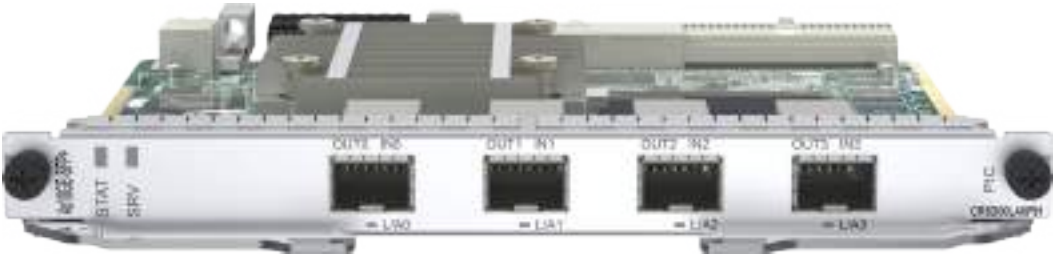
Overview

Table 1-755 Basic information about the CR5D00L4XF91

Item	Details
Description	4-Port 10GE(SFP+)/GE(SFP) MACsec Physical Interface Card(PIC)
Part Number	03033FXF-001
Model	CR5D00L4XF91
Card type	Interface Card
Silkscreen	4x10GE-SFP+

Appearance

Figure 1-156 Appearance of the CR5D00L4XF91



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find

desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03033FXF-001&productModel=CR5D00L4XF91#matchRelation>

Indicators

Table 1-756 Indicators on the CR5D00L4XF91

Silkscreen	Name	Color	Status	Description
STAT	Working status indicator	Green	Steady on	If the indicator is steady on, the PIC is working properly.
		Red	Steady on	If the indicator is steady on, the hardware on the PIC is faulty.
		Orange	Steady on	If the indicator is steady on, the PIC is installed in a slot for a HIC and an alarm is reported or the PIC is not loaded with the logic.
		-	Off	If the indicator is off, the PIC is powered off or is not registered.
SRV	Service status indicator	Green	Steady on	The board has no CRC alarm.
		Orange	Steady on	The board has CRC alarms.
		-	Off	The board is not running or no power is input.

Silkscreen	Name	Color	Status	Description
L/A(0 to 3)	Running status indicator	Green	Steady on	The link is Up.
		Green	Blink	If the indicator blinks, data is being transmitted.
		-	Off	If the indicator is off, the link is Down.

Ports

Table 1-757 Ports on the CR5D00L4XF91

Port	Connector Type	Description	Available Components
OUT0 IN0 to OUT3 IN3	SFP+	Interfaces for 4-channel SFP+/SFP optical signal input and output	1.25/9.953/10.31 25Gbps SFP+ Optical Module 1.25Gbps eSFP BIDI Optical Module 1.25Gbps eSFP CWDM Optical Module 1.25Gbps eSFP Optical Module 10Gbps SFP+ BIDI Optical Module 10Gbps SFP+ CWDM Optical Module 10Gbps SFP+ Optical Module 125M~2.67Gbps eSFP DWDM Optical Module 1Gbps Electrical Module

Table 1-758 Port attributes about the CR5D00L4XF91

Attributes	Details
Working mode	Full-duplex
Standards compliance	IEEE 802.1AE MACsec
Frame format	Ethernet_II, Ethernet_SAP, and Ethernet_SNAP

Functions and Features

Table 1-759 Functions and features of the CR5D00L4XF91

Functions and Features	Description
Basic function	Occupies one flexible interface card slot. Provides four 10GE/GE interfaces for line-rate transmitting and receiving, and supports SFP+/SFP optical modules. Supports MACsec.
Reliability and availability	Hot swappable.
Restrictions and remarks	Ports 0 to 3 belong to a port group. In LAN/WAN switching, the mode of ports in a port group must be switched at the same time.
Breakout supported	No
Interface rate auto-sensing supported	Supports GE/10GE auto-sensing.
FlexE supported	No

Technical Specifications

Table 1-760 Technical specifications of the CR5D00L4XF91

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	19.8 mm x 193.8 mm x 209.3 mm (0.78 in. x 7.63 in. x 8.24 in.)
Dimensions with packaging (H x W x D) [mm(in.)]	70 mm x 285 mm x 300 mm (2.76 in. x 11.22 in. x 11.81 in.)
Weight without packaging [kg(lb)]	0.7 kg (1.54 lb)

Item	Specification
Weight with packaging [kg(lb)]	1.1 kg (2.43 lb)
Typical power consumption [W]	23.9 W
Typical heat dissipation [BTU/hour]	77.38 BTU/hour
Long-term operating temperature [°C(°F)]	–40°C to +65°C (–40°F to +149°F)

1.4.4.34 CR5D00LAXF90 (10-Port 10GBase LAN/WAN-SFP+ Physical Interface Card(PIC))

Overview

Table 1-761 Basic information about the CR5D00LAXF90

Item	Details
Description	10-Port 10GBase LAN/WAN-SFP+ Physical Interface Card(PIC)
Part Number	03032TUB
Model	CR5D00LAXF90
Card type	Interface Card
Silkscreen	10x10GE-SFP+

Appearance

Figure 1-157 Appearance of the CR5D00LAXF90



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find

desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03032TUB&productModel=CR5D00LAXF90#matchRelation>

Indicators

Table 1-762 Indicators on the CR5D00LAXF90

Silkscreen	Name	Color	Status	Description
STAT	Working status indicator	Green	Steady on	The board is working properly.
		Red	Steady on	The hardware on the board is faulty.
		Yellow	Steady on	An alarm about a fault is generated on the board or the board is not loaded with logic.
		-	Off	The board is not powered on or is not registered.
L/A (0-9)	Interface status indicator	Green	Steady on	The link is Up.
		Green	Blinking	Data is being transmitted or received.
		-	Off	The link is Down.

Ports

Table 1-763 Ports on the CR5D00LAXF90

Port	Connector Type	Description	Available Components
OUT0 IN0 to OUT9 IN9	SFP+	10GE/GE service signal input and output interface	10Gbps SFP+ Optical Module 10Gbps SFP+ CWDM Optical Module 10Gbps SFP+ BIDI Optical Module 1.25Gbps eSFP Optical Module 1.25Gbps eSFP CWDM Optical Module 1.25Gbps eSFP BIDI Optical Module 125M~2.67Gbps eSFP DWDM Optical Module 1.25/9.953/10.31 25Gbps SFP+ Optical Module

Table 1-764 Port attributes about the CR5D00LAXF90

Attributes	Details
Working mode	Full-duplex
Standards compliance	IEEE 802.3
Frame format	Ethernet_II, Ethernet_SAP and Ethernet_SNAP

Functions and Features

Table 1-765 Functions and features of the CR5D00LAXF90

Functions and Features	Description
Basic function	Occupies one flexible interface card slot. Provides ten 10GE/GE interfaces for line-rate transmitting and receiving, and supports SFP+/SFP optical modules.
Reliability and availability	Hot swappable.
Restrictions and remarks	The outdoor application is not supported. The board does not support the OTN or FEC mode. When being used with the colored optical module, it does not support the intermediate optical amplifier and supports only point-to-point transmission.
Breakout supported	No
Interface rate auto-sensing supported	Supports GE/10GE auto-sensing.
FlexE supported	No

Technical Specifications

Table 1-766 Technical specifications of the CR5D00LAXF90

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	19.8 mm x 193.8 mm x 209.3 mm (0.78 in. x 7.63 in. x 8.24 in.)
Dimensions with packaging (H x W x D) [mm(in.)]	70 mm x 285 mm x 300 mm (2.76 in. x 11.22 in. x 11.81 in.)
Weight without packaging [kg(lb)]	0.6 kg (1.32 lb)
Typical power consumption [W]	46.7 W
Typical heat dissipation [BTU/hour]	151.51 BTU/hour
Long-term operating temperature [°C(°F)]	0°C to 45°C (32°F to 113°F)
Short-term operating temperature [°C(°F)]	-5°C to +55°C (23°F to 131°F)

1.4.4.35 CR5D00LAXF91 (03033BLN: 10-Port 10GE/GE LAN/WAN-SFP+ MACsec Physical Interface Card(PIC))

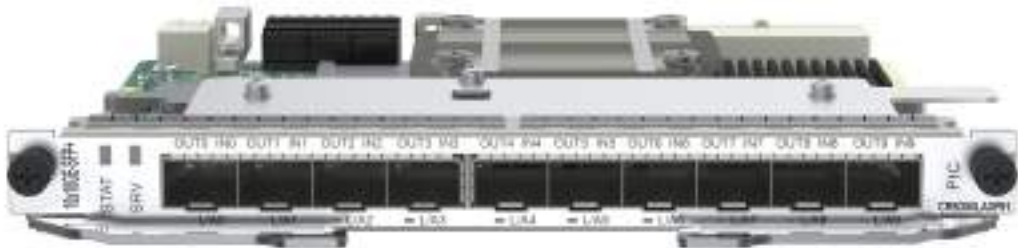
Overview

Table 1-767 Basic information about the CR5D00LAXF91

Item	Details
Description	10-Port 10GE/GE LAN/WAN-SFP+ MACsec Physical Interface Card(PIC)
Part Number	03033BLN
Model	CR5D00LAXF91
Card type	Interface Card
Silkscreen	10x10GE-SFP+

Appearance

Figure 1-158 Appearance of the CR5D00LAXF91



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03033BLN&productModel=CR5D00LAXF91#matchRelation>

Indicators

Table 1-768 Indicators on the CR5D00LAXF91

Silkscreen	Name	Color	Status	Description
STAT	Working status indicator	Green	Steady on	The board is working properly.
		Red	Steady on	The hardware on the board is faulty.
		Orange	Steady on	A high-speed interface card is unexpectedly inserted into a low-speed interface card slot and an alarm is generated, or the logic is not loaded to the board.
		-	Off	The board is not powered on or is not registered.
SRV	Service Status Indicator	Green	Steady on	The board has no CRC alarm.
		Orange	Steady on	The board has CRC alarm.
		-	Off	The board is not running or no power is input.
L/A (0-9)	Interface status indicator	Green	Steady on	The link is Up.
		Green	Blinking	Data is being transmitted or received.
		-	Off	The link is Down.

Ports

Table 1-769 Ports on the CR5D00LAXF91

Port	Connector Type	Description	Available Components
OUT0 IN0-OUT9 IN9	SFP+	10 GE service signal input and output interface	155Mbps eSFP Optical Module 155Mbps eSFP BIDI Optical Module 1.25/9.953/10.31 25Gbps SFP+ Optical Module 1.25Gbps eSFP BIDI Optical Module 1.25Gbps eSFP CWDM Optical Module 1.25Gbps eSFP Optical Module 10Gbps SFP+ BIDI Optical Module 10Gbps SFP+ CWDM Optical Module 10Gbps SFP+ Optical Module 125M~2.67Gbps eSFP DWDM Optical Module 1Gbps SyncE and 1588V2 Electrical Module 1Gbps Electrical Module

Table 1-770 Port attributes about the CR5D00LAXF91

Attributes	Details
Working mode	Full-duplex
Standards compliance	IEEE 802.3

Attributes	Details
Frame format	Ethernet_II, Ethernet_SAP and Ethernet_SNAP

Functions and Features

Table 1-771 Functions and features of the CR5D00LAXF91

Functions and Features	Description
Basic function	Occupies one flexible interface card slot. Provides ten 10GE/GE interfaces for line-rate transmitting and receiving, and supports SFP+/SFP optical modules. Supports MACsec.
Reliability and availability	Hot swappable.
Restrictions and remarks	Ports 2–5 belong to a port group, and ports 6–9 belong to another one. In LAN/WAN switching, the mode of ports in a port group must be switched at the same time.
Breakout supported	No
Interface rate auto-sensing supported	Supports GE/10GE auto-sensing.
FlexE supported	No

Technical Specifications

Table 1-772 Technical specifications of the CR5D00LAXF91

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	19.8 mm x 193.8 mm x 209.3 mm (0.78 in. x 7.63 in. x 8.24 in.)
Dimensions with packaging (H x W x D) [mm(in.)]	70 mm x 285 mm x 300 mm (2.76 in. x 11.22 in. x 11.81 in.)
Weight without packaging [kg(lb)]	0.8 kg (1.76 lb)
Weight with packaging [kg(lb)]	1.2 kg (2.65 lb)
Typical power consumption [W]	32.6 W

Item	Specification
Typical heat dissipation [BTU/hour]	105.8 BTU/hour
Long-term operating temperature [°C(°F)]	–40°C to +65°C (–40°F to +149°F)

1.4.4.36 CR5D00LAXF91 (03033BLN-001: 10-Port 10GE/GE LAN/WAN-SFP+ MACsec Physical Interface Card(PIC))

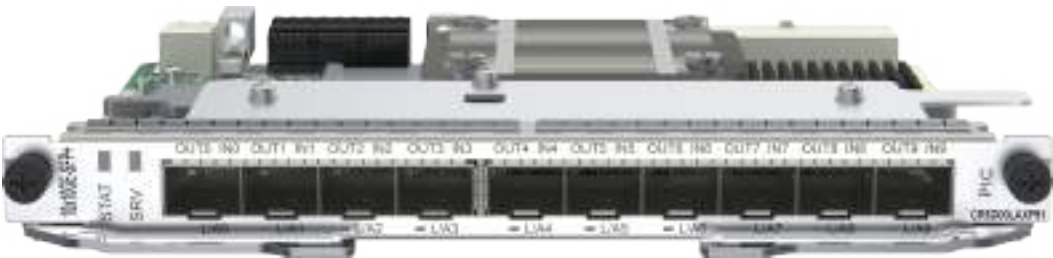
Overview

Table 1-773 Basic information about the CR5D00LAXF91

Item	Details
Description	10-Port 10GE/GE LAN/WAN-SFP+ MACsec Physical Interface Card(PIC)
Part Number	03033BLN-001
Model	CR5D00LAXF91
Card type	Interface Card
Silkscreen	10x10GE-SFP+

Appearance

Figure 1-159 Appearance of the CR5D00LAXF91



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03033BLN-001&productModel=CR5D00LAXF91#matchRelation>

Indicators

Table 1-774 Indicators on the CR5D00LAXF91

Silkscreen	Name	Color	Status	Description
STAT	Working status indicator	Green	Steady on	The board is working properly.
		Red	Steady on	The hardware on the board is faulty.
		Orange	Steady on	A high-speed interface card is unexpectedly inserted into a low-speed interface card slot and an alarm is generated, or the logic is not loaded to the board.
		-	Off	The board is not powered on or is not registered.
SRV	Service Status Indicator	Green	Steady on	The board has no CRC alarm.
		Orange	Steady on	The board has CRC alarm.
		-	Off	The board is not running or no power is input.
L/A (0-9)	Interface status indicator	Green	Steady on	The link is Up.
		Green	Blinking	Data is being transmitted or received.

Silkscreen	Name	Color	Status	Description
		-	Off	The link is Down.

Ports

Table 1-775 Ports on the CR5D00LAXF91

Port	Connector Type	Description	Available Components
OUT0 IN0-OUT9 IN9	SFP+	10 GE service signal input and output interface	155Mbps eSFP Optical Module 155Mbps eSFP BIDI Optical Module 1.25/9.953/10.31 25Gbps SFP+ Optical Module 1.25Gbps eSFP BIDI Optical Module 1.25Gbps eSFP CWDM Optical Module 1.25Gbps eSFP Optical Module 10Gbps SFP+ BIDI Optical Module 10Gbps SFP+ CWDM Optical Module 10Gbps SFP+ Optical Module 125M~2.67Gbps eSFP DWDM Optical Module 1Gbps SyncE and 1588V2 Electrical Module 1Gbps Electrical Module

Table 1-776 Port attributes about the CR5D00LAXF91

Attributes	Details
Working mode	Full-duplex
Standards compliance	IEEE 802.3
Frame format	Ethernet_II, Ethernet_SAP and Ethernet_SNAP

Functions and Features

Table 1-777 Functions and features of the CR5D00LAXF91

Functions and Features	Description
Basic function	Occupies one flexible interface card slot. Provides ten 10GE/GE interfaces for line-rate transmitting and receiving, and supports SFP+/SFP optical modules. Supports MACsec.
Reliability and availability	Hot swappable.
Restrictions and remarks	Ports 2–5 belong to a port group, and ports 6–9 belong to another one. In LAN/WAN switching, the mode of ports in a port group must be switched at the same time.
Breakout supported	No
Interface rate auto-sensing supported	Supports GE/10GE auto-sensing.
FlexE supported	No

Technical Specifications

Table 1-778 Technical specifications of the CR5D00LAXF91

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	19.8 mm x 193.8 mm x 209.3 mm (0.78 in. x 7.63 in. x 8.24 in.)
Dimensions with packaging (H x W x D) [mm(in.)]	70 mm x 285 mm x 300 mm (2.76 in. x 11.22 in. x 11.81 in.)

Item	Specification
Weight without packaging [kg(lb)]	0.8 kg (1.76 lb)
Weight with packaging [kg(lb)]	1.2 kg (2.65 lb)
Typical power consumption [W]	32.6 W
Typical heat dissipation [BTU/hour]	105.8 BTU/hour
Long-term operating temperature [°C(°F)]	–40°C to +65°C (–40°F to +149°F)

1.4.4.37 CR5D00P4CFC1 (4-Port OC-3c/STM-1c POS-SFP Flexible Interface Card(PIC))

Overview

Table 1-779 Basic information about the CR5D00P4CFC1

Item	Details
Description	4-Port OC-3c/STM-1c POS-SFP Flexible Interface Card(PIC)
Part Number	03033GGE
Model	CR5D00P4CFC1
Card type	Interface Card
Silkscreen	4xSTM1-POS

Appearance

Figure 1-160 Appearance of the CR5D00P4CFC1



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03033GGE&productModel=CR5D00P4CFC1#matchRelation>

Indicators

Table 1-780 Indicators on the CR5D00P4CFC1

Silkscreen	Name	Color	Status	Description
STAT	Working status indicator	Green	Steady on	If the indicator is steady on, the PIC is working properly.
		Red	Steady on	If the indicator is steady on, the hardware on the PIC is faulty.
		Orange	Steady on	If the indicator is steady on, the PIC is installed in a slot for a HIC and an alarm is reported or the PIC is not loaded with the logic.
		-	Off	If the indicator is off, the PIC is powered off or is not registered.
SRV	Service status indicator	Green	Steady on	The board has no CRC alarm.
		Orange	Steady on	The board has CRC alarms.

Silkscreen	Name	Color	Status	Description
		-	Off	The board is not running or no power is input.
L/A(0-3)	Running status indicator	Green	Steady on	The link is Up.
		Green	Blink	If the indicator blinks, data is being transmitted.
		-	Off	If the indicator is off, the link is Down.

Ports

Table 1-781 Ports on the CR5D00P4CFC1

Port	Connector Type	Description	Available Components
OUT0 IN0 to OUT3 IN3	SFP	STM-1 service signal input and output interface	155Mbps SFP Electrical Module 155Mbps eSFP Optical Module 155Mbps eSFP BIDI Optical Module

Table 1-782 Port attributes about the CR5D00P4CFC1

Attributes	Details
Working mode	Full-duplex
Standards compliance	ITU-T G707
Frame format	PPP

Functions and Features

Table 1-783 Functions and features of the CR5D00P4CFC1

Functions and Features	Description
Basic function	Provides four 155M POS interfaces.
Reliability and availability	Hot swappable.
Restrictions and remarks	Ports 0 to 3 support clock master mode.
Breakout supported	No
Interface rate auto-sensing supported	No
FlexE supported	No
Link protocol	PPP

Technical Specifications

Table 1-784 Technical specifications of the CR5D00P4CFC1

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	19.8 mm x 193.8 mm x 209.3 mm (0.78 in. x 7.63 in. x 8.24 in.)
Dimensions with packaging (H x W x D) [mm(in.)]	70 mm x 285 mm x 300 mm (2.76 in. x 11.22 in. x 11.81 in.)
Weight without packaging [kg(lb)]	0.5 kg (1.1 lb)
Weight with packaging [kg(lb)]	0.9 kg (1.98 lb)
Typical power consumption [W]	16.8 W
Typical heat dissipation [BTU/hour]	54.5 BTU/hour
Long-term operating temperature [°C(°F)]	-20°C to +65°C (-4°F to +149°F)

1.4.4.38 CR5D04OTU290 (4-Port 10G OTU2-XFP Physical Interface Card(PIC))

Overview

Table 1-785 Basic information about the CR5D04OTU290

Item	Details
Description	4-Port 10G OTU2-XFP Physical Interface Card(PIC)
Part Number	03032RNN
Model	CR5D04OTU290
Card type	Interface Card
Silkscreen	4x10G-XFP

Appearance

Figure 1-161 Appearance of the CR5D04OTU290



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03032RNN&productModel=CR5D04OTU290#matchRelation>

Indicators

Table 1-786 Indicators on the CR5D04OTU290

Silkscreen	Name	Color	Status	Description
STAT	Working status indicator	Green	Steady on	The board is working properly.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	The hardware on the board is faulty.
		Yellow	Steady on	A high-speed interface card is unexpectedly inserted into a low-speed interface card slot and an alarm is generated, or the logic is not loaded to the board.
		-	Off	The board is not powered on or is not registered.
L/A0 to L/A3	Interface status indicator	Green	Steady on	The link is Up.
		Green	Blinking	Data is being transmitted or received.
		-	Off	The link is Down.

Ports

Table 1-787 Ports on the CR5D04OTU290

Port	Connector Type	Description	Available Components
OUT0 IN0-OUT3 IN3	XFP	10 GE service signal input and output interface	10Gbps TXFP DWDM Optical Module 10Gbps XFP DWDM Optical Module

Table 1-788 Port attributes about the CR5D04OTU290

Attributes	Details
Working mode	Full-duplex
Standards compliance	Supports Ethernet IEEE 802.3 standard network protocol
Frame format	OTN

Functions and Features

Table 1-789 Functions and features of the CR5D04OTU290

Functions and Features	Description
Basic function	Supports 4-channel OTU2/OTU2E and 10GE LAN services.
Reliability and availability	Supports the hot swappable function.
Breakout supported	No
Interface rate auto-sensing supported	No
FlexE supported	No

Technical Specifications

Table 1-790 Technical specifications of the CR5D04OTU290

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	19.8 mm x 193.8 mm x 209.3 mm (0.78 in. x 7.63 in. x 8.24 in.)
Dimensions with packaging (H x W x D) [mm(in.)]	70 mm x 285 mm x 300 mm (2.76 in. x 11.22 in. x 11.81 in.)
Weight without packaging [kg(lb)]	0.8 kg (1.76 lb)
Weight with packaging [kg(lb)]	1.2 kg (2.65 lb)
Typical power consumption [W]	52.5 W
Typical heat dissipation [BTU/hour]	170.33 BTU/hour
Long-term operating temperature [°C(°F)]	0°C to 45°C (32°F to 113°F)
Short-term operating temperature [°C(°F)]	-5°C to +50°C (23°F to 122°F)

1.4.4.39 CR5D0E5XMF94 (03033GDR: 2-Port 50GBase/1-Port 100GBase-QSFP28 FlexE MACsec Physical Interface Card(PIC))

Overview

Table 1-791 Basic information about the CR5D0E5XMF94

Item	Details
Description	2-Port 50GBase/1-Port 100GBase-QSFP28 FlexE MACsec Physical Interface Card(PIC)
Part Number	03033GDR
Model	CR5D0E5XMF94
Card type	Interface Card
Silkscreen	2x50GE-QSFP28

Appearance

Figure 1-162 Appearance of the CR5D0E5XMF94



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03033GDR&productModel=CR5D0E5XMF94#matchRelation>

Indicators

Table 1-792 Indicators on the CR5D0E5XMF94

Silkscreen	Name	Color	Status	Description
STAT	Working status indicator	Green	Steady on	The board is working properly.
		Red	Steady on	The hardware on the board is faulty.
		Orange	Steady on	A high-speed interface card is unexpectedly inserted into a low-speed interface card slot and an alarm is generated, or the logic is not loaded to the board.
		-	Off	The board is not powered on or is not registered.
SRV	Service Status Indicator	Green	Steady on	The board has no CRC alarm.
		Orange	Steady on	The board has CRC alarm.
		-	Off	The board is not running or no power is input.
L/A (0-1)	Interface status indicator	Green	Steady on	The link is Up.
		Green	Blinking	Data is being transmitted or received.
		-	Off	The link is Down.

Silkscreen	Name	Color	Status	Description
BREAKOUT (0–3)	Breakout channel indicator	Green	Blink	Each indicator for channels 0, 1, 2, and 3 turns on for 5s in sequence to indicate the status of the corresponding channel. This process repeats.
		-	Off	No data is transmitting or receiving in this channel.

Ports

Table 1-793 Ports on the CR5D0E5XMF94

Port	Connector Type	Description	Available Components
OUT0 IN0-OUT1 IN1	QSFP28	100 GE/50 GE service signal input and output interface	100Gbps QSFP28 Optical Module 40Gbps QSFP+ Optical Module 50Gbps QSFP28 BIDI Optical Module 50Gbps QSFP28 Optical Module Optical Amplifier Module

Table 1-794 Port attributes about the CR5D0E5XMF94

Attributes	Details
Working mode	Full-duplex
Standards compliance	IEEE 802.3

Attributes	Details
Frame format	Ethernet_II, Ethernet_SAP and Ethernet_SNAP

Functions and Features

Table 1-795 Functions and features of the CR5D0E5XMF94

Functions and Features	Description
Basic function	Occupies one flexible interface card slot. Provides two 50GE interfaces or one 100GE interface for line-rate transmitting and receiving, and supports QSFP28 optical modules. Supports FlexE and MACsec.
Reliability and availability	Hot swappable.
Restrictions and remarks	Port 0 supports 100GE, 50GE, 40GE, 4x25GE breakout, and 4x10GE breakout. Port 1 is valid only when port 0 is 50GE. If the switch-mode command is set to auto and port 0 has a 100GE optical module or a 40GE optical module installed, the port automatically works in 100GE/40GE mode. In this case, port 0 takes effect, but port 1 does not take effect. If a 50GE optical module is inserted into port 0, ports 0 and 1 are both valid, and port 0 automatically works in 50GE mode. If port 1 works in 50GE mode and services are running on port 1, port 0 does not support auto-sensing.
Breakout supported	Yes. To enable the breakout function on a port, run the port split command.
Interface rate auto-sensing supported	Supports 40GE/50GE/100GE auto-sensing.

Functions and Features	Description
FlexE supported	Yes Physical ports 0 and 1 can be added to the same group. port-id ranges from 129 to 148 and from 1000 to 3000. A maximum of 20 FlexE clients can be created.

Technical Specifications

Table 1-796 Technical specifications of the CR5D0E5XMF94

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	19.8 mm x 193.8 mm x 209.3 mm (0.78 in. x 7.63 in. x 8.24 in.)
Dimensions with packaging (H x W x D) [mm(in.)]	70 mm x 285 mm x 300 mm (2.76 in. x 11.22 in. x 11.81 in.)
Weight without packaging [kg(lb)]	0.8 kg (1.76 lb)
Weight with packaging [kg(lb)]	1.2 kg (2.65 lb)
Typical power consumption [W]	28.8 W
Typical heat dissipation [BTU/hour]	93.47 BTU/hour
Long-term operating temperature [°C(°F)]	-5°C to +55°C (23°F to 131°F)

1.4.4.40 CR5D0E5XMF94 (03033GDR-001: 2-Port 50GBase/1-Port 100GBase-QSFP28 FlexE MACsec Physical Interface Card(PIC))

Overview

Table 1-797 Basic information about the CR5D0E5XMF94

Item	Details
Description	2-Port 50GBase/1-Port 100GBase-QSFP28 FlexE MACsec Physical Interface Card(PIC)
Part Number	03033GDR-001
Model	CR5D0E5XMF94

Item	Details
Card type	Interface Card
Silkscreen	2x50GE-QSFP28

Appearance

Figure 1-163 Appearance of the CR5D0E5XMF94



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03033GDR-001&productModel=CR5D0E5XMF94#matchRelation>

Indicators

Table 1-798 Indicators on the CR5D0E5XMF94

Silkscreen	Name	Color	Status	Description
STAT	Working status indicator	Green	Steady on	The board is working properly.
		Red	Steady on	The hardware on the board is faulty.

Silkscreen	Name	Color	Status	Description
		Orange	Steady on	A high-speed interface card is unexpectedly inserted into a low-speed interface card slot and an alarm is generated, or the logic is not loaded to the board.
		-	Off	The board is not powered on or is not registered.
SRV	Service Status Indicator	Green	Steady on	The board has no CRC alarm.
		Orange	Steady on	The board has CRC alarm.
		-	Off	The board is not running or no power is input.
L/A (0-1)	Interface status indicator	Green	Steady on	The link is Up.
		Green	Blinking	Data is being transmitted or received.
		-	Off	The link is Down.
BREAKOUT (0-3)	Breakout channel indicator	Green	Blink	Each indicator for channels 0, 1, 2, and 3 turns on for 5s in sequence to indicate the status of the corresponding channel. This process repeats.

Silkscreen	Name	Color	Status	Description
		-	Off	No data is transmitting or receiving in this channel.

Ports

Table 1-799 Ports on the CR5D0E5XMF94

Port	Connector Type	Description	Available Components
OUT0 IN0-OUT1 IN1	QSFP28	100 GE/50 GE service signal input and output interface	100Gbps QSFP28 Optical Module 40Gbps QSFP+ Optical Module 50Gbps QSFP28 BIDI Optical Module 50Gbps QSFP28 Optical Module Optical Amplifier Module

Table 1-800 Port attributes about the CR5D0E5XMF94

Attributes	Details
Working mode	Full-duplex
Standards compliance	IEEE 802.3
Frame format	Ethernet_II, Ethernet_SAP and Ethernet_SNAP

Functions and Features

Table 1-801 Functions and features of the CR5D0E5XMF94

Functions and Features	Description
Basic function	Occupies one flexible interface card slot. Provides two 50GE interfaces or one 100GE interface for line-rate transmitting and receiving, and supports QSFP28 optical modules. Supports FlexE and MACsec.
Reliability and availability	Hot swappable.
Restrictions and remarks	Port 0 supports 100GE, 50GE, 40GE, 4x25GE breakout, and 4x10GE breakout. Port 1 is valid only when port 0 is 50GE. If the switch-mode command is set to auto and port 0 has a 100GE optical module or a 40GE optical module installed, the port automatically works in 100GE/40GE mode. In this case, port 0 takes effect, but port 1 does not take effect. If a 50GE optical module is inserted into port 0, ports 0 and 1 are both valid, and port 0 automatically works in 50GE mode. If port 1 works in 50GE mode and services are running on port 1, port 0 does not support auto-sensing.
Breakout supported	Yes. To enable the breakout function on a port, run the port split command.
Interface rate auto-sensing supported	Supports 40GE/50GE/100GE auto-sensing.
FlexE supported	Yes Physical ports 0 and 1 can be added to the same group. port-id ranges from 129 to 148 and from 1000 to 3000. A maximum of 20 FlexE clients can be created.

Technical Specifications

Table 1-802 Technical specifications of the CR5D0E5XMF94

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	19.8 mm x 193.8 mm x 209.3 mm (0.78 in. x 7.63 in. x 8.24 in.)
Dimensions with packaging (H x W x D) [mm(in.)]	70 mm x 285 mm x 300 mm (2.76 in. x 11.22 in. x 11.81 in.)
Weight without packaging [kg(lb)]	0.8 kg (1.76 lb)
Weight with packaging [kg(lb)]	1.2 kg (2.65 lb)
Typical power consumption [W]	28.8 W
Typical heat dissipation [BTU/hour]	93.47 BTU/hour
Long-term operating temperature [°C(°F)]	-5°C to +55°C (23°F to 131°F)

1.4.4.41 CR5DE2NE4X14 (03033GDS: 2-Port 100GBase/50GBase-QSFP28 FlexE MACsec Interface Card(PIC))

Overview

Table 1-803 Basic information about the CR5DE2NE4X14

Item	Details
Description	2-Port 100GBase/50GBase-QSFP28 FlexE MACsec Interface Card(PIC)
Part Number	03033GDS
Model	CR5DE2NE4X14
Card type	Interface Card
Silkscreen	2x100GE-QSFP28

Appearance

Figure 1-164 Appearance of the CR5DE2NE4X14



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03033GDS&productModel=CR5DE2NE4X14#matchRelation>

Indicators

Table 1-804 Indicators on the CR5DE2NE4X14

Silkscreen	Name	Color	Status	Description
STAT	Working status indicator	Green	Steady on	The board is working properly.
		Red	Steady on	The hardware on the board is faulty.

Silkscreen	Name	Color	Status	Description
		Orange	Steady on	A high-speed interface card is unexpectedly inserted into a low-speed interface card slot and an alarm is generated, or the logic is not loaded to the board.
		-	Off	The board is not powered on or is not registered.
SRV	Service Status Indicator	Green	Steady on	The board has no CRC alarm.
		Orange	Steady on	The board has CRC alarm.
		-	Off	The board is not running or no power is input.
L/A (0-1)	Interface status indicator	Green	Steady on	The link is Up.
		Green	Blinking	Data is being transmitted or received.
		-	Off	The link is Down.
BREAKOUT (0-3)	Breakout channel indicator	Green	Blink	Each indicator for channels 0, 1, 2, and 3 turns on for 5s in sequence to indicate the status of the corresponding channel. This process repeats.

Silkscreen	Name	Color	Status	Description
		-	Off	No data is transmitting or receiving in this channel.

Ports

Table 1-805 Ports on the CR5DE2NE4X14

Port	Connector Type	Description	Available Components
OUT0 IN0-OUT1 IN1	QSFP28	100 GE/50 GE service signal input and output interface	100Gbps QSFP28 Optical Module 40Gbps QSFP+ Optical Module 50Gbps QSFP28 BIDI Optical Module 50Gbps QSFP28 Optical Module Optical Amplifier Module

Table 1-806 Port attributes about the CR5DE2NE4X14

Attributes	Details
Working mode	Full-duplex
Standards compliance	IEEE 802.3
Frame format	Ethernet_II, Ethernet_SAP and Ethernet_SNAP

Functions and Features

Table 1-807 Functions and features of the CR5DE2NE4X14

Functions and Features	Description
Basic function	Occupies one flexible interface card slot. Provides two 100GE/50GE interfaces for line-rate transmitting and receiving, and supports QSFP28 optical modules. FlexE and MACsec are supported.
Reliability and availability	Hot swappable.
Restrictions and remarks	Supports 100GE, 50GE, 40GE, 4x25GE breakout, and 4x10GE breakout.
Breakout supported	Yes. To enable the breakout function on a port, run the port split command.
Interface rate auto-sensing supported	Supports 40GE/50GE/100GE auto-sensing.
FlexE supported	Yes Physical ports 0 and 1 can be added to the same group. port-id ranges from 2 to 21 and from 1000 to 3000. A maximum of 20 FlexE clients can be created. Specifically, the interface card on the CX600-X2-M8A, NE40E-X2-M8A, NetEngine 8000 M8, NetEngine 8000E M8, PTN 6900-2-M8A, and PTN 6900-2-M8C supports the following FlexE specifications: Physical port 0 can be added to one group. port-id ranges from 2 to 11 and from 1000 to 3000. A maximum of 10 FlexE clients can be created. Physical port 1 can be added to one group. port-id ranges from 12 to 21 and from 1000 to 3000. A maximum of 10 FlexE clients can be created.

Technical Specifications

Table 1-808 Technical specifications of the CR5DE2NE4X14

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	19.8 mm x 193.8 mm x 209.3 mm (0.78 in. x 7.63 in. x 8.24 in.)
Dimensions with packaging (H x W x D) [mm(in.)]	70 mm x 285 mm x 300 mm (2.76 in. x 11.22 in. x 11.81 in.)
Weight without packaging [kg(lb)]	0.8 kg (1.76 lb)
Weight with packaging [kg(lb)]	1.2 kg (2.65 lb)
Typical power consumption [W]	32.2 W
Typical heat dissipation [BTU/hour]	104.41 BTU/hour
Long-term operating temperature [°C(°F)]	–5°C to +55°C (23°F to 131°F)

1.4.4.42 CR5DE2NE4X14 (03033GDS-001: 2-Port 100GBase/50GBase-QSFP28 FlexE MACsec Interface Card(PIC))

Overview

Table 1-809 Basic information about the CR5DE2NE4X14

Item	Details
Description	2-Port 100GBase/50GBase-QSFP28 FlexE MACsec Interface Card(PIC)
Part Number	03033GDS-001
Model	CR5DE2NE4X14
Card type	Interface Card
Silkscreen	2x100GE-QSFP28

Appearance

Figure 1-165 Appearance of the CR5DE2NE4X14



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03033GDS-001&productModel=CR5DE2NE4X14#matchRelation>

Indicators

Table 1-810 Indicators on the CR5DE2NE4X14

Silkscreen	Name	Color	Status	Description
STAT	Working status indicator	Green	Steady on	The board is working properly.
		Red	Steady on	The hardware on the board is faulty.
		Orange	Steady on	A high-speed interface card is unexpectedly inserted into a low-speed interface card slot and an alarm is generated, or the logic is not loaded to the board.
		-	Off	The board is not powered on or is not registered.
SRV	Service Status Indicator	Green	Steady on	The board has no CRC alarm.
		Orange	Steady on	The board has CRC alarm.

Silkscreen	Name	Color	Status	Description
		-	Off	The board is not running or no power is input.
L/A (0-1)	Interface status indicator	Green	Steady on	The link is Up.
		Green	Blinking	Data is being transmitted or received.
		-	Off	The link is Down.
BREAKOUT (0-3)	Breakout channel indicator	Green	Blink	Each indicator for channels 0, 1, 2, and 3 turns on for 5s in sequence to indicate the status of the corresponding channel. This process repeats.
		-	Off	No data is transmitting or receiving in this channel.

Ports

Table 1-811 Ports on the CR5DE2NE4X14

Port	Connector Type	Description	Available Components
OUT0 IN0-OUT1 IN1	QSFP28	100 GE/50 GE service signal input and output interface	100Gbps QSFP28 Optical Module 40Gbps QSFP+ Optical Module 50Gbps QSFP28 BIDI Optical Module 50Gbps QSFP28 Optical Module Optical Amplifier Module

Table 1-812 Port attributes about the CR5DE2NE4X14

Attributes	Details
Working mode	Full-duplex
Standards compliance	IEEE 802.3
Frame format	Ethernet_II, Ethernet_SAP and Ethernet_SNAP

Functions and Features

Table 1-813 Functions and features of the CR5DE2NE4X14

Functions and Features	Description
Basic function	Occupies one flexible interface card slot. Provides two 100GE/50GE interfaces for line-rate transmitting and receiving, and supports QSFP28 optical modules. FlexE and MACsec are supported.
Reliability and availability	Hot swappable.
Restrictions and remarks	Supports 100GE, 50GE, 40GE, 4x25GE breakout, and 4x10GE breakout.

Functions and Features	Description
Breakout supported	Yes. To enable the breakout function on a port, run the port split command.
Interface rate auto-sensing supported	Supports 40GE/50GE/100GE auto-sensing.
FlexE supported	Yes Physical ports 0 and 1 can be added to the same group. port-id ranges from 2 to 21 and from 1000 to 3000. A maximum of 20 FlexE clients can be created. Specifically, the interface card on the CX600-X2-M8A, NE40E-X2-M8A, NetEngine 8000 M8, NetEngine 8000E M8, PTN 6900-2-M8A, and PTN 6900-2-M8C supports the following FlexE specifications: Physical port 0 can be added to one group. port-id ranges from 2 to 11 and from 1000 to 3000. A maximum of 10 FlexE clients can be created. Physical port 1 can be added to one group. port-id ranges from 12 to 21 and from 1000 to 3000. A maximum of 10 FlexE clients can be created.

Technical Specifications

Table 1-814 Technical specifications of the CR5DE2NE4X14

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	19.8 mm x 193.8 mm x 209.3 mm (0.78 in. x 7.63 in. x 8.24 in.)
Dimensions with packaging (H x W x D) [mm(in.)]	70 mm x 285 mm x 300 mm (2.76 in. x 11.22 in. x 11.81 in.)
Weight without packaging [kg(lb)]	0.8 kg (1.76 lb)
Weight with packaging [kg(lb)]	1.2 kg (2.65 lb)
Typical power consumption [W]	32.2 W
Typical heat dissipation [BTU/hour]	104.41 BTU/hour

Item	Specification
Long-term operating temperature [°C(°F)]	–5°C to +55°C (23°F to 131°F)

1.4.4.43 CR8D000DE1C2 (16-Port E1 Physical Interface Card(PIC,75ohm/120ohm))

Overview

Table 1-815 Basic information about the CR8D000DE1C2

Item	Details
Description	16-Port E1 Physical Interface Card(PIC, 75ohm/120ohm)
Part Number	03034KEG
Model	CR8D000DE1C2
Card type	Interface Card
Silkscreen	16xE1-75/120

Appearance

Figure 1-166 Appearance of the CR8D000DE1C2



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03034KEG&productModel=CR8D000DE1C2#matchRelation>

Indicators

Table 1-816 Indicators on the CR8D000DE1C2

Silkscreen	Name	Color	Status	Description
STAT	Working status indicator	Green	Steady on	The board is working properly.
		Red	Steady on	The board hardware is faulty.
		Orange	Steady on	An alarm is generated when a high-speed interface card is incorrectly inserted into a low-speed interface card slot, or the board is not loaded with logic.
		-	Steady off	The board is not registered or is not powered on.
SRV	Service status indicator	Green	Steady on	There is no CRC error alarm on the board.
		Orange	Steady on	There is a CRC error alarm on the board.
		-	Steady off	The board is not powered on or is faulty.

Ports

Table 1-817 Ports on the CR8D000DE1C2

Port	Connector Type	Description	Available Components
E1(0-15)	ANEA 96	16 E1 interfaces	75-ohm/120-ohm 16xE1 cable

Table 1-818 Port attributes about the CR8D000DE1C2

Attributes	Details
Working mode	Full-duplex
Standards compliance	IMA 1.1, RFC4385, RFC5086, RFC4553_SATOP, RFC1662
Frame format	Unframed mode, CRC4, and NO-CRC4

Functions and Features

Table 1-819 Functions and features of the CR8D000DE1C2

Functions and Features	Description
Basic function	Supports a maximum of 16 E1 interfaces. Services on each interface can be configured.
Reliability and availability	Hot swappable.
Breakout supported	No
Interface rate auto-sensing supported	No
FlexE supported	No
Link protocol	TDM, PPP, MP, ATM, IMA

Technical Specifications

Table 1-820 Technical specifications of the CR8D000DE1C2

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	19.8 mm x 193.8 mm x 209.3 mm (0.78 in. x 7.63 in. x 8.24 in.)

Item	Specification
Dimensions with packaging (H x W x D) [mm(in.)]	70 mm x 285 mm x 300 mm (2.76 in. x 11.22 in. x 11.81 in.)
Weight without packaging [kg(lb)]	0.52 kg (1.15 lb)
Weight with packaging [kg(lb)]	0.81 kg (1.79 lb)
Typical power consumption [W]	16.78 W
Typical heat dissipation [BTU/hour]	54.45 BTU/hour
Long-term operating temperature [°C(°F)]	–40°C to +65°C (–40°F to +149°F)

1.4.4.44 CR8D00E1KBC0 (1-Port 400GBase-QSFP-DD Physical Interface Card(PIC))

Overview

Table 1-821 Basic information about the CR8D00E1KBC0

Item	Details
Description	1-Port 400GBase-QSFP-DD Physical Interface Card(PIC)
Part Number	03034AAY
Model	CR8D00E1KBC0
Card type	Interface Card
Silkscreen	1x400GE-QSFP-DD

Appearance

Figure 1-167 Appearance of the CR8D00E1KBC0



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03034AAY&productModel=CR8D00E1KBC0#matchRelation>

Indicators and Buttons

Table 1-822 Indicators on the CR8D00E1KBC0

Silkscreen	Name	Color	Status	Description
STAT	Running status indicator	Green	Steady on	The board is working properly.
		Red	Steady on	The hardware on the board is faulty.
		Orange	Steady on	A high-speed interface card is unexpectedly inserted into a low-speed interface card slot and an alarm is generated, or the logic is not loaded to the board.
		-	Off	The board is not powered on or is not registered.
SRV	Service status indicator	Green	Steady on	The board has no CRC alarm.
		Orange	Steady on	The board has CRC alarm.

Silkscreen	Name	Color	Status	Description
		-	Off	The board is not running or no power is input.
L0	Optical Amplifier running status indicator	Green	Steady on	The Optical Amplifier port works normally and there is no LOS.
		-	Off	The Optical Amplifier port does not work normally or LOS occurs.
L/A1	Port running status indicator	Green	Steady on	The link is Up.
		Green	Blink	Data is being transmitted or received.
		-	Off	The link is Down.

Silkscreen	Name	Color	Status	Description
Breakout(0 to 7)	Breakout channel indicator	Green	Steady on	Breakout 0–7: breakout indicators. If breakout indicator X is on, the port indicators are showing the running status of breakout channel X. For example, if breakout indicator 0 is on, port indicators are showing the running status of breakout channel 0 (if a port indicator is blinking, breakout channel 0 on this port is receiving and transmitting data); if breakout indicator 1 is on, port indicators are showing the running status of breakout channel 1. If the breakout indicators 0 to 7 are off, the board has no breakout ports. If the board has breakout ports, each breakout

Silkscreen	Name	Color	Status	Description
				indicator for channels 0 to 7 turns on for 5 seconds in sequence to indicate the status of the corresponding channel. To rapidly identify the running status of a specific breakout channel, press the Breakout button to switch to your desired indicator. For example, if you want to check the running status of breakout channel 1 on each port, press the Breakout button to have breakout indicator 1 turn on.

Table 1-823 Buttons on the CR8D00E1KBC0

Silkscreen	Name	Description
Breakout	Breakout button	When you press the button, the Breakout indicators stop status cycling and remain in the current status. Then, each time you press the button, the Breakout indicators' status switches. If you do not press the button in 60s, the Breakout indicators automatically start status cycling.

Ports

Table 1-824 Ports on the CR8D00E1KBC0

Port	Connector Type	Description	Available Components
OUT0 IN0	QSFP28	Fiber amplifier interface (supported since V800R021C10)	Optical Amplifier Module
OUT1 IN1	QSFP-DD	Inputs and outputs 400GE optical signals.	400Gbps QSFP-DD Optical Module 100Gbps QSFP28 Optical Module

Table 1-825 Port attributes about the CR8D00E1KBC0

Attributes	Details
Working mode	Full-duplex
Standards compliance	IEEE 802.3
Frame format	Ethernet_II, Ethernet_SAP and Ethernet_SNAP

Functions and Features

Table 1-826 Functions and features of the CR8D00E1KBC0

Functions and Features	Description
Basic function	Occupies two flexible card slots. The line side of the board provides one 400GE QSFP-DD interface and one QSFP28 optical amplifier interface. 1. The rate of the QSFP-DD interface is 400 Gbit/s. 2. The QSFP28 fiber amplifier interface supports the fiber amplifier module (reserved). Supports FlexE.
Reliability and availability	Hot swappable.
Restrictions and remarks	1. By default, a QSFP-DD port supports 400G QSFP-DD optical modules. To enable a QSFP-DD port to support 100G QSFP28 optical modules instead, run the port split command in the system view. 2. When the board works with a 400G ZR+ optical module, the short-term operating temperature supported by the optical module ranges from -5°C to +50°C.
Breakout supported	No
Interface rate auto-sensing supported	No
FlexE supported	Yes Ports on the straight-through card can be bonded to form a group. port-id ranges from 1000 to 3000. A maximum of 48 FlexE clients can be created.

Technical Specifications

Table 1-827 Technical specifications of the CR8D00E1KBC0

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	40.14 mm x 193.8 mm x 209.3 mm (1.58 in. x 7.63 in. x 8.24 in.)

Item	Specification
Dimensions with packaging (H x W x D) [mm(in.)]	115 mm x 317.5 mm x 305 mm (4.53 in. x 12.50 in. x 12.01 in.)
Weight without packaging [kg(lb)]	1.1 kg (2.43 lb)
Weight with packaging [kg(lb)]	1.7 kg (3.75 lb)
Typical power consumption [W]	69.8 W
Typical heat dissipation [BTU/hour]	226.34 BTU/hour
Long-term operating temperature [°C(°F)]	–5°C to +55°C (23°F to 131°F)

1.4.4.45 CR8D00E1NBC2 (03034KEF: 2-Port 50GBase/1-Port 100GBase-QSFP28 FlexE MACsec Physical Interface Card(PIC))

Overview

Table 1-828 Basic information about the CR8D00E1NBC2

Item	Details
Description	2-Port 50GBase/1-Port 100GBase-QSFP28 FlexE MACsec Physical Interface Card(PIC)
Part Number	03034KEF
Model	CR8D00E1NBC2
Card type	Interface Card
Silkscreen	2x50GE-QSFP28

Appearance

Figure 1-168 Appearance of the CR8D00E1NBC2



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03034KEF&productModel=CR8D00E1NBC2#matchRelation>

Indicators

Table 1-829 Indicators on the CR8D00E1NBC2

Silkscreen	Name	Color	Status	Description
STAT	Running status indicator	Green	Steady on	The board is working properly.
		Red	Steady on	The board hardware is faulty.
		Orange	Steady on	An alarm is generated when a high-speed interface card is incorrectly inserted into a low-speed interface card slot, or the board is not loaded with logic.
		-	Steady off	The board is not registered or is not powered on.
SRV	Service status indicator	Green	Steady on	There is no CRC error alarm on the board.

Silkscreen	Name	Color	Status	Description
		Orange	Steady on	There is a CRC error alarm on the board.
		-	Off	The board is not powered on or is faulty.
L/A (0-1)	Interface status indicator	Green	Steady on	The link is up.
		Green	Blinking	Data is being transmitted or received.
		-	Steady off	The link is down.
BREAKOUT (0-3)	Breakout channel indicator	Green	Blinking	Each indicator for channels 0, 1, 2, and 3 turns on for 5s in sequence to indicate the status of the corresponding channel. This process repeats.
		-	Off	No data is transmitting or receiving in this channel.

Ports

Table 1-830 Ports on the CR8D00E1NBC2

Port	Connector Type	Description	Available Components
OUT0 IN0-OUT1 IN1	QSFP28	Interface for inputting and outputting 50GE/100GE optical signals	100Gbps QSFP28 Optical Module 40Gbps QSFP+ Optical Module 50Gbps QSFP28 BIDI Optical Module 50Gbps QSFP28 Optical Module Optical Amplifier Module

Table 1-831 Port attributes about the CR8D00E1NBC2

Attributes	Details
Working mode	Full-duplex
Standards compliance	IEEE 802.3
Frame format	Ethernet_II, Ethernet_SAP, Ethernet_SNAP

Functions and Features

Table 1-832 Functions and features of the CR8D00E1NBC2

Functions and Features	Description
Basic function	Occupies one flexible interface card slot. Provides two 50GE interfaces or one 100GE interface for line-rate transmitting and receiving, and supports QSFP28 optical modules. Supports FlexE and MACsec.
Reliability and availability	Hot swappable.

Functions and Features	Description
Restrictions and remarks	Port 0 supports 100GE, 50GE, 40GE, 4x25GE breakout, and 4x10GE breakout. Port 1 is valid only when port 0 is 50GE. If the switch-mode command is set to auto and port 0 has a 100GE optical module or a 40GE optical module installed, the port automatically works in 100GE/40GE mode. In this case, port 0 takes effect, but port 1 does not take effect. If a 50GE optical module is inserted into port 0, ports 0 and 1 are both valid, and port 0 automatically works in 50GE mode. If port 1 works in 50GE mode and services are running on port 1, port 0 does not support auto-sensing.
Breakout supported	Yes. To enable the breakout function on a port, run the port split command.
Interface rate auto-sensing supported	Supports 40GE/50GE/100GE auto-sensing.
FlexE supported	Yes Physical ports 0 and 1 can be added to the same group. port-id ranges from 129 to 148 and from 1000 to 3000. A maximum of 20 FlexE clients can be created.

Technical Specifications

Table 1-833 Technical specifications of the CR8D00E1NBC2

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	19.8 mm x 193.8 mm x 209.3 mm (0.78 in. x 7.63 in. x 8.24 in.)
Dimensions with packaging (H x W x D) [mm(in.)]	70 mm x 285 mm x 300 mm (2.76 in. x 11.22 in. x 11.81 in.)
Weight without packaging [kg(lb)]	0.8 kg (1.76 lb)
Weight with packaging [kg(lb)]	1.2 kg (2.65 lb)
Typical power consumption [W]	28.8 W

Item	Specification
Typical heat dissipation [BTU/hour]	93.47 BTU/hour
Long-term operating temperature [°C(°F)]	-5°C to +55°C (23°F to 131°F)

1.4.4.46 CR8D00E1NBC2 (03035JVJ): 2-Port 50GBase/1-Port 100GBase-QSFP28 FlexE MACsec Physical Interface Card(PIC) for SME

Overview

Table 1-834 Basic information about the CR8D00E1NBC2

Item	Details
Description	2-Port 50GBase/1-Port 100GBase-QSFP28 FlexE MACsec Physical Interface Card(PIC) for SME
Part Number	03035JVJ
Model	CR8D00E1NBC2
Card type	Interface Card
Silkscreen	2x50GE-QSFP28

Appearance

Figure 1-169 Appearance of the CR8D00E1NBC2



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03035JVJ&productModel=CR8D00E1NBC2#matchRelation>

Indicators

Table 1-835 Indicators on the CR8D00E1NBC2

Silkscreen	Name	Color	Status	Description
STAT	Running status indicator	Green	Steady on	The board is working properly.
		Red	Steady on	The hardware on the board is faulty.
		Orange	Steady on	An alarm is generated when a high-speed interface card is incorrectly inserted into a low-speed interface card slot, or the board is not loaded with logic.
		-	Steady off	The board is not powered on or is not registered.
SRV	Service status indicator	Green	Steady on	The board has no CRC alarm.
		Orange	Steady on	The board has CRC alarms.
		-	Off	The board is not powered on or is faulty.
L/A (0-1)	Interface status indicator	Green	Steady on	The link is Up.
		Green	Blinking	Data is being transmitted and received.
		-	Steady off	The link is Down.

Silkscreen	Name	Color	Status	Description
BREAKOUT (0–3)	Breakout channel indicator	Green	Blinking	Each indicator for channels 0, 1, 2, and 3 turns on for 5s in sequence to indicate the status of the corresponding channel. This process repeats.
		-	Off	No data is transmitting or receiving in this channel.

Ports

Table 1-836 Ports on the CR8D00E1NBC2

Port	Connector Type	Description	Available Components
OUT0 IN0-OUT1 IN1	QSFP28	Interface for inputting and outputting 50GE/ 100GE optical signals	100Gbps QSFP28 Optical Module 40Gbps QSFP+ Optical Module 50Gbps QSFP28 BIDI Optical Module 50Gbps QSFP28 Optical Module Optical Amplifier Module

Table 1-837 Port attributes about the CR8D00E1NBC2

Attributes	Details
Working mode	Full-duplex
Standards compliance	IEEE 802.3

Attributes	Details
Frame format	Ethernet_II, Ethernet_SAP, Ethernet_SNAP

Functions and Features

Table 1-838 Functions and features of the CR8D00E1NBC2

Functions and Features	Description
Basic function	Occupies one flexible interface card slot. Provides two 50GE interfaces or one 100GE interface for line-rate transmitting and receiving, and supports QSFP28 optical modules. Supports FlexE and MACsec.
Reliability and availability	Hot swappable
Restrictions and remarks	Port 0 supports 100GE, 50GE, 40GE, 4x25GE breakout, and 4x10GE breakout. Port 1 is valid only when port 0 is 50GE. If the switch-mode command is set to auto and port 0 has a 100GE optical module or a 40GE optical module installed, the port automatically works in 100GE/40GE mode. In this case, port 0 takes effect, but port 1 does not take effect. If a 50GE optical module is inserted into port 0, ports 0 and 1 are both valid, and port 0 automatically works in 50GE mode. If port 1 works in 50GE mode and services are running on port 1, port 0 does not support auto-sensing.
Breakout supported	Yes. To enable the breakout function on a port, run the port split command.
Interface rate auto-sensing supported	Supports 40GE/50GE/100GE auto-sensing.

Functions and Features	Description
FlexE supported	Yes Physical ports 0 and 1 can be added to the same group. port-id ranges from 129 to 148 and from 1000 to 3000. A maximum of 20 FlexE clients can be created.

Technical Specifications

Table 1-839 Technical specifications of the CR8D00E1NBC2

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	19.8 mm x 193.8 mm x 209.3 mm (0.78 in. x 7.63 in. x 8.24 in.)
Dimensions with packaging (H x W x D) [mm(in.)]	70 mm x 285 mm x 300 mm (2.76 in. x 11.22 in. x 11.81 in.)
Weight without packaging [kg(lb)]	0.8 kg (1.76 lb)
Weight with packaging [kg(lb)]	1.2 kg (2.65 lb)
Typical power consumption [W]	28.8 W
Typical heat dissipation [BTU/hour]	93.47 BTU/hour
Long-term operating temperature [°C(°F)]	-5°C to +55°C (23°F to 131°F)

1.4.4.47 CR8D00E1NCC1 (1-Port 100GE/ 1-Port 200GE CFP2 Physical Interface Card(PIC) (Enable 1*100GE by default))

Overview

Table 1-840 Basic information about the CR8D00E1NCC1

Item	Details
Description	1-Port 100GE/ 1-Port 200GE CFP2 Physical Interface Card(PIC) (Enable 1*100GE by default)
Part Number	03034REK
Model	CR8D00E1NCC1

Item	Details
Card type	Interface Card
Silkscreen	1x100/200GE-CFP2

Appearance

Figure 1-170 Appearance of the CR8D00E1NCC1



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03034REK&productModel=CR8D00E1NCC1#matchRelation>

Indicators

Table 1-841 Indicators on the CR8D00E1NCC1

Silkscreen	Name	Color	Status	Description
STAT	Working status indicator	Green	Steady on	The board is working properly.
		Red	Steady on	The board hardware is faulty.

Silkscreen	Name	Color	Status	Description
		Orange	Steady on	An alarm is generated when a high-speed interface card is incorrectly inserted into a low-speed interface card slot, or the board is not loaded with logic.
		-	Off	The board is not registered or is not powered on.
SRV	Service status indicator	Green	Steady on	There is no CRC error alarm on the board.
		Orange	Steady on	There is a CRC error alarm on the board.
		-	Off	The board is not powered on or is faulty.
L/A 0	Running status indicator	Green	Steady on	The link is up.
		Green	Blinking	Data is being transmitted or received.
		-	Off	The link is down.

Ports

Table 1-842 Ports on the CR8D00E1NCC1

Port	Connector Type	Description	Available Components
OUT0 IN0	CFP2	Interfaces for 1-channel CFP2 optical signal input and output	100Gbps CFP2 Optical Module 100Gbps CFP2 DWDM Optical Module

Table 1-843 Port attributes about the CR8D00E1NCC1

Attributes	Details
Working mode	Full-duplex
Standards compliance	IEEE 802.3
Frame format	Ethernet_II, Ethernet_SAP, Ethernet_SNAP

Functions and Features

Table 1-844 Functions and features of the CR8D00E1NCC1

Functions and Features	Description
Basic function	Provides one 100GEBase CFP2 interface. The 200GE interface capability is reserved on the hardware. MACsec is supported.
Reliability and availability	Hot swappable.
Breakout supported	No
Interface rate auto-sensing supported	No
FlexE supported	No

Technical Specifications

Table 1-845 Technical specifications of the CR8D00E1NCC1

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	19.8 mm x 193.8 mm x 209.3 mm (0.78 in. x 7.63 in. x 8.24 in.)
Dimensions with packaging (H x W x D) [mm(in.)]	70 mm x 285 mm x 300 mm (2.76 in. x 11.22 in. x 11.81 in.)
Weight without packaging [kg(lb)]	0.8 kg (1.76 lb)
Weight with packaging [kg(lb)]	1.2 kg (2.65 lb)
Typical power consumption [W]	43.2 W
Typical heat dissipation [BTU/hour]	140.16 BTU/hour
Long-term operating temperature [°C(°F)]	-5°C to +50°C (23°F to 122°F)

1.4.4.48 CR8D00E2NBC4 (03034KEN: 2-Port 100GBase/50GBase-QSFP28 FlexE MACsec Physical Interface Card(PIC))

Overview

Table 1-846 Basic information about the CR8D00E2NBC4

Item	Details
Description	2-Port 100GBase/50GBase-QSFP28 FlexE MACsec Physical Interface Card(PIC)
Part Number	03034KEN
Model	CR8D00E2NBC4
Card type	Interface Card
Silkscreen	2x100GE-QSFP28

Appearance

Figure 1-171 Appearance of the CR8D00E2NBC4



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03034KEN&productModel=CR8D00E2NBC4#matchRelation>

Indicators

Table 1-847 Indicators on the CR8D00E2NBC4

Silkscreen	Name	Color	Status	Description
STAT	Running status indicator	Green	Steady on	The board is working properly.
		Red	Steady on	The board hardware is faulty.

Silkscreen	Name	Color	Status	Description
		Orange	Steady on	An alarm is generated when a high-speed interface card is incorrectly inserted into a low-speed interface card slot, or the board is not loaded with logic.
		-	Steady off	The board is not registered or is not powered on.
SRV	Service status indicator	Green	Steady on	There is no CRC error alarm on the board.
		Orange	Steady on	There is a CRC error alarm on the board.
		-	Off	The board is not powered on or is faulty.
L/A (0-1)	Interface status indicator	Green	Steady on	The link is up.
		Green	Blinking	Data is being transmitted or received.
		-	Steady off	The link is down.

Silkscreen	Name	Color	Status	Description
BREAKOUT (0–3)	Breakout channel indicator	Green	Blinking	Each indicator for channels 0, 1, 2, and 3 turns on for 5s in sequence to indicate the status of the corresponding channel. This process repeats.
		-	Steady off	No data is transmitting or receiving in this channel.

Ports

Table 1-848 Ports on the CR8D00E2NBC4

Port	Connector Type	Description	Available Components
OUT0 IN0-OUT1 IN1	QSFP28	Interface for inputting and outputting 50GE/ 100GE optical signals	100Gbps QSFP28 Optical Module 40Gbps QSFP+ Optical Module 50Gbps QSFP28 BIDI Optical Module 50Gbps QSFP28 Optical Module Optical Amplifier Module

Table 1-849 Port attributes about the CR8D00E2NBC4

Attributes	Details
Working mode	Full-duplex
Standards compliance	IEEE 802.3

Attributes	Details
Frame format	Ethernet_II, Ethernet_SAP, Ethernet_SNAP

Functions and Features

Table 1-850 Functions and features of the CR8D00E2NBC4

Functions and Features	Description
Basic function	Occupies one flexible interface card slot. Provides two 100GE/50GE interfaces for line-rate transmitting and receiving, and supports QSFP28 optical modules. FlexE and MACsec are supported.
Reliability and availability	Hot swappable.
Restrictions and remarks	Supports 100GE, 50GE, 40GE, 4x25GE breakout, and 4x10GE breakout.
Breakout supported	Yes. To enable the breakout function on a port, run the port split command.
Interface rate auto-sensing supported	Supports 40GE/50GE/100GE auto-sensing.
FlexE supported	Yes Physical ports 0 and 1 can be added to the same group. port-id ranges from 2 to 21 and from 1000 to 3000. A maximum of 20 FlexE clients can be created. Specifically, the interface card on the NetEngine 8000E M8 and NetEngine 8000 M8 supports the following FlexE specifications: Physical port 0 can be added to one group. port-id ranges from 2 to 11 and from 1000 to 3000. A maximum of 10 FlexE clients can be created. Physical port 1 can be added to the same group. port-id ranges from 12 to 21 and from 1000 to 3000. A maximum of 10 FlexE clients can be created.

Technical Specifications

Table 1-851 Technical specifications of the CR8D00E2NBC4

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	19.8 mm x 193.8 mm x 209.3 mm (0.78 in. x 7.63 in. x 8.24 in.)
Dimensions with packaging (H x W x D) [mm(in.)]	70 mm x 285 mm x 300 mm (2.76 in. x 11.22 in. x 11.81 in.)
Weight without packaging [kg(lb)]	0.8 kg (1.76 lb)
Weight with packaging [kg(lb)]	1.2 kg (2.65 lb)
Typical power consumption [W]	32.2 W
Typical heat dissipation [BTU/hour]	104.41 BTU/hour
Long-term operating temperature [°C(°F)]	–5°C to +55°C (23°F to 131°F)

1.4.4.49 CR8D00E2NBC4 (03035JVF: 2-Port 100GBase/50GBase-QSFP28 FlexE MACsec Physical Interface Card(PIC) for SME)

Overview

Table 1-852 Basic information about the CR8D00E2NBC4

Item	Details
Description	2-Port 100GBase/50GBase-QSFP28 FlexE MACsec Physical Interface Card(PIC) for SME
Part Number	03035JVF
Model	CR8D00E2NBC4
Card type	Interface Card
Silkscreen	2x100GE-QSFP28

Appearance

Figure 1-172 Appearance of the CR8D00E2NBC4



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03035JVF&productModel=CR8D00E2NBC4#matchRelation>

Indicators

Table 1-853 Indicators on the CR8D00E2NBC4

Silkscreen	Name	Color	Status	Description
STAT	Running status indicator	Green	Steady on	The board is working properly.
		Red	Steady on	The hardware on the board is faulty.

Silkscreen	Name	Color	Status	Description
		Orange	Steady on	An alarm is generated when a high-speed interface card is incorrectly inserted into a low-speed interface card slot, or the board is not loaded with logic.
		-	Steady off	The board is not powered on or is not registered.
SRV	Service status indicator	Green	Steady on	The board has no CRC alarm.
		Orange	Steady on	The board has CRC alarms.
		-	Off	The board is not powered on or is faulty.
L/A (0-1)	Interface status indicator	Green	Steady on	The link is Up.
		Green	Blinking	Data is being transmitted and received.
		-	Steady off	The link is Down.
BREAKOUT (0-3)	Breakout channel indicator	Green	Blinking	Each indicator for channels 0, 1, 2, and 3 turns on for 5s in sequence to indicate the status of the corresponding channel. This process repeats.

Silkscreen	Name	Color	Status	Description
		-	Steady off	No data is transmitting or receiving in this channel.

Ports

Table 1-854 Ports on the CR8D00E2NBC4

Port	Connector Type	Description	Available Components
OUT0 IN0-OUT1 IN1	QSFP28	Interface for inputting and outputting 50GE/100GE optical signals	100Gbps QSFP28 Optical Module 40Gbps QSFP+ Optical Module 50Gbps QSFP28 BIDI Optical Module 50Gbps QSFP28 Optical Module Optical Amplifier Module

Table 1-855 Port attributes about the CR8D00E2NBC4

Attributes	Details
Working mode	Full-duplex
Standards compliance	IEEE 802.3
Frame format	Ethernet_II, Ethernet_SAP, Ethernet_SNAP

Functions and Features

Table 1-856 Functions and features of the CR8D00E2NBC4

Functions and Features	Description
Basic function	Occupies one flexible interface card slot. Provides two 100GE/50GE interfaces for line-rate transmitting and receiving, and supports QSFP28 optical modules. FlexE and MACsec are supported.
Reliability and availability	Hot swappable
Restrictions and remarks	100GE, 50GE, 40GE, 4*25GE, and 4*10GE interfaces are supported.
Breakout supported	Yes. To enable the breakout function on a port, run the port split command.
Interface rate auto-sensing supported	Supports 40GE/50GE/100GE auto-sensing.
FlexE supported	Yes Physical ports 0 and 1 can be added to the same group. port-id ranges from 2 to 21 and from 1000 to 3000. A maximum of 20 FlexE clients can be created. Specifically, the interface card on the NetEngine 8000E M8 and NetEngine 8000 M8 supports the following FlexE specifications: Physical port 0 can be added to one group. port-id ranges from 2 to 11 and from 1000 to 3000. A maximum of 10 FlexE clients can be created. Physical port 1 can be added to the same group. port-id ranges from 12 to 21 and from 1000 to 3000. A maximum of 10 FlexE clients can be created.

Technical Specifications

Table 1-857 Technical specifications of the CR8D00E2NBC4

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	19.8 mm x 193.8 mm x 209.3 mm (0.78 in. x 7.63 in. x 8.24 in.)
Dimensions with packaging (H x W x D) [mm(in.)]	70 mm x 285 mm x 300 mm (2.76 in. x 11.22 in. x 11.81 in.)
Weight without packaging [kg(lb)]	0.8 kg (1.76 lb)
Weight with packaging [kg(lb)]	1.2 kg (2.65 lb)
Typical power consumption [W]	32.2 W
Typical heat dissipation [BTU/hour]	104.41 BTU/hour
Long-term operating temperature [°C(°F)]	–5°C to +55°C (23°F to 131°F)

1.4.4.50 CR8D00E4XFC1 (03034EDU: 4-Port 10GE-SFP+ FlexE MACsec Physical Interface Card(PIC))

Overview

Table 1-858 Basic information about the CR8D00E4XFC1

Item	Details
Description	4-Port 10GE-SFP+ FlexE MACsec Physical Interface Card(PIC)
Part Number	03034EDU
Model	CR8D00E4XFC1
Card type	Interface Card
Silkscreen	4x10GE-SFP+

Appearance

Figure 1-173 Appearance of the CR8D00E4XFC1



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03034EDU&productModel=CR8D00E4XFC1#matchRelation>

Indicators

Table 1-859 Indicators on the CR8D00E4XFC1

Silkscreen	Name	Color	Status	Description
STAT	Working status indicator	Green	Steady on	The board is working properly.
		Red	Steady on	The hardware on the board is faulty.
		Orange	Steady on	A high-speed interface card is unexpectedly inserted into a low-speed interface card slot and an alarm is generated, or the logic is not loaded to the board.
		-	Off	The board is not powered on or is not registered.
SRV	Service Status Indicator	Green	Steady on	The board has no CRC alarm.
		Orange	Steady on	The board has CRC alarm.

Silkscreen	Name	Color	Status	Description
		-	Off	The board is not running or no power is input.
L/A (0-3)	Interface status indicator	Green	Steady on	The link is Up.
		Green	Blinking	Data is being transmitted or received.
		-	Off	The link is Down.

Ports

Table 1-860 Ports on the CR8D00E4XFC1

Port	Connector Type	Description	Available Components
OUT0 IN0-OUT3 IN3	SFP+	10GE/GE service signal input and output interface	155Mbps eSFP Optical Module 155Mbps eSFP BIDI Optical Module 1.25/9.953/10.31 25Gbps SFP+ Optical Module 1.25Gbps eSFP BIDI Optical Module 1.25Gbps eSFP CWDM Optical Module 1.25Gbps eSFP Optical Module 10Gbps SFP+ BIDI Optical Module 10Gbps SFP+ CWDM Optical Module 10Gbps SFP+ Optical Module 125M~2.67Gbps eSFP DWDM Optical Module 1Gbps SyncE and 1588V2 Electrical Module 1Gbps Electrical Module

Table 1-861 Port attributes about the CR8D00E4XFC1

Attributes	Details
Working mode	Full-duplex
Standards compliance	IEEE 802.3

Attributes	Details
Frame format	Ethernet_II, Ethernet_SAP and Ethernet_SNAP

Functions and Features

Table 1-862 Functions and features of the CR8D00E4XFC1

Functions and Features	Description
Basic function	Occupies one flexible interface card slot. Provides four 10GE/GE interfaces for line-rate transmitting and receiving, and supports SFP+/SFP optical modules. Supports FlexE and MACsec.
Reliability and availability	Hot swappable.
Restrictions and remarks	1. By default, a port works in standard Ethernet mode. You can run the flexe enable port port-id command in the system view to switch the port to the FlexE mode. An interface working in GE mode cannot be switched to the FlexE mode. 2. A port working in FlexE mode cannot connect to a standard 10GE port. 3. This FlexE mode supports only GE-granularity hard pipes and interconnection with Huawei FlexE interfaces with the same GE granularity.
Breakout supported	No
Interface rate auto-sensing supported	Supports GE/10GE auto-sensing.

Functions and Features	Description
FlexE supported	Yes Physical port 0 can be added to one group, and port-id ranges from 1000 to 3000. A maximum of six FlexE clients can be created. Physical port 1 can be added to one group, and port-id ranges from 1000 to 3000. A maximum of six FlexE clients can be created. Physical port 2 can be added to one group, and port-id ranges from 1000 to 3000. A maximum of six FlexE clients can be created. Physical port 3 can be added to one group, and port-id ranges from 1000 to 3000. A maximum of six FlexE clients can be created.

Technical Specifications

Table 1-863 Technical specifications of the CR8D00E4XFC1

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	19.8 mm x 193.8 mm x 209.3 mm (0.78 in. x 7.63 in. x 8.24 in.)
Dimensions with packaging (H x W x D) [mm(in.)]	70 mm x 285 mm x 300 mm (2.76 in. x 11.22 in. x 11.81 in.)
Weight without packaging [kg(lb)]	0.7 kg (1.54 lb)
Weight with packaging [kg(lb)]	1.1 kg (2.43 lb)
Typical power consumption [W]	17.4 W
Typical heat dissipation [BTU/hour]	56.40 BTU/hour
Long-term operating temperature [°C(°F)]	–40°C to +65°C (–40°F to +149°F)

1.4.4.51 CR8D00E4XFC1 (03034EDU-001: 4-Port 10GE-SFP+ FlexE MACsec Physical Interface Card(PIC))

Overview

Table 1-864 Basic information about the CR8D00E4XFC1

Item	Details
Description	4-Port 10GE-SFP+ FlexE MACsec Physical Interface Card(PIC)
Part Number	03034EDU-001
Model	CR8D00E4XFC1
Card type	Interface Card
Silkscreen	4x10GE-SFP+

Appearance

Figure 1-174 Appearance of the CR8D00E4XFC1



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03034EDU-001&productModel=CR8D00E4XFC1#matchRelation>

Indicators

Table 1-865 Indicators on the CR8D00E4XFC1

Silkscreen	Name	Color	Status	Description
STAT	Working status indicator	Green	Steady on	The board is working properly.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	The hardware on the board is faulty.
		Orange	Steady on	A high-speed interface card is unexpectedly inserted into a low-speed interface card slot and an alarm is generated, or the logic is not loaded to the board.
		-	Off	The board is not powered on or is not registered.
SRV	Service Status Indicator	Green	Steady on	The board has no CRC alarm.
		Orange	Steady on	The board has CRC alarm.
		-	Off	The board is not running or no power is input.
L/A (0-3)	Interface status indicator	Green	Steady on	The link is Up.
		Green	Blinking	Data is being transmitted or received.
		-	Off	The link is Down.

Ports

Table 1-866 Ports on the CR8D00E4XFC1

Port	Connector Type	Description	Available Components
OUT0 IN0-OUT3 IN3	SFP+	10GE/GE service signal input and output interface	155Mbps eSFP Optical Module 155Mbps eSFP BIDI Optical Module 1.25/9.953/10.31 25Gbps SFP+ Optical Module 1.25Gbps eSFP BIDI Optical Module 1.25Gbps eSFP CWDM Optical Module 1.25Gbps eSFP Optical Module 10Gbps SFP+ BIDI Optical Module 10Gbps SFP+ CWDM Optical Module 10Gbps SFP+ Optical Module 125M~2.67Gbps eSFP DWDM Optical Module 1Gbps SyncE and 1588V2 Electrical Module 1Gbps Electrical Module

Table 1-867 Port attributes about the CR8D00E4XFC1

Attributes	Details
Working mode	Full-duplex
Standards compliance	IEEE 802.3

Attributes	Details
Frame format	Ethernet_II, Ethernet_SAP and Ethernet_SNAP

Functions and Features

Table 1-868 Functions and features of the CR8D00E4XFC1

Functions and Features	Description
Basic function	Occupies one flexible interface card slot. Provides four 10GE/GE interfaces for line-rate transmitting and receiving, and supports SFP+/SFP optical modules. Supports FlexE and MACsec.
Reliability and availability	Hot swappable.
Restrictions and remarks	1. By default, a port works in standard Ethernet mode. You can run the flexe enable port port-id command in the system view to switch the port to the FlexE mode. An interface working in GE mode cannot be switched to the FlexE mode. 2. A port working in FlexE mode cannot connect to a standard 10GE port. 3. This FlexE mode supports only GE-granularity hard pipes and interconnection with Huawei FlexE interfaces with the same GE granularity.
Breakout supported	No
Interface rate auto-sensing supported	Supports GE/10GE auto-sensing.

Functions and Features	Description
FlexE supported	Yes Physical port 0 can be added to one group, and port-id ranges from 1000 to 3000. A maximum of six FlexE clients can be created. Physical port 1 can be added to one group, and port-id ranges from 1000 to 3000. A maximum of six FlexE clients can be created. Physical port 2 can be added to one group, and port-id ranges from 1000 to 3000. A maximum of six FlexE clients can be created. Physical port 3 can be added to one group, and port-id ranges from 1000 to 3000. A maximum of six FlexE clients can be created.

Technical Specifications

Table 1-869 Technical specifications of the CR8D00E4XFC1

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	19.8 mm x 193.8 mm x 209.3 mm (0.78 in. x 7.63 in. x 8.24 in.)
Dimensions with packaging (H x W x D) [mm(in.)]	70 mm x 285 mm x 300 mm (2.76 in. x 11.22 in. x 11.81 in.)
Weight without packaging [kg(lb)]	0.7 kg (1.54 lb)
Weight with packaging [kg(lb)]	1.1 kg (2.43 lb)
Typical power consumption [W]	17.4 W
Typical heat dissipation [BTU/hour]	56.40 BTU/hour
Long-term operating temperature [°C(°F)]	–40°C to +65°C (–40°F to +149°F)

1.4.4.52 CR8D00E4XFC3 (4-Port 10GE(SFP+)/GE(SFP) MACsec Physical Interface Card(PIC))

Overview

Table 1-870 Basic information about the CR8D00E4XFC3

Item	Details
Description	4-Port 10GE(SFP+)/GE(SFP) MACsec Physical Interface Card(PIC)
Part Number	03034PJJ
Model	CR8D00E4XFC3
Card type	Interface Card
Silkscreen	4x10GE-SFP+

Appearance

Figure 1-175 Appearance of the CR8D00E4XFC3



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03034PJJ&productModel=CR8D00E4XFC3#matchRelation>

Indicators

Table 1-871 Indicators on the CR8D00E4XFC3

Silkscreen	Name	Color	Status	Description
STAT	Working status indicator	Green	Steady on	The board is working properly.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	The board hardware is faulty.
		Orange	Steady on	An alarm is generated when a high-speed interface card is incorrectly inserted into a low-speed interface card slot, or the board is not loaded with logic.
		-	Steady off	The board is not registered or is not powered on.
SRV	Service status indicator	Green	Steady on	There is no CRC error alarm on the board.
		Orange	Steady on	There is a CRC error alarm on the board.
		-	Steady off	The board is not powered on or is faulty.
L/A (0-3)	Running status indicator	Green	Steady on	The link is up.
		Green	Blinking	Data is being transmitted or received.
		-	Steady off	The link is down.

Ports

Table 1-872 Ports on the CR8D00E4XFC3

Port	Connector Type	Description	Available Components
OUT0 IN0-OUT3 IN3	SFP+	Interfaces for 4-channel SFP+/SFP optical signal input and output	1.25/9.953/10.31 25Gbps SFP+ Optical Module 1.25Gbps eSFP BIDI Optical Module 1.25Gbps eSFP CWDM Optical Module 1.25Gbps eSFP Optical Module 10Gbps SFP+ BIDI Optical Module 10Gbps SFP+ CWDM Optical Module 10Gbps SFP+ Optical Module 125M~2.67Gbps eSFP DWDM Optical Module 1Gbps Electrical Module

Table 1-873 Port attributes about the CR8D00E4XFC3

Attributes	Details
Working mode	Full-duplex
Standards compliance	IEEE 802.1AE MACsec
Frame format	Ethernet_II, Ethernet_SAP, Ethernet_SNAP

Functions and Features

Table 1-874 Functions and features of the CR8D00E4XFC3

Functions and Features	Description
Basic function	Occupies one flexible interface card slot. Provides four 10GE/GE interfaces for line-rate transmitting and receiving, and supports SFP+/SFP optical modules. Supports MACsec.
Reliability and availability	Hot swappable.
Restrictions and remarks	Ports 0 to 3 belong to a port group. In LAN/WAN switching, the mode of ports in a port group must be switched at the same time.
Breakout supported	No
Interface rate auto-sensing supported	Supports GE/10GE auto-sensing.
FlexE supported	No

Technical Specifications

Table 1-875 Technical specifications of the CR8D00E4XFC3

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	19.8 mm x 193.8 mm x 209.3 mm (0.78 in. x 7.63 in. x 8.24 in.)
Dimensions with packaging (H x W x D) [mm(in.)]	70 mm x 285 mm x 300 mm (2.76 in. x 11.22 in. x 11.81 in.)
Weight without packaging [kg(lb)]	0.7 kg (1.54 lb)
Weight with packaging [kg(lb)]	1.1 kg (2.43 lb)
Typical power consumption [W]	23.9 W
Typical heat dissipation [BTU/hour]	77.38 BTU/hour
Long-term operating temperature [°C(°F)]	–40°C to +65°C (–40°F to +149°F)

1.4.4.53 CR8D00E4XFC4 (4-Port 10GE-SFP+ FlexE MACsec Physical Interface Card(PIC))

Overview

Table 1-876 Basic information about the CR8D00E4XFC4

Item	Details
Description	4-Port 10GE-SFP+ FlexE MACsec Physical Interface Card(PIC)
Part Number	03034SUQ
Model	CR8D00E4XFC4
Card type	Interface Card
Silkscreen	4x10GE-SFP+

Appearance

Figure 1-176 Appearance of the CR8D00E4XFC4



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03034SUQ&productModel=CR8D00E4XFC4#matchRelation>

Indicators

Table 1-877 Indicators on the CR8D00E4XFC4

Silkscreen	Name	Color	Status	Description
STAT	Running status indicator	Green	Steady on	The board is working properly.
		Red	Steady on	The board hardware is faulty.
		Orange	Steady on	An alarm is generated when a high-speed interface card is incorrectly inserted into a low-speed interface card slot, or the board is not loaded with logic.
		-	Off	The board is not registered or is not powered on.
SRV	Service status indicator	Green	Steady on	There is no CRC error alarm on the board.
		Orange	Steady on	There is a CRC error alarm on the board.
		-	Off	The board is not powered on or is faulty.
L/A (0-3)	Interface status indicator	Green	Steady on	The link is up.
		Green	Blinking	Data is being transmitted or received.
		-	Off	The link is down.

Ports

Table 1-878 Ports on the CR8D00E4XFC4

Port	Connector Type	Description	Available Components
OUT0 IN0-OUT3 IN3	SFP+	Interface for inputting and outputting 10GE/GE optical signals	155Mbps eSFP Optical Module 155Mbps eSFP BIDI Optical Module 1.25/9.953/10.31 25Gbps SFP+ Optical Module 1.25Gbps eSFP BIDI Optical Module 1.25Gbps eSFP CWDM Optical Module 1.25Gbps eSFP Optical Module 10Gbps SFP+ BIDI Optical Module 10Gbps SFP+ CWDM Optical Module 10Gbps SFP+ Optical Module 125M~2.67Gbps eSFP DWDM Optical Module 1Gbps SyncE and 1588V2 Electrical Module 1Gbps Electrical Module

Table 1-879 Port attributes about the CR8D00E4XFC4

Attributes	Details
Working mode	Full-duplex

Attributes	Details
Standards compliance	IEEE 802.3
Frame format	Ethernet_II, Ethernet_SAP, Ethernet_SNAP

Functions and Features

Table 1-880 Functions and features of the CR8D00E4XFC4

Functions and Features	Description
Basic function	Occupies one flexible interface card slot. Provides four 10GE/GE interfaces for line-rate transmitting and receiving, and supports SFP+/SFP optical modules. Supports FlexE and MACsec.
Reliability and availability	Hot swappable.
Restrictions and remarks	1. By default, an interface works in standard Ethernet mode. To change the interface mode to FlexE, run the flexe enable port port-id command in the system view. An interface working in GE mode cannot be switched to the FlexE mode. 2. An interface in FlexE mode cannot interconnect with a standard 10GE interface. 3. This FlexE mode supports only GE-granularity hard pipes and can be interconnected only with Huawei's GE-granularity FlexE interfaces that use the same protocol.
Breakout supported	No
Interface rate auto-sensing supported	GE/10GE auto-sensing

Functions and Features	Description
FlexE supported	Yes Physical port 0 can be added to one group, and port-id ranges from 1000 to 3000. A maximum of six FlexE clients can be created. Physical port 1 can be added to one group, and port-id ranges from 1000 to 3000. A maximum of six FlexE clients can be created. Physical port 2 can be added to one group, and port-id ranges from 1000 to 3000. A maximum of six FlexE clients can be created. Physical port 3 can be added to one group, and port-id ranges from 1000 to 3000. A maximum of six FlexE clients can be created.

Technical Specifications

Table 1-881 Technical specifications of the CR8D00E4XFC4

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	19.8 mm x 193.8 mm x 209.3 mm (0.78 in. x 7.63 in. x 8.24 in.)
Dimensions with packaging (H x W x D) [mm(in.)]	70 mm x 285 mm x 300 mm (2.76 in. x 11.22 in. x 11.81 in.)
Weight without packaging [kg(lb)]	0.7 kg (1.54 lb)
Weight with packaging [kg(lb)]	1.1 kg (2.43 lb)
Typical power consumption [W]	17.4 W
Typical heat dissipation [BTU/hour]	56.40 BTU/hour
Long-term operating temperature [°C(°F)]	–40°C to +65°C (–40°F to +149°F)

1.4.4.54 CR8D00E4YBC1 (4-Port 25GE-SFP28 Physical Interface Card (PIC))

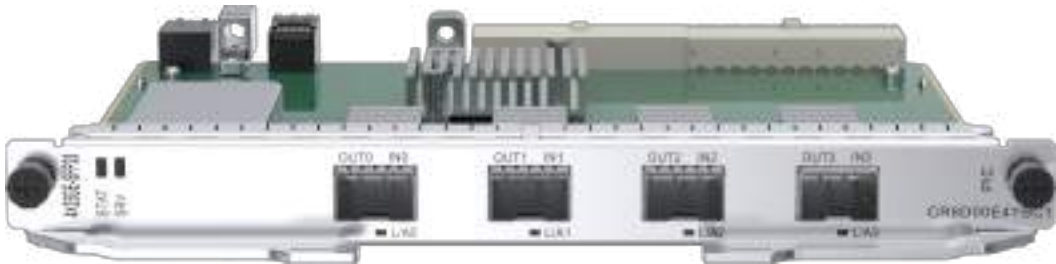
Overview

Table 1-882 Basic information about the CR8D00E4YBC1

Item	Details
Description	4-Port 25GE-SFP28 Physical Interface Card (PIC)
Part Number	03033VME
Model	CR8D00E4YBC1
Card type	Interface Card
Silkscreen	4x25GE-SFP28

Appearance

Figure 1-177 Appearance of the CR8D00E4YBC1



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03033VME&productModel=CR8D00E4YBC1#matchRelation>

Indicators

Table 1-883 Indicators on the CR8D00E4YBC1

Silkscreen	Name	Color	Status	Description
STAT	Working status indicator	Green	Steady on	The board is working properly.
		Red	Steady on	The hardware on the board is faulty.
		Orange	Steady on	A high-speed interface card is unexpectedly inserted into a low-speed interface card slot and an alarm is generated, or the logic is not loaded to the board.
		-	Off	The board is not powered on or is not registered.
SRV	Service Status Indicator	Green	Steady on	The board has no CRC alarm.
		Orange	Steady on	The board has CRC alarm.
		-	Off	The board is not running or no power is input.
L/A (0-3)	Interface status indicator	Green	Steady on	The link is Up.
		Green	Blinking	Data is being transmitted or received.
		-	Off	The link is Down.

Ports

Table 1-884 Ports on the CR8D00E4YBC1

Port	Connector Type	Description	Available Components
OUT0 IN0-OUT3 IN3	SFP28	25GE/10GE service signal input and output interface	10Gbps SFP+ BIDI Optical Module 10Gbps SFP+ CWDM Optical Module 10Gbps SFP+ Optical Module 25Gbps SFP28 Optical Module 25Gbps SFP28 BIDI Optical Module

Table 1-885 Port attributes about the CR8D00E4YBC1

Attributes	Details
Working mode	Full-duplex
Standards compliance	IEEE 802.3
Frame format	Ethernet_II, Ethernet_SAP and Ethernet_SNAP

Functions and Features

Table 1-886 Functions and features of the CR8D00E4YBC1

Functions and Features	Description
Basic function	Occupies one flexible interface card slot. Provides four 25GE/10GE interfaces for line-rate transmitting and receiving, and supports SFP28/SFP+ optical modules.
Reliability and availability	Hot swappable.

Functions and Features	Description
Restrictions and remarks	By default, 25GE is supported. You can run the port-mode group-id command in the slot view to switch the port mode to 10GE.
Breakout supported	No
Interface rate auto-sensing supported	No
FlexE supported	No

Technical Specifications

Table 1-887 Technical specifications of the CR8D00E4YBC1

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	19.8 mm x 193.8 mm x 209.3 mm (0.78 in. x 7.63 in. x 8.24 in.)
Dimensions with packaging (H x W x D) [mm(in.)]	70 mm x 285 mm x 300 mm (2.76 in. x 11.22 in. x 11.81 in.)
Weight without packaging [kg(lb)]	0.61 kg (1.34 lb)
Weight with packaging [kg(lb)]	0.95 kg (2.09 lb)
Typical power consumption [W]	18.20 W
Typical heat dissipation [BTU/hour]	59.05 BTU/hour
Long-term operating temperature [°C(°F)]	-20°C to +65°C (-4°F to +149°F)

1.4.4.55 CR8D00E4YFC0 (4-Port 25GE(SFP28)/10GE(SFP+) MACsec Physical Interface Card (PIC))

Overview

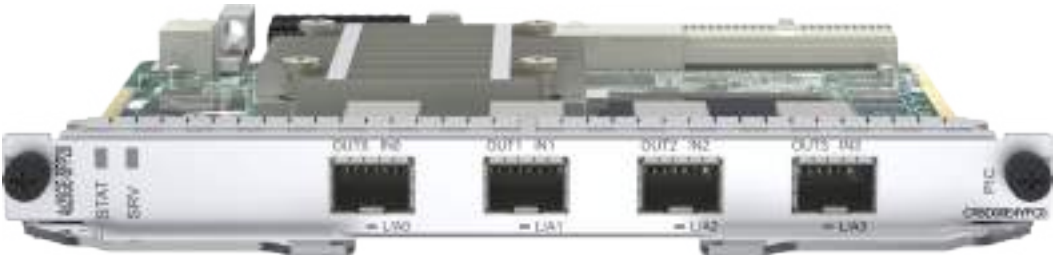
Table 1-888 Basic information about the CR8D00E4YFC0

Item	Details
Description	4-Port 25GE(SFP28)/10GE(SFP+) MACsec Physical Interface Card (PIC)
Part Number	03034PJH

Item	Details
Model	CR8D00E4YFC0
Card type	Interface Card
Silkscreen	4x25GE-SFP28

Appearance

Figure 1-178 Appearance of the CR8D00E4YFC0



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03034PJH&productModel=CR8D00E4YFC0#matchRelation>

Indicators

Table 1-889 Indicators on the CR8D00E4YFC0

Silkscreen	Name	Color	Status	Description
STAT	Running status indicator	Green	Steady on	The board is working properly.
		Red	Steady on	The board hardware is faulty.

Silkscreen	Name	Color	Status	Description
		Orange	Steady on	An alarm is generated when a high-speed interface card is incorrectly inserted into a low-speed interface card slot, or the board is not loaded with logic.
		-	Steady off	The board is not registered or is not powered on.
SRV	Service status indicator	Green	Steady on	There is no CRC error alarm on the board.
		Orange	Steady on	There is a CRC error alarm on the board.
		-	Off	The board is not powered on or is faulty.
L/A (0-3)	Interface status indicator	Green	Steady on	The link is up.
		Green	Blinking	Data is being transmitted or received.
		-	Steady off	The link is down.

Ports

Table 1-890 Ports on the CR8D00E4YFC0

Port	Connector Type	Description	Available Components
OUT0 IN0-OUT3 IN3	SFP28	Interface for inputting and outputting 25GE/10GE optical signals	155Mbps eSFP Optical Module 155Mbps eSFP BIDI Optical Module 1.25/9.953/10.31 25Gbps SFP+ Optical Module 1.25Gbps eSFP BIDI Optical Module 1.25Gbps eSFP CWDM Optical Module 1.25Gbps eSFP Optical Module 10Gbps SFP+ BIDI Optical Module 10Gbps SFP+ CWDM Optical Module 10Gbps SFP+ Optical Module 125M~2.67Gbps eSFP DWDM Optical Module 1Gbps SyncE and 1588V2 Electrical Module 1Gbps Electrical Module 25Gbps SFP28 Optical Module 25Gbps SFP28 BIDI Optical Module

Table 1-891 Port attributes about the CR8D00E4YFC0

Attributes	Details
Working mode	Full-duplex
Standards compliance	IEEE 802.3
Frame format	Ethernet_II, Ethernet_SAP, Ethernet_SNAP

Functions and Features

Table 1-892 Functions and features of the CR8D00E4YFC0

Functions and Features	Description
Basic function	Occupies one flexible interface card slot. Provides four 25GE/10GE interfaces for line-rate transmitting and receiving, and supports SFP28/SFP+ optical modules. MACsec is supported.
Reliability and availability	Hot swappable.
Restrictions and remarks	By default, 25GE/10GE auto-negotiation is supported. You can run the port-mode group-id command in the slot view to switch the mode to 10GE/GE auto-negotiation.
Breakout supported	No
Interface rate auto-sensing supported	In 25GE mode, 10GE/25GE auto-sensing supported; In 10GE mode, GE/10GE auto-sensing supported.
FlexE supported	No

Technical Specifications

Table 1-893 Technical specifications of the CR8D00E4YFC0

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	19.8 mm x 193.8 mm x 209.3 mm (0.78 in. x 7.63 in. x 8.24 in.)

Item	Specification
Dimensions with packaging (H x W x D) [mm(in.)]	70 mm x 285 mm x 300 mm (2.76 in. x 11.22 in. x 11.81 in.)
Weight without packaging [kg(lb)]	0.7 kg (1.54 lb)
Weight with packaging [kg(lb)]	1.1 kg (2.43 lb)
Typical power consumption [W]	25.3 W
Typical heat dissipation [BTU/hour]	82.15 BTU/hour
Long-term operating temperature [°C(°F)]	–40°C to +65°C (–40°F to +149°F)

1.4.4.56 CR8D00E8GEB2 (8-Port 100/1000Base-RJ45 Physical Interface Card(PIC))

Overview

Table 1-894 Basic information about the CR8D00E8GEB2

Item	Details
Description	8-Port 100/1000Base-RJ45 Physical Interface Card(PIC)
Part Number	03034REJ
Model	CR8D00E8GEB2
Card type	Interface Card
Silkscreen	8xFE/GE-RJ

Appearance

Figure 1-179 Appearance of the CR8D00E8GEB2



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03034REJ&productModel=CR8D00E8GEB2#matchRelation>

Indicators

Table 1-895 Indicators on the CR8D00E8GEB2

Silkscreen	Name	Color	Status	Description
STAT	Working status indicator	Green	Steady on	The board is working normally.
		Red	Steady on	The board hardware is faulty.
		Orange	Steady on	The interface board is not loaded with the logic.
		-	Off	The board is not running or no power is input.
SRV	Service status indicator	Green	Steady on	The board has no CRC alarm.
		Orange	Steady on	The board has CRC alarms.
		-	Off	The board is not running or no power is input.

Silkscreen	Name	Color	Status	Description
LINK	Port connection status indicator	Green	Steady on	If the LINK indicator above each FE/GE electrical interface is steady green, the physical connection of the interface is normal.
LINK	Connection status indicator	-	Off	Indicates that the physical port fails to be connected.
ACT	Port data transmit/receive status indicator	Orange	Blinking	If the ACT indicator is blinking orange above each FE/GE electrical service port, the service port is receiving or transmitting data.
ACT	Transmitting/receiving status indicator	-	Off	The service port is not receiving or transmitting data.

Ports

Table 1-896 Ports on the CR8D00E8GEB2

Port	Connector Type	Description	Available Components
FE/GE0 to FE/GE7	RJ45	GE/FE service signal input and output interface	Ethernet Cables

Table 1-897 Port attributes about the CR8D00E8GEB2

Attributes	Details
Working mode	Full-duplex
Standards compliance	IEEE 802.3
Frame format	Ethernet_II, Ethernet_SAP, and Ethernet_SNAP

Functions and Features

Table 1-898 Functions and features of the CR8D00E8GEB2

Functions and Features	Description
Basic function	Occupies one flexible interface board slot. Provides eight GE/FE interfaces for line-rate transmitting and receiving, and supports 10M/100M/1000M electrical interfaces. Supports MACsec.
Reliability and availability	Hot swappable.
Breakout supported	No
Interface rate auto-sensing supported	No
FlexE supported	No

Technical Specifications

Table 1-899 Technical specifications of the CR8D00E8GEB2

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	19.8 mm x 193.8 mm x 209.3 mm (0.78 in. x 7.63 in. x 8.24 in.)
Dimensions with packaging (H x W x D) [mm(in.)]	70 mm x 285 mm x 300 mm (2.76 in. x 11.22 in. x 11.81 in.)
Weight without packaging [kg(lb)]	0.55 kg (1.21 lb)
Weight with packaging [kg(lb)]	0.89 kg (1.96 lb)
Typical power consumption [W]	12.33 W
Typical heat dissipation [BTU/hour]	40.00 BTU/hour

Item	Specification
Long-term operating temperature [°C(°F)]	–40°C to +65°C (–40°F to +149°F)

1.4.4.57 CR8D00E8GEC1 (03033MNN: 8-Port 100/1000Base-RJ45 Physical Interface Card(PIC))

Overview

Table 1-900 Basic information about the CR8D00E8GEC1

Item	Details
Description	8-Port 100/1000Base-RJ45 Physical Interface Card(PIC)
Part Number	03033MNN
Model	CR8D00E8GEC1
Card type	Interface Card
Silkscreen	8xFE/GE-RJ

Appearance

Figure 1-180 Appearance of the CR8D00E8GEC1



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03033MNN&productModel=CR8D00E8GEC1#matchRelation>

Indicators

Table 1-901 Indicators on the CR8D00E8GEC1

Silkscreen	Name	Color	Status	Description
STAT	Working status indicator	Green	Steady on	The board is working normally.
		Red	Steady on	The board hardware is faulty.
		Orange	Steady on	The interface board is not loaded with the logic.
		-	Off	The board is not running or no power is input.
SRV	Service status indicator	Green	Steady on	The board has no CRC alarm.
		Orange	Steady on	The board has CRC alarms.
		-	Off	The board is not running or no power is input.
LINK	Port connection status indicator	Green	Steady on	If the LINK indicator above each FE/GE electrical interface is steady green, the physical connection of the interface is normal.
LINK	Connection status indicator	-	Off	Indicates that the physical port fails to be connected.

Silkscreen	Name	Color	Status	Description
ACT	Port data transmit/receive status indicator	Orange	Blinking	If the ACT indicator is blinking orange above each FE/GE electrical service port, the service port is receiving or transmitting data.
ACT	Transmitting/receiving status indicator	-	Off	The service port is not receiving or transmitting data.

Ports

Table 1-902 Ports on the CR8D00E8GEC1

Port	Connector Type	Description	Available Components
FE/GE0 to FE/GE7	RJ45	GE/FE service signal input and output interface	Ethernet Cables

Table 1-903 Port attributes about the CR8D00E8GEC1

Attributes	Details
Working mode	Full-duplex
Standards compliance	IEEE 802.3
Frame format	Ethernet_II, Ethernet_SAP, and Ethernet_SNAP

Functions and Features

Table 1-904 Functions and features of the CR8D00E8GEC1

Functions and Features	Description
Basic function	Occupies one flexible interface board slot. Provides eight GE/FE interfaces for line-rate transmitting and receiving, and supports 10M/100M/1000M electrical interfaces. Supports MACsec.
Reliability and availability	Hot swappable.
Breakout supported	No
Interface rate auto-sensing supported	No
FlexE supported	No

Technical Specifications

Table 1-905 Technical specifications of the CR8D00E8GEC1

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	19.8 mm x 193.8 mm x 209.3 mm (0.78 in. x 7.63 in. x 8.24 in.)
Dimensions with packaging (H x W x D) [mm(in.)]	70 mm x 285 mm x 300 mm (2.76 in. x 11.22 in. x 11.81 in.)
Weight without packaging [kg(lb)]	0.55 kg (1.21 lb)
Weight with packaging [kg(lb)]	0.89 kg (1.96 lb)
Typical power consumption [W]	12.33 W
Typical heat dissipation [BTU/hour]	40.00 BTU/hour
Long-term operating temperature [°C(°F)]	–40°C to +65°C (–40°F to +149°F)

1.4.4.58 CR8D00E8GEC1 (03033MNN-001: 8-Port 100/1000Base-RJ45 Physical Interface Card(PIC))

Overview

Table 1-906 Basic information about the CR8D00E8GEC1

Item	Details
Description	8-Port 100/1000Base-RJ45 Physical Interface Card(PIC)
Part Number	03033MNN-001
Model	CR8D00E8GEC1
Card type	Interface Card
Silkscreen	8xFE/GE-RJ

Appearance

Figure 1-181 Appearance of the CR8D00E8GEC1



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03033MNN-001&productModel=CR8D00E8GEC1#matchRelation>

Indicators

Table 1-907 Indicators on the CR8D00E8GEC1

Silkscreen	Name	Color	Status	Description
STAT	Working status indicator	Green	Steady on	The board is working normally.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	The board hardware is faulty.
		Orange	Steady on	The interface board is not loaded with the logic.
		-	Off	The board is not running or no power is input.
SRV	Service status indicator	Green	Steady on	The board has no CRC alarm.
		Orange	Steady on	The board has CRC alarms.
		-	Off	The board is not running or no power is input.
LINK	Port connection status indicator	Green	Steady on	If the LINK indicator above each FE/GE electrical interface is steady green, the physical connection of the interface is normal.
LINK	Connection status indicator	-	Off	Indicates that the physical port fails to be connected.

Silkscreen	Name	Color	Status	Description
ACT	Port data transmit/receive status indicator	Orange	Blinking	If the ACT indicator is blinking orange above each FE/GE electrical service port, the service port is receiving or transmitting data.
ACT	Transmitting/receiving status indicator	-	Off	The service port is not receiving or transmitting data.

Ports

Table 1-908 Ports on the CR8D00E8GEC1

Port	Connector Type	Description	Available Components
FE/GE0 to FE/GE7	RJ45	GE/FE service signal input and output interface	Ethernet Cables

Table 1-909 Port attributes about the CR8D00E8GEC1

Attributes	Details
Working mode	Full-duplex
Standards compliance	IEEE 802.3
Frame format	Ethernet_II, Ethernet_SAP, and Ethernet_SNAP

Functions and Features

Table 1-910 Functions and features of the CR8D00E8GEC1

Functions and Features	Description
Basic function	Occupies one flexible interface board slot. Provides eight GE/FE interfaces for line-rate transmitting and receiving, and supports 10M/100M/1000M electrical interfaces. Supports MACsec.
Reliability and availability	Hot swappable.
Breakout supported	No
Interface rate auto-sensing supported	No
FlexE supported	No

Technical Specifications

Table 1-911 Technical specifications of the CR8D00E8GEC1

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	19.8 mm x 193.8 mm x 209.3 mm (0.78 in. x 7.63 in. x 8.24 in.)
Dimensions with packaging (H x W x D) [mm(in.)]	70 mm x 285 mm x 300 mm (2.76 in. x 11.22 in. x 11.81 in.)
Weight without packaging [kg(lb)]	0.55 kg (1.21 lb)
Weight with packaging [kg(lb)]	0.89 kg (1.96 lb)
Typical power consumption [W]	12.33 W
Typical heat dissipation [BTU/hour]	40.00 BTU/hour
Long-term operating temperature [°C(°F)]	–40°C to +65°C (–40°F to +149°F)

1.4.4.59 CR8D00EAGFC3 (10-Port 100/1000Base-X-SFP Physical Interface Card(PIC))

Overview

Table 1-912 Basic information about the CR8D00EAGFC3

Item	Details
Description	10-Port 100/1000Base-X-SFP Physical Interface Card(PIC)
Part Number	03034BAS
Model	CR8D00EAGFC3
Card type	Interface Card
Silkscreen	10xFE/GE-SFP

Appearance

Figure 1-182 Appearance of the CR8D00EAGFC3



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03034BAS&productModel=CR8D00EAGFC3#matchRelation>

Indicators

Table 1-913 Indicators on the CR8D00EAGFC3

Silkscreen	Name	Color	Status	Description
STAT	Working status indicator	Green	Steady on	The board is working properly.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	The hardware on the board is faulty.
		Orange	Steady on	A high-speed interface card is unexpectedly inserted into a low-speed interface card slot and an alarm is generated, or the logic is not loaded to the board.
		-	Off	The board is not powered on or is not registered.
SRV	Service status indicator	Green	Steady on	The board has no CRC alarm.
		Orange	Steady on	The board has CRC alarms.
		-	Off	The board is not running or no power is input.
L/A (0 to 9)	Interface status indicator	Green	Steady on	The link is Up.
		Green	Blinking	Data is being transmitted or received.
		-	Off	The link is Down.

Ports

Table 1-914 Ports on the CR8D00EAGFC3

Port	Connector Type	Description	Available Components
OUT0 IN0-OUT9 IN9	SFP	GE/FE service signal input and output interface	155Mbps SFP Electrical Module 155Mbps eSFP Optical Module 155Mbps eSFP BIDI Optical Module 1Gbps Electrical Module 1.25Gbps eSFP Optical Module 1.25Gbps eSFP CWDM Optical Module 1.25Gbps eSFP BIDI Optical Module 125M~2.67Gbps eSFP DWDM Optical Module

Table 1-915 Port attributes about the CR8D00EAGFC3

Attributes	Details
Working mode	Full-duplex
Standards compliance	IEEE 802.3
Frame format	Ethernet_II, Ethernet_SAP and Ethernet_SNAP

Functions and Features

Table 1-916 Functions and features of the CR8D00EAGFC3

Functions and Features	Description
Basic function	Occupies one flexible interface card slot. Provides 10 GE interfaces for line-rate transmitting and receiving, and supports SFP optical modules.
Reliability and availability	Hot swappable.
Breakout supported	No
Interface rate auto-sensing supported	No
FlexE supported	No

Technical Specifications

Table 1-917 Technical specifications of the CR8D00EAGFC3

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	19.8 mm x 193.8 mm x 209.3 mm (0.78 in. x 7.63 in. x 8.24 in.)
Dimensions with packaging (H x W x D) [mm(in.)]	70 mm x 285 mm x 300 mm (2.76 in. x 11.22 in. x 11.81 in.)
Weight without packaging [kg(lb)]	0.6 kg (1.32 lb)
Weight with packaging [kg(lb)]	0.95 kg (2.09 lb)
Typical power consumption [W]	13.47 W
Typical heat dissipation [BTU/hour]	43.70 BTU/hour
Long-term operating temperature [°C(°F)]	–40°C to +65°C (–40°F to +149°F)

1.4.4.60 CR8D00EAGFCA (10-port 100/1000Base-X-SFP MACsec Physical Interface Card(PIC))

Overview

Table 1-918 Basic information about the CR8D00EAGFCA

Item	Details
Description	10-port 100/1000Base-X-SFP MACsec Physical Interface Card(PIC)
Part Number	03034KEB
Model	CR8D00EAGFCA
Card type	Interface Card
Silkscreen	10xFE/GE-SFP

Appearance

Figure 1-183 Appearance of the CR8D00EAGFCA



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03034KEB&productModel=CR8D00EAGFCA#matchRelation>

Indicators

Table 1-919 Indicators on the CR8D00EAGFCA

Silkscreen	Name	Color	Status	Description
STAT	Working status indicator	Green	Steady on	The board is working properly.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	The board hardware is faulty.
		Orange	Steady on	An alarm is generated when a high-speed interface card is incorrectly inserted into a low-speed interface card slot, or the board is not loaded with logic.
		-	Steady off	The board is not registered or is not powered on.
SRV	Service status indicator	Green	Steady on	There is no CRC error alarm on the board.
		Orange	Steady on	There is a CRC error alarm on the board.
		-	Steady off	The board is not powered on or is faulty.
L/A (0-9)	Running status indicator	Green	Steady on	The link is up.
		Green	Blinking	Data is being transmitted or received.
		-	Steady off	The link is down.

Ports

Table 1-920 Ports on the CR8D00EAGFCA

Port	Connector Type	Description	Available Components
OUT0 IN0-OUT9 IN9	SFP	GE/FE optical signal input/ output port	155Mbps eSFP Optical Module 155Mbps eSFP BIDI Optical Module 1Gbps Electrical Module 1.25Gbps eSFP Optical Module 1.25Gbps eSFP CWDM Optical Module 1.25Gbps eSFP BIDI Optical Module 125M~2.67Gbps eSFP DWDM Optical Module

Table 1-921 Port attributes about the CR8D00EAGFCA

Attributes	Details
Working mode	Full-duplex
Standards compliance	IEEE 802.3
Frame format	Ethernet_II, Ethernet_SAP, Ethernet_SNAP

Functions and Features

Table 1-922 Functions and features of the CR8D00EAGFCA

Functions and Features	Description
Basic function	Occupies one flexible interface card slot. Provides ten FE/GE interfaces for line-rate transmitting and receiving, and supports SFP optical modules. Supports MACsec.
Reliability and availability	Hot swappable.
Breakout supported	No
Interface rate auto-sensing supported	No
FlexE supported	No

Technical Specifications

Table 1-923 Technical specifications of the CR8D00EAGFCA

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	19.8 mm x 193.8 mm x 209.3 mm (0.78 in. x 7.63 in. x 8.24 in.)
Dimensions with packaging (H x W x D) [mm(in.)]	70 mm x 285 mm x 300 mm (2.76 in. x 11.22 in. x 11.81 in.)
Weight without packaging [kg(lb)]	0.7 kg (1.54 lb)
Weight with packaging [kg(lb)]	1.1 kg (2.43 lb)
Typical power consumption [W]	19.3 W
Typical heat dissipation [BTU/hour]	62.54 BTU/hour
Long-term operating temperature [°C(°F)]	–40°C to +65°C (–40°F to +149°F)

1.4.4.61 CR8D00EAXFC1 (03034KEC: 10-Port 10GE/GE LAN/WAN-SFP+ MACsec Physical Interface Card(PIC))

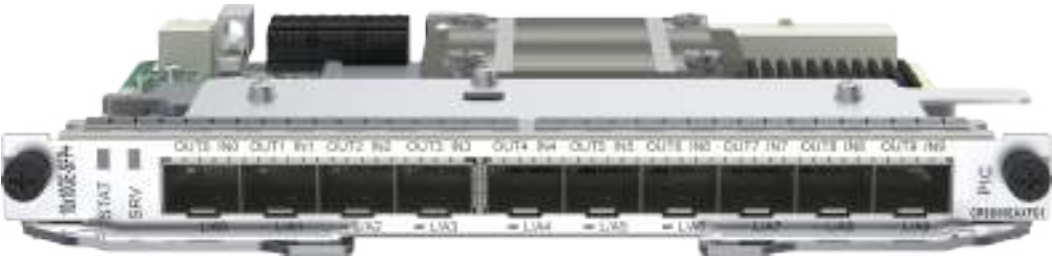
Overview

Table 1-924 Basic information about the CR8D00EAXFC1

Item	Details
Description	10-Port 10GE/GE LAN/WAN-SFP+ MACsec Physical Interface Card(PIC)
Part Number	03034KEC
Model	CR8D00EAXFC1
Card type	Interface Card
Silkscreen	10x10GE-SFP+

Appearance

Figure 1-184 Appearance of the CR8D00EAXFC1



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03034KEC&productModel=CR8D00EAXFC1#matchRelation>

Indicators

Table 1-925 Indicators on the CR8D00EAXFC1

Silkscreen	Name	Color	Status	Description
STAT	Running status indicator	Green	Steady on	The board is working properly.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	The board hardware is faulty.
		Orange	Steady on	An alarm is generated when a high-speed interface card is incorrectly inserted into a low-speed interface card slot, or the board is not loaded with logic.
		-	Steady off	The board is not registered or is not powered on.
SRV	Service status indicator	Green	Steady on	There is no CRC error alarm on the board.
		Orange	Steady on	There is a CRC error alarm on the board.
		-	Off	The board is not powered on or is faulty.
L/A (0-9)	Interface status indicator	Green	Steady on	The link is up.
		Green	Blinking	Data is being transmitted or received.
		-	Steady off	The link is down.

Ports

Table 1-926 Ports on the CR8D00EAXFC1

Port	Connector Type	Description	Available Components
OUT0 IN0-OUT9 IN9	SFP+	Interface for 10GE optical signal input and output	155Mbps eSFP Optical Module 155Mbps eSFP BIDI Optical Module 1.25/9.953/10.31 25Gbps SFP+ Optical Module 1.25Gbps eSFP BIDI Optical Module 1.25Gbps eSFP CWDM Optical Module 1.25Gbps eSFP Optical Module 10Gbps SFP+ BIDI Optical Module 10Gbps SFP+ CWDM Optical Module 10Gbps SFP+ Optical Module 125M~2.67Gbps eSFP DWDM Optical Module 1Gbps SyncE and 1588V2 Electrical Module 1Gbps Electrical Module

Table 1-927 Port attributes about the CR8D00EAXFC1

Attributes	Details
Working mode	Full-duplex
Standards compliance	IEEE 802.3

Attributes	Details
Frame format	Ethernet_II, Ethernet_SAP, Ethernet_SNAP

Functions and Features

Table 1-928 Functions and features of the CR8D00EAXFC1

Functions and Features	Description
Basic function	Occupies one flexible interface card slot. Provides ten 10GE/GE interfaces for line-rate transmitting and receiving, and supports SFP+/SFP optical modules. Supports MACsec.
Reliability and availability	Hot swappable.
Restrictions and remarks	Ports 2–5 belong to a port group, and ports 6–9 belong to another one. In LAN/WAN switching, the mode of ports in a port group must be switched at the same time.
Breakout supported	No
Interface rate auto-sensing supported	Supports GE/10GE auto-sensing.
FlexE supported	No

Technical Specifications

Table 1-929 Technical specifications of the CR8D00EAXFC1

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	19.8 mm x 193.8 mm x 209.3 mm (0.78 in. x 7.63 in. x 8.24 in.)
Dimensions with packaging (H x W x D) [mm(in.)]	70 mm x 285 mm x 300 mm (2.76 in. x 11.22 in. x 11.81 in.)
Weight without packaging [kg(lb)]	0.8 kg (1.76 lb)
Weight with packaging [kg(lb)]	1.2 kg (2.65 lb)
Typical power consumption [W]	32.6 W

Item	Specification
Typical heat dissipation [BTU/hour]	105.8 BTU/hour
Long-term operating temperature [°C(°F)]	–40°C to +65°C (–40°F to +149°F)

1.4.4.62 CR8D00EAXFC1 (03035JVG: 10-Port 10GE/GE LAN/WAN-SFP+ MACsec Physical Interface Card(PIC) for SME)

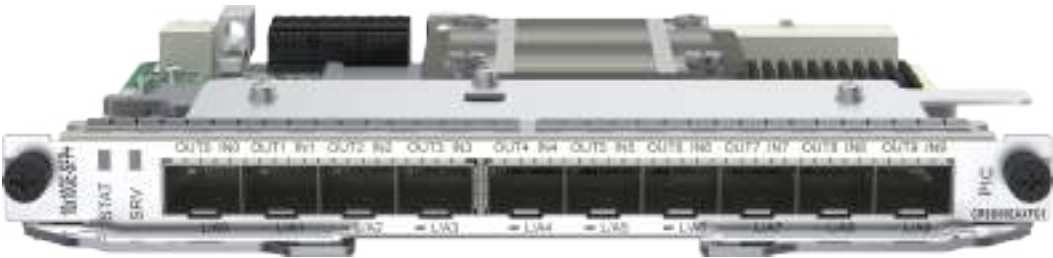
Overview

Table 1-930 Basic information about the CR8D00EAXFC1

Item	Details
Description	10-Port 10GE/GE LAN/WAN-SFP+ MACsec Physical Interface Card(PIC) for SME
Part Number	03035JVG
Model	CR8D00EAXFC1
Card type	Interface Card
Silkscreen	10x10GE-SFP+

Appearance

Figure 1-185 Appearance of the CR8D00EAXFC1



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03035JVG&productModel=CR8D00EAXFC1#matchRelation>

Indicators

Table 1-931 Indicators on the CR8D00EAXFC1

Silkscreen	Name	Color	Status	Description
STAT	Running status indicator	Green	Steady on	The board is working properly.
		Red	Steady on	The hardware on the board is faulty.
		Orange	Steady on	An alarm is generated when a high-speed interface card is incorrectly inserted into a low-speed interface card slot, or the board is not loaded with logic.
		-	Steady off	The board is not powered on or is not registered.
SRV	Service status indicator	Green	Steady on	The board has no CRC alarm.
		Orange	Steady on	The board has CRC alarms.
		-	Off	The board is not powered on or is faulty.
L/A (0-9)	Interface status indicator	Green	Steady on	The link is Up.
		Green	Blinking	Data is being transmitted and received.
		-	Steady off	The link is Down.

Ports

Table 1-932 Ports on the CR8D00EAXFC1

Port	Connector Type	Description	Available Components
OUT0 IN0-OUT9 IN9	SFP+	10GE/GE optical signal input and output interface	155Mbps eSFP Optical Module 155Mbps eSFP BIDI Optical Module 1.25/9.953/10.31 25Gbps SFP+ Optical Module 1.25Gbps eSFP BIDI Optical Module 1.25Gbps eSFP CWDM Optical Module 1.25Gbps eSFP Optical Module 10Gbps SFP+ BIDI Optical Module 10Gbps SFP+ CWDM Optical Module 10Gbps SFP+ Optical Module 125M~2.67Gbps eSFP DWDM Optical Module 1Gbps SyncE and 1588V2 Electrical Module 1Gbps Electrical Module

Table 1-933 Port attributes about the CR8D00EAXFC1

Attributes	Details
Working mode	Full-duplex

Attributes	Details
Standards compliance	IEEE 802.3
Frame format	Ethernet_II, Ethernet_SAP, Ethernet_SNAP

Functions and Features

Table 1-934 Functions and features of the CR8D00EAXFC1

Functions and Features	Description
Basic function	Occupies one flexible interface card slot. Provides ten 10GE/GE interfaces for line-rate transmitting and receiving, and supports SFP+/SFP optical modules. Supports MACsec.
Reliability and availability	Hot swappable
Restrictions and remarks	Ports 2–5 belong to a port group, and ports 6–9 belong to another one. In LAN/WAN switching, the mode of ports in a port group must be switched at the same time.
Breakout supported	No
Interface rate auto-sensing supported	Supports GE/10GE auto-sensing.
FlexE supported	No

Technical Specifications

Table 1-935 Technical specifications of the CR8D00EAXFC1

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	19.8 mm x 193.8 mm x 209.3 mm (0.78 in. x 7.63 in. x 8.24 in.)
Dimensions with packaging (H x W x D) [mm(in.)]	70 mm x 285 mm x 300 mm (2.76 in. x 11.22 in. x 11.81 in.)
Weight without packaging [kg(lb)]	0.8 kg (1.76 lb)
Weight with packaging [kg(lb)]	1.2 kg (2.65 lb)

Item	Specification
Typical power consumption [W]	32.6 W
Typical heat dissipation [BTU/hour]	105.8 BTU/hour
Long-term operating temperature [°C(°F)]	–40°C to +65°C (–40°F to +149°F)

1.4.4.63 CR8D00EEGFXA (20-Port 100/1000Base-X-CSFP/10-Port 100/1000Base-X-SFP Physical Interface Card(PIC))

Overview

Table 1-936 Basic information about the CR8D00EEGFXA

Item	Details
Description	20-Port 100/1000Base-X-CSFP/10-Port 100/1000Base-X-SFP Physical Interface Card(PIC)
Part Number	03034QXF
Model	CR8D00EEGFXA
Card type	Interface Card
Silkscreen	20xGE-CSFP

Appearance

Figure 1-186 Appearance of the CR8D00EEGFXA



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find

desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03034QXF&productModel=CR8D00EEGFXA#matchRelation>

Indicators

Table 1-937 Indicators on the CR8D00EEGFXA

Silkscreen	Name	Color	Status	Description
STAT	Running status indicator	Green	Steady on	The board is working properly.
		Red	Steady on	The board hardware is faulty.
		Orange	Steady on	An alarm is generated when a high-speed interface card is incorrectly inserted into a low-speed interface card slot, or the board is not loaded with logic.
		-	Steady off	The board is not registered or is not powered on.
OUT0 IN0 to OUT19 IN19	Interface status indicator	Green	Steady on	The link is up.
		Green	Blinking	Data is being transmitted or received.
		-	Steady off	The link is down.
SRV	Service status indicator	Green	Steady on	There is no CRC error alarm on the board.

Silkscreen	Name	Color	Status	Description
		Orange	Steady on	There is a CRC error alarm on the board.
		-	Off	The board is not powered on or is faulty.

Ports

Table 1-938 Ports on the CR8D00EEGFXA

Port	Connector Type	Description	Available Components
OUT0 IN0-OUT19 IN19	CSFP/SFP	GE/FE optical signal input/output port	1.25Gbps eSFP Optical Module 1.25Gbps eSFP CWDM Optical Module 1.25Gbps eSFP BIDI Optical Module 1.25Gbps CSFP BIDI Optical Module 125M~1.25Gbps CSFP BIDI Optical Module 125M~2.67Gbps eSFP DWDM Optical Module 155Mbps eSFP Optical Module 155Mbps eSFP BIDI Optical Module 1Gbps Electrical Module

Table 1-939 Port attributes about the CR8D00EEGFXA

Attributes	Details
Working mode	Full-duplex
Standards compliance	IEEE 802.3
Frame format	Ethernet_II, Ethernet_SAP, Ethernet_SNAP

Functions and Features

Table 1-940 Functions and features of the CR8D00EEGFXA

Functions and Features	Description
Basic function	Occupies one flexible interface card slot. Provides 20 GE interfaces for line-rate transmitting and receiving, and supports CSFP optical modules on interfaces 0 to 19. Provides 10 GE interfaces for line-rate transmitting and receiving, and supports SFP modules on even-numbered interfaces (0, 2, 4, ..., 18). Supports CSFP and SFP modules used separately and in combination. Supports MACsec.
Reliability and availability	Hot swappable.
Breakout supported	No
Interface rate auto-sensing supported	No
FlexE supported	No

Technical Specifications

Table 1-941 Technical specifications of the CR8D00EEGFXA

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	19.8 mm x 193.8 mm x 209.3 mm (0.78 in. x 7.63 in. x 8.24 in.)
Dimensions with packaging (H x W x D) [mm(in.)]	70 mm x 285 mm x 300 mm (2.76 in. x 11.22 in. x 11.81 in.)

Item	Specification
Weight without packaging [kg(lb)]	0.8 kg (1.76 lb)
Weight with packaging [kg(lb)]	1.3 kg (2.87 lb)
Typical power consumption [W]	32.2 W
Typical heat dissipation [BTU/hour]	104.34 BTU/hour
Long-term operating temperature [°C(°F)]	-20°C to +65°C (-4°F to +149°F)

1.4.4.64 CR8D00L4XFC1 (4-Port 10G-SFP+ Physical Interface Card(PIC))

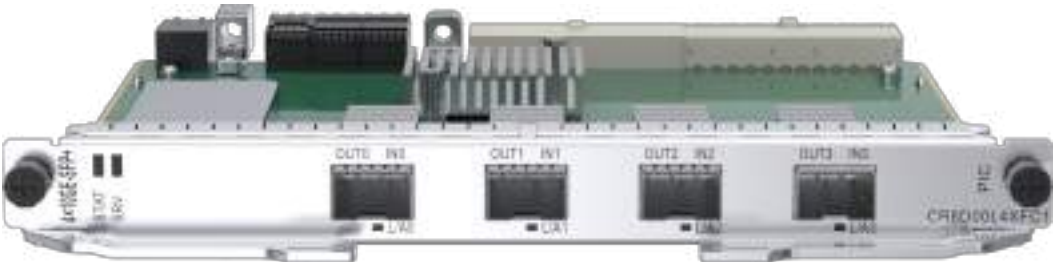
Overview

Table 1-942 Basic information about the CR8D00L4XFC1

Item	Details
Description	4-Port 10G-SFP+ Physical Interface Card(PIC)
Part Number	03033VMB
Model	CR8D00L4XFC1
Card type	Interface Card
Silkscreen	4x10GE-SFP+

Appearance

Figure 1-187 Appearance of the CR8D00L4XFC1



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find

desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03033VMB&productModel=CR8D00L4XFC1#matchRelation>

Indicators

Table 1-943 Indicators on the CR8D00L4XFC1

Silkscreen	Name	Color	Status	Description
STAT	Board status indicator	Green	Steady on	The board is working normally.
		Red	Steady on	The board hardware is faulty.
		Orange	Steady on	The board is not loaded with the logic.
		-	Off	The board is not running or no power is input.
SRV	Service status indicator	Green	Steady on	The board has no CRC alarm.
		Orange	Steady on	The board has CRC alarms.
		-	Off	The board is not running or no power is input.
L/A 0 to 3	Port connection/data transceiving status indicator	Green	Steady on	The service port is properly connected to an optical fiber or cable.

Silkscreen	Name	Color	Status	Description
		Orange or green	Blinking	The service port is properly connected to the optical fiber or cable, and the port is receiving or transmitting data.
		-	Off	The connection between the service port and the optical fiber or cable fails.

Ports

Table 1-944 Ports on the CR8D00L4XFC1

Port	Connector Type	Description	Available Components
OUT0 IN0 to OUT3 IN3	SFP+	10GE service signal input and output interface	10Gbps SFP+ Optical Module 10Gbps SFP+ BIDI Optical Module 10Gbps SFP+ CWDM Optical Module

Table 1-945 Port attributes about the CR8D00L4XFC1

Attributes	Details
Working mode	Full-duplex
Standards compliance	Supports Ethernet IEEE 802.3 standard network protocol

Functions and Features

Table 1-946 Functions and features of the CR8D00L4XFC1

Functions and Features	Description
Basic function	Occupies one flexible interface board slot. Provides four 10GE interfaces for line-rate transmitting and receiving, and supports SFP+ optical modules.
Reliability and availability	Hot swappable.
Breakout supported	No
Interface rate auto-sensing supported	No
FlexE supported	No

Technical Specifications

Table 1-947 Technical specifications of the CR8D00L4XFC1

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	19.8 mm x 193.8 mm x 209.3 mm (0.78 in. x 7.63 in. x 8.24 in.)
Dimensions with packaging (H x W x D) [mm(in.)]	70 mm x 285 mm x 300 mm (2.76 in. x 11.22 in. x 11.81 in.)
Weight without packaging [kg(lb)]	0.6 kg (1.32 lb)
Weight with packaging [kg(lb)]	1.0 kg (2.2 lb)
Typical power consumption [W]	14.27 W
Typical heat dissipation [BTU/hour]	46.30 BTU/hour
Long-term operating temperature [°C(°F)]	–20°C to +65°C (–4°F to +149°F)

1.4.4.65 CR8D00L8YFC1 (8-Port 25GE(SFP28)/10GE(SFP+) MACsec Physical Interface Card (PIC))

Overview

Table 1-948 Basic information about the CR8D00L8YFC1

Item	Details
Description	8-Port 25GE(SFP28)/10GE(SFP+) MACsec Physical Interface Card (PIC)
Part Number	03034SUP
Model	CR8D00L8YFC1
Card type	Interface Card
Silkscreen	8x25GE-SFP28

Appearance

Figure 1-188 Appearance of the CR8D00L8YFC1



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03034SUP&productModel=CR8D00L8YFC1#matchRelation>

Indicators

Table 1-949 Indicators on the CR8D00L8YFC1

Silkscreen	Name	Color	Status	Description
STAT	Working status indicator	Green	Steady on	The board is working properly.
		Red	Steady on	The board hardware is faulty.
		Orange	Steady on	An alarm is generated when a high-speed interface card is incorrectly inserted into a low-speed interface card slot, or the board is not loaded with logic.
		-	Off	The board is not registered or is not powered on.
SRV	Service status indicator	Green	Steady on	There is no CRC error alarm on the board.
		Orange	Steady on	There is a CRC error alarm on the board.
		-	Off	The board is not powered on or is faulty.
L/A (0-7)	Running status indicator	Green	Steady on	The link is up.
		Green	Blinking	Data is being transmitted or received.
		-	Off	The link is down.

Ports

Table 1-950 Ports on the CR8D00L8YFC1

Port	Connector Type	Description	Available Components
OUT0 IN0-OUT7 IN7	SFP28	Interface for inputting and outputting 25GE/10GE optical signals	155Mbps eSFP Optical Module 155Mbps eSFP BIDI Optical Module 1.25/9.953/10.31 25Gbps SFP+ Optical Module 1.25Gbps eSFP BIDI Optical Module 1.25Gbps eSFP CWDM Optical Module 1.25Gbps eSFP Optical Module 10Gbps SFP+ BIDI Optical Module 10Gbps SFP+ CWDM Optical Module 10Gbps SFP+ Optical Module 125M~2.67Gbps eSFP DWDM Optical Module 1Gbps Electrical Module 25Gbps SFP28 Optical Module

Table 1-951 Port attributes about the CR8D00L8YFC1

Attributes	Details
Working mode	Full-duplex

Attributes	Details
Standards compliance	IEEE 802.3
Frame format	Ethernet_II, Ethernet_SAP, Ethernet_SNAP

Functions and Features

Table 1-952 Functions and features of the CR8D00L8YFC1

Functions and Features	Description
Basic function	Provides eight 25GE/10GE ports to transmit and receive optical signals. Supports MACsec.
Reliability and availability	Hot swappable.
Restrictions and remarks	Ports 0 to 3 belong to a port group, and ports 4 to 7 belong to another one. By default, 25GE/10GE auto-negotiation is supported. You can run the port-mode group-id command in the slot view to switch the mode of a port group to 10GE/GE auto-negotiation. In LAN/WAN switching, the mode of ports in a port group must be switched at the same time.
Breakout supported	No
Interface rate auto-sensing supported	In 25GE mode, 10GE/25GE auto-sensing supported; In 10GE mode, GE/10GE auto-sensing supported.
FlexE supported	No

Technical Specifications

Table 1-953 Technical specifications of the CR8D00L8YFC1

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	19.8 mm x 193.8 mm x 209.3 mm (0.78 in. x 7.63 in. x 8.24 in.)
Dimensions with packaging (H x W x D) [mm(in.)]	70 mm x 285 mm x 300 mm (2.76 in. x 11.22 in. x 11.81 in.)

Item	Specification
Weight without packaging [kg(lb)]	0.8 kg (1.76 lb)
Weight with packaging [kg(lb)]	1.2 kg (2.65 lb)
Typical power consumption [W]	37.5 W
Typical heat dissipation [BTU/hour]	121.57 BTU/hour
Long-term operating temperature [°C(°F)]	–40°C to +65°C (–40°F to +149°F)

1.4.4.66 CR8D00P8CFC1 (8-Port STM-1c/8-Port STM-4c POS Card (PIC) (Default Enable 4*STM-1c))

Overview

Table 1-954 Basic information about the CR8D00P8CFC1

Item	Details
Description	8-Port STM-1c/8-Port STM-4c POS Card (PIC) (Default Enable 4*STM-1c)
Part Number	03033YLL
Model	CR8D00P8CFC1
Card type	Interface Card
Silkscreen	8xSTM1/8xSTM4-POS

Appearance

Figure 1-189 Appearance of the CR8D00P8CFC1



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03033YLL&productModel=CR8D00P8CFC1#matchRelation>

Indicators

Table 1-955 Indicators on the CR8D00P8CFC1

Silkscreen	Name	Color	Status	Description
STAT	Working status indicator	Green	Steady on	The board is working normally.
		Red	Steady on	The board hardware is faulty.
		Orange	Steady on	The interface board is not loaded with the logic.
		-	Off	The board is not running or no power is input.
SRV	Service status indicator	Green	Steady on	The board has no CRC alarm.
		Orange	Steady on	The board has CRC alarms.
		-	Steady off	The board is not powered on or is faulty.
L/A (0-7)	Running status indicator	Green	Steady on	The link is Up.
L/A(0-7)		Green	Blinking	Data is being transmitted and received.
		-	Steady off	The link is Down.

Ports

Table 1-956 Ports on the CR8D00P8CFC1

Port	Connector Type	Description	Available Components
OUT0 IN0 to OUT7 IN7	SFP	Interface for inputting and outputting STM-1/STM-4/STM-16 optical signals	155Mbps SFP Electrical Module 155Mbps eSFP Optical Module 155Mbps eSFP BIDI Optical Module 622Mbps eSFP Optical Module 2.5Gbps eSFP Optical Module

Table 1-957 Port attributes about the CR8D00P8CFC1

Attributes	Details
Working mode	Full-duplex
Standards compliance	ITU-T G707
Frame format	PPP, HDLC

Functions and Features

Table 1-958 Functions and features of the CR8D00P8CFC1

Functions and Features	Description
Basic function	Provides eight 155 Mbit/s or 622 Mbit/s POS interfaces. From V800R023C00, the first four ports support the 2.5G POS mode. The port mode can be switched among 8 x 155M, 8 x 622M, and 4 x 2.5G using the set service mode command.
Reliability and availability	Hot swappable.

Functions and Features	Description
Restrictions and remarks	1. Ports 0–7 support the master clock mode, and ports 0–3 support the slave clock mode. 2. A subcard supports only one of the three modes. To switch the mode, all interfaces must switch the mode at the same time.
Breakout supported	No
Interface rate auto-sensing supported	No
FlexE supported	No
Link protocol	PPP,HDLC

Technical Specifications

Table 1-959 Technical specifications of the CR8D00P8CFC1

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	19.8 mm x 193.8 mm x 209.3 mm (0.78 in. x 7.63 in. x 8.24 in.)
Dimensions with packaging (H x W x D) [mm(in.)]	70 mm x 285 mm x 300 mm (2.76 in. x 11.22 in. x 11.81 in.)
Weight without packaging [kg(lb)]	0.7 kg (1.54 lb)
Weight with packaging [kg(lb)]	1.1 kg (2.43 lb)
Typical power consumption [W]	24.79 W
Typical heat dissipation [BTU/hour]	80.43 BTU/hour
Long-term operating temperature [°C(°F)]	–20°C to +65°C (–4°F to +149°F)

1.4.4.67 CR8DE1KB00M1 (1-Port 400GBase-QSFP-DD Single-layer Physical Interface Card(PIC))

Overview

Table 1-960 Basic information about the CR8DE1KB00M1

Item	Details
Description	1-Port 400GBase-QSFP-DD Single-layer Physical Interface Card(PIC)
Part Number	03034YNM
Model	CR8DE1KB00M1
Card type	Interface Card
Silkscreen	1x400GE-QSFPDD

Appearance

Figure 1-190 Appearance of the CR8DE1KB00M1



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03034YNM&productModel=CR8DE1KB00M1#matchRelation>

Indicators and Buttons

Table 1-961 Indicators on the CR8DE1KB00M1

Silkscreen	Name	Color	Status	Description
STAT	Running status indicator	Green	Steady on	The board is working properly.
		Red	Steady on	The hardware on the board is faulty.
		Orange	Steady on	A high-speed interface card is unexpectedly inserted into a low-speed interface card slot and an alarm is generated, or the logic is not loaded to the board.
		-	Off	The board is not powered on or is not registered.
SRV	Service status indicator	Green	Steady on	The board has no CRC alarm.
		Orange	Steady on	The board has CRC alarm.
		-	Off	The board is not running or no power is input.
L0	Optical Amplifier running status indicator	Green	Steady on	The Optical Amplifier port works normally and there is no LOS.

Silkscreen	Name	Color	Status	Description
		-	Off	The Optical Amplifier port does not work normally or LOS occurs.
L/A1	Port running status indicator	Green	Steady on	The link is Up.
		Green	Blink	Data is being transmitted or received.
		-	Off	The link is Down.

Silkscreen	Name	Color	Status	Description
BREAKOUT (0-7)	Breakout channel indicator	Green	Steady on	Breakout 0-7: breakout indicators. If breakout indicator X is on, the port indicators are showing the running status of breakout channel X. For example, if breakout indicator 0 is on, port indicators are showing the running status of breakout channel 0 (if a port indicator is blinking, breakout channel 0 on this port is receiving and transmitting data); if breakout indicator 1 is on, port indicators are showing the running status of breakout channel 1. If the breakout indicators 0 to 7 are off, the board has no breakout ports. If the board has breakout ports, each breakout

Silkscreen	Name	Color	Status	Description
				indicator for channels 0 to 7 turns on for 5 seconds in sequence to indicate the status of the corresponding channel. To rapidly identify the running status of a specific breakout channel, press the Breakout button to switch to your desired indicator. For example, if you want to check the running status of breakout channel 1 on each port, press the Breakout button to have breakout indicator 1 turn on.

Table 1-962 Buttons on the CR8DE1KB00M1

Silkscreen	Name	Description
BREAKOUT	Breakout button	When you press the button, the Breakout indicators stop status cycling and remain in the current status. Then, each time you press the button, the Breakout indicators' status switches. If you do not press the button in 60s, the Breakout indicators automatically start status cycling.

Ports

Table 1-963 Ports on the CR8DE1KB00M1

Port	Connector Type	Description	Available Components
OUT0 IN0	QSFP28	Optical Amplifier Module	Optical Amplifier Module
OUT1 IN1	QSFP-DD	Inputs and outputs 400GE optical signals.	400Gbps QSFP-DD Optical Module 100Gbps QSFP28 Optical Module

Table 1-964 Port attributes about the CR8DE1KB00M1

Attributes	Details
Working mode	Full-duplex
Standards compliance	IEEE 802.3
Frame format	Ethernet_II, Ethernet_SAP and Ethernet_SNAP

Functions and Features

Table 1-965 Functions and features of the CR8DE1KB00M1

Functions and Features	Description
Basic function	The line side of the board provides one 400GE QSFP-DD interface and one QSFP28 optical amplifier interface. 1. The rate of the QSFP-DD interface is 400 Gbit/s. 2. The QSFP28 fiber amplifier interface supports the fiber amplifier module (reserved). The FlexE feature is supported.
Reliability and availability	Hot swappable.
Restrictions and remarks	1. QSFP-DD supports 400G QSFP-DD optical modules by default. You need to run the port split command in the system view to switch QSFP-DD optical modules to 100G QSFP28 optical modules. 2. When the board is used with a 400G ZR+ optical module, the operating temperature supported by the optical module ranges from -5°C to 48°C. When the board is used with a 400G ZR/ER optical module, the operating temperature supported by the optical module ranges from -5°C to 50°C.
Breakout supported	Yes. To enable the breakout function on a port, run the port split command.
Interface rate auto-sensing supported	No
FlexE supported	Yes Ports on straight-through cards can be bonded to form a group. Physical port 1 can be added to a group. port-id ranges from 1000 to 3000. A maximum of 80 FlexE clients can be created.

Technical Specifications

Table 1-966 Technical specifications of the CR8DE1KB00M1

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	19.8 mm x 193.8 mm x 209.3 mm (0.78 in. x 7.63 in. x 8.24 in.)
Dimensions with packaging (H x W x D) [mm(in.)]	70 mm x 285 mm x 300 mm (2.76 in. x 11.22 in. x 11.81 in.)
Weight without packaging [kg(lb)]	0.8 kg (1.76 lb)
Weight with packaging [kg(lb)]	1.3 kg (2.87 lb)
Typical power consumption [W]	59 W
Typical heat dissipation [BTU/hour]	191.29 BTU/hour
Long-term operating temperature [°C(°F)]	-5°C to +55°C (23°F to 131°F)

1.4.4.68 CR8DE1KB00M2 (1-Port 400GBase-QSFP-DD Single-layer Physical Interface Card CM(Enable 1*100G)(PIC))

Overview

Table 1-967 Basic information about the CR8DE1KB00M2

Item	Details
Description	1-Port 400GBase-QSFP-DD Single-layer Physical Interface Card CM(Enable 1*100G)(PIC)
Part Number	03035EXD
Model	CR8DE1KB00M2
Card type	Interface Card
Silkscreen	1x400GE-QSFPDD

Appearance

Figure 1-191 Appearance of the CR8DE1KB00M2



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03035EXD&productModel=CR8DE1KB00M2#matchRelation>

Indicators and Buttons

Table 1-968 Indicators on the CR8DE1KB00M2

Silkscreen	Name	Color	Status	Description
STAT	Running status indicator	Green	Steady on	The board is working properly.
		Red	Steady on	The hardware on the board is faulty.
		Orange	Steady on	An alarm is generated when a high-speed interface card is incorrectly inserted into a low-speed interface card slot, or the board is not loaded with logic.

Silkscreen	Name	Color	Status	Description
		-	Steady off	The board is not powered on or is not registered.
SRV	Service status indicator	Green	Steady on	The board has no CRC alarm.
		Orange	Steady on	The board has CRC alarms.
		-	Off	The board is not powered on or is faulty.
LO	Optical Amplifier running status indicator	Green	Steady on	The Optical Amplifier port works normally and there is no LOS.
		-	Off	The Optical Amplifier port does not work normally or LOS occurs.
L/A1	Interface status indicator	Green	Steady on	The link is Up.
		Green	Blinking	Data is being transmitted and received.
		-	Steady off	The link is Down.

Silkscreen	Name	Color	Status	Description
BREAKOUT (0-7)	Breakout indicator	Green	Steady on	Breakout 0-7: breakout indicators. If breakout indicator X is on, the port indicators are showing the running status of breakout channel X. For example, if breakout indicator 0 is on, port indicators are showing the running status of breakout channel 0 (if a port indicator is blinking, breakout channel 0 on this port is receiving and transmitting data); if breakout indicator 1 is on, port indicators are showing the running status of breakout channel 1. If the breakout indicators 0 to 7 are off, the board has no breakout ports. If the board has breakout ports, each breakout

Silkscreen	Name	Color	Status	Description
				indicator for channels 0 to 7 turns on for 5 seconds in sequence to indicate the status of the corresponding channel. To rapidly identify the running status of a specific breakout channel, press the Breakout button to switch to your desired indicator. For example, if you want to check the running status of breakout channel 1 on each port, press the Breakout button to have breakout indicator 1 turn on.

Table 1-969 Buttons on the CR8DE1KB00M2

Silkscreen	Name	Description
BREAKOUT	Interface splitting button	When you press the button, the breakout indicators stop status cycling and remain in the current status. Then, each time you press the button, the breakout indicators' status switches. If you do not press the button in 60 seconds, the indicators automatically start status cycling.

Ports

Table 1-970 Ports on the CR8DE1KB00M2

Port	Connector Type	Description	Available Components
OUT0 IN0	QSFP28	Optical Amplifier Module	Optical Amplifier Module
OUT1 IN1	QSFP-DD	Interface for inputting and outputting 400GE optical signals	400Gbps QSFP-DD Optical Module 100Gbps QSFP28 Optical Module

Table 1-971 Port attributes about the CR8DE1KB00M2

Attributes	Details
Working mode	Full-duplex
Standards compliance	IEEE 802.3
Frame format	Ethernet_II, Ethernet_SAP, Ethernet_SNAP

Functions and Features

Table 1-972 Functions and features of the CR8DE1KB00M2

Functions and Features	Description
Basic function	The line side of the board provides one 400GE QSFP-DD interface and one QSFP28 optical amplifier interface. 1. The default rate of a QSFP-DD port is 100 Gbit/s. The initial 100G capability can be upgraded to 200G/400G through the 50G RTU. 2. The QSFP28 fiber amplifier interface supports the fiber amplifier module (reserved). The FlexE feature is supported.
Reliability and availability	Hot swappable
Restrictions and remarks	1. QSFP-DD supports 400G QSFP-DD optical modules by default. You need to run the port split command in the system view to switch QSFP-DD optical modules to 100G QSFP28 optical modules. 2. When the board is used with a 400G ZR+ optical module, the operating temperature supported by the optical module ranges from -5°C to 48°C. When the board is used with a 400G ZR/ER optical module, the operating temperature supported by the optical module ranges from -5°C to 50°C.
Breakout supported	Yes. To enable the breakout function on a port, run the port split command.
Interface rate auto-sensing supported	No
FlexE supported	Yes Ports on straight-through cards can be bonded to form a group. Physical port 1 can be added to a group. port-id ranges from 1000 to 3000. A maximum of 80 FlexE clients can be created.

Technical Specifications

Table 1-973 Technical specifications of the CR8DE1KB00M2

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	19.8 mm x 193.8 mm x 209.3 mm (0.78 in. x 7.63 in. x 8.24 in.)
Dimensions with packaging (H x W x D) [mm(in.)]	70 mm x 285 mm x 300 mm (2.76 in. x 11.22 in. x 11.81 in.)
Weight without packaging [kg(lb)]	0.8 kg (1.76 lb)
Weight with packaging [kg(lb)]	1.3 kg (2.87 lb)
Typical power consumption [W]	59 W
Typical heat dissipation [BTU/hour]	191.29 BTU/hour
Long-term operating temperature [°C(°F)]	-5°C to +55°C (23°F to 131°F)

1.4.4.69 CR8DE1N8XFM0 (1-Port 100GBase/50GBase-QSFP28 FlexE MACsec +8-Port 10GE(SFP+)/GE(SFP) MACsec Physical Interface Card(PIC))

Overview

Table 1-974 Basic information about the CR8DE1N8XFM0

Item	Details
Description	1-Port 100GBase/50GBase-QSFP28 FlexE MACsec+8-Port 10GE(SFP+)/GE(SFP) MACsec Physical Interface Card(PIC)
Part Number	03034XQW
Model	CR8DE1N8XFM0
Card type	Interface Card
Silkscreen	1x100G-QSFP28+8x10GE-SFP+

Appearance

Figure 1-192 Appearance of the CR8DE1N8XFM0



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03034XQW&productModel=CR8DE1N8XFM0#matchRelation>

Indicators

Table 1-975 Indicators on the CR8DE1N8XFM0

Silkscreen	Name	Color	Status	Description
STAT	Working status indicator	Green	Steady on	The board is working properly.
		Red	Steady on	The hardware on the board is faulty.
		Orange	Steady on	An alarm is generated when a high-speed interface card is incorrectly inserted into a low-speed interface card slot, or the board is not loaded with logic.

Silkscreen	Name	Color	Status	Description
		-	Steady off	The board is not powered on or is not registered.
SRV	Service status indicator	Green	Steady on	The board has no CRC alarm.
		Orange	Steady on	The board has CRC alarms.
		-	Steady off	The board is not powered on or is faulty.
L/A(0 to 8)	Running status indicator	Green	Steady on	The link is Up.
		Green	Blinking	Data is being transmitted and received.
		-	Steady off	The link is Down.
BREAKOUT (0-3)	Breakout channel indicator	Green	Blinking	Each indicator for channels 0, 1, 2, and 3 turns on for 5s in sequence to indicate the status of the corresponding channel. This process repeats.
		-	Steady off	No data is transmitting or receiving in this channel.

Ports

Table 1-976 Ports on the CR8DE1N8XFM0

Port	Connector Type	Description	Available Components
OUT0 IN0	QSFP28	Interface for inputting and outputting 50GE/100GE optical signals	100Gbps QSFP28 Optical Module 40Gbps QSFP+ Optical Module 50Gbps QSFP28 BIDI Optical Module 50Gbps QSFP28 Optical Module Optical Amplifier Module
OUT1 IN1-OUT8 IN8	SFP+	10GE/GE optical signal input and output interface	155Mbps eSFP Optical Module 155Mbps eSFP BIDI Optical Module 1.25/9.953/10.31 25Gbps SFP+ Optical Module 1.25Gbps eSFP BIDI Optical Module 1.25Gbps eSFP CWDM Optical Module 1.25Gbps eSFP Optical Module 10Gbps SFP+ BIDI Optical Module 10Gbps SFP+ CWDM Optical Module 10Gbps SFP+ Optical Module 125M~2.67Gbps eSFP DWDM Optical Module 1Gbps Electrical Module

Table 1-977 Port attributes about the CR8DE1N8XFM0

Attributes	Details
Working mode	Full-duplex
Standards compliance	IEEE 802.3
Frame format	Ethernet_II, Ethernet_SAP, Ethernet_SNAP

Functions and Features

Table 1-978 Functions and features of the CR8DE1N8XFM0

Functions and Features	Description
Basic function	Occupies one flexible card slot. Provides one 100GE/50GE interface for line-rate transmitting and receiving (using QSFP28 optical modules) and eight 10GE/GE interfaces for line-rate transmitting and receiving (using SFP +/SFP optical modules). Supports MACsec.
Reliability and availability	Hot swappable
Restrictions and remarks	8 x 10G ports: Ports 1 to 4 form a group, and ports 5 to 8 form another group. To switch between LAN and WAN, all ports in a group must be switched at the same time.
Breakout supported	Yes. To enable the breakout function on a port, run the port split command.
Interface rate auto-sensing supported	SFP+ ports support GE/10GE auto-sensing. QSFP28 ports support 40GE/50GE/100GE auto-sensing.
FlexE supported	Yes Physical port 0 can be added to the one group. port-id ranges from 1000 to 3000. A maximum of 20 FlexE clients can be created.

Technical Specifications

Table 1-979 Technical specifications of the CR8DE1N8XFM0

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	19.8 mm x 193.8 mm x 209.3 mm (0.78 in. x 7.63 in. x 8.24 in.)
Dimensions with packaging (H x W x D) [mm(in.)]	70 mm x 285 mm x 300 mm (2.76 in. x 11.22 in. x 11.81 in.)
Weight without packaging [kg(lb)]	0.8 kg (1.76 lb)
Weight with packaging [kg(lb)]	1.2 kg (2.65 lb)
Typical power consumption [W]	65.6 W
Typical heat dissipation [BTU/hour]	212.83 BTU/hour
Long-term operating temperature [°C(°F)]	–40°C to +65°C (–40°F to +149°F)

1.4.4.70 CR8DE1NE2VC1 (2-Port 50GBase/1-Port 100GBase-QSFP28 Physical Interface Card(PIC))

Overview

Table 1-980 Basic information about the CR8DE1NE2VC1

Item	Details
Description	2-Port 50GBase/1-Port 100GBase-QSFP28 Physical Interface Card(PIC)
Part Number	03033VMC
Model	CR8DE1NE2VC1
Card type	Interface Card
Silkscreen	2x50GE-QSFP28

Appearance

Figure 1-193 Appearance of the CR8DE1NE2VC1



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03033VMC&productModel=CR8DE1NE2VC1#matchRelation>

Indicators

Table 1-981 Indicators on the CR8DE1NE2VC1

Silkscreen	Name	Color	Status	Description
STAT	Working status indicator	Green	Steady on	The board is working properly.
		Red	Steady on	The hardware on the board is faulty.
		Orange	Steady on	A high-speed interface card is unexpectedly inserted into a low-speed interface card slot and an alarm is generated, or the logic is not loaded to the board.
		-	Off	The board is not powered on or is not registered.
SRV	Service Status Indicator	Green	Steady on	The board has no CRC alarm.
		Orange	Steady on	The board has CRC alarm.

Silkscreen	Name	Color	Status	Description
		-	Off	The board is not running or no power is input.
L/A (0-1)	Interface status indicator	Green	Steady on	The link is Up.
		Green	Blinking	Data is being transmitted or received.
		-	Off	The link is Down.
BREAKOUT (0-3)	Breakout channel indicator	Green	Blink	Each indicator for channels 0, 1, 2, and 3 turns on for 5s in sequence to indicate the status of the corresponding channel. This process repeats.
		-	Off	No data is transmitting or receiving in this channel.

Ports

Table 1-982 Ports on the CR8DE1NE2VC1

Port	Connector Type	Description	Available Components
OUT0 IN0 to OUT1 IN1	QSFP28	100GE/50GE service signal input and output interface	100Gbps QSFP28 Optical Module 50Gbps QSFP28 BIDI Optical Module 50Gbps QSFP28 Optical Module 40Gbps QSFP+ Optical Module Optical Amplifier Module

Table 1-983 Port attributes about the CR8DE1NE2VC1

Attributes	Details
Working mode	Full-duplex
Standards compliance	IEEE 802.3
Frame format	Ethernet_II, Ethernet_SAP and Ethernet_SNAP

Functions and Features

Table 1-984 Functions and features of the CR8DE1NE2VC1

Functions and Features	Description
Basic function	Occupies one flexible interface board slot. Provides two 50GE interfaces or one 100GE interface for line-rate transmitting and receiving, and supports QSFP28 optical modules. Supports FlexE.
Reliability and availability	Hot swappable.

Functions and Features	Description
Restrictions and remarks	Port 0 supports 100GE, 50GE, 40GE, 4x25GE breakout, and 4x10GE breakout. Port 1 is valid only when port 0 is a 50GE port. If the switch-mode command is set to auto and port 0 has a 100GE optical module or a 40GE optical module installed, the port automatically works in 100GE/40GE mode. In this case, port 0 takes effect, but port 1 does not take effect. If a 50GE optical module is inserted into port 0, ports 0 and 1 are both valid, and port 0 works in 50GE mode through auto-sensing. If port 1 works in 50GE mode and services are running on port 1, port 0 does not support auto-sensing.
Breakout supported	Yes. To enable the breakout function on a port, run the port split command.
Interface rate auto-sensing supported	Supports 40GE/50GE/100GE auto-sensing.

Functions and Features	Description
FlexE supported	Yes On the NetEngine 8000 M14, NetEngine 8000E M14, NE40E-X2-M14 and PTN 6900-2-M14: 1. When IPUUs such as CR5D0IPU2TC1, CR5D0IPU2TC2, CR5DIPU1T2C3, CR5DIPU1T2C4, CR5D0IPU2TC3, CR5DIPU1T2C1, CR5D0IPU2TC5, CR5DIPU1T2C5, CR5D0IPU2T91, CR5DIPU1T292, CR8D0IPU2TC1, CR8D0IPU2TC2, CR8DIPU1T2C3, CR8DIPU1T2C4, CR8D0IPU2TC3, CR8D0IPU2TC4, CR8DIPU1T2C7, CR8DIPU1T2C8, CR8D0IPU2TC5, CR8DIPU1T2CJ, CR8D0IPU2TD2, CR8D0IPU2TM1, CR8DIPU1T2M6, CR8D0IPU2TM2, CR8D0IPU2TM3, CR8DIPU1T2M9, TNRDIPU1T2C3, TNRD0IPU2TC1, TNRD0IPU2TC2, TNRDIPU1T2C4, TNRD0IPU2T90 and TNRDIPU1T292 are used, slots 7 to 14 support FlexE, and other slots do not support FlexE. 2. When IPUUs such as CR5DIPU1T2CA, CR5DIPU1T2CB, CR5DIPU1T291, CR8DIPU1T2C5, CR8DIPU1T2C6, CR8DIPU1T2C9, CR8DIPU1T2CA, CR8DIPU1T2CD, CR8DIPU1T2CE, CR8DIPU1T2M0, CR8DIPU1T2CN, CR8DIPU1T2M3, CR8DIPU1T2M2, CR8DIPU1T2MK, CR8DIPU1T2ME, CR8DIPU1T2MS, TNRDIPU1T2C5 and TNRDIPU1T290 are used, slots 3 to 10 support FlexE, and other slots do not support FlexE. Ports on straight-through cards can be bound to form a group. The port ID ranges from 0 to 47 and 1000 to 3000. A maximum of 48 FlexE clients can be created.

Technical Specifications

Table 1-985 Technical specifications of the CR8DE1NE2VC1

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	19.8 mm x 193.8 mm x 209.3 mm (0.78 in. x 7.63 in. x 8.24 in.)
Dimensions with packaging (H x W x D) [mm(in.)]	70 mm x 285 mm x 300 mm (2.76 in. x 11.22 in. x 11.81 in.)
Weight without packaging [kg(lb)]	0.62 kg (1.37 lb)
Weight with packaging [kg(lb)]	0.96 kg (2.12 lb)
Typical power consumption [W]	23.13 W
Typical heat dissipation [BTU/hour]	75.04 BTU/hour
Long-term operating temperature [°C(°F)]	-5°C to +55°C (23°F to 131°F)

1.4.4.71 CR8DE1NEKB11 (1-Port 400GBase-QSFP-DD Physical Interface Card CM(Enable 1*100G)(PIC))

Overview

Table 1-986 Basic information about the CR8DE1NEKB11

Item	Details
Description	1-Port 400GBase-QSFP-DD Physical Interface Card CM(Enable 1*100G)(PIC)
Part Number	03034EDT
Model	CR8DE1NEKB11
Card type	Interface Card
Silkscreen	1x400GE-QSFP-DD

Appearance

Figure 1-194 Appearance of the CR8DE1NEKB11



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03034EDT&productModel=CR8DE1NEKB11#matchRelation>

Indicators and Buttons

Table 1-987 Indicators on the CR8DE1NEKB11

Silkscreen	Name	Color	Status	Description
STAT	Running status indicator	Green	Steady on	The board is working properly.
		Red	Steady on	The hardware on the board is faulty.

Silkscreen	Name	Color	Status	Description
		Orange	Steady on	A high-speed interface card is unexpectedly inserted into a low-speed interface card slot and an alarm is generated, or the logic is not loaded to the board.
		-	Off	The board is not powered on or is not registered.
SRV	Service status indicator	Green	Steady on	The board has no CRC alarm.
		Orange	Steady on	The board has CRC alarm.
		-	Off	The board is not running or no power is input.
LO	Optical Amplifier running status indicator	Green	Steady on	The Optical Amplifier port works normally and there is no LOS.
		-	Off	The Optical Amplifier port does not work normally or LOS occurs.
L/A1	Port running status indicator	Green	Steady on	The link is Up.
		Green	Blink	Data is being transmitted or received.
		-	Off	The link is Down.

Silkscreen	Name	Color	Status	Description
Breakout(0 to 7)	Breakout channel indicator	Green	Steady on	Breakout 0–7: breakout indicators. If breakout indicator X is on, the port indicators are showing the running status of breakout channel X. For example, if breakout indicator 0 is on, port indicators are showing the running status of breakout channel 0 (if a port indicator is blinking, breakout channel 0 on this port is receiving and transmitting data); if breakout indicator 1 is on, port indicators are showing the running status of breakout channel 1. If the breakout indicators 0 to 7 are off, the board has no breakout ports. If the board has breakout ports, each breakout

Silkscreen	Name	Color	Status	Description
				indicator for channels 0 to 7 turns on for 5 seconds in sequence to indicate the status of the corresponding channel. To rapidly identify the running status of a specific breakout channel, press the Breakout button to switch to your desired indicator. For example, if you want to check the running status of breakout channel 1 on each port, press the Breakout button to have breakout indicator 1 turn on.

Table 1-988 Buttons on the CR8DE1NEKB11

Silkscreen	Name	Description
Breakout	Breakout button	When you press the button, the Breakout indicators stop status cycling and remain in the current status. Then, each time you press the button, the Breakout indicators' status switches. If you do not press the button in 60s, the Breakout indicators automatically start status cycling.

Ports

Table 1-989 Ports on the CR8DE1NEKB11

Port	Connector Type	Description	Available Components
OUT0 IN0	QSFP28	Fiber amplifier interface (supported since V800R021C10)	Optical Amplifier Module
OUT1 IN1	QSFP-DD	Inputs and outputs 400GE optical signals.	400Gbps QSFP-DD Optical Module 100Gbps QSFP28 Optical Module

Table 1-990 Port attributes about the CR8DE1NEKB11

Attributes	Details
Working mode	Full-duplex
Standards compliance	IEEE 802.3
Frame format	Ethernet_II, Ethernet_SAP and Ethernet_SNAP

Functions and Features

Table 1-991 Functions and features of the CR8DE1NEKB11

Functions and Features	Description
Basic function	Occupies two flexible card slots. The line side of the board provides one 400GE QSFP-DD interface and one QSFP28 optical amplifier interface. 1. The default rate of a QSFP-DD port is 100 Gbit/s. The initial 100G capability can be upgraded to 200G/400G through 50G RTU. 2. The QSFP28 fiber amplifier interface supports the fiber amplifier module (reserved). Supports FlexE.
Reliability and availability	Hot swappable.
Restrictions and remarks	1. By default, a QSFP-DD port supports 400G QSFP-DD optical modules. To enable a QSFP-DD port to support 100G QSFP28 optical modules instead, run the port split command in the system view. 2. When the board works with a 400G ZR+ optical module, the short-term operating temperature supported by the optical module ranges from -5°C to +50°C.
Breakout supported	No
Interface rate auto-sensing supported	No
FlexE supported	Yes Ports on the straight-through card can be bonded to form a group. port-id ranges from 1000 to 3000. A maximum of 48 FlexE clients can be created.

Technical Specifications

Table 1-992 Technical specifications of the CR8DE1NEKB11

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	40.14 mm x 193.8 mm x 209.3 mm (1.58 in. x 7.63 in. x 8.24 in.)
Dimensions with packaging (H x W x D) [mm(in.)]	115 mm x 317.5 mm x 305 mm (4.53 in. x 12.50 in. x 12.01 in.)
Weight without packaging [kg(lb)]	1.1 kg (2.43 lb)
Weight with packaging [kg(lb)]	1.7 kg (3.75 lb)
Typical power consumption [W]	69.8 W
Typical heat dissipation [BTU/hour]	226.34 BTU/hour
Long-term operating temperature [°C(°F)]	–5°C to +55°C (23°F to 131°F)

1.4.4.72 CR8DE2NB00M0 (2-Port 100GBase/50GBase-QSFP28 Interface Card(PIC))

Overview

Table 1-993 Basic information about the CR8DE2NB00M0

Item	Details
Description	2-Port 100GBase/50GBase-QSFP28 Interface Card(PIC)
Part Number	03035FBD
Model	CR8DE2NB00M0
Card type	Interface Card
Silkscreen	2x100GE-QSFP28

Appearance

Figure 1-195 Appearance of the CR8DE2NB00M0



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03035FBD&productModel=CR8DE2NB00M0#matchRelation>

Indicators

Figure 1-196 Indicators on the CR8DE2NB00M0



Table 1-994 Indicators on the CR8DE2NB00M0

Silkscreen	Name	Color	Status	Description
STAT	Working status indicator	Green	Steady on	The board is working properly.
		Red	Steady on	The hardware on the board is faulty.
		Orange	Steady on	A high-speed interface card is unexpectedly inserted into a low-speed interface card slot and an alarm is generated, or the logic is not loaded to the board.
		-	Off	The board is not powered on or is not registered.

Silkscreen	Name	Color	Status	Description
SRV	Service Status Indicator	Green	Steady on	The board has no CRC alarm.
		Orange	Steady on	The board has CRC alarm.
		-	Off	The board is not running or no power is input.
L/A (0-1)	Interface status indicator	Green	Steady on	The link is Up.
		Green	Blinking	Data is being transmitted or received.
		-	Off	The link is Down.
BREAKOUT (0-3)	Breakout channel indicator	Green	Blink	Each indicator for channels 0, 1, 2, and 3 turns on for 5s in sequence to indicate the status of the corresponding channel. This process repeats.
		-	Off	No data is transmitting or receiving in this channel.

Ports

Table 1-995 Ports on the CR8DE2NB00M0

Port	Connector Type	Description	Available Components
OUT0 IN0-OUT1 IN1	QSFP28	100 GE/50 GE service signal input and output interface	100Gbps QSFP28 Optical Module 40Gbps QSFP+ Optical Module 50Gbps QSFP28 BIDI Optical Module 50Gbps QSFP28 Optical Module Optical Amplifier Module

Table 1-996 Port attributes about the CR8DE2NB00M0

Attributes	Details
Working mode	Full-duplex
Standards compliance	IEEE 802.3
Frame format	Ethernet_II, Ethernet_SAP and Ethernet_SNAP

Functions and Features

Table 1-997 Functions and features of the CR8DE2NB00M0

Functions and Features	Description
Basic function	Occupies one flexible interface card slot. Provides two 100GE/50GE interfaces for line-rate transmitting and receiving, and the interfaces support QSFP28 optical modules.
Reliability and availability	Hot swappable.
Restrictions and remarks	100GE, 50GE, 40GE, 4*25GE, and 4*10GE interfaces are supported.

Functions and Features	Description
Breakout supported	Yes. To enable the breakout function on a port, run the port split command.
Interface rate auto-sensing supported	03035FBD

Functions and Features	Description
FlexE supported	Yes Ports on straight-through cards can be bonded to form a group. 1. If this interface card is installed on a PTN 6900-2-M8C or NetEngine 8000 M8 and works with IPU's such as TNRDIPU1T2C1, TNRDIPU480C1, TNRDIPU1T2C2, TNRDIPU480C2, TNRDIPU1T293, TNRDIPU48091, CR8DIPU1T2C1, CR8DIPU1T2C2, CR8DIPU480C1, CR8DIPU480C2, CR8DIPU240C1, CR8DIPU240C2, CR8DIPU480CT, CR8DIPU1T2CB, CR8DIPU1T2CC, CR8DIPU480C3, CR8DIPU480C4, CR8DIPU240C3, CR8DIPU240C4, CR8DIPU480C5, CR8DIPU480C8, CR8DIPU1T2CG, CR8DIPU480C7, CR8DIPU1T2CU, CR8DIPU480C9, CR8DIPU640C9, CR8PM9BASACQ, CR8DIPU480MA, CR8DIPU480MB, CR8DIPU1T2M5, CR8DIPU480M3, CR8DIPU1T2MG, CR8DIPU480MH, CR8DIPU1T2MT, or CR8DIPU480MG, it supports FlexE and the specifications are as follows: Physical port 0 can be added to a group. port-id ranges from 1000 to 3000. A maximum of 48 FlexE clients can be created. 2. If this interface card is installed on a PTN 6900-2-M4 or NetEngine 8000 M4, it supports the following specifications: Physical ports 0 and 1 can be added to the same group. port-id ranges from 1000 to 3000. A maximum of 48 FlexE clients can be created. 3. If this interface card is installed on a PTN 6900-2-M14, NetEngine 8000 M14, or NE40E-X2-M14 and works with IPU's such as TNRD0IPU2TC1, TNRDIPU1T2C3, TNRD0IPU2TC2, TNRDIPU1T2C4, TNRD0IPU2T90, TNRDIPU1T292, CR8D0IPU2TC1, CR8D0IPU2TC5, CR8DIPU1T2CJ, CR8D0IPU2TC2, CR8DIPU1T2C3, CR8DIPU1T2C4, CR8D0IPU2TC3, CR8D0IPU2TC4, CR8DIPU1T2C7,

Functions and Features	Description
	CR8DIPU1T2C8, CR8D0IPU2TD2, CR8D0IPU2TM1, CR8DIPU1T2M6, CR8D0IPU2TM2, CR5D0IPU2TC1, CR5D0IPU2TC5, CR5DIPU1T2C5, CR5DIPU1T2C3, CR5D0IPU2TC2, CR5DIPU1T2C4, CR5D0IPU2TC3, CR5DIPU1T2C1, CR5D0IPU2T91, or CR5DIPU1T292, it supports FlexE and the specifications are as follows: Physical ports 0 and 1 can be added to the same group. port-id ranges from 1000 to 3000. A maximum of 48 FlexE clients can be created. 4. If this interface card is installed on a PTN 6900-2-M14, NetEngine 8000 M14, or NE40E-X2-M14 and works with IPU's such as TNRD0IPU3T91, TNRD0IPU3T92, TNRDUIPU2T91, TNRDUIPU2T92, CR8D0IPU3TM1, CR8D0IPU3TM2, CR8DUIPU2TM1, CR8DUIPU2TM2, CR8D0IPU3TM3, CR8DUIPU2TM3, CR5D0IPU3TM1, CR5D0IPU3TM2, CR5DUIPU2TM1, or CR5DUIPU2TM2, it supports FlexE and the specifications are as follows: Physical ports 0 and 1 can be added to the same group. port-id ranges from 1000 to 3000. A maximum of 80 FlexE clients can be created.

Technical Specifications

Table 1-998 Technical specifications of the CR8DE2NB00M0

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	19.8 mm x 193.8 mm x 209.3 mm (0.78 in. x 7.63 in. x 8.24 in.)
Dimensions with packaging (H x W x D) [mm(in.)]	70 mm x 285 mm x 300 mm (2.76 in. x 11.22 in. x 11.81 in.)
Weight without packaging [kg(lb)]	0.7 kg (1.54 lb)
Weight with packaging [kg(lb)]	1.2 kg (2.65 lb)
Typical power consumption [W]	41.5 W
Typical heat dissipation [BTU/hour]	134.55 BTU/hour

Item	Specification
Long-term operating temperature [°C(°F)]	–5°C to +55°C (23°F to 131°F)

1.4.4.73 CR8DE4NB00M1 (4-Port 100GBase/50GBase-QSFP28 Physical Interface Card(PIC))

Overview

Table 1-999 Basic information about the CR8DE4NB00M1

Item	Details
Description	4-Port 100GBase/50GBase-QSFP28 Physical Interface Card(PIC)
Part Number	03034YNN
Model	CR8DE4NB00M1
Card type	Interface Card
Silkscreen	4x100GE-QSFP28

Appearance

Figure 1-197 Appearance of the CR8DE4NB00M1



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03034YNN&productModel=CR8DE4NB00M1#matchRelation>

Indicators

Table 1-1000 Indicators on the CR8DE4NB00M1

Silkscreen	Name	Color	Status	Description
STAT	Working status indicator	Green	Steady on	The board is working properly.
		Red	Steady on	The hardware on the board is faulty.
		Orange	Steady on	A high-speed interface card is unexpectedly inserted into a low-speed interface card slot and an alarm is generated, or the logic is not loaded to the board.
		-	Off	The board is not powered on or is not registered.
SRV	Service Status Indicator	Green	Steady on	The board has no CRC alarm.
		Orange	Steady on	The board has CRC alarm.
		-	Off	The board is not running or no power is input.
L/A (0-3)	Interface status indicator	Green	Steady on	The link is Up.
		Green	Blinking	Data is being transmitted or received.

Silkscreen	Name	Color	Status	Description
		-	Off	The link is Down.
BREAKOUT (0-3)	Breakout channel indicator	Green	Blink	Each indicator for channels 0, 1, 2, and 3 turns on for 5s in sequence to indicate the status of the corresponding channel. This process repeats.
		-	Off	No data is transmitting or receiving in this channel.

Ports

Table 1-1001 Ports on the CR8DE4NB00M1

Port	Connector Type	Description	Available Components
OUT0 IN0-OUT3 IN3	QSFP28	100GE/50GE service signal input and output interface	100Gbps QSFP28 Optical Module 50Gbps QSFP28 BIDI Optical Module 50Gbps QSFP28 Optical Module 40Gbps QSFP+ Optical Module Optical Amplifier Module

Table 1-1002 Port attributes about the CR8DE4NB00M1

Attributes	Details
Working mode	Full-duplex

Attributes	Details
Standards compliance	IEEE 802.3
Frame format	Ethernet_II, Ethernet_SAP and Ethernet_SNAP

Functions and Features

Table 1-1003 Functions and features of the CR8DE4NB00M1

Functions and Features	Description
Basic function	Occupies one flexible card slot. Provides four 100GE/50GE interfaces for line-rate transmitting and receiving, and supports QSFP28 optical modules.
Reliability and availability	Hot swappable.
Restrictions and remarks	100GE, 50GE, 40GE, 4*25GE, and 4*10GE interfaces are supported.
Breakout supported	Yes. To enable the breakout function on a port, run the port split command.
Interface rate auto-sensing supported	Supports 40GE/50GE/100GE auto-sensing.
FlexE supported	Yes Ports on straight-through cards can be bonded to form a group. Physical ports 0, 1, 2, 3 can be added to the same group. port-id ranges from 1000 to 3000. A maximum of 80 FlexE clients can be created.

Technical Specifications

Table 1-1004 Technical specifications of the CR8DE4NB00M1

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	19.8 mm x 193.8 mm x 209.3 mm (0.78 in. x 7.63 in. x 8.24 in.)

Item	Specification
Dimensions with packaging (H x W x D) [mm(in.)]	70 mm x 285 mm x 300 mm (2.76 in. x 11.22 in. x 11.81 in.)
Weight without packaging [kg(lb)]	0.7 kg (1.54 lb)
Weight with packaging [kg(lb)]	1.2 kg (2.65 lb)
Typical power consumption [W]	50.0 W
Typical heat dissipation [BTU/hour]	162.35 BTU/hour
Long-term operating temperature [°C(°F)]	-5°C to +55°C (23°F to 131°F)

1.4.5 VSUP

1.4.5.1 CR8DGSUPA0C0 (Multi-function Integrated Service Card(VSUP-100))

Overview

Table 1-1005 Basic information about the CR8DGSUPA0C0

Item	Details
Description	Multi-function Integrated Service Card(VSUP-100)
Part Number	03035AAY
Model	CR8DGSUPA0C0
Card type	Service Card
Silkscreen	VSUP-100

Appearance

Figure 1-198 Appearance of the CR8DGSUPA0C0



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03035AAY&productModel=CR8DGSUPA0C0#matchRelation>

Indicators

Table 1-1006 Indicators on the CR8DGSUPA0C0

Silkscreen	Name	Color	Status	Description
STAT	Running status indicator	Green	Steady on	The board is working properly.
		Green	Blinking	The board is registering. In normal cases, the indicator starts to blink 6s after the board is powered on.
		Red	Steady on	The hardware on the board is faulty.

Functions and Features

Table 1-1007 Functions and features of the CR8DGSUPA0C0

Functions and Features	Description
Basic function	Occupies one flexible card slot. The memory size of the processor is 32 GB.
Reliability and availability	Support for hot swap.

Technical Specifications

Table 1-1008 Technical specifications of the CR8DGSUPA0C0

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	19.8 mm x 193.8 mm x 209.3 mm (0.78 in. x 7.63 in. x 8.24 in.)
Dimensions with packaging (H x W x D) [mm(in.)]	115 mm x 317.5 mm x 305 mm (4.53 in. x 12.50 in. x 12.01 in.)
Weight without packaging [kg(lb)]	0.8 kg (1.76 lb)
Weight with packaging [kg(lb)]	1.4 kg (3.09 lb)
Typical power consumption [W]	78.5 W
Typical heat dissipation [BTU/hour]	254.74 BTU/hour
Long-term operating temperature [°C(°F)]	–20°C to +55°C (–4°F to +131°F)
SDRAM	32GB

1.4.5.2 CR8DVSUPA0C0 (Integrated Service Card VSUP-100)

Overview

Table 1-1009 Basic information about the CR8DVSUPA0C0

Item	Details
Description	Integrated Service Card VSUP-100
Part Number	03034EDS
Model	CR8DVSUPA0C0
Card type	Service Card
Silkscreen	VSUP-100

Appearance

Figure 1-199 Appearance of the CR8DVSUPA0C0



Version Mapping

The huge hardware mapping data is migrated to the Info-Finder hardware center, where you can easily obtain hardware mapping information.

The Info-Finder hardware center incorporates comprehensive hardware information and mapping data by sales regions, allowing you to quickly find desired information and filter the mapping data for more fine-grained data display.

enterprise: <https://info.support.huawei.com/info-finder/search-center/en/enterprise/routers/netengine-8000-pid-252772223/hardwarecenter?keyword=03034EDS&productModel=CR8DVSUPA0C0#matchRelation>

Indicators

Table 1-1010 Indicators on the CR8DVSUPA0C0

Silkscreen	Name	Color	Status	Description
STAT	Running status indicator	Green	Steady on	The board is working properly.
		Green	Blinking	The board is registering. In normal cases, the indicator starts to blink 6s after the board is powered on.
		Red	Steady on	The hardware on the board is faulty.
		-	Steady off	The board is not running or no power is input.

Functions and Features

Table 1-1011 Functions and features of the CR8DVSUPA0C0

Functions and Features	Description
Basic function	Occupies one flexible interface card slot. Supports the CGN universal service processing capability. CGN is supported since V800R021C00. The memory size of the processor is 32 GB.
Reliability and availability	Support for hot swap.

Technical Specifications

Table 1-1012 Technical specifications of the CR8DVSUPA0C0

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	19.8 mm x 193.8 mm x 209.3 mm (0.78 in. x 7.63 in. x 8.24 in.)
Dimensions with packaging (H x W x D) [mm(in.)]	115 mm x 317.5 mm x 305 mm (4.53 in. x 12.50 in. x 12.01 in.)
Weight without packaging [kg(lb)]	0.8 kg (1.76 lb)
Weight with packaging [kg(lb)]	1.4 kg (3.09 lb)
Typical power consumption [W]	78.5 W
Typical heat dissipation [BTU/hour]	254.74 BTU/hour
Long-term operating temperature [°C(°F)]	-20°C to +55°C (-4°F to +131°F)
SDRAM	32GB

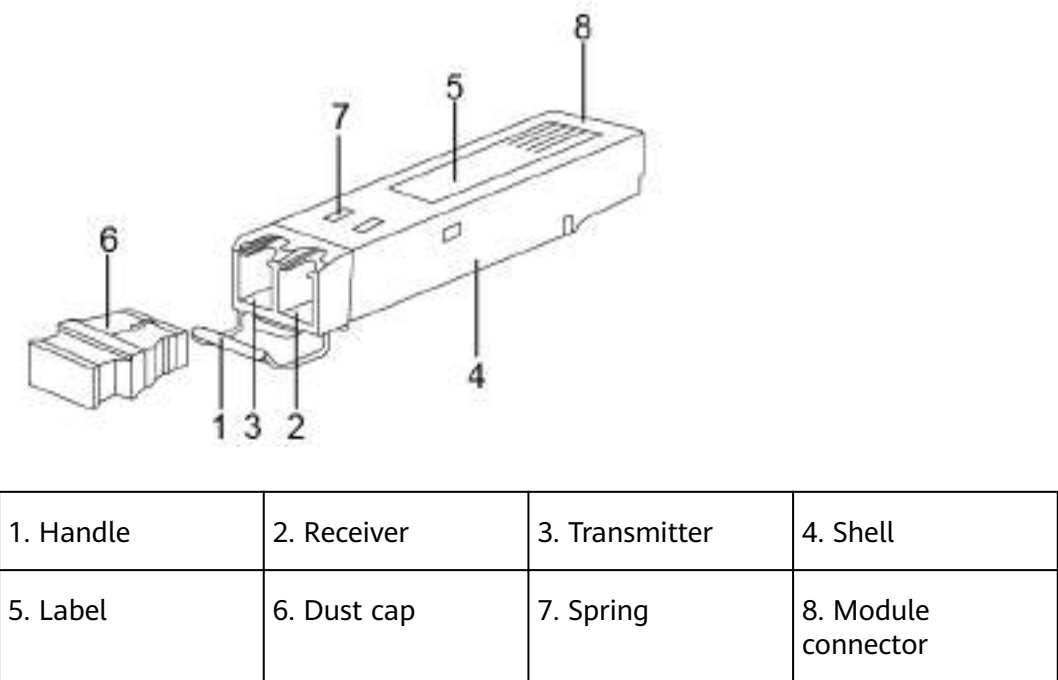
1.5 Optical Module

1.5.1 Understanding Pluggable Optical Modules

1.5.1.1 Optical Module Structure

Figure1 shows the structure of an optical module.

Figure 1-200 Optical module structure



1.5.1.2 Optical Module Classification

Optical modules are available in various types to meet diversified requirements.

- **Classified by transmission rate**
Currently, the transmission rates of optical modules cover a wide range. According to different transmission rates, optical modules can be classified into 400 Gbit/s optical modules, 200 Gbit/s optical modules, 100 Gbit/s optical modules, 40 Gbit/s optical modules, 25 Gbit/s optical modules, and 10 Gbit/s optical modules, 2.5 Gbit/s optical modules, 1.25 Gbit/s optical modules, 1000 Mbit/s optical modules, 155 Mbit/s optical modules, and 100 Mbit/s optical modules.
- **Classified by encapsulation type**
The higher transmission rate an optical module provides, the more complex structure it has. According to the encapsulation type, optical modules are classified into SFP, eSFP, SFP+, XFP, SFP28, QSFP28, QSFP+, CXP, CFP and CSFP.
 - SFP: small form-factor pluggable.
 - eSFP: enhanced small form-factor pluggable. An eSFP optical module is an SFP optical module that supports monitoring of voltage, temperature, bias current, transmit optical power, and receive optical power. Currently, SFP modules also have the preceding functions. Therefore, eSFP and SFP optical modules are both called SFP optical modules.
 - SFP+: small form-factor pluggable plus. An SFP+ optical module is an SFP module with a higher rate. SFP+ optical modules are more sensitive to electromagnetic interference (EMI) because they have a higher rate. Compared with SFP modules, SFP+ optical modules use more springs and tighter SFP+ optical module cages.

- XFP: 10G small form-factor pluggable. X is the Roman numeral 10, meaning that all XFP optical modules provide a transmission rate of 10 Gbit/s. XFP optical modules are wider and longer than SFP+ optical modules.
 - SFP28: small form-factor pluggable 28. The size of an SFP28 optical module is the same as that of an SFP+ optical module. An SFP28 port can use a 25G SFP28 optical module or 10G SFP+ optical module.
 - QSFP28: quad small form-factor pluggable 28. The size of a QSFP28 optical module is the same as that of a QSFP+ optical module. A QSFP28 port can use either a 100G QSFP28 optical module or a 40G QSFP+ optical module.
 - QSFP+: quad small form-factor pluggable plus. QSFP+ optical modules support MPO fiber connectors and are larger than SFP+ optical modules.
 - QSFP-DD: quad small form factor pluggable-double density. A QSFP-DD optical module is a high-speed pluggable module defined by the QSFP-DD MSA group.
 - CXP: extended-capability form-factor pluggable. A CXP optical module is a hot-pluggable high-density parallel optical module, which provides 12 channels of traffic in each of the Tx and Rx directions. It applies only to short-distance multimode links.
 - CFP: centum form-factor pluggable, a new standard for high-speed, hot-pluggable optical transceivers that support data communication and telecommunication applications. The dimensions of a CFP optical module are 13.6 mm x 144.75 mm x 82 mm (0.54 in. x 5.70 in. x 3.23 in.).
 - CSFP: compact small form-factor pluggable. A CSFP optical module is a compact SFP transceiver with two 100 Mbit/s or 1 Gbit/s single-fiber bidirectional transceivers inside a standard SFP form factor. Its compact and low-power design allows the system supplier to double port density and access channel quantity for the LPU. This type of optical module is mainly used in scenarios where one CSFP optical module connects to two BIDI SFP optical modules. It is essential to ensure that the transmit and receive wavelengths are consistent and the transmission distances are the same.
- **Classified by physical layer standard**

Different physical layer standards are defined to allow data transmission in different modes. Therefore, different types of optical modules are produced to comply with these standards. For details, see **Standards compliance** of the specific optical module.
 - **Classified by mode**

Optical fibers are classified into single-mode and multimode fibers. Therefore, optical modules are also classified into single-mode and multimode modules to support different optical fibers.

 - Single-mode optical modules are used with single-mode fibers. Single-mode fibers support a wide band and large transmission capacity, and are used for long-distance transmission.
 - Multimode optical modules are used with multimode fibers. Multimode fibers have lower transmission performance than single-mode fibers because of modal dispersion, but their costs are also lower. They are used for small-capacity, short-distance transmission.

WDM modules differ from other types of optical modules in center wavelengths. A common optical module has a center wavelength of 850 nm, 1310 nm, or 1550 nm, whereas a WDM module has different center wavelengths. WDM modules are classified into two types: coarse wavelength division multiplexing (CWDM) and dense wavelength division multiplexing (DWDM). Within the same band, DWDM modules are available in more types and use wavelength resources more efficiently than CWDM modules. DWDM and CWDM modules allow lights with different center wavelengths to be transmitted on one fiber without interfering each other. Therefore, a passive multiplexer can be used to combine the lights into one channel, which is then split into multiple channels by a demultiplexer on the remote end. This reduces the optical fibers required. DWDM and CWDM modules are used for long-distance transmission.

The transmit power of a long-distance optical module is often larger than its overload power. Therefore, when using such optical modules, select optical fibers of an appropriate length to ensure that the actual receive power is smaller than the overload power. If the optical fibers connected to a long-distance optical module are too short, use an optical attenuator to reduce the receive power on the remote optical module. Otherwise, the remote optical module may be burnt.

1.5.1.3 Optical Module Appearance

The following lists some common optical modules, which may not be supported by this product. The figures are for reference only.

Table 1-1013 Commonly used optical modules

Encapsulation type	Interface type	Appearance
SFP	LC	Single-fiber bidirectional 
	RJ45	1 Gbps electrical module 

Encapsulation type	Interface type	Appearance
eSFP	LC	Two-fiber bidirectional  Single-fiber bidirectional 
SFP+	LC	Two-fiber bidirectional  Single-fiber bidirectional 

Encapsulation type	Interface type	Appearance
XFP	LC	Two-fiber bidirectional  Single-fiber bidirectional 
SFP28	LC	Two-fiber bidirectional  Single-fiber bidirectional 
QSFP28	LC/MPO	Two-fiber bidirectional  Single-fiber bidirectional 

Encapsulation type	Interface type	Appearance
QSFP+	LC/MPO	
CXP	MPO	

Encapsulation type	Interface type	Appearance
CFP	LC/MPO	CFP  CFP2  CFP4  CFP8 

Encapsulation type	Interface type	Appearance
CSFP	LC	
QSFP-DD	LC	

1.5.1.4 How to Use an Optical Module

This section provides instructions on how to use an optical module.

 NOTE

Only optical modules matching Huawei products can be used. These optical modules are strictly tested by Huawei. If non-matching optical modules are used, device requirements may fail to be met, and services may fail to run properly. To replace optical modules, see [Parts Replacement > Replacing an Optical Module](#).

ESD Measures

Before touching any optical module, wear an ESD wrist strap or ESD gloves. Take full ESD measures when installing optical apparatus such as optical modules indoors or outdoors.

Figure 1-201 Method for wearing ESD gloves



Figure 1-202 Method for wearing the ESD wrist strap

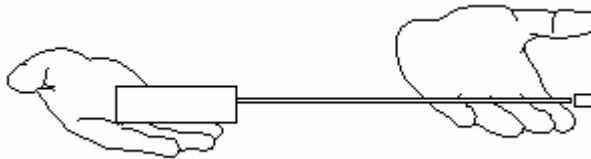


Placing Optical Apparatus and Fibers

Do not touch the pins or edge connectors of optical modules with bare hands. Handle optical fibers gently. Use two fingers to hold the fiber connector instead of grasping the fiber or fiber cover.

Avoid applying axial or lateral pressure to a fiber. Refrain from folding or twisting a fiber or crushing its pigtail. Do not drag a fiber or apply pressure to the coupling point of its pigtail. [Figure 1-203](#) shows how to properly place optical apparatus and fibers.

Figure 1-203 Methods for placing optical apparatus and fibers



NOTE

During installation, ensure that an optical fiber is coiled into a loop with a diameter of at least 6 cm.

Uninstalling Optical Apparatus

- Release the latch and slowly take out the optical apparatus. Do not drag the optical fiber to forcibly take out the optical module. Ensure that the optical fiber is connected to and removed from the optical interface horizontally. Before taking out an optical module, remove its optical fiber and release its latch first.

Figure 1-204 Closed latch

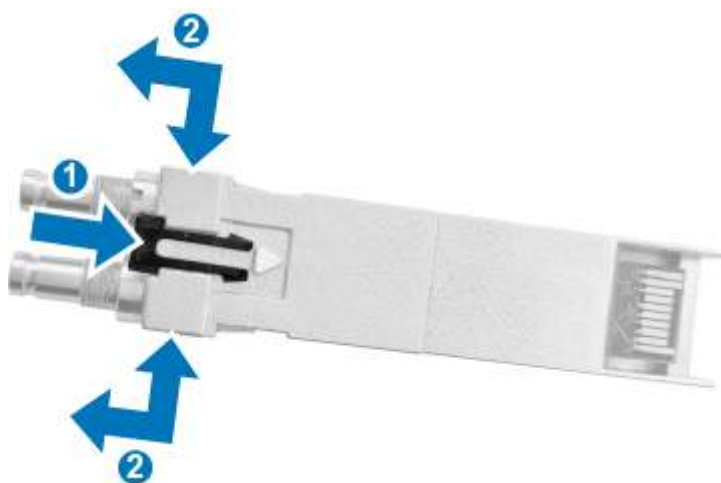


Figure 1-205 Released latch



- The black plastic latch shown by (1) in [Figure 1-206](#) is the unlocking device for the 155 Mbit/s electrical transceiver. You can press the latch to unlock the optical module. Hold the two sides of the optical module to remove it, as shown by (2) in [Figure 1-206](#).

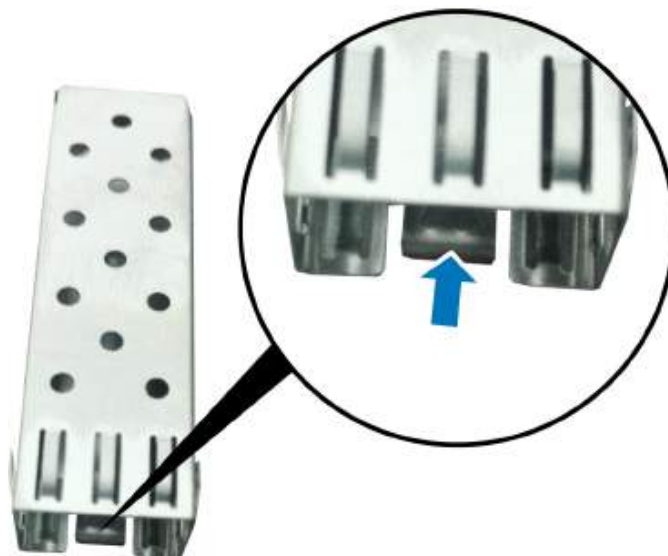
Figure 1-206 Removing a 155 Mbit/s electrical transceiver



Do not remove the black plastic latch when removing the 155 Mbit/s electrical transceiver.



If the black plastic latch falls off, use an auxiliary tool, such as a pair of tweezers, to press the cage latch, as shown in the following figure. Then, hold the two sides of the optical module to remove it.

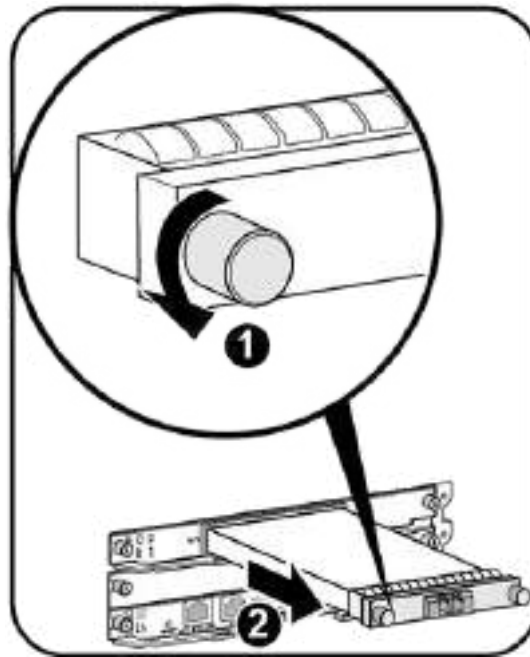


- If the CFP2 optical module is used, its unlocking device is a spring clip, as shown by (1) in [Figure 1-207](#). Before removing such an optical module, release the latch first. Then, hold the latch to horizontally remove the optical module.

Figure 1-207 Removing a CFP2 optical module



- When removing a CFP optical module, loosen the two screw rods of the module and then remove the module slowly. Do not directly drag the optical fiber to pull out the optical module or forcibly pull out the optical module.



CAUTION

The QSFP28 and QSFP-DD modules will get very hot during operation. To prevent injuries, do not touch the module shells when removing the modules.

Precautions for Loose Optical Modules

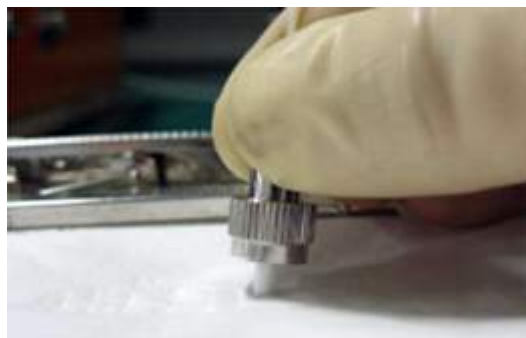
- When installing an optical module, firmly push it into position until you hear a click or feel a slight tremor, signaling that the optical module is securely locked. If the optical module is not securely locked, its edge connector may be loosely connected to the board connector, and the link may still be up. However, if the optical module experiences tremors or collisions with other objects, it may come loose or temporarily lose optical signals.
- When inserting the optical module, ensure that the latch is closed (the latch locks the optical module). After the optical module is inserted, try pulling it out to see if it is installed in position. If the optical module cannot be pulled out, it is secured.
- If you cannot push the optical module into an optical module cage any further, the optical module is in good contact with the board connector.
- When installing a CFP optical module, push the module panel horizontally into the connector using even force with both thumbs. After the module is inserted, push the module slightly to ensure that it has reached the stop position.
- When inserting a CFP optical module into a board, ensure that the module is in full contact with the connector. At the same time, fasten the captive screws at both ends of the CFP optical module. If you fasten the captive screws one by one, the CFP optical module may fail to be securely installed. You are advised to use a screwdriver to tighten the screws on the optical module.

Precautions for Receptacle Contamination

- Prepare cleaning tissues for onsite use. Clean an optical connector before inserting it into the receptacle to protect the receptacle against contamination.

NOTE

Use at least three cleaning tissues. Wipe the end of an optical connector horizontally in one direction, and then shift the connector to the unused part of the cleaning tissue to continue. Generally, one cleaning tissue is used for cleaning one optical connector.



- Cover optical modules that are temporarily unused with dust-proof caps to prevent dust pollution. If no dust-proof caps are available, inserting fibers can effectively prevent dust from entering these optical modules.

Covering temporarily unused optical modules with dust-proof caps



Using a fiber instead of a dust-proof cap to prevent dust pollution



- Cover optical fiber jumpers that are temporarily unused with dust-proof caps and place them on the optical distribution frame (ODF) or coil them up in a fiber management tray. Ensure that they are not compressed or squeezed.



- If optical modules or fibers remain unused for a long time without being covered with dust-proof caps, clean their receptacles or optical connectors before using them again. Use cotton swabs to clean receptacles and cleaning tissues to clean optical connectors.

 NOTE

When cleaning a receptacle, gently insert a cotton swab and rotate it without applying excessive force. Otherwise, the receptacle may be damaged.



- If optical signals are unexpectedly lost during device operation, use the previously mentioned method to clean the receptacle or optical connector. This can help exclude contamination as the cause of signal loss.

Precautions for Overload-caused Burnt Optical Modules

- When using an OTDR to test the connectivity or attenuation of a fiber channel, disconnect the optical fiber from the optical module. Otherwise, the optical module may be burnt.
- When performing a self-loop test, use an optical attenuator if necessary. Do not loosen the optical connector instead.
- It is required that a long-distance optical module have an input optical power of less than -7 dBm. If the input optical power is greater than -7 dBm, you need to add an optical attenuator. For example, if the transmit optical power is X dBm and the optical attenuation is Y dB, the receive optical power is $X - Y$, which must be smaller than -7 dBm ($X - Y < -7$ dBm).

Inspecting Optical Fiber Connectors

As the Ethernet port rate continues to rise, the quality standards for optical fibers and cables rise accordingly. [Table 1-1014](#) describes requirements for the loss of optical fiber connectors according to the national standard (GBT50312-2016).

Table 1-1014 Maximum attenuation of optical fiber connectors

Type	Maximum Attenuation of an Optical Fiber Connector (dB)
Fiber splicing connector	0.3
Optical mechanical connector	0.3
Optical connector	0.75

 NOTE

Fiber cores are connected through connectors, such as the ODF, optical attenuator, and flange, in splicing and mechanical modes.

[Table 1-1015](#) describes requirements in IEEE Std 802.3-2018 for the reflection of the optical fiber connector when Ethernet ports (such as 200G and 50G) use PAM4 encoding to double the rate. More connectors bring lower requirements for the reflection.

Table 1-1015 Maximum reflection of connectors

Number of Optical Fiber Connectors	Maximum Reflection of Each Connector (dB)
1	-22
2	-29
4	-33
6	-35
8	-37
10	-39

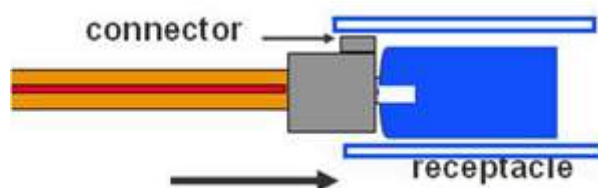
The loss and reflectance values of optical fiber connectors are tested and processed as follows:

1. Disconnect the peer optical fiber and use the OTDR to test the loss and reflection of each link and node at the local end. Check whether the loss and reflection of each link and node are normal (splicing connector loss < 0.3 dB/node, connector loss < 0.75 dB/node, and connector reflection < -30 dB/node). If the loss and reflection exceed the limits, handle the abnormal interface.

2. Based on the distance of the abnormal point in the OTDR test result, locate the equipment room where the interface is located and preliminarily determine the interface location. Disconnect the interface and perform the OTDR test on the interface that reports the alarm to check whether the distance is consistent with the previous test. If not, continue to test other interfaces.
3. After locating the abnormal interface, use a fiberscope to check the interface. If the interface is contaminated, clean it.
4. After the interface is cleaned, restore the interface and tighten the connector. Perform an OTDR test on the interface that reports alarms to check whether the loss and reflection of each link and node are normal.
5. If the fault persists, replace the flange plate and perform an OTDR test on the interface that reports the alarm to check whether the loss and reflection of each link and node are normal.
6. If the fault persists, replace the optical fiber and perform an OTDR test on the interface that reports the alarm to check whether the loss and reflection of each link and node are normal.
7. If there are multiple abnormal points on the link, repeat steps 2 to 6.

Other Precautions

- An optical connector must be horizontally inserted into a receptacle to avoid damages to the receptacle.



- Mixed use of multimode and single-mode optical fibers is prohibited. Otherwise, faults such as signal loss may occur.
The following table describes methods for distinguishing single-mode optical modules from multimode optical modules.

Table 1-1016 Method for distinguishing single-mode optical modules from multimode optical modules

Item	Single-Mode	Multimode
Transmission distance	10 km or longer	Below 0.5 km
Wavelength	Non-850 nm	850 nm
Information on the label	SM	MM

50G Optical Module Installation

1. Precautions for optical module installation

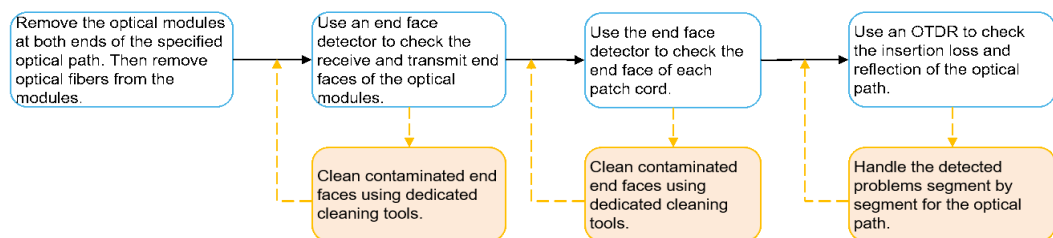
(1) If a cabinet with a door is used, a sufficient distance must be reserved between the optical module and the cabinet door to prevent the latch or pigtail from bumping on the door.

(2) Long-distance optical modules must be equipped with optical attenuators for self-loops. For a 50GBase-ER (40 km) long-distance optical module, the receive optical power damage threshold is lower than the average minimum transmit optical power, making the module prone to damage caused by self-loops. Therefore, the module must be equipped with an optical attenuator for self-loops.

(3) When optical path quality is tested using an OTDR, optical fibers must be removed from the associated optical module. This is because the OTDR's transmit optical power is far greater than the optical power damage threshold at the receive end of an optical module.

2. Method for checking an optical path

Figure 1-208 Method for checking an optical path



3. Method for cleaning the end faces of an optical fiber

Before cleaning the end faces of an optical fiber that is in use, ensure that the optical fiber has no optical signals. To achieve this, shut down the ports at both ends of the fiber. Then, clean the end faces and insert the optical fiber back into the corresponding port.

To clean the end faces of an optical fiber that is not in use, remove the dust-proof cap from the fiber connector (or the pigtail connector of the involved optical component), and put the dust-proof cap into a dedicated cleaning kit. After the cleaning is complete, re-install the dust-proof cap.

- Use the untouched part of a lint-free wipe to wipe the connector end face along one direction.
- If the end face of an optical fiber cannot be cleaned due to serious contamination, use a lint-free wipe dipped with cleanser to wipe the end face along one direction. Then, use a dry lint-free wipe to clean the end face. Ensure that the end face is dry before using the optical fiber.
- After the cleaning is complete, immediately install a dust-proof cap for any optical fiber connector that is not in use.

4. Precautions for using a lint-free wipe to clean the end face of an optical fiber

- Use the smooth surface of the lint-free wipe for cleaning.
- Ensure that the optical fiber connector is vertical to the lint-free wipe during cleaning.

- Wipe the end face along the direction of the lint-free wipe's grain.
- Wipe the end face along one direction only.
- Any part of a lint-free wipe can be used only once, and a small piece of lint-free wipe can be used to clean only one connector.

5. Method for using lint-free swabs to clean the optical port of an optical module

Remove the dust-proof cap from the optical port of the optical module, and put the dust-proof cap into a dedicated cleaning kit.

- Select a proper lint-free swab based on the type of the optical port to be cleaned. (For SC optical ports, use lint-free swabs with a diameter of 2.5 mm; for LC and MTRJ optical ports, use lint-free swabs with a diameter of 1.25 mm.) Dip the lint-free swab into cleanser, insert it into the inside of the optical port, and clean the optical port by rotating the swab 360 degrees in one direction along the inner wall of the optical port.
- Insert a dry lint-free swab of the same type into the optical port and clean the optical port by rotating the swab 360 degrees in one direction along the inner wall of the optical port.
- Cover the optical port with a dust-proof cap after the cleaning is complete.

6. Precautions for using lint-free swabs to clean the optical port of an optical module

- When cleaning the optical port of an optical module, clean the end faces of associated optical fibers to prevent the optical fibers from contaminating the optical port.
- In general, each lint-free swab can be used for cleaning only once. If a used lint-free swab is confirmed clean and can be reused, it can be used for a maximum of three times. For example, a lint-free swab that is ever used to dry an optical port can be used for a maximum of three times.

7. Safety precautions

- Electrostatic protection: Active optical and electrical components are extremely sensitive to electrostatic. Therefore, take strict measures to protect against electrostatic. For example, wear ESD gloves during operations and touch only the shell of the involved component.
- Laser protection: Do not look into optical ports without eye protection when reseating a module.

8. Discrete reflectance

During link deployment, focus on the reflection indicators of each node. The discrete reflection indicators must meet the IEEE specifications.

Table 1-1017 Maximum discrete reflectances of QSFP28 50G defined by IEEE

Number of Discrete Reflectances Greater Than -55 dB	QSFP28 50G-FR (Maximum Value of Each Discrete Reflectance Coefficient)	QSFP28 50G-LR (Maximum Value of Each Discrete Reflectance Coefficient)	QSFP28 50G-ER (Maximum Value of Each Discrete Reflectance Coefficient)
1	-25 dB	-22 dB	-19 dB
2	-31 dB	-29 dB	-27 dB
4	-35 dB	-33 dB	-32 dB
6	-38 dB	-35 dB	-35 dB
8	-40 dB	-37 dB	-37 dB
10	-41 dB	-39 dB	-39 dB

1.5.1.5 Optical Attenuator Configuration

This section describes how to configure an optical attenuator.

Calculating the Optical Attenuation

You can calculate the optical attenuation based on the actual optical power.

Table 1-1018 Parameters for calculating optical attenuation

Name	Description
P(in)min	Worst sensitivity (maximum value of the worst sensitivity).
P(out)max	Maximum transmit optical power.
S	Transmission distance.
A	Attenuation coefficient. Note that the attenuation coefficient is related to optical fiber types and wavelengths. Generally, the attenuation coefficient is 0.45 dBm/km or 0.4 dBm/km. The attenuation coefficient of a 1550-nm wavelength is 0.235 dBm/km or 0.25 dBm/km.
P(in)max	Maximum receive optical power, that is, minimum overload point.

The principle for determining whether an attenuator needs to be configured at a transmission point is as follows:

If $P(\text{out})_{\text{max}} - S \times A > P(\text{in})_{\text{max}}$, an attenuator needs to be configured. The optical attenuation is calculated in the following formula: $T = P(\text{out})_{\text{max}} - S \times A - P(\text{in})_{\text{max}}$.

Table 1-1019 Reference for configuring an attenuator

Part Number	Description	P(out) max	P(out) min	P(in) min	P(in) max
34060276	eSFP, 1310nm,STM1,LC,SM, 15km	-8 dBm	-15 dBm	-31 dBm	-8 dbm

 NOTE

- If $P(\text{in})_{\text{max}}$ of an optical module equals $P(\text{out})_{\text{max}}$, you do not need to configure an attenuator.
- You can choose the 5 dBm and 10 dBm attenuators for optical modules on the device.

Part Numbers and Descriptions of Attenuators

Table 1-1020 Part numbers and descriptions of attenuators

Part Number	Description
45030021	Fixed Optical Attenuator, 1260nm~1620nm-5dB-LC/PC-45dB
45030022	Fixed Optical Attenuator, 1260nm~1620nm-10dB-LC/PC-45dB

 NOTE

This table is for reference only. The part numbers of attenuators vary according to configuration documents.

1.5.2 OA Overview

1.5.2.1 Functions

An optical amplifier (OA) is a C-band pluggable optical amplification module, which can be configured at the transmit or receive end of a device according to the actual scenario.

Table 1-1021 OA functions

Function	Description
Basic function	Amplifies optical signals over C-band wavelengths in the range from 1535 nm to 1547 nm.
Pluggable	Is pluggable.
Gain adjustment	Adjusts the gain based on the input optical power to achieve continuous gain adjustment in the range from 9 dB to 23 dB.
Power adjustment	Fixes the output power at 7 dBm according to the input power range.
Working mode	Indicates the working mode. In power locking and gain locking modes, the EDFA module's gain is adjustable and its actual gain can be queried.
Performance and alarm monitoring	- Monitors and reports optical power. - Controls the temperature of the pump laser. - Monitors the pump drive current, cooling current, pump laser temperature, and ambient temperature of the board.

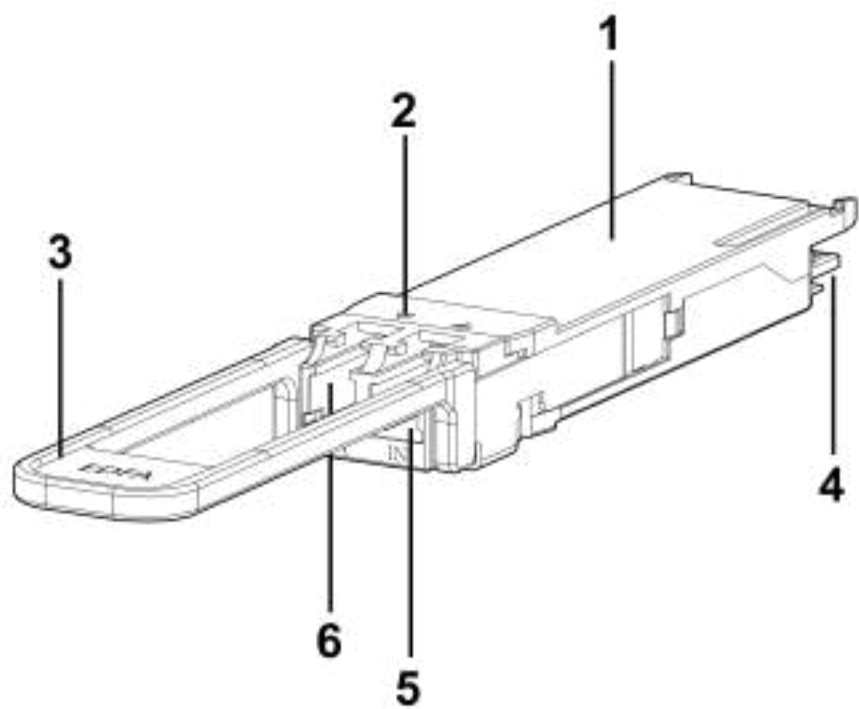
1.5.2.2 Appearance and Structure

Figure 1-209 Appearance of an OA



Figure 1-210 shows the structure of an OA.

Figure 1-210 Structure of an OA



1. Shell	2. Signal direction indication	3. Pull tab
4. Connector	5. Receive port	6. Transmit port

Table 1-1022 Type and function of OA ports

Port Silkscreen	Port Type	Function
IN/OUT	LC	Amplifies multiplexed signals.

1.5.2.3 Usage Scenarios

When a router uses an 80 km QSFP-DD optical module, an OA is required. OAs are deployed at the transmit or receive end based on the transmission distance between two interconnected routers and the insertion loss of optical links, as shown in the following figures.



NOTE

1. When the transmission distance ranges from 40 km to 80 km and the insertion loss of optical links is less than 22 dB, OAs need to be deployed only at the transmit end.
2. When the transmission distance ranges from 80 km to 120 km:
 - If the insertion loss of an optical link is less than or equal to 24 dB, OAs need to be deployed only at the transmit end.
 - If the insertion loss of an optical link is greater than 24 dB but less than or equal to 28 dB, OAs must be deployed at both the transmit and receive ends, and the receive optical power threshold of the QSFP-DD optical module must be manually modified.
 - If the insertion loss of an optical link is greater than 28 dB, reduce the insertion loss to a value less than 28 dB.

Figure 1-211 shows how to deploy an OA at the transmit end.

Figure 1-211 Deploying an OA at the transmit end

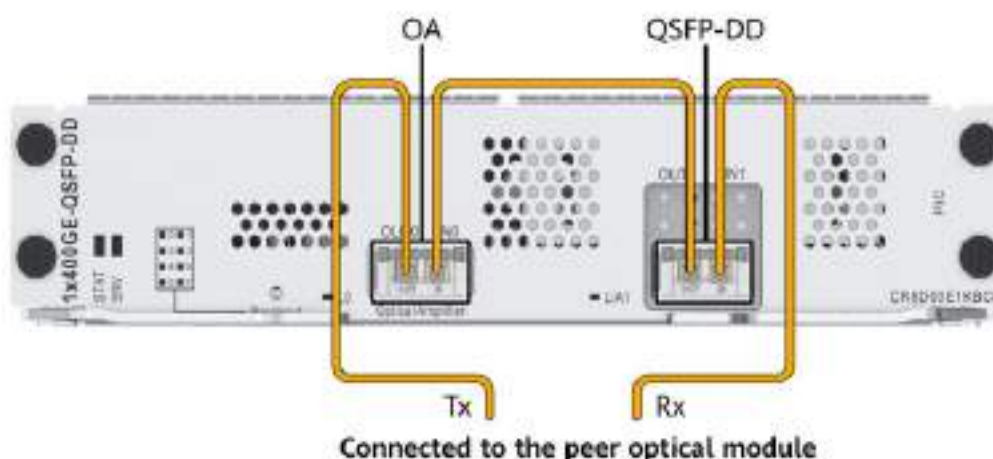
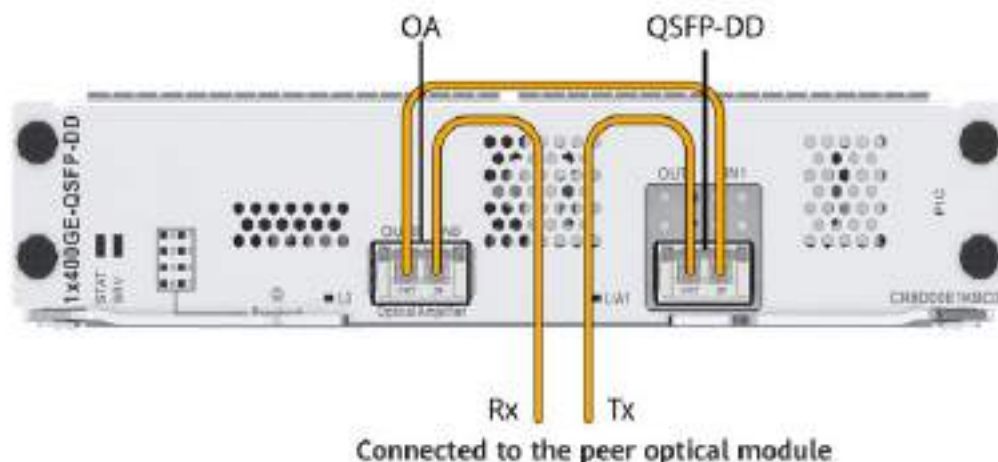


Figure 1-211 shows how to deploy an OA at the receive end.

Figure 1-212 Deploying an OA at the receive end



1.5.2.4 Guide to Using OAs

NOTE

- OAs need to be used in indoor scenarios with thermal power exchange or in environments that can guarantee the working conditions of OAs.
- An OA can be inserted into a port for installing the QSFP-DD/QSFP28/QSFP+ optical module or a dedicated power supply port on a router.
- An OA can amplify the optical signals of only one fiber.
- An OA can be connected only to a 400G ZR/ZR+ optical module, either on the same board or on a different board, through an optical fiber.
- Only OAs matching Huawei products can be used. If non-matching OAs are used, device requirements may fail to be met, and services may fail to run properly.

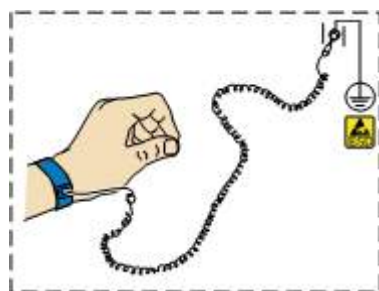
ESD Measures

Before touching an OA, wear ESD gloves or an ESD wrist strap. Take ESD measures when installing an OA indoors or outdoors.

Figure 1-213 Methods of wearing an ESD glove



Figure 1-214 Methods of wearing an ESD wrist strap



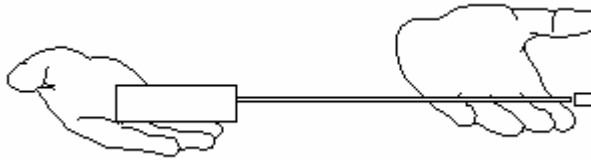
Placing Optical Apparatus and Fibers

Avoid touching pins or edge connectors with bare hands. Handle optical fibers delicately. Use two fingers to grip the fiber connector instead of directly holding the fiber or its cover.

Avoid applying axial or lateral pressure to a fiber. Refrain from folding or twisting a fiber or crushing its pigtail. Do not drag a fiber or apply pressure to the coupling

point of its pigtail. **Figure 1-215** shows how to properly place optical apparatus and fibers.

Figure 1-215 Methods of placing optical apparatus and fibers



 NOTE

During installation, ensure that an optical fiber is coiled into a loop with a diameter of at least 6 cm.

Uninstalling Optical Apparatus

When removing an OA, first disconnect its optical fiber, then pull the tab latch, and slowly take out the OA. Avoid forcefully pulling out an OA by directly dragging its optical fiber. Ensure that the optical fiber is connected to and removed from an OA horizontally.

Precautions for Loose OAs

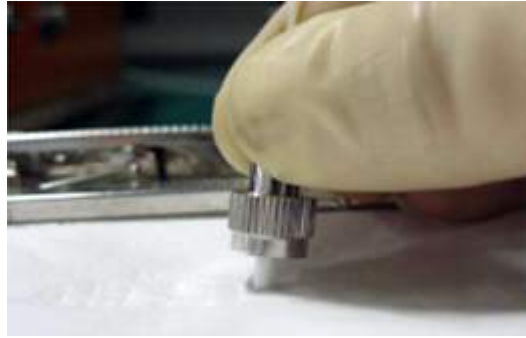
- When installing an OA, firmly push it into position until you hear a click or feel a slight tremor, signaling that the OA is securely locked. If the OA is not securely locked, its edge connector may be loosely connected to the board connector, and the link may still be up. However, if the OA experiences tremors or collisions with other objects, it may come loose or temporarily lose optical signals.
- When installing an OA into an optical module cage, push it firmly into position until it cannot be pushed any further. This ensures that the OA has good contact with the board connector.

Precautions for Receptacle Contamination

- Prepare clean tissues for onsite use. Clean an optical connector before inserting it into the receptacle to protect the receptacle against contamination.

 NOTE

Use at least three cleaning tissues. Wipe the end of an optical connector horizontally in one direction, and then shift the connector to the unused part of the cleaning tissue to continue. Generally, one cleaning tissue is used for cleaning one optical connector.



- Cover OAs that are temporarily unused with dust-proof caps to prevent dust pollution. If no dust-proof caps are available, inserting fibers can effectively prevent dust from entering these OAs.
- Place optical fiber jumpers that are temporarily unused on the optical distribution frame (ODF) or coil them up in a fiber management tray. Ensure that they are not compressed or squeezed.



- If OAs or optical fibers remain unused for a long time without being covered with dust-proof caps, clean their receptacles or optical connectors before using them again. Use cotton swabs to clean receptacles and cleaning tissues to clean optical connectors.

 NOTE

When cleaning a receptacle, gently insert a cotton swab and rotate it without applying excessive force. Otherwise, the receptacle may be damaged.

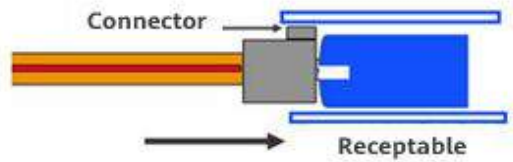
- If optical signals are unexpectedly lost during device operation, use the previously mentioned method to clean the receptacle or optical connector. This can help exclude contamination as the cause of signal loss.

Precautions for Overload-caused Burnt OAs

- When using an OTDR to test the connectivity or attenuation of a fiber channel, disconnect the optical fiber from the OA. Otherwise, the OA may be burnt.
- When performing a self-loop test, use an optical attenuator if necessary. Do not loosen the optical connector instead.
- If an OA is directly connected to an optical module, ensure that the output power of the OA does not exceed the overload threshold (0 dBm) of the optical module.

Other Precautions

An optical connector must be horizontally inserted into a receptacle to avoid damages to the receptacle.



1.5.3 2.048Mbps SFP Optical Module

1.5.3.1 2Mbps-eSFP-SMF-1310nm-2km-commercial

Table 1-1023 2Mbps-eSFP-SMF-1310nm-2km-commercial specifications

Item	Value
Basic Information	
Module name	2Mbps-eSFP-SMF-1310nm-2km-commercial
Part Number	34060588
Model	OSC002B01
Form factor	eSFP
Application standard	C37.94
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	-10°C to 70°C(14°F to 158°F)
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	2.048 Mbit/s
Target transmission distance [km]	2 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1310 nm

Item	Value
Tx operating wavelength range [nm]	1260 nm - 1360 nm
Maximum Tx optical power (AVG) [dBm]	-8 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	-15 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	8.2 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1100 nm - 1600 nm
Rx sensitivity (AVG) [dBm]	-26 dBm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	-8 dBm
Overload power (OMA) [dBm]	-

1.5.4 155Mbps SFP Electrical Module

1.5.4.1 155Mbps-SFP-120m-extended

Table 1-1024 155Mbps-SFP-120m-extended specifications

Item	Value
Basic Information	
Module name	155Mbps-SFP-120m-extended
Part Number	34100104
Model	STM1E-SFP02
Form factor	SFP
Application standard	ITU-T G.703/G.783, STM-1e
Connector type	SMB
Bit error ratio (BER)	<1x10E-12

Item	Value
Working case temperature [°C(°F)]	-20°C to 85°C(-4°F to 185°F)
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	155 Mbit/s
Target transmission distance [km]	0.1 km

1.5.5 1Gbps SyncE and 1588V2 Electrical Module

1.5.5.1 1Gbps-SFP-100m-industry-Support SyncE and 1588V2

Table 1-1025 1Gbps-SFP-100m-industry-Support SyncE and 1588V2 specifications

Item	Value
Basic Information	
Module name	1Gbps-SFP-100m-industry-Support SyncE and 1588V2
Part Number	34100255
Model	OEGD01N02
Form factor	SFP
Application standard	IEEE 802.3, 1000Base-T
Connector type	RJ45
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	-40°C to +85°C(-40°F to +185°F)
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	1000 Mbit/s
Target transmission distance [km]	0.1 km

1.5.6 155Mbps eSFP Optical Module

1.5.6.1 155Mbps-eSFP-SMF-1550nm-80km-commercial (34060282)

Table 1-1026 155Mbps-eSFP-SMF-1550nm-80km-commercial specifications

Item	Value
Basic Information	
Module name	155Mbps-eSFP-SMF-1550nm-80km-commercial
Part Number	34060282
Model	eSFP-FE-LH80-SM1550
Form factor	eSFP
Application standard	ITU-T G.957, STM-1
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	155 Mbit/s
Target transmission distance [km]	80 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1550 nm
Tx operating wavelength range [nm]	1480 nm - 1580 nm
Maximum Tx optical power (AVG) [dBm]	0 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	-5 dBm

Item	Value
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	10.5 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1263 nm - 1580 nm
Rx sensitivity (AVG) [dBm]	-34 dBm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	-10 dBm
Overload power (OMA) [dBm]	-

1.5.6.2 155Mbps-eSFP-SMF-1310nm-15km-industry

Table 1-1027 155Mbps-eSFP-SMF-1310nm-15km-industry specifications

Item	Value
Basic Information	
Module name	155Mbps-eSFP-SMF-1310nm-15km-industry
Part Number	34060307
Model	eSFP-1310nm-I-1
Form factor	eSFP
Application standard	ITU-T G.957, STM-1
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	-40°C to +85°C(-40°F to +185°F)
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	155 Mbit/s
Target transmission distance [km]	15 km

Item	Value
Transmitter Optical Characteristics	
Center wavelength [nm]	1310 nm
Tx operating wavelength range [nm]	1261 nm - 1360 nm
Maximum Tx optical power (AVG) [dBm]	-8 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	-15 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	8.2 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1260 nm - 1580 nm
Rx sensitivity (AVG) [dBm]	-31 dBm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	-8 dBm
Overload power (OMA) [dBm]	-

1.5.6.3 155Mbps-eSFP-SMF-1310nm-40km-industry

Table 1-1028 155Mbps-eSFP-SMF-1310nm-40km-industry specifications

Item	Value
Basic Information	
Module name	155Mbps-eSFP-SMF-1310nm-40km-industry
Part Number	34060308
Model	eSFP-1310nm-L-1.1
Form factor	eSFP
Application standard	ITU-T G.957, STM-1
Connector type	LC
Optical fiber type	SMF

Item	Value
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	-40°C to +85°C(-40°F to +185°F)
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	155 Mbit/s
Target transmission distance [km]	40 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1310 nm
Tx operating wavelength range [nm]	1263 nm - 1360 nm
Maximum Tx optical power (AVG) [dBm]	0 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	-5 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	10.5 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1263 nm - 1360 nm
Rx sensitivity (AVG) [dBm]	-34 dBm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	-10 dBm
Overload power (OMA) [dBm]	-

1.5.6.4 155Mbps-eSFP-SMF-1550nm-80km-industry

Table 1-1029 155Mbps-eSFP-SMF-1550nm-80km-industry specifications

Item	Value
Basic Information	

Item	Value
Module name	155Mbps-eSFP-SMF-1550nm-80km-industry
Part Number	34060309
Model	eSFP-1550nm-L-1.2
Form factor	eSFP
Application standard	ITU-T G.957, STM-1
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	-40°C to +85°C(-40°F to +185°F)
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	155 Mbit/s
Target transmission distance [km]	80 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1550 nm
Tx operating wavelength range [nm]	1480 nm - 1580 nm
Maximum Tx optical power (AVG) [dBm]	0 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	-5 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	10.5 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1263 nm - 1580 nm
Rx sensitivity (AVG) [dBm]	-34 dBm
Rx sensitivity (OMA) [dBm]	-

Item	Value
Overload power (AVG) [dBm]	-10 dBm
Overload power (OMA) [dBm]	-

1.5.6.5 155Mbps-eSFP-SMF-1310nm-40km-commercial

Table 1-1030 155Mbps-eSFP-SMF-1310nm-40km-commercial specifications

Item	Value
Basic Information	
Module name	155Mbps-eSFP-SMF-1310nm-40km-commercial
Part Number	S4015715
Model	eSFP-FE-LH40-SM1310
Form factor	eSFP
Application standard	ITU-T G.957, STM-1
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	155 Mbit/s
Target transmission distance [km]	40 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1310 nm
Tx operating wavelength range [nm]	1263 nm - 1360 nm
Maximum Tx optical power (AVG) [dBm]	0 dBm
Maximum Tx optical power (OMA) [dBm]	-

Item	Value
Minimum Tx optical power (AVG) [dBm]	-5 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	10.5 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1263 nm - 1360 nm
Rx sensitivity (AVG) [dBm]	-34 dBm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	-10 dBm
Overload power (OMA) [dBm]	-10 dBm

1.5.6.6 155Mbps-eSFP-SMF-1550nm-80km-commercial (S4015716)

Table 1-1031 155Mbps-eSFP-SMF-1550nm-80km-commercial specifications

Item	Value
Basic Information	
Module name	155Mbps-eSFP-SMF-1550nm-80km-commercial
Part Number	S4015716
Model	eSFP-FE-LH80-SM1550
Form factor	eSFP
Application standard	ITU-T G.957, STM-1
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V

Item	Value
Transmission rate [bit/s]	155 Mbit/s
Target transmission distance [km]	80 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1550 nm
Tx operating wavelength range [nm]	1480 nm - 1580 nm
Maximum Tx optical power (AVG) [dBm]	0 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	-5 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	10.5 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1263 nm - 1580 nm
Rx sensitivity (AVG) [dBm]	-34 dBm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	-10 dBm
Overload power (OMA) [dBm]	-10 dBm

1.5.6.7 155Mbps-eSFP-MMF-1310nm-2km-commercial

Table 1-1032 155Mbps-eSFP-MMF-1310nm-2km-commercial specifications

Item	Value
Basic Information	
Module name	155Mbps-eSFP-MMF-1310nm-2km-commercial
Part Number	S4015731
Model	SFP-FE-SX-MM1310
Form factor	eSFP
Application standard	ITU-T G.957, STM-1

Item	Value
Connector type	LC
Optical fiber type	MMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	-
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	155 Mbit/s
Target transmission distance [km]	2 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1310 nm
Tx operating wavelength range [nm]	1270 nm - 1380 nm
Maximum Tx optical power (AVG) [dBm]	-14 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	-19 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	10 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1270 nm - 1380 nm
Rx sensitivity (AVG) [dBm]	-30 dBm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	-14 dBm
Overload power (OMA) [dBm]	-14 dBm

1.5.6.8 155Mbps-eSFP-SMF-1310nm-15km-commercial

Table 1-1033 155Mbps-eSFP-SMF-1310nm-15km-commercial specifications

Item	Value
Basic Information	
Module name	155Mbps-eSFP-SMF-1310nm-15km-commercial
Part Number	S4015755
Model	eSFP-FE-LX-SM1310
Form factor	eSFP
Application standard	ITU-T G.957, STM-1
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	155 Mbit/s
Target transmission distance [km]	15 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1310 nm
Tx operating wavelength range [nm]	1261 nm - 1360 nm
Maximum Tx optical power (AVG) [dBm]	-8 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	-15 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	8.2 dB
Receiver Optical Characteristics	

Item	Value
Rx operating wavelength range [nm]	1260 nm - 1580 nm
Rx sensitivity (AVG) [dBm]	-28 dBm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	-8 dBm
Overload power (OMA) [dBm]	-8 dBm

1.5.7 155Mbps eSFP BIDI Optical Module

1.5.7.1 155Mbps-eSFP-SM-1310nm(Tx)/1550nm(Rx)-15km-commercial (02310QNG)

Table 1-1034 155Mbps-eSFP-SM-1310nm(Tx)/1550nm(Rx)-15km-commercial specifications

Item	Value
Basic Information	
Module name	155Mbps-eSFP-SM-1310nm(Tx)/1550nm(Rx)-15km-commercial
Part Number	02310QNG
Model	OSC015B01
Form factor	eSFP
Application standard	IEEE 802.3, 100BASE-BX10-U
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	155 Mbit/s
Target transmission distance [km]	15 km

Item	Value
Transmitter Optical Characteristics	
Center wavelength [nm]	1310 nm
Tx operating wavelength range [nm]	1260 nm - 1360 nm
Maximum Tx optical power (AVG) [dBm]	-8 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	-14 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	8.5 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1480 nm - 1580 nm
Rx sensitivity (AVG) [dBm]	-32 dBm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	-8 dBm
Overload power (OMA) [dBm]	-

1.5.7.2 155Mbps-eSFP-SM-1550nm(Tx)/1310nm(Rx)-15km-commercial (02310QNH)

Table 1-1035 155Mbps-eSFP-SM-1550nm(Tx)/1310nm(Rx)-15km-commercial specifications

Item	Value
Basic Information	
Module name	155Mbps-eSFP-SM-1550nm(Tx)/1310nm(Rx)-15km-commercial
Part Number	02310QNH
Model	OSC015B02
Form factor	eSFP
Application standard	IEEE 802.3, 100BASE-BX10-D
Connector type	LC

Item	Value
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	155 Mbit/s
Target transmission distance [km]	15 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1550 nm
Tx operating wavelength range [nm]	1480 nm - 1580 nm
Maximum Tx optical power (AVG) [dBm]	-8 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	-14 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	8.5 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1260 nm - 1360 nm
Rx sensitivity (AVG) [dBm]	-32 dBm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	-8 dBm
Overload power (OMA) [dBm]	-

1.5.8 622Mbps eSFP Optical Module

1.5.8.1 622Mbps-eSFP-SMF-1550nm-80km-commercial (02310QXT)

Table 1-1036 622Mbps-eSFP-SMF-1550nm-80km-commercial specifications

Item	Value
Basic Information	
Module name	622Mbps-eSFP-SMF-1550nm-80km-commercial
Part Number	02310QXT
Model	OSH080N01
Form factor	eSFP
Application standard	ITU-T G.957, STM-4
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	622 Mbit/s
Target transmission distance [km]	80 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1550 nm
Tx operating wavelength range [nm]	1480 nm - 1580 nm
Maximum Tx optical power (AVG) [dBm]	2 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	-3 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	10.5 dB
Receiver Optical Characteristics	

Item	Value
Rx operating wavelength range [nm]	1260 nm - 1580 nm
Rx sensitivity (AVG) [dBm]	-28 dBm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	-8 dBm
Overload power (OMA) [dBm]	-

1.5.8.2 622Mbps-eSFP-SMF-1310nm-40km-commercial (34060280)

Table 1-1037 622Mbps-eSFP-SMF-1310nm-40km-commercial specifications

Item	Value
Basic Information	
Module name	622Mbps-eSFP-SMF-1310nm-40km-commercial
Part Number	34060280
Model	eSFP-1310nm-L-4.1
Form factor	eSFP
Application standard	ITU-T G.957, STM-4
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	622 Mbit/s
Target transmission distance [km]	40 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1310 nm
Tx operating wavelength range [nm]	1296 nm - 1330 nm

Item	Value
Maximum Tx optical power (AVG) [dBm]	2 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	-3 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	10.5 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1296 nm - 1330 nm
Rx sensitivity (AVG) [dBm]	-28 dBm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	-8 dBm
Overload power (OMA) [dBm]	-

1.5.8.3 622Mbps-eSFP-SMF-1550nm-80km-commercial (34060284)

Table 1-1038 622Mbps-eSFP-SMF-1550nm-80km-commercial specifications

Item	Value
Basic Information	
Module name	622Mbps-eSFP-SMF-1550nm-80km-commercial
Part Number	34060284
Model	eSFP-1550nm-L-4.2
Form factor	eSFP
Application standard	ITU-T G.957, STM-4
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-8472

Item	Value
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	622 Mbit/s
Target transmission distance [km]	80 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1550 nm
Tx operating wavelength range [nm]	1480 nm - 1580 nm
Maximum Tx optical power (AVG) [dBm]	2 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	-3 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	10.5 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1260 nm - 1580 nm
Rx sensitivity (AVG) [dBm]	-28 dBm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	-8 dBm
Overload power (OMA) [dBm]	-

1.5.8.4 622Mbps-eSFP-SMF-1310nm-40km-commercial (S4015713)

Table 1-1039 622Mbps-eSFP-SMF-1310nm-40km-commercial specifications

Item	Value
Basic Information	
Module name	622Mbps-eSFP-SMF-1310nm-40km-commercial
Part Number	S4015713

Item	Value
Model	eSFP-1310nm-L-4.1
Form factor	eSFP
Application standard	ITU-T G.957, STM-4
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	622 Mbit/s
Target transmission distance [km]	40 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1310 nm
Tx operating wavelength range [nm]	1296 nm - 1330 nm
Maximum Tx optical power (AVG) [dBm]	2 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	-3 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	10.5 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1296 nm - 1330 nm
Rx sensitivity (AVG) [dBm]	-28 dBm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	-8 dBm
Overload power (OMA) [dBm]	-

1.5.8.5 622Mbps-eSFP-SMF-1310nm-15km-commercial

Table 1-1040 622Mbps-eSFP-SMF-1310nm-15km-commercial specifications

Item	Value
Basic Information	
Module name	622Mbps-eSFP-SMF-1310nm-15km-commercial
Part Number	S4015760
Model	OSH015N05
Form factor	eSFP
Application standard	ITU-T G.957, STM-4
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	622 Mbit/s
Target transmission distance [km]	15 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1310 nm
Tx operating wavelength range [nm]	1274 nm - 1356 nm
Maximum Tx optical power (AVG) [dBm]	-8 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	-15 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	8.5 dB
Receiver Optical Characteristics	

Item	Value
Rx operating wavelength range [nm]	1260 nm - 1580 nm
Rx sensitivity (AVG) [dBm]	-28 dBm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	-8 dBm
Overload power (OMA) [dBm]	-8 dBm

1.5.9 1Gbps Electrical Module

1.5.9.1 1Gbps-SFP-100m-industry (02310RAV)

Table 1-1041 1Gbps-SFP-100m-industry specifications

Item	Value
Basic Information	
Module name	1Gbps-SFP-100m-industry
Part Number	02310RAV
Model	OEGD01N01
Form factor	SFP
Application standard	IEEE 802.3, 1000Base-T
Connector type	RJ45
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	-40°C to +85°C(-40°F to +185°F)
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	10 Mbit/s 100 Mbit/s 1000 Mbit/s
Target transmission distance [km]	0.1 km

1.5.9.2 1Gbps-SFP-100m-industry (02310VPT)

Table 1-1042 1Gbps-SFP-100m-industry specifications

Item	Value
Basic Information	
Module name	1Gbps-SFP-100m-industry
Part Number	02310VPT
Model	OEGD01N02
Form factor	SFP
Application standard	IEEE 802.3, 1000Base-T
Connector type	RJ45
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	–40°C to +85°C(–40°F to +185°F)
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	10 Mbit/s 100 Mbit/s 1000 Mbit/s
Target transmission distance [km]	0.1 km

1.5.9.3 1Gbps-SFP-100m-industry (02314FNP)

Table 1-1043 1Gbps-SFP-100m-industry specifications

Item	Value
Basic Information	
Module name	1Gbps-SFP-100m-industry
Part Number	02314FNP
Model	OEGD01N03
Form factor	SFP
Application standard	IEEE 802.3, 1000Base-T
Connector type	RJ45

Item	Value
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	-40°C to +85°C(-40°F to +185°F)
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	1000 V
Transmission rate [bit/s]	10 Mbit/s 100 Mbit/s 1000 Mbit/s
Target transmission distance [km]	0.1 km

1.5.10 1.25Gbps eSFP Optical Module

1.5.10.1 1.25Gbps-eSFP-SMF-1550nm-80km-commercial (02310RAW)

Table 1-1044 1.25Gbps-eSFP-SMF-1550nm-80km-commercial specifications

Item	Value
Basic Information	
Module name	1.25Gbps-eSFP-SMF-1550nm-80km-commercial
Part Number	02310RAW
Model	OSG080N01
Form factor	eSFP
Application standard	IEEE 802.3, 1000BASE-ZX
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1

Item	Value
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	1.25 Gbit/s
Target transmission distance [km]	80 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1550 nm
Tx operating wavelength range [nm]	1500 nm - 1580 nm
Maximum Tx optical power (AVG) [dBm]	5 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	-2 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	9 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1260 nm - 1580 nm
Rx sensitivity (AVG) [dBm]	-23 dBm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	-3 dBm
Overload power (OMA) [dBm]	-
NOTE The interface standard is Huawei-specific.	

1.5.10.2 1.25Gbps-eSFP-SMF-1310nm-10km-industry (02310WQW)

Table 1-1045 1.25Gbps-eSFP-SMF-1310nm-10km-industry specifications

Item	Value
Basic Information	
Module name	1.25Gbps-eSFP-SMF-1310nm-10km-industry
Part Number	02310WQW
Model	OSG010N03

Item	Value
Form factor	eSFP
Application standard	IEEE 802.3, 1000BASE-LX10
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	-40°C to +85°C(-40°F to +185°F)
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	1.25 Gbit/s
Target transmission distance [km]	10 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1310 nm
Tx operating wavelength range [nm]	1260 nm - 1360 nm
Maximum Tx optical power (AVG) [dBm]	-3 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	-9 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	9.5 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1260 nm - 1360 nm
Rx sensitivity (AVG) [dBm]	-19 dBm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	-3 dBm
Overload power (OMA) [dBm]	-

1.5.10.3 1.25Gbps-eSFP-SMF-1310nm-40km-industry (02310WQX)

Table 1-1046 1.25Gbps-eSFP-SMF-1310nm-40km-industry specifications

Item	Value
Basic Information	
Module name	1.25Gbps-eSFP-SMF-1310nm-40km-industry
Part Number	02310WQX
Model	OSG040N03
Form factor	eSFP
Application standard	IEEE 802.3, 1000BASE-EX
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	−40°C to +85°C(−40°F to +185°F)
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	1.25 Gbit/s
Target transmission distance [km]	40 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1310 nm
Tx operating wavelength range [nm]	1275 nm - 1350 nm
Maximum Tx optical power (AVG) [dBm]	0 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	-5 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	9.5 dB
Receiver Optical Characteristics	

Item	Value
Rx operating wavelength range [nm]	1275 nm - 1350 nm
Rx sensitivity (AVG) [dBm]	-23 dBm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	-3 dBm
Overload power (OMA) [dBm]	-

1.5.10.4 1.25Gbps-eSFP-MMF-850nm-500m-extended (34060286)

Table 1-1047 1.25Gbps-eSFP-MMF-850nm-500m-extended specifications

Item	Value
Basic Information	
Module name	1.25Gbps-eSFP-MMF-850nm-500m-extended
Part Number	34060286
Model	eSFP-850nm-1000Base-Sx/FC200MM
Form factor	eSFP
Application standard	IEEE 802.3, 1000BASE-SX
Connector type	LC
Optical fiber type	MMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	-20°C to 85°C(-4°F to 185°F)
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	1.25 Gbit/s
Target transmission distance [km]	0.5 km(OM1)
Transmitter Optical Characteristics	
Center wavelength [nm]	850 nm
Tx operating wavelength range [nm]	770 nm - 860 nm

Item	Value
Maximum Tx optical power (AVG) [dBm]	0 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	-9.5 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	9 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	760 nm - 860 nm
Rx sensitivity (AVG) [dBm]	-17 dBm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	0 dBm
Overload power (OMA) [dBm]	-

1.5.10.5 1.25Gbps-eSFP-SMF-1310nm-10km-industry (34060290)

Table 1-1048 1.25Gbps-eSFP-SMF-1310nm-10km-industry specifications

Item	Value
Basic Information	
Module name	1.25Gbps-eSFP-SMF-1310nm-10km-industry
Part Number	34060290
Model	eSFP(S)-1310nm-1000Base-Lx
Form factor	eSFP
Application standard	IEEE 802.3, 1000BASE-LX10
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	-40°C to +85°C(-40°F to +185°F)
DDM options	SFF-8472

Item	Value
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	1.25 Gbit/s
Target transmission distance [km]	10 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1310 nm
Tx operating wavelength range [nm]	1260 nm - 1360 nm
Maximum Tx optical power (AVG) [dBm]	-3 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	-9 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	9.5 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1260 nm - 1360 nm
Rx sensitivity (AVG) [dBm]	-19 dBm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	-3 dBm
Overload power (OMA) [dBm]	-

1.5.10.6 1.25Gbps-eSFP-SMF-1550nm-100km-commercial

Table 1-1049 1.25Gbps-eSFP-SMF-1550nm-100km-commercial specifications

Item	Value
Basic Information	
Module name	1.25Gbps-eSFP-SMF-1550nm-100km-commercial
Part Number	34060295

Item	Value
Model	eSFP-GE-ZX100-SM1550
Form factor	eSFP
Application standard	IEEE 802.3, 1000BASE-ZX
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	1.25 Gbit/s
Target transmission distance [km]	100 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1550 nm
Tx operating wavelength range [nm]	1500 nm - 1580 nm
Maximum Tx optical power (AVG) [dBm]	5 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	0 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	9.5 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1260 nm - 1580 nm
Rx sensitivity (AVG) [dBm]	-30 dBm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	-9 dBm
Overload power (OMA) [dBm]	-

Item	Value
NOTE The interface standard is Huawei-specific.	

1.5.10.7 1.25Gbps-eSFP-SMF-1310nm-40km-industry (34060320)

Table 1-1050 1.25Gbps-eSFP-SMF-1310nm-40km-industry specifications

Item	Value
Basic Information	
Module name	1.25Gbps-eSFP-SMF-1310nm-40km-industry
Part Number	34060320
Model	eSFP-1310nm-Lx-40Km
Form factor	eSFP
Application standard	IEEE 802.3, 1000BASE-EX
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	-40°C to +85°C(-40°F to +185°F)
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	1.25 Gbit/s
Target transmission distance [km]	40 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1310 nm
Tx operating wavelength range [nm]	1275 nm - 1350 nm
Maximum Tx optical power (AVG) [dBm]	0 dBm
Maximum Tx optical power (OMA) [dBm]	-

Item	Value
Minimum Tx optical power (AVG) [dBm]	-5 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	9.5 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1275 nm - 1350 nm
Rx sensitivity (AVG) [dBm]	-23 dBm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	-3 dBm
Overload power (OMA) [dBm]	-

1.5.10.8 1.25Gbps-eSFP-MMF-850nm-500m-industry

Table 1-1051 1.25Gbps-eSFP-MMF-850nm-500m-industry specifications

Item	Value
Basic Information	
Module name	1.25Gbps-eSFP-MMF-850nm-500m-industry
Part Number	34060321
Model	eSFP-850nm-1000Base-Sx
Form factor	eSFP
Application standard	IEEE 802.3, 1000BASE-SX
Connector type	LC
Optical fiber type	MMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	-40°C to +85°C(-40°F to +185°F)
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V

Item	Value
Transmission rate [bit/s]	1.25 Gbit/s
Target transmission distance [km]	0.5 km
Transmitter Optical Characteristics	
Center wavelength [nm]	850 nm
Tx operating wavelength range [nm]	770 nm - 860 nm
Maximum Tx optical power (AVG) [dBm]	0 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	-9.5 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	9 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	760 nm - 860 nm
Rx sensitivity (AVG) [dBm]	-17 dBm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	0 dBm
Overload power (OMA) [dBm]	-

1.5.10.9 1.25Gbps-eSFP-SMF-1550nm-80km-commercial (34060360)

Table 1-1052 1.25Gbps-eSFP-SMF-1550nm-80km-commercial specifications

Item	Value
Basic Information	
Module name	1.25Gbps-eSFP-SMF-1550nm-80km-commercial
Part Number	34060360
Model	eSFP-1550nm-1000Base-Zx/FC100
Form factor	eSFP
Application standard	IEEE 802.3, 1000BASE-ZX

Item	Value
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	1.25 Gbit/s
Target transmission distance [km]	80 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1550 nm
Tx operating wavelength range [nm]	1500 nm - 1580 nm
Maximum Tx optical power (AVG) [dBm]	5 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	-2 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	9 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1260 nm - 1580 nm
Rx sensitivity (AVG) [dBm]	-23 dBm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	-3 dBm
Overload power (OMA) [dBm]	-
NOTE The interface standard is Huawei-specific.	

1.5.10.10 1.25Gbps-eSFP-SMF-1310nm-10km-commercial

Table 1-1053 1.25Gbps-eSFP-SMF-1310nm-10km-commercial specifications

Item	Value
Basic Information	
Module name	1.25Gbps-eSFP-SMF-1310nm-10km-commercial
Part Number	S4016067
Model	OSG010N05
Form factor	eSFP
Application standard	IEEE 802.3, 1000BASE-LX10
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	1.25 Gbit/s
Target transmission distance [km]	10 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1310 nm
Tx operating wavelength range [nm]	1270 nm - 1355 nm
Maximum Tx optical power (AVG) [dBm]	-3 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	-9 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	9.5 dB
Receiver Optical Characteristics	

Item	Value
Rx operating wavelength range [nm]	1270 nm - 1355 nm
Rx sensitivity (AVG) [dBm]	-20 dBm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	-3 dBm
Overload power (OMA) [dBm]	-3 dBm

1.5.10.11 1.25Gbps-eSFP-SMF-1310nm-40km-commercial (S4016954)

Table 1-1054 1.25Gbps-eSFP-SMF-1310nm-40km-commercial specifications

Item	Value
Basic Information	
Module name	1.25Gbps-eSFP-SMF-1310nm-40km-commercial
Part Number	S4016954
Model	OSG040002
Form factor	eSFP
Application standard	IEEE 802.3, 1000BASE-EX
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	1.25 Gbit/s
Target transmission distance [km]	40 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1310 nm
Tx operating wavelength range [nm]	1275 nm - 1350 nm

Item	Value
Maximum Tx optical power (AVG) [dBm]	0 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	-5 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	9.5 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1260 nm - 1580 nm
Rx sensitivity (AVG) [dBm]	-23 dBm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	-3 dBm
Overload power (OMA) [dBm]	-3 dBm

1.5.10.12 1.25Gbps-eSFP-MMF-850nm-500m-extended (S4017307)

Table 1-1055 1.25Gbps-eSFP-MMF-850nm-500m-extended specifications

Item	Value
Basic Information	
Module name	1.25Gbps-eSFP-MMF-850nm-500m-extended
Part Number	S4017307
Model	OMGD50N02
Form factor	eSFP
Application standard	IEEE 802.3, 1000BASE-SX
Connector type	LC
Optical fiber type	MMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	-20°C to 85°C(-4°F to 185°F)
DDM options	SFF-8472

Item	Value
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	1.25 Gbit/s
Target transmission distance [km]	0.5 km(OM1)
Transmitter Optical Characteristics	
Center wavelength [nm]	850 nm
Tx operating wavelength range [nm]	770 nm - 860 nm
Maximum Tx optical power (AVG) [dBm]	0 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	-9.5 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	9 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	760 nm - 860 nm
Rx sensitivity (AVG) [dBm]	-17 dBm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	0 dBm
Overload power (OMA) [dBm]	0 dBm

1.5.10.13 1.25Gbps-eSFP-SMF-1310nm-40km-commercial (S4017309)

Table 1-1056 1.25Gbps-eSFP-SMF-1310nm-40km-commercial specifications

Item	Value
Basic Information	
Module name	1.25Gbps-eSFP-SMF-1310nm-40km-commercial
Part Number	S4017309

Item	Value
Model	OSG040N02
Form factor	eSFP
Application standard	IEEE 802.3, 1000BASE-EX
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	1.25 Gbit/s
Target transmission distance [km]	40 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1310 nm
Tx operating wavelength range [nm]	1275 nm - 1350 nm
Maximum Tx optical power (AVG) [dBm]	0 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	-5 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	9.5 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1260 nm - 1580 nm
Rx sensitivity (AVG) [dBm]	-23 dBm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	-3 dBm
Overload power (OMA) [dBm]	-3 dBm

1.5.11 1.25Gbps eSFP BIDI Optical Module

1.5.11.1 1.25Gbps-eSFP-SMF-1310nm(Tx)/1490nm(Rx)-10km-commercial(34060470)

Table 1-1057 1.25Gbps-eSFP-SMF-1310nm(Tx)/1490nm(Rx)-10km-commercial(34060470) specifications

Item	Value
Basic Information	
Module name	1.25Gbps-eSFP-SMF-1310nm(Tx)/1490nm(Rx)-10km-commercial(34060470)
Part Number	34060470
Model	SFP-GE-LX-SM1310-BIDI
Form factor	eSFP
Application standard	IEEE 802.3ah, 1000Base-BX10-U
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	1.25 Gbit/s
Target transmission distance [km]	10 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1310 nm
Tx operating wavelength range [nm]	1260 nm - 1360 nm
Maximum Tx optical power (AVG) [dBm]	-3 dBm
Maximum Tx optical power (OMA) [dBm]	-

Item	Value
Minimum Tx optical power (AVG) [dBm]	-9 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	6 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1480 nm - 1500 nm
Rx sensitivity (AVG) [dBm]	-19.5 dBm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	-3 dBm
Overload power (OMA) [dBm]	-
NOTE Used in pair with 34060475.	

1.5.11.2 1.25Gbps-eSFP-SMF-1490nm(Tx)/1310nm(Rx)-10km-commercial

Table 1-1058 1.25Gbps-eSFP-SMF-1490nm(Tx)/1310nm(Rx)-10km-commercial specifications

Item	Value
Basic Information	
Module name	1.25Gbps-eSFP-SMF-1490nm(Tx)/1310nm(Rx)-10km-commercial
Part Number	34060475
Model	SFP-GE-LX-SM1490-BIDI
Form factor	eSFP
Application standard	IEEE 802.3ah, 1000Base-BX10-D
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-8472
Environment standard	RoHS

Item	Value
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	1.25 Gbit/s
Target transmission distance [km]	10 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1490 nm
Tx operating wavelength range [nm]	1480 nm - 1500 nm
Maximum Tx optical power (AVG) [dBm]	-3 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	-9 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	6 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1260 nm - 1360 nm
Rx sensitivity (AVG) [dBm]	-19.5 dBm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	-3 dBm
Overload power (OMA) [dBm]	-
NOTE Used in pair with 34060470.	

1.5.11.3 1.25Gbps-eSFP-SMF-1310nm(Tx)/1490nm(Rx)-40km-commercial

Table 1-1059 1.25Gbps-eSFP-SMF-1310nm(Tx)/1490nm(Rx)-40km-commercial specifications

Item	Value
Basic Information	
Module name	1.25Gbps-eSFP-SMF-1310nm(Tx)/1490nm(Rx)-40km-commercial

Item	Value
Part Number	34060539
Model	OGEBIDI41
Form factor	eSFP
Application standard	IEEE 802.3ah, 1000Base-BX40-U
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	1.25 Gbit/s
Target transmission distance [km]	40 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1310 nm
Tx operating wavelength range [nm]	1260 nm - 1360 nm
Maximum Tx optical power (AVG) [dBm]	3 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	-2 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	9 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1480 nm - 1500 nm
Rx sensitivity (AVG) [dBm]	-
Rx sensitivity (OMA) [dBm]	-23 dBm
Overload power (AVG) [dBm]	-3 dBm
Overload power (OMA) [dBm]	-

Item	Value
NOTE Used in pair with 34060540.	

1.5.11.4 1.25Gbps-eSFP-SMF-1490nm(Tx)/1310nm(Rx)-40km-commercial

Table 1-1060 1.25Gbps-eSFP-SMF-1490nm(Tx)/1310nm(Rx)-40km-commercial specifications

Item	Value
Basic Information	
Module name	1.25Gbps-eSFP-SMF-1490nm(Tx)/1310nm(Rx)-40km-commercial
Part Number	34060540
Model	OGEBIDI40
Form factor	eSFP
Application standard	IEEE 802.3ah, 1000Base-BX40-D
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	1.25 Gbit/s
Target transmission distance [km]	40 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1490 nm
Tx operating wavelength range [nm]	1480 nm - 1500 nm
Maximum Tx optical power (AVG) [dBm]	3 dBm
Maximum Tx optical power (OMA) [dBm]	-

Item	Value
Minimum Tx optical power (AVG) [dBm]	-2 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	9 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1260 nm - 1360 nm
Rx sensitivity (AVG) [dBm]	-
Rx sensitivity (OMA) [dBm]	-23 dBm
Overload power (AVG) [dBm]	-3 dBm
Overload power (OMA) [dBm]	-
NOTE Used in pair with 34060539.	

1.5.11.5 1.25Gbps-eSFP-SMF-1570nm(Tx)/1490nm(Rx)-80km-commercial

Table 1-1061 1.25Gbps-eSFP-SMF-1570nm(Tx)/1490nm(Rx)-80km-commercial specifications

Item	Value
Basic Information	
Module name	1.25Gbps-eSFP-SMF-1570nm(Tx)/1490nm(Rx)-80km-commercial
Part Number	34060595
Model	OGEBIDI80
Form factor	eSFP
Application standard	IEEE 802.3ah, 1000Base-BX80-D
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-8472
Environment standard	RoHS

Item	Value
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	1.25 Gbit/s
Target transmission distance [km]	80 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1570 nm
Tx operating wavelength range [nm]	1560 nm - 1580 nm
Maximum Tx optical power (AVG) [dBm]	4 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	-2 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	9 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1480 nm - 1500 nm
Rx sensitivity (AVG) [dBm]	-26 dBm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	-3 dBm
Overload power (OMA) [dBm]	-
NOTE Used in pair with 34060596.	

1.5.11.6 1.25Gbps-eSFP-SMF-1490nm(Tx)/1570nm(Rx)-80km-commercial

Table 1-1062 1.25Gbps-eSFP-SMF-1490nm(Tx)/1570nm(Rx)-80km-commercial specifications

Item	Value
Basic Information	
Module name	1.25Gbps-eSFP-SMF-1490nm(Tx)/1570nm(Rx)-80km-commercial

Item	Value
Part Number	34060596
Model	OGE8000M14
Form factor	eSFP
Application standard	IEEE 802.3ah, 1000Base-BX80-U
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	1.25 Gbit/s
Target transmission distance [km]	80 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1490 nm
Tx operating wavelength range [nm]	1480 nm - 1500 nm
Maximum Tx optical power (AVG) [dBm]	4 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	-2 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	9 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1560 nm - 1580 nm
Rx sensitivity (AVG) [dBm]	-26 dBm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	-3 dBm
Overload power (OMA) [dBm]	-

Item	Value
NOTE Used in pair with 34060595.	

1.5.11.7 0.1~1.25Gbps-eSFP-SMF-1310nm(Tx)/1550nm(Rx)-40km-commercial

Table 1-1063 0.1~1.25Gbps-eSFP-SMF-1310nm(Tx)/1550nm(Rx)-40km-commercial specifications

Item	Value
Basic Information	
Module name	0.1~1.25Gbps-eSFP-SMF-1310nm(Tx)/1550nm(Rx)-40km-commercial
Part Number	34060638
Model	eSFP-1310/1550-L1.1-BIDI
Form factor	eSFP
Application standard	IEEE 802.3ah, 1000Base-BX40-U
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	-5°C to 70°C(23°F to 158°F)
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	0.1~1.25 Gbit/s
Target transmission distance [km]	40 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1310 nm
Tx operating wavelength range [nm]	1260 nm - 1360 nm
Maximum Tx optical power (AVG) [dBm]	2 dBm

Item	Value
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	-3 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	9 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1480 nm - 1580 nm
Rx sensitivity (AVG) [dBm]	-25 dBm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	-3 dBm
Overload power (OMA) [dBm]	-
NOTE When an optical module is used on the OptiX PTN equipment, the rate of the optical module cannot exceed 155 Mbit/s.	

1.5.11.8 0.1~1.25Gbps-eSFP-SMF-1550nm(Tx)/1310nm(Rx)-40km-commercial

Table 1-1064 0.1~1.25Gbps-eSFP-SMF-1550nm(Tx)/1310nm(Rx)-40km-commercial specifications

Item	Value
Basic Information	
Module name	0.1~1.25Gbps-eSFP-SMF-1550nm(Tx)/1310nm(Rx)-40km-commercial
Part Number	34060639
Model	eSFP-1550/1310-L1.1-BIDI
Form factor	eSFP
Application standard	IEEE 802.3ah, 1000Base-BX40-D
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12

Item	Value
Working case temperature [°C(°F)]	-5°C to 70°C(23°F to 158°F)
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	0.1~1.25 Gbit/s
Target transmission distance [km]	40 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1550 nm
Tx operating wavelength range [nm]	1530 nm - 1580 nm
Maximum Tx optical power (AVG) [dBm]	2 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	-3 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	9 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1260 nm - 1360 nm
Rx sensitivity (AVG) [dBm]	-25 dBm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	-3 dBm
Overload power (OMA) [dBm]	-
NOTE When an optical module is used on the OptiX PTN equipment, the rate of the optical module cannot exceed 155 Mbit/s.	

1.5.11.9 1.25Gbps-eSFP-SMF-1550nm(Tx)/1490nm(Rx)-80km-commercial

Table 1-1065 1.25Gbps-eSFP-SMF-1550nm(Tx)/1490nm(Rx)-80km-commercial specifications

Item	Value
Basic Information	
Module name	1.25Gbps-eSFP-SMF-1550nm(Tx)/1490nm(Rx)-80km-commercial
Part Number	02314RDH
Model	SFP-GE-BIDI-80km-SM1550
Form factor	eSFP
Application standard	IEEE 802.3ah, 1000Base-BX80-D
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-8472
Environment standard	RoHS
Security standard	TUV; EN 60825-1, EN 62368-1; NRTL: CAN/CSA C22.2 NO.62368-1-12 * UL 62368-1; FDA: 21CFR 1040.10 and 1040.11, Notice NO.56.
ESD(HBM1) [V]	1000 V
Transmission rate [bit/s]	1.25 Gbit/s
Target transmission distance [km]	80 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1550 nm
Tx operating wavelength range [nm]	1540 nm - 1560 nm
Maximum Tx optical power (AVG) [dBm]	5 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	-2 dBm

Item	Value
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	9 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1480 nm - 1500 nm
Rx sensitivity (AVG) [dBm]	-26 dBm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	-3 dBm
Overload power (OMA) [dBm]	-
NOTE It is used with 02314RDK in pairs.	

1.5.11.10 1.25Gbps-eSFP-SMF-1490nm(Tx)/1550nm(Rx)-80km-commercial

Table 1-1066 1.25Gbps-eSFP-SMF-1490nm(Tx)/1550nm(Rx)-80km-commercial specifications

Item	Value
Basic Information	
Module name	1.25Gbps-eSFP-SMF-1490nm(Tx)/1550nm(Rx)-80km-commercial
Part Number	02314RDK
Model	SFP-GE-BIDI-80km-SM1490
Form factor	eSFP
Application standard	IEEE 802.3ah, 1000Base-BX80-D
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	0°C to 70°C (32°F to 158°F)
DDM options	SFF-8472
Environment standard	RoHS

Item	Value
Security standard	TUV; EN 60825-1, EN 62368-1; NRTL: CAN/CSA C22.2 NO.62368-1-12 * UL 62368-1; FDA: 21CFR 1040.10 and 1040.11, Notice NO.56.
ESD(HBM1) [V]	1000 V
Transmission rate [bit/s]	1.25 Gbit/s
Target transmission distance [km]	80 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1490 nm
Tx operating wavelength range [nm]	1480 nm - 1500 nm
Maximum Tx optical power (AVG) [dBm]	5 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	-2 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	9 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1540 nm - 1560 nm
Rx sensitivity (AVG) [dBm]	-26 dBm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	-3 dBm
Overload power (OMA) [dBm]	-
NOTE It is used with 02314RDH in pairs.	

1.5.12 1.25Gbps CSFP BIDI Optical Module

1.5.12.1 1.25Gbps-CSFP-SMF-1490nm(Tx)/1310nm(Rx)-40km-industry
(02312WHB)

Table 1-1067 1.25Gbps-CSFP-SMF-1490nm(Tx)/1310nm(Rx)-40km-industry specifications

Item	Value
Basic Information	
Module name	1.25Gbps-CSFP-SMF-1490nm(Tx)/1310nm(Rx)-40km-industry
Part Number	02312WHB
Model	OSG040008
Form factor	CSFP
Application standard	IEEE 802.3ah, 1000BASE-BX
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	-40°C to +85°C(-40°F to +185°F)
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	1.25 Gbit/s
Target transmission distance [km]	40 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1490 nm(Tx)/1310 nm(Rx)
Tx operating wavelength range [nm]	1480 nm - 1500 nm
Maximum Tx optical power (AVG) [dBm]	-5 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	0 dBm
Minimum Tx optical power (OMA) [dBm]	-

Item	Value
Minimum extinction ratio [dB]	6.6 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1260 nm - 1360 nm
Rx sensitivity (AVG) [dBm]	-25 dBm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	0 dBm
Overload power (OMA) [dBm]	-

1.5.12.2 1.25Gbps-CSFP-SMF-1310nm(Tx)/1490nm(Rx)-10km-industry (02312WHC)

Table 1-1068 1.25Gbps-CSFP-SMF-1310nm(Tx)/1490nm(Rx)-10km-industry specifications

Item	Value
Basic Information	
Module name	1.25Gbps-CSFP-SMF-1310nm(Tx)/1490nm(Rx)-10km-industry
Part Number	02312WHC
Model	OSG010007
Form factor	CSFP
Application standard	IEEE 802.3ah, 1000BASE-BX
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	-40°C to +85°C(-40°F to +185°F)
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	1.25 Gbit/s
Target transmission distance [km]	10 km

Item	Value
Transmitter Optical Characteristics	
Center wavelength [nm]	1310 nm(Tx)/1490 nm(Rx)
Tx operating wavelength range [nm]	1260 nm - 1360 nm
Maximum Tx optical power (AVG) [dBm]	-3 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	-9 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	6.6 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1480 nm - 1500 nm
Rx sensitivity (AVG) [dBm]	-24 dBm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	-3 dBm
Overload power (OMA) [dBm]	-

1.5.12.3 1.25Gbps-CSFP-SMF-1490nm(Tx)/1310nm(Rx)-10km-industry (02312WHD)

Table 1-1069 1.25Gbps-CSFP-SMF-1490nm(Tx)/1310nm(Rx)-10km-industry specifications

Item	Value
Basic Information	
Module name	1.25Gbps-CSFP-SMF-1490nm(Tx)/1310nm(Rx)-10km-industry
Part Number	02312WHD
Model	O00CSFP25
Form factor	CSFP
Application standard	IEEE 802.3ah, 1000BASE-BX10-D
Connector type	LC

Item	Value
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	-40°C to +85°C(-40°F to +185°F)
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	1.25 Gbit/s
Target transmission distance [km]	10 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1490 nm
Tx operating wavelength range [nm]	1480 nm - 1500 nm
Maximum Tx optical power (AVG) [dBm]	-3 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	-9 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	6 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1260 nm - 1360 nm
Rx sensitivity (AVG) [dBm]	-19.5 dBm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	-3 dBm
Overload power (OMA) [dBm]	-

1.5.12.4 1.25Gbps-CSFP-SMF-1490nm(Tx)/1310nm(Rx)-10km-industry
(34060525)

Table 1-1070 1.25Gbps-CSFP-SMF-1490nm(Tx)/1310nm(Rx)-10km-industry specifications

Item	Value
Basic Information	
Module name	1.25Gbps-CSFP-SMF-1490nm(Tx)/1310nm(Rx)-10km-industry
Part Number	34060525
Model	O00CSFP25
Form factor	CSFP
Application standard	IEEE 802.3ah, 1000BASE-BX10-D
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	-40°C to +85°C(-40°F to +185°F)
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	1.25 Gbit/s
Target transmission distance [km]	10 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1490 nm
Tx operating wavelength range [nm]	1480 nm - 1500 nm
Maximum Tx optical power (AVG) [dBm]	-3 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	-9 dBm
Minimum Tx optical power (OMA) [dBm]	-

Item	Value
Minimum extinction ratio [dB]	6 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1260 nm - 1360 nm
Rx sensitivity (AVG) [dBm]	-19.5 dBm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	-3 dBm
Overload power (OMA) [dBm]	-

1.5.12.5 1.25Gbps-CSFP-SMF-1490nm(Tx)/1310nm(Rx)-40km-industry (34060894)

Table 1-1071 1.25Gbps-CSFP-SMF-1490nm(Tx)/1310nm(Rx)-40km-industry specifications

Item	Value
Basic Information	
Module name	1.25Gbps-CSFP-SMF-1490nm(Tx)/1310nm(Rx)-40km-industry
Part Number	34060894
Model	OSG040008
Form factor	CSFP
Application standard	IEEE 802.3ah, 1000BASE-BX
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	-40°C to +85°C(-40°F to +185°F)
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	1.25 Gbit/s
Target transmission distance [km]	40 km

Item	Value
Transmitter Optical Characteristics	
Center wavelength [nm]	1490 nm(Tx)/1310 nm(Rx)
Tx operating wavelength range [nm]	1480 nm - 1500 nm
Maximum Tx optical power (AVG) [dBm]	-5 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	0 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	6.6 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1260 nm - 1360 nm
Rx sensitivity (AVG) [dBm]	-25 dBm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	0 dBm
Overload power (OMA) [dBm]	-

1.5.12.6 1.25Gbps-CSFP-SMF-1310nm(Tx)/1490nm(Rx)-10km-industry (34061329)

Table 1-1072 1.25Gbps-CSFP-SMF-1310nm(Tx)/1490nm(Rx)-10km-industry specifications

Item	Value
Basic Information	
Module name	1.25Gbps-CSFP-SMF-1310nm(Tx)/1490nm(Rx)-10km-industry
Part Number	34061329
Model	OSG010007
Form factor	CSFP
Application standard	IEEE 802.3ah, 1000BASE-BX
Connector type	LC

Item	Value
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	-40°C to +85°C(-40°F to +185°F)
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	1.25 Gbit/s
Target transmission distance [km]	10 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1310 nm(Tx)/1490 nm(Rx)
Tx operating wavelength range [nm]	1260 nm - 1360 nm
Maximum Tx optical power (AVG) [dBm]	-3 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	-9 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	6.6 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1480 nm - 1500 nm
Rx sensitivity (AVG) [dBm]	-24 dBm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	-3 dBm
Overload power (OMA) [dBm]	-3 dBm

1.5.13 125M~1.25Gbps CSFP BIDI Optical Module

1.5.13.1 125M~1.25Gbps-CSFP-SMF-1490nm(Tx)/1310nm(Rx)-10km-industry (02312WHA)

Table 1-1073 125M~1.25Gbps-CSFP-SMF-1490nm(Tx)/1310nm(Rx)-10km-industry specifications

Item	Value
Basic Information	
Module name	125M~1.25Gbps-CSFP-SMF-1490nm(Tx)/1310nm(Rx)-10km-industry
Part Number	02312WHA
Model	OSG010005
Form factor	CSFP
Application standard	IEEE 802.3ah, 1000BASE-BX
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	-40°C to +85°C(-40°F to +185°F)
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	125 Mbit/s ~ 1.25 Gbit/s
Target transmission distance [km]	10 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1490 nm(Tx)/1310 nm(Rx)
Tx operating wavelength range [nm]	1480 nm - 1500 nm
Maximum Tx optical power (AVG) [dBm]	-9 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	-3 dBm
Minimum Tx optical power (OMA) [dBm]	-

Item	Value
Minimum extinction ratio [dB]	6.6 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1260 nm - 1360 nm
Rx sensitivity (AVG) [dBm]	-24 dBm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	-3 dBm
Overload power (OMA) [dBm]	-

1.5.13.2 125M~1.25Gbps-CSFP-SMF-1490nm(Tx)/1310nm(Rx)-10km-industry (34060803)

Table 1-1074 125M~1.25Gbps-CSFP-SMF-1490nm(Tx)/1310nm(Rx)-10km-industry specifications

Item	Value
Basic Information	
Module name	125M~1.25Gbps-CSFP-SMF-1490nm(Tx)/1310nm(Rx)-10km-industry
Part Number	34060803
Model	OSG010005
Form factor	CSFP
Application standard	IEEE 802.3ah, 1000BASE-BX
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	-40°C to +85°C(-40°F to +185°F)
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	125 Mbit/s ~ 1.25 Gbit/s

Item	Value
Target transmission distance [km]	10 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1490 nm(Tx)/1310 nm(Rx)
Tx operating wavelength range [nm]	1480 nm - 1500 nm
Maximum Tx optical power (AVG) [dBm]	-9 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	-3 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	6.6 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1260 nm - 1360 nm
Rx sensitivity (AVG) [dBm]	-24 dBm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	-3 dBm
Overload power (OMA) [dBm]	-

1.5.13.3 125M~1.25Gbps-CSFP-SMF-1490nm(Tx)/1310nm(Rx)-10km-commercial

Table 1-1075 125M~1.25Gbps-CSFP-SMF-1490nm(Tx)/1310nm(Rx)-10km-commercial specifications

Item	Value
Basic Information	
Module name	125M~1.25Gbps-CSFP-SMF-1490nm(Tx)/1310nm(Rx)-10km-commercial
Part Number	34060805
Model	OSG010006
Form factor	CSFP

Item	Value
Application standard	IEEE 802.3ah, 1000BASE-BX
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	125 Mbit/s ~ 1.25 Gbit/s
Target transmission distance [km]	10 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1490 nm(Tx)/1310 nm(Rx)
Tx operating wavelength range [nm]	1480 nm - 1500 nm
Maximum Tx optical power (AVG) [dBm]	-9 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	-3 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	6.6 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1260 nm - 1360 nm
Rx sensitivity (AVG) [dBm]	-23 dBm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	-3 dBm
Overload power (OMA) [dBm]	-

1.5.14 1.25Gbps eSFP CWDM Optical Module

1.5.14.1 1.25Gbps-eSFP-SMF-1571nm-80km-commercial

Table 1-1076 1.25Gbps-eSFP-SMF-1571nm-80km-commercial specifications

Item	Value
Basic Information	
Module name	1.25Gbps-eSFP-SMF-1571nm-80km-commercial
Part Number	34060476
Model	eSFP-LH80-SM1571
Form factor	eSFP
Application standard	ITU-T G.957, STM-16
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	1.25 Gbit/s
Target transmission distance [km]	80 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1571 nm
Tx operating wavelength range [nm]	1564.5 nm - 1577.5 nm
Maximum Tx optical power (AVG) [dBm]	5 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	0 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	8.5 dB
Receiver Optical Characteristics	

Item	Value
Rx operating wavelength range [nm]	1260 nm - 1620 nm
Rx sensitivity (AVG) [dBm]	-28 dBm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	-9 dBm
Overload power (OMA) [dBm]	-

1.5.14.2 1.25Gbps-eSFP-SMF-1591nm-80km-commercial

Table 1-1077 1.25Gbps-eSFP-SMF-1591nm-80km-commercial specifications

Item	Value
Basic Information	
Module name	1.25Gbps-eSFP-SMF-1591nm-80km-commercial
Part Number	34060477
Model	eSFP-LH80-SM1591
Form factor	eSFP
Application standard	ITU-T G.957, STM-16
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	1.25 Gbit/s
Target transmission distance [km]	80 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1591 nm
Tx operating wavelength range [nm]	1584.5 nm - 1597.5 nm

Item	Value
Maximum Tx optical power (AVG) [dBm]	5 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	0 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	8.5 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1260 nm - 1620 nm
Rx sensitivity (AVG) [dBm]	-28 dBm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	-9 dBm
Overload power (OMA) [dBm]	-

1.5.14.3 1.25Gbps-eSFP-SMF-1551nm-80km-commercial

Table 1-1078 1.25Gbps-eSFP-SMF-1551nm-80km-commercial specifications

Item	Value
Basic Information	
Module name	1.25Gbps-eSFP-SMF-1551nm-80km-commercial
Part Number	34060478
Model	eSFP-LH80-SM1551
Form factor	eSFP
Application standard	ITU-T G.957, STM-16
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-8472

Item	Value
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	1.25 Gbit/s
Target transmission distance [km]	80 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1551 nm
Tx operating wavelength range [nm]	1544.5 nm - 1557.5 nm
Maximum Tx optical power (AVG) [dBm]	5 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	0 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	8.5 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1260 nm - 1620 nm
Rx sensitivity (AVG) [dBm]	-28 dBm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	-9 dBm
Overload power (OMA) [dBm]	-

1.5.14.4 1.25Gbps-eSFP-SMF-1511nm-80km-commercial

Table 1-1079 1.25Gbps-eSFP-SMF-1511nm-80km-commercial specifications

Item	Value
Basic Information	
Module name	1.25Gbps-eSFP-SMF-1511nm-80km-commercial
Part Number	34060479

Item	Value
Model	eSFP-LH80-SM1511
Form factor	eSFP
Application standard	ITU-T G.957, STM-16
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	1.25 Gbit/s
Target transmission distance [km]	80 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1511 nm
Tx operating wavelength range [nm]	1504.5 nm - 1517.5 nm
Maximum Tx optical power (AVG) [dBm]	5 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	0 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	8.5 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1260 nm - 1620 nm
Rx sensitivity (AVG) [dBm]	-28 dBm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	-9 dBm
Overload power (OMA) [dBm]	-

1.5.14.5 1.25Gbps-eSFP-SMF-1611nm-80km-commercial

Table 1-1080 1.25Gbps-eSFP-SMF-1611nm-80km-commercial specifications

Item	Value
Basic Information	
Module name	1.25Gbps-eSFP-SMF-1611nm-80km-commercial
Part Number	34060480
Model	eSFP-LH80-SM1611
Form factor	eSFP
Application standard	ITU-T G.957, STM-16
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	1.25 Gbit/s
Target transmission distance [km]	80 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1611 nm
Tx operating wavelength range [nm]	1604.5 nm - 1617.5 nm
Maximum Tx optical power (AVG) [dBm]	5 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	0 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	8.5 dB
Receiver Optical Characteristics	

Item	Value
Rx operating wavelength range [nm]	1260 nm - 1620 nm
Rx sensitivity (AVG) [dBm]	-28 dBm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	-9 dBm
Overload power (OMA) [dBm]	-

1.5.14.6 1.25Gbps-eSFP-SMF-1491nm-80km-commercial

Table 1-1081 1.25Gbps-eSFP-SMF-1491nm-80km-commercial specifications

Item	Value
Basic Information	
Module name	1.25Gbps-eSFP-SMF-1491nm-80km-commercial
Part Number	34060481
Model	eSFP-LH80-SM1491
Form factor	eSFP
Application standard	ITU-T G.957, STM-16
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	1.25 Gbit/s
Target transmission distance [km]	80 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1491 nm
Tx operating wavelength range [nm]	1484.5 nm - 1497.5 nm

Item	Value
Maximum Tx optical power (AVG) [dBm]	5 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	0 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	8.5 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1260 nm - 1620 nm
Rx sensitivity (AVG) [dBm]	-28 dBm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	-9 dBm
Overload power (OMA) [dBm]	-

1.5.14.7 1.25Gbps-eSFP-SMF-1531nm-80km-commercial

Table 1-1082 1.25Gbps-eSFP-SMF-1531nm-80km-commercial specifications

Item	Value
Basic Information	
Module name	1.25Gbps-eSFP-SMF-1531nm-80km-commercial
Part Number	34060482
Model	eSFP-LH80-SM1531
Form factor	eSFP
Application standard	ITU-T G.957, STM-16
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-8472

Item	Value
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	1.25 Gbit/s
Target transmission distance [km]	80 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1531 nm
Tx operating wavelength range [nm]	1524.5 nm - 1537.5 nm
Maximum Tx optical power (AVG) [dBm]	5 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	0 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	8.5 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1260 nm - 1620 nm
Rx sensitivity (AVG) [dBm]	-28 dBm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	-9 dBm
Overload power (OMA) [dBm]	-

1.5.14.8 1.25Gbps-eSFP-SMF-1471nm-80km-commercial

Table 1-1083 1.25Gbps-eSFP-SMF-1471nm-80km-commercial specifications

Item	Value
Basic Information	
Module name	1.25Gbps-eSFP-SMF-1471nm-80km-commercial
Part Number	34060483

Item	Value
Model	eSFP-LH80-SM1471
Form factor	eSFP
Application standard	ITU-T G.957, STM-16
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	1.25 Gbit/s
Target transmission distance [km]	80 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1471 nm
Tx operating wavelength range [nm]	1464.5 nm - 1477.5 nm
Maximum Tx optical power (AVG) [dBm]	5 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	0 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	8.5 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1260 nm - 1620 nm
Rx sensitivity (AVG) [dBm]	-28 dBm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	-9 dBm
Overload power (OMA) [dBm]	-

1.5.15 125M~2.67Gbps eSFP DWDM Optical Module

1.5.15.1 125M~2.67Gbps-eSFP-SMF-1560.61nm-120km-commercial

Table 1-1084 125M~2.67Gbps-eSFP-SMF-1560.61nm-120km-commercial specifications

Item	Value
Basic Information	
Module name	125M~2.67Gbps-eSFP-SMF-1560.61nm-120km-commercial
Part Number	34060366
Model	eSFP-LH120-SM192.10
Form factor	eSFP
Application standard	SONET OC-48 LR-2, Gigabit Ethernet
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	155 Mbit/s ~ 2.67 Gbit/s
Target transmission distance [km]	120 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1560.61 nm
Maximum Tx optical power (AVG) [dBm]	4 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	0 dBm
Minimum Tx optical power (OMA) [dBm]	-

Item	Value
Minimum extinction ratio [dB]	8.2 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1520 nm - 1570 nm
Rx sensitivity (AVG) [dBm]	-28 dBm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	-8 dBm
Overload power (OMA) [dBm]	-

1.5.15.2 125M~2.67Gbps-eSFP-SMF-1559.79nm-120km-commercial

Table 1-1085 125M~2.67Gbps-eSFP-SMF-1559.79nm-120km-commercial specifications

Item	Value
Basic Information	
Module name	125M~2.67Gbps-eSFP-SMF-1559.79nm-120km-commercial
Part Number	34060372
Model	eSFP-LH120-SM192.20
Form factor	eSFP
Application standard	SONET OC-48 LR-2, Gigabit Ethernet
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	155 Mbit/s ~ 2.67 Gbit/s
Target transmission distance [km]	120 km
Transmitter Optical Characteristics	

Item	Value
Center wavelength [nm]	1559.79 nm
Maximum Tx optical power (AVG) [dBm]	4 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	0 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	8.2 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1520 nm - 1570 nm
Rx sensitivity (AVG) [dBm]	-28 dBm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	-8 dBm
Overload power (OMA) [dBm]	-

1.5.15.3 125M~2.67Gbps-eSFP-SMF-1558.98nm-120km-commercial

Table 1-1086 125M~2.67Gbps-eSFP-SMF-1558.98nm-120km-commercial specifications

Item	Value
Basic Information	
Module name	125M~2.67Gbps-eSFP-SMF-1558.98nm-120km-commercial
Part Number	34060373
Model	eSFP-LH120-SM192.30
Form factor	eSFP
Application standard	SONET OC-48 LR-2, Gigabit Ethernet
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12

Item	Value
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	155 Mbit/s ~ 2.67 Gbit/s
Target transmission distance [km]	120 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1558.98 nm
Maximum Tx optical power (AVG) [dBm]	4 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	0 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	8.2 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1520 nm - 1570 nm
Rx sensitivity (AVG) [dBm]	-28 dBm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	-8 dBm
Overload power (OMA) [dBm]	-

1.5.15.4 125M~2.67Gbps-eSFP-SMF-1558.17nm-120km-commercial

Table 1-1087 125M~2.67Gbps-eSFP-SMF-1558.17nm-120km-commercial specifications

Item	Value
Basic Information	

Item	Value
Module name	125M~2.67Gbps-eSFP-SMF-1558.17nm-120km-commercial
Part Number	34060374
Model	eSFP-LH120-SM192.40
Form factor	eSFP
Application standard	SONET OC-48 LR-2, Gigabit Ethernet
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	155 Mbit/s ~ 2.67 Gbit/s
Target transmission distance [km]	120 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1558.17 nm
Maximum Tx optical power (AVG) [dBm]	4 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	0 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	8.2 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1520 nm - 1570 nm
Rx sensitivity (AVG) [dBm]	-28 dBm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	-8 dBm

Item	Value
Overload power (OMA) [dBm]	-

1.5.15.5 125M~2.67Gbps-eSFP-SMF-1557.36nm-120km-commercial

Table 1-1088 125M~2.67Gbps-eSFP-SMF-1557.36nm-120km-commercial specifications

Item	Value
Basic Information	
Module name	125M~2.67Gbps-eSFP-SMF-1557.36nm-120km-commercial
Part Number	34060375
Model	eSFP-LH120-SM192.50
Form factor	eSFP
Application standard	SONET OC-48 LR-2, Gigabit Ethernet
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	155 Mbit/s ~ 2.67 Gbit/s
Target transmission distance [km]	120 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1557.36 nm
Maximum Tx optical power (AVG) [dBm]	4 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	0 dBm

Item	Value
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	8.2 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1520 nm - 1570 nm
Rx sensitivity (AVG) [dBm]	-28 dBm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	-8 dBm
Overload power (OMA) [dBm]	-

1.5.15.6 125M~2.67Gbps-eSFP-SMF-1556.55nm-120km-commercial

Table 1-1089 125M~2.67Gbps-eSFP-SMF-1556.55nm-120km-commercial specifications

Item	Value
Basic Information	
Module name	125M~2.67Gbps-eSFP-SMF-1556.55nm-120km-commercial
Part Number	34060376
Model	eSFP-LH120-SM192.60
Form factor	eSFP
Application standard	SONET OC-48 LR-2, Gigabit Ethernet
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	155 Mbit/s ~ 2.67 Gbit/s

Item	Value
Target transmission distance [km]	120 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1556.55 nm
Maximum Tx optical power (AVG) [dBm]	4 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	0 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	8.2 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1520 nm - 1570 nm
Rx sensitivity (AVG) [dBm]	-28 dBm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	-8 dBm
Overload power (OMA) [dBm]	-

1.5.15.7 125M~2.67Gbps-eSFP-SMF-1555.75nm-120km-commercial

Table 1-1090 125M~2.67Gbps-eSFP-SMF-1555.75nm-120km-commercial specifications

Item	Value
Basic Information	
Module name	125M~2.67Gbps-eSFP-SMF-1555.75nm-120km-commercial
Part Number	34060377
Model	eSFP-LH120-SM192.70
Form factor	eSFP
Application standard	SONET OC-48 LR-2, Gigabit Ethernet
Connector type	LC

Item	Value
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	155 Mbit/s ~ 2.67 Gbit/s
Target transmission distance [km]	120 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1555.75 nm
Maximum Tx optical power (AVG) [dBm]	4 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	0 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	8.2 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1520 nm - 1570 nm
Rx sensitivity (AVG) [dBm]	-28 dBm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	-8 dBm
Overload power (OMA) [dBm]	-

1.5.15.8 125M~2.67Gbps-eSFP-SMF-1554.94nm-120km-commercial

Table 1-1091 125M~2.67Gbps-eSFP-SMF-1554.94nm-120km-commercial specifications

Item	Value
Basic Information	
Module name	125M~2.67Gbps-eSFP-SMF-1554.94nm-120km-commercial
Part Number	34060378
Model	eSFP-LH120-SM192.80
Form factor	eSFP
Application standard	SONET OC-48 LR-2, Gigabit Ethernet
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	155 Mbit/s ~ 2.67 Gbit/s
Target transmission distance [km]	120 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1554.94 nm
Maximum Tx optical power (AVG) [dBm]	4 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	0 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	8.2 dB
Receiver Optical Characteristics	

Item	Value
Rx operating wavelength range [nm]	1520 nm - 1570 nm
Rx sensitivity (AVG) [dBm]	-28 dBm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	-8 dBm
Overload power (OMA) [dBm]	-

1.5.15.9 125M~2.67Gbps-eSFP-SMF-1554.13nm-120km-commercial

Table 1-1092 125M~2.67Gbps-eSFP-SMF-1554.13nm-120km-commercial specifications

Item	Value
Basic Information	
Module name	125M~2.67Gbps-eSFP-SMF-1554.13nm-120km-commercial
Part Number	34060379
Model	eSFP-LH120-SM192.90
Form factor	eSFP
Application standard	SONET OC-48 LR-2, Gigabit Ethernet
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	155 Mbit/s ~ 2.67 Gbit/s
Target transmission distance [km]	120 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1554.13 nm

Item	Value
Maximum Tx optical power (AVG) [dBm]	4 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	0 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	8.2 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1520 nm - 1570 nm
Rx sensitivity (AVG) [dBm]	-28 dBm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	-8 dBm
Overload power (OMA) [dBm]	-

1.5.15.10 125M~2.67Gbps-eSFP-SMF-1553.33nm-120km-commercial

Table 1-1093 125M~2.67Gbps-eSFP-SMF-1553.33nm-120km-commercial specifications

Item	Value
Basic Information	
Module name	125M~2.67Gbps-eSFP-SMF-1553.33nm-120km-commercial
Part Number	34060380
Model	eSFP-LH120-SM193.00
Form factor	eSFP
Application standard	SONET OC-48 LR-2, Gigabit Ethernet
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)

Item	Value
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	155 Mbit/s ~ 2.67 Gbit/s
Target transmission distance [km]	120 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1553.33 nm
Maximum Tx optical power (AVG) [dBm]	4 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	0 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	8.2 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1520 nm - 1570 nm
Rx sensitivity (AVG) [dBm]	-28 dBm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	-8 dBm
Overload power (OMA) [dBm]	-

1.5.15.11 125M~2.67Gbps-eSFP-SMF-1552.52nm-120km-commercial

Table 1-1094 125M~2.67Gbps-eSFP-SMF-1552.52nm-120km-commercial specifications

Item	Value
Basic Information	
Module name	125M~2.67Gbps-eSFP-SMF-1552.52nm-120km-commercial

Item	Value
Part Number	34060381
Model	eSFP-LH120-SM193.10
Form factor	eSFP
Application standard	SONET OC-48 LR-2, Gigabit Ethernet
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	155 Mbit/s ~ 2.67 Gbit/s
Target transmission distance [km]	120 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1552.52 nm
Maximum Tx optical power (AVG) [dBm]	4 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	0 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	8.2 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1520 nm - 1570 nm
Rx sensitivity (AVG) [dBm]	-28 dBm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	-8 dBm
Overload power (OMA) [dBm]	-

1.5.15.12 125M~2.67Gbps-eSFP-SMF-1551.72nm-120km-commercial

Table 1-1095 125M~2.67Gbps-eSFP-SMF-1551.72nm-120km-commercial specifications

Item	Value
Basic Information	
Module name	125M~2.67Gbps-eSFP-SMF-1551.72nm-120km-commercial
Part Number	34060382
Model	eSFP-LH120-SM193.20
Form factor	eSFP
Application standard	SONET OC-48 LR-2, Gigabit Ethernet
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	155 Mbit/s ~ 2.67 Gbit/s
Target transmission distance [km]	120 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1551.72 nm
Maximum Tx optical power (AVG) [dBm]	4 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	0 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	8.2 dB
Receiver Optical Characteristics	

Item	Value
Rx operating wavelength range [nm]	1520 nm - 1570 nm
Rx sensitivity (AVG) [dBm]	-28 dBm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	-8 dBm
Overload power (OMA) [dBm]	-

1.5.15.13 125M~2.67Gbps-eSFP-SMF-1550.92nm-120km-commercial

Table 1-1096 125M~2.67Gbps-eSFP-SMF-1550.92nm-120km-commercial specifications

Item	Value
Basic Information	
Module name	125M~2.67Gbps-eSFP-SMF-1550.92nm-120km-commercial
Part Number	34060383
Model	eSFP-LH120-SM193.30
Form factor	eSFP
Application standard	SONET OC-48 LR-2, Gigabit Ethernet
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	155 Mbit/s ~ 2.67 Gbit/s
Target transmission distance [km]	120 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1550.92 nm

Item	Value
Maximum Tx optical power (AVG) [dBm]	4 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	0 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	8.2 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1520 nm - 1570 nm
Rx sensitivity (AVG) [dBm]	-28 dBm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	-8 dBm
Overload power (OMA) [dBm]	-

1.5.15.14 125M~2.67Gbps-eSFP-SMF-1550.12nm-120km-commercial

Table 1-1097 125M~2.67Gbps-eSFP-SMF-1550.12nm-120km-commercial specifications

Item	Value
Basic Information	
Module name	125M~2.67Gbps-eSFP-SMF-1550.12nm-120km-commercial
Part Number	34060384
Model	eSFP-LH120-SM193.40
Form factor	eSFP
Application standard	SONET OC-48 LR-2, Gigabit Ethernet
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)

Item	Value
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	155 Mbit/s ~ 2.67 Gbit/s
Target transmission distance [km]	120 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1550.12 nm
Maximum Tx optical power (AVG) [dBm]	4 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	0 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	8.2 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1520 nm - 1570 nm
Rx sensitivity (AVG) [dBm]	-28 dBm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	-8 dBm
Overload power (OMA) [dBm]	-

1.5.15.15 125M~2.67Gbps-eSFP-SMF-1549.32nm-120km-commercial

Table 1-1098 125M~2.67Gbps-eSFP-SMF-1549.32nm-120km-commercial specifications

Item	Value
Basic Information	
Module name	125M~2.67Gbps-eSFP-SMF-1549.32nm-120km-commercial

Item	Value
Part Number	34060385
Model	eSFP-LH120-SM193.50
Form factor	eSFP
Application standard	SONET OC-48 LR-2, Gigabit Ethernet
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	155 Mbit/s ~ 2.67 Gbit/s
Target transmission distance [km]	120 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1549.32 nm
Maximum Tx optical power (AVG) [dBm]	4 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	0 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	8.2 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1520 nm - 1570 nm
Rx sensitivity (AVG) [dBm]	-28 dBm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	-8 dBm
Overload power (OMA) [dBm]	-

1.5.15.16 125M~2.67Gbps-eSFP-SMF-1548.51nm-120km-commercial

Table 1-1099 125M~2.67Gbps-eSFP-SMF-1548.51nm-120km-commercial specifications

Item	Value
Basic Information	
Module name	125M~2.67Gbps-eSFP-SMF-1548.51nm-120km-commercial
Part Number	34060386
Model	eSFP-LH120-SM193.60
Form factor	eSFP
Application standard	SONET OC-48 LR-2, Gigabit Ethernet
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	155 Mbit/s ~ 2.67 Gbit/s
Target transmission distance [km]	120 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1548.51 nm
Maximum Tx optical power (AVG) [dBm]	4 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	0 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	8.2 dB
Receiver Optical Characteristics	

Item	Value
Rx operating wavelength range [nm]	1520 nm - 1570 nm
Rx sensitivity (AVG) [dBm]	-28 dBm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	-8 dBm
Overload power (OMA) [dBm]	-

1.5.15.17 125M~2.67Gbps-eSFP-SMF-1547.72nm-120km-commercial

Table 1-1100 125M~2.67Gbps-eSFP-SMF-1547.72nm-120km-commercial specifications

Item	Value
Basic Information	
Module name	125M~2.67Gbps-eSFP-SMF-1547.72nm-120km-commercial
Part Number	34060387
Model	eSFP-LH120-SM193.70
Form factor	eSFP
Application standard	SONET OC-48 LR-2, Gigabit Ethernet
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	155 Mbit/s ~ 2.67 Gbit/s
Target transmission distance [km]	120 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1547.72 nm

Item	Value
Maximum Tx optical power (AVG) [dBm]	4 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	0 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	8.2 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1520 nm - 1570 nm
Rx sensitivity (AVG) [dBm]	-28 dBm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	-8 dBm
Overload power (OMA) [dBm]	-

1.5.15.18 125M~2.67Gbps-eSFP-SMF-1546.92nm-120km-commercial

Table 1-1101 125M~2.67Gbps-eSFP-SMF-1546.92nm-120km-commercial specifications

Item	Value
Basic Information	
Module name	125M~2.67Gbps-eSFP-SMF-1546.92nm-120km-commercial
Part Number	34060388
Model	eSFP-LH120-SM193.80
Form factor	eSFP
Application standard	SONET OC-48 LR-2, Gigabit Ethernet
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)

Item	Value
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	155 Mbit/s ~ 2.67 Gbit/s
Target transmission distance [km]	120 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1546.92 nm
Maximum Tx optical power (AVG) [dBm]	4 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	0 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	8.2 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1520 nm - 1570 nm
Rx sensitivity (AVG) [dBm]	-28 dBm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	-8 dBm
Overload power (OMA) [dBm]	-

1.5.15.19 125M~2.67Gbps-eSFP-SMF-1546.12nm-120km-commercial

Table 1-1102 125M~2.67Gbps-eSFP-SMF-1546.12nm-120km-commercial specifications

Item	Value
Basic Information	
Module name	125M~2.67Gbps-eSFP-SMF-1546.12nm-120km-commercial

Item	Value
Part Number	34060389
Model	eSFP-LH120-SM193.90
Form factor	eSFP
Application standard	SONET OC-48 LR-2, Gigabit Ethernet
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	155 Mbit/s ~ 2.67 Gbit/s
Target transmission distance [km]	120 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1546.12 nm
Maximum Tx optical power (AVG) [dBm]	4 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	0 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	8.2 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1520 nm - 1570 nm
Rx sensitivity (AVG) [dBm]	-28 dBm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	-8 dBm
Overload power (OMA) [dBm]	-

1.5.15.20 125M~2.67Gbps-eSFP-SMF-1545.32nm-120km-commercial

Table 1-1103 125M~2.67Gbps-eSFP-SMF-1545.32nm-120km-commercial specifications

Item	Value
Basic Information	
Module name	125M~2.67Gbps-eSFP-SMF-1545.32nm-120km-commercial
Part Number	34060390
Model	eSFP-LH120-SM194.00
Form factor	eSFP
Application standard	SONET OC-48 LR-2, Gigabit Ethernet
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	155 Mbit/s ~ 2.67 Gbit/s
Target transmission distance [km]	120 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1545.32 nm
Maximum Tx optical power (AVG) [dBm]	4 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	0 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	8.2 dB
Receiver Optical Characteristics	

Item	Value
Rx operating wavelength range [nm]	1520 nm - 1570 nm
Rx sensitivity (AVG) [dBm]	-28 dBm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	-8 dBm
Overload power (OMA) [dBm]	-

1.5.15.21 125M~2.67Gbps-eSFP-SMF-1544.53nm-120km-commercial

Table 1-1104 125M~2.67Gbps-eSFP-SMF-1544.53nm-120km-commercial specifications

Item	Value
Basic Information	
Module name	125M~2.67Gbps-eSFP-SMF-1544.53nm-120km-commercial
Part Number	34060391
Model	eSFP-LH120-SM194.10
Form factor	eSFP
Application standard	SONET OC-48 LR-2, Gigabit Ethernet
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	155 Mbit/s ~ 2.67 Gbit/s
Target transmission distance [km]	120 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1544.53 nm

Item	Value
Maximum Tx optical power (AVG) [dBm]	4 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	0 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	8.2 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1520 nm - 1570 nm
Rx sensitivity (AVG) [dBm]	-28 dBm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	-8 dBm
Overload power (OMA) [dBm]	-

1.5.15.22 125M~2.67Gbps-eSFP-SMF-1543.73nm-120km-commercial

Table 1-1105 125M~2.67Gbps-eSFP-SMF-1543.73nm-120km-commercial specifications

Item	Value
Basic Information	
Module name	125M~2.67Gbps-eSFP-SMF-1543.73nm-120km-commercial
Part Number	34060392
Model	eSFP-LH120-SM194.20
Form factor	eSFP
Application standard	SONET OC-48 LR-2, Gigabit Ethernet
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)

Item	Value
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	155 Mbit/s ~ 2.67 Gbit/s
Target transmission distance [km]	120 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1543.73 nm
Maximum Tx optical power (AVG) [dBm]	4 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	0 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	8.2 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1520 nm - 1570 nm
Rx sensitivity (AVG) [dBm]	-28 dBm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	-8 dBm
Overload power (OMA) [dBm]	-

1.5.15.23 125M~2.67Gbps-eSFP-SMF-1542.94nm-120km-commercial

Table 1-1106 125M~2.67Gbps-eSFP-SMF-1542.94nm-120km-commercial specifications

Item	Value
Basic Information	
Module name	125M~2.67Gbps-eSFP-SMF-1542.94nm-120km-commercial

Item	Value
Part Number	34060393
Model	eSFP-LH120-SM194.30
Form factor	eSFP
Application standard	SONET OC-48 LR-2, Gigabit Ethernet
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	155 Mbit/s ~ 2.67 Gbit/s
Target transmission distance [km]	120 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1542.94 nm
Maximum Tx optical power (AVG) [dBm]	4 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	0 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	8.2 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1520 nm - 1570 nm
Rx sensitivity (AVG) [dBm]	-28 dBm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	-8 dBm
Overload power (OMA) [dBm]	-

1.5.15.24 125M~2.67Gbps-eSFP-SMF-1542.14nm-120km-commercial

Table 1-1107 125M~2.67Gbps-eSFP-SMF-1542.14nm-120km-commercial specifications

Item	Value
Basic Information	
Module name	125M~2.67Gbps-eSFP-SMF-1542.14nm-120km-commercial
Part Number	34060394
Model	eSFP-LH120-SM194.40
Form factor	eSFP
Application standard	SONET OC-48 LR-2, Gigabit Ethernet
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	155 Mbit/s ~ 2.67 Gbit/s
Target transmission distance [km]	120 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1542.14 nm
Maximum Tx optical power (AVG) [dBm]	4 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	0 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	8.2 dB
Receiver Optical Characteristics	

Item	Value
Rx operating wavelength range [nm]	1520 nm - 1570 nm
Rx sensitivity (AVG) [dBm]	-28 dBm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	-8 dBm
Overload power (OMA) [dBm]	-

1.5.15.25 125M~2.67Gbps-eSFP-SMF-1541.35nm-120km-commercial

Table 1-1108 125M~2.67Gbps-eSFP-SMF-1541.35nm-120km-commercial specifications

Item	Value
Basic Information	
Module name	125M~2.67Gbps-eSFP-SMF-1541.35nm-120km-commercial
Part Number	34060395
Model	eSFP-LH120-SM194.50
Form factor	eSFP
Application standard	SONET OC-48 LR-2, Gigabit Ethernet
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	155 Mbit/s ~ 2.67 Gbit/s
Target transmission distance [km]	120 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1541.35 nm

Item	Value
Maximum Tx optical power (AVG) [dBm]	4 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	0 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	8.2 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1520 nm - 1570 nm
Rx sensitivity (AVG) [dBm]	-28 dBm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	-8 dBm
Overload power (OMA) [dBm]	-

1.5.15.26 125M~2.67Gbps-eSFP-SMF-1540.56nm-120km-commercial

Table 1-1109 125M~2.67Gbps-eSFP-SMF-1540.56nm-120km-commercial specifications

Item	Value
Basic Information	
Module name	125M~2.67Gbps-eSFP-SMF-1540.56nm-120km-commercial
Part Number	34060396
Model	eSFP-LH120-SM194.60
Form factor	eSFP
Application standard	SONET OC-48 LR-2, Gigabit Ethernet
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)

Item	Value
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	155 Mbit/s ~ 2.67 Gbit/s
Target transmission distance [km]	120 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1540.56 nm
Maximum Tx optical power (AVG) [dBm]	4 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	0 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	8.2 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1520 nm - 1570 nm
Rx sensitivity (AVG) [dBm]	-28 dBm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	-8 dBm
Overload power (OMA) [dBm]	-

1.5.15.27 125M~2.67Gbps-eSFP-SMF-1539.77nm-120km-commercial

Table 1-1110 125M~2.67Gbps-eSFP-SMF-1539.77nm-120km-commercial specifications

Item	Value
Basic Information	
Module name	125M~2.67Gbps-eSFP-SMF-1539.77nm-120km-commercial

Item	Value
Part Number	34060397
Model	eSFP-LH120-SM194.70
Form factor	eSFP
Application standard	SONET OC-48 LR-2, Gigabit Ethernet
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	155 Mbit/s ~ 2.67 Gbit/s
Target transmission distance [km]	120 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1539.77 nm
Maximum Tx optical power (AVG) [dBm]	4 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	0 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	8.2 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1520 nm - 1570 nm
Rx sensitivity (AVG) [dBm]	-28 dBm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	-8 dBm
Overload power (OMA) [dBm]	-

1.5.15.28 125M~2.67Gbps-eSFP-SMF-1538.98nm-120km-commercial

Table 1-1111 125M~2.67Gbps-eSFP-SMF-1538.98nm-120km-commercial specifications

Item	Value
Basic Information	
Module name	125M~2.67Gbps-eSFP-SMF-1538.98nm-120km-commercial
Part Number	34060398
Model	eSFP-LH120-SM194.80
Form factor	eSFP
Application standard	SONET OC-48 LR-2, Gigabit Ethernet
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	155 Mbit/s ~ 2.67 Gbit/s
Target transmission distance [km]	120 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1538.98 nm
Maximum Tx optical power (AVG) [dBm]	4 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	0 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	8.2 dB
Receiver Optical Characteristics	

Item	Value
Rx operating wavelength range [nm]	1520 nm - 1570 nm
Rx sensitivity (AVG) [dBm]	-28 dBm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	-8 dBm
Overload power (OMA) [dBm]	-

1.5.15.29 125M~2.67Gbps-eSFP-SMF-1538.19nm-120km-commercial

Table 1-1112 125M~2.67Gbps-eSFP-SMF-1538.19nm-120km-commercial specifications

Item	Value
Basic Information	
Module name	125M~2.67Gbps-eSFP-SMF-1538.19nm-120km-commercial
Part Number	34060399
Model	eSFP-LH120-SM194.90
Form factor	eSFP
Application standard	SONET OC-48 LR-2, Gigabit Ethernet
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	155 Mbit/s ~ 2.67 Gbit/s
Target transmission distance [km]	120 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1538.19 nm

Item	Value
Maximum Tx optical power (AVG) [dBm]	4 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	0 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	8.2 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1520 nm - 1570 nm
Rx sensitivity (AVG) [dBm]	-28 dBm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	-8 dBm
Overload power (OMA) [dBm]	-

1.5.15.30 125M~2.67Gbps-eSFP-SMF-1537.40nm-120km-commercial

Table 1-1113 125M~2.67Gbps-eSFP-SMF-1537.40nm-120km-commercial specifications

Item	Value
Basic Information	
Module name	125M~2.67Gbps-eSFP-SMF-1537.40nm-120km-commercial
Part Number	34060400
Model	eSFP-LH120-SM195.00
Form factor	eSFP
Application standard	SONET OC-48 LR-2, Gigabit Ethernet
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)

Item	Value
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	155 Mbit/s ~ 2.67 Gbit/s
Target transmission distance [km]	120 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1537.4 nm
Maximum Tx optical power (AVG) [dBm]	4 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	0 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	8.2 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1520 nm - 1570 nm
Rx sensitivity (AVG) [dBm]	-28 dBm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	-8 dBm
Overload power (OMA) [dBm]	-

1.5.15.31 125M~2.67Gbps-eSFP-SMF-1536.61nm-120km-commercial

Table 1-1114 125M~2.67Gbps-eSFP-SMF-1536.61nm-120km-commercial specifications

Item	Value
Basic Information	
Module name	125M~2.67Gbps-eSFP-SMF-1536.61nm-120km-commercial

Item	Value
Part Number	34060401
Model	eSFP-LH120-SM195.10
Form factor	eSFP
Application standard	SONET OC-48 LR-2, Gigabit Ethernet
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	155 Mbit/s ~ 2.67 Gbit/s
Target transmission distance [km]	120 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1536.61 nm
Maximum Tx optical power (AVG) [dBm]	4 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	0 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	8.2 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1520 nm - 1570 nm
Rx sensitivity (AVG) [dBm]	-28 dBm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	-8 dBm
Overload power (OMA) [dBm]	-

1.5.15.32 125M~2.67Gbps-eSFP-SMF-1535.82nm-120km-commercial

Table 1-1115 125M~2.67Gbps-eSFP-SMF-1535.82nm-120km-commercial specifications

Item	Value
Basic Information	
Module name	125M~2.67Gbps-eSFP-SMF-1535.82nm-120km-commercial
Part Number	34060402
Model	eSFP-LH120-SM195.20
Form factor	eSFP
Application standard	SONET OC-48 LR-2, Gigabit Ethernet
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	155 Mbit/s ~ 2.67 Gbit/s
Target transmission distance [km]	120 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1535.82 nm
Maximum Tx optical power (AVG) [dBm]	4 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	0 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	8.2 dB
Receiver Optical Characteristics	

Item	Value
Rx operating wavelength range [nm]	1520 nm - 1570 nm
Rx sensitivity (AVG) [dBm]	-28 dBm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	-8 dBm
Overload power (OMA) [dBm]	-

1.5.15.33 125M~2.67Gbps-eSFP-SMF-1535.04nm-120km-commercial

Table 1-1116 125M~2.67Gbps-eSFP-SMF-1535.04nm-120km-commercial specifications

Item	Value
Basic Information	
Module name	125M~2.67Gbps-eSFP-SMF-1535.04nm-120km-commercial
Part Number	34060403
Model	eSFP-LH120-SM195.30
Form factor	eSFP
Application standard	SONET OC-48 LR-2, Gigabit Ethernet
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	155 Mbit/s ~ 2.67 Gbit/s
Target transmission distance [km]	120 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1535.04 nm

Item	Value
Maximum Tx optical power (AVG) [dBm]	4 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	0 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	8.2 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1520 nm - 1570 nm
Rx sensitivity (AVG) [dBm]	-28 dBm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	-8 dBm
Overload power (OMA) [dBm]	-

1.5.15.34 125M~2.67Gbps-eSFP-SMF-1534.25nm-120km-commercial

Table 1-1117 125M~2.67Gbps-eSFP-SMF-1534.25nm-120km-commercial specifications

Item	Value
Basic Information	
Module name	125M~2.67Gbps-eSFP-SMF-1534.25nm-120km-commercial
Part Number	34060404
Model	eSFP-LH120-SM195.40
Form factor	eSFP
Application standard	SONET OC-48 LR-2, Gigabit Ethernet
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)

Item	Value
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	155 Mbit/s ~ 2.67 Gbit/s
Target transmission distance [km]	120 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1534.25 nm
Maximum Tx optical power (AVG) [dBm]	4 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	0 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	8.2 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1520 nm - 1570 nm
Rx sensitivity (AVG) [dBm]	-28 dBm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	-8 dBm
Overload power (OMA) [dBm]	-

1.5.15.35 125M~2.67Gbps-eSFP-SMF-1533.47nm-120km-commercial

Table 1-1118 125M~2.67Gbps-eSFP-SMF-1533.47nm-120km-commercial specifications

Item	Value
Basic Information	
Module name	125M~2.67Gbps-eSFP-SMF-1533.47nm-120km-commercial

Item	Value
Part Number	34060405
Model	eSFP-LH120-SM195.50
Form factor	eSFP
Application standard	SONET OC-48 LR-2, Gigabit Ethernet
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	155 Mbit/s ~ 2.67 Gbit/s
Target transmission distance [km]	120 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1533.47 nm
Maximum Tx optical power (AVG) [dBm]	4 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	0 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	8.2 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1520 nm - 1570 nm
Rx sensitivity (AVG) [dBm]	-28 dBm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	-8 dBm
Overload power (OMA) [dBm]	-

1.5.15.36 125M~2.67Gbps-eSFP-SMF-1532.68nm-120km-commercial

Table 1-1119 125M~2.67Gbps-eSFP-SMF-1532.68nm-120km-commercial specifications

Item	Value
Basic Information	
Module name	125M~2.67Gbps-eSFP-SMF-1532.68nm-120km-commercial
Part Number	34060406
Model	eSFP-LH120-SM195.60
Form factor	eSFP
Application standard	SONET OC-48 LR-2, Gigabit Ethernet
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	155 Mbit/s ~ 2.67 Gbit/s
Target transmission distance [km]	120 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1532.68 nm
Maximum Tx optical power (AVG) [dBm]	4 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	0 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	8.2 dB
Receiver Optical Characteristics	

Item	Value
Rx operating wavelength range [nm]	1520 nm - 1570 nm
Rx sensitivity (AVG) [dBm]	-28 dBm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	-8 dBm
Overload power (OMA) [dBm]	-

1.5.15.37 125M~2.67Gbps-eSFP-SMF-1531.90nm-120km-commercial

Table 1-1120 125M~2.67Gbps-eSFP-SMF-1531.90nm-120km-commercial specifications

Item	Value
Basic Information	
Module name	125M~2.67Gbps-eSFP-SMF-1531.90nm-120km-commercial
Part Number	34060407
Model	eSFP-LH120-SM195.70
Form factor	eSFP
Application standard	SONET OC-48 LR-2, Gigabit Ethernet
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	155 Mbit/s ~ 2.67 Gbit/s
Target transmission distance [km]	120 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1531.9 nm

Item	Value
Maximum Tx optical power (AVG) [dBm]	4 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	0 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	8.2 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1520 nm - 1570 nm
Rx sensitivity (AVG) [dBm]	-28 dBm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	-8 dBm
Overload power (OMA) [dBm]	-

1.5.15.38 125M~2.67Gbps-eSFP-SMF-1531.12nm-120km-commercial

Table 1-1121 125M~2.67Gbps-eSFP-SMF-1531.12nm-120km-commercial specifications

Item	Value
Basic Information	
Module name	125M~2.67Gbps-eSFP-SMF-1531.12nm-120km-commercial
Part Number	34060408
Model	eSFP-LH120-SM195.80
Form factor	eSFP
Application standard	SONET OC-48 LR-2, Gigabit Ethernet
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)

Item	Value
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	155 Mbit/s ~ 2.67 Gbit/s
Target transmission distance [km]	120 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1531.12 nm
Maximum Tx optical power (AVG) [dBm]	4 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	0 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	8.2 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1520 nm - 1570 nm
Rx sensitivity (AVG) [dBm]	-28 dBm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	-8 dBm
Overload power (OMA) [dBm]	-

1.5.15.39 125M~2.67Gbps-eSFP-SMF-1530.33nm-120km-commercial

Table 1-1122 125M~2.67Gbps-eSFP-SMF-1530.33nm-120km-commercial specifications

Item	Value
Basic Information	
Module name	125M~2.67Gbps-eSFP-SMF-1530.33nm-120km-commercial

Item	Value
Part Number	34060409
Model	eSFP-LH120-SM195.90
Form factor	eSFP
Application standard	SONET OC-48 LR-2, Gigabit Ethernet
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	155 Mbit/s ~ 2.67 Gbit/s
Target transmission distance [km]	120 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1530.33 nm
Maximum Tx optical power (AVG) [dBm]	4 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	0 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	8.2 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1520 nm - 1570 nm
Rx sensitivity (AVG) [dBm]	-28 dBm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	-8 dBm
Overload power (OMA) [dBm]	-

1.5.15.40 125M~2.67Gbps-eSFP-SMF-1529.55nm-120km-commercial

Table 1-1123 125M~2.67Gbps-eSFP-SMF-1529.55nm-120km-commercial specifications

Item	Value
Basic Information	
Module name	125M~2.67Gbps-eSFP-SMF-1529.55nm-120km-commercial
Part Number	34060410
Model	eSFP-LH120-SM196.00
Form factor	eSFP
Application standard	SONET OC-48 LR-2, Gigabit Ethernet
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	155 Mbit/s ~ 2.67 Gbit/s
Target transmission distance [km]	120 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1529.55 nm
Maximum Tx optical power (AVG) [dBm]	4 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	0 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	8.2 dB
Receiver Optical Characteristics	

Item	Value
Rx operating wavelength range [nm]	1520 nm - 1570 nm
Rx sensitivity (AVG) [dBm]	-28 dBm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	-8 dBm
Overload power (OMA) [dBm]	-

1.5.16 2.5Gbps eSFP Optical Module

1.5.16.1 2.5Gbps-eSFP-SMF-1310nm-40km-commercial (34060289)

Table 1-1124 2.5Gbps-eSFP-SMF-1310nm-40km-commercial specifications

Item	Value
Basic Information	
Module name	2.5Gbps-eSFP-SMF-1310nm-40km-commercial
Part Number	34060289
Model	eSFP-1310nm-L-16.1
Form factor	eSFP
Application standard	ITU-T G.957, STM-16
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	2.5 Gbit/s
Target transmission distance [km]	40 km
Transmitter Optical Characteristics	

Item	Value
Center wavelength [nm]	1310 nm
Tx operating wavelength range [nm]	1280 nm - 1335 nm
Maximum Tx optical power (AVG) [dBm]	3 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	-2 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	8.2 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1260 nm - 1580 nm
Rx sensitivity (AVG) [dBm]	-27 dBm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	-9 dBm
Overload power (OMA) [dBm]	-

1.5.16.2 2.5Gbps-eSFP-SMF-1310nm-2km-commercial (34060484)

Table 1-1125 2.5Gbps-eSFP-SMF-1310nm-2km-commercial specifications

Item	Value
Basic Information	
Module name	2.5Gbps-eSFP-SMF-1310nm-2km-commercial
Part Number	34060484
Model	eSFP-1310nm-155M-2.5G-LC-SM-2km
Form factor	eSFP
Application standard	ITU-T G.957, STM-16
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12

Item	Value
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	2.5 Gbit/s
Target transmission distance [km]	2 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1310 nm
Tx operating wavelength range [nm]	1266 nm - 1360 nm
Maximum Tx optical power (AVG) [dBm]	-3 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	-10 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	8.5 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1200 nm - 1650 nm
Rx sensitivity (AVG) [dBm]	-18 dBm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	-3 dBm
Overload power (OMA) [dBm]	-

1.5.16.3 2.5Gbps-eSFP-SMF-1310nm-15km-commercial (34060485)

Table 1-1126 2.5Gbps-eSFP-SMF-1310nm-15km-commercial specifications

Item	Value
Basic Information	

Item	Value
Module name	2.5Gbps-eSFP-SMF-1310nm-15km-commercial
Part Number	34060485
Model	eSFP-SM1310-155M~2.5G-15km
Form factor	eSFP
Application standard	ITU-T G.957, STM-16
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	2.5 Gbit/s
Target transmission distance [km]	15 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1310 nm
Tx operating wavelength range [nm]	1260 nm - 1360 nm
Maximum Tx optical power (AVG) [dBm]	0 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	-5 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	8.5 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1250 nm - 1620 nm
Rx sensitivity (AVG) [dBm]	-18 dBm
Rx sensitivity (OMA) [dBm]	-

Item	Value
Overload power (AVG) [dBm]	0 dBm
Overload power (OMA) [dBm]	-

1.5.16.4 2.5Gbps-eSFP-SMF-1550nm-80km-commercial (34060488)

Table 1-1127 2.5Gbps-eSFP-SMF-1550nm-80km-commercial specifications

Item	Value
Basic Information	
Module name	2.5Gbps-eSFP-SMF-1550nm-80km-commercial
Part Number	34060488
Model	eSFP-1550nm-L-16.2
Form factor	eSFP
Application standard	ITU-T G.957, STM-16
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	2.5 Gbit/s
Target transmission distance [km]	80 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1550 nm
Tx operating wavelength range [nm]	1500 nm - 1580 nm
Maximum Tx optical power (AVG) [dBm]	3 dBm
Maximum Tx optical power (OMA) [dBm]	-

Item	Value
Minimum Tx optical power (AVG) [dBm]	-2 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	8.2 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1200 nm - 1650 nm
Rx sensitivity (AVG) [dBm]	-28 dBm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	-9 dBm
Overload power (OMA) [dBm]	-

1.5.16.5 2.5Gbps-eSFP-SMF-1310nm-15km-commercial (02310QXP)

Table 1-1128 2.5Gbps-eSFP-SMF-1310nm-15km-commercial specifications

Item	Value
Basic Information	
Module name	2.5Gbps-eSFP-SMF-1310nm-15km-commercial
Part Number	02310QXP
Model	OSU015N02
Form factor	eSFP
Application standard	ITU-T G.957, STM-16
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V

Item	Value
Transmission rate [bit/s]	2.5 Gbit/s
Target transmission distance [km]	15 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1310 nm
Tx operating wavelength range [nm]	1260 nm - 1360 nm
Maximum Tx optical power (AVG) [dBm]	0 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	-5 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	8.5 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1250 nm - 1620 nm
Rx sensitivity (AVG) [dBm]	-18 dBm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	0 dBm
Overload power (OMA) [dBm]	-

1.5.16.6 2.5Gbps-eSFP-SMF-1310nm-40km-commercial (02310QXQ)

Table 1-1129 2.5Gbps-eSFP-SMF-1310nm-40km-commercial specifications

Item	Value
Basic Information	
Module name	2.5Gbps-eSFP-SMF-1310nm-40km-commercial
Part Number	02310QXQ
Model	OSU040N01
Form factor	eSFP
Application standard	ITU-T G.957, STM-16

Item	Value
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	2.5 Gbit/s
Target transmission distance [km]	40 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1310 nm
Tx operating wavelength range [nm]	1280 nm - 1335 nm
Maximum Tx optical power (AVG) [dBm]	3 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	-2 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	8.2 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1260 nm - 1580 nm
Rx sensitivity (AVG) [dBm]	-27 dBm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	-9 dBm
Overload power (OMA) [dBm]	-

1.5.16.7 2.5Gbps-eSFP-SMF-1550nm-80km-commercial (02310QXR)

Table 1-1130 2.5Gbps-eSFP-SMF-1550nm-80km-commercial specifications

Item	Value
Basic Information	
Module name	2.5Gbps-eSFP-SMF-1550nm-80km-commercial
Part Number	02310QXR
Model	OSU080N02
Form factor	eSFP
Application standard	ITU-T G.957, STM-16
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	2.5 Gbit/s
Target transmission distance [km]	80 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1550 nm
Tx operating wavelength range [nm]	1500 nm - 1580 nm
Maximum Tx optical power (AVG) [dBm]	3 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	-2 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	8.2 dB
Receiver Optical Characteristics	

Item	Value
Rx operating wavelength range [nm]	1200 nm - 1650 nm
Rx sensitivity (AVG) [dBm]	-28 dBm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	-9 dBm
Overload power (OMA) [dBm]	-

1.5.16.8 2.5Gbps-eSFP-SMF-1310nm-2km-commercial (02310QXW)

Table 1-1131 2.5Gbps-eSFP-SMF-1310nm-2km-commercial specifications

Item	Value
Basic Information	
Module name	2.5Gbps-eSFP-SMF-1310nm-2km-commercial
Part Number	02310QXW
Model	OSU002N02
Form factor	eSFP
Application standard	ITU-T G.957, STM-16
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	2.5 Gbit/s
Target transmission distance [km]	2 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1310 nm
Tx operating wavelength range [nm]	1266 nm - 1360 nm

Item	Value
Maximum Tx optical power (AVG) [dBm]	-3 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	-10 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	8.5 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1200 nm - 1650 nm
Rx sensitivity (AVG) [dBm]	-18 dBm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	-3 dBm
Overload power (OMA) [dBm]	-

1.5.17 10Gbps SFP+ Optical Module

1.5.17.1 10Gbps-SFP+-SMF-1550nm-80km-commercial (02310PVT)

Table 1-1132 10Gbps-SFP+-SMF-1550nm-80km-commercial specifications

Item	Value
Basic Information	
Module name	10Gbps-SFP+-SMF-1550nm-80km-commercial
Part Number	02310PVT
Model	OSX080N03
Form factor	SFP+
Application standard	IEEE 802.3ae, 10GBASE-ZR/ZW
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12

Item	Value
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	9.953 Gbit/s 10.3125 Gbit/s
Target transmission distance [km]	80 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1550 nm
Tx operating wavelength range [nm]	1530 nm - 1565 nm
Maximum Tx optical power (AVG) [dBm]	4 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	0 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	9 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1260 nm - 1565 nm
Rx sensitivity (AVG) [dBm]	-24 dBm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	-7 dBm
Overload power (OMA) [dBm]	-
NOTE The interface standard is Huawei-specific. Self-loop is not supported. An optical attenuator must be added if self-loop is required.	

1.5.17.2 10Gbps-SFP+-SMF-1550nm-80km-commercial (02310PVU)

Table 1-1133 10Gbps-SFP+-SMF-1550nm-80km-commercial specifications

Item	Value
Basic Information	
Module name	10Gbps-SFP+-SMF-1550nm-80km-commercial
Part Number	02310PVU
Model	OSX080N04
Form factor	SFP+
Application standard	IEEE 802.3ae, 10GBASE-ZR/ZW
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	9.953 Gbit/s 10.3125 Gbit/s
Target transmission distance [km]	80 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1550 nm
Tx operating wavelength range [nm]	1530 nm - 1565 nm
Maximum Tx optical power (AVG) [dBm]	4 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	0 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	9 dB

Item	Value
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1260 nm - 1565 nm
Rx sensitivity (AVG) [dBm]	-24 dBm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	-7 dBm
Overload power (OMA) [dBm]	-
NOTE The interface standard is Huawei-specific. Self-loop is not supported. An optical attenuator must be added if self-loop is required.	

1.5.17.3 10Gbps-SFP+-SMF-1310nm-10km-industry

Table 1-1134 10Gbps-SFP+-SMF-1310nm-10km-industry specifications

Item	Value
Basic Information	
Module name	10Gbps-SFP+-SMF-1310nm-10km-industry
Part Number	34060599
Model	OSX010N05
Form factor	SFP+
Application standard	IEEE 802.3ae, 10GBASE-LR/LW
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	-40°C to +85°C(-40°F to +185°F)
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	9.953 Gbit/s 10.3125 Gbit/s
Target transmission distance [km]	10 km

Item	Value
Transmitter Optical Characteristics	
Center wavelength [nm]	1310 nm
Tx operating wavelength range [nm]	1260 nm - 1355 nm
Maximum Tx optical power (AVG) [dBm]	0.5 dBm
Maximum Tx optical power (OMA) [dBm]	-5.2 dBm
Minimum Tx optical power (AVG) [dBm]	-8.2 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	3.5 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1260 nm - 1355 nm
Rx sensitivity (AVG) [dBm]	-14.4 dBm
Rx sensitivity (OMA) [dBm]	-12.6 dBm
Overload power (AVG) [dBm]	0.5 dBm
Overload power (OMA) [dBm]	-

1.5.17.4 10Gbps-SFP+-SMF-1550nm-40km-industry

Table 1-1135 10Gbps-SFP+-SMF-1550nm-40km-industry specifications

Item	Value
Basic Information	
Module name	10Gbps-SFP+-SMF-1550nm-40km-industry
Part Number	34060684
Model	OSX040N05
Form factor	SFP+
Application standard	IEEE 802.3ae, 10GBASE-ER/EW
Connector type	LC
Optical fiber type	SMF

Item	Value
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	-40°C to +85°C(-40°F to +185°F)
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	9.953 Gbit/s 10.3125 Gbit/s
Target transmission distance [km]	40 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1550 nm
Tx operating wavelength range [nm]	1530 nm - 1565 nm
Maximum Tx optical power (AVG) [dBm]	4 dBm
Maximum Tx optical power (OMA) [dBm]	-1.7 dBm
Minimum Tx optical power (AVG) [dBm]	-4.7 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	3 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1530 nm - 1565 nm
Rx sensitivity (AVG) [dBm]	-15.8 dBm
Rx sensitivity (OMA) [dBm]	-14.1 dBm
Overload power (AVG) [dBm]	-1 dBm
Overload power (OMA) [dBm]	-
NOTE Self-loop is not supported. An optical attenuator must be added if self-loop is required.	

1.5.17.5 10Gbps-SFP+-SMF-1310nm-10km-commercial (S4017405)

Table 1-1136 10Gbps-SFP+-SMF-1310nm-10km-commercial specifications

Item	Value
Basic Information	
Module name	10Gbps-SFP+-SMF-1310nm-10km-commercial
Part Number	S4017405
Model	OSX010N10
Form factor	SFP+
Application standard	IEEE 802.3ae, 10GBASE-LR/LW
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	9.953 Gbit/s 10.3125 Gbit/s
Target transmission distance [km]	10 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1310 nm
Tx operating wavelength range [nm]	1260 nm - 1355 nm
Maximum Tx optical power (AVG) [dBm]	0.5 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	-8.2 dBm
Minimum Tx optical power (OMA) [dBm]	-5.2 dBm
Minimum extinction ratio [dB]	3.5 dB

Item	Value
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1260 nm - 1355 nm
Rx sensitivity (AVG) [dBm]	-14.4 dBm
Rx sensitivity (OMA) [dBm]	-12.6 dBm
Overload power (AVG) [dBm]	0.5 dBm
Overload power (OMA) [dBm]	0.5 dBm

1.5.17.6 10Gbps-SFP+-SMF-1550nm-40km-commercial (S4017406)

Table 1-1137 10Gbps-SFP+-SMF-1550nm-40km-commercial specifications

Item	Value
Basic Information	
Module name	10Gbps-SFP+-SMF-1550nm-40km-commercial
Part Number	S4017406
Model	OSX040N10
Form factor	SFP+
Application standard	IEEE 802.3ae, 10GBASE-ER/EW
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	9.953 Gbit/s 10.3125 Gbit/s
Target transmission distance [km]	40 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1550 nm

Item	Value
Tx operating wavelength range [nm]	1530 nm - 1565 nm
Maximum Tx optical power (AVG) [dBm]	4 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	-4.7 dBm
Minimum Tx optical power (OMA) [dBm]	-1.7 dBm
Minimum extinction ratio [dB]	3 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1530 nm - 1565 nm
Rx sensitivity (AVG) [dBm]	-15.8 dBm
Rx sensitivity (OMA) [dBm]	-14.1 dBm
Overload power (AVG) [dBm]	-1 dBm
Overload power (OMA) [dBm]	-1 dBm
NOTE Self-loop is not supported. An optical attenuator must be added if self-loop is required.	

1.5.17.7 10Gbps-SFP+-MMF-850nm-0.3km-commercial

Table 1-1138 10Gbps-SFP+-MMF-850nm-0.3km-commercial specifications

Item	Value
Basic Information	
Module name	10Gbps-SFP+-MMF-850nm-0.3km-commercial
Part Number	S4017482
Model	OSX040N03
Form factor	SFP+
Application standard	IEEE 802.3ae, 10GBASE-SR/SW
Connector type	LC
Optical fiber type	MMF
Bit error ratio (BER)	<1x10E-12

Item	Value
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	9.953 Gbit/s 10.3125 Gbit/s
Target transmission distance [km]	0.3 km(OM3)
Transmitter Optical Characteristics	
Center wavelength [nm]	850 nm
Tx operating wavelength range [nm]	840 nm - 860 nm
Maximum Tx optical power (AVG) [dBm]	-1 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	-7.3 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	3 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	840 nm - 860 nm
Rx sensitivity (AVG) [dBm]	-9.9 dBm
Rx sensitivity (OMA) [dBm]	-11.1 dBm
Overload power (AVG) [dBm]	-1 dBm
Overload power (OMA) [dBm]	-1 dBm

1.5.17.8 10Gbps-SFP+-SMF-1310nm-10km-commercial (S4017483)

Table 1-1139 10Gbps-SFP+-SMF-1310nm-10km-commercial specifications

Item	Value
Basic Information	

Item	Value
Module name	10Gbps-SFP+-SMF-1310nm-10km-commercial
Part Number	S4017483
Model	OSX001002
Form factor	SFP+
Application standard	IEEE 802.3ae, 10GBASE-LR/LW
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	9.953 Gbit/s 10.3125 Gbit/s
Target transmission distance [km]	10 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1310 nm
Tx operating wavelength range [nm]	1260 nm - 1355 nm
Maximum Tx optical power (AVG) [dBm]	0.5 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	-8.2 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	3.5 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1260 nm - 1355 nm
Rx sensitivity (AVG) [dBm]	-14.4 dBm

Item	Value
Rx sensitivity (OMA) [dBm]	-12.6 dBm
Overload power (AVG) [dBm]	0.5 dBm
Overload power (OMA) [dBm]	0.5 dBm

1.5.17.9 10Gbps-SFP+-SMF-1550nm-40km-commercial (S4017484)

Table 1-1140 10Gbps-SFP+-SMF-1550nm-40km-commercial specifications

Item	Value
Basic Information	
Module name	10Gbps-SFP+-SMF-1550nm-40km-commercial
Part Number	S4017484
Model	OMXD30002
Form factor	SFP+
Application standard	IEEE 802.3ae, 10GBASE-ER/EW
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	9.953 Gbit/s 10.3125 Gbit/s
Target transmission distance [km]	40 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1550 nm
Tx operating wavelength range [nm]	1530 nm - 1565 nm
Maximum Tx optical power (AVG) [dBm]	4 dBm

Item	Value
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	-4.7 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	3 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1530 nm - 1565 nm
Rx sensitivity (AVG) [dBm]	-15.8 dBm
Rx sensitivity (OMA) [dBm]	-14.1 dBm
Overload power (AVG) [dBm]	-1 dBm
Overload power (OMA) [dBm]	-1 dBm
NOTE Self-loop is not supported. An optical attenuator must be added if self-loop is required.	

1.5.17.10 SFP+ with CDR-1550nm-10G--1dBm-2dBm--16dBm-LC-SMF-40km-industry

Table 1-1141 SFP+ with CDR-1550nm-10G--1dBm-2dBm--16dBm-LC-SMF-40km-industry specifications

Item	Value
Basic Information	
Module name	SFP+ with CDR-1550nm-10G--1dBm-2dBm--16dBm-LC-SMF-40km-industry
Part Number	02314UWM
Model	OSX040N13
Form factor	SFP+
Application standard	IEEE 802.3ae, 10GBASE-ZR/ZW
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12

Item	Value
Working case temperature [°C(°F)]	-40°C to 85°C(-40°F to 185°F)
DDM options	SFF-8472
Environment standard	RoHS
ESD(HBM1) [V]	1000 V
Transmission rate [bit/s]	10.3125 Gbit/s
Target transmission distance [km]	40 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1550 nm
Tx operating wavelength range [nm]	1530 nm - 1565 nm
Maximum Tx optical power (AVG) [dBm]	2 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	-1 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	8.2 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1260 nm - 1605 nm
Rx sensitivity (AVG) [dBm]	-16(11.1G, FEC enable); -15.8(10.3125G)
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	-1 dBm
Overload power (OMA) [dBm]	-
NOTE The interface standard is Huawei-specific. Self-loop is not supported. An optical attenuator must be added if self-loop is required.	

1.5.18 1.25/9.953/10.3125Gbps SFP+ Optical Module

1.5.18.1 1.25/9.953/10.3125Gbps-SFP+-SMF-1310nm-10km-commercial

Table 1-1142 1.25/9.953/10.3125Gbps-SFP+-SMF-1310nm-10km-commercial specifications

Item	Value
Basic Information	
Module name	1.25/9.953/10.3125Gbps-SFP+-SMF-1310nm-10km-commercial
Part Number	34061042
Model	OSX010N13
Form factor	SFP+
Application standard	IEEE 802.3ae, 10GBASE-LR/LW, 1000BASE-LX
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	1.25 Gbit/s 9.953 Gbit/s 10.3125 Gbit/s
Target transmission distance [km]	GE: 10 km 10GE: 10 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1310 nm
Tx operating wavelength range [nm]	1260 nm - 1360 nm
Maximum Tx optical power (AVG) [dBm]	GE: 0.5 dBm 10GE: 0.5 dBm
Maximum Tx optical power (OMA) [dBm]	-

Item	Value
Minimum Tx optical power (AVG) [dBm]	GE: -8.2 dBm 10GE: -8.2 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	GE: 9 dB 10GE: 3.5 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1260 nm - 1360 nm
Rx sensitivity (AVG) [dBm]	GE: -19 dBm 10GE: -14.4 dBm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	0.5 dBm
Overload power (OMA) [dBm]	0.5 dBm

1.5.18.2 1.25/9.953/10.3125Gbps-SFP+-SMF-1550nm-40km-commercial

Table 1-1143 1.25/9.953/10.3125Gbps-SFP+-SMF-1550nm-40km-commercial specifications

Item	Value
Basic Information	
Module name	1.25/9.953/10.3125Gbps-SFP+-SMF-1550nm-40km-commercial
Part Number	34061043
Model	OSX040N12
Form factor	SFP+
Application standard	IEEE 802.3ae, 10GBASE-ER/EW, 1000BASE-LX
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-8472

Item	Value
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	1.25 Gbit/s 9.953 Gbit/s 10.3125 Gbit/s
Target transmission distance [km]	GE: 40 km 10GE: 40 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1550 nm
Tx operating wavelength range [nm]	1530 nm - 1565 nm
Maximum Tx optical power (AVG) [dBm]	GE: 4 dBm 10GE: 4 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	GE: -4.7 dBm 10GE: -4.7 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	GE: 9 dB 10GE: 3.0 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1530 nm - 1565 nm
Rx sensitivity (AVG) [dBm]	GE: -15.8 dBm 10GE: -15.8 dBm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	-1 dBm
Overload power (OMA) [dBm]	-1 dBm

1.5.19 10Gbps SFP+ CWDM Optical Module

1.5.19.1 10Gbps-SFP+-SMF-1511nm-70km-commercial

Table 1-1144 10Gbps-SFP+-SMF-1511nm-70km-commercial specifications

Item	Value
Basic Information	
Module name	10Gbps-SFP+-SMF-1511nm-70km-commercial
Part Number	34060686
Model	OSX070001
Form factor	SFP+
Application standard	IEEE 802.3ae, 10GBASE-X
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	9.953 Gbit/s 10.3125 Gbit/s
Target transmission distance [km]	70 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1511 nm
Tx operating wavelength range [nm]	1504.5 nm - 1517.5 nm
Maximum Tx optical power (AVG) [dBm]	4 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	0 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	8.2 dB

Item	Value
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1460 nm - 1620 nm
Rx sensitivity (AVG) [dBm]	-23 dBm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	-7 dBm
Overload power (OMA) [dBm]	-

1.5.19.2 10Gbps-SFP+-SMF-1471nm-70km-commercial

Table 1-1145 10Gbps-SFP+-SMF-1471nm-70km-commercial specifications

Item	Value
Basic Information	
Module name	10Gbps-SFP+-SMF-1471nm-70km-commercial
Part Number	34060687
Model	OSX070002
Form factor	SFP+
Application standard	IEEE 802.3ae, 10GBASE-X
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	9.953 Gbit/s 10.3125 Gbit/s
Target transmission distance [km]	70 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1471 nm

Item	Value
Tx operating wavelength range [nm]	1464.5 nm - 1477.5 nm
Maximum Tx optical power (AVG) [dBm]	4 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	0 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	8.2 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1460 nm - 1620 nm
Rx sensitivity (AVG) [dBm]	-23 dBm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	-7 dBm
Overload power (OMA) [dBm]	-

1.5.19.3 10Gbps-SFP+-SMF-1491nm-70km-commercial

Table 1-1146 10Gbps-SFP+-SMF-1491nm-70km-commercial specifications

Item	Value
Basic Information	
Module name	10Gbps-SFP+-SMF-1491nm-70km-commercial
Part Number	34060688
Model	OSX070003
Form factor	SFP+
Application standard	IEEE 802.3ae, 10GBASE-X
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)

Item	Value
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	9.953 Gbit/s 10.3125 Gbit/s
Target transmission distance [km]	70 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1491 nm
Tx operating wavelength range [nm]	1484.5 nm - 1497.5 nm
Maximum Tx optical power (AVG) [dBm]	4 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	0 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	8.2 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1460 nm - 1620 nm
Rx sensitivity (AVG) [dBm]	-23 dBm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	-7 dBm
Overload power (OMA) [dBm]	-

1.5.19.4 10Gbps-SFP+-SMF-1531nm-70km-commercial

Table 1-1147 10Gbps-SFP+-SMF-1531nm-70km-commercial specifications

Item	Value
Basic Information	

Item	Value
Module name	10Gbps-SFP+-SMF-1531nm-70km-commercial
Part Number	34060689
Model	OSX070004
Form factor	SFP+
Application standard	IEEE 802.3ae, 10GBASE-X
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	9.953 Gbit/s 10.3125 Gbit/s
Target transmission distance [km]	70 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1531 nm
Tx operating wavelength range [nm]	1524.5 nm - 1537.5 nm
Maximum Tx optical power (AVG) [dBm]	4 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	0 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	8.2 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1460 nm - 1620 nm
Rx sensitivity (AVG) [dBm]	-23 dBm

Item	Value
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	-7 dBm
Overload power (OMA) [dBm]	-

1.5.19.5 10Gbps-SFP+-SMF-1551nm-70km-commercial

Table 1-1148 10Gbps-SFP+-SMF-1551nm-70km-commercial specifications

Item	Value
Basic Information	
Module name	10Gbps-SFP+-SMF-1551nm-70km-commercial
Part Number	34060690
Model	OSX070005
Form factor	SFP+
Application standard	IEEE 802.3ae, 10GBASE-X
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	9.953 Gbit/s 10.3125 Gbit/s
Target transmission distance [km]	70 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1551 nm
Tx operating wavelength range [nm]	1544.5 nm - 1557.5 nm
Maximum Tx optical power (AVG) [dBm]	4 dBm

Item	Value
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	0 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	8.2 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1460 nm - 1620 nm
Rx sensitivity (AVG) [dBm]	-23 dBm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	-7 dBm
Overload power (OMA) [dBm]	-

1.5.19.6 10Gbps-SFP+-SMF-1571nm-70km-commercial

Table 1-1149 10Gbps-SFP+-SMF-1571nm-70km-commercial specifications

Item	Value
Basic Information	
Module name	10Gbps-SFP+-SMF-1571nm-70km-commercial
Part Number	34060691
Model	OSX070006
Form factor	SFP+
Application standard	IEEE 802.3ae, 10GBASE-X
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1

Item	Value
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	9.953 Gbit/s 10.3125 Gbit/s
Target transmission distance [km]	70 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1571 nm
Tx operating wavelength range [nm]	1564.5 nm - 1577.5 nm
Maximum Tx optical power (AVG) [dBm]	4 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	0 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	8.2 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1460 nm - 1620 nm
Rx sensitivity (AVG) [dBm]	-23 dBm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	-7 dBm
Overload power (OMA) [dBm]	-

1.5.19.7 10Gbps-SFP+-SMF-1591nm-70km-commercial

Table 1-1150 10Gbps-SFP+-SMF-1591nm-70km-commercial specifications

Item	Value
Basic Information	
Module name	10Gbps-SFP+-SMF-1591nm-70km-commercial
Part Number	34060692
Model	OSX070007

Item	Value
Form factor	SFP+
Application standard	IEEE 802.3ae, 10GBASE-X
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	9.953 Gbit/s 10.3125 Gbit/s
Target transmission distance [km]	70 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1591 nm
Tx operating wavelength range [nm]	1584.5 nm - 1597.5 nm
Maximum Tx optical power (AVG) [dBm]	4 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	0 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	8.2 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1460 nm - 1620 nm
Rx sensitivity (AVG) [dBm]	-21 dBm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	-7 dBm
Overload power (OMA) [dBm]	-

1.5.19.8 10Gbps-SFP+-SMF-1611nm-70km-commercial

Table 1-1151 10Gbps-SFP+-SMF-1611nm-70km-commercial specifications

Item	Value
Basic Information	
Module name	10Gbps-SFP+-SMF-1611nm-70km-commercial
Part Number	34060693
Model	OSX070008
Form factor	SFP+
Application standard	IEEE 802.3ae, 10GBASE-X
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	9.953 Gbit/s 10.3125 Gbit/s
Target transmission distance [km]	70 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1611 nm
Tx operating wavelength range [nm]	1604.5 nm - 1617.4 nm
Maximum Tx optical power (AVG) [dBm]	4 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	0 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	8.2 dB

Item	Value
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1460 nm - 1620 nm
Rx sensitivity (AVG) [dBm]	-21 dBm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	-7 dBm
Overload power (OMA) [dBm]	-

1.5.20 10Gbps SFP+ BIDI Optical Module

1.5.20.1 10Gbps-SFP+-SMF-1270nm(Tx)/1330nm(Rx)-40km-commercial (02311JNF)

Table 1-1152 10Gbps-SFP+-SMF-1270nm(Tx)/1330nm(Rx)-40km-commercial specifications

Item	Value
Basic Information	
Module name	10Gbps-SFP+-SMF-1270nm(Tx)/1330nm(Rx)-40km-commercial
Part Number	02311JNF
Model	OSX040B10
Form factor	SFP+
Application standard	IEEE 802.3ae, 10GBASE-BX-U
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	0°C to 70°C (32°F to 158°F)
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V

Item	Value
Transmission rate [bit/s]	9.953 Gbit/s 10.3125 Gbit/s
Target transmission distance [km]	40 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1270 nm
Tx operating wavelength range [nm]	1260 nm - 1280 nm
Maximum Tx optical power (AVG) [dBm]	5 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	0 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	3.5 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1320 nm - 1340 nm
Rx sensitivity (AVG) [dBm]	-18 dBm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	-9 dBm
Overload power (OMA) [dBm]	-
NOTE 1. Ensure that the optical power on the receive side is less than or equal to -5 dBm. If the optical power does not meet the requirement, add an optical attenuator. Do not use short-distance optical fibers for direct connection.	

1.5.20.2 10Gbps-SFP+-SMF-1330nm(Tx)/1270nm(Rx)-40km-commercial (02311JNQ)

Table 1-1153 10Gbps-SFP+-SMF-1330nm(Tx)/1270nm(Rx)-40km-commercial specifications

Item	Value
Basic Information	
Module name	10Gbps-SFP+-SMF-1330nm(Tx)/1270nm(Rx)-40km-commercial

Item	Value
Part Number	02311JNQ
Model	OSX040B11
Form factor	SFP+
Application standard	IEEE 802.3ae, 10GBASE-BX40-D
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	9.953 Gbit/s 10.3125 Gbit/s
Target transmission distance [km]	40 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1330 nm
Tx operating wavelength range [nm]	1320 nm - 1340 nm
Maximum Tx optical power (AVG) [dBm]	5 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	0 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	3.5 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1260 nm - 1280 nm
Rx sensitivity (AVG) [dBm]	-18 dBm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	-9 dBm

Item	Value
Overload power (OMA) [dBm]	-
NOTE 1. Ensure that the optical power on the receive side is less than or equal to -5 dBm. If the optical power does not meet the requirement, add an optical attenuator. Do not use short-distance optical fibers for direct connection.	

1.5.20.3 10Gbps-SFP+-SMF-1270nm(Tx)/1330nm(Rx)-10km-industry
(02311RXL)

Table 1-1154 10Gbps-SFP+-SMF-1270nm(Tx)/1330nm(Rx)-10km-industry specifications

Item	Value
Basic Information	
Module name	10Gbps-SFP+-SMF-1270nm(Tx)/1330nm(Rx)-10km-industry
Part Number	02311RXL
Model	OSX010B12
Form factor	SFP+
Application standard	IEEE 802.3ae, 10GBASE-BX10-U
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	-40°C to +85°C(-40°F to +185°F)
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	9.953 Gbit/s 10.3125 Gbit/s
Target transmission distance [km]	10 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1270 nm
Tx operating wavelength range [nm]	1260 nm - 1280 nm

Item	Value
Maximum Tx optical power (AVG) [dBm]	0.5 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	-8.2 dBm
Minimum Tx optical power (OMA) [dBm]	-5.2 dBm
Minimum extinction ratio [dB]	3.5 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1320 nm - 1340 nm
Rx sensitivity (AVG) [dBm]	-14.4 dBm
Rx sensitivity (OMA) [dBm]	-10.3 dBm
Overload power (AVG) [dBm]	0.5 dBm
Overload power (OMA) [dBm]	-

1.5.20.4 10Gbps-SFP+-SMF-1330nm(Tx)/1270nm(Rx)-10km-industry (02311RXM)

Table 1-1155 10Gbps-SFP+-SMF-1330nm(Tx)/1270nm(Rx)-10km-industry specifications

Item	Value
Basic Information	
Module name	10Gbps-SFP+-SMF-1330nm(Tx)/1270nm(Rx)-10km-industry
Part Number	02311RXM
Model	OSX010B13
Form factor	SFP+
Application standard	IEEE 802.3ae, 10GBASE-BX10-D
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	-40°C to +85°C(-40°F to +185°F)

Item	Value
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	9.953 Gbit/s 10.3125 Gbit/s
Target transmission distance [km]	10 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1330 nm
Tx operating wavelength range [nm]	1320 nm - 1340 nm
Maximum Tx optical power (AVG) [dBm]	0.5 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	-8.2 dBm
Minimum Tx optical power (OMA) [dBm]	-5.2 dBm
Minimum extinction ratio [dB]	3.5 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1260 nm - 1280 nm
Rx sensitivity (AVG) [dBm]	-14.4 dBm
Rx sensitivity (OMA) [dBm]	-12.6 dBm
Overload power (AVG) [dBm]	0.5 dBm
Overload power (OMA) [dBm]	-

1.5.20.5 10Gbps-SFP+-SMF-1270nm(Tx)/1330nm(Rx)-10km-industry (34060544-001)

Table 1-1156 10Gbps-SFP+-SMF-1270nm(Tx)/1330nm(Rx)-10km-industry specifications

Item	Value
Basic Information	

Item	Value
Module name	10Gbps-SFP+-SMF-1270nm(Tx)/1330nm(Rx)-10km-industry
Part Number	34060544-001
Model	SFP-GE-LX-SM1270-BIDI
Form factor	SFP+
Application standard	IEEE 802.3ae, 10GBASE-BX10-U
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	-40°C to +85°C(-40°F to +185°F)
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	9.953 Gbit/s 10.3125 Gbit/s
Target transmission distance [km]	10 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1270 nm
Tx operating wavelength range [nm]	1260 nm - 1280 nm
Maximum Tx optical power (AVG) [dBm]	0.5 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	-8.2 dBm
Minimum Tx optical power (OMA) [dBm]	-5.2 dBm
Minimum extinction ratio [dB]	3.5 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1320 nm - 1340 nm
Rx sensitivity (AVG) [dBm]	-14.4 dBm

Item	Value
Rx sensitivity (OMA) [dBm]	-10.3 dBm
Overload power (AVG) [dBm]	0.5 dBm
Overload power (OMA) [dBm]	-

1.5.20.6 10Gbps-SFP+-SMF-1270nm(Tx)/1330nm(Rx)-10km-industry (34060544-002)

Table 1-1157 10Gbps-SFP+-SMF-1270nm(Tx)/1330nm(Rx)-10km-industry specifications

Item	Value
Basic Information	
Module name	10Gbps-SFP+-SMF-1270nm(Tx)/1330nm(Rx)-10km-industry
Part Number	34060544-002
Model	OSX010B10
Form factor	SFP+
Application standard	IEEE 802.3ae, 10GBASE-BX10-U
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	-40°C to +85°C(-40°F to +185°F)
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	9.953 Gbit/s 10.3125 Gbit/s
Target transmission distance [km]	10 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1270 nm
Tx operating wavelength range [nm]	1260 nm - 1280 nm

Item	Value
Maximum Tx optical power (AVG) [dBm]	0.5 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	-8.2 dBm
Minimum Tx optical power (OMA) [dBm]	-5.2 dBm
Minimum extinction ratio [dB]	3.5 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1320 nm - 1340 nm
Rx sensitivity (AVG) [dBm]	-14.4 dBm
Rx sensitivity (OMA) [dBm]	-10.3 dBm
Overload power (AVG) [dBm]	0.5 dBm
Overload power (OMA) [dBm]	-

1.5.20.7 10Gbps-SFP+-SMF-1330nm(Tx)/1270nm(Rx)-10km-industry
(34060546-001)

Table 1-1158 10Gbps-SFP+-SMF-1330nm(Tx)/1270nm(Rx)-10km-industry specifications

Item	Value
Basic Information	
Module name	10Gbps-SFP+-SMF-1330nm(Tx)/1270nm(Rx)-10km-industry
Part Number	34060546-001
Model	SFP-GE-LX-SM1330-BIDI
Form factor	SFP+
Application standard	IEEE 802.3ae, 10GBASE-BX10-D
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	-40°C to +85°C(-40°F to +185°F)

Item	Value
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	9.953 Gbit/s 10.3125 Gbit/s
Target transmission distance [km]	10 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1330 nm
Tx operating wavelength range [nm]	1320 nm - 1340 nm
Maximum Tx optical power (AVG) [dBm]	0.5 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	-8.2 dBm
Minimum Tx optical power (OMA) [dBm]	-5.2 dBm
Minimum extinction ratio [dB]	3.5 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1260 nm - 1280 nm
Rx sensitivity (AVG) [dBm]	-14.4 dBm
Rx sensitivity (OMA) [dBm]	-12.6 dBm
Overload power (AVG) [dBm]	0.5 dBm
Overload power (OMA) [dBm]	-

1.5.20.8 10Gbps-SFP+-SMF-1330nm(Tx)/1270nm(Rx)-10km-industry (34060546-002)

Table 1-1159 10Gbps-SFP+-SMF-1330nm(Tx)/1270nm(Rx)-10km-industry specifications

Item	Value
Basic Information	

Item	Value
Module name	10Gbps-SFP+-SMF-1330nm(Tx)/1270nm(Rx)-10km-industry
Part Number	34060546-002
Model	OSX010B11
Form factor	SFP+
Application standard	IEEE 802.3ae, 10GBASE-BX10-D
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	-40°C to +85°C(-40°F to +185°F)
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	9.953 Gbit/s 10.3125 Gbit/s
Target transmission distance [km]	10 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1330 nm
Tx operating wavelength range [nm]	1320 nm - 1340 nm
Maximum Tx optical power (AVG) [dBm]	0.5 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	-8.2 dBm
Minimum Tx optical power (OMA) [dBm]	-5.2 dBm
Minimum extinction ratio [dB]	3.5 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1260 nm - 1280 nm
Rx sensitivity (AVG) [dBm]	-14.4 dBm

Item	Value
Rx sensitivity (OMA) [dBm]	-12.6 dBm
Overload power (AVG) [dBm]	0.5 dBm
Overload power (OMA) [dBm]	-

1.5.20.9 10Gbps-SFP+-SMF-1490nm(Tx)/1550nm(Rx)-80km-commercial

Table 1-1160 10Gbps-SFP+-SMF-1490nm(Tx)/1550nm(Rx)-80km-commercial specifications

Item	Value
Basic Information	
Module name	10Gbps-SFP+-SMF-1490nm(Tx)/1550nm(Rx)-80km-commercial
Part Number	02314REC
Model	OSX080B00
Form factor	SFP+
Application standard	IEEE 802.3ae, 10GBASE-BX80-D
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	1000 V
Transmission rate [bit/s]	9.953 Gbit/s 10.3125 Gbit/s
Target transmission distance [km]	80 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1490 nm
Tx operating wavelength range [nm]	1480 nm - 1500 nm

Item	Value
Maximum Tx optical power (AVG) [dBm]	4 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	0 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	9 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1540 nm - 1560 nm
Rx sensitivity (AVG) [dBm]	-24 dBm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	-7 dBm
Overload power (OMA) [dBm]	-

1.5.20.10 10Gbps-SFP+-SMF-1550nm(Tx)/1490nm(Rx)-80km-commercial

Table 1-1161 10Gbps-SFP+-SMF-1550nm(Tx)/1490nm(Rx)-80km-commercial specifications

Item	Value
Basic Information	
Module name	10Gbps-SFP+-SMF-1550nm(Tx)/1490nm(Rx)-80km-commercial
Part Number	02314RED
Model	OSX080B01
Form factor	SFP+
Application standard	IEEE 802.3ae, 10GBASE-BX80-D
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	0°C to 70°C (32°F to 158°F)

Item	Value
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	1000 V
Transmission rate [bit/s]	9.953 Gbit/s 10.3125 Gbit/s
Target transmission distance [km]	80 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1550 nm
Tx operating wavelength range [nm]	1540 nm - 1560 nm
Maximum Tx optical power (AVG) [dBm]	4 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	0 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	9 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1480 nm - 1500 nm
Rx sensitivity (AVG) [dBm]	-24 dBm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	-7 dBm
Overload power (OMA) [dBm]	-

1.5.21 10Gbps SFP+ DWDM Optical Module

1.5.21.1 10Gbps-SFP+-SMF-1528nm~1568nm-40km-commercial

Table 1-1162 10Gbps-SFP+-SMF-1528nm~1568nm-40km-commercial specifications

Item	Value
Basic Information	
Module name	10Gbps-SFP+-SMF-1528nm~1568nm-40km-commercial
Part Number	02314MED
Model	OSX040C01
Form factor	SFP+
Application standard	IEEE 802.3ae, 10GBASE-ER/EW, ITUT G. 709
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12(10GE) <1x10E-4(OTU2, OTU2e)
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-INF-8077i
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	9.953 Gbit/s 10.3125 Gbit/s 11.1 Gbit/s
Target transmission distance [km]	40 km
Transmitter Optical Characteristics	
Center wavelength [nm]	-
Tx operating wavelength range [nm]	1529.163 nm - 1567.133 nm
Maximum Tx optical power (AVG) [dBm]	3 dBm
Maximum Tx optical power (OMA) [dBm]	-

Item	Value
Minimum Tx optical power (AVG) [dBm]	-1 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	8.2 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1260 nm - 1600 nm
Rx sensitivity (AVG) [dBm]	-16 dBm(EOL)(@ BER 1E-12, 9.95Gbps~10.7Gbps) -19 dBm(EOL)(@ BER 2E-03, 11.3Gbps, dispersion 800ps/nm, at room temperature and OSNR > 31dB)
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	-1 dBm
Overload power (OMA) [dBm]	-

1.5.22 10Gbps TXFP DWDM Optical Module

1.5.22.1 9.95G~11.35Gbps-XFP-SMF-1529.16~1560.61nm-40km-commercial

Table 1-1163 9.95G~11.35Gbps-XFP-SMF-1529.16~1560.61nm-40km-commercial specifications

Item	Value
Basic Information	
Module name	9.95G~11.35Gbps-XFP-SMF-1529.16~1560.61nm-40km-commercial
Part Number	34060568
Model	ODX0480T1
Form factor	XFP
Application standard	IEEE 802.3ae, 10GBASE-ZR/ZW, ITU-T G.709
Connector type	LC
Optical fiber type	SMF

Item	Value
Bit error ratio (BER)	<1x10E-12(10GE) <1x10E-4(OTU2, OTU2e)
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-INF-8077i
Environment standard	RoHS
Security standard	FCC Class B, IEC61000-4-2
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	9.95 bit/s ~ 11.1 Gbit/s
Target transmission distance [km]	40 km
Transmitter Optical Characteristics	
Center wavelength [nm]	C band
Tx operating wavelength range [nm]	1529.16 nm - 1560.61 nm
Maximum Tx optical power (AVG) [dBm]	2 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	-1 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	10.5 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1260 nm - 1600 nm
Rx sensitivity (AVG) [dBm]	-
Rx sensitivity (OMA) [dBm]	-16 dBm
Overload power (AVG) [dBm]	0 dBm
Overload power (OMA) [dBm]	-

1.5.22.2 9.95G~11.35Gbps-XFP-SMF-1529.16~1560.61nm-80km-commercial

Table 1-1164 9.95G~11.35Gbps-XFP-SMF-1529.16~1560.61nm-80km-commercial specifications

Item	Value
Basic Information	
Module name	9.95G~11.35Gbps-XFP-SMF-1529.16~1560.61nm-80km-commercial
Part Number	34060800
Model	OSX080D01
Form factor	XFP
Application standard	IEEE 802.3ae, 10GBASE-ZR/ZW, ITU-T G.709
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12(10GE) <1x10E-4(OTU2, OTU2e)
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-INF-8077i
Environment standard	RoHS
Security standard	FCC Class B, IEC61000-4-2
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	9.95 bit/s ~ 11.1 Gbit/s
Target transmission distance [km]	80 km
Transmitter Optical Characteristics	
Center wavelength [nm]	C band
Tx operating wavelength range [nm]	1529.16 nm - 1560.61 nm
Maximum Tx optical power (AVG) [dBm]	3 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	0.5 dBm

Item	Value
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	9 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1260 nm - 1600 nm
Rx sensitivity (AVG) [dBm]	-
Rx sensitivity (OMA) [dBm]	-24 dBm
Overload power (AVG) [dBm]	-7 dBm
Overload power (OMA) [dBm]	-

1.5.23 10Gbps XFP DWDM Optical Module

1.5.23.1 9.95G~11.1Gbps-XFP-SMF-1550.92nm-80km-commercial

Table 1-1165 9.95G~11.1Gbps-XFP-SMF-1550.92nm-80km-commercial specifications

Item	Value
Basic Information	
Module name	9.95G~11.1Gbps-XFP-SMF-1550.92nm-80km-commercial
Part Number	34060501
Model	XFP-LH80-193.30
Form factor	XFP
Application standard	IEEE 802.3ae, 10GBASE-ZR/ZW, ITU-T G.709
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12(10GE) <1x10E-4(OTU2, OTU2e)
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-INF-8077i
Environment standard	RoHS

Item	Value
Security standard	FCC Class B, CE Class B, IEC61000-4-2
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	9.95 bit/s ~ 11.1 Gbit/s
Target transmission distance [km]	80 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1550.92 nm
Tx operating wavelength range [nm]	1550.90 nm - 1550.94 nm
Maximum Tx optical power (AVG) [dBm]	3 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	-1 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	8.2 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1270 nm - 1600 nm
Rx sensitivity (AVG) [dBm]	-
Rx sensitivity (OMA) [dBm]	-24 dBm
Overload power (AVG) [dBm]	-9 dBm
Overload power (OMA) [dBm]	-

1.5.23.2 9.95G~11.1Gbps-XFP-SMF-1550.12nm-80km-commercial

Table 1-1166 9.95G~11.1Gbps-XFP-SMF-1550.12nm-80km-commercial specifications

Item	Value
Basic Information	
Module name	9.95G~11.1Gbps-XFP-SMF-1550.12nm-80km-commercial
Part Number	34060502

Item	Value
Model	XFP-LH80-193.40
Form factor	XFP
Application standard	IEEE 802.3ae, 10GBASE-ZR/ZW, ITU-T G.709
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12(10GE) <1x10E-4(OTU2, OTU2e)
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-INF-8077i
Environment standard	RoHS
Security standard	FCC Class B, CE Class B, IEC61000-4-2
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	9.95 bit/s ~ 11.1 Gbit/s
Target transmission distance [km]	80 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1550.12 nm
Tx operating wavelength range [nm]	1550.10 nm - 1550.14 nm
Maximum Tx optical power (AVG) [dBm]	3 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	-1 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	8.2 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1270 nm - 1600 nm
Rx sensitivity (AVG) [dBm]	-
Rx sensitivity (OMA) [dBm]	-24 dBm
Overload power (AVG) [dBm]	-9 dBm

Item	Value
Overload power (OMA) [dBm]	-

1.5.23.3 9.95G~11.1Gbps-XFP-SMF-1531.90nm-80km-commercial

Table 1-1167 9.95G~11.1Gbps-XFP-SMF-1531.90nm-80km-commercial specifications

Item	Value
Basic Information	
Module name	9.95G~11.1Gbps-XFP-SMF-1531.90nm-80km-commercial
Part Number	34060503
Model	XFP-LH80-195.70
Form factor	XFP
Application standard	IEEE 802.3ae, 10GBASE-ZR/ZW, ITU-T G.709
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12(10GE) <1x10E-4(OTU2, OTU2e)
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-INF-8077i
Environment standard	RoHS
Security standard	FCC Class B, CE Class B, IEC61000-4-2
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	9.95 bit/s ~ 11.1 Gbit/s
Target transmission distance [km]	80 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1531.9 nm
Tx operating wavelength range [nm]	1531.88 nm - 1531.92 nm
Maximum Tx optical power (AVG) [dBm]	3 dBm

Item	Value
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	-1 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	8.2 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1270 nm - 1600 nm
Rx sensitivity (AVG) [dBm]	-
Rx sensitivity (OMA) [dBm]	-24 dBm
Overload power (AVG) [dBm]	-9 dBm
Overload power (OMA) [dBm]	-

1.5.23.4 9.95G~11.1Gbps-XFP-SMF-1531.12nm-80km-commercial

Table 1-1168 9.95G~11.1Gbps-XFP-SMF-1531.12nm-80km-commercial specifications

Item	Value
Basic Information	
Module name	9.95G~11.1Gbps-XFP-SMF-1531.12nm-80km-commercial
Part Number	34060504
Model	XFP-LH80-195.80
Form factor	XFP
Application standard	IEEE 802.3ae, 10GBASE-ZR/ZW, ITU-T G.709
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12(10GE) <1x10E-4(OTU2, OTU2e)
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-INF-8077i

Item	Value
Environment standard	RoHS
Security standard	FCC Class B, CE Class B, IEC61000-4-2
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	9.95 bit/s ~ 11.1 Gbit/s
Target transmission distance [km]	80 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1531.12 nm
Tx operating wavelength range [nm]	1531.10 nm - 1531.14 nm
Maximum Tx optical power (AVG) [dBm]	3 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	-1 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	8.2 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1270 nm - 1600 nm
Rx sensitivity (AVG) [dBm]	-
Rx sensitivity (OMA) [dBm]	-24 dBm
Overload power (AVG) [dBm]	-9 dBm
Overload power (OMA) [dBm]	-

1.5.23.5 9.95G~11.1Gbps-XFP-SMF-1549.32nm-80km-commercial

Table 1-1169 9.95G~11.1Gbps-XFP-SMF-1549.32nm-80km-commercial specifications

Item	Value
Basic Information	
Module name	9.95G~11.1Gbps-XFP-SMF-1549.32nm-80km-commercial

Item	Value
Part Number	34060514
Model	XFP-LH80-193.50
Form factor	XFP
Application standard	IEEE 802.3ae, 10GBASE-ZR/ZW, ITU-T G.709
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12(10GE) <1x10E-4(OTU2, OTU2e)
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-INF-8077i
Environment standard	RoHS
Security standard	FCC Class B, CE Class B, IEC61000-4-2
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	9.95 bit/s ~ 11.1 Gbit/s
Target transmission distance [km]	80 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1549.32 nm
Tx operating wavelength range [nm]	1549.30 nm - 1549.34 nm
Maximum Tx optical power (AVG) [dBm]	3 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	-1 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	8.2 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1270 nm - 1600 nm
Rx sensitivity (AVG) [dBm]	-
Rx sensitivity (OMA) [dBm]	-24 dBm

Item	Value
Overload power (AVG) [dBm]	-9 dBm
Overload power (OMA) [dBm]	-

1.5.23.6 9.95G~11.1Gbps-XFP-SMF-1530.33nm-80km-commercial

Table 1-1170 9.95G~11.1Gbps-XFP-SMF-1530.33nm-80km-commercial specifications

Item	Value
Basic Information	
Module name	9.95G~11.1Gbps-XFP-SMF-1530.33nm-80km-commercial
Part Number	34060515
Model	XFP-LH80-195.90
Form factor	XFP
Application standard	IEEE 802.3ae, 10GBASE-ZR/ZW, ITU-T G.709
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12(10GE) <1x10E-4(OTU2, OTU2e)
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-INF-8077i
Environment standard	RoHS
Security standard	FCC Class B, CE Class B, IEC61000-4-2
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	9.95 bit/s ~ 11.1 Gbit/s
Target transmission distance [km]	80 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1530.33 nm
Tx operating wavelength range [nm]	1530.31 nm - 1530.35 nm
Maximum Tx optical power (AVG) [dBm]	3 dBm

Item	Value
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	-1 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	8.2 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1270 nm - 1600 nm
Rx sensitivity (AVG) [dBm]	-
Rx sensitivity (OMA) [dBm]	-24 dBm
Overload power (AVG) [dBm]	-9 dBm
Overload power (OMA) [dBm]	-

1.5.23.7 9.95G~11.1Gbps-XFP-SMF-1548.51nm-80km-commercial

Table 1-1171 9.95G~11.1Gbps-XFP-SMF-1548.51nm-80km-commercial specifications

Item	Value
Basic Information	
Module name	9.95G~11.1Gbps-XFP-SMF-1548.51nm-80km-commercial
Part Number	34060531
Model	XFP-LH80-193.60
Form factor	XFP
Application standard	IEEE 802.3ae, 10GBASE-ZR/ZW, ITU-T G.709
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12(10GE) <1x10E-4(OTU2, OTU2e)
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-INF-8077i

Item	Value
Environment standard	RoHS
Security standard	FCC Class B, CE Class B, IEC61000-4-2
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	9.95 bit/s ~ 11.1 Gbit/s
Target transmission distance [km]	80 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1548.51 nm
Tx operating wavelength range [nm]	1548.49 nm - 1548.53 nm
Maximum Tx optical power (AVG) [dBm]	3 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	-1 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	8.2 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1270 nm - 1600 nm
Rx sensitivity (AVG) [dBm]	-
Rx sensitivity (OMA) [dBm]	-24 dBm
Overload power (AVG) [dBm]	-9 dBm
Overload power (OMA) [dBm]	-

1.5.23.8 9.95G~11.1Gbps-XFP-SMF-1529.55nm-80km-commercial

Table 1-1172 9.95G~11.1Gbps-XFP-SMF-1529.55nm-80km-commercial specifications

Item	Value
Basic Information	
Module name	9.95G~11.1Gbps-XFP-SMF-1529.55nm-80km-commercial

Item	Value
Part Number	34060532
Model	XFP-LH80-196.00
Form factor	XFP
Application standard	IEEE 802.3ae, 10GBASE-ZR/ZW, ITU-T G.709
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12(10GE) <1x10E-4(OTU2, OTU2e)
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-INF-8077i
Environment standard	RoHS
Security standard	FCC Class B, CE Class B, IEC61000-4-2
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	9.95 bit/s ~ 11.1 Gbit/s
Target transmission distance [km]	80 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1529.55 nm
Tx operating wavelength range [nm]	1529.53 nm - 1529.57 nm
Maximum Tx optical power (AVG) [dBm]	3 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	-1 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	8.2 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1270 nm - 1600 nm
Rx sensitivity (AVG) [dBm]	-
Rx sensitivity (OMA) [dBm]	-24 dBm

Item	Value
Overload power (AVG) [dBm]	-9 dBm
Overload power (OMA) [dBm]	-

1.5.23.9 9.95G~11.1Gbps-XFP-SMF-1551.72nm-80km-commercial

Table 1-1173 9.95G~11.1Gbps-XFP-SMF-1551.72nm-80km-commercial specifications

Item	Value
Basic Information	
Module name	9.95G~11.1Gbps-XFP-SMF-1551.72nm-80km-commercial
Part Number	34060533
Model	XFP-LH80-193.20
Form factor	XFP
Application standard	IEEE 802.3ae, 10GBASE-ZR/ZW, ITU-T G.709
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12(10GE) <1x10E-4(OTU2, OTU2e)
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-INF-8077i
Environment standard	RoHS
Security standard	FCC Class B, CE Class B, IEC61000-4-2
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	9.95 bit/s ~ 11.1 Gbit/s
Target transmission distance [km]	80 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1551.72 nm
Tx operating wavelength range [nm]	1551.70 nm - 1551.74 nm
Maximum Tx optical power (AVG) [dBm]	3 dBm

Item	Value
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	-1 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	8.2 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1270 nm - 1600 nm
Rx sensitivity (AVG) [dBm]	-
Rx sensitivity (OMA) [dBm]	-24 dBm
Overload power (AVG) [dBm]	-9 dBm
Overload power (OMA) [dBm]	-

1.5.23.10 9.95G~11.1Gbps-XFP-SMF-1532.68nm-80km-commercial

Table 1-1174 9.95G~11.1Gbps-XFP-SMF-1532.68nm-80km-commercial specifications

Item	Value
Basic Information	
Module name	9.95G~11.1Gbps-XFP-SMF-1532.68nm-80km-commercial
Part Number	34060534
Model	XFP-LH80-195.60
Form factor	XFP
Application standard	IEEE 802.3ae, 10GBASE-ZR/ZW, ITU-T G.709
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12(10GE) <1x10E-4(OTU2, OTU2e)
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-INF-8077i

Item	Value
Environment standard	RoHS
Security standard	FCC Class B, CE Class B, IEC61000-4-2
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	9.95 bit/s ~ 11.1 Gbit/s
Target transmission distance [km]	80 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1532.68 nm
Tx operating wavelength range [nm]	1532.66 nm - 1532.70 nm
Maximum Tx optical power (AVG) [dBm]	3 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	-1 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	8.2 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1270 nm - 1600 nm
Rx sensitivity (AVG) [dBm]	-
Rx sensitivity (OMA) [dBm]	-24 dBm
Overload power (AVG) [dBm]	-9 dBm
Overload power (OMA) [dBm]	-

1.5.23.11 9.95G~11.1Gbps-XFP-SMF-1554.13nm-80km-commercial

Table 1-1175 9.95G~11.1Gbps-XFP-SMF-1554.13nm-80km-commercial specifications

Item	Value
Basic Information	
Module name	9.95G~11.1Gbps-XFP-SMF-1554.13nm-80km-commercial

Item	Value
Part Number	34060623
Model	XFP-LH80-192.90
Form factor	XFP
Application standard	IEEE 802.3ae, 10GBASE-ZR/ZW, ITU-T G.709
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12(10GE) <1x10E-4(OTU2, OTU2e)
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-INF-8077i
Environment standard	RoHS
Security standard	FCC Class B, CE Class B, IEC61000-4-2
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	9.95 bit/s ~ 11.1 Gbit/s
Target transmission distance [km]	80 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1554.13 nm
Tx operating wavelength range [nm]	1554.11 nm - 1554.15 nm
Maximum Tx optical power (AVG) [dBm]	3 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	-1 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	8.2 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1270 nm - 1600 nm
Rx sensitivity (AVG) [dBm]	-
Rx sensitivity (OMA) [dBm]	-24 dBm

Item	Value
Overload power (AVG) [dBm]	-9 dBm
Overload power (OMA) [dBm]	-

1.5.23.12 9.95G~11.1Gbps-XFP-SMF-1553.33nm-80km-commercial

Table 1-1176 9.95G~11.1Gbps-XFP-SMF-1553.33nm-80km-commercial specifications

Item	Value
Basic Information	
Module name	9.95G~11.1Gbps-XFP-SMF-1553.33nm-80km-commercial
Part Number	34060624
Model	XFP-LH80-193.00
Form factor	XFP
Application standard	IEEE 802.3ae, 10GBASE-ZR/ZW, ITU-T G.709
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12(10GE) <1x10E-4(OTU2, OTU2e)
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-INF-8077i
Environment standard	RoHS
Security standard	FCC Class B, CE Class B, IEC61000-4-2
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	9.95 bit/s ~ 11.1 Gbit/s
Target transmission distance [km]	80 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1553.33 nm
Tx operating wavelength range [nm]	1553.31 nm - 1553.35 nm
Maximum Tx optical power (AVG) [dBm]	3 dBm

Item	Value
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	-1 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	8.2 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1270 nm - 1600 nm
Rx sensitivity (AVG) [dBm]	-
Rx sensitivity (OMA) [dBm]	-24 dBm
Overload power (AVG) [dBm]	-9 dBm
Overload power (OMA) [dBm]	-

1.5.23.13 9.95G~11.1Gbps-XFP-SMF-1552.52nm-80km-commercial

Table 1-1177 9.95G~11.1Gbps-XFP-SMF-1552.52nm-80km-commercial specifications

Item	Value
Basic Information	
Module name	9.95G~11.1Gbps-XFP-SMF-1552.52nm-80km-commercial
Part Number	34060625
Model	XFP-LH80-193.10
Form factor	XFP
Application standard	IEEE 802.3ae, 10GBASE-ZR/ZW, ITU-T G.709
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12(10GE) <1x10E-4(OTU2, OTU2e)
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-INF-8077i

Item	Value
Environment standard	RoHS
Security standard	FCC Class B, CE Class B, IEC61000-4-2
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	9.95 bit/s ~ 11.1 Gbit/s
Target transmission distance [km]	80 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1552.52 nm
Tx operating wavelength range [nm]	1552.50 nm - 1552.54 nm
Maximum Tx optical power (AVG) [dBm]	3 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	-1 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	8.2 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1270 nm - 1600 nm
Rx sensitivity (AVG) [dBm]	-
Rx sensitivity (OMA) [dBm]	-24 dBm
Overload power (AVG) [dBm]	-9 dBm
Overload power (OMA) [dBm]	-

1.5.23.14 9.95G~11.1Gbps-XFP-SMF-1535.04nm-80km-commercial

Table 1-1178 9.95G~11.1Gbps-XFP-SMF-1535.04nm-80km-commercial specifications

Item	Value
Basic Information	
Module name	9.95G~11.1Gbps-XFP-SMF-1535.04nm-80km-commercial

Item	Value
Part Number	34060626
Model	XFP-LH80-195.30
Form factor	XFP
Application standard	IEEE 802.3ae, 10GBASE-ZR/ZW, ITU-T G.709
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12(10GE) <1x10E-4(OTU2, OTU2e)
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-INF-8077i
Environment standard	RoHS
Security standard	FCC Class B, CE Class B, IEC61000-4-2
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	9.95 bit/s ~ 11.1 Gbit/s
Target transmission distance [km]	80 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1535.04 nm
Tx operating wavelength range [nm]	1535.02 nm - 1535.06 nm
Maximum Tx optical power (AVG) [dBm]	3 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	-1 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	8.2 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1270 nm - 1600 nm
Rx sensitivity (AVG) [dBm]	-
Rx sensitivity (OMA) [dBm]	-24 dBm

Item	Value
Overload power (AVG) [dBm]	-9 dBm
Overload power (OMA) [dBm]	-

1.5.23.15 9.95G~11.1Gbps-XFP-SMF-1534.25nm-80km-commercial

Table 1-1179 9.95G~11.1Gbps-XFP-SMF-1534.25nm-80km-commercial specifications

Item	Value
Basic Information	
Module name	9.95G~11.1Gbps-XFP-SMF-1534.25nm-80km-commercial
Part Number	34060627
Model	XFP-LH80-195.40
Form factor	XFP
Application standard	IEEE 802.3ae, 10GBASE-ZR/ZW, ITU-T G.709
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12(10GE) <1x10E-4(OTU2, OTU2e)
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-INF-8077i
Environment standard	RoHS
Security standard	FCC Class B, CE Class B, IEC61000-4-2
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	9.95 bit/s ~ 11.1 Gbit/s
Target transmission distance [km]	80 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1534.25 nm
Tx operating wavelength range [nm]	1534.25 nm - 1534.27 nm
Maximum Tx optical power (AVG) [dBm]	3 dBm

Item	Value
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	-1 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	8.2 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1270 nm - 1600 nm
Rx sensitivity (AVG) [dBm]	-
Rx sensitivity (OMA) [dBm]	-24 dBm
Overload power (AVG) [dBm]	-9 dBm
Overload power (OMA) [dBm]	-

1.5.23.16 9.95G~11.1Gbps-XFP-SMF-1533.47nm-80km-commercial

Table 1-1180 9.95G~11.1Gbps-XFP-SMF-1533.47nm-80km-commercial specifications

Item	Value
Basic Information	
Module name	9.95G~11.1Gbps-XFP-SMF-1533.47nm-80km-commercial
Part Number	34060628
Model	XFP-LH80-195.50
Form factor	XFP
Application standard	IEEE 802.3ae, 10GBASE-ZR/ZW, ITU-T G.709
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12(10GE) <1x10E-4(OTU2, OTU2e)
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-INF-8077i

Item	Value
Environment standard	RoHS
Security standard	FCC Class B, CE Class B,IEC61000-4-2
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	9.95 bit/s ~ 11.1 Gbit/s
Target transmission distance [km]	80 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1533.47 nm
Tx operating wavelength range [nm]	1533.45 nm - 1533.48 nm
Maximum Tx optical power (AVG) [dBm]	3 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	-1 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	8.2 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1270 nm - 1600 nm
Rx sensitivity (AVG) [dBm]	-
Rx sensitivity (OMA) [dBm]	-24 dBm
Overload power (AVG) [dBm]	-9 dBm
Overload power (OMA) [dBm]	-

1.5.24 25Gbps SFP28 Optical Module

1.5.24.1 25Gbps-SFP28-MMF-850nm-0.1km-commercial (02312PDK)

Table 1-1181 25Gbps-SFP28-MMF-850nm-0.1km-commercial specifications

Item	Value
Basic Information	

Item	Value
Module name	25Gbps-SFP28-MMF-850nm-0.1km-commercial
Part Number	02312PDK
Model	SFP28-25G-850nm-0.1km-MM
Form factor	SFP28
Application standard	IEEE802.3-2012, 25GBASE-SR
Connector type	LC
Optical fiber type	MMF
Bit error ratio (BER)	<5x10E-5
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	1000 V
Transmission rate [bit/s]	25.78125 Gbit/s
Target transmission distance [km]	0.1 km
Transmitter Optical Characteristics	
Center wavelength [nm]	850 nm
Tx operating wavelength range [nm]	840 nm - 860 nm
Maximum Tx optical power (AVG) [dBm]	2.4 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	-8.4 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	2 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	840 nm - 860 nm
Rx sensitivity (AVG) [dBm]	-10.3 dBm
Rx sensitivity (OMA) [dBm]	-

Item	Value
Overload power (AVG) [dBm]	2.4 dBm
Overload power (OMA) [dBm]	-

1.5.24.2 25Gbps-SFP28-SMF-1310nm-10km-industry (02312PDL)

Table 1-1182 25Gbps-SFP28-SMF-1310nm-10km-industry specifications

Item	Value
Basic Information	
Module name	25Gbps-SFP28-SMF-1310nm-10km-industry
Part Number	02312PDL
Model	SFP28-25G-1310nm-10km-SM
Form factor	SFP28
Application standard	IEEE 802.3 cc, 25GBASE-LR, 10GBASE-LR
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<5x10E-5(24.33024G,25.78125G) <1x10E-12(9.8304G,10.1376G, 10.3125G)
Working case temperature [°C(°F)]	-40°C to +85°C(-40°F to +185°F)
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	1000 V
Transmission rate [bit/s]	25.78125 Gbit/s
Target transmission distance [km]	10 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1310 nm
Tx operating wavelength range [nm]	1295 nm - 1325 nm
Maximum Tx optical power (AVG) [dBm]	2 dBm

Item	Value
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	-7 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	3.5 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1295 nm - 1325 nm
Rx sensitivity (AVG) [dBm]	-
Rx sensitivity (OMA) [dBm]	-11.3 dBm
Overload power (AVG) [dBm]	2 dBm
Overload power (OMA) [dBm]	-

1.5.24.3 25Gbps-SFP28-MMF-850nm-0.1km-commercial (34061254)

Table 1-1183 25Gbps-SFP28-MMF-850nm-0.1km-commercial specifications

Item	Value
Basic Information	
Module name	25Gbps-SFP28-MMF-850nm-0.1km-commercial
Part Number	34061254
Model	OMXD30011
Form factor	SFP28
Application standard	IEEE802.3-2012, 25GBASE-SR
Connector type	LC
Optical fiber type	MMF
Bit error ratio (BER)	<5x10E-5
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1

Item	Value
ESD(HBM1) [V]	1000 V
Transmission rate [bit/s]	25.78125 Gbit/s
Target transmission distance [km]	0.1 km
Transmitter Optical Characteristics	
Center wavelength [nm]	850 nm
Tx operating wavelength range [nm]	840 nm - 860 nm
Maximum Tx optical power (AVG) [dBm]	2.4 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	-8.4 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	2 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	840 nm - 860 nm
Rx sensitivity (AVG) [dBm]	-10.3 dBm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	2.4 dBm
Overload power (OMA) [dBm]	-

1.5.24.4 25Gbps-SFP28-SMF-1310nm-10km-industry (34061618)

Table 1-1184 25Gbps-SFP28-SMF-1310nm-10km-industry specifications

Item	Value
Basic Information	
Module name	25Gbps-SFP28-SMF-1310nm-10km-industry
Part Number	34061618
Model	SFP28-25G-1310nm-10km-SM
Form factor	SFP28

Item	Value
Application standard	IEEE 802.3 by-2016, 25GBASE-LR, 10GBASE-LR
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<5x10E-5(24.33024G,25.78125G) <1x10E-12(9.8304G,10.1376G, 10.3125G)
Working case temperature [°C(°F)]	-40°C to +85°C(-40°F to +185°F)
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	1000 V
Transmission rate [bit/s]	25.78125 Gbit/s
Target transmission distance [km]	10 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1310 nm
Tx operating wavelength range [nm]	1295 nm - 1325 nm
Maximum Tx optical power (AVG) [dBm]	2 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	-7 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	3.5 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1295 nm - 1325 nm
Rx sensitivity (AVG) [dBm]	-
Rx sensitivity (OMA) [dBm]	-11.3 dBm
Overload power (AVG) [dBm]	2 dBm
Overload power (OMA) [dBm]	-

1.5.24.5 25Gbps-SFP28-MMF-850nm-0.1km-extended

Table 1-1185 25Gbps-SFP28-MMF-850nm-0.1km-extended specifications

Item	Value
Basic Information	
Module name	25Gbps-SFP28-MMF-850nm-0.1km-extended
Part Number	34061631
Model	SFP28-25G-850nm-0.1km-MM
Form factor	SFP28
Application standard	IEEE 802.3 by-2016, 25GBASE-SR, 10GBASE-SR
Connector type	LC
Optical fiber type	MMF
Bit error ratio (BER)	<5x10E-5(24.33024G,25.78125G) <1x10E-12(9.8304G,10.1376G, 10.3125G)
Working case temperature [°C(°F)]	-20°C to 75°C(-4°F to 167°F)
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	1000 V
Transmission rate [bit/s]	25.78125 Gbit/s
Target transmission distance [km]	0.1 km
Transmitter Optical Characteristics	
Center wavelength [nm]	850 nm
Tx operating wavelength range [nm]	840 nm - 860 nm
Maximum Tx optical power (AVG) [dBm]	2.4 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	-8.4 dBm
Minimum Tx optical power (OMA) [dBm]	-

Item	Value
Minimum extinction ratio [dB]	2 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	840 nm - 860 nm
Rx sensitivity (AVG) [dBm]	-10.3 dBm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	2.4 dBm
Overload power (OMA) [dBm]	-

1.5.24.6 25Gbps-SFP28-SMF-1310nm-0.3km-industry

Table 1-1186 25Gbps-SFP28-SMF-1310nm-0.3km-industry specifications

Item	Value
Basic Information	
Module name	25Gbps-SFP28-SMF-1310nm-0.3km-industry
Part Number	34061940-001
Model	SFP28-25G-1310nm-300m-SM
Form factor	SFP28
Application standard	IEEE 802.3 by-2016, 25GBASE-SR, 10GBASE-SR
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<5x10E-5(24.33024G,25.78125G) <1x10E-12(9.8304G,10.1376G, 10.3125G)
Working case temperature [°C(°F)]	-40°C to +85°C(-40°F to +185°F)
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	1000 V
Transmission rate [bit/s]	25.78125 Gbit/s

Item	Value
Target transmission distance [km]	0.3 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1310 nm
Tx operating wavelength range [nm]	1260 nm - 1360 nm
Maximum Tx optical power (AVG) [dBm]	2 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	-7 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	3.5 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1260 nm - 1360 nm
Rx sensitivity (AVG) [dBm]	-8.5 dBm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	2 dBm
Overload power (OMA) [dBm]	2 dBm

1.5.24.7 25Gbps-SFP28-SMF-1310nm-40km-commercial

Table 1-1187 25Gbps-SFP28-SMF-1310nm-40km-commercial specifications

Item	Value
Basic Information	
Module name	25Gbps-SFP28-SMF-1310nm-40km-commercial
Part Number	34062754
Model	SFP28-25G-1310nm-40km-SM
Form factor	SFP28
Application standard	IEEE 802.3 cc
Connector type	LC

Item	Value
Optical fiber type	SMF
Bit error ratio (BER)	<5x10E-5(25.78125G) <1x10E-12(10.3125G)
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	1000 V
Transmission rate [bit/s]	25.78125 Gbit/s
Target transmission distance [km]	40 km
Transmitter Optical Characteristics	
Tx operating wavelength range [nm]	1295 nm - 1310 nm
Maximum Tx optical power (AVG) [dBm]	6.0 dBm
Maximum Tx optical power (OMA) [dBm]	6.0 dBm
Minimum Tx optical power (AVG) [dBm]	-3.0 dBm
Minimum Tx optical power (OMA) [dBm]	0 dBm
Minimum extinction ratio [dB]	4 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1295 nm - 1325 nm
Minimum Rx optical power (AVG) [dBm]	-21.0 dBm
Maximum Rx optical power (AVG) [dBm]	-4.0 dBm
Minimum Rx optical power (OMA) [dBm]	-
Maximum Rx optical power (OMA) [dBm]	-4.0 dBm
Rx sensitivity (AVG) [dBm]	-18.35 dBm
Rx sensitivity (OMA) [dBm]	-19.0 dBm

Item	Value
Overload power (AVG) [dBm]	-4.0 dBm
Overload power (OMA) [dBm]	-

1.5.25 25Gbps SFP28 BIDI Optical Module

1.5.25.1 25Gbps-SFP28-SMF-1270nm(Tx)/1330nm(Rx)-10km-commercial

Table 1-1188 25Gbps-SFP28-SMF-1270nm(Tx)/1330nm(Rx)-10km-commercial specifications

Item	Value
Basic Information	
Module name	25Gbps-SFP28-SMF-1270nm(Tx)/1330nm(Rx)-10km-commercial
Part Number	02312TVC
Model	SFP28-25G-TX1270/RX1330-10km-SM-BIDI
Form factor	SFP28
Application standard	IEEE 802.3 by-2016, 25GBASE-LR, 10GBASE-LR
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<5x10E-5(24.33024G,25.78125G) <1x10E-12(9.8304G,10.1376G,10.3125G)
Working case temperature [°C(°F)]	-40°C to +85°C(-40°F to +185°F)
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	1000 V
Transmission rate [bit/s]	25.78125 Gbit/s
Target transmission distance [km]	10 km
Transmitter Optical Characteristics	

Item	Value
Center wavelength [nm]	1270 nm
Tx operating wavelength range [nm]	1260 nm - 1280 nm
Maximum Tx optical power (AVG) [dBm]	2 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	-4 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	3 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1320 nm - 1340 nm
Rx sensitivity (AVG) [dBm]	-
Rx sensitivity (OMA) [dBm]	-12 dBm
Overload power (AVG) [dBm]	2 dBm
Overload power (OMA) [dBm]	-

1.5.25.2 25Gbps-SFP28-SMF-1330nm(Tx)/1270nm(Rx)-10km-commercial

Table 1-1189 25Gbps-SFP28-SMF-1330nm(Tx)/1270nm(Rx)-10km-commercial specifications

Item	Value
Basic Information	
Module name	25Gbps-SFP28-SMF-1330nm(Tx)/1270nm(Rx)-10km-commercial
Part Number	02312TXG
Model	SFP28-25G-TX1330/RX1270-10km-SM-BIDI
Form factor	SFP28
Application standard	IEEE 802.3 by-2016, 25GBASE-LR, 10GBASE-LR
Connector type	LC

Item	Value
Optical fiber type	SMF
Bit error ratio (BER)	<5x10E-5(24.33024G,25.78125G) <1x10E-12(9.8304G,10.1376G, 10.3125G)
Working case temperature [°C(°F)]	-40°C to +85°C(-40°F to +185°F)
DDM options	SFF-8472
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	1000 V
Transmission rate [bit/s]	25.78125 Gbit/s
Target transmission distance [km]	10 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1330 nm
Tx operating wavelength range [nm]	1320 nm - 1340 nm
Maximum Tx optical power (AVG) [dBm]	2 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	-4 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	3 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1260 nm - 1280 nm
Rx sensitivity (AVG) [dBm]	-
Rx sensitivity (OMA) [dBm]	-12 dBm
Overload power (AVG) [dBm]	2 dBm
Overload power (OMA) [dBm]	-

1.5.26 40Gbps QSFP+ Optical Module

1.5.26.1 40Gbps(4*10.3)-QSFP+-SMF-1271~1331nm-10km-commercial

Table 1-1190 40Gbps(4*10.3)-QSFP+-SMF-1271~1331nm-10km-commercial specifications

Item	Value
Basic Information	
Module name	40Gbps(4*10.3)-QSFP+-SMF-1271~1331nm-10km-commercial
Part Number	02310WUT
Model	OMXD30009
Form factor	QSFP+
Application standard	IEEE 802.3ba, 40GBASE-LR4
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-8436
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	41.25 Gbit/s
Target transmission distance [km]	10 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1271 nm 1291 nm 1311 nm 1331 nm
Tx operating wavelength range [nm]	1264.5 nm - 1277.5 nm 1284.5 nm - 1297.5 nm 1304.5 nm - 1317.5 nm 1324.5 nm - 1337.5 nm
Maximum Tx optical power (AVG) [dBm]	per lane: 2.3 dBm
Maximum Tx optical power (OMA) [dBm]	per lane: 3.5 dBm

Item	Value
Minimum Tx optical power (AVG) [dBm]	per lane: -7 dBm
Minimum Tx optical power (OMA) [dBm]	per lane: -4 dBm
Minimum extinction ratio [dB]	3.5 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1264.5 nm - 1277.5 nm 1284.5 nm - 1297.5 nm 1304.5 nm - 1317.5 nm 1324.5 nm - 1337.5 nm
Rx sensitivity (AVG) [dBm]	-
Rx sensitivity (OMA) [dBm]	per lane: -11.5 dBm
Overload power (AVG) [dBm]	per lane: 2.3 dBm
Overload power (OMA) [dBm]	-
NOTE The optical power calculation is based on the OMA value.	

1.5.26.2 40Gbps(4*10.3)-QSFP+-MMF-850nm-0.1km-commercial

Table 1-1191 40Gbps(4*10.3)-QSFP+-MMF-850nm-0.1km-commercial specifications

Item	Value
Basic Information	
Module name	40Gbps(4*10.3)-QSFP+-MMF-850nm-0.1km-commercial
Part Number	02310WUU
Model	OMXD30010
Form factor	QSFP+
Application standard	IEEE 802.3ba, 40GBASE-SR4
Connector type	MPO-12
Optical fiber type	MMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)

Item	Value
DDM options	SFF-8436
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	41.25 Gbit/s
Target transmission distance [km]	0.1 km(OM3) 0.15 km(OM4)
Transmitter Optical Characteristics	
Center wavelength [nm]	850 nm
Tx operating wavelength range [nm]	840 nm - 860 nm
Maximum Tx optical power (AVG) [dBm]	per lane: 0.5 dBm
Maximum Tx optical power (OMA) [dBm]	per lane: 3 dBm
Minimum Tx optical power (AVG) [dBm]	per lane: -7.6 dBm
Minimum Tx optical power (OMA) [dBm]	per lane: -5.6 dBm
Minimum extinction ratio [dB]	3 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	840 nm - 860 nm
Rx sensitivity (AVG) [dBm]	-
Rx sensitivity (OMA) [dBm]	per lane: -9.5 dBm
Overload power (AVG) [dBm]	per lane: 0.5 dBm
Overload power (OMA) [dBm]	-
NOTE The optical power calculation is based on the OMA value.	

1.5.26.3 40Gbps(4*10.3)-QSFP+-SMF-1310nm-10km-commercial

Table 1-1192 40Gbps(4*10.3)-QSFP+-SMF-1310nm-10km-commercial specifications

Item	Value
Basic Information	
Module name	40Gbps(4*10.3)-QSFP+-SMF-1310nm-10km-commercial
Part Number	02311NUA
Model	OSM010N11
Form factor	QSFP+
Application standard	IEEE 802.3ba, 40GBASE-LR4
Connector type	MPO-12
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-8436
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	4x9.95328 Gbit/s 4x10.3125 Gbit/s
Target transmission distance [km]	10 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1310 nm
Tx operating wavelength range [nm]	1260 nm - 1355 nm
Maximum Tx optical power (AVG) [dBm]	per lane: 0.5 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	per lane: -8.2 dBm
Minimum Tx optical power (OMA) [dBm]	-

Item	Value
Minimum extinction ratio [dB]	3.5 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1260 nm - 1355 nm
Rx sensitivity (AVG) [dBm]	-
Rx sensitivity (OMA) [dBm]	per lane: -12.6 dBm
Overload power (AVG) [dBm]	per lane: 0.5 dBm
Overload power (OMA) [dBm]	-

1.5.26.4 QSFP-40G-LX4-MM (02313NUG)

Table 1-1193 QSFP-40G-LX4-MM specifications

Item	Value
Basic Information	
Module name	QSFP-40G-LX4-MM
Part Number	02313NUG
Model	QSFP-40G-LX4-MM
Form factor	QSFP+
Application standard	40GBASE-LX4
Connector type	LC
Optical fiber type	MMF
Working case temperature [°C(°F)]	0°C to 70°C (32°F to 158°F)
Transmission rate [bit/s]	40Gbit/s
Target transmission distance [km]	150m
Transmitter Optical Characteristics	
Center wavelength [nm]	1271nm,1291nm,1311nm,1331nm
Maximum Tx optical power (AVG) [dBm]	2.3 dBm
Minimum Tx optical power (OMA) [dBm]	-7.0 dBm
Minimum extinction ratio [dB]	3.5 dB
Receiver Optical Characteristics	

Item	Value
Rx sensitivity (OMA) [dBm]	-10.5 dBm
Overload power (AVG) [dBm]	3.5 dBm
NOTE Application limitations: - In actual applications, the number of conversion connectors for the optical fiber link cannot exceed 4. - This module is sensitive to fiber link contamination. During deployment, ensure that the fiber endface meets the fiber application standard. For details, see the single-mode connector requirements under the ceramic ferrule endface requirements of fibers in section "Cable -Optical Jumper."	

1.5.26.5 QSFP-40G-LX4-MM (02314MWG)

Table 1-1194 QSFP-40G-LX4-MM specifications

Item	Value
Basic Information	
Module name	QSFP-40G-LX4-MM
Part Number	02314MWG
Model	OMMD15N01
Form factor	QSFP+
Application standard	40GBASE-LX4
Connector type	LC
Optical fiber type	MMF
Working case temperature [°C(°F)]	0°C to 70°C (32°F to 158°F)
Transmission rate [bit/s]	40Gbit/s
Target transmission distance [km]	150m
Transmitter Optical Characteristics	
Center wavelength [nm]	1271nm,1291nm,1311nm,1331nm
Maximum Tx optical power (AVG) [dBm]	2.3 dBm
Minimum Tx optical power (OMA) [dBm]	-7.0 dBm
Minimum extinction ratio [dB]	3.5 dB
Receiver Optical Characteristics	

Item	Value
Rx sensitivity (OMA) [dBm]	-10.5 dBm
Overload power (AVG) [dBm]	3.5 dBm
NOTE Application limitations: - In actual applications, the number of conversion connectors for the optical fiber link cannot exceed 4. - This module is sensitive to fiber link contamination. During deployment, ensure that the fiber endface meets the fiber application standard. For details, see the single-mode connector requirements under the ceramic ferrule endface requirements of fibers in section "Cable -Optical Jumper."	

1.5.27 50Gbps QSFP28 Optical Module

1.5.27.1 50Gbps-QSFP28-SMF-1311nm-10km-commercial

Table 1-1195 50Gbps-QSFP28-SMF-1311nm-10km-commercial specifications

Item	Value
Basic Information	
Module name	50Gbps-QSFP28-SMF-1311nm-10km-commercial
Part Number	02311YNR
Model	OSL010N01
Form factor	QSFP28
Application standard	IEEE 802.3cd, 50GBASE-LR
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-8636
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1 CE Class B
ESD(HBM1) [V]	1000 V

Item	Value
Transmission rate [bit/s]	53.125 Gbit/s
Target transmission distance [km]	10 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1311 nm
Tx operating wavelength range [nm]	1304.5 nm - 1317.5 nm
Maximum Tx optical power (AVG) [dBm]	4.2 dBm
Maximum Tx optical power (OMA) [dBm]	4 dBm
Minimum Tx optical power (AVG) [dBm]	-4.5 dBm
Minimum Tx optical power (OMA) [dBm]	-1.5 dBm
Minimum extinction ratio [dB]	3.5 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1304.5 nm - 1317.5 nm
Rx sensitivity (AVG) [dBm]	-8.9 dBm
Rx sensitivity (OMA) [dBm]	-8.9 dBm
Overload power (AVG) [dBm]	4.2 dBm
Overload power (OMA) [dBm]	-
NOTE 1. The optical power calculation is based on the OMA value. If the module needs to reach the nominal data, the board FEC function must be enabled. 2. The optical power read by the device is the average optical power, not the OMA optical power. 3. When this type of optical module is used to interconnect with a WDM device, the 1+1 protection switching duration on the client side of the WDM device is longer than 50 ms.	

1.5.27.2 50Gbps-QSFP28-SMF-1311nm-40km-commercial

Table 1-1196 50Gbps-QSFP28-SMF-1311nm-40km-commercial specifications

Item	Value
Basic Information	
Module name	50Gbps-QSFP28-SMF-1311nm-40km-commercial

Item	Value
Part Number	02312AXF
Model	QSFP28-50G-1311nm-40km-SM
Form factor	QSFP28
Application standard	IEEE 802.3cd, 50GBASE-ER
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-8636
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1 CE Class B
ESD(HBM1) [V]	1000 V
Transmission rate [bit/s]	53.125 Gbit/s
Target transmission distance [km]	40 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1311 nm
Tx operating wavelength range [nm]	1304.5 nm - 1317.5 nm
Maximum Tx optical power (AVG) [dBm]	8 dBm
Maximum Tx optical power (OMA) [dBm]	9 dBm
Minimum Tx optical power (AVG) [dBm]	1.5 dBm
Minimum Tx optical power (OMA) [dBm]	4.5 dBm
Minimum extinction ratio [dB]	6 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1304.5 nm - 1317.5 nm
Rx sensitivity (AVG) [dBm]	-
Rx sensitivity (OMA) [dBm]	-15 dBm
Overload power (AVG) [dBm]	-3 dBm

Item	Value
Overload power (OMA) [dBm]	-
NOTE The optical power calculation is based on the OMA value. If the module needs to reach the nominal data, the board FEC function must be enabled. 1. To keep the optical module running stably for a long time, set the receive optical power less than -4 dBm. (According to IEEE 802.3, if the receive optical power exceeds -2.3 dBm, the optical module may be permanently damaged.) 2. Before connecting the optical module, you are advised to use the optical power meter to measure the receive optical power (P). If P is less than -4 dBm, the optical module can be directly connected. If P is greater than -4 dBm, add an appropriate attenuator at the receive end to ensure that P is less than -4 dBm. Alternatively, add an appropriate attenuator (the recommended value is no less than 10 dB) before the optical module is connected, and then adjust the attenuator according to the actual situation to prevent the module from being damaged. 3. If the pigtail loopback or short-distance connection of the optical module is used, the attenuator must be added. It is recommended that the attenuator be greater than or equal to 10 dB. 4. When this type of optical module is used to interconnect with a WDM device, the 1+1 protection switching duration on the client side of the WDM device is longer than 50 ms.	

1.5.27.3 50Gbps-QSFP28-SMF-1295.56~1300.05nm-80km-commercial

Table 1-1197 50Gbps-QSFP28-SMF-1295.56~1300.05nm-80km-commercial specifications

Item	Value
Basic Information	
Module name	50Gbps-QSFP28-SMF-1295.56~1300.05nm-80km-commercial
Part Number	02312MLF
Model	QSFP28-50G-1310nm-80km-SM-01
Form factor	QSFP28
Application standard	Huawei Define
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-8636
Environment standard	RoHS

Item	Value
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	1000 V
Transmission rate [bit/s]	51.5625 Gbit/s
Target transmission distance [km]	80 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1295.56 nm 1300.05 nm
Tx operating wavelength range [nm]	1294.53 nm - 1296.59 nm 1299.02 nm - 1301.09 nm
Maximum Tx optical power (AVG) [dBm]	per lane: 6.5 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	per lane: 2 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	6 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1294.53 nm - 1296.59 nm 1299.02 nm - 1301.09 nm
Rx sensitivity (AVG) [dBm]	per lane: -28 dBm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	per lane: -3.5 dBm
Overload power (OMA) [dBm]	-
NOTE 1. The optical power is subject to the OMA. To achieve the nominal optical power, the FEC function must be enabled on the board and G.652 fibers must be used. 2. The optical power of the optical module read on the device is the average optical power, not the OMA optical power. 3. This module can be connected only to optical modules with the same BOM number.	

1.5.28 50Gbps QSFP28 BIDI Optical Module

1.5.28.1 50Gbps-QSFP28-SMF-1331nm(Tx)/1271nm(Rx)-10km-commercial

Table 1-1198 50Gbps-QSFP28-SMF-1331nm(Tx)/1271nm(Rx)-10km-commercial specifications

Item	Value
Basic Information	
Module name	50Gbps-QSFP28-SMF-1331nm(Tx)/1271nm(Rx)-10km-commercial
Part Number	02312EVV
Model	QSFP28-50G-BIDI-10km-SM-1
Form factor	QSFP28
Application standard	IEEE 802.3bm-2015, 50GBASE-BR10
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12(after FEC) <2x10E-4(before FEC)
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-8636
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	1000 V
Transmission rate [bit/s]	53.125 Gbit/s
Target transmission distance [km]	10 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1331 nm
Tx operating wavelength range [nm]	1324.5 nm - 1337.5 nm
Maximum Tx optical power (AVG) [dBm]	4.2 dBm
Maximum Tx optical power (OMA) [dBm]	4 dBm
Minimum Tx optical power (AVG) [dBm]	-4.5 dBm
Minimum Tx optical power (OMA) [dBm]	-1.5 dBm

Item	Value
Minimum extinction ratio [dB]	3.5 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1264.5 nm - 1277.5 nm
Rx sensitivity (AVG) [dBm]	-
Rx sensitivity (OMA) [dBm]	-8.9 dBm
Overload power (AVG) [dBm]	4.2 dBm
Overload power (OMA) [dBm]	-
NOTE 1. The optical power calculation is based on the OMA value. If the module needs to reach the nominal data, the board FEC function must be enabled. 2. The optical power read by the device is the average optical power, not the OMA optical power. 3. When this type of optical module is used to interconnect with a WDM device, the 1+1 protection switching duration on the client side of the WDM device is longer than 50 ms.	

1.5.28.2 50Gbps-QSFP28-SMF-1271nm(Tx)/1331nm(Rx)-10km-commercial

Table 1-1199 50Gbps-QSFP28-SMF-1271nm(Tx)/1331nm(Rx)-10km-commercial specifications

Item	Value
Basic Information	
Module name	50Gbps-QSFP28-SMF-1271nm(Tx)/1331nm(Rx)-10km-commercial
Part Number	02312EVW
Model	QSFP28-50G-BIDI-10km-SM-2
Form factor	QSFP28
Application standard	IEEE 802.3bm-2015, 50GBASE-BR10
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12(after FEC) <2x10E-4(before FEC)
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-8636
Environment standard	RoHS

Item	Value
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	1000 V
Transmission rate [bit/s]	53.125 Gbit/s
Target transmission distance [km]	10 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1271 nm
Tx operating wavelength range [nm]	1264.5 nm - 1277.5 nm
Maximum Tx optical power (AVG) [dBm]	4.2 dBm
Maximum Tx optical power (OMA) [dBm]	4 dBm
Minimum Tx optical power (AVG) [dBm]	-4.5 dBm
Minimum Tx optical power (OMA) [dBm]	-1.5 dBm
Minimum extinction ratio [dB]	3.5 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1324.5 nm - 1337.5 nm
Rx sensitivity (AVG) [dBm]	-
Rx sensitivity (OMA) [dBm]	-8.9 dBm
Overload power (AVG) [dBm]	4.2 dBm
Overload power (OMA) [dBm]	-
NOTE 1. The optical power calculation is based on the OMA value. If the module needs to reach the nominal data, the board FEC function must be enabled. 2. The optical power read by the device is the average optical power, not the OMA optical power. 3. When this type of optical module is used to interconnect with a WDM device, the 1+1 protection switching duration on the client side of the WDM device is longer than 50 ms.	

1.5.28.3 50Gbps-QSFP28-SMF-1295nm(Tx)/1309nm(Rx)-40km-commercial

Table 1-1200 50Gbps-QSFP28-SMF-1295nm(Tx)/1309nm(Rx)-40km-commercial specifications

Item	Value
Basic Information	
Module name	50Gbps-QSFP28-SMF-1295nm(Tx)/1309nm(Rx)-40km-commercial
Part Number	02312EVX
Model	QSFP28-50G-BIDI-40km-SM-2
Form factor	QSFP28
Application standard	IEEE 802.3bm-2015, 50GBASE-BR40
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12(after FEC) <2x10E-4(before FEC)
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-8636
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	1000 V
Transmission rate [bit/s]	53.125 Gbit/s
Target transmission distance [km]	40 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1295 nm
Tx operating wavelength range [nm]	1294.53 nm - 1296.59 nm
Maximum Tx optical power (AVG) [dBm]	8 dBm
Maximum Tx optical power (OMA) [dBm]	9 dBm
Minimum Tx optical power (AVG) [dBm]	1.5 dBm
Minimum Tx optical power (OMA) [dBm]	4.5 dBm

Item	Value
Minimum extinction ratio [dB]	6 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1308.09 nm - 1310.19 nm
Rx sensitivity (AVG) [dBm]	-15 dBm
Rx sensitivity (OMA) [dBm]	-13.5 dBm
Overload power (AVG) [dBm]	-3 dBm
Overload power (OMA) [dBm]	-
NOTE The optical power calculation is based on the OMA value. If the module needs to reach the nominal data, the board FEC function must be enabled. The optical power read by the device is the average optical power, not the OMA optical power. 1. To keep the optical module running stably for a long time, set the receive optical power less than -4 dBm. (According to IEEE 802.3, if the receive optical power exceeds -2.3 dBm, the optical module may be permanently damaged.) 2. Before connecting the optical module, you are advised to use the optical power meter to measure the receive optical power (P). If P is less than -4 dBm, the optical module can be directly connected. If P is greater than -4 dBm, add an appropriate attenuator at the receive end to ensure that P is less than -4 dBm. Alternatively, add an appropriate attenuator (the recommended value is no less than 10 dB) before the optical module is connected, and then adjust the attenuator according to the actual situation to prevent the module from being damaged. 3. If the pigtail loopback or short-distance connection of the optical module is used, the attenuator must be added. It is recommended that the attenuator be greater than or equal to 10 dB. 4. When this type of optical module is used to interconnect with a WDM device, the 1+1 protection switching duration on the client side of the WDM device is longer than 50 ms.	

1.5.28.4 50Gbps-QSFP28-SMF-1309nm(Tx)/1295nm(Rx)-40km-commercial

Table 1-1201 50Gbps-QSFP28-SMF-1309nm(Tx)/1295nm(Rx)-40km-commercial specifications

Item	Value
Basic Information	
Module name	50Gbps-QSFP28-SMF-1309nm(Tx)/1295nm(Rx)-40km-commercial
Part Number	02312EVY
Model	QSFP28-50G-BIDI-40km-SM-1
Form factor	QSFP28

Item	Value
Application standard	IEEE 802.3bm-2015, 50GBASE-BR40
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12(after FEC) <2x10E-4(before FEC)
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-8636
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	1000 V
Transmission rate [bit/s]	53.125 Gbit/s
Target transmission distance [km]	40 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1309 nm
Tx operating wavelength range [nm]	1308.09 nm - 1310.19 nm
Maximum Tx optical power (AVG) [dBm]	8 dBm
Maximum Tx optical power (OMA) [dBm]	9 dBm
Minimum Tx optical power (AVG) [dBm]	1.5 dBm
Minimum Tx optical power (OMA) [dBm]	4.5 dBm
Minimum extinction ratio [dB]	6 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1294.53 nm - 1296.59 nm
Rx sensitivity (AVG) [dBm]	-15 dBm
Rx sensitivity (OMA) [dBm]	-13.5 dBm
Overload power (AVG) [dBm]	-3 dBm
Overload power (OMA) [dBm]	-

Item	Value
NOTE The optical power calculation is based on the OMA value. If the module needs to reach the nominal data, the board FEC function must be enabled. The optical power read by the device is the average optical power, not the OMA optical power. 1. To keep the optical module running stably for a long time, set the receive optical power less than -4 dBm. (According to IEEE 802.3, if the receive optical power exceeds -2.3 dBm, the optical module may be permanently damaged.) 2. Before connecting the optical module, you are advised to use the optical power meter to measure the receive optical power (P). If P is less than -4 dBm, the optical module can be directly connected. If P is greater than -4 dBm, add an appropriate attenuator at the receive end to ensure that P is less than -4 dBm. Alternatively, add an appropriate attenuator (the recommended value is no less than 10 dB) before the optical module is connected, and then adjust the attenuator according to the actual situation to prevent the module from being damaged. 3. If the pigtail loopback or short-distance connection of the optical module is used, the attenuator must be added. It is recommended that the attenuator be greater than or equal to 10 dB. 4. When this type of optical module is used to interconnect with a WDM device, the 1+1 protection switching duration on the client side of the WDM device is longer than 50 ms.	

1.5.29 100Gbps CFP2 DWDM Optical Module

1.5.29.1 CFP2-Tunable C-band 191.3-196.05THz---10dBm-1dBm--30dBm-LC

Table 1-1202 CFP2-Tunable C-band 191.3-196.05THz---10dBm-1dBm--30dBm-LC specifications

Item	Value
Basic Information	
Module name	CFP2-Tunable C-band 191.3-196.05THz---10dBm-1dBm--30dBm-LC
Part Number	02312NDL
Model	CFP2-Coherent-DCO-100G-SM
Form factor	CFP2
Application standard	ITU-T G.709/G.798
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)

Item	Value
DDM options	CFP MSA
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	134.698 bit/s(SDFEC)/111.81 bit/s(GFEC, HFEC)
Target transmission distance [km]	80km
Transmitter Optical Characteristics	
Center wavelength [nm]	C band
Center frequency [THz]	191.3 THz ~ 196.05 THz
Tx operating wavelength range [nm]	1529.16 nm - 1560.61 nm
Maximum Tx optical power (AVG) [dBm]	1 dBm
Minimum Tx optical power (AVG) [dBm]	-10 dBm
Minimum extinction ratio [dB]	-
Modulation mode	DP-QPSK/DP-DQPSK
Number of tunable channels	80
Channel spacing [GHz]	50 GHz
Center frequency deviation [GHz]	±1.5 GHz
Tx channel tuning time [s]	60 s
Output optical power in case of laser shutdown [dBm]	≤-35 dBm
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1529.16 nm - 1560.61 nm
Rx sensitivity (AVG) [dBm]	QPSK: -30 dBm DQPSK: -27 dBm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	10 dBm
Overload power (OMA) [dBm]	-
FEC mode	SDFEC StaircaseFEC

Item	Value
Rx optical power range [dBm]	-16 dBm ~ 0 dBm(EOL) 0.5dB penalty@input power: -3 dBm ~ 0 dBm; 0dB penalty@input power: -14 dBm ~ -3 dBm; 0.5dB penalty@input power: -16 dBm ~ -14 dBm 0.5dB penalty@input power: -16 dBm ~ -14 dBm;
Minimum OSNR (back-to-back&EOL) [dB]	12.0 dB for SDFEC&QPSK 13.5 dB for SDFEC&DQPSK 15 dB for StaircaseFEC&DQPSK
CD tolerance [ps/nm]	-10000 ps/nm ~ 40000 ps/nm
PMD tolerance [ps]	25 ps
PDL tolerance [dB]	SD FEC/SC FEC 1 dB penalty@2dB PDL 2 dB penalty@4dB PDL 4 dB penalty@6dB PDL GFEC: 1.5 dB penalty@2dB PDL 2.5 dB penalty@4dB PDL 4.5 dB penalty@6dB PDL
SOP tolerance [krad/s]	0.5 dB penalty@50KHz

1.5.30 100Gbps CFP2 Optical Module

1.5.30.1 100Gbps(4*25.7)-CFP2-SMF-1295.56~1309.14nm-10km-commercial

Table 1-1203 100Gbps(4*25.7)-CFP2-SMF-1295.56~1309.14nm-10km-commercial specifications

Item	Value
Basic Information	
Module name	100Gbps(4*25.7)-CFP2-SMF-1295.56~1309.14nm-10km-commercial

Item	Value
Part Number	02310WUR
Model	OSN010N09
Form factor	CFP2
Application standard	IEEE 802.3ba, 100GBASE-LR4
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	CFP MSA
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	103.125 Gbit/s
Target transmission distance [km]	10 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1295.56 nm 1300.05 nm 1304.58 nm 1309.14 nm
Tx operating wavelength range [nm]	1294.53 nm - 1296.59 nm 1299.02 nm - 1301.09 nm 1303.54 nm - 1305.63 nm 1308.09 nm - 1310.19 nm
Maximum Tx optical power (AVG) [dBm]	per lane: 4.5 dBm
Maximum Tx optical power (OMA) [dBm]	per lane: 4.5 dBm
Minimum Tx optical power (AVG) [dBm]	per lane: -4.3 dBm
Minimum Tx optical power (OMA) [dBm]	per lane: -1.3 dBm
Minimum extinction ratio [dB]	4 dB
Receiver Optical Characteristics	

Item	Value
Rx operating wavelength range [nm]	1294.53 nm - 1296.59 nm 1299.02 nm - 1301.09 nm 1303.54 nm - 1305.63 nm 1308.09 nm - 1310.19 nm
Rx sensitivity (AVG) [dBm]	-
Rx sensitivity (OMA) [dBm]	per lane: -10.6 dBm
Overload power (AVG) [dBm]	per lane: 4.5 dBm
Overload power (OMA) [dBm]	-
NOTE The optical power calculation is based on the OMA value. The optical power read by the device is the average optical power, not the OMA optical power.	

1.5.30.2 100Gbps(4*25.7)-CFP2-SMF-1295.56~1309.14nm-40km-commercial

Table 1-1204 100Gbps(4*25.7)-CFP2-SMF-1295.56~1309.14nm-40km-commercial specifications

Item	Value
Basic Information	
Module name	100Gbps(4*25.7)-CFP2-SMF-1295.56~1309.14nm-40km-commercial
Part Number	02311FAP
Model	OSN040N03
Form factor	CFP2
Application standard	IEEE 802.3ba, 100GBASE-ER4
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	CFP MSA
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V

Item	Value
Transmission rate [bit/s]	103.125 Gbit/s
Target transmission distance [km]	40 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1295.56 nm 1300.05 nm 1304.58 nm 1309.14 nm
Tx operating wavelength range [nm]	1294.53 nm - 1296.59 nm 1299.02 nm - 1301.09 nm 1303.54 nm - 1305.63 nm 1308.09 nm - 1310.19 nm
Maximum Tx optical power (AVG) [dBm]	per lane: 2.9 dBm
Maximum Tx optical power (OMA) [dBm]	per lane: 4.5 dBm
Minimum Tx optical power (AVG) [dBm]	per lane: -2.9 dBm
Minimum Tx optical power (OMA) [dBm]	per lane: 0.1 dBm
Minimum extinction ratio [dB]	8 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1294.53 nm - 1296.59 nm 1299.02 nm - 1301.09 nm 1303.54 nm - 1305.63 nm 1308.09 nm - 1310.19 nm
Rx sensitivity (AVG) [dBm]	-
Rx sensitivity (OMA) [dBm]	per lane: -21.4 dBm
Overload power (AVG) [dBm]	per lane: 4.5 dBm
Overload power (OMA) [dBm]	-
NOTE The optical power calculation is based on the OMA value. The optical power read by the device is the average optical power, not the OMA optical power.	

1.5.30.3 100Gbps(4*25.7)-CFP2-SMF-1295.56~1309.14nm-80km-commercial

Table 1-1205 100Gbps(4*25.7)-CFP2-SMF-1295.56~1309.14nm-80km-commercial specifications

Item	Value
Basic Information	
Module name	100Gbps(4*25.7)-CFP2-SMF-1295.56~1309.14nm-80km-commercial
Part Number	02312AXE
Model	CFP2-100G-1310nm-80km-SM
Form factor	CFP2
Application standard	Huawei Define
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	CFP MSA
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1M
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	103.125 Gbit/s
Target transmission distance [km]	80 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1295.56 nm 1300.05 nm 1304.58 nm 1309.14 nm
Tx operating wavelength range [nm]	1294.53 nm - 1296.59 nm 1299.02 nm - 1301.09 nm 1303.54 nm - 1305.63 nm 1308.09 nm - 1310.19 nm
Maximum Tx optical power (AVG) [dBm]	per lane: 6.5 dBm

Item	Value
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	per lane: 3 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	6 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1294.53 nm - 1296.59 nm 1299.02 nm - 1301.09 nm 1303.54 nm - 1305.63 nm 1308.09 nm - 1310.19 nm
Rx sensitivity (AVG) [dBm]	per lane: -30 dBm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	per lane: -7 dBm
Overload power (OMA) [dBm]	-
NOTE 1. The interface standard is Huawei-specific. 2. The optical power calculation is based on the OMA value. 3. If the module needs to reach the nominal data, the board FEC function must be enabled. 4. This module can be interconnected only with Huawei 100Gbps-80km optical modules. 5. The optical power read by the device is the average optical power, not the OMA optical power.	

1.5.31 100Gbps QSFP28 Optical Module

1.5.31.1 100Gbps(4*25.7)-QSFP28-MMF-850nm-0.1km-commercial

Table 1-1206 100Gbps(4*25.7)-QSFP28-MMF-850nm-0.1km-commercial specifications

Item	Value
Basic Information	
Module name	100Gbps(4*25.7)-QSFP28-MMF-850nm-0.1km-commercial
Part Number	02311NTY

Item	Value
Model	OMND10N13
Form factor	QSFP28
Application standard	IEEE 802.3bm, 100GBASE-SR4
Connector type	MPO-12
Optical fiber type	MMF
Bit error ratio (BER)	<5x10E-5
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-8636
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	103.125 Gbit/s
Target transmission distance [km]	0.07 km(OM3) 0.1 km(OM4)
Transmitter Optical Characteristics	
Center wavelength [nm]	850 nm
Tx operating wavelength range [nm]	840 nm - 860 nm
Maximum Tx optical power (AVG) [dBm]	per lane: 2.4 dBm
Maximum Tx optical power (OMA) [dBm]	per lane: 3 dBm
Minimum Tx optical power (AVG) [dBm]	per lane: -8.4 dBm
Minimum Tx optical power (OMA) [dBm]	per lane: -6.4 dBm
Minimum extinction ratio [dB]	2 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	840 nm - 860 nm
Rx sensitivity (AVG) [dBm]	-
Rx sensitivity (OMA) [dBm]	per lane: -10.3 dBm
Overload power (AVG) [dBm]	per lane: 2.4 dBm
Overload power (OMA) [dBm]	-

Item	Value
NOTE 1. If the module needs to reach the nominal data, the board FEC function must be enabled. 2. The BER 5x10E - 5 is the data that is not enabled by FEC, so that 1x10E - 12 can be reached after FEC . 3. The optical power calculation is based on the OMA value. The optical power read by the device is the average optical power, not the OMA optical power.	

1.5.31.2 100Gbps(4*25.7)-QSFP28-SMF-1271~1331nm-2km-commercial

Table 1-1207 100Gbps(4*25.7)-QSFP28-SMF-1271~1331nm-2km-commercial specifications

Item	Value
Basic Information	
Module name	100Gbps(4*25.7)-QSFP28-SMF-1271~1331nm-2km-commercial
Part Number	02311QDH
Model	OSN020N15
Form factor	QSFP28
Application standard	100G CWDM4 MSA
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<5x10E-5
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-8636
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	103.125 Gbit/s
Target transmission distance [km]	2 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1271 nm 1291 nm 1311 nm 1331 nm

Item	Value
Tx operating wavelength range [nm]	1264.5 nm - 1277.5 nm 1284.5 nm - 1297.5 nm 1304.5 nm - 1317.5 nm 1324.5 nm - 1337.5 nm
Maximum Tx optical power (AVG) [dBm]	per lane: 2.5 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	per lane: -6.5 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	3.5 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1264.5 nm - 1277.5 nm 1284.5 nm - 1297.5 nm 1304.5 nm - 1317.5 nm 1324.5 nm - 1337.5 nm
Rx sensitivity (AVG) [dBm]	-
Rx sensitivity (OMA) [dBm]	per lane: -10 dBm
Overload power (AVG) [dBm]	per lane: 2.5 dBm
Overload power (OMA) [dBm]	-
NOTE 1. If the module needs to reach the nominal data, the board FEC function must be enabled. 2. The BER 5x10E - 5 is the data that is not enabled by FEC, so that 1x10E - 12 can be reached after FEC .	

1.5.31.3 100Gbps(4*25.7)-QSFP28-SMF-1310nm-30km(NO FEC)-40km(FEC)-commercial

Table 1-1208 100Gbps(4*25.7)-QSFP28-SMF-1310nm-30km(NO FEC)-40km(FEC)-commercial specifications

Item	Value
Basic Information	

Item	Value
Module name	100Gbps(4*25.7)-QSFP28-SMF-1310nm-30km(NO FEC)-40km(FEC)-commercial
Part Number	02312AUE
Model	OSN030N05
Form factor	QSFP28
Application standard	IEEE 802.3ba, 100GBASE-ER4
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<5x10E-5
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-8636
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	103.125 Gbit/s
Target transmission distance [km]	40 km(FEC) 30 km(NO FEC)
Transmitter Optical Characteristics	
Center wavelength [nm]	1295.56 nm 1300.05 nm 1304.58 nm 1309.14 nm
Tx operating wavelength range [nm]	1294.53 nm - 1296.59 nm 1299.02 nm - 1301.09 nm 1303.54 nm - 1305.63 nm 1308.09 nm - 1310.19 nm
Maximum Tx optical power (AVG) [dBm]	per lane: 2.9 dBm
Maximum Tx optical power (OMA) [dBm]	per lane: 4.5 dBm
Minimum Tx optical power (AVG) [dBm]	per lane: -2.5 dBm

Item	Value
Minimum Tx optical power (OMA) [dBm]	per lane: 0.1 dBm
Minimum extinction ratio [dB]	8 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1294.53 nm - 1296.59 nm 1299.02 nm - 1301.09 nm 1303.54 nm - 1305.63 nm 1308.09 nm - 1310.19 nm
Rx sensitivity (AVG) [dBm]	-
Rx sensitivity (OMA) [dBm]	per lane: -21.4 dBm
Overload power (AVG) [dBm]	per lane: -3.5 dBm
Overload power (OMA) [dBm]	-
NOTE 1. If the module needs to reach 40 km, the board FEC function must be enabled. Otherwise, it can reach only 30 km. 2. The BER 5×10^{-5} is the data obtained with FEC not enabled. After FEC is enabled, 1×10^{-12} can be reached. 3. In 30 km application, the BER can reach 1×10^{-12} . 4. The optical power calculation is based on the OMA value. The optical power read by the device is the average value, not the OMA optical power.	

1.5.31.4 100Gbps(4*25.7)-QSFP28-SMF-1295.56~1309.14nm-10km-commercial (02312BSS)

Table 1-1209 100Gbps(4*25.7)-QSFP28-SMF-1295.56~1309.14nm-10km-commercial specifications

Item	Value
Basic Information	
Module name	100Gbps(4*25.7)-QSFP28-SMF-1295.56~1309.14nm-10km-commercial
Part Number	02312BSS
Model	OSN010N24
Form factor	QSFP28
Application standard	IEEE 802.3ba, 100GBASE-LR4
Connector type	LC

Item	Value
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-8636
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	103.125 Gbit/s
Target transmission distance [km]	10 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1295.56 nm 1300.05 nm 1304.58 nm 1309.14 nm
Tx operating wavelength range [nm]	1294.53 nm - 1296.59 nm 1299.02 nm - 1301.09 nm 1303.54 nm - 1305.63 nm 1308.09 nm - 1310.19 nm
Maximum Tx optical power (AVG) [dBm]	per lane: 4.5 dBm
Maximum Tx optical power (OMA) [dBm]	per lane: 4.5 dBm
Minimum Tx optical power (AVG) [dBm]	per lane: -4.3 dBm
Minimum Tx optical power (OMA) [dBm]	per lane: -1.3 dBm
Minimum extinction ratio [dB]	4 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1294.53 nm - 1296.59 nm 1299.02 nm - 1301.09 nm 1303.54 nm - 1305.63 nm 1308.09 nm - 1310.19 nm
Rx sensitivity (AVG) [dBm]	-
Rx sensitivity (OMA) [dBm]	per lane: -8.6 dBm

Item	Value
Overload power (AVG) [dBm]	per lane: 4.5 dBm
Overload power (OMA) [dBm]	-
NOTE The optical power calculation is based on the OMA value. The optical power read by the device is the average optical power, not the OMA optical power.	

1.5.31.5 100Gbps(4*25.7)-QSFP28-SMF-1295.56~1309.14nm-80km-commercial

Table 1-1210 100Gbps(4*25.7)-QSFP28-SMF-1295.56~1309.14nm-80km-commercial specifications

Item	Value
Basic Information	
Module name	100Gbps(4*25.7)-QSFP28-SMF-1295.56~1309.14nm-80km-commercial
Part Number	02312NCX
Model	QSFP28-100G-1310nm-80km-SM-01
Form factor	QSFP28
Application standard	Huawei Define
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-8636
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	1000 V
Transmission rate [bit/s]	103.125 Gbit/s
Target transmission distance [km]	80 km
Transmitter Optical Characteristics	

Item	Value
Center wavelength [nm]	1295.56 nm 1300.05 nm 1304.58 nm 1309.14 nm
Tx operating wavelength range [nm]	1294.53 nm - 1296.59 nm 1299.02 nm - 1301.09 nm 1303.54 nm - 1305.63 nm 1308.09 nm - 1310.19 nm
Maximum Tx optical power (AVG) [dBm]	per lane: 6.5 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	per lane: 2 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	6 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1294.53 nm - 1296.59 nm 1299.02 nm - 1301.09 nm 1303.54 nm - 1305.63 nm 1308.09 nm - 1310.19 nm
Rx sensitivity (AVG) [dBm]	per lane: -28 dBm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	per lane: -3.5 dBm
Overload power (OMA) [dBm]	-
NOTE 1. To achieve the nominal value, the FEC function must be enabled on the board and G.652 fibers must be used. 2. The BER 5x10E-5 is the data when FEC is disabled. After FEC is enabled, the BER can reach 1x10E-12. 3. When the optical module is used for 80 km transmission, it must be interconnected with Huawei optical modules of the same model.	

1.5.31.6 100Gbps(4*25.7)-QSFP28-SMF-1295.56~1309.14nm-40km-commercial

Table 1-1211 100Gbps(4*25.7)-QSFP28-SMF-1295.56~1309.14nm-40km-commercial specifications

Item	Value
Basic Information	
Module name	100Gbps(4*25.7)-QSFP28-SMF-1295.56~1309.14nm-40km-commercial
Part Number	02312NVQ
Model	QSFP28-100G-1310-40km-SM
Form factor	QSFP28
Application standard	IEEE 802.3 100GBASE-ER4
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-8636
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	1000 V
Transmission rate [bit/s]	103.125 Gbit/s
Target transmission distance [km]	40 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1295.56 nm 1300.05 nm 1304.58 nm 1309.14 nm
Tx operating wavelength range [nm]	1294.53 nm - 1296.59 nm 1299.02 nm - 1301.09 nm 1303.54 nm - 1305.63 nm 1308.09 nm - 1310.19 nm
Maximum Tx optical power (AVG) [dBm]	per lane: 2.9 dBm

Item	Value
Maximum Tx optical power (OMA) [dBm]	per lane: 4.5 dBm
Minimum Tx optical power (AVG) [dBm]	per lane: -2.9 dBm
Minimum Tx optical power (OMA) [dBm]	per lane: 0.1 dBm
Minimum extinction ratio [dB]	8 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1294.53 nm - 1296.59 nm 1299.02 nm - 1301.09 nm 1303.54 nm - 1305.63 nm 1308.09 nm - 1310.19 nm
Rx sensitivity (AVG) [dBm]	per lane: -20.9 dBm
Rx sensitivity (OMA) [dBm]	per lane: -21.4 dBm
Overload power (AVG) [dBm]	per lane: -3.5 dBm
Overload power (OMA) [dBm]	-
NOTE When an optical module is installed on an interface, the FEC function on the interface is disabled by default. Pay attention to the FEC status of the peer interface during interconnection.	

1.5.31.7 100Gbps(4*25.7)-QSFP28-SMF-1295.56~1309.14nm-10km-commercial (02313SWA)

Table 1-1212 100Gbps(4*25.7)-QSFP28-SMF-1295.56~1309.14nm-10km-commercial specifications

Item	Value
Basic Information	
Module name	100Gbps(4*25.7)-QSFP28-SMF-1295.56~1309.14nm-10km-commercial
Part Number	02313SWA
Model	OSN010N24
Form factor	QSFP28
Application standard	IEEE 802.3ba, 100GBASE-LR4

Item	Value
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-8636
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	500 V
Transmission rate [bit/s]	103.125 Gbit/s
Target transmission distance [km]	10 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1295.56 nm 1300.05 nm 1304.58 nm 1309.14 nm
Tx operating wavelength range [nm]	1294.53 nm - 1296.59 nm 1299.02 nm - 1301.09 nm 1303.54 nm - 1305.63 nm 1308.09 nm - 1310.19 nm
Maximum Tx optical power (AVG) [dBm]	per lane: 4.5 dBm
Maximum Tx optical power (OMA) [dBm]	per lane: 4.5 dBm
Minimum Tx optical power (AVG) [dBm]	per lane: -4.3 dBm
Minimum Tx optical power (OMA) [dBm]	per lane: -1.3 dBm
Minimum extinction ratio [dB]	4 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1294.53 nm - 1296.59 nm 1299.02 nm - 1301.09 nm 1303.54 nm - 1305.63 nm 1308.09 nm - 1310.19 nm
Rx sensitivity (AVG) [dBm]	-

Item	Value
Rx sensitivity (OMA) [dBm]	per lane: -8.6 dBm
Overload power (AVG) [dBm]	per lane: 4.5 dBm
Overload power (OMA) [dBm]	-
NOTE The optical power calculation is based on the OMA value. The optical power read by the device is the average optical power, not the OMA optical power.	

1.5.31.8 100Gbps(4*25.7)-QSFP28-SMF-1310nm-10km-industry

Table 1-1213 100Gbps(4*25.7)-QSFP28-SMF-1310nm-10km-industry specifications

Item	Value
Basic Information	
Module name	100Gbps(4*25.7)-QSFP28-SMF-1310nm-10km-industry
Part Number	02313VFM
Model	QSFP28-100G-10km-SM-I
Form factor	QSFP28
Application standard	IEEE 802.3ba, 100GBASE-LR4
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	-20°C to +85°C (-4°F to +185°F)
DDM options	SFF-8636
Environment standard	RoHS
Security standard	CISPR32 FCC Part15, IEC 61000-4-3,IEC 61000-4-2,EN/IEC60825-1:2014
ESD(HBM1) [V]	1000 V
Transmission rate [bit/s]	103.125 Gbit/s
Target transmission distance [km]	10 km
Transmitter Optical Characteristics	

Item	Value
Center wavelength [nm]	1295.56 nm 1300.055 nm 1304.585 nm 1309.14 nm
Tx operating wavelength range [nm]	1294.53-1296.59 nm 1299.02-1301.09 nm 1303.54-1305.63 nm 1308.09-1310.19 nm
Maximum Tx optical power (AVG) [dBm]	per lane: 4.5 dBm
Maximum Tx optical power (OMA) [dBm]	per lane: 4.5 dBm
Minimum Tx optical power (AVG) [dBm]	per lane: -4.3 dBm
Minimum Tx optical power (OMA) [dBm]	per lane: -1.3 dBm
Minimum extinction ratio [dB]	4 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1294.53-1296.59 nm 1299.02-1301.09 nm 1303.54-1305.63 nm 1308.09-1310.19 nm
Rx sensitivity (AVG) [dBm]	-
Rx sensitivity (OMA) [dBm]	per lane: -8.6 dBm
Overload power (AVG) [dBm]	per lane: 4.5 dBm
Overload power (OMA) [dBm]	-
NOTE 1. The FEC function of the board must be enabled to ensure that the optical module reaches the nominal value. 2. The BER of 5x10E-5 is the data when FEC is disabled. After FEC is enabled, the BER can reach 1x10E-12. 3. See OMA for the actual optical power. The optical power read from the device is the average optical power, not the OMA optical power.	

1.5.31.9 100Gbps-QSFP28-1310nm-10km-commercial

Table 1-1214 100Gbps-QSFP28-1310nm-10km-commercial specifications

Item	Value
Basic Information	
Module name	100Gbps-QSFP28-1310nm-10km-commercial
Part Number	02314GYE
Model	OSN010N22
Form factor	QSFP28
Application standard	IEEE 802.3cd, 100GBASE-LR1
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-8636
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1 CE Class B
ESD(HBM1) [V]	1000 V
Transmission rate [bit/s]	106.25 Gbit/s(PAM4)
Target transmission distance [km]	10 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1310 nm
Tx operating wavelength range [nm]	1304.5 nm - 1317.5 nm
Maximum Tx optical power (AVG) [dBm]	4.8 dBm
Maximum Tx optical power (OMA) [dBm]	4.8 dBm
Minimum Tx optical power (AVG) [dBm]	-1.9 dBm
Minimum Tx optical power (OMA) [dBm]	-1.9 dBm
Minimum extinction ratio [dB]	3.5 dB

Item	Value
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1304.5 nm - 1317.5 nm
Rx sensitivity (AVG) [dBm]	-6.1 dBm
Rx sensitivity (OMA) [dBm]	-6.1 dBm
Overload power (AVG) [dBm]	4.8 dBm
Overload power (OMA) [dBm]	-
NOTE 1. The optical power is subject to the OMA optical power. To reach the nominal value, the FEC function of the optical module must be enabled. 2. The optical power of an optical module displayed on the device is the average optical power, not the OMA optical power. 3. When this type of optical module is used to interconnect with WDM equipment, the WDM client-side 1+1 protection switching time is greater than 50 ms.	

1.5.31.10 100Gbps-QSFP28-SMF-1271nm(Tx)/1331nm(Rx)-10km-commercial

Table 1-1215 100Gbps-QSFP28-SMF-1271nm(Tx)/1331nm(Rx)-10km-commercial specifications

Item	Value
Basic Information	
Module name	100Gbps-QSFP28-SMF-1271nm(Tx)/1331nm(Rx)-10km-commercial
Part Number	02314REE
Model	OSN010B00
Form factor	QSFP28
Application standard	IEEE 802.3bm-2015,IEEE 802.3cu, 100GBASE-BR10
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12(after FEC) <2x10E-4(before FEC)
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	SFF-8636
Environment standard	RoHS

Item	Value
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	1000 V
Transmission rate [bit/s]	106.25 Gbit/s
Target transmission distance [km]	10 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1271 nm
Tx operating wavelength range [nm]	1264.5 nm - 1277.5 nm
Maximum Tx optical power (AVG) [dBm]	4.8 dBm
Maximum Tx optical power (OMA) [dBm]	5 dBm
Minimum Tx optical power (AVG) [dBm]	-1.9 dBm
Minimum Tx optical power (OMA) [dBm]	1.1dBm(TDECQ<1.4dB), -0.3+TDECQ(1.4dB<=TDECQ<=3.4dB)
Minimum extinction ratio [dB]	3.5 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1324.5 nm - 1337.5 nm
Minimum Rx optical power (AVG) [dBm]	-8.2 dBm
Maximum Rx optical power (AVG) [dBm]	4.8 dBm
Maximum Rx optical power (OMA) [dBm]	5 dBm
Rx sensitivity (AVG) [dBm]	-8.2 dBm
Rx sensitivity (OMA) [dBm]	Max(-6.1,TECQ-7.5)
Overload power (AVG) [dBm]	4.8 dBm
Overload power (OMA) [dBm]	-
NOTE 1. The optical power is subject to the OMA. 2. The optical power of an optical module displayed on the device is the average optical power, not the OMA optical power. 3. When this type of optical module is used to connect to a WDM device, the WDM client-side 1+1 protection switching time is greater than 50 ms.	

1.5.31.11 100Gbps-QSFP28-SMF-1331nm(Tx)/1271nm(Rx)-10km-commercial

Table 1-1216 100Gbps-QSFP28-SMF-1331nm(Tx)/1271nm(Rx)-10km-commercial specifications

Item	Value
Basic Information	
Module name	100Gbps-QSFP28-SMF-1331nm(Tx)/1271nm(Rx)-10km-commercial
Part Number	02314REF
Model	OSN010B01
Form factor	QSFP28
Application standard	IEEE 802.3bm-2015, IEEE 802.3cu, 100GBASE-BR10
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12(after FEC) <2x10E-4(before FEC)
Working case temperature [°C(°F)]	0°C to 70°C (32°F to 158°F)
DDM options	SFF-8636
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	1000 V
Transmission rate [bit/s]	106.25 Gbit/s
Target transmission distance [km]	10 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1331 nm
Tx operating wavelength range [nm]	1324.5 nm - 1337.5 nm
Maximum Tx optical power (AVG) [dBm]	4.8 dBm
Maximum Tx optical power (OMA) [dBm]	5 dBm
Minimum Tx optical power (AVG) [dBm]	-1.9 dBm
Minimum Tx optical power (OMA) [dBm]	1.1dBm(TDECQ<1.4dB), -0.3+TDECQ(1.4dB<=TDECQ<=3.4dB)

Item	Value
Minimum extinction ratio [dB]	3.5 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1264.5 nm - 1277.5 nm
Minimum Rx optical power (AVG) [dBm]	-8.2 dBm
Maximum Rx optical power (AVG) [dBm]	4.8 dBm
Maximum Rx optical power (OMA) [dBm]	5 dBm
Rx sensitivity (AVG) [dBm]	-8.2 dBm
Rx sensitivity (OMA) [dBm]	Max(-6.1,TECQ-7.5)
Overload power (AVG) [dBm]	4.8 dBm
Overload power (OMA) [dBm]	-
NOTE 1. The optical power is subject to the OMA. 2. The optical power of an optical module displayed on the device is the average optical power, not the OMA optical power. 3. When this type of optical module is used to connect to a WDM device, the WDM client-side 1+1 protection switching time is greater than 50 ms.	

1.5.32 400Gbps QSFP-DD Optical Module

1.5.32.1 400Gbps(8*56Gb/s PAM4)-QSFP-DD-SMF-1310nm-10km-commercial

Table 1-1217 400Gbps(8*56Gb/s PAM4)-QSFP-DD-SMF-1310nm-10km-commercial specifications

Item	Value
Basic Information	
Module name	400Gbps(8*56Gb/s PAM4)-QSFP-DD-SMF-1310nm-10km-commercial
Part Number	02312HCM
Model	OSK010N02
Form factor	QSFP-DD
Application standard	IEEE802.3 bs-2017

Item	Value
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12(after FEC) <2.4E-4(before FEC)
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	QSFP-DD MSA
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	1000 V
Transmission rate [bit/s]	425 Gbit/s
Target transmission distance [km]	10 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1273.54 nm 1277.89 nm 1282.26 nm 1286.66 nm 1295.56 nm 1300.05 nm 1304.58 nm 1309.14 nm
Tx operating wavelength range [nm]	1272.55 nm ~ 1274.54 nm 1276.89 nm ~ 1278.89 nm 1281.25 nm ~ 1283.27 nm 1285.65 nm ~ 1287.68 nm 1294.53 nm ~ 1296.59 nm 1299.02 nm ~ 1301.09 nm 1303.54 nm ~ 1305.63 nm 1308.09 nm ~ 1310.19 nm
Maximum Tx optical power (AVG) [dBm]	5.3 dBm
Maximum Tx optical power (OMA) [dBm]	5.7 dBm
Minimum Tx optical power (AVG) [dBm]	-2.8 dBm

Item	Value
Minimum Tx optical power (OMA) [dBm]	0.2 dBm
Minimum extinction ratio [dB]	3.5 dB
Modulation mode	-
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1272.55 nm ~ 1274.54 nm 1276.89 nm ~ 1278.89 nm 1281.25 nm ~ 1283.27 nm 1285.65 nm ~ 1287.68 nm 1294.53 nm ~ 1296.59 nm 1299.02 nm ~ 1301.09 nm 1303.54 nm ~ 1305.63 nm 1308.09 nm ~ 1310.19 nm
Rx sensitivity (AVG) [dBm]	-10.1 dBm
Rx sensitivity (OMA) [dBm]	-7.1 dBm
Overload power (AVG) [dBm]	5.3 dBm
Overload power (OMA) [dBm]	-
NOTE When this type of optical module is used to interconnect with WDM equipment, the protection switching time of WDM client-side 1+1 protection is greater than 50 ms.	

1.5.32.2 400Gbps(8*56Gb/s PAM4)-QSFP-DD-SMF-1310nm-40km-commercial

Table 1-1218 400Gbps(8*56Gb/s PAM4)-QSFP-DD-SMF-1310nm-40km-commercial specifications

Item	Value
Basic Information	
Module name	400Gbps(8*56Gb/s PAM4)-QSFP-DD-SMF-1310nm-40km-commercial
Part Number	02312YQG
Model	OSK040N02
Form factor	QSFP-DD
Application standard	IEEE802.3 bs-2017
Connector type	LC

Item	Value
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-12(after FEC) <2.4E-4(before FEC)
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	QSFP-DD MSA
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	1000 V
Transmission rate [bit/s]	425 Gbit/s
Target transmission distance [km]	40 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1273.54 nm 1277.89 nm 1282.26 nm 1286.66 nm 1295.56 nm 1300.05 nm 1304.58 nm 1309.14 nm
Tx operating wavelength range [nm]	1272.55 nm ~ 1274.54 nm 1276.89 nm ~ 1278.89 nm 1281.25 nm ~ 1283.27 nm 1285.65 nm ~ 1287.68 nm 1294.53 nm ~ 1296.59 nm 1299.02 nm ~ 1301.09 nm 1303.54 nm ~ 1305.63 nm 1308.09 nm ~ 1310.19 nm
Maximum Tx optical power (AVG) [dBm]	5.6 dBm
Maximum Tx optical power (OMA) [dBm]	6.4 dBm
Minimum Tx optical power (AVG) [dBm]	-0.6 dBm
Minimum Tx optical power (OMA) [dBm]	2.4 dBm

Item	Value
Minimum extinction ratio [dB]	6 dB
Modulation mode	-
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1272.55 nm ~ 1274.54 nm 1276.89 nm ~ 1278.89 nm 1281.25 nm ~ 1283.27 nm 1285.65 nm ~ 1287.68 nm 1294.53 nm ~ 1296.59 nm 1299.02 nm ~ 1301.09 nm 1303.54 nm ~ 1305.63 nm 1308.09 nm ~ 1310.19 nm
Minimum Rx optical power (AVG) [dBm]	-18.6 dBm
Maximum Rx optical power (AVG) [dBm]	-4.4 dBm
Maximum Rx optical power (OMA) [dBm]	-3.6 dBm
Rx sensitivity (AVG) [dBm]	-18.6 dBm
Rx sensitivity (OMA) [dBm]	-16.1 dBm
Overload power (AVG) [dBm]	-4.4 dBm
Overload power (OMA) [dBm]	-
NOTE When the board is used for 40 km transmission, it must be interconnected with a Huawei optical module of the same model. When this type of optical module is used to interconnect with WDM equipment, the protection switching time of WDM client-side 1+1 protection is greater than 50 ms.	

1.5.32.3 400Gbps-QSFP-DD-SMF-1528.773~1567.133nm-commercial

Table 1-1219 400Gbps-QSFP-DD-SMF-1528.773~1567.133nm-commercial specifications

Item	Value
Basic Information	
Module name	400Gbps-QSFP-DD-SMF-1528.773~1567.133nm-commercial

Item	Value
Part Number	02313GSB
Model	OSK080N03
Form factor	QSFP-DD
Application standard	IEEE802.3 cw
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-15(after FEC) <1.25E-2(before FEC)
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	QSFP-DD MSA
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	1000 V
Transmission rate [bit/s]	478.75 Gbit/s
Target transmission distance [km]	80 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1528.773nm~1567.133nm
Tx operating wavelength range [nm]	1528.773nm~1567.133nm
Maximum Tx optical power (AVG) [dBm]	-6 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	-10 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	-
Modulation mode	400G DP-16QAM
Number of tunable channels	48/64
Channel spacing [GHz]	75GHz/100GHz
Center frequency deviation [GHz]	±1.8 GHz
Tx channel tuning time [s]	180s

Item	Value
Output optical power in case of laser shutdown [dBm]	≤-20 dBm
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1528.773nm~1567.133nm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	0 dBm
Overload power (OMA) [dBm]	-
FEC mode	CFEC RS-FEC
Rx optical power range [dBm]	-12 dBm ~ 0 dBm
Minimum OSNR (back-to-back&EOL) [dB]	26 dB
CD tolerance [ps/nm]	0 ps/nm ~ 2400 ps/nm
PMD tolerance [ps]	10 ps@0.5 dB OSNR penalty
PDL tolerance [dB]	3.5 with ≤ 1.3 dB OSNR penalty
SOP tolerance [krad/s]	50 krad/s@0.3dB penalty
NOTE 1. After being switched to the 400G ZR mode, modules with part numbers 02314MEF and 02314PXP cannot interconnect with this optical module. 2. Only the single-span application is supported. This optical module needs to be used with the fiber amplifier module. 3. When this type of optical module is used to interconnect with WDM equipment, the WDM client-side 1+1 protection switching time is greater than 50 ms. 4. According to the 400G ZR protocol, the hot start time of a port with a 400G ZR optical module is less than 180s, and the cold start time is less than 200s.	

1.5.32.4 400Gbps(4*100Gb/s PAM4)-QSFP-DD-SMF-1310nm-2km-commercial

Table 1-1220 400Gbps(4*100Gb/s PAM4)-QSFP-DD-SMF-1310nm-2km-commercial specifications

Item	Value
Basic Information	
Module name	400Gbps(4*100Gb/s PAM4)-QSFP-DD-SMF-1310nm-2km-commercial
Part Number	02313KLT

Item	Value
Model	OSK002N01
Form factor	QSFP-DD
Application standard	IEEE802.3 bs-2017
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1E-4(before FEC)
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	QSFP-DD MSA
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	1000 V
Transmission rate [bit/s]	425 Gbit/s
Target transmission distance [km]	2 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1271 nm 1291 nm 1311 nm 1331 nm
Tx operating wavelength range [nm]	1264.5 nm ~ 1277.5 nm 1284.5 nm ~ 1297.5 nm 1304.5 nm ~ 1317.5 nm 1324.5 nm ~ 1337.5 nm
Maximum Tx optical power (AVG) [dBm]	per lane: 3.5 dBm
Maximum Tx optical power (OMA) [dBm]	per lane: 3.7 dBm
Minimum Tx optical power (AVG) [dBm]	per lane: -3.3 dBm
Minimum Tx optical power (OMA) [dBm]	per lane: -0.3 dBm
Minimum extinction ratio [dB]	3.5 dB
Modulation mode	-
Receiver Optical Characteristics	

Item	Value
Rx operating wavelength range [nm]	1264.5 nm ~ 1277.5 nm 1284.5 nm ~ 1297.5 nm 1304.5 nm ~ 1317.5 nm 1324.5 nm ~ 1337.5 nm
Minimum Rx optical power (AVG) [dBm]	per lane: -7.3 dBm
Maximum Rx optical power (AVG) [dBm]	per lane: 3.5 dBm
Maximum Rx optical power (OMA) [dBm]	per lane: -7.6 dBm
Rx sensitivity (AVG) [dBm]	per lane: -7.6 dBm
Rx sensitivity (OMA) [dBm]	-4.6dBm, SECQ – 6.0dB
Overload power (AVG) [dBm]	per lane: 3.5 dBm
Overload power (OMA) [dBm]	-
NOTE During link deployment, focus on the reflection indicators of each node. The discrete reflection indicators must meet the IEEE specifications. IEEE 802.3cu standard discrete reflection requirements: For a 400G FR4 link with six reflection points, each reflection point must be less than or equal to -38 dB. For a 400G FR4 link with eight reflection points, each reflection point must be less than or equal to -40 dB. For a 400G FR4 link with 10 reflection points, each reflection point must be less than or equal to -41 dB. When this type of optical module is used to interconnect with WDM equipment, the protection switching time of WDM client-side 1+1 protection is greater than 50 ms.	

1.5.32.5 400Gbps-QSFP-DD-SMF-1547nm-80km-commercial

Table 1-1221 400Gbps-QSFP-DD-SMF-1547nm-80km-commercial specifications

Item	Value
Basic Information	
Module name	400Gbps-QSFP-DD-SMF-1547nm-80km-commercial
Part Number	02313LHH
Model	OSK080N04
Form factor	QSFP-DD
Application standard	IEEE802.3 cw

Item	Value
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1E-15(after FEC) <1.25E-2(before FEC)
Working case temperature [°C(°F)]	0°C to 70°C (32°F to 158°F)
DDM options	QSFP-DD MSA
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	1000 V
Transmission rate [bit/s]	478.75 Gbit/s
Target transmission distance [km]	80 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1547.715 nm
Tx operating wavelength range [nm]	1547.715 nm
Maximum Tx optical power (AVG) [dBm]	-6 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	-10 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	-
Modulation mode	400G DP-16QAM
Number of tunable channels	-
Channel spacing [GHz]	-
Center frequency deviation [GHz]	±1.8 GHz
Tx channel tuning time [s]	180s
Output optical power in case of laser shutdown [dBm]	≤-20 dBm
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1547.715 nm

Item	Value
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	0 dBm
Overload power (OMA) [dBm]	-
FEC mode	CFEC RS-FEC
Rx optical power range [dBm]	-12 dBm ~ 0 dBm: OSNR $\geq$ 26dB -20 dBm ~ 0 dBm: OSNR $\geq$ 34dB
Minimum OSNR (back-to-back&EOL) [dB]	26 dB
PMD tolerance [ps]	10 ps@0.5 dB OSNR penalty
PDL tolerance [dB]	2.5 with $\leq$ 0.8 dB OSNR penalty
SOP tolerance [krad/s]	50 krad/s@0.3dB penalty
NOTE After being switched to the 400G ZR mode, modules with part numbers 02314MEF and 02314PXP cannot interconnect with this optical module. Only single-span applications are supported. 1. If the link loss is less than 22 dB (standard-defined), site survey is not required and an OA only needs to be configured at the transmit end. 2. If the transmission distance ranges from 80 km to 120 km, site survey is required to determine the OA configuration scheme: (1) If the link loss is less than or equal to 24 dB, an OA needs to be configured only at the transmit end. (2) If the link loss is greater than 24 dB but less than or equal to 28 dB, an OA needs to be configured at both the transmit and receive ends, and the receive optical power threshold of the involved module needs to be switched using a command. (3) If the link loss is greater than 28 dB, optimize the link to ensure that the loss is less than 28 dB. (4) When this type of optical module is used to interconnect with WDM equipment, the protection switching time of WDM client-side 1+1 protection is greater than 50 ms. (5) According to the 400G ZR protocol, the hot start time of a port with a 400G ZR optical module is less than 180s, and the cold start time is less than 200s.	

1.5.32.6 400Gbps(4*100Gb/s PAM4)-QSFP-DD-SMF-1310nm-6km-commercial

Table 1-1222 400Gbps(4*100Gb/s PAM4)-QSFP-DD-SMF-1310nm-6km-commercial specifications

Item	Value
Basic Information	

Item	Value
Module name	400Gbps(4*100Gb/s PAM4)-QSFP-DD-SMF-1310nm-6km-commercial
Part Number	02314LKG
Model	OSK006NO1
Form factor	QSFP-DD
Application standard	IEEE802.3 bs-2017
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	1E-6(Pre-FEC)
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	QSFP-DD MSA
Environment standard	RoHS
Security standard	FCC Class B, IEC 60825-1 Class 1
ESD(HBM1) [V]	1000 V
Transmission rate [bit/s]	425 Gbit/s
Target transmission distance [km]	6 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1271nm 1291nm 1311nm 1331nm
Tx operating wavelength range [nm]	1264.5nm~1277.5nm 1284.5nm~1297.5nm 1304.5nm~1317.5nm 1324.5nm~1337.5nm
Maximum Tx optical power (AVG) [dBm]	5.1 dBm
Maximum Tx optical power (OMA) [dBm]	4.4 dBm
Minimum Tx optical power (AVG) [dBm]	-2.7 dBm
Minimum Tx optical power (OMA) [dBm]	0.3 dBm

Item	Value
Minimum extinction ratio [dB]	3.5 dB
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1264.5nm~1277.5nm 1284.5nm~1297.5nm 1304.5nm~1317.5nm 1324.5nm~1337.5nm
Minimum Rx optical power (AVG) [dBm]	-9 dBm
Maximum Rx optical power (AVG) [dBm]	5.1 dBm
Maximum Rx optical power (OMA) [dBm]	4.4 dBm
Rx sensitivity (AVG) [dBm]	-9.8 dBm
Rx sensitivity (OMA) [dBm]	-6.8 dBm
Overload power (AVG) [dBm]	5.1 dBm
Overload power (OMA) [dBm]	-
NOTE The optical module needs to be interconnected with the optical module of the same model. When this type of optical module is used to interconnect with WDM equipment, the WDM client-side 1+1 protection switching time is greater than 50 ms. Reflectance indicators of each node must be given attention during link deployment. Discrete reflectance indicators must comply with IEEE specifications. Discrete reflectance requirements of the IEEE 802.3cu standard: For a 400G LR4-6 link with six reflection points, the discrete reflectance of each reflection point must be less than or equal to -35 dB. For a 400G LR4-6 link with eight reflection points, the discrete reflectance of each reflection point must be less than or equal to -37 dB. For a 400G LR4-6 link with 10 reflection points, the discrete reflectance of each reflection point must be less than or equal to -39 dB.	

1.5.32.7 High Speed Transceiver,QSFP-DD,Tunable C-band 191.3-196.1THz, 478.75Gb/s,-10dBm,-6dBm,-12dBm,LC,SMF

Table 1-1223 High Speed Transceiver,QSFP-DD,Tunable C-band 191.3-196.1THz, 478.75Gb/s,-10dBm,-6dBm,-12dBm,LC,SMF specifications

Item	Value
Basic Information	

Item	Value
Module name	High Speed Transceiver,QSFP-DD,Tunable C-band 191.3-196.1THz, 478.75Gb/s,-10dBm,-6dBm,-12dBm,LC,SMF
Part Number	02314MEF
Model	OSK080N11
Form factor	QSFP-DD
Application standard	IEEE802.3 cw / IEEE 802.3 bs
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-15(after FEC)
Working case temperature [°C(°F)]	0°C to 70°C(32°F to 158°F)
DDM options	QSFP-DD MSA
Environment standard	RoHS
Security standard	TUV: EN 62368-1, EN 60825-1; CB: IEC60825-1, IEC60825-2; NRTL: CAN/CSA C22.2 NO.62368-1-12 * UL 62368-1; FDA: 21CFR 1040.10 and 1040.11, Notice NO.56;
ESD(HBM1) [V]	1000 V
Transmission rate [bit/s]	478.75 Gbit/s
Target transmission distance [km]	400G ZR:80km(with OA) 400G ZR+:80km(single-span)/ 480km(six-span)
Transmitter Optical Characteristics	
Center wavelength [nm]	1528.773nm~1567.133nm
Tx operating wavelength range [nm]	1528.773nm~1567.133nm
Maximum Tx optical power (AVG) [dBm]	-6 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	-10 dBm

Item	Value
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	-
Modulation mode	400G DP-16QAM
Number of tunable channels	48 channels (100 GHz spacing)/64 channels (75 GHz spacing)
Channel spacing [GHz]	75GHz/100GHz/3.125GHz/6.25GHz/12.5GHz/25GHz/50GHz/100GHz/33GHz
Center frequency deviation [GHz]	±1.8 GHz
Tx channel tuning time [s]	180s
Output optical power in case of laser shutdown [dBm]	≤-20 dBm
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1528.773nm~1567.133nm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	0 dBm
Overload power (OMA) [dBm]	-
FEC mode	CFEC for ZR OFEC for ZR+
Rx optical power range [dBm]	-12 dBm ~ 0 dBm (Colored light mode) -20 dBm ~ 0 dBm (Gray light mode)
Minimum OSNR (back-to-back&EOL) [dB]	24 dB
CD tolerance [ps/nm]	400ZR 2400 ps/nm 400ZR+ 20000 ps/nm 300ZR+ 40000 ps/nm 200ZR+ 50000 ps/nm 100ZR+ 100000 ps/nm
PMD tolerance [ps]	20 ps@0.5 dB OSNR penalty
PDL tolerance [dB]	3.5 with ≤ 1.3 dB OSNR penalty
SOP tolerance [krad/s]	50 krad/s@0.5dB penalty

Item	Value
NOTE 1. The 400G ZR mode supports only single-span applications. This optical module must be used with an OA module. 2. The 400G ZR+ mode supports multi-span applications (six spans at most). 3. When working in 400G ZR mode, the module cannot interconnect with modules with part numbers 02313GRH, 02313LHH, and 02313GSB. 4. The optical module supports eight working modes. The two ends must work in the same mode. If the two ends work in different modes, the working status of the interconnected link is abnormal. When this type of optical module is used to interconnect with WDM equipment, the WDM client-side 1+1 protection switching time is greater than 50 ms. Appendix: Eight modes supported by the optical module: 400G ZR colored light, 400G ZR gray light, 4 x 100G ZR colored light, 400G ZR+, 4 x 100G ZR+, 3 x 100G ZR+, 2 x 100G ZR+, and 1 x 100G ZR+	

1.5.32.8 400G Transceiver,QSFP-DD,Tunable C-band 191.3-196.1THz, 478.75Gb/s,0dBm,4dBm,-20dBm,LC,ITU-T G.652(SMF)

Table 1-1224 400G Transceiver,QSFP-DD,Tunable C-band 191.3-196.1THz, 478.75Gb/s,0dBm,4dBm,-20dBm,LC,ITU-T G.652(SMF) specifications

Item	Value
Basic Information	
Module name	400G Transceiver,QSFP-DD,Tunable C-band 191.3-196.1THz,478.75Gb/s, 0dBm,4dBm,-20dBm,LC,ITU-T G. 652(SMF)
Part Number	02314PXP
Model	OSK080N12
Form factor	QSFP-DD
Application standard	IEEE802.3 cw / IEEE 802.3 bs
Connector type	LC
Optical fiber type	SMF
Bit error ratio (BER)	<1x10E-15(after FEC) <1.8E-2(before FEC)
Working case temperature [°C(°F)]	0°C to 75°C (32°F to 167°F)
DDM options	QSFP-DD MSA
Environment standard	RoHS

Item	Value
Security standard	TUV: EN 62368-1, EN 60825-1; CB: IEC60825-1, IEC60825-2; NRTL: CAN/CSA C22.2 NO.62368-1-12 * UL 62368-1; FDA: 21CFR 1040.10 and 1040.11, Notice NO.56;
ESD(HBM1) [V]	1000 V
Transmission rate [bit/s]	478.75 Gbit/s
Target transmission distance [km]	400G ZR:80km/120km(with OA) 400G ZR+:80km(single-span)/ 480km(six-span)
Transmitter Optical Characteristics	
Center wavelength [nm]	1528.773nm~1567.133nm
Tx operating wavelength range [nm]	1528.773nm~1567.133nm
Maximum Tx optical power (AVG) [dBm]	1 dBm
Maximum Tx optical power (OMA) [dBm]	-
Minimum Tx optical power (AVG) [dBm]	-10 dBm
Minimum Tx optical power (OMA) [dBm]	-
Minimum extinction ratio [dB]	-
Modulation mode	400G DP-16QAM
Number of tunable channels	48 channels (100 GHz spacing)/64 channels (75 GHz spacing)
Channel spacing [GHz]	75GHz/100GHz
Center frequency deviation [GHz]	±1.8 GHz
Tx channel tuning time [s]	180s
Output optical power in case of laser shutdown [dBm]	≤-20 dBm
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1528.773nm~1567.133nm
Rx sensitivity (OMA) [dBm]	-
Overload power (AVG) [dBm]	1 dBm

Item	Value
Overload power (OMA) [dBm]	-
FEC mode	CFEC for ZR OFEC for ZR+
Rx optical power range [dBm]	Gray light mode: 400G ZR -20 dBm ~ 1 dBm 4*100G ZR -20 dBm ~ 1 dBm Colored light mode: 400G ZR -12~1 dBm 4*100G ZR -12~1 dBm 400G ZR+ -12~1 dBm 4*100G ZR+ -12~1 dBm 3*100G ZR+ -15~1 dBm 2*100G ZR+ -18~1 dBm 1*100G ZR+ -18~1 dBm
Minimum OSNR (back-to-back&EOL) [dB]	400ZR 26 dB/0.1nm 400ZR+ 24 dB/0.1nm 300ZR+ 21 dB/0.1nm 200ZR+ 16 dB/0.1nm 100ZR+ 12.5 dB/0.1nm
CD tolerance [ps/nm]	400ZR 2400 ps/nm 400ZR+ 20000 ps/nm 300ZR+ 40000 ps/nm 200ZR+ 50000 ps/nm 100ZR+ 100000 ps/nm
PMD tolerance [ps]	20 ps@0.5 dB OSNR penalty
PDL tolerance [dB]	3.5 with ≤ 1.3 dB OSNR penalty
SOP tolerance [krad/s]	50 krad/s@0.5dB penalty

Item	Value
NOTE 1. 400G ZR modes support only single-span applications. Use the 400G ZR single mode for single-span applications. 2. The 400G ZR+ mode supports multi-span applications (six spans at most). 3. When working in 400G ZR mode, the module cannot interconnect with modules with part numbers 02313GRH, 02313LHH, and 02313GSB. 4. The optical module supports nine working modes (see below). The two ends must work in the same mode. If the two ends work in different modes, the working status of the interconnected link is abnormal. When this type of optical module is used to interconnect with WDM equipment, the WDM client-side 1+1 protection switching time is greater than 50 ms. Appendix: The nine modes supported by the optical module: 400G ZR colored light, 400G ZR gray light, 4 x 100G ZR colored light, 4 x 100G ZR gray light, 400G ZR+, 4 x 100G ZR+, 3 x 100G ZR+, 2 x 100G ZR+, and 1 x 100G ZR+	

1.5.33 Optical Amplifier Module

1.5.33.1 Optical Amplifier-QSFP28-SMF-1535nm~1547nm

Table 1-1225 Optical Amplifier-QSFP28-SMF-1535nm~1547nm specifications

Item	Value
Basic Information	
Module name	Optical Amplifier-QSFP28-SMF-1535nm~1547nm
Part Number	02313JCM
Model	QSFP28-Amplifier-SM
Form factor	QSFP28
Connector type	LC
Optical fiber type	SMF
Working case temperature [°C(°F)]	-5°C to +75°C (23°F to 167°F)
Environment standard	RoHS
Security standard	EN/IEC60825-1:2014;EN/IEC62368-1:2014;UL 62368-1 Ed.2
ESD(HBM1) [V]	2000 V
Transmitter Optical Characteristics	
Center wavelength [nm]	1547.715nm

Item	Value
Maximum Tx optical power (AVG) [dBm]	+7 dBm
Maximum Tx optical power (OMA) [dBm]	-
Receiver Optical Characteristics	
Rx operating wavelength range [nm]	1535.82 nm~1547.72 nm
Minimum Rx optical power (AVG) [dBm]	-29 dbm
NOTE Works with 400G ZR optical modules. 1. When the transmission distance ranges from 40 km to 80 km and the insertion loss of the optical link is less than 22 dB, the optical amplifier needs to be configured only at the transmit end. 2. When the transmission distance ranges from 80 km to 120 km: 1. When the insertion loss of an optical link is less than or equal to 24 dB, an optical amplifier needs to be configured only at the transmit end. 2. When the insertion loss of an optical link is greater than 24 dB but less than or equal to 28 dB, optical amplifiers must be configured at both the transmit and receive ends, and the receive optical power threshold of the QSFP-DD optical module must be manually modified. 3. If the insertion loss of an optical link is greater than 28 dB, the link needs to be optimized to reduce the insertion loss to a value less than 28 dB. Note: The optical amplifier module works at the physical layer. If the corresponding physical port is steady on, it indicates that the optical amplifier module works normally and no LOS alarm is reported. If the corresponding physical port is off, it indicates that the optical amplifier module does not work normally or a LOS alarm is reported. If the status of an amplifier module port is up, the amplifier module board is working properly and no LOS alarm is reported. If the status of an amplifier module port is down, the amplifier module board is not working properly or a LOS alarm is reported.	

1.5.34 AE 905S B Module

Description

Existing core network devices, which do not support 1588, cannot obtain clock signals from BITS servers. Upgrading core network devices to support 1588 is both complex and costly. To address this issue, the AE 905S B module is developed. After having an AE 905S B module equipped, the device will be able to support 1588v2. [Figure 1-216](#) and [Figure 1-217](#) illustrate the appearance of an AE 905S B module.

Figure 1-216 AE 905S B module (front view)



Figure 1-217 AE 905S B module (rear view)



 NOTE

An AE 905S B module must be inserted into a GE optical interface of SFP type.

The AE 905S B module has a STAT indicator under the **HUAWEI** logo. [Table 1-1226](#) describes STAT indicator states.

Table 1-1226 Description of STAT indicator states

State	Description
On (green)	The AE 905S B module is operating properly.
On (red)	The AE 905S B module has experienced a hardware fault or is overheated.

State	Description
Blinking green once every second	No connection has been established on the GE interface.
Blinking red once every second	The GPS frequency or time is out of lock.
Off	The AE 905S B module is powered off or not operating.

The AE 905S B module meets industrial-grade requirements. [Table 1-1227](#) lists its interface specifications.

Table 1-1227 AE 905S B (03032SWQ) module interface specifications

Item	Specification
BOM Number	03032SWQ
Board Name for Order	ANPM000GPS02
Interface type	SMA
Encapsulation type	SFP
Input signal	An AE 905S B module uses the SMA interface to receive GPS/Beidou/GLONASS satellite signals.
Output signal	An AE 905S B module uses the GE interface to provide synchronous Ethernet and 1588v2

Installation

NOTICE

To meet heat dissipation requirements, leave interfaces around the AE 905S B module empty. A maximum of two AE 905S B modules can be installed on a device.

[Figure 1-218](#) illustrates connections between the AE 905S B module, surge protector, and antenna.

Figure 1-218 Connections between the AE 905S B module, surge protector, and antenna



NOTE

For the convenience of description, the GPS antenna in the description generally refers to the GPS/GLONASS/BEIDOU antenna.

The AE 905S B module must be used with the GPS antenna and GPS surge protector and is connected to the GPS antenna and GPS surge protector through the GPS feeder.

The GPS antenna receives satellite signals from the GPS. A GPS surge protector can protect a device against the lightning strikes induced by the antenna feeder. Without the protection of a surge protector, a device may be damaged by surge currents or voltage in a lightning weather. The GPS feeder transmits GPS signals.

The image shown here is indicative only. If there is any inconsistency between the image and the actual product, the actual product shall govern.

Before inserting an AE 905S B module into an interface, lock the latch. Before removing an AE 905S B module from an optical interface, unlock the latch. For details, see [Figure 1-219](#).

Figure 1-219 Latch

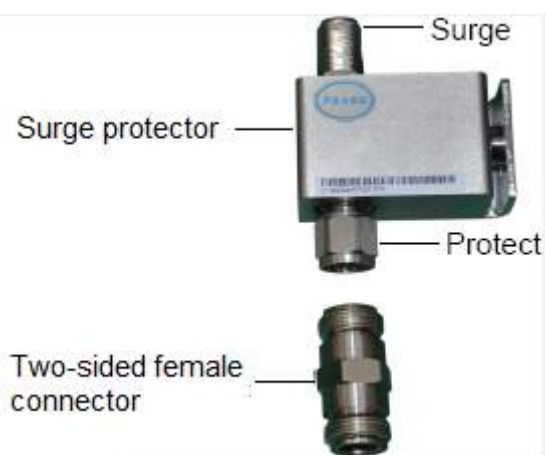


Installation Procedure

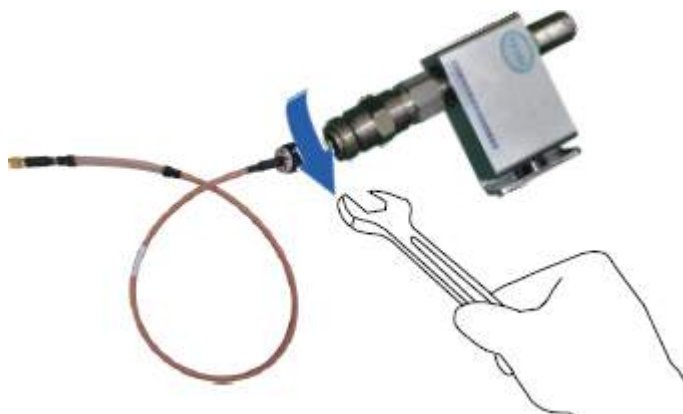
NOTE

For details about how to install the GPS satellite antenna system, see the related [GPS quick installation guide](#).

1. Connect a two-sided female connector to the Protect interface of the surge protector.



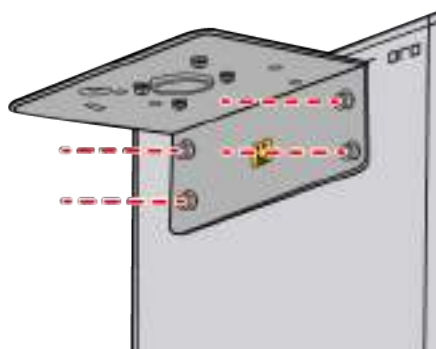
2. Connect one end of the coaxial cable to the Protect interface of the surge protector and fasten the joint with a wrench.



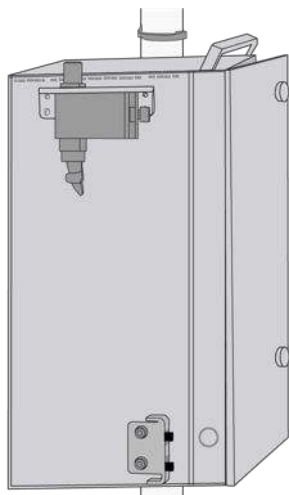
3. Implement 1+3+3 waterproof protection where the coaxial cable and the Protect interface of the surge protector are connected, and fasten cable ties at both ends.



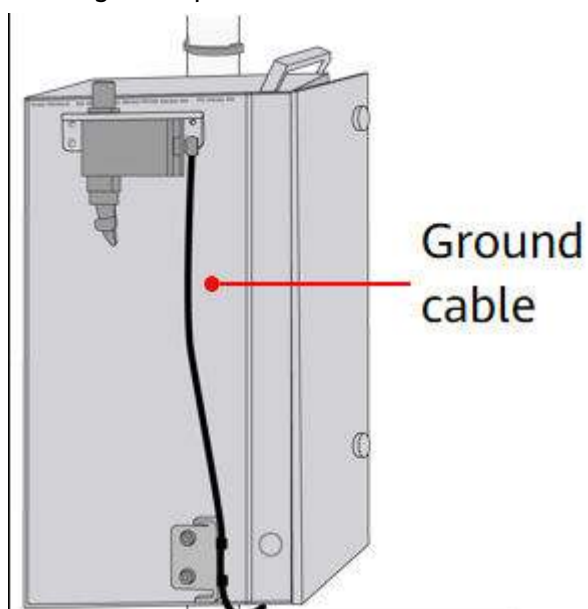
4. Install the GPS support on the left or right side of the cabinet and tighten the four screws.



5. Secure the surge protector on the GPS support.



6. Install the ground cable of the surge protector. Connect one end of the ground cable to the GND interface of the surge protector and the other end to the ground point inside the cabinet.



 NOTE

The surge protector must use OT M8 terminals.

7. Route the jumper through the cable inlet on the right side of the floor of the cabinet, and connect the GPS feeder to the AE 905S.
8. Insert the AE 905S B module into an optical interface of the device with the latch of the AE 905S B module locked.
9. Connect one end of the GPS feeder to the Surge interface of the surge protector and the other end of the GPS feeder to the GPS antenna. Fasten each connector with a wrench.
10. Implement 1+3+3 waterproof protection, and fasten cable ties at both ends.
 - Implement waterproof protection where the feeder connector and the Surge interface of the surge protector are connected.
 - Implement waterproof protection where the feeder connector and the N joint of the GPS antenna are connected.

 NOTE

For details about how to make a GPS feeder connector, see the related GPS quick installation guide.

1.5.35 AE 905S Module

Description

Existing core network devices, which do not support 1588, cannot obtain clock signals from BITS servers. Upgrading core network devices to support 1588 is both complex and costly. To address this issue, the AE 905S module is developed. After having an AE 905S module equipped, the device will be able to support 1588v2. **Figure 1-220** and **Figure 1-221** illustrate the appearance of an AE 905S module.

Figure 1-220 AE 905S module (front view)



Figure 1-221 AE 905S module (rear view)



 NOTE

An AE 905S module must be inserted into a GE optical interface of SFP type.

The AE 905S module has a STAT indicator under the **HUAWEI** logo. [Table 1-1228](#) describes STAT indicator states.

Table 1-1228 Description of STAT indicator states

State	Description
On (green)	The AE 905S module is operating properly.
On (red)	The AE 905S module has experienced a hardware fault or is overheated.
Blinking green once every second	No connection has been established on the GE interface.
Blinking red once every second	The GPS frequency or time is out of lock.
Off	The AE 905S module is powered off or not operating.

The AE 905S module meets industrial-grade requirements. [Table 1-1229](#) lists its interface specifications.

Table 1-1229 AE 905S (03031TUX) module interface specifications

Item	Specification
BOM Number	03031TUX
Board Name for Order	ANPM000GPS01
Interface type	SMA
Encapsulation type	SFP
Input signal	An AE 905S module uses the SMA interface to receive GPS satellite signals.
Output signal	An AE 905S module uses the GE interface to provide synchronous Ethernet and 1588v2 for device.

Installation

NOTICE

To meet heat dissipation requirements, leave interfaces around the AE 905S module empty. A maximum of two AE 905S modules can be installed on a device.

Figure 1-222 illustrates connections between the AE 905S module, surge protector, and antenna.

Figure 1-222 Connections between the AE 905S module, surge protector, and antenna



NOTE

For the convenience of description, the GPS antenna in the description generally refers to the GPS/GLONASS/BEIDOU antenna.

The AE 905S module must be used with the GPS antenna and GPS surge protector and is connected to the GPS antenna and GPS surge protector through the GPS feeder.

The GPS antenna receives satellite signals from the GPS. A GPS surge protector can protect a device against the lightning strikes induced by the antenna feeder. Without the protection of a surge protector, a device may be damaged by surge currents or voltage in a lightning weather. The GPS feeder transmits GPS signals.

The image shown here is indicative only. If there is any inconsistency between the image and the actual product, the actual product shall govern.

Before inserting an AE 905S module into an interface, lock the latch. Before removing an AE 905S module from an optical interface, unlock the latch. For details, see [Figure 1-223](#).

Figure 1-223 Latch

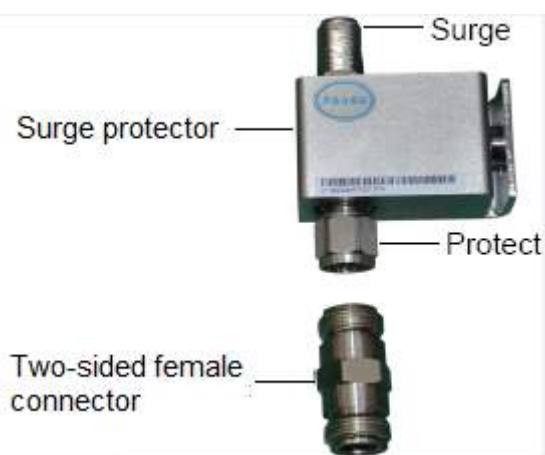


Installation Procedure

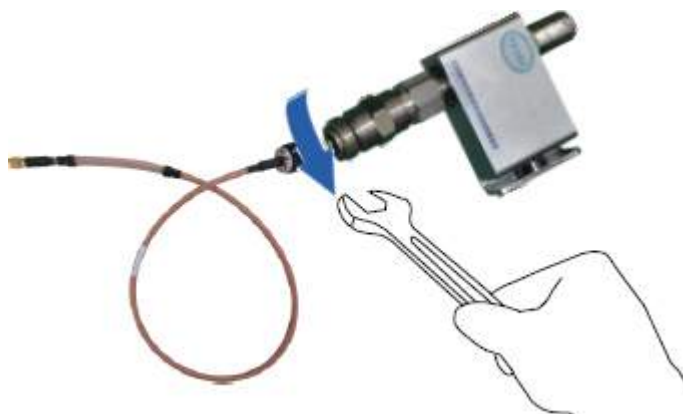
NOTE

For details about how to install the GPS satellite antenna system, see the related [GPS quick installation guide](#).

1. Connect a two-sided female connector to the Protect interface of the surge protector.



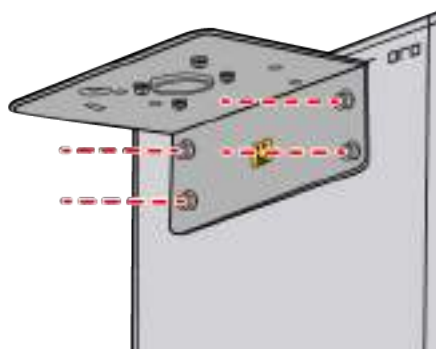
2. Connect one end of the coaxial cable to the Protect interface of the surge protector and fasten the joint with a wrench.



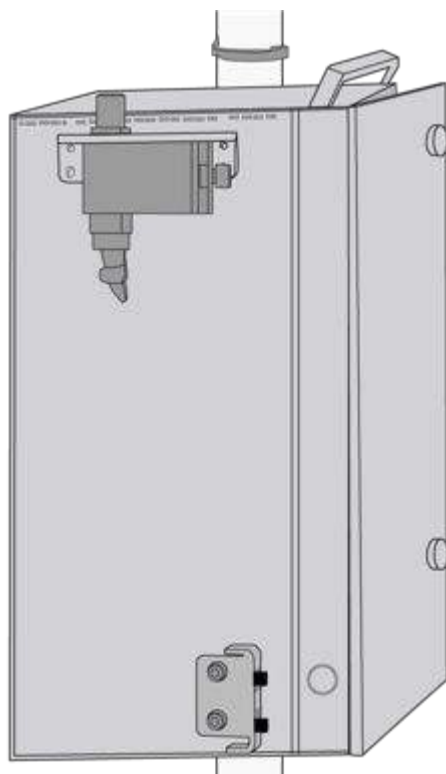
3. Implement 1+3+3 waterproof protection where the coaxial cable and the Protect interface of the surge protector are connected, and fasten cable ties at both ends.



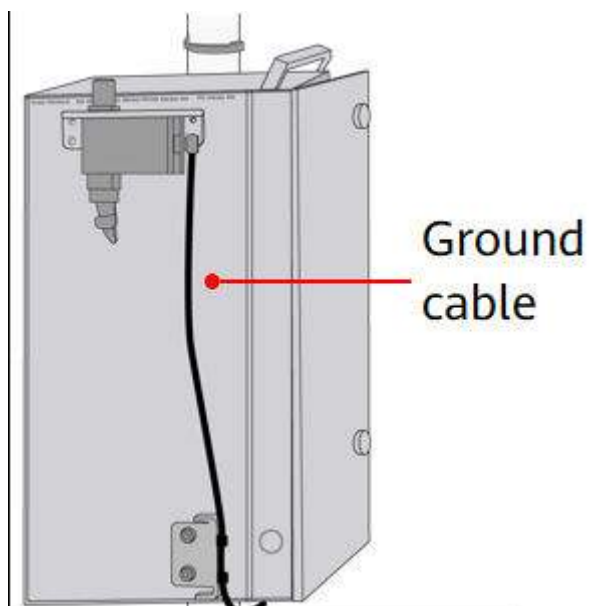
4. Install the GPS support on the left or right side of the cabinet and tighten the four screws.



5. Secure the surge protector on the GPS support.



6. Install the ground cable of the surge protector. Connect one end of the ground cable to the GND interface of the surge protector and the other end to the ground point inside the cabinet.



NOTE

The surge protector must use OT M8 terminals.

7. Route the jumper through the cable inlet on the right side of the floor of the cabinet, and connect the GPS feeder to the AE 905S.
8. Insert the AE 905S module into an optical interface of the device with the latch of the AE 905S module locked.

9. Connect one end of the GPS feeder to the Surge interface of the surge protector and the other end of the GPS feeder to the GPS antenna. Fasten each connector with a wrench.
10. Implement 1+3+3 waterproof protection, and fasten cable ties at both ends.
 - Implement waterproof protection where the feeder connector and the Surge interface of the surge protector are connected.
 - Implement waterproof protection where the feeder connector and the N joint of the GPS antenna are connected.

 NOTE

For details about how to make a GPS feeder connector, see the related GPS quick installation guide.

1.5.36 AE 905M Module

Description

Existing core network devices, which do not support 1588, cannot obtain clock signals from BITS servers. Upgrading core network devices to support 1588 is both complex and costly. To address this issue, the AE 905M module is developed. After having an AE 905M module equipped, the device will be able to support 1588v2. [Figure 1-224](#) and [Figure 1-225](#) illustrate the appearance of an AE 905M module.

Figure 1-224 AE 905M module (front view)



Figure 1-225 AE 905M module (rear view)



NOTE

- The AE 905M module applies only to SFP/SFP+/SFP28 ports.

The AE 905M module has a STAT indicator under the **HUAWEI** logo. [Table 1-1230](#) describes STAT indicator states.

Table 1-1230 Description of STAT indicator states

Status	Description
Blinking green every 2s	The AE 905M module is waiting to be loaded.
Blinking red every 0.5s	The AE 905M module is being upgraded or loaded.
Blinking red every 1s	Satellite synchronization is unavailable for the AE 905M module.
Blinking green every 1s	Satellite synchronization is available for the AE 905M module.
Off	The AE 905M module is powered off or not operating.

The AE 905M module meets industrial-grade requirements. [Table 1-1231](#) lists its interface specifications.

Table 1-1231 AE 905M (03034CGQ) module interface specifications

Item	Specification
BOM Number	03034CGQ

Item	Specification
Board Name for Order	ANPM000GPS03
Interface type	SMA
Encapsulation type	SFP/SFP+/SFP28
Input signal	An AE 905M module uses the SMA interface to receive GPS satellite signals.
Output signal	Time information is provided for devices through SFP/SFP+/SFP28 ports.

Installation

NOTICE

To meet heat dissipation requirements, leave interfaces around the AE 905M module empty. A maximum of two AE 905M modules can be installed on a device.

Figure 1-226 illustrates connections between the AE 905M module, surge protector, and antenna.

Figure 1-226 Connections between the AE 905M module, surge protector, and antenna



NOTE

For the convenience of description, the GPS antenna in the description generally refers to the GPS/GLONASS/GALILEO/BEIDOU antenna.

The AE 905M module must be used with the GPS antenna and GPS surge protector and is connected to the GPS antenna and GPS surge protector through the GPS feeder.

The GPS antenna receives satellite signals from the GPS. A GPS surge protector can protect a device against the lightning strikes induced by the antenna feeder. Without the protection of a surge protector, a device may be damaged by surge currents or voltage in a lightning weather. The GPS feeder transmits GPS signals.

The image shown here is indicative only. If there is any inconsistency between the image and the actual product, the actual product shall govern.

Before inserting an AE 905M module into an interface, lock the latch. Before removing an AE 905M module from an optical interface, unlock the latch. For details, see [Figure 1-227](#).

Figure 1-227 Latch

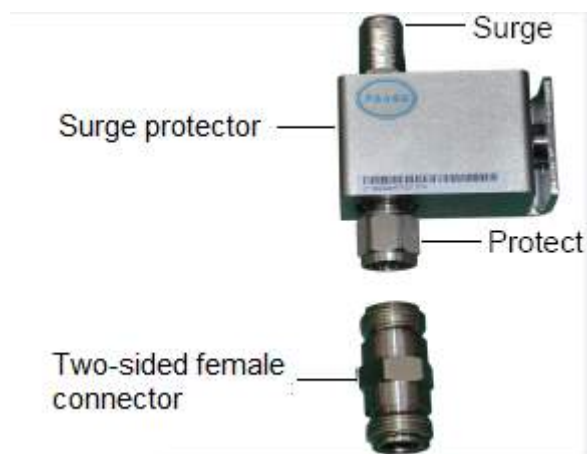


Installation Procedure

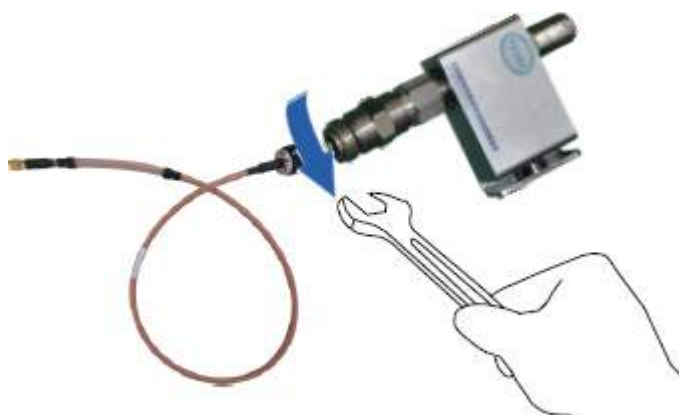
NOTE

For details about how to install the GPS satellite antenna system, see the related [GPS quick installation guide](#).

1. Connect a two-sided female connector to the Protect interface of the surge protector.



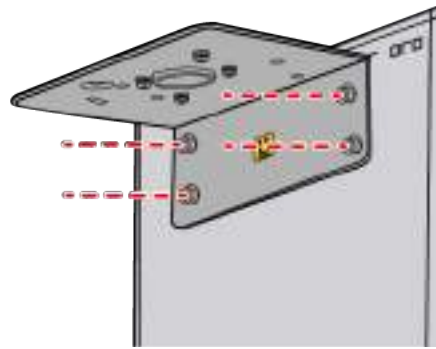
2. Connect one end of the coaxial cable to the Protect interface of the surge protector and fasten the joint with a wrench.



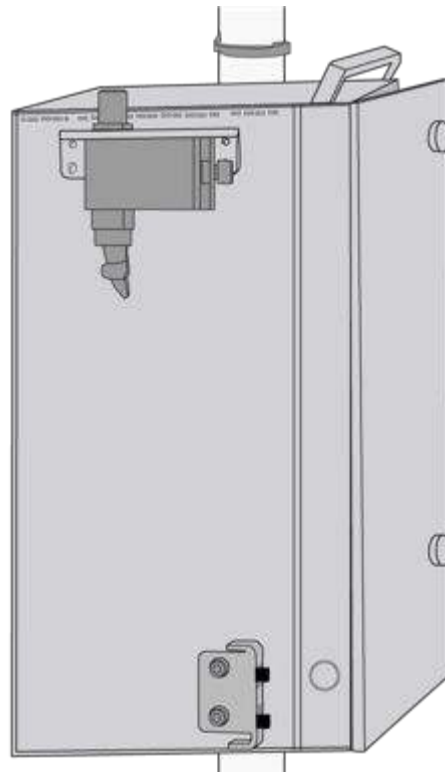
3. Implement 1+3+3 waterproof protection where the coaxial cable and the Protect interface of the surge protector are connected, and fasten cable ties at both ends.



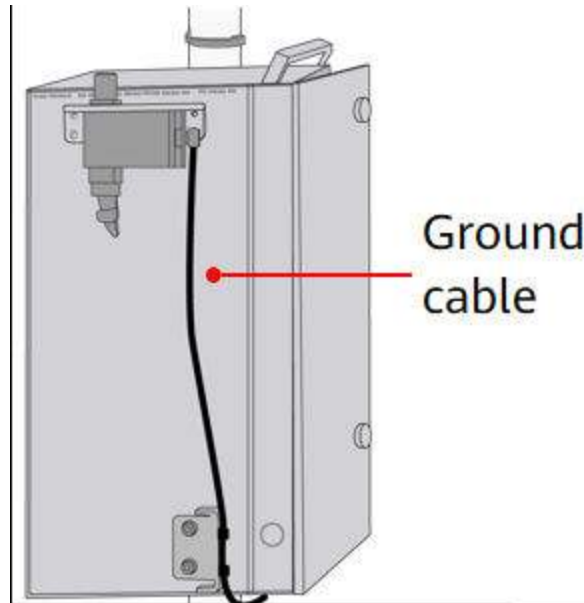
4. Install the GPS support on the left or right side of the cabinet and tighten the four screws.



5. Secure the surge protector on the GPS support.



6. Install the ground cable of the surge protector. Connect one end of the ground cable to the GND interface of the surge protector and the other end to the ground point inside the cabinet.



 NOTE

The surge protector must use OT M8 terminals.

7. Route the jumper through the cable inlet on the right side of the floor of the cabinet, and connect the GPS feeder to the AE 905M.
8. Insert the AE 905M module into an optical interface of the device with the latch of the AE 905M module locked.

 NOTE

After the AE 905M module is inserted into the device, check whether the LINK indicator of the optical module interface is steady on. If the indicator is steady on, the module is properly inserted. If the indicator is off, the module is not properly inserted. In this case, you need to remove the module and re-insert it.

9. Connect one end of the GPS feeder to the Surge interface of the surge protector and the other end of the GPS feeder to the GPS antenna. Fasten each connector with a wrench.
10. Implement 1+3+3 waterproof protection, and fasten cable ties at both ends.
 - Implement waterproof protection where the feeder connector and the Surge interface of the surge protector are connected.
 - Implement waterproof protection where the feeder connector and the N joint of the GPS antenna are connected.

 NOTE

For details about how to make a GPS feeder connector, see the related GPS quick installation guide.

1.6 Power distribution box

1.6.1 DPD63-8-8 DC PDB

The DPD63-8-8 is installed in the upper part of a cabinet to supply power to devices in the cabinet.

Overview

The DPD63-8-8 provides two independent power areas A and B as a backup for each other. Power areas A and B each have four -48 V/-60 V power inputs and four power outputs.

Table 1-1232 lists DPD63-8-8's typical power input configurations, which vary according to the current of power supply devices in different equipment rooms.

Table 1-1232 Typical power input configurations of the DPD63-8-8

N o.	Input Current	Output Current	Circuit Breaker Specifications	Copper Fitting Configuration
1	8 x 63 A	8 x 63 A	8 x 63 A	None

Figure 1-228 Appearance of the DPD63-8-8

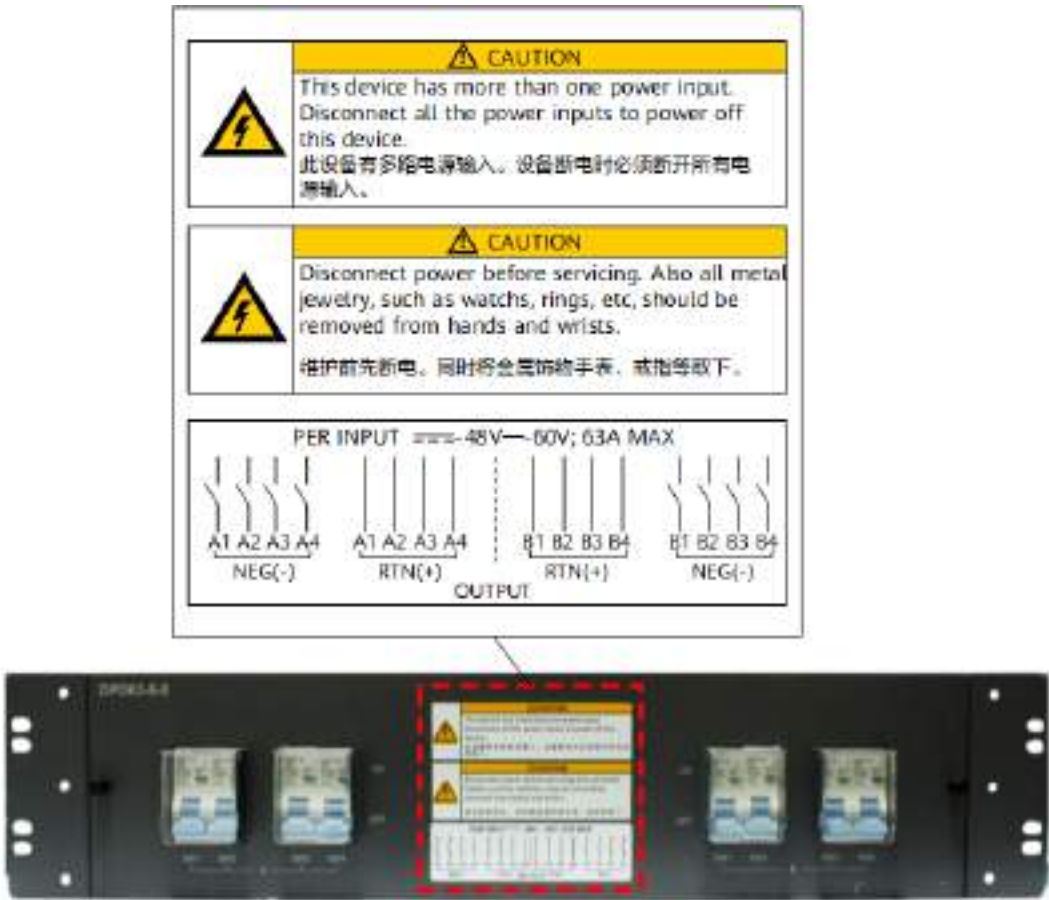
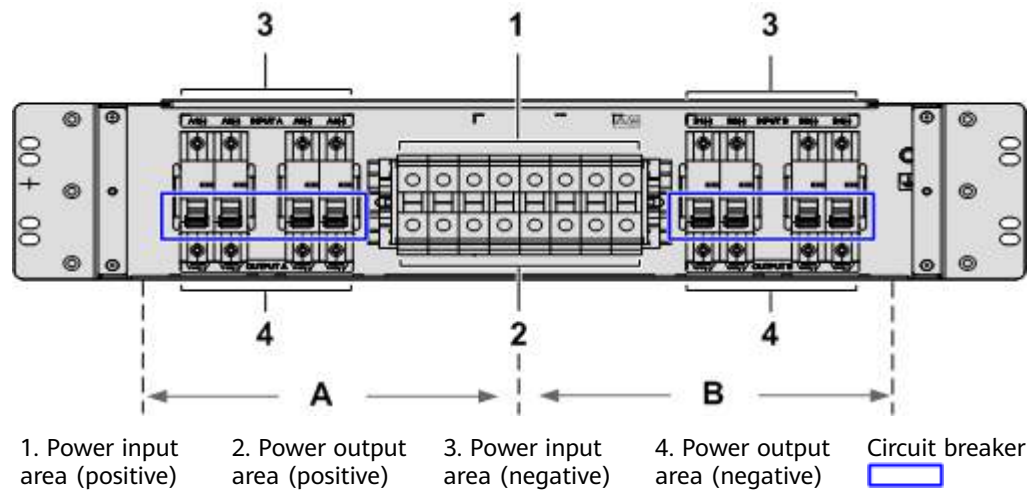


Figure 1-229 Components of the DPD63-8-8



The DPD63-8-8 consists of the following parts:

Power input area: connected to external power sources through power cables.

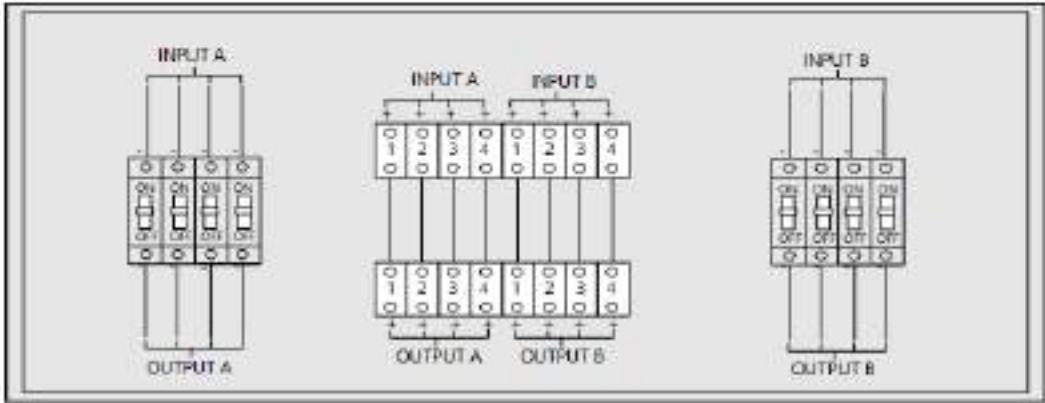
- Power input area (positive): Each power area has four input cable terminals for connecting to the positive pole of external power cables. From left to right, the terminals in the two areas are in turn marked from RTN(+) A1 to RTN(+) A4 and from RTN(+) B1 to RTN(+) B4, respectively.
- Power input area (negative): Each power area has four input cable terminals for connecting to the negative pole of external power cables. From left to right, the terminals are in turn marked from INPUT A A1(-) to A4(-) and from INPUT B B1(-) to B4(-).

Power output area: supplies power to devices through power cables.

- Power output area (positive): Each power area has four output cable terminals for connecting to the positive pole of devices' power cables. From left to right, the terminals in the two areas are in turn marked from RTN(+) A1 to RTN(+) A4 and from RTN(+) B1 to RTN(+) B4, respectively.
- Power output area (negative): Each power area has four output cable terminals for connecting to the negative pole of devices' power cables. From left to right, the terminals in the two areas are in turn marked from OUTPUT A A1(-) to A4(-) and from OUTPUT B B1(-) to B4(-), respectively.

Power areas A and B each have four circuit breakers that correspond to the output cable terminals and control power supply to devices in the cabinet.

Figure 1-230 Internal connections of the DPD63-8-8



Physical Specifications

Table 1-1233 Physical specifications

Item		Description
Dimensions (H x W x D)		110 mm x 482.6 mm x 89.2 mm (4.33 in. x 19.00 in. x 3.51 in.)
Part number		02120724
Installation platform		ETSI 21-inch; IEC 19-inch
Weight		3.26 kg (7.19 lb)
DC input voltage	Rated voltage	-48 V/-60 V
	Maximum voltage range	-40 V to -72 V
Power input	Maximum current	4 x 63 A + 4 x 63 A
	Terminal specifications	Cord end terminal
	Maximum cross-sectional area	35 mm ²
Power output	Maximum current	4 x 63 A + 4 x 63 A
	Terminal specifications	Cord end terminal
	Maximum cross-sectional area	16 mm ²

Item		Description
Circuit breaker	Replaceable	63 A

1.7 Cables

1.7.1 Power Cable

This section describes the structure and technical specifications of the power cable.

1.7.1.1 NetEngine 8000 M4 DC Power Cable

This section describes the structure and technical specifications of the DC-input power cable.

NOTICE

If no special requirements are imposed on power cables, power cables are delivered according to default configurations. Otherwise, power cables need to be purchased locally.

If the Power Distribution Frame (PDF) is more than 35 m (114.83 ft.) away from the device, install the PDF as near as possible.

Overview

A DC power cable is used to connect to the power modules to supply power to the device.

NOTE

Cable colors below are for reference only. The actual cable colors depend on the requirements of the target country or customer.

Appearance

Figure 1-231 DC power cable





Figure 1-232 DC power connector (straight female, 4-10mm²)



Figure 1-233 OT terminal



Figure 1-234 Heat shrink tubing



Technical Specifications

Table 1-1234 Technical specifications of DC power cables

Distance from the PDF to the device	Part Number	Description	Fire Rating
Shorter than or	25030701	Electronic Electric Cable,450V/750V,H07Z-K UL3386,4mm ² Blue, 45A,LSZH Cable,VDE,UL	IEC 60332-1, VM-1, Dca-s2a2d2

Distance from the PDF to the device	Part Number	Description	Fire Rating
equal to 15 m	25030700	Electronic Electric Cable,450V/750V,H07Z-K UL3386,4mm ² ,Black,45A,LSZH Cable,VDE,UL	IEC 60332-1, VM-1, Dca-s2a2d2
	25030735	Electronic Electric Cable,450V/750V,H07Z-K UL3386,4mm ² ,Gray,45A,LSZH Cable,VDE,UL	IEC 60332-1, VM-1, Dca-s2a2d2
	25030730	Electronic Electric Cable,450V/750V,H07Z-K UL3386,4mm ² ,Red,45A,LSZH Cable,VDE,UL	IEC 60332-1, VM-1, Dca-s2a2d2
	25032787-001	Electronic Electric Cable,1100V,Y(FR-LSH),4mm ² ,Blue,45A,For India,BIS	IEC 60332-1,IEC 60332-3-22
	25032787	Electronic Electric Cable,1100V,Y(FR-LSH),4mm ² ,Black,45A,For India,BIS	IEC 60332-1,IEC 60332-3-22
	14191480	Power Connector,2pin,72V,63A,Straight Female,4~10mm ²	-
Longer than 15 m and shorter than or equal to 20 m	25030697	Electronic Electric Cable,450V/750V,H07Z-K UL3386,6mm ² ,Blue,58A,LSZH Cable,VDE,UL	IEC 60332-1, VM-1, Dca-s2a2d2
	25030703	Electronic Electric Cable,450V/750V,H07Z-K UL3386,6mm ² ,Black,58A,LSZH Cable,VDE,UL	IEC 60332-1, VM-1, Dca-s2a2d2
	25030717	Electronic Electric Cable,450V/750V,H07Z-K UL3386,6mm ² ,Gray,58A,LSZH Cable,VDE,UL	IEC 60332-1, VM-1, Dca-s2a2d2
	25030731	Electronic Electric Cable,450V/750V,H07Z-K UL3386,6mm ² ,Red,58A,LSZH Cable,VDE,UL	IEC 60332-1, VM-1, Dca-s2a2d2
	25032788-001	Electronic Electric Cable,1100V,Y(FR-LSH),6mm ² ,Blue,58A,For India,BIS	IEC 60332-1,IEC 60332-3-22
	25032788	Electronic Electric Cable,1100V,Y(FR-LSH),6mm ² ,Black,58A,For India,BIS	IEC 60332-1,IEC 60332-3-22
	14191480	Power Connector,2pin,72V,63A,Straight Female,4~10mm ²	-

Distance from the PDF to the device	Part Number	Description	Fire Rating
Longer than 20 m and shorter than or equal to 35 m	25030710	Electronic Electric Cable,450V/750V,H07Z-K UL3386,10mm ² ,Blue,80A,LSZH Cable,VDE,UL	IEC 60332-1, VM-1, Dca-s2a2d2
	25030711	Electronic Electric Cable,450V/750V,H07Z-K UL3386,10mm ² ,Black,80A,LSZH Cable,VDE,UL	IEC 60332-1, VM-1, Dca-s2a2d2
	25030736	Electronic Electric Cable,450V/750V,H07Z-K UL3386,10mm ² ,Gray,80A,LSZH Cable,VDE,UL	IEC 60332-1, VM-1, Dca-s2a2d2
	25030732	Electronic Electric Cable,450V/750V,H07Z-K UL3386,10mm ² ,Red,80A,LSZH Cable,VDE,UL	IEC 60332-1, VM-1, Dca-s2a2d2
	25032789-001	Electronic Electric Cable,1100V,Y(FR-LSH),10mm ² ,Blue,For India,BIS	IEC 60332-1,IEC 60332-3-22
	25032789-002	Electronic Electric Cable,1100V,Y(FR-LSH),10mm ² ,Black,For India,BIS	IEC 60332-1,IEC 60332-3-22
	14191480	Power Connector,2pin,72V,63A,Straight Female,4~10mm ²	-

Table 1-1235 Technical specifications of ground cables

Part Number	Description	Connector X1	Connector X2	Fire Rating
25030699	Electronic Electric Cable,450V/750V,H07Z-K UL3386,6mm ² ,Yellow/Green,58A,LSZH Cable,VDE,UL	OT Terminal	OT Terminal	IEC 60332-1, VM-1, Dca-s2a2d2
25032788-002	Electronic Electric Cable,1100V,Y(FR-LSH),6mm ² ,Yellow/Green,58A,For India,BIS	OT Terminal	OT Terminal	IEC 60332-1,IEC 60332-3-22

Table 1-1236 Technical specifications of power cable terminals

Distance from the PDF to the device	Part Number	Description	Terminal Type
Shorter than or equal to 20 m	0223 0JAB	Subassembly Accessories,Package of Terminal for Single Input 6mm ² DC Power Cable(Exclude Backup)	Naked Crimping Terminal,OT, 6mm ² ,M6,Tin Plating,Insulated Ring Terminal,12~10AWG,yellow Naked Crimping Terminal,OT, 6mm ² ,M8,Tin Plating,Insulated Ring Terminal,12~10AWG,yellow Common Terminal,Conductor Cross Section 6mm ² ,Length 20mm,Insertion Depth 12mm,Black
Longer than 20 m and shorter than or equal to 35 m	0223 0HY Y	Subassembly Accessories,Package of Terminal for Single Input 10mm ² DC Power Cable (Exclude Backup)	Naked Crimping Terminal,OT, 10mm ² ,M8,Tin Plating,Naked Ring Terminal Naked Crimping Terminal,OT, 10mm ² ,M6,Tin Plating,Naked Ring Terminal Common Terminal,Single Cord End Terminal,Conductor Cross Section 10mm ² ,Length 22mm,Insertion Depth 12mm,Ivory

Table 1-1237 Technical specifications of power cable terminals

Part Number	Description	Terminal Type
0223 0JAC	Subassembly Accessories,Package of Terminal for Single Input 6mm^2 Grounding Cable(Include Backup)	Naked Crimping Terminal,OT, 6mm^2,M4,Tin Plating,Insulated Ring Terminal,12~10AWG,yellow Naked Crimping Connector,OT2,6mm^2,M4,Tin Plating Naked Crimping Terminal,OT, 6mm^2,M8,Tin Plating,Insulated Ring Terminal,12~10AWG,yellow

1.7.1.2 NetEngine 8000 M8 DC Power Cable

This section describes the structure and technical specifications of the DC-input power cable.

NOTICE

If no special requirements are imposed on power cables, power cables are delivered according to default configurations. Otherwise, power cables need to be purchased locally.

If the Power Distribution Frame (PDF) is more than 20 m (65.62 ft.) away from the device, install the PDF as near as possible.

Overview

A DC power cable is used to connect to the power modules to supply power to the device.

 NOTE

Cable colors below are for reference only. The actual cable colors depend on the requirements of the target country or customer.

Appearance

Figure 1-235 DC power cable





Figure 1-236 DC power connector (straight female, 4-10mm^2)



Figure 1-237 OT terminal



Figure 1-238 Heat shrink tubing



Technical Specifications

Table 1-1238 Technical specifications of DC power cables

Part Number	Description	Fire Rating
25030710	Electronic Electric Cable,450V/750V,H07Z-K UL3386,10mm^2,Blue,80A,LSZH Cable,VDE,UL	IEC 60332-1, VM-1, Dca-s2a2d2
25030711	Electronic Electric Cable,450V/750V,H07Z-K UL3386,10mm^2,Black,80A,LSZH Cable,VDE,UL	IEC 60332-1, VM-1, Dca-s2a2d2

Part Number	Description	Fire Rating
25030736	Electronic Electric Cable,450V/750V,H07Z-K UL3386,10mm^2,Gray,80A,LSZH Cable,VDE,UL	IEC 60332-1, VM-1, Dca-s2a2d2
25030732	Electronic Electric Cable,450V/750V,H07Z-K UL3386,10mm^2,Red,80A,LSZH Cable,VDE,UL	IEC 60332-1, VM-1, Dca-s2a2d2
25032789-001	Electronic Electric Cable,1100V,Y(FR-LSH), 10mm^2,Blue,For India,BIS	IEC 60332-1,IEC 60332-3-22
25032789-002	Electronic Electric Cable,1100V,Y(FR-LSH), 10mm^2,Blcack,For India,BIS	IEC 60332-1,IEC 60332-3-22
25030698	Electronic Electric Cable,450V/750V,H07Z-K UL3386,10mm^2,Yellow/Green,80A,LSZH Cable,VDE,UL	IEC 60332-1, VM-1, Dca-s2a2d2
25032789	Electronic Electric Cable,1100V,Y(FR-LSH), 10mm^2,Yellow/Green,80A,For India,BIS	IEC 60332-1,IEC 60332-3-22
14191480	Power Connector,2pin,72V,63A,Straight Female,4~10mm^2	-

Table 1-1239 Technical specifications of ground cables

Part Number	Description	Connect or X1	Connec tor X2	Fire Rating
25030698	Electronic Electric Cable,450V/750V,H07Z-K UL3386,10mm^2,Yellow/Green, 80A,LSZH Cable,VDE,UL	OT Terminal	OT Termin al	IEC 60332-1, VM-1, Dca-s2a2d2

Table 1-1240 Technical specifications of power cable terminals

Part Number	Description	Terminal Type
0223 3ABC	MPE Site Materials Kit, Double Power Supply Cable 10mm ² End Terminal Set	Naked Crimping Terminal,OT, 10mm ² ,M8,Tin Plating,Naked Ring Terminal Naked Crimping Terminal,OT, 10mm ² ,M6,Tin Plating,Naked Ring Terminal

Table 1-1241 Technical specifications of power cable terminals

Part Number	Description	Terminal Type
0223 0JAA	Subassembly Accessories, Package of Terminal for Single Input 10mm ² Grounding Cable(Include Backup)	Naked Crimping Terminal,OT, 10mm ² ,M8,Tin Plating,Naked Ring Terminal Naked Crimping Connector,OT2,10mm ² ,M6,Tin Plating Naked Crimping Terminal,OT, 10mm ² ,M6,Tin Plating,Naked Ring Terminal

1.7.1.3 NetEngine 8000 M14 DC Power Cable

This section describes the structure and technical specifications of the DC-input power cable.

NOTICE

If no special requirements are imposed on power cables, power cables are delivered according to default configurations. Otherwise, power cables need to be purchased locally.

If the Power Distribution Frame (PDF) is more than 20 m (65.62 ft.) away from the device, install the PDF as near as possible.

Overview

A DC power cable is used to connect to the power modules to supply power to the device.

 NOTE

Cable colors below are for reference only. The actual cable colors depend on the requirements of the target country or customer.

Appearance

Figure 1-239 DC power cable



Figure 1-240 DC power connector (straight female, 6-16mm²)



Figure 1-241 OT terminal



Figure 1-242 Heat-shrink tubing



Technical Specifications

Table 1-1242 Technical specifications of DC power cables

Distance from the PDF to the device	Part Number	Description	Fire Rating
Shorter than or equal to 10 m	25030710	Electronic Electric Cable,450V/750V,H07Z-K UL3386,10mm ² ,Blue,80A,LSZH Cable,VDE,UL	IEC 60332-1, VM-1, Dca-s2a2d2
	25030711	Electronic Electric Cable,450V/750V,H07Z-K UL3386,10mm ² ,Black,80A,LSZH Cable,VDE,UL	IEC 60332-1, VM-1, Dca-s2a2d2
	25030736	Electronic Electric Cable,450V/750V,H07Z-K UL3386,10mm ² ,Gray,80A,LSZH Cable,VDE,UL	IEC 60332-1, VM-1, Dca-s2a2d2
	25030732	Electronic Electric Cable,450V/750V,H07Z-K UL3386,10mm ² ,Red,80A,LSZH Cable,VDE,UL	IEC 60332-1, VM-1, Dca-s2a2d2
	25032789-001	Electronic Electric Cable,1100V,Y(FR-LSH),10mm ² ,Blue,For India,BIS	IEC 60332-1,IEC 60332-3-22
	25032789-002	Electronic Electric Cable,1100V,Y(FR-LSH),10mm ² ,Black,For India,BIS	IEC 60332-1,IEC 60332-3-22
	14191482	Power Connector,2pin,72V,80A,Straight Female,6~16mm ²	-
Longer than 10 m and shorter than or	25030586	Electronic Electric Cable,450V/750V,H07Z-K UL3386,16mm ² ,Blue,107A,LSZH Cable,VDE,UL	IEC 60332-1, VM-1, Dca-s2a2d2
	25030722	Electronic Electric Cable,450V/750V,H07Z-K UL3386,16mm ² ,Red,107A,LSZH Cable,VDE,UL	IEC 60332-1, VM-1, Dca-s2a2d2

Distance from the PDF to the device	Part Number	Description	Fire Rating
equal to 20 m	25030639	Electronic Electric Cable,450V/750V,H07Z-K UL3386,16mm ² ,Black,107A,LSZH Cable,VDE,UL	IEC 60332-1, VM-1, Dca-s2a2d2
	25030737	Electronic Electric Cable,450V/750V,H07Z-K UL3386,16mm ² ,Gray,107A,LSZH Cable,VDE,UL	IEC 60332-1, VM-1, Dca-s2a2d2
	25032790	Electronic Electric Cable,1100V,Y(FR-LSH),16mm ² ,Black,107A,For India,BIS	IEC 60332-1,IEC 60332-3-22
	25032790-001	Electronic Electric Cable,1100V,Y(FR-LSH),16mm ² ,Blue,107A,For India,BIS	IEC 60332-1,IEC 60332-3-22
	14191482	Power Connector,2pin,72V,80A,Straight Female,6~16mm ²	-

Table 1-1243 Technical specifications of ground cables

Part Number	Description	Connector X1	Connector X2	Fire Rating
25030698	Electronic Electric Cable,450V/750V,H07Z-K UL3386,10mm ² ,Yellow/Green,80A,LSZH Cable,VDE,UL	OT Terminal	OT Terminal	IEC 60332-1, VM-1, Dca-s2a2d2
25032789	Electronic Electric Cable,1100V,Y(FR-LSH),10mm ² ,Yellow/Green,80A,For India,BIS	OT Terminal	OT Terminal	IEC 60332-1,IEC 60332-3-22

Table 1-1244 Technical specifications of power cable terminals

Distance from the PDF to the device	Part Number	Description	Terminal Type
Shorter than or equal to 10 m	0223 2YPF	MPE Site Materials Kit, Power Supply Cable 10mm ² End Terminal Set	Common Terminal,Single Cord End Terminal,Conductor Cross Section 10mm ² ,Length 22mm,Insertion Depth 12mm,Ivory Naked Crimping Terminal,OT, 10mm ² ,M6,Tin Plating,Naked Ring Terminal Naked Crimping Terminal,OT, 10mm ² ,M8,Tin Plating,Naked Ring Terminal
Longer than 10 m and shorter than or equal to 20 m	0223 2YP G	MPE Site Materials Kit, Power Supply Cable 16mm ² End Terminal	Common Terminal,Single Cord End Terminal,Conductor Cross Section 16mm ² ,Length 24mm,Insertion Depth 12mm,Green Naked Crimping Terminal,OT, 16mm ² ,M8,Tin Plating,Naked Ring Terminal Naked Crimping Terminal,OT, 16mm ² ,M6,Tin Plating,Naked Ring Terminal

Table 1-1245 Technical specifications of power cable terminals

Part Number	Description	Terminal Type
0223 0JAA	Subassembly Accessories,Package of Terminal for Single Input 10mm^2 Grounding Cable(Include Backup)	Naked Crimping Terminal,OT, 10mm^2,M8,Tin Plating,Naked Ring Terminal Naked Crimping Connector,OT2,10mm^2,M6,Tin Plating Naked Crimping Terminal,OT, 10mm^2,M6,Tin Plating,Naked Ring Terminal

1.7.1.4 AC Power Cable

This section describes the structure and technical specifications of the AC-input power cable.

NOTICE

If no special requirements are imposed on power cables, power cables are delivered according to default configurations. Otherwise, power cables need to be purchased locally.

Overview

An AC power cable is used to connect to the AC power module of a device to supply power to the device.

 NOTE

Cables must be in compliance with standards of the destination country or region. The actual cable type depends on the requirements of the target country or customer.

Appearance

Figure 1-243 Connector C13 (PDU)



Figure 1-244 Connector C13 (wall-mounted)



Figure 1-245 Heat shrink tubing



Technical Specifications

Table 1-1246 Technical specifications of AC power cables in different countries or regions (PDU)

Part Number	Model	Description	Connector X1	Connector X2	Length	Number of Cores	Fire Rating
04050188	-	Power Cords Cable,China AC Power 250V10A, 1.5m,C14SM, 227IEC53(RVV)1.0mm ² (3C),C13SF,PDU Cable	C14SM	C13SF	1.5m	3 cores	-
0405G019	-	Power Cords Cable,Europe AC 250V10A, 1.8m,C14SM,H05VV-F-3*1.00mm ² ,C13SF,PDU Cable	C14SM	C13SF	1.8m	3 cores	-

Part Number	Model	Description	Connector X1	Connector X2	Length	Number of Cores	Fire Rating
0405G019-002	-	Power cord,Europe AC 250V10A, 3.0m,C14SM,H05VV-F-3*1.00 ² ,C13SF,250V, 10A,PDU Cable	C14SM	C13SF	3 m	3 cores	-
0405G029	-	Power Cords Cable,North America AC Power 250V10A,1.8m,C14SM,SJT 18AWG(3C),C13SF,PDU Cable	C14SM	C13SF	1.8 m	3 cores	-
0405G02D	-	Power Cords Cable,Japan AC Power 250V12A, 1.8m,C14SM,HVCTF 1.25mm ² (3C),C13SF,PDU Cable	C14SM	C13SF	1.8 m	3 cores	-
0405G02F	-	Power Cords Cable,Australia AC Power 250V10A, 1.8m,C14SM,H05VV-F-1.0mm ² (3C),C13SF,PDU Cable	C14SM	C13SF	1.8 m	3 cores	-
0405G02H	-	Power Cords Cable,Korea AC Power 250V10A, 1.8m,C14SM,H05VV-F-1.0mm ² (3C),C13SF,PDU Cable	C14SM	C13SF	1.8 m	3 cores	-

Table 1-1247 Technical specifications of AC power cables in different countries or regions (wall-mounted)

Part Number	Model	Description	Connector X1	Connector X2	Length	Number of Cores	Fire Rating
04041104	-	Power Cords Cable,China AC Power 250V10A, 3.0m,PISM, 227IEC53-1.0mm ² (3C),C13SF,Black	PISM	C13SF	3m	3 cores	-

Part Number	Model	Description	Connector X1	Connector X2	Length	Number of Cores	Fire Rating
04020728	-	Power Cable,America AC Power Cable,125V10A, 3.0m,PBSM, 18SJT(3C),C13SF,Black	PBSM	C13SF	3m	3 cores	-
04041056	-	Power cord,Europe AC Power Cable,250V10A, 3.0m,PFSM,(H05VVF 1.0mm ² (3C)),C13SF,250V, 10A,Black	PFSM	C13SF	3m	3 cores	-
04040890	-	Power Cable,Britain AC Power Cable 250V10A, 3.0m,PGAM ,H05VV-F-1.0mm ² (3C),C13SF,Black	PGAM	C13SF	3m	3 cores	-
04040887	-	Power Cable,Japan AC Power Cable 125V12A, 3.0m,PBSM,HVCTF-1.25mm ² (3C),C13SF,Black	PBSM	C13SF	3m	3 cores	-
04040889	-	Power cord,BS546 250V10A,3.0m,PM-IAM,H05VV-F-1.5mm ² (3C),C13SF,250V, 10A,Black	PM-IAM	C13SF	3m	3 cores	-
04040888	-	Power Cords Cable,Australia AC Power Cable,250V 10A, 3.0m,PISM,H05VV-F-1.0mm ² (3C),C13SF,Black	PISM	C13SF	3m	3 cores	-
04041119	-	Power Cable,Switzerland AC Power Cable 250V10A, 3.0m,PJSM ,H05VV-F-1.0mm ² (3C),C13SF,Black	PJSM	C13SF	3m	3 cores	-
04041120	-	Power Cable,Italy AC Power Cable 250V10A, 3.0m,PLSM,H05VV-F-1.0mm ² (3C),C13SF,Black	PLSM	C13SF	3m	3 cores	-
04047785	-	Power Cords Cable,Argentina AC Power 250V10A, 3.0m,PISM,H05VV-F-1.0mm ² (3C),C13SF,Black	PISM	C13SF	3m	3 cores	-

Part Number	Model	Description	Connector X1	Connector X2	Length	Number of Cores	Fire Rating
04150258	-	Power Cable,Brazil AC Power Cable 250V10A, 3.0m,PNSM ,H05VV-F-1.0mm ² (3C),C13SF,Black	PNSM	C13SF	3m	3 cores	-
0405G028	-	Power Cords Cable,Korea AC Power 250V10A, 3m,PFSM,H05VV-F 3*1.0mm ² (3C),C13SF,Black	PFSM	C13SF	3m	3 cores	-
0405G02K	-	Power Cords Cable,Denmark AC Power 250V10A, 3m,PKSM,H05VV-F-3*1.0mm ² (3C),C13SF,Black	PKSM	C13SF	3m	3 cores	-
04051035	-	Power Cords Cable,India AC Power 250V10A, 3.0m,PM-IIAM,IS 694-1.0mm ² (3C), C13 SF, 250V,10A,Black	PM-IIAM	C13SF	3m	3 cores	-
04051080	-	Power cord,South Africa AC Power 250V10A, 3m,PMAM,H05VV-F-1.0mm ² (3C),C13SF,250V, 10A,Black	PMAM	C13SF	3m	3 cores	-
04052137	-	Power cord,Taiwan, China AC 125V11A, 3.0m,PBSM,HVCTF 3*1.25mm ² ,C13SF,125V, 11A,Black,BSMI	PBSM	C13SF	3m	3 cores	-

1.7.1.5 2400 W AC Power Cable

This section describes the structure and technical specifications of the AC-input power cable.

NOTICE

The 2400 W AC device uses Huawei-customized HVDC connectors. The cables must be delivered by Huawei and cannot be purchased from a third party.

Overview

An AC power cable is used to connect to the AC power module of a device to supply power to the device.

NOTE

Cables must be in compliance with standards of the destination country or region. The actual cable type depends on the requirements of the target country or customer.

Appearance

Figure 1-246 HVDC-C20 PDU



Figure 1-247 Schematic diagram of PDU HVDC-C20 power cables

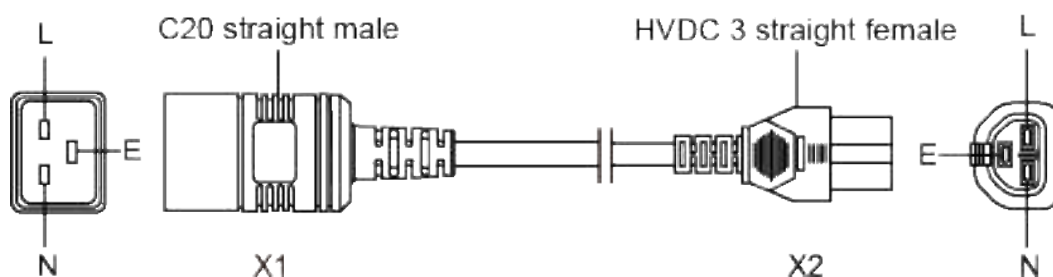
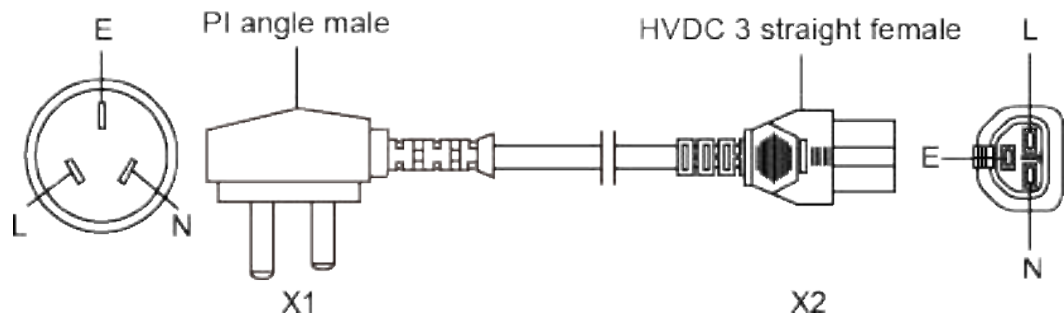


Figure 1-248 HVDC-Wall-mounted



Figure 1-249 Schematic diagram of wall-mounted HVDC power cables



Technical Specifications

Table 1-1248 Technical specifications of AC power cables in different countries or regions (PDU)

Part Number	Description	Connector X1	Connector X2	Length	Number of Cores
0415 2987	Power Cable,3m, 3*14AWG,Black,C20SM, 14SJT3(3C),HVDC 3SF,American PDU AC Power 250V16A	C20 straight male	HVDC 3 straight female	3 m	3 cores
0415 3071	Power Cable,3m, 3*2.0mm ² ,Black,C20SM,HVCTF3* 2.0mm ² (3C),HVDC 3SF,Taiwan, China AC PDU Power 250V16A	C20 straight male	HVDC 3 straight female	3 m	3 cores
0415 2991	Power Cable,2.5m, 3*2.0mm ² ,Black,C20SM,HVCTF-2 .0mm ² (3C),HVDC 3SF,Japan AC Power 250V15A	C20 straight male	HVDC 3 straight female	2.5 m	3 cores
0415 3072	Power Cable,3m, 3*1.5mm ² ,Black,C20SM,H05V2V 2-F-1.5mm ² (3C),HVDC 3SF,Europe AC PDU Power 250V16A	C20 straight male	HVDC 3 straight female	3 m	3 cores
0415 3073	Power Cable,3m, 3*1.5mm ² ,Black,C20SM,H05V2V 2-F-1.5mm ² (3C),HVDC 3SF,Australia AC PDU Power 250V16A	C20 straight male	HVDC 3 straight female	3 m	3 cores
0415 3075	Power Cable,3m, 3*1.5mm ² ,Black,C20SM,IS 694-3*1.5mm ² (3C),HVDC 3SF,India AC PDU Power 250V16A	C20 straight male	HVDC 3 straight female	3 m	3 cores

Part Number	Description	Connector X1	Connector X2	Length	Number of Cores
0415 3074	Power Cable,3m, 3*1.5mm ² ,Black,C20SM,H05V2V 2-F-1.5mm ² (3C),HVDC 3SF,Kroea AC PDU Power 250V16A	C20 straight male	HVDC 3 straight female	3 m	3 cores
0415 2988	Power Cable,3m, 3*1.5mm ² ,Black,C20SM, 227IEC57-1.5 ² (3C),HVDC 3SF,China AC PDU Power 250V16A	C20 straight male	HVDC 3 straight female	3 m	3 cores

Table 1-1249 Technical specifications of AC power cables in different countries or regions (wall-mounted)

Part Number	Description	Connector X1	Connector X2	Length	Number of Cores
0415 2719	Power Cable,2.5m,HVDC 3SF, 12SJT(3C),NEMA5-20PAM,North America AC Power 125V20A	NEMA5-20P angle male	HVDC 3 straight female	2.5 m	3 cores
0415 3043	Power Cable,3m, 3*2.0mm ² ,Black,PBSM,HVCTF3* 2.0mm ² (3C),HVDC 3SF,Taiwan, China AC Power 125V15A	PB straight male	HVDC 3 straight female	3 m	3 cores
0415 2722	Power Cable,2.5m,HVDC 3SF,HVCTF-3.5mm ² (3C),NEMA5-20PAM,Japan AC Power Cable 125V20A	NEMA5-20P angle male	HVDC 3 straight female	2.5 m	3 cores
0415 3063	Power Cable,3m, 3*1.5mm ² ,Black,PGAM,H05V2V 2-F-1.5mm ² (3C),HVDC 3SF,Britain AC Power 250V13A	PG angle male	HVDC 3 straight female	3 m	3 cores
0415 2980	Power Cable,3m, 3*1.5mm ² ,Black,PFAM,H05V2V 2-F-1.5mm ² (3C),HVDC 3SF,Europe AC Power 250V16A	PF angle male	HVDC 3 straight female	3 m	3 cores

Part Number	Description	Connector X1	Connector X2	Length	Number of Cores
0415 2981	Power Cable,3m, 3*1.5mm ² ,Black,PISM,HO5V2V2-F-1.5mm ² (3C),HVDC 3SF,Australia AC Power 250V16A	PI straight male	HVDC 3 straight female	3 m	3 cores
0415 2983	Power Cable,3m, 3*1.5mm ² ,Black,PJ-IISM,HO5V2V2-F-1.5mm ² (3C),HVDC 3SF,Switzerland AC Power 250V16A	PJ-II straight male	HVDC 3 straight female	3 m	3 cores
0415 2982	Power Cable,3m, 3*1.5mm ² ,Black,PM-IIIAM,IS 694-3*1.5mm ² (3C),HVDC 3SF,India AC Power 250V16A	PM-III angle male	HVDC 3 straight female	3 m	3 cores
0415 2984	Power Cable,3m, 3*1.5mm ² ,Black,PMAM,HO5V2V2-F-1.5mm ² (3C),HVDC 3SF,South Africa AC Power 250V16A	PM angle male	HVDC 3 straight female	3 m	3 cores
0415 3062	Power Cable,3m, 3*1.5mm ² ,Black,PFAM,HO5V2V2-F-1.5mm ² (3C),HVDC 3SF,Korea AC Power 250V16A	PF angle male	HVDC 3 straight female	3 m	3 cores
0415 2985	Power Cable,3m, 3*1.5mm ² ,Black,PN-ISM, 227IEC57-1.5 ² (3C),HVDC 3SF,Brazil AC Power 250V16A	PN-I straight male	HVDC 3 straight female	3 m	3 cores
0415 2986	Power Cable,3m, 3*1.5mm ² ,Black,PIAM, 227IEC57-1.5 ² (3C),HVDC 3SF,China AC Power 250V16A	PI angle male	HVDC 3 straight female	3 m	3 cores

1.7.2 Chassis Ground Cable

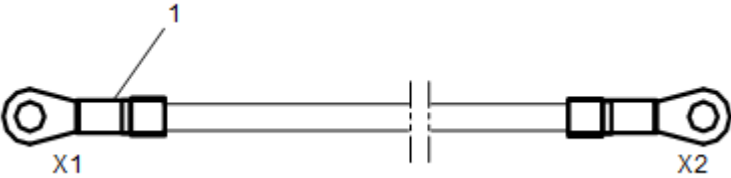
Overview

One end of a chassis ground cable is connected to the ground screw on the right-side cabinet column, and the other end is connected to the ground screw on the chassis.

Appearance

Figure1 shows the appearance of chassis ground cable.

Figure 1-250 Appearance of chassis ground cable.



1. OT one-hole naked crimping connector

Technical Specifications

Table 1-1250 Technical specifications of chassis ground cable

Part Number	Model	Description	Connector X1	Connector X2	Length	Number of Cores	Fire Rating
25030698	CPGN01001	Electronic Electric Cable, 450V/750V,H07Z-K UL3386,10mm ² ,Yellow/Green,80A,LSZH Cable,VDE,UL	14170016: Naked Crimping Terminal,OT, 10mm ² ,M6,Tin Plating, Naked Ring Terminal	14170056: Naked Crimping Terminal,OT, 10mm ² ,M8,Tin Plating, Naked Ring Terminal	10m	-	IEC 60332-1 &VW-1

1.7.3 ETH/OAM Management Cable

The ETH/OAM management interface, integrating the Ethernet NM interface and console interface, uses the RJ45 connector. Tables below list the functions of connector pins.

Table 1-1251 ETH/OAM interface

Connector Pin	Function
1	Transmit positive of the Ethernet NM interface
2	Transmit negative of the Ethernet NM interface
3	Receive positive of the Ethernet NM interface
4	Console ground end
5	Console receive end
6	Receive negative of the Ethernet NM interface
7	Unspecified
8	Console transmit end

Management Network Interface Cables

An ETH/OAM interface that functions only as an Ethernet NM interface uses Ethernet cables to ensure communication between an device and the NMS. The ETH/OAM interface can connect to straight-through and crossover cables in autosensing mode. For details, see [Ethernet Cable](#).

When the ETH/OAM port is used as both a management Ethernet port and a management console port or is used only as a management console port, the ETH/OAM 1-into-2 transfer cable is used. For details, see [1.7.4 ETH/OAM 1-to-2 Adapter Cable](#).

1.7.4 ETH/OAM 1-to-2 Adapter Cable

Overview

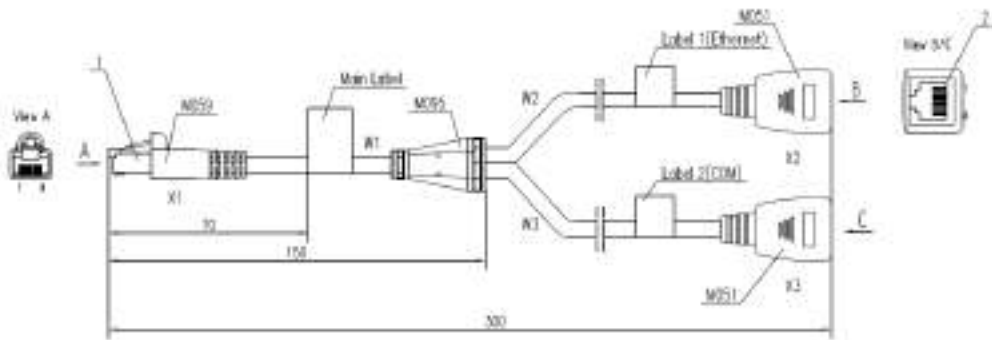
An ETH/OAM 1-to-2 adapter cable features an RJ45 connector at one end and two RJ45 female connectors at the other end. As shown in [Figure 1-251](#), X2 is an MGMT-ETH port, and X3 is a management console port. X2 can be connected to an MGMT-ETH cable, and X3 can be connected to a standard console cable.

Table 1-1252 Configuration principles of the ETH/OAM 1-to-2 adapter cable

MGMT-ETH Port	Management Console Port	Management Network Port + Management Console Port
N	Y	Y

Appearance

Figure 1-251 Structure of an ETH/OAM 1-to-2 adapter cable



Pin Assignments

Table 1-1253 ETH/OAM interface

Connector Pin	Connector X1 Function	Connector X2 Function	Connector X3 Function
1	Transmit positive of the MGMT-ETH port	Transmit positive of the MGMT-ETH port	Not defined
2	Transmit negative of the MGMT-ETH port	Transmit negative of the MGMT-ETH port	Not defined
3	Receive positive of the MGMT-ETH port	Receive positive of the MGMT-ETH port	Transmit end of the console port
6	Receive negative of the MGMT-ETH port	Receive negative of the MGMT-ETH port	Receive end of the console port
4	Ground end of the console port	Not defined	Not defined
5	Receive end of the console port	Not defined	Ground end of the console port

Connector Pin	Connector X1 Function	Connector X2 Function	Connector X3 Function
7	Not defined	Not defined	Not defined
8	Transmit end of the console port	Not defined	Not defined

Table 1-1254 Pin assignments of an ETH/OAM 1-to-2 adapter cable

Wire No.	Start Pin	End Pin	Core Wire Color	Relationship	Wire No.
W1	X1.1	X2.1	White/ Green	Twisted	W2
	X1.2	X2.2	Green		
	X1.3	X2.3	White/Blue	Twisted	
	X1.6	X2.6	Blue		
	X1.4	X3.5	White/ Orange	Twisted	W3
	X1.5	X3.6	Orange		
	X1.7	Not defined	White/ Brown	Twisted	
	X1.8	X3.3	Brown		

Technical Specifications

Table 1-1255 Technical specifications of the ETH/OAM 1-to-2 adapter cable

Part Number	Model	Description	Connector X1	Connector X2	Connector X3	Cable Type	Cable Length	Number of Cores	Fireproof Level
04072045	-	Traditional Signal Cable, 0.3m, MP8-II, CC4 P0.5G Y(S), 2*MP8(S)-III, Serial port and network port integrated cable	RJ45 connector	RJ45 female connector	RJ45 female connector	Twisted-Pair Cable, 100ohm, Category 5e, FTP, 0.52mm, 24AWG, 8Cores, 4Pairs, PANTONE 430U	0.3 m	2 cores/pair	VW-1

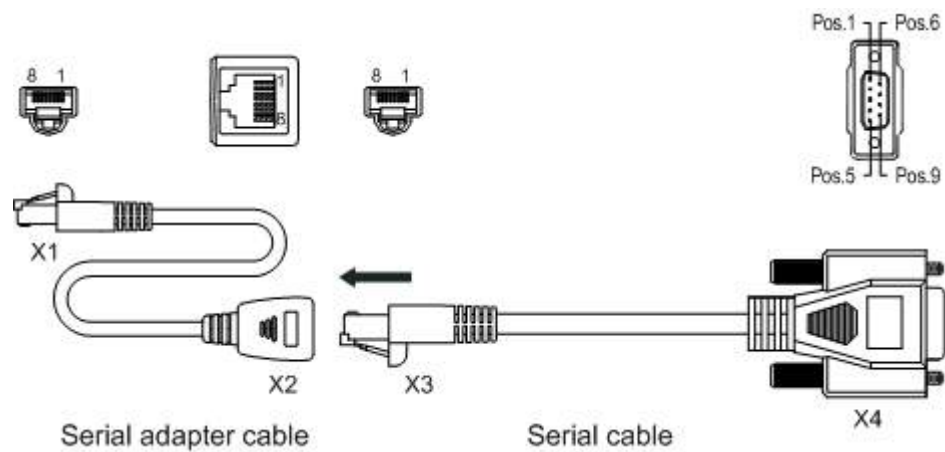
1.7.5 Console Cable

This section describes the structure and technical specifications of the console cable.

Overview

A console cable consists of a standard serial cable and a serial adapter cable, as shown in [Figure 1-252](#). A console cable is used to connect the serial port of a computer to the console port of a device to transmit device configuration data.

Figure 1-252 Console cable



A standard console cable features a DB9 connector (used to connect to the serial port of a computer) at one end and an RJ45 connector (used to connect to the standard console port of a device) at the other end. If a device uses a non-standard console port, a serial adapter cable is required between the RJ45 connector and the device.

Appearance

Figure 1-253 shows a standard serial cable, and Table 1-1256 describes its connections.

Figure 1-253 Standard serial cable (04040838: Traditional Signal,Serial Port Cable, 3m,D9F,CC2P0.32PWG1U,MP8-VI,S3026V)

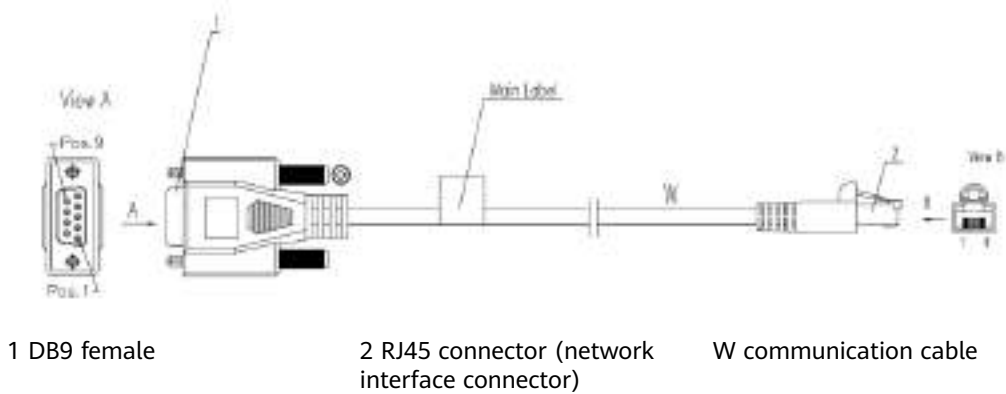


Table 1-1256 Connections of a standard serial cable

Start	End	Color
X1.2	X2.3	Black
X1.3	X2.6	Brown
X1.5	X2.5	Red

A serial adapter cable features a network interface (used to connect to a standard console cable) at one end and an RJ45 connector (used to connect to a router) at the other end.

Figure 1-254 shows a serial adapter cable, and **Table 1-1257** describes its connections.

Figure 1-254 Serial adapter cable (04070730: Traditional Signal Cable,0.3m,MP8-II,CC4P0.5GY(S),MP8(S)-III,NO,ATN Serial adapter cable)

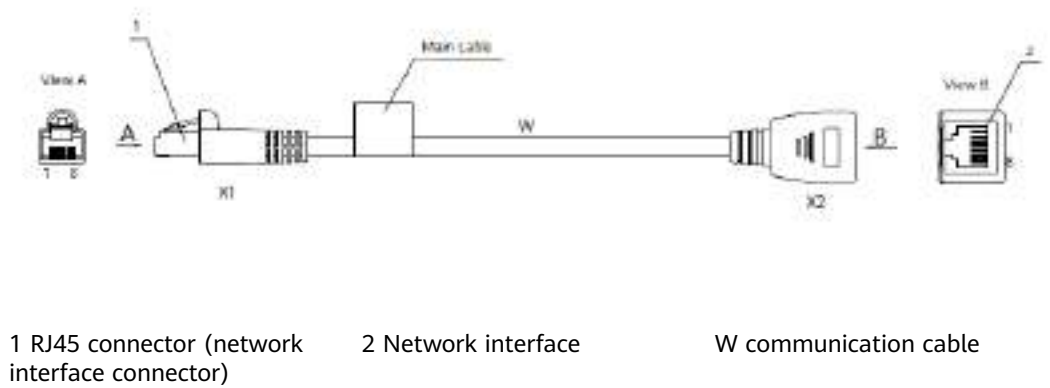


Table 1-1257 Connections of a serial adapter cable

Start	End	Color	Relationship
X1.4	X2.5	White/Green	Twisted
X1.5	X2.6	Green	
X1.8	X2.3	White/Blue	-

Technical Specifications

Table 1-1258 Technical specifications of the standard serial cable and serial adapter cable

Part Number	Description	Connector X1	Connector X2	Cable Type	Cable Length	Number of Cores	Fireproof Level
04040838	Traditional Signal,Serial Port Cable, 3m,D9F,CC2P0.32PWG1U,MP8-VI,S3026V	D9 female	MP8-VI	Twisted-Pair Cable,UL2464,0.32m,28AWG,2 Pairs,PANTONE WARM GRAY 1U	3 m	2 cores/pair	-

Part Number	Description	Connector X1	Connector X2	Cable Type	Cable Length	Number of Cores	Fireproof Level
04070730	Traditional Signal Cable,0.3m,MP8-II,CC4P0.5GY(S),MP8(S)-III,NO,ATN Serial adapter cable	MP8-II	MP8-III	Twisted-Pair Cable, 100ohm,Category 5e,FTP,0.52mm, 24AWG,8Cores, 4Pairs,PANTONE 430U,Use with Plug: 14080097	0.3m	2 cores/ pair	CM

1.7.6 Alarm Input/Output Cable

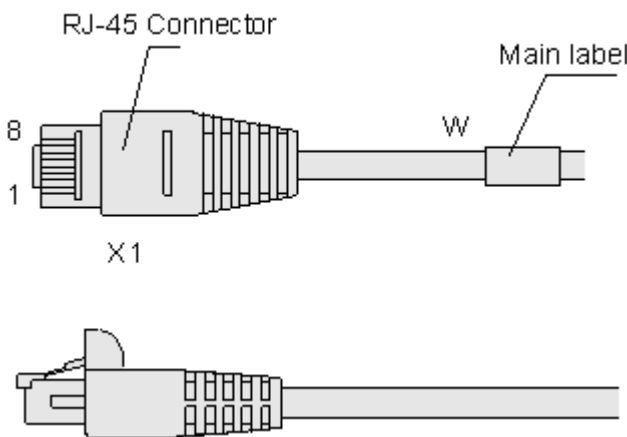
Overview

The alarm input/output cable uses an RJ45 connector to connect to the ALMI/ALMO interface of a device at one end and a connector tailored as required onsite to connect to an external device or central alarm monitoring device at the other end.

Appearance

Figure 1-255 shows the structure of the alarm input/output cable.

Figure 1-255 Structure of the alarm input/output cable



Pin Assignments

Table 1-1259 Pin assignments of the alarm input/output cable (ALMI/ALMO)

Connector X1 Pin	Color	Relationship	Description
1	White/Orange	Twisted	Alarm input 1
2	Orange		Ground end for alarm input 1
3	White/Green	Twisted	Alarm input 2
6	Green		Ground end for alarm input 2
4	Blue	Twisted	Alarm input 3
5	White/Blue		Ground end for alarm input 3
7	White/Brown	Twisted	Positive for alarm output
8	Brown		Negative for alarm output

Technical Specifications

Table 1-1260 Technical specifications of the alarm input/output cable

Part Number	Model	Description	Connector X1	Cable Length	Inner Diameter	Number of Cores	Fireproof Level
25050057	E-Cat. 5e-FTP-4pairs	Twisted-Pair Cable, 100ohm,Category 5e,FTP, 0.52mm,24AWG,8Cores, 4Pairs,PANTONE 430U,Use with Plug:14080097	Network Interface Connector, 8PIN, 8-Bit, Shielded, Crystal Plug,AWG24-26, Solid-Core Cable with Inner Conductor	-	0.52mm	2 cores/pair	CM

1.7.7 Alarm 4 Input Cable (Supported Only by NetEngine 8000 M4s)

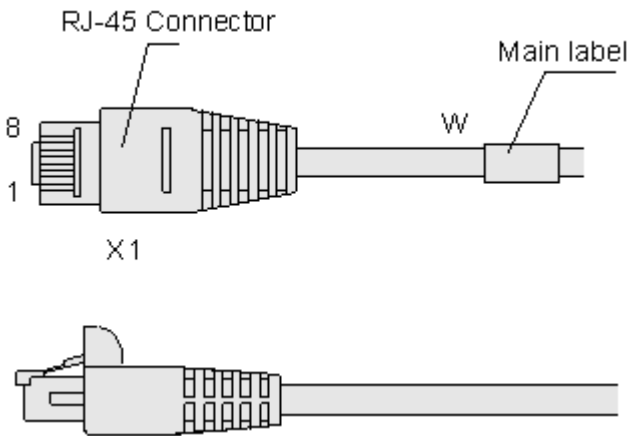
Overview

The alarm input cable uses an RJ45 connector to connect to the ALMI interface of a device at one end and a connector tailored as required onsite to connect to an external device or central alarm monitoring device at the other end.

Structure

Figure 1-256 shows the structure of the alarm input cable.

Figure 1-256 Structure of the alarm input cable



Pin Assignments

Table 1-1261 Pin assignments of the alarm input cable (ALMI/ALMO)

Connector X1 Pin	Color	Relationship	Description
1	White/Orange	Twisted	Alarm input 1
2	Orange		Ground end for alarm input 1
3	White/Green	Twisted	Alarm input 2
6	Green		Ground end for alarm input 2
4	Blue	Twisted	Alarm input 3
5	White/Blue		Ground end for alarm input 3

Connector X1 Pin	Color	Relationship	Description
7	White/Brown	Twisted	Alarm input 4
8	Brown		Ground end for alarm input 4

Technical Specifications

Table 1-1262 Technical specifications of the alarm input/output cable

Part Number	Model	Description	Connector X1	Cable Length	Inner Diameter	Number of Cores	Fireproof Level
25050057	E-Cat. 5e-FTP-4pairs	Twisted-Pair Cable, 100ohm,Category 5e,FTP,0.52mm, 24AWG,8Cores, 4Pairs,PANTONE 430U,Use with Plug: 14080097	Network Interface Connector, 8PIN,8-Bit,Shielded, Crystal Plug,AWG24-26,Solid-Core Cable with Inner Conductor	-	0.52 mm	2 cores/pair	CM

1.7.8 16xE1 Cable

This section describes the structure and technical specifications of the 16xE1 cable.

Overview

The E1 interfaces on a device use 75-ohm or 120-ohm trunk cables. The 16xE1 cable uses an Anea 96 connector to connect to the E1 electrical interface on a board at one end and uses a connector made as required onsite to connect to the digital distribution frame (DDF) at the other end.

Appearance

The 16xE1 cable uses an Anea 96 connector at one end. [Figure 1-257](#) and [Figure 1-258](#) show the appearance of the 16xE1 cable.

Figure 1-257 Appearance of the 75-ohm 16xE1 cable

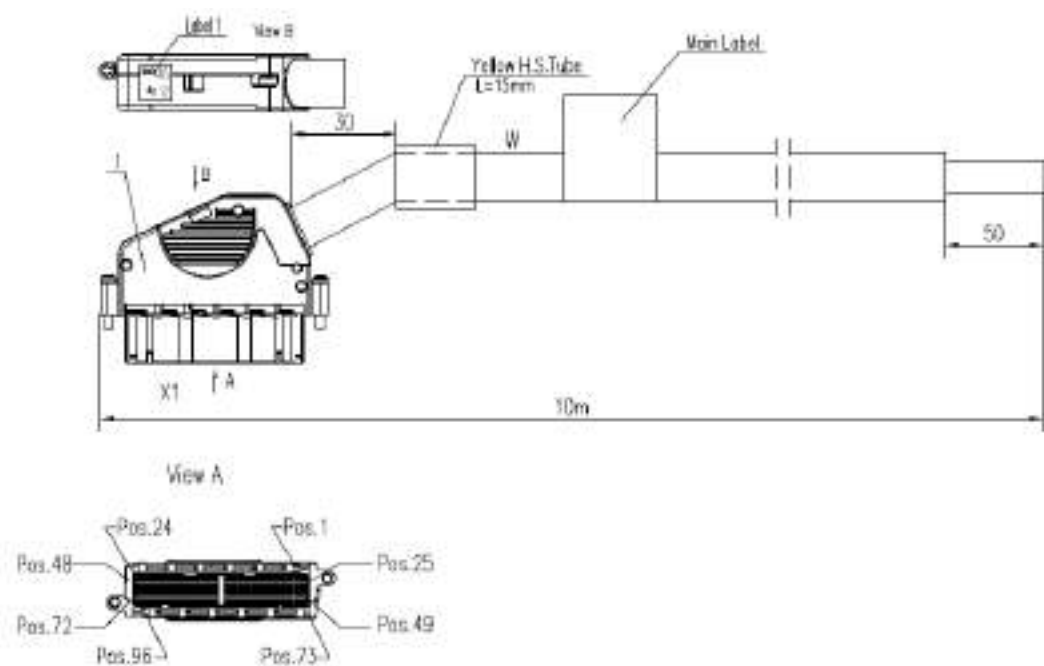


Figure 1-258 Appearance of the 120-ohm 16xE1 cable



Figure 1-259 shows the structure of the 16xE1 cable.

Figure 1-259 Structure of the 16xE1 cable



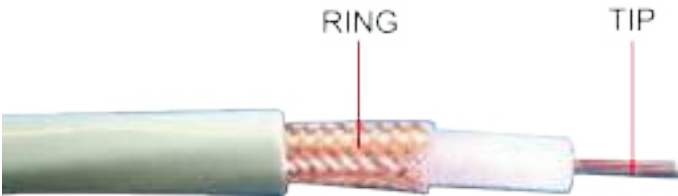
- Pin assignments of the 75-ohm 16xE1 cable
Table 1-1263 describes the pin assignments of the 75-ohm 16xE1 cable. In this table, Tip indicates the positive pole, and Ring indicates the negative pole, and the pin pairs are numbered from 1 to 32.

Table 1-1263 Pin assignments of the 75-ohm 16xE1 cable

Connector Pin	Cable		Remarks	Connector Pin	Cable		Remarks
	Core	No.			Core	No.	
1	Tip	1	R0	25	Tip	2	T0
2	Ring			26	Ring		
3	Tip	3	R1	27	Tip	4	T1
4	Ring			28	Ring		
5	Tip	5	R2	29	Tip	6	T2
6	Ring			30	Ring		
7	Tip	7	R3	31	Tip	8	T3
8	Ring			32	Ring		
9	Tip	9	R4	33	Tip	10	T4
10	Ring			34	Ring		
11	Tip	11	R5	35	Tip	12	T5
12	Ring			36	Ring		
13	Tip	13	R6	37	Tip	14	T6
14	Ring			38	Ring		
15	Tip	15	R7	39	Tip	16	T7
16	Ring			40	Ring		
17	Tip	17	R8	41	Tip	18	T8
18	Ring			42	Ring		
19	Tip	19	R9	43	Tip	20	T9
20	Ring			44	Ring		
21	Tip	21	R10	45	Tip	22	T10
22	Ring			46	Ring		
23	Tip	23	R11	47	Tip	24	T11
24	Ring			48	Ring		
49	Tip	25	R12	73	Tip	26	T12
50	Ring			74	Ring		
51	Tip	27	R13	75	Tip	28	T13
52	Ring			76	Ring		

Conne ctor Pin	Cable		Remar ks	Conne ctor Pin	Cable		Remar ks
	Core	No.			Core	No.	
53	Tip	29	R14	77	Tip	30	T14
54	Ring			78	Ring		
55	Tip	31	R15	79	Tip	32	T15
56	Ring			80	Ring		
Externa l layer	External woven shield						

Figure 1-260 Core



- Pin assignments of the 120-ohm 16xE1 cable
[Table 1-1264](#) describes the pin assignments of the 120-ohm 16xE1 cable.

Table 1-1264 Pin assignments of the 120-ohm 16xE1 cable

Connector Pin	Cable		Remarks	Connector Pin	Cable		Remarks
	Core	Cable Tie			Core	Cable Tie	
1	White	Blue	R0	25	White	Blue	T0
2	Blue			26	Orange		
3	White		R1	27	White		T1
4	Green			28	Brown		
5	White		R2	29	Red		T2
6	Gray			30	Blue		
7	Red		R3	31	Red		T3

Con nec tor Pin	Cable		Re mar ks	Con nec tor Pin	Cable		Remarks
	Cor e	Cab le Tie			Cor e	Cab le Tie	
8	Ora nge		R4	32	Gre en		T4
9	Red			33	Red		
10	Bro wn			34	Gra y		
11	Blac k		R5	35	Blac k		T5
12	Blue			36	Ora nge		
13	Blac k		R6	37	Blac k		T6
14	Gre en			38	Bro wn		
15	Blac k		R7	39	Yell ow		T7
16	Gra y			40	Blue		
17	Whi te	Ora nge	R8	41	Whi te	Ora nge	T8
18	Blue			42	Ora nge		
19	Whi te		R9	43	Whi te		T9
20	Gre en			44	Bro wn		
21	Whi te		R10	45	Red		T10
22	Gra y			46	Blue		
23	Red		R11	47	Red		T11
24	Ora nge			48	Gre en		
49	Red		R12	73	Red		T12

Con nec tor Pin	Cable		Re mar ks	Con nec tor Pin	Cable		Remarks
	Cor e	Cab le Tie			Cor e	Cab le Tie	
50	Bro wn		R13	74	Gra y		T13
51	Blac k			75	Blac k		
52	Blue			76	Ora nge		
53	Blac k		R14	77	Blac k	T14	
54	Gre en			78	Bro wn		
55	Blac k		R15	79	Yell ow	T15	
56	Gra y			80	Blue		
Exte rnal laye r	External woven shield						

Technical Specifications of the 75-ohm 16xE1 Cable

Table 1-1265 Technical specifications of the 75-ohm 16xE1 cable

Part Number	Description	Connector X1	Cable Type	Shield Layer Diameter	Inner Insulation Diameter	Inner Conductor Diameter	Cable Length	Number of Cores	Fireproof Level
0412 0638	Trunk Cable,5m, 75ohm, 16E1,1.6mm,Anea 96F-III,SYFVZP75-1.1/0.26*32(S),-45deg	An ea 96 F-III	Coaxial Cable, SYFVZP-MC 75-1-1x32, 75ohm, 12.4 mm, 1.1mm, 0.26 mm,P antone Warm Gray 1U	12.4 mm	1.6 mm	0.26 mm	5 m	32 cores	IEC 60332-1
0412 0638 -001	Trunk Cable,10m, 75ohm, 16E1,1.6mm,Anea 96F-III,SYFVZP75-1.1/0.26*32(S),-45deg	An ea 96 F-III		12.4 mm	1.6 mm	0.26 mm	10 m	32 cores	IEC 60332-1
0412 0638 -002	Trunk Cable,15m, 75ohm, 16E1,1.6mm,Anea 96F-III,SYFVZP75-1.1/0.26*32(S),-45deg	An ea 96 F-III		12.4 mm	1.6 mm	0.26 mm	15 m	32 cores	IEC 60332-1
0412 0638 -003	Trunk Cable,20m, 75ohm, 16E1,1.6mm,Anea 96F-III,SYFVZP75-1.1/0.26*32(S),-45deg	An ea 96 F-III		12.4 mm	1.6 mm	0.26 mm	20 m	32 cores	IEC 60332-1
0412 0638 -004	Trunk Cable,30m, 75ohm, 16E1,1.6mm,Anea 96F-III,SYFVZP75-1.1/0.26*32(S),-45deg	An ea 96 F-III		12.4 mm	1.6 mm	0.26 mm	30 m	32 cores	IEC 60332-1

Technical Specifications of the 120-ohm 16xE1 Cable

Table 1-1266 Technical specifications of the 120-ohm 16xE1 cable

Part Number	Description	Connector X1	Cable Type	Inner Conductor Diameter	Cable Length	Number of Cores	Fireproof Level
0412 0639	Trunk Cable,5m, 120ohm, 16E1,0.4mm,Anea 96F-II, 120CC32P0.4P430 U(S),-45deg	Anea 96F-II	Twisted-Pair Cable, 120ohm,SE YVP, 0.4mm, 26AWG, 32Pairs,Pan tone 430U	0.4 mm	5 m	2 core s/ pair	IEC 60332 -1
0412 0639-001	Trunk Cable,10m, 120ohm, 16E1,0.4mm,Anea 96F-II, 120CC32P0.4P430 U(S),-45deg	Anea 96F-II		0.4 mm	10 m	2 core s/ pair	IEC 60332 -1
0412 0639-002	Trunk Cable,15m, 120ohm, 16E1,0.4mm,Anea 96F-II, 120CC32P0.4P430 U(S),-45deg	Anea 96F-II		0.4 mm	15 m	2 core s/ pair	IEC 60332 -1
0412 0639-003	Trunk Cable,20m, 120ohm, 16E1,0.4mm,Anea 96F-II, 120CC32P0.4P430 U(S),-45deg	Anea 96F-II		0.4 mm	20 m	2 core s/ pair	IEC 60332 -1
0412 0639-004	Trunk Cable,30m, 120ohm, 16E1,0.4mm,Anea 96F-II, 120CC32P0.4P430 U(S),-45deg	Anea 96F-II		0.4 mm	30 m	2 core s/ pair	IEC 60332 -1

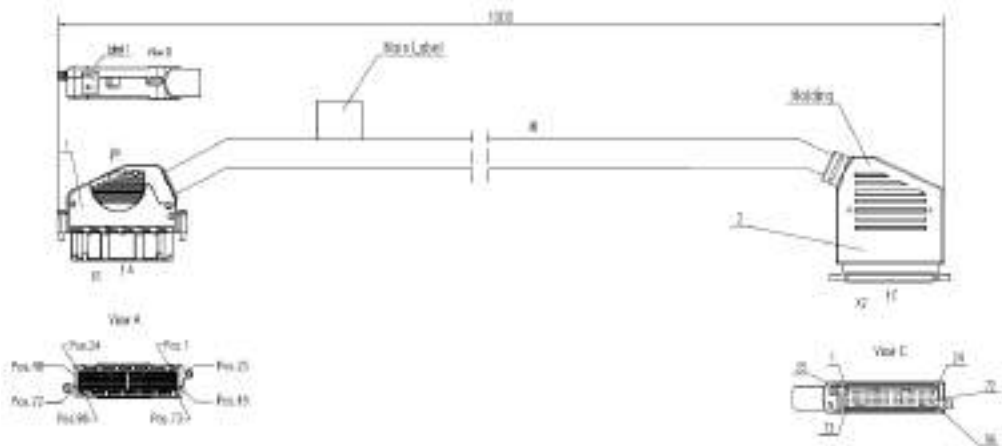
1.7.9 16xE1 Adapter Cable

Overview

The 16xE1 cable required by a chassis is one that uses a negative 45-degree connector. If a 16xE1 cable with a positive 45-degree connector has been used in the equipment room, the cable cannot be directly connected to the chassis. Instead, the cable needs to be connected to the X2 terminal of an adapter cable, and then the X1 terminal of the adapter cable can be connected to the chassis. **Figure 1-261** shows the X1 and X2 terminals of such an adapter cable. The adapter cable can be connected to either a 75-ohm or a 120-ohm 16xE1 cable.

Structure

Figure 1-261 Structure of the 16xE1 adapter cable



Pin Assignments

Table 1-1267 Pin assignments of the 16xE1 adapter cable

Connector X1 Pin	Connector X2 Pin	Cable		Remarks	Connector X1 Pin	Connector X2 Pin	Cable		Remarks
		Core	Cable Tie				Core	Cable Tie	
1	1	White	Blue	R0	25	25	White	Blue	T0
2	2	Blue			26	26	Orange		
3	3	White		R1	27	27	White		T1
4	4	Green			28	28	Brown		
5	5	White		R2	29	29	Red		T2
6	6	Gray			30	30	Blue		
7	7	Red		R3	31	31	Red		T3

Connector X1 Pin	Connector X2 Pin	Cable		Remarks	Connector X1 Pin	Connector X2 Pin	Cable		Remarks
		Core	Cable Tie				Core	Cable Tie	
8	8	Orange			32	32	Green		
9	9	Red		R4	33	33	Red		T4
10	10	Brown			34	34	Gray		
11	11	Black		R5	35	35	Black		T5
12	12	Blue			36	36	Orange		
13	13	Black		R6	37	37	Black		T6
14	14	Green			38	38	Brown		
15	15	Black		R7	39	39	Yellow		T7
16	16	Gray			40	40	Blue		
17	17	White	Orange	R8	41	41	White	Orange	T8
18	18	Blue			42	42	Orange		
19	19	White		R9	43	43	White		T9
20	20	Green			44	44	Brown		
21	21	White		R10	45	45	Red		T10
22	22	Gray			46	46	Blue		
23	23	Red		R11	47	47	Red		T11
24	24	Orange			48	48	Green		
49	49	Red		R12	73	73	Red		T12
50	50	Brown			74	74	Gray		
51	51	Black		R13	75	75	Black		T13
52	52	Blue			76	76	Orange		
53	53	Black		R14	77	77	Black		T14

Conne ctor X1 Pin	Conne ctor X2 Pin	Cable		Remar ks	Connect or X1 Pin	Connect or X2 Pin	Cable		Remarks
		Core	Cable Tie				Core	Cable Tie	
54	54	Gree n			78	78	Brown		
55	55	Black		R15	79	79	Yellow		T15
56	56	Gray			80	80	Blue		
External layer	External woven shield								

Technical Specifications

Table 1-1268 Technical specifications of the 16xE1 adapter cable

Part Number	Model	Description	Connector X1	Connector X2	Cable Type	Inner Conductor Core Diameter	Cable Length	Number of Cores	Fir eproof Level
04120697	-	Trunk Cable, 1.0m,120ohm, 16E1,1.6mm, Anea 96F-II, 120CC32P0.4P 430U(S)),Anea 96M,For Reuse Scenario	Cable Connector ,Anea, 96PIN,IDC Type,Female Connector and Shielding Case(meet ClassB),For 24~26 AWG Solid Wire ,Matching Cable Outer Diameter 12.0mm~14.0mm, 1A	Cable Connector, ANEA, 96PIN,Right Angle Male Socket,PC B PressFit Type,With Shielding(Meet Class B), 1A	Twisted-Pair Cable, 120ohm,S EYVP, 0.4mm, 26A WG, 32Pairs,Pantone 430U	0.4mm	1m	32 pairs	IEC 60332-1

1.7.10 75-ohm 155M Electrical Interface Module Cable

This section describes the structure and technical specifications of the 75-ohm 155M electrical interface optical module cable.

Overview

The 155M electrical interface module uses the 75-ohm trunk cable. This cable is connected to a 155M electrical interface optical module through an SAA connector at one end, and connected to a digital distribution frame (DDF) through a connector made as required onsite at the other end.

Appearance

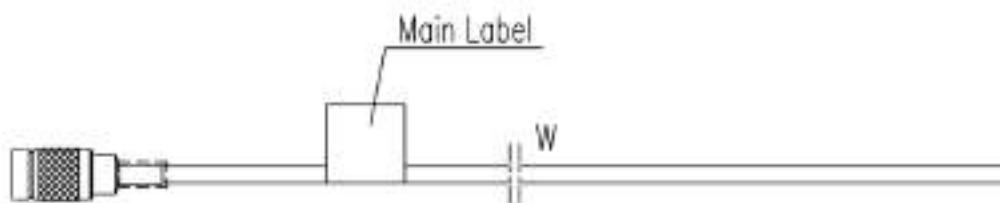
One end of the 75-ohm 155M electrical interface optical module cable uses the SAA connector. [Figure 1-262](#) shows the cable appearance.

Figure 1-262 Appearance of the 75-ohm 155M electrical interface optical module cable



[Figure 1-263](#) shows the structure of the 75-ohm 155M electrical interface optical module cable.

Figure 1-263 Structure of the 75-ohm 155M electrical interface optical module cable



Technical Specifications

Table 1-1269 Technical specifications of the 75-ohm 155M electrical interface optical module cable

Part Number	Model	Description	Connector X1	Connector X2	Cable Type	Shield Layer Diameter	Inner Insulation Diameter	Inner Conductor Diameter	Cable Length	Number of Cores	Fireproof Level
04120309	-	Trunk Cable, 10m,75ohm, 3.9mm,SAA75SM-I,SYV75-2/0.34(S)	SAA75SM-I	-	Coaxial Cable, SYV-75-2-2(4.0Z)-1/0.34mm,OD3.9mm,Double-Layer Copper Braid Shielded	3.9mm	2.1mm	0.34mm	10m	Single	IEC 60332-1

1.7.11 Clock Cable (External Clock Mode, Delivery-Ready)

This section describes the structure and technical specifications of the clock cable.

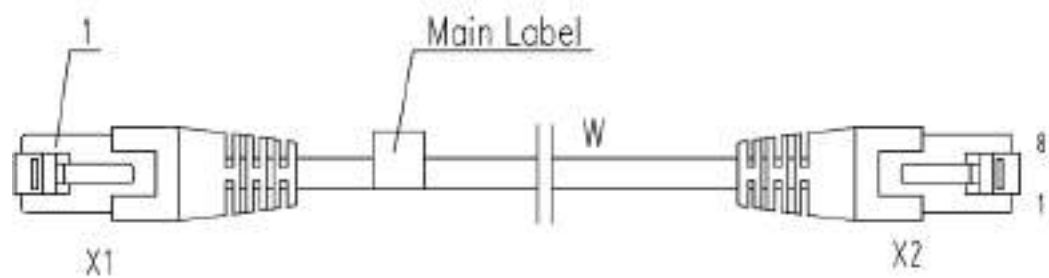
Overview

The external clock cable can be connected to the CLK or CLK/TOD interface on an device.

Structure

The 120-ohm trunk cable adopts the RJ45 connector. The following figure shows the structure of the cable.

Figure 1-264 120-ohm trunk cable structure



Pin Assignments

Table 1-1270 Pin assignments of the 120-ohm trunk cable

Connector X1	Connector X2	Color	Relationship
1	4	White	Twisted
2	5	Blue	
3	6	Orange	-
4	1	White	Twisted
5	2	Green	
6	3	Brown	-

Technical Specifications

Table 1-1271 Technical specifications of the 120-ohm clock cable

Part Number	Description	Connector X1	Connector X2	Cable Type	Cable Length	Number of Cores	Fireproof Level
04040497	Trunk Cable,3m, 120ohm, 1E1,0.4mm,MP8-II, 120CC4P0.4P430U(S) ,MP8-II	MP8-II	MP8-II	Twisted-Pair Cable, 120ohm,SEYVP,0.4mm, 26AWG, 4Pairs,Pantone 430U	3m	2 core s/ pair	IEC60332-1
04040261	Trunk Cable,15m, 120ohm, 1E1,0.4mm,MP8-II, 120CC4P0.4P430U(S) ,MP8-II	MP8-II	MP8-II		15m	2 core s/ pair	IEC60332-1

1.7.12 Clock Cable (External Clock/External Time Mode, Prepared Onsite)

Overview

A clock cable is used to connect the clock interface on the device to the clock interface on another device. In the case of the device be equipped with one IPU, the device can receive one channel 2-Mbit/s clock signals, one channel 2-MHz clock signals, one channel 1 PPS +TOD time signals, or one channel of input and one channel of output DCLS time signals from the upstream device and provides one channel 2-Mbit/s clock signals, one channel 2-MHz clock signals, one channel 1 PPS +TOD time signals or one channel of input and one channel of output DCLS time signals to the downstream device. In the case of the device be equipped with tow IPUs, the device can receive two channels 2-Mbit/s clock signals, two channels 2-MHz clock signals, two channels 1 PPS +TOD time signals, or two channels of input and two channels of output DCLS time signals from the upstream device and provides two channels 2-Mbit/s clock signals, two channels 2-MHz clock signals, two channels of 1 PPS +TOD time signals,, or two channels of input and two channels of output DCLS time signals to the downstream device.

One end of the clock cable is an RJ45 connector connected to the clock interface on the IPU and the other end is connected to the clock interface on an external device.

An RJ45 connector is used together with the 120-ohm trunk cable.

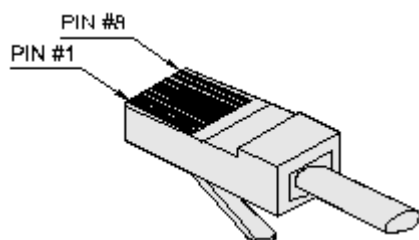
NOTE

The wire sequence of a trunk cable used as a clock cable differs from the wire sequence of an ordinary twisted cable.

Appearance

The 120-ohm clock cable uses the RJ45 connector. [Figure 1-265](#) shows the structure of the 120-ohm trunk cable.

Figure 1-265 Appearance of the 120-ohm clock cable connector



Cable Pinouts

Table 1-1272 Pin assignment of the 120-ohm clock cable

Connector Pin	Relationship	Description		
		External Clock Mode	External Time Mode (1PPS + Time Information)	External Time Mode (DCLS)
X1.1	Twisted pair	Negative receive end of CLK	Unspecified	Unspecified
X1.2		Positive receive end of CLK	Unspecified	Unspecified
X1.3	Twisted pair	Unspecified	Negative of 1PPS signals	Negative of DCLS signals
X1.6		Unspecified	Positive of 1PPS signals	Positive of DCLS signals
X1.4	Twisted pair	Negative transmit end of CLK	Grounding end	Grounding end
X1.5		Positive transmit end of CLK	Grounding end	Grounding end
X1.7	Twisted pair	Unspecified	Negative of time information	Unspecified
X1.8		Unspecified	Positive of time information	Unspecified

Technical Specifications

Table 1-1273 Technical specifications of External Clock Cable

Part Number	Model	Description	Connector X1	Connector X2	Number of Cores	Fire Rating
14080097	-	Network Interface Connector, 8-Bit 8PIN,Shielded,Crystal Model Connector, 24-26AWG,Leads Single Solide Cable,For OEM Matching 25050057	-	-	-	-
25050057	-	Twisted-Pair Cable, 100ohm,Category 5e,FTP, 0.52mm,24AWG,8Cores, 4Pairs,PANTONE 430U,Use with Plug:14080097	-	-	2 Cores/ Pairs	IEC60332-1

1.7.13 Clock Bridging Cable

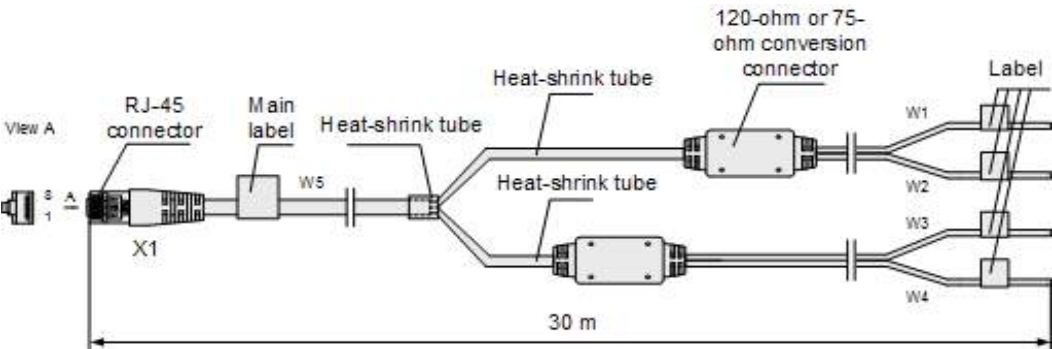
Overview

The clock bridging cable used by a device is the 120-ohm-to-75-ohm clock cable.

Structure

Figure 1-266 shows the structure of the 120-ohm-to-75-ohm clock cable.

Figure 1-266 Clock bridging cable structure



Pin Assignments

Table 1-1274 Pin assignments of clock bridging cables

120-ohm Cable				75-ohm Cable
Connector Pin	Core Wire Color	Cable Description	Relationship	Core Wire No.
X1.1	Orange	Receive negative for the 120-ohm external clock	Twisted	W1
X1.2	White	Receive positive for the 120-ohm external clock		
X1.4	Blue	Transmit negative for the 120-ohm external clock	Twisted	W2
X1.5	White	Transmit positive for the 120-ohm external clock		
X1.3	Green	Not defined	Twisted	W3
X1.6	White	Not defined		
X1.7	White	Not defined	Twisted	W4
X1.8	Brown	Not defined		

Technical Specifications

Table 1-1275 Technical specifications of a clock bridging cable

Part Number	Model	Description	Connector X1	Connector X2	Cable Length	Number of Cores	Fireproof Level
04044726	T-1-Sh-RJ45-30	Signal Cable,120ohm To 75ohm Clock Cable, 30m,MP8-II, (120CC4P0.4P430U(S) +4*(SYV75-2/0.34(S))	MP8-II	-	30m	-	-

1.7.14 Ethernet Cable

Overview

Ethernet cables are also referred to as network cables and can be classified into straight-through cables and crossover cables according to the connection sequence of the copper cores in the cables.

The Ethernet service interfaces on the equipment support Auto MDI-X to the straight-through cables and crossover cables. Hence, you can connect either type of the network cables to the Ethernet service interfaces as required.

 NOTE

Ethernet cables need to be made on site.

Appearance

Figure 1-267 Appearance of the network cable



Cable Pinouts

Table 1-1276 Pin assignment of the straight-through cable connector

Connector X1 Pin	Connector X2 Pin	Color	Relation
X1.1	X2.1	White-orange	Twisted pair
X1.2	X2.2	Orange	
X1.3	X2.3	White-green	Twisted pair
X1.6	X2.6	Green	
X1.4	X2.4	Blue	Twisted pair
X1.5	X2.5	White-blue	
X1.7	X2.7	White-brown	Twisted pair
X1.8	X2.8	Brown	

Table 1-1277 Pin assignment of the crossover cable connector

Connector X1 Pin	Connector X2 Pin	Color	Relation
X1.1	X2.3	White-orange	Twisted pair
X1.2	X2.6	Orange	
X1.3	X2.1	White-green	Twisted pair

Connector X1 Pin	Connector X2 Pin	Color	Relation
X1.6	X2.2	Green	Twisted pair
X1.4	X2.4	Blue	
X1.5	X2.5	White-blue	
X1.7	X2.7	White-brown	Twisted pair
X1.8	X2.8	Brown	

Technical Specifications

Table 1-1278 Technical specifications of the straight-through cable

Descr iption	Twisted-Pair Cable, 100ohm, Category 5e, STP, 0.52mm, 24AWG, 8Cores, 4Pairs, PANTONE 430U
Conn ector X1	Network Interface Connector, 8-Bit 8PIN, Shielded, Crystal Model Connector, 24-26AWG, Leads Single Solide Cable
Conn ector X2	Network Interface Connector, 8-Bit 8PIN, Shielded, Crystal Model Connector, 24-26AWG, Leads Single Solide Cable
Leng th	1m,3m,5m,10m,20m,30m
Inner Diam eter	0.52mm
Num ber of Cores	8 Cores

Table 1-1279 Technical specifications of the crossover cable

Descr iption	Twisted-Pair Cable, 100ohm, Category 5e, STP, 0.52mm, 24AWG, 8Cores, 4Pairs, PANTONE 430U
Conn ector X1	Network Interface Connector, 8-Bit 8PIN, Shielded, Crystal Model Connector, 24-26AWG, Leads Single Solide Cable

Connector X2	Network Interface Connector, 8-Bit 8PIN, Shielded, Crystal Model Connector, 24-26AWG, Leads Single Solide Cable
Length	2m,3m,5m,10m,20m,30m
Inner Diameter	0.52mm
Number of Cores	8 Cores

1.7.15 1-into-2 Clock Cable

Overview

One end of a 1-to-2 cable is an RJ45 connector, and the other end is split into two RJ45 female connectors, as shown in [Figure 1-268](#). The X2 and X3 ports are a clock port and a time port, respectively. The X2 port uses 120-ohm clock cables or 120-to-75-ohm clock cables, while the X3 port uses network cables.

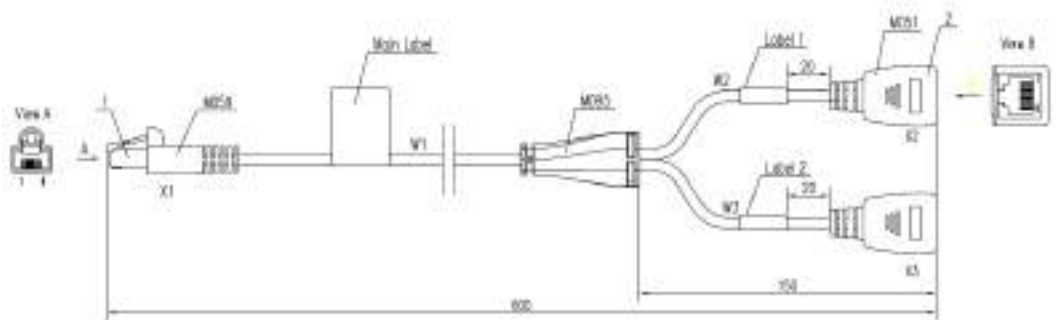
[Table 1-1280](#) shows the usage scenario of such cables.

Table 1-1280 Configuration rules for 1-to-2 cables

Clock (120-ohm) + Time	Clock (75-ohm) + Time	Time	Clock (120-ohm)	Clock (75-ohm)
Y	Y	Y	N	N

Appearance

Figure 1-268 Structure of a 1-into-2 clock cable



Pin Assignment

Table 1-1281 Pin assignment of the 1-into-2 clock cable

Wire	Start Pin	End Pin	Color	Relationship	Wire
W1	X1.1	X2.1	Blue	Pair	W2
	X1.2	X2.2	White/Blue		
	X1.4	X2.4	Orange	Pair	
	X1.5	X2.5	White/Orange		
	X1.3	X3.3	Green	Pair	W3
	X1.6	X3.6	White/Green		
	X1.7	X3.7	Brown	Pair	
	X1.8	X3.8	White/Brown		

Technical Specifications

Table 1-1282 Technical specifications of the 1-into-2 clock cable

Part Number	Model	Description	Connector X1	Connector X2/X3	Cable Type	Length	Number of Cores	Flame Spread Rating
04071532	-	Signal Cable,Clock Cable,0.6m,MP8-II, 3*CC4P0.5P430U(S)-I,2*MP8-III,LSZH	MP8-II	MP8-III	Twisted-Pair Cable, 100ohm,FTP,CAT5E, 0.52mm, 24AWG, 4Pairs,PANTONE 430U,Low Smoke Zero Halogen Cable	0.6m	2 cores/pair	CM

1.7.16 USB-to-Ethernet Cable

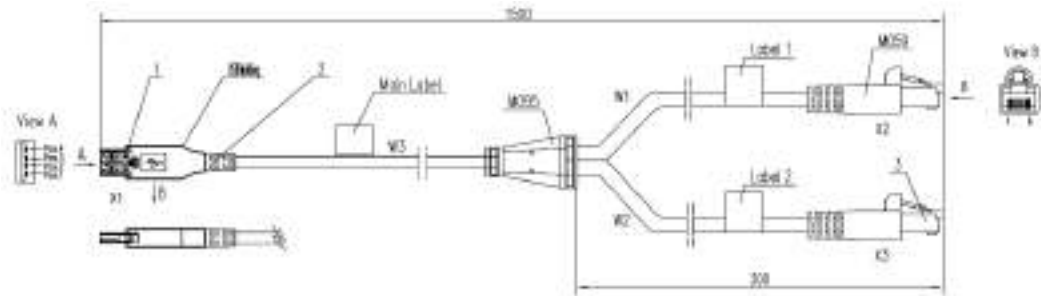
This section describes the structure and technical specifications of the USB-to-Ethernet cable.

Overview

One end of a USB-to-Ethernet cable is a USB port used to connect to a client, and the other end is split into two RJ45 connectors used to connect to routers. The two RJ45 connectors include one standard console port and one non-standard console port. Pin assignment for the two ports is different. For details, see [Table 1](#).

Appearance

Figure 1-269 Structure of the USB-to-Ethernet cable



Pin Assignment

Table 1-1283 Pin assignment of the USB-to-Ethernet cable

Wire	Relation	Color	Description	Start Pin	Converter	End Pin	Description	Color	Relation	Wire
W3-Main Label	Pair	White	+5V DC	X1.1	USB to RS232 conversion	X2.6	RS232_RX	Blue	Pair	W1-Label 1 (RS232)
		Blue	Data -	X1.2		X2.3	RS232_TX	White		
	Pair	Orange	Data +	X1.3		X2.5	GND	Orange	-	W2-Label 2 (Special RS232)
		White	GND	X1.4		X3.5	RS232_RX	Blue	Pair	
-	-	-	-	-		X3.8	RS232_TX	White		
-	-	-	-	-		X3.4	GND	Orange	-	

Technical Specifications

Table 1-1284 Technical specifications of the USB-to-Ethernet cable

Part Number	Model	Description	Connector X1	Connector X2	Length	Number of Cores	Fire Rating
04071851	-	Signal Cable,USB-to-Ethernet cable, 1.5m,USB-A(Male),CC2P0.48B(S), 2*MP8-II	USB-A(Male)	MP8-II	1.5 m	-	-

NOTICE

To use the USB-to-Ethernet serial cable, download the driver from the website and configure the driver as required. Download the driver from <http://www.wch-ic.com/search?q=CH340&t=downloads>.

1.7.17 Fiber Jumper

Overview

A fiber jumper consists of one or more optical fibers of a certain length and the optical connectors at both ends. A fiber jumper connects an optical module to a fiber terminal box.

NOTE

The MPO-MPO optical fibers for routers use type B connectors (Key Up/Key Up).

Comply with the following rules when selecting fiber jumpers:

1. Determine the length of fiber jumpers based on the onsite cabling distance.
2. Determine the fiber type based on the optical module type.
 - Use a multimode fiber jumper for a multimode optical module.
 - Use a single-mode fiber jumper for a single-mode optical module.
3. Determine the optical connector type based on the port type.

Ensure that the optical connector at each end of a fiber jumper is of the same type as the port to which it will be connected.

Optical Fiber Types

Optical fibers are generally classified into single-mode fibers and multimode fibers.

- Single-mode fibers have a diameter of 5 to 10 μm and transmit laser in a single mode with a specified wavelength. These fibers support a wide frequency band and a large transmission capacity, so they are used for long-distance transmission. Most single-mode fibers are yellow, as shown in [Figure 1-271](#).
- Multimode fibers have a diameter of 50 or 62.5 μm and transmit laser in multiple modes with a specified wavelength. They have a small capacity and their performance is inferior to that of single-mode fibers, making them suitable to short-distance transmission.

In the latest cabling infrastructure of ISO/IEC 11801, multimode fibers are classified into four categories: OM1, OM2, OM3, and OM4.

- OM1: traditional 62.5/125 μm multimode fibers. OM1 fibers have a large core diameter and numerical aperture, and provide high light gathering ability and bending resistance.
- OM2: traditional 50/125 μm multimode fibers. OM2 fibers have a small core diameter and numerical aperture. Compared with OM1 fibers, OM2 fibers provide higher bandwidth because they significantly reduce the modal dispersion. When transmitting data at 1 Gbit/s with 850 nm wavelength, OM1 and OM2 fibers support maximum link lengths of 220 m and 550 m, respectively. These two types of fibers can provide sufficient bandwidth within a distance of 300 m. Generally, OM1 and OM2 fibers are orange, as shown in [Figure 1-272](#).
- OM3: next-generation multimode fibers, with longer transmission distances than OM1 and OM2 fibers.
- OM4: laser optimized multimode fibers with 50 μm core diameter. OM4 is an improvement to OM3 and only increases the modal bandwidth. OM4 fibers provide 4700 MHz*km of modal bandwidth, whereas OM3 fibers provide only 2000 MHz*km of modal bandwidth. Generally, OM3 and OM4 fibers are light green, as shown in [Figure 1-273](#). You can identify OM3 and OM4 fibers by their labels or printed marks.

MPO fibers are used for 40G and 100G optical modules. An MPO fiber consists of multiple multimode fiber cores, and each multimode fiber core provides one laser transmission channel. Some fiber suppliers produce 8-strand MPO fibers, while some suppliers produce 12-strand or 24-strand MPO fibers.

- A 40G optical module uses four channels to transmit laser and four channels to receive laser. That is, a total of eight channels are required for a 40G optical module. 8-strand and 12-strand MPO fibers use the same definition of fiber channels. Therefore, they are equivalent in functionality when connecting to 40G optical modules.
- When 100G optical modules are used, choose MPO fibers according to the optical module form factor. Choose 8-strand or 12-strand fibers for QSFP28 optical modules with MPO connectors.

Optical Connector

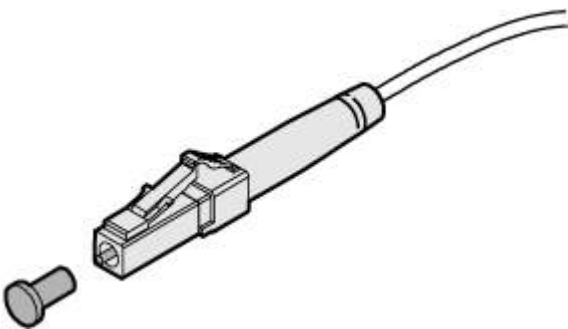
Optical connectors are used to connect optical fibers of the same type. [Table 1-1285](#) lists common optical connectors.

Table 1-1285 Common optical connectors

Connector Type	Optical Connector			
Square connector	SC/PC connector 	LC/PC connector 	MTRJ/PC connector 	MPO connector 
Round connector	FC/PC connector 	ST/PC connector 	-	-

Figure 1-270 shows an LC/PC optical connector.

Figure 1-270 LC/PC connector



NOTICE

- When connecting or removing an LC/PC optical connector, align the connector with the optical port and do not rotate the fiber. Pay attention to the following points:
- When inserting an optical fiber, align the optical connector with the optical port and gently insert the optical fiber into the port.
 - When removing an optical fiber, press the clip on the connector, push the connector inward slightly, and then pull the fiber out.

Appearance

Figure 1-271 shows the appearance of an LC single-mode fiber.

Figure 1-271 Appearance of an LC single-mode fiber

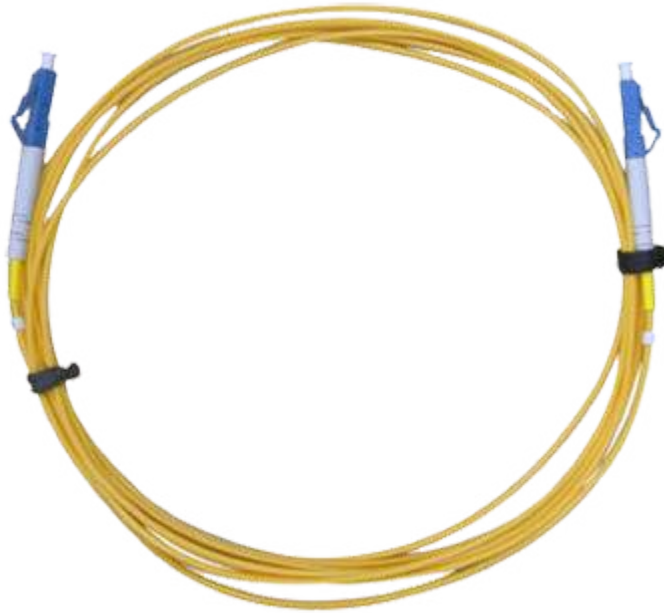


Figure 1-272 shows the appearance of an LC multimode fiber.

Figure 1-272 Appearance of an LC multimode fiber



Figure 1-273 shows the appearance of an MPO-MPO fiber.

Figure 1-273 Appearance of an MPO-MPO fiber



1.8 Fiber Breakout

1.8.1 Breakout Fibers

Positioning

Breakout fibers are used to flexibly allocate bandwidth resources to routers. They cooperate with a breakout optical module to convert an MPO interface into multiple LC interfaces to facilitate fiber layout.

Usage Scenario

The breakout function applies to a scenario where bandwidth resources are unevenly allocated between two routers at different levels. In comparison with breakout boxes, breakout fibers implement the following functions:

- Loosen fiber layout.
- Extend the cabling distance.

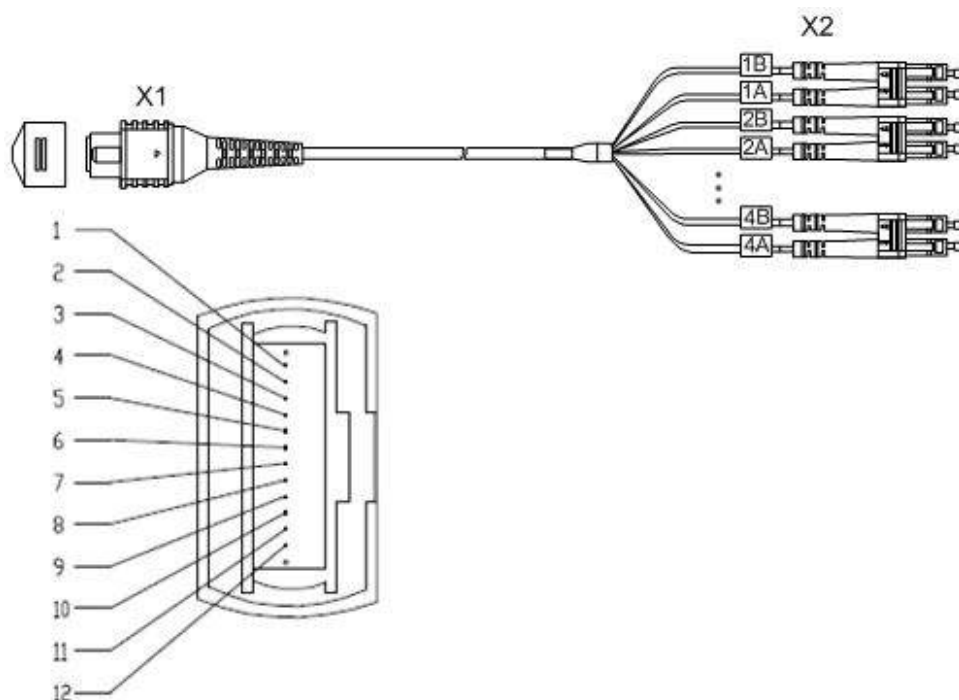
Fiber Structure

To insert a breakout optical module (multimode and short transmission distance) into a port, use an MPO-LC fiber to directly connect the optical module. [Figure 1-274](#) shows the appearance of an MPO-4*DLC breakout fiber.

Figure 1-274 MPO-4*DLC fiber appearance



Figure 1-275 MPO-4*DLC fiber structure



NOTE

When using a breakout optical fiber, insert the X1 end into a 40GE QSFP+ optical module and the X2 end into four 10GE interfaces. If the 40GE interface is Down but the four 10GE interfaces are Up, check the 40GE interface status.

- Check whether the 40GE interface supports the breakout function. If the 40GE interface does not support the breakout function, replace it with another interface that supports the breakout function.
- Check whether the 40GE interface is in the breakout state. If the 40GE interface is not in the breakout state, configure the 40GE interface to be in the breakout state.

Table 1-1286 Pin assignments of an MPO-4*DLC breakout fiber

X1 Pin	X2 Pin
1	1B
2	2B
3	3B
4	4B
9	4A
10	3A
11	2A
12	1A

Technical Specifications

Table 1-1287 lists the breakout fibers.

Table 1-1287 Breakout fibers supported

Type	BOM Number	Official Name	Mode
Optical Cable Parts,MPO/APC,MPO/APC,Single mode,10m,8 cores,GJFH-8G.657A2,3.5mm,LSZH,43mm Short MPO,Bending insensitive	14134759	MPO12-MPO12-SM-10	Single mode
Optical Cable Assembly,MPO/APC,4*DLC/PC,Singlemode,GJFH 8G.657A2(LSZH),15m,3.5mm,8 Cores,0m/1m,2mm	14132538	MPO12-4DLC-SM-15	Single mode
Optical Cable Parts,MPO/PC,4DLC/PC,Multi-mode,5m,8 cores,0/1m,GJFH-8A1a.2(OM3),3.5mm,2mm,LSZH,43mm Short MPO,Bending insensitive	14132537	SS-OP-MPO12-4*DLC-M-5	Multimode
Optical Cable Parts,MPO/PC,4DLC/PC,Multi-mode,15m,8 cores,0/1m,GJFH-8A1a.2(OM3),3.5mm,2mm,LSZH,43mm Short MPO,Bending insensitive	14132537-005	SS-OP-MPO12-4*DLC-M-15B	Multimode

Type	BOM Number	Official Name	Mode
Optical Cable Parts,MPO/PC,4DLC/PC,MULTI-MODE,30m,8 CORES,0/1m,GJFH-8A1A.2(OM3),3.5MM,2mm,LSZH,43MM SHORT MPO,BENDING INSENSITIVE	14132537-006	SS-OP-MPO12-4*DLC-M-30	Multimode
Optical Cable Parts,MPO/PC,4DLC/PC,Multi-mode,100m,8 cores,0/1m,GJFH-8A1a.2(OM3),3.5mm,2mm,LSZH,43mm Short MPO,Bending insensitive	14132537-001	SS-OP-MPO12-4*DLC-M-100	Multimode
Optical Cable Parts,MPO/APC,8FC/UPC,SINGLE MODE,15m,8 CORES,0/1m,GJFH-8G.657A2,3.5mm,2mm,LSZH,43MM SHORT MPO,BENDING INSENSITIVE	14136626	MPO12-4DFC-SM-15	Single mode
Optical Cable Parts,MPO/APC,8SC/UPC,SINGLE MODE,15m,8 CORES,0/1m,GJFH-8G.657A2,3.5mm,2mm,LSZH,43MM SHORT MPO,BENDING INSENSITIVE	14136627	MPO12-4DSC-SM-15	Single mode

Table 1-1288 lists the optical modules and ports mapping breakout fibers.

Table 1-1288 Optical modules and boards mapping breakout fibers

Breakout Fiber	Optical Module	Board
[14134759/MPO12-MPO12-SM-10] Optical Cable Parts,MPO/APC,MPO/APC,Single mode,10m,8 cores,GJFH-8G.657A2,3.5mm,LSZH,43mm Short MPO,Bending insensitive	[02311NUA/OSM010N11] High Speed Transceiver,QSFP+,1310,41.25Gbps,-8.2dBm,0.5dBm,-12.6dBm,MPO,S M,10km	Check the <i>Hardware Description</i> . If the description of a port contains QSFP28 or QSFP+ and supports the breakout function, the optical module and breakout fiber can be used together with the port.
[14132538/MPO12-4DLC-SM-15] Optical Cable Assembly,MPO/APC,4*DLC/PC,Singlemode,GJFH 8G.657A2(LSZH),15m,3.5mm,8 Cores,0m/1m,2mm		

Breakout Fiber	Optical Module	Board
[14136626/ MPO12-4DFC-SM-15] Optical Cable Parts,MPO/APC,8FC/ UPC,SINGLE MODE,15m, 8 CORES,0/1m,GJFH-8G. 657A2,3.5mm, 2mm,LSZH,43MM SHORT MPO,BENDING INSENSITIVE		
[14136627/ MPO12-4DSC-SM-15] Optical Cable Parts,MPO/APC,8SC/ UPC,SINGLE MODE,15m, 8 CORES,0/1m,GJFH-8G. 657A2,3.5mm, 2mm,LSZH,43MM SHORT MPO,BENDING INSENSITIVE		
[14132537/SS-OP- MPO12-4*DLC-M-5] Optical Cable Parts,MPO/PC,4DLC/ PC,Multi-mode,5m,8 cores,0/1m,GJFH-8A1a. 2(OM3),3.5mm, 2mm,LSZH,43mm Short MPO,Bending insensitive	[02310WUU/ OMXD30010] Optical Transceiver,QSFP+, 850nm, 41.25Gbps,-7.6dBm,-1dB m,-9.5dBm,MPO,MMF, 0.15km	
[14132537-005/SS-OP- MPO12-4*DLC-M-15B] Optical Cable Parts,MPO/PC,4DLC/ PC,Multi-mode,15m,8 cores,0/1m,GJFH-8A1a. 2(OM3),3.5mm, 2mm,LSZH,43mm Short MPO,Bending insensitive		
[14132537-006/SS-OP- MPO12-4*DLC-M-30] Optical Cable Parts,MPO/PC,4DLC/ PC,MULTI-MODE,30m,8 CORES,0/1m,GJFH-8A1A. 2(OM3),3.5MM, 2mm,LSZH,43MM SHORT MPO,BENDING INSENSITIVE		

Breakout Fiber	Optical Module	Board
[14132537-001/SS-OP-MPO12-4*DLC-M-100] Optical Cable Parts,MPO/PC,4DLC/ PC,Multi-mode,100m,8 cores,0/1m,GJFH-8A1a. 2(OM3),3.5mm, 2mm,LSZH,43mm Short MPO,Bending insensitive		
[14132537/SS-OP-MPO12-4*DLC-M-5] Optical Cable Parts,MPO/PC,4DLC/ PC,Multi-mode,5m,8 cores,0/1m,GJFH-8A1a. 2(OM3),3.5mm, 2mm,LSZH,43mm Short MPO,Bending insensitive	[02311NTY/ OMND10N13] High Speed Transceiver,QSFP28,850n m,100G,-8.4dBm, 2.4dBm,-10.3dBm,MPO, MM,0.1km,OM4	
[14132537-005/SS-OP-MPO12-4*DLC-M-15B] Optical Cable Parts,MPO/PC,4DLC/ PC,Multi-mode,15m,8 cores,0/1m,GJFH-8A1a. 2(OM3),3.5mm, 2mm,LSZH,43mm Short MPO,Bending insensitive		
[14132537-006/SS-OP-MPO12-4*DLC-M-30] Optical Cable Parts,MPO/PC,4DLC/ PC,MULTI-MODE,30m,8 CORES,0/1m,GJFH-8A1A. 2(OM3),3.5MM, 2mm,LSZH,43MM SHORT MPO,BENDING INSENSITIVE		

 NOTE

Characters in the brackets indicate the BOM number and official name that are separated using a slash ("/") in the [BOM number/Official name] format.

1.8.2 Breakout Boxes

1.8.2.1 Product Overview

Positioning

Breakout boxes are used to flexibly allocate bandwidth resources to routers. They cooperate with a breakout optical module and optical jumper to convert an MPO interface into multiple LC interfaces to facilitate fiber layout.

Product Features

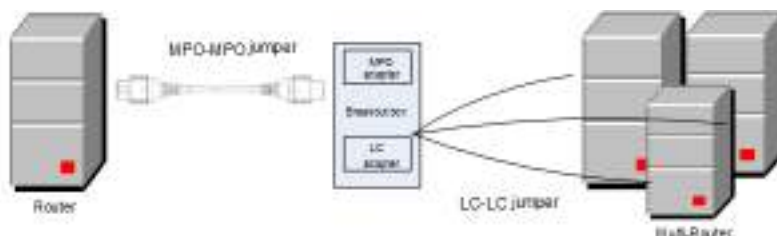
- LC interfaces support automatic dust-proof covers, and MPO interfaces support dust-proof plugs.
- In comparison with breakout fibers (MPO-LC fibers), breakout boxes feature more flexible fiber layout.
- There is no restriction on the installation direction of breakout boxes, so that the breakout boxes can be installed positively or negatively as needed.

Usage Scenario

The breakout function applies to a scenario where bandwidth resources are unevenly allocated between two routers at different levels. In comparison with breakout fibers, breakout boxes implement the following functions:

- Increase the density of fiber layout.
- Shorten the cabling distance.

Figure 1-276 Breakout box cabling



1.8.2.2 10-Port-MPO-12-40-Port-LC-Breakout Box(single-mode)

Appearance

Figure 1-277 10-Port-MPO-12-40-Port-LC-Breakout Box(single-mode)



Components

Figure 1-278 10-Port-MPO-12-40-Port-LC-Breakout Box(single-mode)



1. MPO adapter	2. LC adapter	3. Mounting ear
----------------	---------------	-----------------

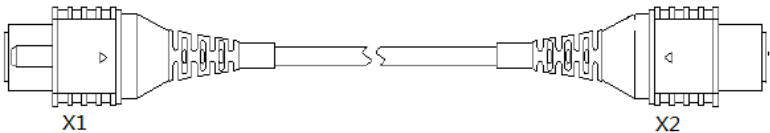
MPO adapters map LC adapters according to their numbers displayed on the panel. [Figure 1-279](#) shows the mapping between MPO and LC adapters.

Figure 1-279 Mapping between MPO and LC adapters



MPO-12 Jumper

The TYPE-B MPO jumper is used. The following figure shows the MPO-12 jumper structure and line orders of X1 and X2 ports.



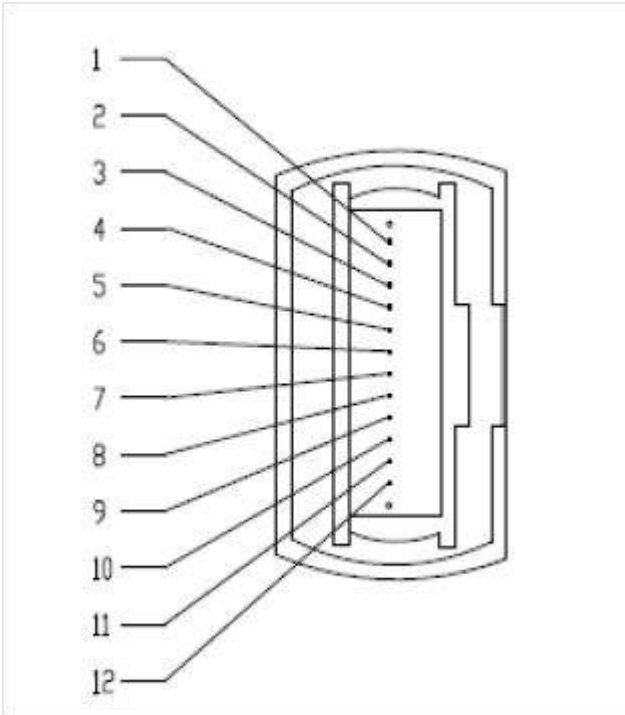


Table 1-1289 describes the mappings between the lanes of the X1 and X2 ports.

Table 1-1289 Mappings between the lanes of the X1 and X2 ports

Start	End	Start	End
X1-1	X2-12	X1-7	/
X1-2	X2-11	X1-8	/
X1-3	X2-10	X1-9	X2-4
X1-4	X2-9	X1-10	X2-3
X1-5	/	X1-11	X2-2
X1-6	/	X1-12	X2-1

Technical Specifications

Table 1-1290 Technical specifications of the 10-Port-MPO-12-40-Port-LC-Breakout Box (single-mode)

Item	Description
BOM Number	02082890
Official Name	ODBS10040
Dimensions (H x W x D)	43.6 mm x 442 mm x 164 mm

Item	Description
Mounting ear	19-inch supported
Weight	2.3 kg
Insertion loss	≤ 0.75 db
Return loss	Single-mode: ≥ 30 db
Operating temperature	-40°C to +65°C
Storage temperature	-40°C to +70°C
Relative operating humidity	- Long term: 5% RH to 85% RH, non-condensing - Short term: 5% RH to 95% RH, non-condensing
Relative storage humidity	0% RH to 95% RH, non-condensing

Table 1-1291 Optical modules, jumpers, and ports supported by the 10-Port-MPO-12-40-Port-LC-Breakout Box (single-mode)

Optical Module	MPO-MPO Fiber	Port
[02311NUA/OSM010N11] Function Module,OSM010N11,High Speed Transceiver,QSFP+,1310,41.25Gbps,-8.2dBm,0.5dBm,-12.6dBm,MPO,S M,10km	[14134759/MPO12-MPO12-SM-10] Optical Cable Parts,MPO/APC,MPO/APC,Single mode,10m,8 cores,GJFH-8G.657A2,3.5mm,LSZH,43mm Short MPO,Bending insensitive	Supported boards: Check the <i>Hardware Description</i> . If the description of a port contains QSFP28 or QSFP+ and supports the breakout function, the optical module and breakout fiber can be used together with the port.

1.9 Air deflectors

1.9.1 CR8M000ADFC1 (Air Deflector for Front-to-Back Airflow)

Overview

Table 1-1292 Basic information about the CR8M000ADFC1

Item	Description
Description	Air deflector (front-to-back airflow, for an IEC 19-inch cabinet with a depth of 600 mm or more)
Part number	21042341
Model	CR8M000ADFC1
Remarks	NOTE A group of air deflectors are used to convert the airflow direction of 300 mm deep fixed routers from left-to-right to front-to-back.

Appearance

A group of air deflectors contain an upper air deflector and a lower air deflector, as shown in [Figure 1-280](#).

Figure 1-280 CR8M000ADFC1 appearance

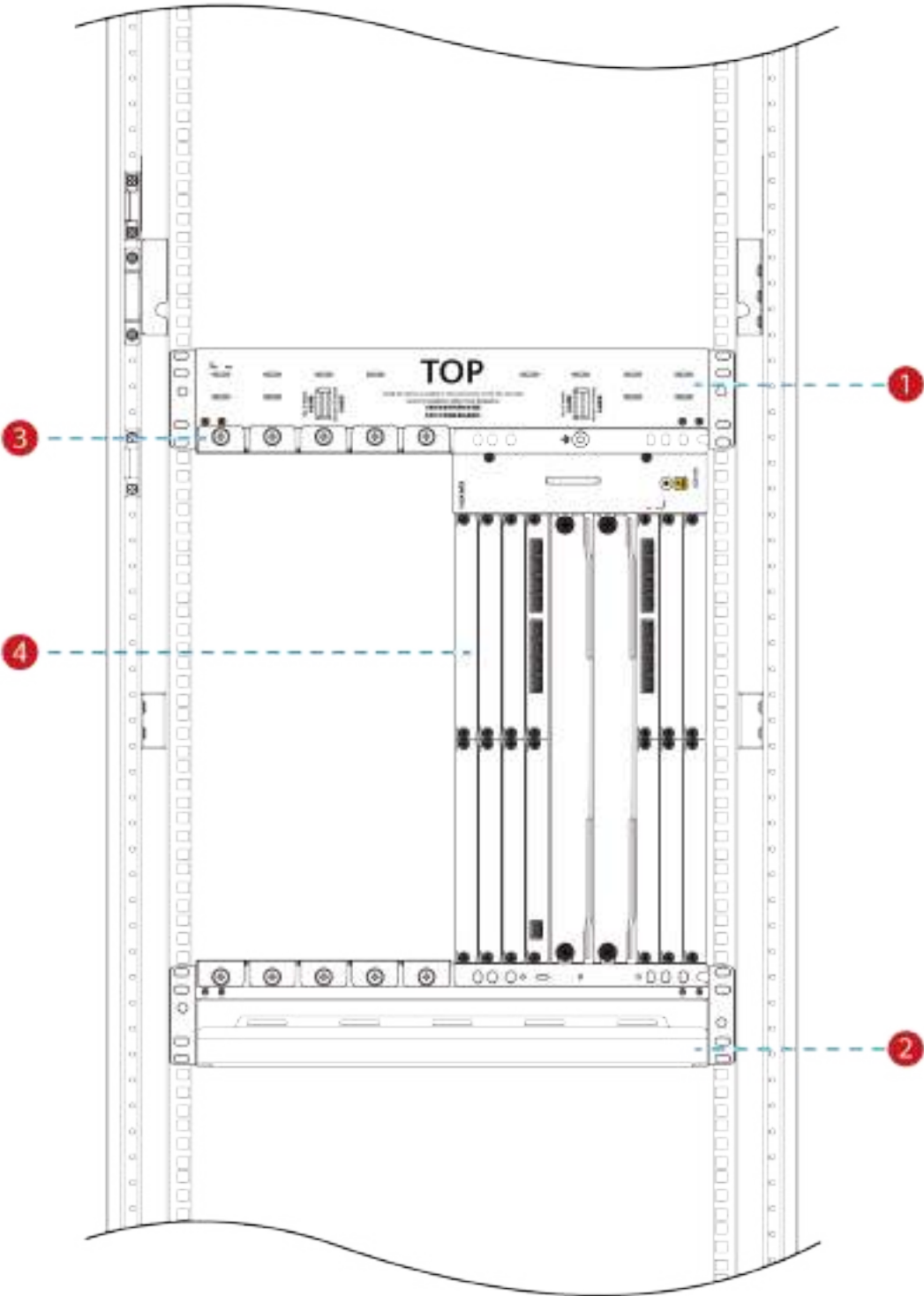


Table 1-1293 Cabinet with air deflectors installed

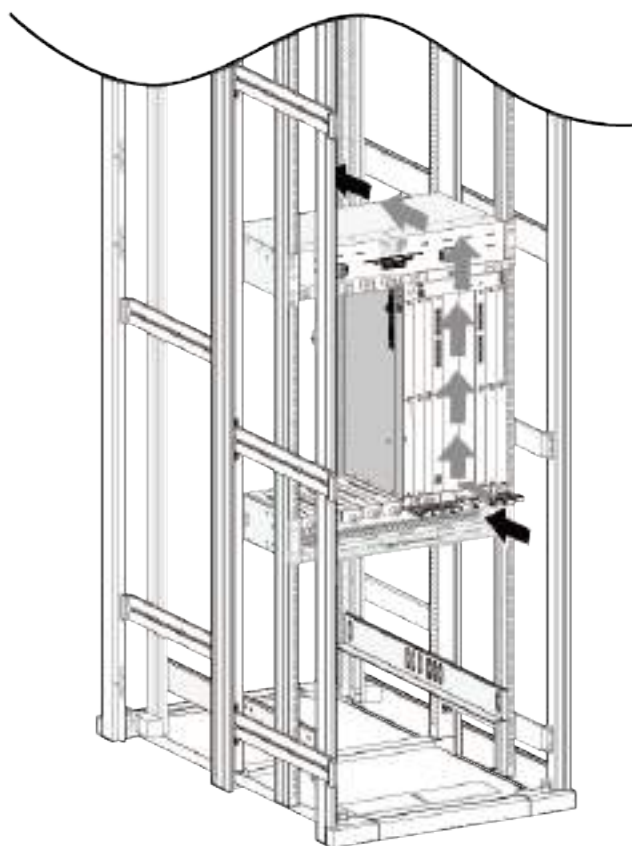
No.	Name
1	Upper air deflector

No.	Name
2	Lower air deflector
3	Air baffle
4	Device

Airflow

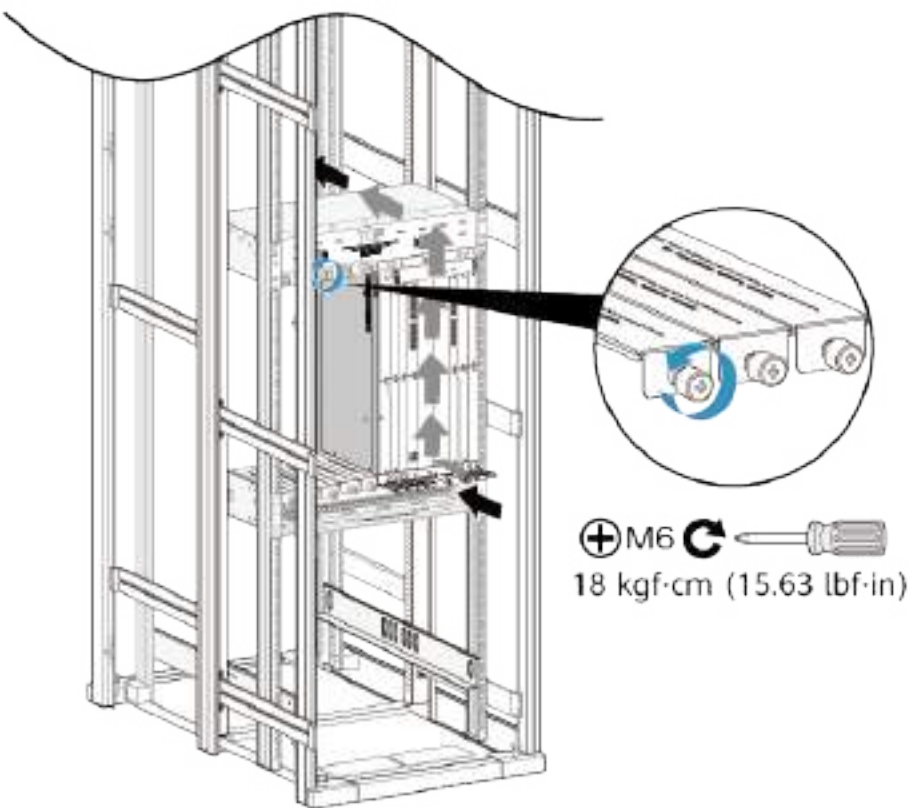
Figure 1-281 shows the airflow direction after devices and air reflectors are properly installed.

Figure 1-281 Airflow direction



By default, air deflectors are equipped with air baffles. Before installing a device, remove air baffles in the corresponding position, as shown in **Figure 1-282**.

Figure 1-282 Removing an air baffle



Technical Specifications

Table 1-1294 Technical specifications of the CR8M000ADFC1

Item	Specification
Dimensions of air deflectors installed in a cabinet (H x W x D)	622.3 mm x 443 mm x 306.5 mm, 14 U
Dimensions of uninstalled air deflectors (H x W x D, excluding packaging)	Upper air deflector: 90.6 mm (including air baffles) x 443 mm x 221 mm Lower air deflector: 92.4 mm (including air baffles) x 443 mm x 306.5 mm (including the cable tray)
Air baffle dimensions (H x W x D)	24 mm x 43.5 mm x 226.5 mm (including captive screws)
Weight (excluding packaging)	Upper air deflector: 4.0 kg Lower air deflector: 4.0 kg

Item	Specification
Vertical available space between upper and lower air deflectors	10 U (no clearance is required between devices)
Remarks	- By default, air deflectors are equipped with air baffles. Each air baffle is 1 U wide. To prevent air leakage, do not remove air baffles in the position where no device is installed. - Devices are vertically installed between upper and lower air deflectors. Considering this, pay attention to the insertion direction of modules when replacing or inserting modules in a device.

 **CAUTION**

You are not advised to install devices from other vendors between Huawei's upper and lower air deflectors. This is because the airflow direction and mounting bracket installation positions of these devices may be incompatible with the air deflectors.

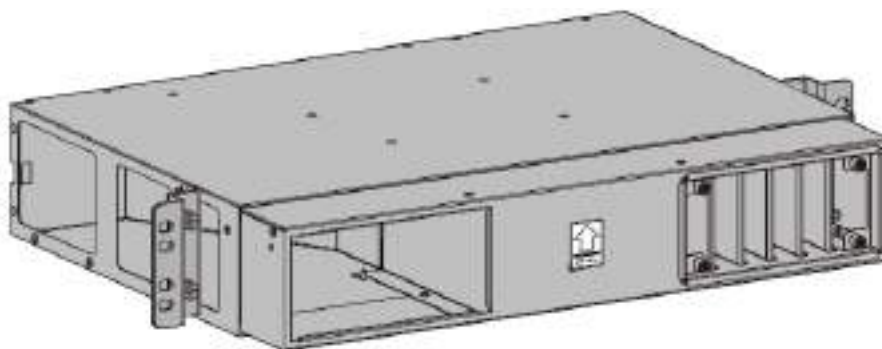
1.9.2 1800 Air Guide Frame (Simple Box-shaped Air Deflector)

Overview

Table 1-1295 Basic information

Item	Description
Description	Simple box-shaped air deflector
Part number	21032341
Model	1800 Air Guide Frame

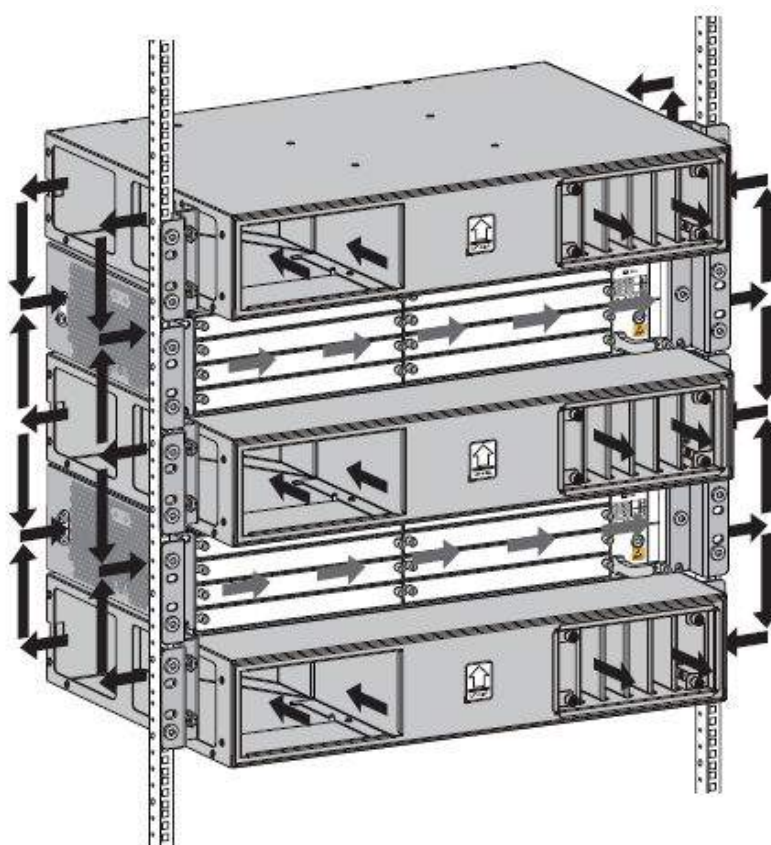
Appearance



Airflow

Figure 1-283 shows the airflow direction after devices and air reflectors are properly installed.

Figure 1-283 Airflow direction



Technical Specifications

Table 1-1296 Technical specifications

Item	Specification
Cabinet installation standards	ETSI 21-inch
Dimensions without packaging (H x W x D) [mm (in.)]	84.1 mm x 442 mm x 335 mm (3.31 in. x 17.40 in. x 13.19 in.)
Weight (without packaging) [kg (lb)]	3.9 kg (8.60 lb)
Air deflector height [U]	2 U